

Krishna's

TEXT BOOK on

ABSTRACT ALGEBRA

(For B.A. and B.Sc. IIIrd year students of All Colleges affiliated to Lucknow University, Lucknow)

As per Latest Syllabus of Lucknow University

(w.e.f. 2014-2015)

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Jai Shri Radhey Shyam

Dedicated
to
Lord
Krishna

Authors & Publishers

Preface

This book on **ABSTRACT ALGEBRA** has been specially written according to the **Latest Unified Syllabus** to meet the requirements of the **B.A. and B.Sc. Part-III Students** of all colleges affiliated to **Lucknow University, Lucknow**.

The subject matter has been discussed in such a simple way that the students will find no difficulty to understand it. The proofs of various theorems and examples have been given with minute details. Each chapter of this book contains complete theory and a fairly large number of solved examples. Sufficient problems have also been selected from various university examination papers. At the end of each chapter an exercise containing objective questions has been given.

We have tried our best to keep the book free from misprints. The authors shall be grateful to the readers who point out errors and omissions which, inspite of all care, might have been there.

The authors, in general, hope that the present book will be warmly received by the students and teachers. We shall **indeed** be very thankful to our colleagues for their recommending this book to their students.

The authors wish to express their thanks to **Mr. S.K. Rastogi, M.D., Mr. Sugam Rastogi, Executive Director, Mrs. Kanupriya Rastogi, Director** and **entire team of KRISHNA Prakashan Media (P) Ltd., Meerut** for bringing out this book in the present nice form.

The authors will feel amply rewarded if the book serves the purpose for which it is meant. Suggestions for the improvement of the book are always welcome.

July, 2014

—*Authors*

Syllabus

ABSTRACT ALGEBRA

Lucknow University, Lucknow
(w.e.f. 2014-15)

B.A./B.Sc. Paper-II

Unit-I

Automorphism, inner automorphism, automorphism groups and their computations. Conjugacy relations, Normaliser, Counting principle and the class equation of a finite group, Center of group of prime power order, Sylow's theorems, Sylow p-subgroup.

Unit-II

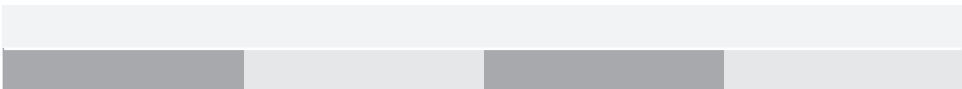
Prime and maximal ideals, Euclidean Rings, Principal ideal rings, Polynomial Rings, Polynomial over the Rational Field, The Eisenstein Criterion, Polynomial Rings over Commutative Rings, unique factorization domain, R is unique factorization domain implies so is $R[x_1, x_2, \dots, x_n]$.

Unit-III

Direct sum, Quotient space, Linear transformations and their representation as matrices, The Algebra of linear transformations, rank nullity theorem, change of basis, linear functional, Dual space, Bidual space and natural isomorphism, transpose of a linear transformation, Characteristic values, annihilating polynomials, diagonalisation, Cayley Hamilton Theorem, Invariant subspaces, Primary decomposition theorem.

Unit-IV

Inner product spaces, Cauchy-Schwarz inequality, orthogonal vectors, Orthogonal complements, Orthonormal sets and bases, Bessel's inequality for finite dimensional spaces, Gram-Schmidt orthogonalization process, Bilinear, Quadratic and Hermitian forms.



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ABSTRACT ALGEBRA

Chapters



1. Group Automorphisms

2. Rings

3. Linear Transformations

4. Matrices and Linear Transformations

5. Linear Functionals



6. Characteristic Values and
Annihilating Polynomials

7. Inner Product Spaces

8. Bilinear, Quadratic and
Hermitian Forms

Chapter

1



Group Automorphisms

1.1 Automorphisms of a Group

Definition: An isomorphic mapping of a group G onto itself is called an automorphism of G .

Thus $f : G \xrightarrow[\text{one-one}]{\text{onto}} G$ is an automorphism of G if

$$f(ab) = f(a)f(b) \quad \forall \quad a, b \in G.$$

If G is any group, then the identity mapping $I : G \rightarrow G$ such that $I(x) = x$, $\forall \quad x \in G$ is an automorphism of G . Obviously, I is one-one onto and

$$I(xy) = xy = I(x)I(y), \quad \forall \quad x, y \in G.$$

The identity mapping I of G is called a *trivial automorphism* of G .

Illustrative Examples

Example 1: Show that the mapping $f : \mathbf{I} \rightarrow \mathbf{I}$ such that $f(x) = -x \quad \forall \quad x \in \mathbf{I}$ is an automorphism of the additive group of integers $\mathbf{I}$.

Solution: Obviously the mapping f is one-one onto.

Let x_1, x_2 be any two elements of $\mathbf{I}$. Then

$$f(x_1 + x_2) = -(x_1 + x_2) = (-x_1) + (-x_2) = f(x_1) + f(x_2).$$

Hence f is an automorphism of $\mathbf{I}$.

Example 2: Show that $a \rightarrow a^{-1}$ is an automorphism of a group G iff G is abelian.

Solution: Let $f : G \rightarrow G$ be such that $f(x) = x^{-1} \forall x \in G$.

The function f is one-one because

$$f(x) = f(y) \Rightarrow x^{-1} = y^{-1} \Rightarrow (x^{-1})^{-1} = (y^{-1})^{-1} \Rightarrow x = y.$$

Also if $x \in G$, then $x^{-1} \in G$ and we have $f(x^{-1}) = (x^{-1})^{-1} = x$.

$\therefore f$ is onto.

Now suppose G is abelian. Let a, b be any two elements of G . Then

$$\begin{aligned} f(ab) &= (ab)^{-1} && [\text{by def. of } f] \\ &= b^{-1} a^{-1} = a^{-1} b^{-1} && [\because G \text{ is abelian}] \\ &= f(a) f(b) && [\text{by def. of } f] \end{aligned}$$

$\therefore f$ is an automorphism of G .

Conversely suppose that f is an automorphism of G . Let $a, b \in G$.

$$\begin{aligned} \text{We have } f(ab) &= (ab)^{-1} && [\text{by def. of } f] \\ &= b^{-1} a^{-1} = f(b) f(a) && [\text{by def. of } f] \\ &= f(ba). && [\because f \text{ is an automorphism}] \end{aligned}$$

Since f is one-one, therefore $f(ab) = f(ba) \Rightarrow ab = ba \Rightarrow G$ is abelian.

Example 3: Let G be a non-abelian group. Show that the mapping $f : G \rightarrow G$ such that $f(x) = x^{-1}, \forall x \in G$ is not an automorphism.

Solution: Let G be a non-abelian group.

Then to prove that the mapping $f : G \rightarrow G$ such that $f(x) = x^{-1}, \forall x \in G$ is not an automorphism of G .

Suppose the mapping f is an automorphism of G .

Let x, y be any two elements of G . Then

$$f(xy) = (xy)^{-1}. \quad [\text{By def. of the mapping } f]$$

$$\begin{aligned} \text{Also } f(xy) &= f(x) f(y) && [\because f \text{ is an automorphism}] \\ &= x^{-1} y^{-1} && [\text{By def. of the mapping } f] \\ &= (yx)^{-1}. \end{aligned}$$

$$\begin{aligned} \text{Thus, } f(xy) &= (xy)^{-1} \text{ and } f(yx) = (yx)^{-1} \\ \Rightarrow (xy)^{-1} &= (yx)^{-1} \Rightarrow [(xy)^{-1}]^{-1} = [(yx)^{-1}]^{-1} \Rightarrow xy = yx. \end{aligned}$$

Thus $xy = yx, \forall x, y \in G$ i.e., G is abelian which is a contradiction.

Hence, the mapping f is not an automorphism of G .

Example 4: Let G be a finite abelian group of order n , where n is odd and > 1 . Then show that G has a non-trivial automorphism.

Solution: Define a mapping $f : G \rightarrow G$ such that $f(x) = x^{-1}$, $\forall x \in G$.

Then f is an automorphism of G .

[See Ex. 3]

We shall show that the mapping f is a non-trivial automorphism of G i.e., $f \neq I$, where I is the identity mapping of G i.e., $I : G \rightarrow G$ such that $I(x) = x$, $\forall x \in G$.

Suppose $f = I$.

Then $f(x) = I(x)$, $\forall x \in G$

$$\Rightarrow x^{-1} = x, \forall x \in G \Rightarrow x^{-1}x = x^2, \forall x \in G$$

$$\Rightarrow x^2 = e, \forall x \in G, \text{ where } e \text{ is the identity of } G$$

$$\Rightarrow o(x) \mid 2, \forall x \in G \Rightarrow o(x) = 1 \text{ or } 2, \forall x \in G.$$

Since $o(G) > 1$, therefore G must have an element x such that $x \neq e$.

$$\text{Then } x \neq e \Rightarrow o(x) = 2.$$

But in a group G , $o(x) \mid o(G)$, $\forall x \in G$.

$$\therefore o(x) = 2 \Rightarrow 2 \mid o(G) \Rightarrow o(G) \text{ is even,}$$

which is a contradiction.

$\therefore$ our assumption that $f = I$ is wrong.

Hence, $f \neq I$ i.e., f is a non-trivial automorphism of G .

Example 5: Let G be a group, H a subgroup of G , f an automorphism of G . Let

$$f(H) = \{f(h) : h \in H\}. \quad (\text{Lucknow 2007})$$

Prove that $f(H)$ is a subgroup of G .

Solution: Let a, b be any two elements of $f(H)$. Then

$$a = f(h_1) \text{ and } b = f(h_2) \text{ where } h_1, h_2 \in H.$$

$$\text{Now } h_1, h_2 \in H \Rightarrow h_1 h_2^{-1} \in H \quad [\because H \text{ is a subgroup}]$$

$$\Rightarrow f(h_1 h_2^{-1}) \in f(H)$$

$$\Rightarrow f(h_1) f(h_2^{-1}) \in f(H) \quad [\because f \text{ is an automorphism}]$$

$$\Rightarrow f(h_1) [f(h_2)]^{-1} \in f(H) \Rightarrow ab^{-1} \in f(H).$$

$\therefore f(H)$ is a subgroup of G .

Note: Some authors use the symbol hf in place of $f(h)$ to denote the image of an element.

Example 6: Let G be a group, f an automorphism of G , N a normal subgroup of G . Prove that $f(N)$ is a normal subgroup of G . (Lucknow 2010)

Solution: First show as in Ex. 5 that $f(N)$ is a subgroup of G .

Now to show that $f(N)$ is a normal subgroup of G .

Let $x \in G$ and $k \in f(N)$. Then $x = f(y)$ where $y \in G$ because f is a function of G onto G . Also $k = f(n)$ where $n \in N$.

We have $xkx^{-1} = f(y) f(n) [f(y)]^{-1} = f(y) f(n) f(y^{-1}) = f(yny^{-1})$.

Since N is normal in G , therefore $yny^{-1} \in N$. Consequently

$$f(yny^{-1}) \in f(N).$$

Thus $xkx^{-1} \in f(N)$.

$\therefore f(N)$ is a normal subgroup of G .

1.2 Group of Automorphisms of a Group

Theorem 1: *The set of all automorphisms of a group forms a group with respect to composite of functions as the composition.* (Lucknow 2010)

Proof: Let $A(G)$ be the collection of all automorphisms of a group G . Then

$$A(G) = \{ f : f \text{ is an automorphism of } G \}.$$

We shall prove that $A(G)$ is a group with respect to composite of functions as composition.

Closure Property: Let $f, g \in A(G)$. Then f, g are one-one mappings of G onto itself. Therefore gf is also a one-one mapping of G onto itself. If a, b be any two elements of G , we have

$$\begin{aligned} (gf)(ab) &= g[f(ab)] = g[f(a)f(b)] \\ &= g[f(a)]g[f(b)] = [(gf)(a)][(gf)(b)]. \end{aligned}$$

$\therefore gf$ is also an automorphism of G . Thus $A(G)$ is closed with respect to composite composition.

Associativity: We know that composite of arbitrary mappings is associative. Therefore composite of automorphisms is also associative.

Existence of Identity: The identity function i on G is also an automorphism of G . Obviously i is one-one onto and if $a, b \in G$, then $i(ab) = ab = i(a)i(b)$. Thus $i \in A(G)$ and if $f \in A(G)$, we have $if = f = fi$.

Existence of Inverse: Let $f \in A(G)$. Since f is a one-one mapping of G onto itself, therefore f^{-1} exists and is also a one-one mapping of G onto itself. We shall show that f^{-1} is also an automorphism of G . Let $a, b \in G$. Then there exist $a', b' \in G$ such that

$$\begin{aligned} f^{-1}(a) &= a' \Leftrightarrow f(a') = a \\ f^{-1}(b) &= b' \Leftrightarrow f(b') = b. \end{aligned}$$

We have $f^{-1}(ab) = f^{-1}[f(a')f(b')]$

$$= f^{-1}[f(a'b')] = a'b' = f^{-1}(a)f^{-1}(b).$$

$\therefore f^{-1}$ is an automorphism of G and thus

$$f \in A(G) \Rightarrow f^{-1} \in A(G).$$

Therefore each element of $A(G)$ possesses inverse.

Therefore $A(G)$ is a group with respect to composite composition.

Theorem 2: Let G be a group. Let $A(G)$ denote the set of all automorphisms of G and $P(G)$ be the group of all permutations of G . Then $A(G)$ is a subgroup of $P(G)$.

Proof: We have

$P(G) = \{f : f \text{ is a permutation of } G \text{ i.e., } f : G \rightarrow G \text{ is one-one onto}\}.$

We know that $P(G)$ is a group for the composite of two permutations as the binary operation.

Also, $A(G) = \{f : f \text{ is an automorphism of } G\}.$

The mapping $f : G \rightarrow G$ is said to be an automorphism of G if f is one-one onto and f is a homomorphism i.e., $f(ab) = f(a)f(b)$, $\forall a, b \in G$.

Obviously, $f \in A(G) \Rightarrow f \in P(G)$.

Thus $A(G) \subseteq P(G)$.

Also, $A(G) \neq \emptyset$, because $I \in A(G)$, where I is the identity mapping of the group G .

In order to show that $A(G)$ is a subgroup of $P(G)$ it is sufficient to show that

$$f \in A(G), g \in A(G) \Rightarrow fg^{-1} \in A(G),$$

where g^{-1} is the inverse of g in the group $P(G)$.

First we shall show that $g \in A(G) \Rightarrow g^{-1} \in A(G)$.

Now $g \in A(G) \Rightarrow g$ is an automorphism of G

$\Rightarrow g$ is one-one onto and $g(ab) = g(a)g(b)$, $\forall a, b \in G$.

Since the mapping $g : G \rightarrow G$ is one-one onto, therefore its inverse mapping $g^{-1} : G \rightarrow G$ exists and is also one-one onto.

Let a, b be any two elements of G . Then there exist elements $a', b' \in G$ such that

$$g^{-1}(a) = a', g^{-1}(b) = b' \quad \dots(1)$$

$$\text{and } g(a') = a, g(b') = b. \quad \dots(2)$$

We have $g^{-1}(ab) = g^{-1}[g(a')g(b')]$ [From (2)]

$$= g^{-1}[g(a'b')] \quad [\because g \text{ is a homomorphism}]$$

$$= a'b' \quad [\text{By def. of } g^{-1}]$$

$$= g^{-1}(a)g^{-1}(b) \quad [\text{From (1)}]$$

$\therefore$ the mapping $g^{-1} : G \rightarrow G$ is a homomorphism of G .

Since $g^{-1} : G \rightarrow G$ is one-one onto and is a homomorphism of G , therefore g^{-1} is an automorphism of G i.e., $g^{-1} \in A(G)$.

Thus $g \in A(G) \Rightarrow g^{-1} \in A(G)$.

Now we shall show that

$$f \in A(G), g \in A(G) \Rightarrow f g^{-1} \in A(G),$$

where $f g^{-1}$ is the composite mapping of the mappings f and g^{-1} .

Since the mappings $f : G \rightarrow G$ and $g^{-1} : G \rightarrow G$ are one-one onto, therefore their composite mapping $f g^{-1} : G \rightarrow G$ is also one-one onto.

Now we shall show that the mapping $f g^{-1} : G \rightarrow G$ is a homomorphism.

Let a, b be any two elements of G .

$$\begin{aligned} \text{Then } f g^{-1}(ab) &= f [g^{-1}(ab)] \quad [\text{By def. of the composite of two mappings}] \\ &= f [g^{-1}(a) g^{-1}(b)] \quad [\because g^{-1} \text{ is a homomorphism of } G] \\ &= f [g^{-1}(a)] f [g^{-1}(b)] \quad [\because f \text{ is a homomorphism}] \\ &= f g^{-1}(a) f g^{-1}(b). \\ &\quad [\text{By def. of the composite of two mappings}] \end{aligned}$$

$\therefore$ the mapping $f g^{-1} : G \rightarrow G$ is a homomorphism of G .

$\therefore f g^{-1}$ is an automorphism of G i.e., $f g^{-1} \in A(G)$.

$$\text{Thus } f \in A(G), g \in A(G) \Rightarrow f g^{-1} \in A(G).$$

Hence, $A(G)$ is a subgroup of $P(G)$.

Theorem 3: Let G be a group and f an automorphism of G . If $a \in G$ is of order greater than zero, then prove that $o[f(a)] = o(a)$.

Proof: Suppose the mapping f is an automorphism of a group G . Let a be any element of G such that $o(a) = n$, where $n > 0$.

$$\text{Let } o[f(a)] = m.$$

Let e be the identity of G . Then $f(e) = e$.

$$\text{Now } o(a) = n \Rightarrow a^n = e \Rightarrow f(a^n) = f(e)$$

$$\Rightarrow f(aaa \dots n \text{ times}) = e \Rightarrow f(a) f(a) f(a) \dots n \text{ times} = e$$

$$\Rightarrow [f(a)]^n = e \Rightarrow o[f(a)] \leq n \Rightarrow m \leq n.$$

$$\text{Again } o[f(a)] = m \Rightarrow [f(a)]^m = e$$

$$\Rightarrow f(a) f(a) f(a) \dots m \text{ times} = e \Rightarrow f(aaa \dots m \text{ times}) = e$$

$$\Rightarrow f(a^m) = f(e) \quad [\because f(e) = e]$$

$$\Rightarrow a^m = e \quad [\because f \text{ is one-one}]$$

$$\Rightarrow o(a) \leq m \Rightarrow n \leq m.$$

$$\text{Thus, } m \leq n \text{ and } n \leq m \Rightarrow m = n.$$

$$\text{Hence, } o[f(a)] = o(a).$$

1.3 Inner Automorphisms

We shall now study a special type of automorphisms known as **inner automorphisms**. First we shall prove a preliminary theorem.

Theorem 1: Let a be a fixed element of a group G . Then the mapping $f_a : G \rightarrow G$ defined by $f_a(x) = a^{-1}xa \ \forall x \in G$ is an automorphism of G .

Proof: The mapping f_a is one-one: Let x, y be any two elements of G . Then

$$f_a(x) = f_a(y) \Rightarrow a^{-1}xa = a^{-1}ya \Rightarrow x = y,$$

by cancellation laws in G .

Therefore the mapping f_a is one-one.

The mapping f_a is also onto G : If y is any element of G , then $aya^{-1} \in G$ and we have $f_a(aya^{-1}) = a^{-1}(aya^{-1})a = y$.

$\therefore f_a$ is onto G .

Finally if $x, y \in G$ then

$$f_a(xy) = a^{-1}(xy)a = (a^{-1}xa)(a^{-1}ya) = f_a(x)f_a(y).$$

Hence f_a is an automorphism of G .

Inner Automorphism:

Definition: If G is a group, the mapping

$$f_a : G \rightarrow G \text{ defined by } f_a(x) = a^{-1}xa \ \forall x \in G$$

is an automorphism of G known as inner automorphism.

Also an automorphism which is not inner is called an **outer automorphism**.

Theorem 2: For an abelian group the only inner automorphism is the identity mapping whereas for non-abelian groups there exist non-trivial inner automorphisms.

Proof: Suppose G is an abelian group and f_a is an inner automorphism of G . If $x \in G$, we have

$$\begin{aligned} f_a(x) &= a^{-1}xa = a^{-1}ax & [\because G \text{ is abelian}] \\ &= ex = x. \end{aligned}$$

Thus $f_a(x) = x \ \forall x \in G$.

$\therefore f_a$ is the identity mapping of G .

Let now G be non-abelian. Then there exist at least two elements say $a, b \in G$ such that

$$ba \neq ab \Rightarrow a^{-1}ba \neq b \Rightarrow f_a(b) \neq b.$$

Hence f_a is not the identity mapping of G . Thus for non-abelian groups there always exist non-trivial inner automorphisms.

Theorem 3: The set $I(G)$ of all inner automorphisms of a group G is a normal subgroup of the group of its automorphisms isomorphic to the quotient group G/Z of G where Z is the centre of G .

Proof: Let $A(G)$ denote the group of all automorphisms of G . Then $I(G) \subseteq A(G)$.

Let $a, b \in G$. We shall first prove the following two results :

(i) $f_{a^{-1}} = f_a^{-1}$ i.e., the inner automorphism $f_{a^{-1}}$ is the inverse function of the inner automorphism f_a .

(ii) $f_a f_b = f_{ba}$.

Proof of (i): If $x \in G$, then we have

$$\begin{aligned} (f_a f_{a^{-1}})(x) &= f_a [f_{a^{-1}}(x)] = f_a [(a^{-1})^{-1} x a^{-1}] = f_a [a x a^{-1}] \\ &= a^{-1} (a x a^{-1}) a = x. \end{aligned}$$

$\therefore f_a f_{a^{-1}}$ is the identity function on G .

$\therefore f_{a^{-1}} = (f_a)^{-1}$.

Proof of (ii): If $x \in G$, then we have

$$\begin{aligned} (f_a f_b)(x) &= f_a [f_b(x)] = f_a (b^{-1} x b) = a^{-1} (b^{-1} x b) a = (a^{-1} b^{-1}) x (b a) \\ &= (b a)^{-1} x (b a) = f_{ba}(x). \end{aligned}$$

$\therefore f_a f_b = f_{ba}$.

Now we shall prove that $I(G)$ is a subgroup of $A(G)$. Let f_a, f_b be any two elements of $I(G)$. Then

$$f_a (f_b)^{-1} = f_a f_{b^{-1}} = f_{b^{-1} a} \in I(G), \text{ since } b^{-1} a \in G.$$

Thus $f_a, f_b \in I(G) \Rightarrow f_a (f_b)^{-1} \in I(G)$.

$\therefore I(G)$ is a subgroup of $A(G)$.

Now we shall prove that $I(G)$ is a normal subgroup of $A(G)$.

Let $f \in A(G)$ and $f_a \in I(G)$. If $x \in G$, then we have

$$\begin{aligned} (f f_a f^{-1})(x) &= (f f_a) [f^{-1}(x)] = f [f_a (f^{-1}(x))] \\ &= f [a^{-1} f^{-1}(x) a] \\ &= f(a^{-1}) f[f^{-1}(x)] f(a) && [\because f \text{ is composition preserving}] \\ &= f(a^{-1}) x f(a) && [\because f[f^{-1}(x)] = x] \\ &= [f(a)]^{-1} x f(a) \\ &= c^{-1} x c \text{ where } f(a) = c \in G \\ &= f_c(x). \end{aligned}$$

$\therefore f f_a f^{-1} = f_c \in I(G)$, since $c \in G$.

$\therefore I(G)$ is a normal subgroup of $A(G)$.

Now we shall show that $I(G)$ is isomorphic to G/Z . For this we shall show that $I(G)$ is a homomorphic image of G and Z is the kernel of the corresponding homomorphism.

Then by the fundamental theorem on homomorphism of groups we shall have

$$G/Z \cong I(G).$$

Consider the mapping $\phi : G \rightarrow I(G)$ defined by

$$\phi(a) = f_{a^{-1}} \quad \forall \quad a \in G.$$

Obviously ϕ is onto $I(G)$ because $f_a \in I(G) \Rightarrow a \in G$ and this implies $a^{-1} \in G$.

$$\text{Now } \phi(a^{-1}) = f_{(a^{-1})^{-1}} = f_a.$$

$\therefore \phi$ is onto $I(G)$.

Now to prove that

$$\phi(ab) = \phi(a) \phi(b) \quad \forall \quad a, b \in G.$$

$$\text{We have } \phi(ab) = f_{(ab)^{-1}} = f_{b^{-1}a^{-1}} = f_{a^{-1}} f_{b^{-1}} = \phi(a) \phi(b).$$

Now to show that Z is the kernel of ϕ .

The identity function i on G is the identity of the group $I(G)$.

Let K be the kernel of ϕ .

$$\text{Then we have } z \in K \Leftrightarrow \phi(z) = i \Leftrightarrow f_{z^{-1}} = i \Leftrightarrow f_{z^{-1}}(x) = i(x) \quad \forall \quad x \in G$$

$$\Leftrightarrow (z^{-1})^{-1} x z^{-1} = x \quad \forall \quad x \in G \Leftrightarrow z x z^{-1} = x \quad \forall \quad x \in G$$

$$\Leftrightarrow z x = x z \quad \forall \quad x \in G \Leftrightarrow z \in Z.$$

$$\therefore K = Z.$$

Hence the theorem.

1.4 Group of Automorphisms of a Cyclic Group

Suppose $G = \{a\}$ is a cyclic group generated by a . An automorphism f of G is completely defined by a relation of the form

$$f(a) = a^m, \quad \dots(1)$$

where m is some suitable integer.

For if k is any integer, then for f to be an automorphism of G , we have

$$f(a^k) = [f(a)]^k = (a^m)^k. \quad \dots(2)$$

The relation (2) gives the f -image of each element of G and the mapping f is thus completely defined.

Now let b be any element of G . Since f is a mapping of G onto itself, therefore there must exist an element $a^k \in G$ such that

$$b = f(a^k) = (a^m)^k. \quad \dots(3)$$

From (3) we conclude that for the mapping f defined in (1) to be an automorphism of G , a^m must be a generator of G .

Now if G is infinite, the only generators of G are a and a^{-1} . So in this case the only automorphisms of G are,

(i) the identity mapping I for which

$$I(a) = a \Rightarrow I(a^k) = a^k \quad \forall \quad k \in \mathbf{I}$$

and, (ii) the mapping f defined by $f(a) = a^{-1}$.

Therefore the group of automorphisms of an infinite cyclic group is of order 2.

On the other hand if G is of finite order n , then a^m is a generator of G if and only if m is prime to n and less than n . We shall show that for each such m , the mapping f defined in (1) is an automorphism of G .

f is one-one: Let a^{k_1} and a^{k_2} be any two elements of G where

$$1 \leq k_1 \leq n, 1 \leq k_2 \leq n \quad \text{and} \quad k_1 \geq k_2.$$

$$\text{Then} \quad f(a^{k_1}) = f(a^{k_2}) \Rightarrow (a^m)^{k_1} = (a^m)^{k_2} \Rightarrow a^{mk_1} = a^{mk_2}$$

$$\Rightarrow a^{m(k_1 - k_2)} = e \Rightarrow n \mid m(k_1 - k_2).$$

But m is prime to n and $0 \leq (k_1 - k_2) < n$. Therefore

$$n \mid m(k_1 - k_2) \Rightarrow k_1 - k_2 = 0 \Rightarrow k_1 = k_2 \Rightarrow a^{k_1} = a^{k_2}.$$

$$\text{Thus} \quad f(a^{k_1}) = f(a^{k_2}) \Rightarrow a^{k_1} = a^{k_2}.$$

Therefore the mapping f is one-one.

f is onto: Since G is finite and f is one-one, therefore f must be onto G .

Finally if a^{k_1}, a^{k_2} are any two elements of G , then

$$f(a^{k_1} a^{k_2}) = f(a^{k_1 + k_2})$$

$$= f(a^{nu+k}), \text{ where } u \text{ is some integer and } 0 \leq k < n$$

$$\left[\text{Note that we can write } \frac{k_1 + k_2}{n} = u + \frac{k}{n} \right]$$

$$= f(a^{nu} a^k) = f[(a^n)^u a^k] = f(a^k) \quad [\because a^n = e \Rightarrow (a^n)^u = e]$$

$$= a^{mk} = a^{m(k_1 + k_2 - nu)} \quad [\because k = k_1 + k_2 - nu]$$

$$= a^{m(k_1 + k_2)} a^{-mnu} = a^{m(k_1 + k_2)} \quad [\because a^{-mnu} = (a^n)^{-mu} = e^{-mu} = e]$$

$$= a^{mk_1} a^{mk_2} = (a^m)^{k_1} (a^m)^{k_2} = f(a^{k_1}) f(a^{k_2}).$$

Therefore the mapping f denoted in (1) is an automorphism of G for each positive integer m less than n and prime to n .

Hence the group of automorphisms of a finite cyclic group of order n is of order $\phi(n)$ where $\phi(n)$ denotes the number of integers less than n and prime to n .

In the end we shall show that the group of automorphisms of a cyclic group is abelian.

Let f_{m_1}, f_{m_2} be two automorphisms defined by

$$f_{m_1}(a) = a^{m_1}, \quad f_{m_2}(a) = a^{m_2}.$$

$$\text{Then} \quad (f_{m_1} \circ f_{m_2})(a) = f_{m_1}[f_{m_2}(a)] = f_{m_1}(a^{m_2})$$

$$= [f_{m_1}(a)]^{m_2} = (a^{m_1})^{m_2} = a^{m_1 m_2}$$

$$= (a^{m_2})^{m_1} = [f_{m_2}(a)]^{m_1} = f_{m_2}(a^{m_1})$$

$$= f_{m_2}[f_{m_1}(a)] = (f_{m_2} \circ f_{m_1})(a).$$

Now two automorphisms of a cyclic group are equal if the image of a generator of the group under each of them is the same.

Hence $f_{m_1} \circ f_{m_2} = f_{m_2} \circ f_{m_1}$.

Therefore the group of automorphisms of a cyclic group is abelian.

Illustrative Examples

Example 7: Let G be a finite abelian additive group and n be a positive integer relatively prime to $|G|$. Prove that the mapping $\sigma : G \rightarrow G$ given by $\sigma(x) = nx$ is an automorphism of G .

Solution: The mapping σ is one-one: Let x, y be any two elements of G . Then

$$\begin{aligned} \sigma(x) = \sigma(y) &\Rightarrow nx = ny \\ \Rightarrow n(x - y) = 0, &\text{ where } 0 \text{ is the identity of the group } G \\ \Rightarrow |x - y| \mid n \\ \Rightarrow |x - y| \mid n &\text{ and } |x - y| \mid |G| \quad [\because |x - y| \mid |G|] \\ \Rightarrow |x - y| = 1 &\quad [\because \text{if } |x - y| \neq 1, \text{ then } |x - y| > 1 \Rightarrow n \\ &\text{ and } |G| \text{ are not relatively prime}] \\ \Rightarrow x - y = 0 &\text{ i.e., the identity of } G \\ \Rightarrow x = y. \end{aligned}$$

Therefore the mapping σ is one-one.

The mapping σ is also onto G : Since G is finite and σ is one-one, therefore σ must be onto G .

Finally if $x, y \in G$, then $\sigma(x + y) = n(x + y) = nx + ny = \sigma(x) + \sigma(y)$.

Hence, σ is an automorphism of G .

Example 8: Let $\mathbf{R}^+$ be the multiplicative group of all positive real numbers. Show that the mapping $f : \mathbf{R}^+ \rightarrow \mathbf{R}^+$ defined by $f(x) = x^2, \forall x \in \mathbf{R}^+$ is an automorphism.

(Lucknow 2009)

Solution: We have $\mathbf{R}^+ = \{x : x \in \mathbf{R} \text{ and } x > 0\}$.

The mapping f is one-one : Let $x, y \in \mathbf{R}^+$.

$$\text{Then } f(x) = f(y) \Rightarrow x^2 = y^2 \Rightarrow x = y. \quad [\because x > 0 \text{ and } y > 0]$$

$\therefore f$ is one-one.

The mapping f is onto : Let x be any element of $\mathbf{R}^+$. Then there exists $\sqrt{x} \in \mathbf{R}^+$ such that

$$f(\sqrt{x}) = (\sqrt{x})^2 = x.$$

$\therefore f$ is onto.

The mapping f is a homomorphism : Let x, y be any two elements of $\mathbf{R}^+$. Then

$$f(xy) = (xy)^2 = x^2 y^2 = f(x) f(y).$$

$\therefore f$ is a homomorphism.

Hence, f is an automorphism of $\mathbf{R}^+$.

Example 9: Find the automorphism group of A_3 , where A_3 is the alternating group of degree 3 on three symbols.

Solution. Let A_3 be the alternating group on three symbols a, b, c . Then

$$A_3 = \{e, f, g\},$$

where e = the identity permutation, $f = (a\ b\ c)$ and $g = (a\ c\ b)$.

Let I be the identity mapping of A_3 i.e.,

$$I(e) = e, I(f) = f \text{ and } I(g) = g.$$

Then obviously I is an automorphism of A_3 .

Now consider the mapping $T : A_3 \rightarrow A_3$ defined by

$$T(e) = e, T(f) = g, T(g) = f.$$

Obviously T is a one-one mapping of A_3 onto A_3 .

We have $T(ef) = T(f) = g = eg = T(e)T(f)$,

$$T(fe) = T(f) = g = ge = T(f) = T(e),$$

$$T(eg) = T(g) = f = ef = T(e)T(g),$$

$$T(ge) = T(g)T(e).$$

$$\text{Also, } T(fg) = T(e) \left[\because fg = (abc)(acb) = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix} \begin{pmatrix} a & c & b \\ c & b & a \end{pmatrix} = \begin{pmatrix} a & b & c \\ a & b & c \end{pmatrix} = e \right]$$

$$= e$$

$$\text{and } T(f)T(g) = gf = (acb)(abc) = e.$$

$$\therefore T(fg) = T(f)T(g).$$

Similarly $T(gf) = T(g)T(f)$. [Note that A_3 is an abelian group]

$\therefore$ the mapping T is a homomorphism. Thus the mapping T is also an automorphism of A_3 .

Hence, the automorphism group $\text{Aut}(A_3)$ of A_3 contains only two elements, namely I and T .

Thus $\text{Aut}(A_3) = \{I, T\}$.

Example 10: If $f : G \rightarrow G$ such that $f(a) = a^n, \forall a \in G$ is an automorphism of G , show that $a^{n-1} \in Z$ for all $a \in G$, where Z is the centre of the group G .

Solution: Let x, a be any two elements of G . We have

$$\begin{aligned} f(a^{-n}xa^n) &= (a^{-n}xa^n)^n && [\text{By def. of the mapping } f] \\ &= a^{-n}x^na^n && [\because \text{if } a, b \text{ are any elements of a group } G, \\ &&& \text{then } (b^{-1}ab)^n = b^{-1}a^nb \text{ for any integer } n] \\ &= (a^{-1})^n x^n a^n \\ &= f(a^{-1}) f(x) f(a) && [\text{By def. of the mapping } f] \\ &= f(a^{-1}xa). && [\because \text{the mapping } f \text{ is a homomorphism}] \end{aligned}$$

Thus $f(a^{-n}xa^n) = f(a^{-1}xa)$.

Since the mapping f is one-one, therefore $a^{-n}xa^n = a^{-1}xa$

$$\Rightarrow xa^{n-1} = a^{n-1}x, \forall x, a \in G$$

$$\Rightarrow a^{n-1} \in Z, \forall a \in G.$$

Example 11: Let $f : G \rightarrow G$ be a homomorphism i.e., f is an endomorphism of G . Suppose f commutes with every inner automorphism of G . Show that

(i) $K = \{x \in G : f^2(x) = f(x)\}$ is a normal subgroup of G .

(ii) G / K is abelian.

Solution: (i) Let e be the identity of the group G .

We have $f^2(e) = f(f(e)) = f(e)$.

$\therefore e \in K$ and so $K \neq \emptyset$.

Now let x, y be any two elements of K . Then $f(x) = f^2(x)$ and $f(y) = f^2(y)$. We have

$$\begin{aligned} f^2(xy^{-1}) &= f(f(xy^{-1})) \\ &= f[f(x)f(y^{-1})] && [\because f \text{ is a homomorphism}] \\ &= f(f(x)[f(y)]^{-1}) && [\because f \text{ is a homomorphism} \Rightarrow f(y^{-1}) = [f(y)]^{-1}] \\ &= f(f(x))f([f(y)]^{-1}) && [\because f \text{ is a homomorphism}] \\ &= f^2(x)[f(f(y))]^{-1} && [\because f \text{ is a homomorphism} \\ &\Rightarrow f([f(y)]^{-1}) = [f(f(y))]^{-1}] \\ &= f^2(x)[f^2(y)]^{-1} \\ &= f(x)[f(y)]^{-1} && [\because x, y \in K \Rightarrow f^2(x) = f(x) \text{ and } f^2(y) = f(y)] \\ &= f(x)f(y^{-1}) && \{\because f \text{ is a homomorphism} \\ &\Rightarrow [f(y)]^{-1} = f(y^{-1})\} \\ &= f(xy^{-1}). && [\because f \text{ is a homomorphism}] \end{aligned}$$

$\therefore xy^{-1} \in K$.

Thus $K \neq \emptyset$ and $x, y \in K \Rightarrow xy^{-1} \in K$.

$\therefore K$ is a subgroup of G .

Now to show that K is normal in G . Let $g \in G$ and $x \in K$. Then

$$\begin{aligned} f^2(gxg^{-1}) &= f[f(gxg^{-1})] \\ &= f[ff_g(x)], \quad \text{where } f_g \text{ is the inner automorphism} \\ &\hspace{15em} \text{corresponding to } g \end{aligned}$$

$$\begin{aligned}
 &= f [f_g f (x)] && [\because \text{By hypothesis } f \text{ commutes with every inner automorphism of } G \text{ and so } f f_g = f_g f] \\
 &= f [g f (x) g^{-1}] && [\text{By def. of the inner automorphism } f_g] \\
 &= f (g) f (f (x)) f (g^{-1}) && [\because f \text{ is a homomorphism}] \\
 &= f (g) f^2 (x) f (g^{-1}) \\
 &= f (g) f (x) f (g^{-1}) && [\because x \in K \Rightarrow f^2 (x) = f (x)] \\
 &= f (g x g^{-1}). && [\because f \text{ is a homomorphism}]
 \end{aligned}$$

$\therefore g x g^{-1} \in K$ for all $x \in K, g \in G$.

$\therefore K$ is a normal subgroup of G .

(ii) To show that G / K is abelian.

By definition of a quotient group, we have $G / K = \{Kx : x \in G\}$.

We have G / K is abelian $\Leftrightarrow Kx Ky = Ky Kx, \forall x, y \in G$

$$\Leftrightarrow Kx y = Ky x, \forall x, y \in G$$

$$\Leftrightarrow x y (y x)^{-1} \in K, \forall x, y \in G$$

$$\Leftrightarrow x y x^{-1} y^{-1} \in K, \forall x, y \in G. \quad \dots(1)$$

Now for all $x, y \in G$, we have

$$\begin{aligned}
 f^2 (x y x^{-1} y^{-1}) &= f [f (x y x^{-1} y^{-1})] \\
 &= f [f (x y x^{-1}) f (y^{-1})] && [\because f \text{ is a homomorphism}] \\
 &= f [f (f_x (y)) f (y^{-1})], && \text{where } f_x \text{ is the inner automorphism corresponding to } x \\
 &= f [f_x (f (y)) f (y^{-1})] && [\because f f_x = f_x f] \\
 &= f (x f (y) x^{-1} [f (y)]^{-1}) \\
 &= f [x f_{f(y)} (x^{-1})], && \text{where } f_{f(y)} \text{ is the inner automorphism corresponding to } f (y) \\
 &= f (x) f [f_{f(y)} (x^{-1})] && [\because f \text{ is a homomorphism}] \\
 &= f (x) f_{f(y)} f (x^{-1}) && [\because f f_{f(y)} = f_{f(y)} f] \\
 &= f (x) f (y) f (x^{-1}) [f (y)]^{-1} \\
 &= f (x) f (y) f (x^{-1}) f (y^{-1}) \\
 &= f (x y x^{-1} y^{-1}).
 \end{aligned}$$

$\therefore x y x^{-1} y^{-1} \in K$.

Hence, by virtue of (1), G / K is abelian.

Example 12: Let G be a group and f an automorphism of G . If, for $a \in G$, we have $N(a) = \{x \in G : xa = ax\}$, prove that $N(f(a)) = f(N(a))$.

Solution: If $a \in G$, then by the definition of $N(a)$, we have

$$N(a) = \{x \in G : xa = ax\}.$$

Let y be any arbitrary element of $f(N(a))$. Then

$$\begin{aligned} y \in f(N(a)) &\Rightarrow \exists x \in N(a) \text{ such that } y = f(x) \text{ and } xa = ax \\ \Rightarrow f(xa) &= f(ax) \text{ and } y = f(x) \text{ for some } x \in N(a) \\ \Rightarrow f(x)f(a) &= f(a)f(x) \text{ and } y = f(x) \text{ for some } x \in N(a) \\ &\quad [\because f \text{ is a homomorphism}] \\ \Rightarrow yf(a) &= f(a)y, \forall y \in f(N(a)) \\ \Rightarrow y \in N(f(a)), \forall y \in f(N(a)). \\ \therefore f(N(a)) &\subseteq N(f(a)). \end{aligned} \quad \dots(1)$$

Again let y' be any arbitrary element of $N(f(a))$. Then

$$\begin{aligned} y' \in N(f(a)) &\Rightarrow y' \in G \text{ and } y'f(a) = f(a)y' \\ \Rightarrow \exists x' \in G \text{ such that } y' &= f(x') \text{ and } y'f(a) = f(a)y' \\ &\quad [\because \text{the mapping } f : G \rightarrow G \text{ is onto}] \\ \Rightarrow f(x')f(a) &= f(a)f(x') \text{ for some } x' \in G \\ \Rightarrow f(x'a) &= f(ax') \text{ for some } x' \in G \quad [\because f \text{ is a homomorphism}] \\ \Rightarrow x'a &= ax' \text{ for some } x' \in G \quad [\because f \text{ is one-one}] \\ \Rightarrow x' &\in N(a) \\ \Rightarrow f(x') &\in f(N(a)) \\ \Rightarrow y' &\in f(N(a)), \forall y' \in N(f(a)). \\ \therefore N(f(a)) &\subseteq f(N(a)). \end{aligned} \quad \dots(2)$$

From (1) and (2), we have

$$N(f(a)) = f(N(a)).$$

Example 13: Let G be an infinite cyclic group. Determine $\text{Aut } G$, the group of all automorphisms of G .

Solution: Let $G = (a)$ be an infinite cyclic group generated by a .

Let $f \in \text{Aut } G$ i.e., let f be an automorphism of G .

First we shall show that $f(a)$ is also a generator of G i.e., $G = (f(a))$, the cyclic group generated by $f(a)$.

Let x be any element of G .

Since the mapping $f : G \rightarrow G$ is onto, therefore there exists $y \in G$ such that $x = f(y)$.

But $y \in G \Rightarrow y = a^r$ for some integer r [$\because a$ is a generator of G]

$$\therefore x = f(y) = f(a^r) = (f(a))^r.$$

$\therefore f(a)$ is a generator of G i.e., $G = (f(a))$.

But the infinite cyclic group $G = (a)$ has only two generators, namely a and a^{-1} .

$$\therefore f(a) = a \quad \text{or} \quad f(a) = a^{-1}.$$

Thus f has only two choices and so

$$o(\text{Aut } G) \leq 2. \quad \dots(1)$$

Define a mapping $T : G \rightarrow G$ such that

$$T(x) = x^{-1}, \forall x \in G.$$

Then $T \in \text{Aut } G$.

$$\text{Also } T \neq I \text{ as } T = I \Rightarrow T(x) = x, \forall x \in G$$

$$\Rightarrow x^{-1} = x, \forall x \in G$$

$$\Rightarrow a^{-1} = a \Rightarrow a^2 = e \Rightarrow o(a) \text{ is finite, which is a contradiction because the generator } a \text{ of an infinite cyclic group cannot be of finite order.}$$

$$\therefore T \neq I.$$

Thus G has at least two automorphisms.

$$\therefore o(\text{Aut } G) \geq 2. \quad \dots(2)$$

From (1) and (2), we have $o(\text{Aut } G) = 2$.

In fact, we have $\text{Aut } G = \{ I, T : T(x) = x^{-1}, \forall x \in G \}$.

Since $o(\text{Aut } G) = 2$, therefore $\text{Aut } G$ is a cyclic group of order 2.

We know that any cyclic group of order n is isomorphic to the group

$$Z_n = \{0, 1, 2, \dots, n-1\} \text{ under addition modulo } n.$$

Hence, $\text{Aut } G \cong Z_2$, where $Z_2 = \{0, 1\}$ is a group under addition modulo 2.

Example 14: Let G be a finite cyclic group of order n . Determine $\text{Aut } G$, the group of all automorphisms of G .

Solution: Let $G = \langle a \rangle$ be a finite cyclic group of order n generated by a .

We have $o(a) = o(G) = n$ so that $a^n = e$.

Let $f \in \text{Aut } G$ i.e., let f be an automorphism of G .

First we shall show that $f(a)$ is also a generator of G i.e., $G = \langle f(a) \rangle$, the cyclic group generated by a .

Let x be any element of G .

Since the mapping $f : G \rightarrow G$ is onto, therefore $\exists y \in G$ such that $x = f(y)$.

But $y \in G \Rightarrow y = a^r$ for some integer r . [$\because a$ is a generator of G]

$$\therefore x = f(y) = f(a^r) = (f(a))^r.$$

$$\therefore f(a) \text{ is a generator of } G \text{ i.e., } G = \langle f(a) \rangle.$$

But the finite cyclic group G has only $\phi(n)$ generators, where $\phi(n)$ is Euler's ϕ -function i.e., $\phi(n)$ denotes the number of integers less than n and prime to n .

$\therefore$ if f is an automorphism of G , then f has only $\phi(n)$ choices.

$$\therefore o(\text{Aut } G) \leq \phi(n). \quad \dots(1)$$

Define a mapping $f_m : G \rightarrow G$ such that

$$f_m(x) = x^m, (m, n) = 1, 1 \leq m < n.$$

Here (m, n) denotes the H.C.F. of m and n and $(m, n) = 1$ means that m and n are co-prime.

We claim that the mapping f_m is an automorphism of G .

f_m is one-one : Let $a^r, a^s \in G$, where $1 \leq r \leq n, 1 \leq s \leq n$ and $r \geq s$.

$$\text{Then } f_m(a^r) = f_m(a^s) \Rightarrow (a^r)^m = (a^s)^m$$

$$\Rightarrow a^{rm} = a^{sm} \Rightarrow a^{(r-s)m} = e$$

$$\Rightarrow n \mid (r-s)m.$$

But m is prime to n and $0 \leq r-s < n$.

$$\therefore n \mid (r-s)m \Rightarrow r-s = 0 \Rightarrow r = s \Rightarrow a^r = a^s.$$

$$\text{Thus } f_m(a^r) = f_m(a^s) \Rightarrow a^r = a^s.$$

$$\therefore f_m \text{ is one-one.}$$

f_m is onto: Since G is finite and f_m is one-one, therefore f must be onto G .

f_m is a homomorphism : Let $a^r, a^s \in G$, where $1 \leq r \leq n, 1 \leq s \leq n$. Then

$$f_m(a^r a^s) = f_m(a^{r+s}) = f_m(a^{nu+k}),$$

where $r+s = nu+k$, u is some integer and $0 \leq k < n$

$$= f_m(a^{nu} a^k) = f_m(a^k) \quad [\because a^{nu} = (a^n)^u = e^u = e]$$

$$= (a^k)^m = a^{mk} = a^{m(r+s-nu)}$$

$$= a^{m(r+s)} a^{-mnu}$$

$$= a^{m(r+s)} \quad [\because a^{-mnu} = (a^n)^{-mu} = e^{-mu} = e]$$

$$= a^{mr} a^{ms} = (a^r)^m (a^s)^m$$

$$= f_m(a^r) f_m(a^s).$$

$$\therefore f_m \text{ is a homomorphism.}$$

$$\therefore f_m \text{ is an automorphism of } G \text{ i.e., } f_m \in \text{Aut } G.$$

Thus $f_m \in \text{Aut } G$, where $(m, n) = 1, 1 \leq m < n$.

We now show that $f_r \neq f_s$, for all r, s ($r \neq s$), $1 \leq r, s < n$, where $(r, n) = 1$ and $(s, n) = 1$.

Assume that $r > s$. Suppose, if possible, $f_r = f_s$. Then

$$f_r = f_s \Rightarrow f_r(a) = f_s(a)$$

$$\Rightarrow a^r = a^s \Rightarrow a^{r-s} = e$$

$$\Rightarrow o(a) \mid (r-s) \Rightarrow n \mid (r-s)$$

$$\Rightarrow n \leq r-s < n.$$

This is a contradiction.

$$\therefore f_r \neq f_s, \text{ for all } r, s (r \neq s), 1 \leq r, s < n, \text{ where } r \text{ and } s \text{ are relatively prime to } n.$$

This shows that $\text{Aut } G$ has at least $\phi(n)$ automorphisms of G .

$$\therefore o(\text{Aut } G) \geq \phi(n). \quad \dots(2)$$

From (1) and (2), we have $o(\text{Aut } G) = \phi(n)$.

In fact, we have

$$\text{Aut } G = \left\{ f_m : \begin{array}{l} f_m(x) = x^m, (m, n) = 1 \\ 1 \leq m < n \end{array} \right\}.$$

So far we have shown that $o(\text{Aut } G) = \phi(n)$.

Now we shall show that $\text{Aut}(G) \cong G_n$, where

$$G_n = \{x : x \in I, 1 \leq x < n, (x, n) = 1\}$$

is the group of integers with respect to multiplication modulo n i.e., $\times_n$ as the composition.

Define a mapping $\psi : \text{Aut } G \rightarrow G_n$ such that

$$\psi(f_m) = m, 1 \leq m < n, (m, n) = 1.$$

ψ is one-one : Let $f_r, f_s \in \text{Aut } G$. Then

$$\psi(f_r) = \psi(f_s) \Rightarrow r = s \Rightarrow f_r = f_s.$$

$\therefore \psi$ is one-one.

ψ is onto : Let m be any element of G_n , where $1 \leq m < n, (m, n) = 1$.

Then $\exists f_m \in \text{Aut}(G)$ such that $\psi(f_m) = m$.

$\therefore \psi$ is onto.

ψ is a homomorphism : Let $f_r, f_s \in \text{Aut } G$. Then $1 \leq r, s < n, (r, n) = 1$ and $(s, n) = 1$.

For any $x \in G$, we have $f_r f_s(x) = f_r(x^s) = (x^s)^r = x^{rs}$

$$= x^{nu+k}, \text{ where } u \text{ is some integer and } k \text{ is an integer such that } 0 \leq k < n$$

$$[\because x^{nu} = (x^n)^u = e^u = e]$$

$$= x^k$$

$$= x^{r \times_n s} \quad [\because r \times_n s = k, \text{ where } k \text{ is the least non-negative remainder obtained on dividing } rs \text{ by } n]$$

$$= f_{r \times_n s}(x).$$

$$\therefore f_r f_s = f_{r \times_n s}$$

$$\Rightarrow \psi(f_r f_s) = \psi(f_{r \times_n s}) = r \times_n s = \psi(f_r) \times_n \psi(f_s).$$

$\therefore \psi$ is a homomorphism.

Hence ψ is an isomorphism of $\text{Aut } G$ onto G_n .

$\therefore \text{Aut } G \cong G_n$.

This completely determines $\text{Aut } G$.

Example 15: Show that the group of automorphisms of a cyclic group is abelian.

Solution: Let $G = \langle a \rangle$ be a cyclic group generated by a . Then the following two cases

arise.

Case I: If G is an infinite cyclic group, then its group of automorphisms is given by

$$\text{Aut } G = \{I, T : T(x) = x^{-1}, \forall x \in G\}.$$

Since $o(\text{Aut } G) = 2$, therefore $\text{Aut } G$ is abelian.

Case II. If G is a cyclic group of finite order n , then

$$\text{Aut } G = \{f_m : f_m(x) = x^m, 1 \leq m < n, (m, n) = 1\}.$$

Let $f_r, f_s \in \text{Aut } G$. Then $f_r(a) = a^r, f_s(a) = a^s$.

$$\begin{aligned} \text{Now } (f_r f_s)(a) &= f_r(f_s(a)) = f_r(a^s) = (a^s)^r = a^{sr} = a^{rs} \\ &= (a^r)^s = f_s(a^r) = f_s(f_r(a)) = (f_s f_r)(a). \end{aligned}$$

Now two automorphisms of a cyclic group are equal if the image of a generator of the group under each of them is the same.

Hence $f_r f_s = f_s f_r, \forall f_r, f_s \in \text{Aut } G$.

$\therefore$ in this case also $\text{Aut } G$ is abelian.

Hence, the group of automorphisms of a cyclic group is abelian.

Example 16: Let $\phi(n)$ be the Euler ϕ -function. For any integers $a > 1, n > 0$, show that

$$n \mid \phi(a^n - 1).$$

Solution: Let $G = \langle b \rangle$ be the cyclic group generated by b , where

$$o(b) = o(G) = a^n - 1.$$

Consider the mapping $f_a : G \rightarrow G$ defined by

$$f_a(x) = x^a, \forall x \in G.$$

Since $(a, a^n - 1) = 1$, therefore $f_a \in \text{Aut } G$.

If $x \in G$, then $f_a^2(x) = f_a(f_a(x)) = f_a(x^a) = (x^a)^a = x^{a^2}$.

In general, $f_a^r(x) = x^{a^r}$, for every positive integer r .

$$\begin{aligned} \therefore f_a^n(x) &= x^{a^n} = x x^{a^n - 1} = x x^{o(G)} = xe = x \quad [\because x^{o(G)} = e, \forall x \in G] \\ &= I(x), \forall x \in G. \end{aligned}$$

$$\therefore f_a^n = I. \quad \dots(1)$$

Again if $f_a^m = I$, then

$$\begin{aligned} f_a^m(x) &= I(x) = x, \forall x \in G \Rightarrow f_a^m(b) = b \quad [\because b \in G] \\ \Rightarrow b^{a^m} &= b \Rightarrow b^{a^m - 1} = e \Rightarrow o(b) \mid (a^m - 1) \\ \Rightarrow (a^n - 1) &\mid (a^m - 1) \Rightarrow a^n - 1 \leq a^m - 1 \\ \Rightarrow a^n &\leq a^m \Rightarrow n \leq m \Rightarrow m \geq n \quad \dots(2) \end{aligned}$$

From (1) and (2), we conclude that $o(f_a) = n$.

Also, $o(\text{Aut } G) = \phi(a^n - 1)$.

$$\therefore f_a \in \text{Aut } G \Rightarrow o(f_a) \mid o(\text{Aut } G) \Rightarrow n \mid \phi(a^n - 1).$$

Comprehensive Exercise 1

1. Verify the following statement for being true or false :
If $G = \langle a \rangle$ is a cyclic group of order 10 then the mapping $\sigma : G \rightarrow G$ such that $\sigma(a^k) = a^{2k}$ for all k , is an automorphism of G .
2. Give an example of a group in which (i) the inner automorphisms corresponding to any two elements are the same, (ii) the inner automorphisms corresponding to no two elements are the same.
3. If for group G , the mapping $f : G \rightarrow G$ given by $f(x) = x^3$, $x \in G$, is an automorphism, then prove that G is abelian.
4. Show that the group of all automorphisms of a cyclic group G of order r is isomorphic to the group of integers less than and relatively prime to r under multiplication modulo r . (Lucknow 2006)
5. Let f be an automorphism of a group G . If $a \in G$, then prove that $f(a^r) = [f(a)]^r$, for all integers r .
6. Let f be an automorphism of a group G . Show that $o(f(a)) = o(a)$, for every $a \in G$. Deduce that $o(bab^{-1}) = o(a)$ for all $a, b \in G$.
7. Prove that the group of automorphisms of an infinite cyclic group is of order 2. (Lucknow 2010)
8. Let G be a cyclic group of order 4. Show that $\text{Aut } G = \{I, T\}$ where $T(x) = x^3$, for all $x \in G$.
9. If $o(\text{Aut } G) > 1$, then show that $o(G) > 2$.
10. If G is any group in which $g^2 \neq e$ for some $g \in G$, then G has a non-trivial automorphism.
11. If $G = S_3$, then show that $\text{Aut } G \cong S_3$.

Answers 1

1. False.
2. (i) Every abelian group, (ii) the symmetric group P_3 .

1.5 Conjugate Elements, Relation of Conjugacy and Classes of Conjugate Elements

Definition: If a, b be two elements of a group G , then b is said to be conjugate to a if there exists an element $x \in G$ such that $b = x^{-1}ax$.

If $b = x^{-1}ax$, then b is also called the transform of a by x .

If b is conjugate to a then symbolically we shall write $b \sim a$ and this relation in G will be called the **relation of conjugacy**. Thus $b \sim a$ iff $b = x^{-1}ax$ for some $x \in G$.

Theorem: The relation of conjugacy is an equivalence relation on G .

(Lucknow 2008,10)

Proof: Reflexivity: If a is any element of G , then we have

$$a = e^{-1}ae \Rightarrow a \sim a.$$

Thus $a \sim a \quad \forall \quad a \in G$. Therefore the relation is reflexive.

Symmetry: We have $a \sim b \Rightarrow a = x^{-1}bx$ for some $x \in G$

$$\Rightarrow xax^{-1} = x(x^{-1}bx)x^{-1} \Rightarrow xax^{-1} = b$$

$$\Rightarrow b = (x^{-1})^{-1}ax^{-1}, \text{ where } x^{-1} \in G$$

$$\Rightarrow b \sim a.$$

Therefore the relation is symmetric.

Transitivity: Let $a \sim b, b \sim c$. Then $a = x^{-1}bx, b = y^{-1}cy$ for some $x, y \in G$.

From this we get

$$\begin{aligned} a &= x^{-1}(y^{-1}cy)x & [\because b = y^{-1}cy] \\ &= (yx)^{-1}c(yx), \text{ where } yx \in G. \end{aligned}$$

$\therefore a \sim c$ and thus the relation is transitive.

Hence the relation of conjugacy in a group G is an equivalence relation. Therefore it will partition G into disjoint equivalence classes called **classes of conjugate elements**. These classes will be such that

- (i) any two elements of the same class are conjugate.
- (ii) no two elements of different classes are conjugate.

The collection of all elements conjugate to an element $a \in G$ will be symbolically denoted by $C(a)$ or by $\bar{a}$. Thus $C(a) = \{x \in G : x \sim a\}$.

$C(a)$ will be called the *conjugate class of a in G* . We have $(y^{-1}ay) \sim a$ for all $y \in G$.

Also if $b \sim a$ then b must be equal to $y^{-1}ay$ for some $y \in G$. Therefore

$$C(a) = \{y^{-1}ay : y \in G\}.$$

If G is a finite group, then the number of distinct elements in $C(a)$ will be denoted by c_a .

1.6 Self-Conjugate Elements

Definition: An element $a \in G$ is said to be self-conjugate if a is the only member of the class $C(a)$ of elements conjugate to a i.e., if $C(a) = \{a\}$.

Thus, a , is self-conjugate if and only if

$$a = x^{-1}ax \quad \forall \quad x \in G \quad \text{or} \quad xa = ax \quad \forall \quad x \in G.$$

Thus a self-conjugate element is one which commutes with each element of the group.

If a is a self-conjugate element, then we have $a = x^{-1}ax \quad \forall \quad x \in G$.

Thus the transform of a by every element of G remains equal to a . Therefore sometimes a self-conjugate element is also called an **invariant element**.

1.7 Normalizer or Centralizer of an Element of a Group

Definition: If $a \in G$, then $N(a)$, the normalizer of a in G is the set of all those elements of G which commute with a . Symbolically $N(a) = \{x \in G : ax = xa\}$.

Theorem: The normalizer $N(a)$ of $a \in G$ is a subgroup of G . (Lucknow 2009)

Proof: We have $N(a) = \{x \in G : ax = xa\}$.

Let $x_1, x_2 \in N(a)$. Then $ax_1 = x_1a, ax_2 = x_2a$.

First we show that $x_2^{-1} \in N(a)$.

$$\begin{aligned} \text{We have } ax_2 = x_2a &\Rightarrow x_2^{-1}(ax_2)x_2^{-1} = x_2^{-1}(x_2a)x_2^{-1} \\ &\Rightarrow x_2^{-1}a = ax_2^{-1} \Rightarrow x_2^{-1} \in N(a). \end{aligned}$$

Now we shall show that $x_1x_2^{-1} \in N(a)$.

$$\text{We have } a(x_1x_2^{-1}) = (ax_1)x_2^{-1} = (x_1a)x_2^{-1} = x_1(ax_2^{-1}) = x_1(x_2^{-1}a) = (x_1x_2^{-1})a.$$

$$\therefore x_1x_2^{-1} \in N(a).$$

$$\text{Thus } x_1, x_2 \in N(a) \Rightarrow x_1x_2^{-1} \in N(a).$$

$$\therefore N(a) \text{ is a subgroup of } G.$$

Note 1: It should be noted that $N(a)$ is not necessarily a normal subgroup of G .

Note 2: Since $ex = xe \quad \forall \quad x \in G$, therefore $N(e) = G$.

Note 3: If G is an abelian group and $a \in G$, then $xa = ax \quad \forall \quad x \in G$. Therefore $N(a) = G$.

1.8 Counting Principle and the Class Equation of a Finite Group

Theorem: Let a be any element of a group G . Then two elements $x, y \in G$ give rise to the same conjugate of a if and only if they belong to the same right coset of the normalizer of a in G .

Hence show that if G is a finite group, then $c_a = \frac{o(G)}{o[N(a)]}$, i.e., the number of elements

conjugate to a in G is the index of the normalizer of a in G .

Proof: We have $x, y \in G$ are in the same right coset of

$$N(a) \text{ in } G \Leftrightarrow N(a)x = N(a)y \quad [\because x \in N(a)x, y \in N(a)y]$$

Note that if H is a subgroup, then $x \in Hx$

$$\Leftrightarrow xy^{-1} \in N(a) \quad [\because \text{if } H \text{ is a subgroup, then } Ha = Hb \Leftrightarrow ab^{-1} \in H]$$

$$\Leftrightarrow axy^{-1} = xy^{-1}a \quad [\text{by def. of } N(a)]$$

$$\Leftrightarrow x^{-1}(axy^{-1})y = x^{-1}(xy^{-1}a)y \Leftrightarrow x^{-1}ax = y^{-1}ay$$

$$\Leftrightarrow x, y \text{ give rise to the same conjugate of } a.$$

Hence the first result follows.

Now consider the right coset decomposition of G with respect to the subgroup $N(a)$. We have just proved that if $x, y \in G$ are in the same right coset of $N(a)$ in G , then they give the same conjugate of a . Further if x, y are in different right cosets of $N(a)$ in G , then they give rise to different conjugates of a . The reason is that if x, y give the same conjugate of a , then they must belong to the same right coset of $N(a)$ in G . Thus **there is a one-to-one correspondence between the right cosets of $N(a)$ in G and the conjugates of a .** So if G is a finite group, then

$$\begin{aligned} c_a &= \text{the number of distinct elements in } C(a) \\ &= \text{the number of distinct right cosets of } N(a) \text{ in } G \\ &= \text{the index of } N(a) \text{ in } G = \frac{o(G)}{o[N(a)]}. \end{aligned}$$

Corollary: If G is a finite group, then $o(G) = \sum \frac{o(G)}{o[N(a)]}$ where this sum runs over one element a in each conjugate class.

Proof: We know that the relation of conjugacy is an equivalence relation on G . Therefore it partitions G into disjoint conjugate classes. The union of all disjoint conjugate classes will be equal to G and two distinct conjugate classes of G will have no common element. Since G is a finite group, therefore the number of distinct conjugate classes of G will be finite, say equal to k . Suppose $C(a)$ denotes the conjugate class of a in G and c_a denotes the number of elements in this class. If $C(a_1), C(a_2), \dots, C(a_k)$ are the k distinct conjugate classes of G , then

$$G = C(a_1) \cup C(a_2) \cup \dots \cup C(a_k)$$

$\Rightarrow$ the number of elements in G = the number of elements in $C(a_1)$
 + the number of elements in $C(a_2) + \dots$ + the number of elements in $C(a_k)$
 [$\because$ two distinct conjugate classes have no common element]

$\Rightarrow o(G) = \sum c_a$, in each conjugate class

$\Rightarrow o(G) = \sum \frac{o(G)}{o[N(a)]}$, by previous theorem.

Note: The equation in this corollary is often called the **class equation** of G .

1.9 The Centre of a Group

Definition: The set Z of all self-conjugate elements of a group G is called the centre of G .
 Symbolically

$$Z = \{z \in G : zx = xz \ \forall x \in G\}. \quad (\text{Lucknow 2007})$$

Theorem 1: The centre Z of a group G is a normal subgroup of G . (Lucknow 2007)

Proof: We have $Z = \{z \in G : zx = xz \ \forall x \in G\}$.

First we shall prove that Z is a subgroup of G .

Let $z_1, z_2 \in Z$. Then $z_1 x = xz_1$ and $z_2 x = xz_2$ for all $x \in G$.

We have $z_2 x = xz_2 \ \forall x \in G$

$$\Rightarrow z_2^{-1} (z_2 x) z_2^{-1} = z_2^{-1} (xz_2) z_2^{-1} \Rightarrow xz_2^{-1} = z_2^{-1} x \ \forall x \in G$$

$$\Rightarrow z_2^{-1} \in Z.$$

$$\begin{aligned} \text{Now } (z_1 z_2^{-1}) x &= z_1 (z_2^{-1} x) = z_1 (xz_2^{-1}) = (z_1 x) z_2^{-1} = (xz_1) z_2^{-1} \\ &= x (z_1 z_2^{-1}) \ \forall x \in G. \end{aligned}$$

$$\therefore z_1 z_2^{-1} \in Z.$$

$$\text{Thus } z_1, z_2 \in Z \Rightarrow z_1 z_2^{-1} \in Z.$$

$\therefore Z$ is a subgroup of G .

Now we shall show that Z is a normal subgroup of G . Let $x \in G$ and $z \in Z$. Then

$$zx x^{-1} = (xz) x^{-1} = (zx) x^{-1} = z \in Z.$$

$$\text{Thus } x \in G, z \in Z \Rightarrow zx x^{-1} \in Z.$$

$\therefore Z$ is a normal subgroup of G .

Theorem 2: $a \in Z$ if and only if $N(a) = G$. If G is finite, $a \in Z$ if and only if

$$o[N(a)] = o(G).$$

Proof: Let $a \in Z$. Then by def. of Z , we have

$$ax = xa \ \forall x \in G.$$

$$\text{Also } N(a) = \{x \in G : ax = xa\}.$$

$$\text{Now } a \in Z \Leftrightarrow ax = xa \ \forall x \in G \quad [\text{by def. of } Z]$$

$$\Leftrightarrow x \in N(a) \quad \forall x \in G \quad [\text{by def. of } N(a)]$$

$$\Leftrightarrow N(a) = G \quad [\because N(a) \subseteq G \text{ and each element of } G \text{ is in } N(a)]$$

If the group G is finite, then $N(a) = G \Leftrightarrow o(G) = o[N(a)]$.

Therefore if the group G is finite, then $a \in Z$ if and only if $o[N(a)] = o(G)$.

Theorem 3: Let G be a finite group and Z be the centre of G . Then the class equation of G can be written as

$$o(G) = o(Z) + \sum_{a \notin Z} \frac{o(G)}{o[N(a)]},$$

where the summation runs over one element a in each conjugate class containing more than one element.

Proof: The class equation of G is

$$o(G) = o(Z) + \sum_{a \notin Z} \frac{o(G)}{o[N(a)]},$$

the summation being extended over one element a in each conjugate class.

Now $a \in Z \Leftrightarrow o[N(a)] = o(G) \Leftrightarrow o(G) / o[N(a)] = 1 \Leftrightarrow$ the conjugate class of a in G contains only one element. Thus the number of conjugate classes each having only one element is equal to $o(Z)$. If a is an element of any one of these conjugate classes, we have $o(G) / o[N(a)] = 1$.

Hence, the class equation of G takes the desired form

$$o(G) = o(Z) + \sum_{a \notin Z} \frac{o(G)}{o[N(a)]}.$$

1.10 Centre for Group of Prime Power Order

Theorem 1: If $o(G) = p^n$ where p is a prime number, then the centre $Z \neq \{e\}$.

(Lucknow 2009)

Proof: By the class equation of G , we have

$$o(G) = o(Z) + \sum_{a \notin Z} \frac{o(G)}{o[N(a)]}, \quad \dots(1)$$

where the summation runs over one element a in each conjugate class containing more than one element.

Now $\forall a \in G$, $N(a)$ is a subgroup of G . Therefore by Lagrange's theorem $o[N(a)]$ is a divisor of $o(G)$. Also $a \notin Z \Rightarrow N(a) \neq G \Rightarrow o[N(a)] < o(G)$. Therefore if $a \notin Z$, then $o[N(a)]$ must be of the form p^{n_a} where n_a is some integer such that $1 \leq n_a < n$. Suppose there are exactly z elements in Z i.e., let $o(Z) = z$. Then the class equation (1) gives

$$p^n = z + \sum \frac{p^n}{p^{n_a}}, \text{ where each } n_a \text{ is some integer such that } 1 \leq n_a < n.$$

$$\therefore z = p^n - \sum \frac{p^n}{p^{n_a}}, \quad \dots(2)$$

where n_a 's are some positive integers each less than n .

Now $p \mid p^n$. Also p divides each term in the $\sum$ of the right hand side of (2) because each $n_a < n$. Thus we see that p is a divisor of the right hand side of (2). Therefore p is a divisor of z . Now $e \in Z$. Therefore $z \neq 0$. Therefore z is a positive integer divisible by the prime p . Therefore $z > 1$. Hence Z must contain an element besides e . Therefore $Z \neq \{e\}$.

Corollary: If $o(G) = p^2$ where p is a prime number, then G is abelian.

(Lucknow 2007)

Proof: We shall show that the centre Z of G is equal to G itself. Then obviously G will be an abelian group.

Since p is a prime number, therefore by the previous theorem $Z \neq \{e\}$. Therefore $o(Z) > 1$. But Z is a subgroup of G , therefore $o(Z)$ must be a divisor of $o(G)$ i.e., $o(Z)$ must be a divisor of p^2 . Since p is prime, therefore either $o(Z) = p$ or p^2 .

If $o(Z) = p^2$, then $Z = G$ and our proof is complete.

Now suppose that $o(Z) = p$. Then $o(Z) < o(G)$ because $p < p^2$. Therefore there must be an element which is in G but which is not in Z . Let $a \in G$ and $a \notin Z$. Now $N(a)$ is a subgroup of G and $a \in N(a)$. Also $x \in Z \Rightarrow xa = ax$ and this implies $x \in N(a)$. Thus $Z \subseteq N(a)$. Since $a \notin Z$, therefore the number of elements in $N(a)$ is $> p$ i.e., $o[N(a)] > p$. But order of $N(a)$ must be a divisor of p^2 . Therefore $o[N(a)]$ must be equal to p^2 . Then $N(a) = G$. Therefore $a \in Z$ and thus we get a contradiction.

Therefore it is not possible that $o(Z) = p$. Hence the only possibility is that

$$o(Z) = p^2 \Rightarrow Z = G \Rightarrow G \text{ is abelian.}$$

Example: Is a group of order 121 abelian?

Ans: Yes.

1.11 Conjugate Subgroups

Definition: If A, B be two subgroups of a group G , then B is said to be conjugate to A if there exists an element $x \in G$ such that $B = x^{-1}Ax$.

If $B = x^{-1}Ax$, then B is also called the transform of A by x .

If B is conjugate to A , then symbolically we shall write $B \sim A$.

Theorem: *The relation of being conjugate is an equivalence relation on the set of subgroups of a group G .*

Proof: Proceed as in theorem of article 1.5.

The relation of conjugacy in the family of subgroups of a group G will partition the family into disjoint equivalence classes. The collection of all subgroups conjugate to a subgroup A of G will be symbolically denoted by $C(A)$. Obviously

$$C(A) = \{x^{-1}Ax : x \in G\}.$$

1.12 Normalizer of a Subgroup of a Group

Definition: *If A is a subgroup of a group G , then $N(A)$, the normalizer of A in G is the set of all those elements of G which commute with A . Symbolically*

$$N(A) = \{x \in G : xA = Ax\}.$$

Theorem 1: *The normalizer $N(A)$ of a subgroup A of a group G is a subgroup of G .*

Proof: Proceed as in Theorem of article 1.7.

Theorem 2: *Suppose A is a subgroup of a group G . Then there is a one-to-one correspondence between the right cosets of $N(A)$ in G and the conjugates of A .*

Proof: Proceed as in Theorem of article 1.8.

1.13 Self-Conjugate Subgroup or an Invariant Subgroup

Definition: *A subgroup A of a group G is said to be self-conjugate if A is the only member of the class $C(A)$ of subgroups conjugate to A .*

Thus, A , is self conjugate iff

$$A = x^{-1}Ax \quad \forall x \in G \quad \text{or} \quad xA = Ax \quad \forall x \in G$$

or A is a normal subgroup of G .

If A is a self-conjugate subgroup of a group G , then we have $A = x^{-1}Ax \quad \forall x \in G$.

Thus the transform of A by every element of G remains equal to A . Therefore sometimes a self-conjugate subgroup is also called an **invariant subgroup**. It is quite obvious that a subgroup of a group G is invariant if and only if it is normal. Therefore sometimes a normal subgroup is also called an invariant subgroup.

Illustrative Examples

Example 17: *Write all the conjugate classes in S_3 , find the c_a 's and verify the class equation.*

Solution: The symmetric group on 3 symbols 1, 2, 3 is given by

$$S_3 = \{(1), (1, 2), (2, 3), (3, 1), (1, 2, 3), (1, 3, 2)\}.$$

The three conjugate classes of S_3 are

$$C(a) = \{(1)\}, \text{ where } a = (1) = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix};$$

$$C(b) = \{(123), (132)\}, \text{ where } b = (123) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix};$$

$$C(c) = \{(12), (23), (31)\}, \text{ where } c = (12) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix};$$

$$\therefore c_a = 1, c_b = 2, c_c = 3.$$

Also, we have $Z(S_3) = \{(1)\} = C(a)$.

Here $o(Z(S_3)) = 1$.

Hence, the class equation of S_3 is

$$o(S_3) = o(Z(S_3)) + \sum_a \frac{o(S_3)}{o(C(a))},$$

where the summation is taken over a set of representatives for the distinct conjugacy classes having more than one member

$$\text{i.e., } o(S_3) = o(Z(S_3)) + \frac{o(S_3)}{o(C(b))} + \frac{o(S_3)}{o(C(c))} = 1 + \frac{6}{2} + \frac{6}{3} = 1 + 3 + 2 = 6.$$

Example 18: Let Z be the centre of a group G . If $a \in Z$, then prove that the cyclic subgroup $\{a\}$ of G generated by a is a normal subgroup of G .

Solution: We have

$Z = \{z \in G : zx = xz \ \forall \ x \in G\}$. Let $a \in Z$ and let $H = \{a\}$ be the cyclic subgroup of G generated by a . Let h be any element of H . Then $h = a^n$ for some integer n .

Let x be any element of G . We have

$$\begin{aligned} xhx^{-1} &= xa^n x^{-1} = a^n x x^{-1} & [\because a \in Z \Rightarrow a^n \in Z \in a^n x = xa^n] \\ &= a^n e = a^n \in H. \end{aligned}$$

$\therefore H$ is a normal subgroup of G .

Example 19: Let a be any element of G . Show that the cyclic subgroup of G generated by a is a normal subgroup of the normalizer of a .

Solution: We have the normalizer of $a = N(a) = \{x \in G : xa = ax\}$.

Let H be the cyclic subgroup of G generated by a . Also let h be any element of H . Then $h = a^n$ where n is some integer. We have

$$a^n a = a^{n+1} = aa^n.$$

$$\therefore a^n = h \in N(a).$$

Now $N(a)$ and H are subgroups of G . Also $h \in H \Rightarrow h \in N(a)$.

Therefore $H \subseteq N(a)$. Hence H is a subgroup of $N(a)$.

Now to prove that H is a normal subgroup of $N(a)$. Let x be any element of $N(a)$ and $h = a^n$ be any element of H . We have

$$\begin{aligned} xhx^{-1} &= xa^n x^{-1} = (xax^{-1})^n = (axx^{-1})^n & [\because x \in N(a) \Rightarrow ax = xa] \\ &= (ae)^n = a^n \in H. \end{aligned}$$

$\therefore H$ is a normal subgroup of $N(a)$.

Example 20: Show that two elements are conjugate if and only if they can be put in the form xy and yx respectively where x and y are suitable elements of G .

Solution: Let a, b be two conjugate elements of a group G .

Then $a = c^{-1}bc$ for some $c \in G$.

Let $c^{-1}b = x$ and $c = y$. Then $a = xy$.

Also $yx = c(c^{-1}b) = (cc^{-1})b = eb = b$.

Conversely suppose that $a = xy$ and $b = yx$. We have

$$b = yx \Rightarrow y^{-1}b = y^{-1}yx \Rightarrow y^{-1}b = x.$$

Now $a = xy \Rightarrow a = y^{-1}by \Rightarrow a$ and b are conjugate elements.

Example 21: Give an example to show that in a group G the normalizer of an element is not necessarily a normal subgroup of G .

Solution: Consider the group S_3 , the symmetric group of permutations on three symbols a, b, c . We have $S_3 = \{I, (ab), (bc), (ca), (abc), (acb)\}$. Let $N(ab)$ denote the normalizer of the element $(ab) \in S_3$. We shall show that $N(ab)$ is not a normal subgroup of S_3 . Let us calculate the elements of $N(ab)$. Obviously $(ab) \in N(ab)$. Also $I \in N(ab)$ because $I(ab) = (ab)I$.

Now $(bc)(ab) = (abc)$ and $(ab)(bc) = (acb)$. Thus (bc) does not commute with (ab) . Therefore $(bc) \notin N(ab)$.

Again $(ca)(ab) = (acb)$ and $(ab)(ca) = (abc)$.

Thus $(ca)(ab) \neq (ab)(ca)$ and therefore $(ca) \notin N(ab)$. Similarly we can verify that $(abc) \notin N(ab)$ and $(acb) \notin N(ab)$.

Hence $N(ab) = \{I, (ab)\}$.

Now we shall show that $N(ab)$ is not a normal subgroup of S_3 . Take the element $(bc) \in S_3$ and the element $(ab) \in N(ab)$. We have

$$(bc)(ab)(bc)^{-1} = (bc)(ab)(cb) = (abc)(cb) = (ac) \notin N(ab).$$

Therefore $N(ab)$ is not a normal subgroup of S_3 .

Example 22: Let Z denote the centre of a group G . If G/Z is cyclic prove that G is abelian.

Solution: It is given that G/Z is cyclic. Let Zg be a generator of the cyclic group G/Z where g is some element of G .

Let $a, b \in G$. Then to prove that $ab = ba$. Since $a \in G$, therefore $Za \in G/Z$. But G/Z is cyclic having Zg as a generator. Therefore there exists some integer m such that $Za = (Zg)^m = Zg^m$, because Z is a normal subgroup of G . Now $a \in Za$.

Therefore

$$Za = Zg^m \Rightarrow a \in Zg^m \Rightarrow a = z_1 g^m \text{ for some } z_1 \in Z.$$

Similarly $b = z_2 g^n$ where $z_2 \in Z$ and n is some integer.

$$\begin{aligned}
 \text{Now } ab &= (z_1 g^m) (z_2 g^n) = z_1 g^m z_2 g^n \\
 &= z_1 z_2 g^m g^n \quad [\because z_2 \in Z \Rightarrow z_2 g^m = g^m z_2] \\
 &= z_1 z_2 g^{m+n}.
 \end{aligned}$$

$$\begin{aligned}
 \text{Again } ba &= z_2 g^n z_1 g^m = z_2 z_1 g^n g^m = z_2 z_1 g^{n+m} = z_1 z_2 g^{m+n}. \\
 &[\because z_1 \in Z \Rightarrow z_1 z_2 = z_2 z_1]
 \end{aligned}$$

$$\therefore ab = ba.$$

Since $ab = ba \forall a, b \in G$, therefore G is abelian.

Example 23: If p is a prime number and G is a non-abelian group of order p^3 , show that the centre of G has exactly p elements.

Solution: Let Z denote the centre of G . Since $o(G) = p^3$ where p is a prime number, therefore $Z \neq \{e\}$ i.e., $o(Z) > 1$. But Z is a subgroup of G , therefore $o(Z)$ must be a divisor of $o(G)$ i.e., $o(Z)$ must be a divisor of p^3 . Since p is prime, therefore either $o(Z) = p$ or p^2 or p^3 .

If $o(Z) = p^3 = o(G)$, then $Z = G$ and so G is abelian which contradicts the hypothesis that G is non-abelian. So $o(Z)$ cannot be p^3 .

If $o(Z) = p^2$, then $o(G/Z) = o(G)/o(Z) = p^3/p^2 = p$ i.e., G/Z is a group of prime order p and so is cyclic. But if G/Z is cyclic, then G is abelian which again contradicts the hypothesis. So $o(Z)$ cannot be p^2 .

Hence the only possibility is that $o(Z) = p$ i.e., the centre of G has exactly p elements.

1.14 p-Sylow Subgroup

Definition: Suppose G is a finite group and $o(G) = p^m n$, where p is a prime number and p is not a divisor of n . Then a subgroup H of G is said to be a p -Sylow subgroup of G iff $o(H) = p^m$.

Sylow's Theorem: Suppose G is a group of finite order and p is a prime number. If $p^m \mid o(G)$ and p^{m+1} is not a divisor of $o(G)$, then G has a subgroup of order p^m .

Proof: We shall prove the theorem by induction on $o(G)$.

Assuming that the theorem is true for groups of order less than that of G , we shall show that it is also true for G . To start the induction we see that the theorem is obviously true if $o(G) = 1$.

Let $o(G) = p^m n$, where p is not a divisor of n . If $m = 0$, the theorem is obviously true. If $m = 1$, the theorem is true by Cauchy's theorem. So let $m > 1$. Then G is a group of

composite order and so G must possess a subgroup H such that $H \neq G$.

If p is not a divisor of $\frac{o(G)}{o(H)}$, then $p^m \nmid o(H)$ because

$$o(G) = p^m n = o(H) \frac{o(G)}{o(H)}.$$

Also p^{m+1} cannot be a divisor of $o(H)$ because then p^{m+1} will be a divisor of $o(G)$ of which $o(H)$ is a divisor. Further $o(H) < o(G)$. Therefore by our induction hypothesis, the theorem is true for H . Therefore H has a subgroup of order p^m and this will also be a subgroup of G . So let us assume that for every subgroup H of G where $H \neq G$, p is a divisor of $\frac{o(G)}{o(H)}$.

Consider the class equation,

$$o(G) = o(Z) + \sum_{a \notin Z} \frac{o(G)}{o[N(a)]} \quad \dots(1)$$

Since $a \notin Z \Rightarrow N(a) \neq G$, therefore according to our assumption p is a divisor of

$$\sum_{a \notin Z} \frac{o(G)}{o[N(a)]}.$$

Also $p \mid o(G)$.

Therefore from (1), we conclude that p is a divisor of $o(Z)$. Then by Cauchy's theorem, Z has an element b of order p . Z is the centre of G . Also $N = \{b\}$ is a cyclic subgroup of Z of order p . Therefore N is a cyclic subgroup of G of order p . Since $b \in Z$, therefore N is a normal subgroup of G of order p .

Now consider the quotient group $G' = G/N$.

We have $o(G') = o(G)/o(N) = p^m n / p = p^{m-1} n$.

Thus $o(G') < o(G)$. Also $p^{m-1} \mid o(G')$ but p^m is not a divisor of $o(G')$. Therefore by our induction hypothesis G' has a subgroup, say S' of order p^{m-1} . We know that the natural mapping $\phi: G \rightarrow G/N$ defined by $\phi(x) = Nx \forall x \in G$ is a homomorphism of G onto G/N with kernel N . Let $S = \{x \in G : \phi(x) \in S'\}$.

Then S is a subgroup of G and $S' \cong S/N$.

$$\therefore o(S') = o(S/N) = \frac{o(S)}{o(N)}.$$

Therefore $o(S) = o(S') \cdot o(N) = p^{m-1} p = p^m$.

Thus S is a subgroup of G of order p^m .

This completes the proof of the theorem.

Illustrative Examples

Example 24: If H is a p -Sylow subgroup of G and $x \in G$, then $x^{-1}Hx$ is also a p -Sylow subgroup of G .

Solution: Suppose G is a finite group and $o(G) = p^m n$ where p is a prime number and p is not a divisor of n . If H is a p -Sylow subgroup of G , then $o(H) = p^m$.

Let $x \in G$ be arbitrary. Then $x^{-1}Hx$ will be a p -Sylow subgroup of G if $x^{-1}Hx$ is a subgroup of G and if $o(x^{-1}Hx) = p^m$.

First we shall prove that $x^{-1}Hx$ is a subgroup of G . Let $x^{-1}h_1x, x^{-1}h_2x$ be any two elements of $x^{-1}Hx$. Then $h_1, h_2 \in H$. Also we have

$$\begin{aligned}(x^{-1}h_1x)(x^{-1}h_2x)^{-1} &= x^{-1}h_1xx^{-1}h_2^{-1}(x^{-1})^{-1} = x^{-1}h_1h_2^{-1}x \\ &= x^{-1}h_1h_2^{-1}x \\ &\in x^{-1}Hx,\end{aligned}$$

since $h_1h_2^{-1} \in H$, H being a subgroup of G .

$\therefore x^{-1}Hx$ is a subgroup of G .

Now let ψ be a mapping from H into $x^{-1}Hx$ defined as

$$\psi(h) = x^{-1}hx \quad \forall h \in H.$$

ψ is onto: Let $x^{-1}hx$ be any element of $x^{-1}Hx$. Then $h \in H$ and we have

$$\psi(h) = x^{-1}hx.$$

Therefore ψ is onto.

ψ is one-one: Let $h_1, h_2 \in H$. Then

$$\begin{aligned}\psi(h_1) = \psi(h_2) &\Rightarrow x^{-1}h_1x = x^{-1}h_2x \\ \Rightarrow h_1 = h_2 &\quad \text{[by cancellation laws]} \\ \Rightarrow \psi &\text{ is one-one.}\end{aligned}$$

Thus ψ is a one-to-one correspondence between the elements of H and the elements of $x^{-1}Hx$. Therefore $o(x^{-1}Hx) = o(H) = p^m$. Hence $x^{-1}Hx$ is a p -Sylow subgroup of G .

Example 25: If a group G has only one p -Sylow subgroup H , then H is normal in G .

Solution: Suppose a group G has only one p -Sylow subgroup H . Let x be any element of G . Then by previous example, $x^{-1}Hx$ is also a p -Sylow subgroup of G .

But H is the only p -Sylow subgroup of G . Therefore

$$x^{-1}Hx = H \quad \forall x \in G \Rightarrow H \text{ is a normal subgroup of } G.$$

Comprehensive Exercise 2

1. Let G be a non-abelian group. Show that for all $x \in G$, $Z(G)$ is a subgroup of $N(x)$.
2. Let G be a finite group and an element $a \in G$ has exactly two conjugates.
Prove that G has a normal subgroup $N \neq (e), G$.
3. Let G be a group, and let $a \in G$ has only two conjugates in G . Show that $N(a)$ is a normal subgroup of G .
4. Let N be a normal subgroup of G and $a \in N$. Show that every conjugate of a in G is also in N .
5. Write down the elements of the symmetric group P_3 and determine the classes of conjugate elements.
6. Show that any two conjugate classes of a group are either disjoint or identical.
7. If the order of a group G is a power of a prime p , show that the centre of G has at least p elements.
8. If $o(G) = p^n$, where p is a prime number, and $H \neq G$ is a subgroup of G , show that there exists an $x \in G$, $x \notin H$ such that $x^{-1}Hx = H$.
9. If N be a normal subgroup of G , show that either $C(a) \cap N = \emptyset$ or $C(a) \subseteq N$ for all $a \in G$.
10. Let G be a finite non-abelian group such that $G/Z(G)$ is abelian and $k(G)$ denotes the number of conjugate classes in G . Show that $k(G) \geq o(G/Z(G)) + o(Z(G)) - 1$, where $Z(G)$ is the centre of G .

Answers 2

5. $f_1 = I, f_2 = (12), f_3 = (23), f_4 = (31), f_5 = (123), f_6 = (132);$
 $C(f_1) = \{f_1\}, C(f_2) = C(f_3) = C(f_4) = \{f_2, f_3, f_4\},$
 $C(f_5) = C(f_6) = \{f_5, f_6\}.$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- If p is a prime number and G is a non-abelian group of order p^3 , then the number of elements in the centre of G is
 - p
 - p^2
 - p^3
 - none of these.

Fill in the Blank(s)

Fill in the blanks ".....", so that the following statements are complete and correct.

- The set Z of all self-conjugate elements of a group G is called the of G .
- If $o(G) = p^n$ where p is a prime number, then the centre $Z \neq$
- If $a \in G$, then $N(a)$, the set of all those elements of G which commute with a is called the of a in G .

True or False

Write 'T' for true and 'F' for false statement.

- The relation of conjugacy is an equivalence relation on the group G .
- If $a \in G$, then the set $N(a) = \{x \in G : ax = xa\}$ is always a normal subgroup of G .
- If $o(G) = p^n$ where p is a prime number, then the centre $Z = \{e\}$.
- A group of order 121 is abelian.

Answers

Multiple Choice Questions

- (a).

Fill in the Blank(s)

- centre.
- $\{e\}$.
- normalizer.

True or False

- T.
- F.
- F.
- T.



Chapter

2



Rings

2.1 Principal Ideal Ring

Definition: A commutative ring R without zero divisors and with unity element is a principal ideal ring if every ideal S in R is a principal ideal i.e., if every ideal S in R is of the form $S = (a)$ for some $a \in S$.

Theorem 1: The ring of integers is a principal ideal ring.

Proof: Let $(\mathbf{I}, +, \cdot)$ be the ring of integers. Obviously $\mathbf{I}$ is a commutative ring with unity and without zero divisors. Therefore $\mathbf{I}$ will be a principal ideal ring if every ideal in $\mathbf{I}$ is a principal ideal.

Let S be any ideal of the ring of integers. If S is the null ideal then $S = (0)$ so that S is a principal ideal.

So let us suppose that $S \neq (0)$.

Now S contains at least one non-zero integer, say a . Since S is a subgroup of R under addition, therefore $a \in S \Rightarrow -a \in S$. This shows that S contains at least one positive integer, because if $0 \neq a$, then one of a and $-a$ must be positive.

Let S_+ be the set of all positive integers in S . Since S_+ is not empty, therefore by the well ordering principle S_+ must possess a least positive integer. Let s be this least element. We will now show that S is the principal ideal generated by s i.e., $S = (s)$.

Suppose now that n is any integer in S . Then by division algorithm, there exist integers q and r such that $n = qs + r$ with $0 \leq r < s$.

$$\text{Now } s \in S, q \in \mathbf{I} \quad \Rightarrow \quad qs \in S \quad [\because S \text{ is an ideal}]$$

$$\text{and } n \in S, qs \in S \quad \Rightarrow \quad n - qs \in S \quad [\because S \text{ is a subgroup of the additive group of } \mathbf{I}]$$

$$\Rightarrow \quad r \in S \quad [\because n - qs = r]$$

But $0 \leq r < s$ and s is the least positive integer such that $s \in S$. Hence r must be 0.

$$\therefore \quad n = qs.$$

Thus $n \in S \Rightarrow n = qs$ for some $q \in \mathbf{I}$.

Hence S is a principal ideal of $\mathbf{I}$ generated by s .

Since S was an arbitrary ideal in the ring of integers, therefore the ring of integers is a principal ideal ring.

Theorem 2: Every field is a principal ideal ring.

Proof: A field has no proper ideals. The only ideals of a field are (i) the null ideal which is a principal ideal generated by 0 and (ii) the field itself which is also a principal ideal generated by 1. Thus a field is always a principal ideal ring.

2.2 Polynomial Rings

While studying algebra in high school classes we are introduced with polynomials.

We know that expressions of the type $3x^2 - 4x + 5$, $x + 7$, $9x^3 - \frac{2}{3}x^2 + 4x + 5$

etc. are called polynomials in the *indeterminate* x . In place of x we can use other letters like y , z etc. to denote these polynomials. Now we shall define polynomials over an arbitrary ring.

Definition: Let R be an arbitrary ring and let x , called an **indeterminate**, be any symbol not an element of R . By a polynomial in x over R is meant an expression of the form

$$f(x) = a_0 x^0 + a_1 x + a_2 x^2 + \dots$$

where $a_0, a_1, a_2, \dots$, are elements of R and only a finite number of them are not equal to 0, the zero element of R .

Here x is an indeterminate. We could have used any other letter, say, y in place of x . Also $a_0 x^0, a_1 x, a_2 x^2$ etc. are called terms of the polynomial and a_0, a_1, a_2 etc. are called coefficients of these terms. All these coefficients are elements of R . The number of terms in the polynomial $f(x)$ will be infinite but except a finite number of terms, the coefficients of all the remaining terms will be equal to 0, the zero

element of the ring. The symbol '+' connecting various terms in $f(x)$ has no connection with the addition of the ring. This symbol has been used here only to connect different terms. Also x is not an element of R . The powers of x are nothing to do with the powers of an element of R . The different powers of x only tell us the ordered place of different coefficients. There is no harm if we represent this polynomial $f(x)$ by the infinite ordered set $(a_0, a_1, a_2, \dots)$, where $a_0, a_1, a_2, \dots$ are elements of R and only a finite number of them are not equal to zero. Since from high school classes we represent a polynomial with an indeterminate x , therefore we have preferred this way to represent polynomials.

Remark: The polynomial $a_0x^0 + a_1x + a_2x^2 + \dots$ over a ring R can also be written as

$$a_0 + a_1x + a_2x^2 + \dots$$

Set of all polynomials over a ring: Let R be an arbitrary ring and x an indeterminate. The set of all polynomials $f(x)$,

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = a_0x^0 + a_1x + a_2x^2 + \dots$$

where the a 's are elements of the ring R and only a finite number of them are not equal to zero, is called $R[x]$.

We shall make a ring out of $R[x]$. Then $R[x]$ will be called the ring of all polynomials over the ring R . For this we shall define **equality**, **addition** and **multiplication** of two elements of $R[x]$.

Definition:

Suppose R is an arbitrary ring and $f(x) = a_0x^0 + a_1x + a_2x^2 + a_3x^3 + \dots$ and $g(x) = b_0x^0 + b_1x + b_2x^2 + b_3x^3 + \dots$ are any elements of $R[x]$. Then

- (a) $f(x) = g(x)$ if and only if $a_n = b_n \quad \forall$ non-negative integer n . Thus two polynomials are equal iff their corresponding coefficients are equal.
- (b) $f(x) + g(x) = c_0x^0 + c_1x + c_2x^2 + c_3x^3 + \dots$

where $c_n = a_n + b_n$ for every non-negative integer n . Thus in order to add two polynomials we should add the coefficients of like powers of x .

Since $c_n \in R$ and only a finite number of c 's can be not equal to zero, therefore $f(x) + g(x)$ is an element of $R[x]$. Thus $R[x]$ is closed with respect to addition of polynomials as defined above.

- (c) $f(x)g(x) = d_0x^0 + d_1x + d_2x^2 + d_3x^3 + \dots$

where $d_n = a_0b_n + a_1b_{n-1} + a_2b_{n-2} + \dots + a_nb_0$
for every non-negative integer n .

We can write $d_n = \sum_{i+j=n} a_i b_j$, where by this summation we mean the sum of all the products of the type $a_i b_j$ with i and j non-negative integers whose sum is n .

Since $d_n \in R$ and only a finite number of d 's can be not equal to zero, therefore $f(x)g(x)$ is an element of $R[x]$. Thus $R[x]$ is closed with respect to multiplication of polynomials as defined above.

We have $d_0 = a_0 b_0, d_1 = a_0 b_1 + a_1 b_0, d_2 = a_0 b_2 + a_1 b_1 + a_2 b_0,$

$$d_3 = a_0 b_3 + a_1 b_2 + a_2 b_1 + a_3 b_0 \text{ and so on.}$$

Therefore in order to multiply two polynomials $f(x)$ and $g(x)$, we should first write

$$f(x)g(x) = (a_0 x^0 + a_1 x + a_2 x^2 + \dots)(b_0 x^0 + b_1 x + b_2 x^2 + \dots).$$

Now we should multiply different powers of the indeterminate x and using the relation $x^i x^j = x^{i+j}$ we should collect coefficients of different powers of x .

Zero Polynomial: The polynomial

$$f(x) = \sum a_n x^n = a_0 x^0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

in which all the coefficients $a_0, a_1, a_2, \dots$ are equal to 0 is called the zero polynomial over the ring R .

Degree of a polynomial:

Let
$$f(x) = a_0 x^0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots + a_n x^n + \dots$$

be a polynomial over an arbitrary ring R . We say that n is the degree of the polynomial $f(x)$ if and only if $a_n \neq 0$ and $a_m = 0$ for all $m > n$. We shall write $\deg f(x)$ to denote the degree of $f(x)$. Thus the degree of $f(x)$ is the largest non-negative integer i for which the i^{th} coefficient of $f(x)$ is not 0. If in the polynomial $f(x)$, a_0 (i.e., the coefficient of x^0) is not 0 and all the other coefficients are 0, then according to our definition, the degree of $f(x)$ will be zero. Also according to our definition, if there is no non-zero coefficient in $f(x)$, then its degree will remain undefined. Thus we do not define the degree of the zero polynomial. Also it is obvious that every non-zero polynomial will possess a unique degree.

Note: If $f(x) = a_0 x^0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$ is a polynomial of degree n i.e., if $a_n \neq 0$ and $a_m = 0$ for all $m > n$, then it is convenient to write

$$f(x) = \sum_{i=0}^n a_i x^i = a_0 x^0 + a_1 x + a_2 x^2 + \dots + a_n x^n.$$

It will remain understood that all the terms in $f(x)$ which follow the term $a_n x^n$, have zero coefficients. Also we shall call $a_n x^n$ as the **leading term** and a_n as the **leading coefficient** of the polynomial. The term $a_0 x^0$ is called the **constant term** and a_0 is called the zeroth coefficient of $f(x)$.

For example $f(x) = 2x^0 + 3x - 4x^2 + 4x^3 - 8x^4$ is a polynomial of degree 4 over the ring of integers. Here -8 is the leading coefficient and 2 is the zeroth coefficient. The coefficients of all terms which contain powers of x greater than 4 will be regarded as zero. Similarly $g(x) = 3x^0$ is a polynomial of degree zero over the ring of integers. In this polynomial the coefficients of $x, x^2, x^3, \dots$ are all equal to

zero. The zero polynomial over an arbitrary ring R will be represented by $0x^0$.

Set of constant polynomials over a ring: Let R be an arbitrary ring and $R[x]$ the set of all polynomials over R . Let R' denote the set of all polynomials over R whose coefficients are all zero except for the constant term, which may be either zero or non-zero. That is,

$$R' = \{ax^0 : a \in R\}.$$

Then R' will be called as the set of constant polynomials in $R[x]$.

Thus all the polynomials of degree 0 as well as the zero polynomial will be called as constant polynomials.

Illustration 1: Add and multiply the following polynomials over the ring of integers:

$$f(x) = 2x^0 + 5x + 3x^2 - 4x^3, \quad g(x) = 3x^0 + 4x - x^3 + 5x^4.$$

Solution: By our definition of the sum of two polynomials, we have

$$\begin{aligned} f(x) + g(x) &= (2 + 3)x^0 + (5 + 4)x + (3 + 0)x^2 \\ &\quad + (-4 - 1)x^3 + (0 + 5)x^4 \\ &= 5x^0 + 9x + 3x^2 - 5x^3 + 5x^4. \end{aligned}$$

$$\begin{aligned} \text{Also } f(x)g(x) &= (2x^0 + 5x + 3x^2 - 4x^3)(3x^0 + 4x - x^3 + 5x^4) \\ &= 6x^0 + (8 + 15)x + (20 + 9)x^2 + (-2 + 12 - 12)x^3 \\ &\quad + (10 - 5 - 16)x^4 + (25 - 3)x^5 + (15 + 4)x^6 - 20x^7 \\ &= 6x^0 + 23x + 29x^2 - 2x^3 - 11x^4 + 22x^5 + 19x^6 - 20x^7 \end{aligned}$$

Illustration 2: Add and multiply the following polynomials over the ring $(\mathbb{I}_6, +_6, \times_6)$:

$$f(x) = 2x^0 + 5x + 3x^2, \quad g(x) = 1x^0 + 4x + 2x^3.$$

$$\begin{aligned} \text{Solution: } f(x) + g(x) &= (2 +_6 1)x^0 + (5 +_6 4)x + (3 +_6 0)x^2 + (0 +_6 2)x^3 \\ &= 3x^0 + 3x + 3x^2 + 2x^3. \end{aligned}$$

$$\begin{aligned} \text{Also } f(x)g(x) &= (2x^0 + 5x + 3x^2)(1x^0 + 4x + 2x^3) \\ &= (2 \times_6 1)x^0 + [(2 \times_6 4) +_6 (5 \times_6 1)]x + [(5 \times_6 4) +_6 (3 \times_6 1)]x^2 \\ &\quad + [(2 \times_6 2) +_6 (3 \times_6 4)]x^3 + (5 \times_6 2)x^4 + (3 \times_6 2)x^5 \\ &= 2x^0 + (2 +_6 5)x + (2 +_6 3)x^2 + (4 +_6 0)x^3 + 4x^4 + 0x^5 \\ &= 2x^0 + 1x + 5x^2 + 4x^3 + 4x^4. \end{aligned}$$

Note: Here degree of $f(x) = 2$, degree of $g(x) = 3$ and degree of $f(x)g(x) = 4$. The point to note is that degree $f(x)g(x)$ may be less than the sum of the degrees of $f(x)$ and $g(x)$.

Illustration 3: Add and multiply the following polynomials over the ring $(\mathbb{I}_5, +_5, \times_5)$:

$$f(x) = 3 + 4x + 2x^2, \quad g(x) = 1 + 3x + 4x^2 + 2x^3.$$

$$\begin{aligned} \text{Solution: } f(x) + g(x) &= 4 + 2x + x^2 + 2x^3, \\ f(x)g(x) &= 3 + 3x + x^2 + 3x^3 + x^4 + 4x^5. \end{aligned}$$

2.3 Degree of the Sum and the Product of Two Polynomials

Theorem: Let $f(x)$ and $g(x)$ be two non-zero polynomials over an arbitrary ring R .

Then

(i) $\deg[f(x) + g(x)] \leq \max[\deg f(x), \deg g(x)]$, if $f(x) + g(x) \neq 0$.

(ii) $\deg[f(x)g(x)] \leq \deg f(x) + \deg g(x)$, if $f(x)g(x) \neq 0$.

Proof: Let $f(x) = a_0x^0 + a_1x + a_2x^2 + \dots + a_nx^n$, $a_n \neq 0$

and $g(x) = b_0x^0 + b_1x + b_2x^2 + \dots + b_mx^m$, $b_m \neq 0$

be two elements of $R[x]$.

Here $\deg f(x) = n$ and $\deg g(x) = m$.

From our definition of the sum of two polynomials, it is obvious that if

$f(x) + g(x) \neq 0$, then

$$\deg[f(x) + g(x)] = \begin{cases} \max(n, m) & \text{if } n \neq m \\ n & \text{if } n = m \text{ and } a_n + b_m \neq 0 \\ < n & \text{if } n = m \text{ and } a_n + b_m = 0. \end{cases}$$

Again $f(x)g(x) = (a_0b_0)x^0 + (a_0b_1 + a_1b_0)x + \dots + a_nb_mx^{n+m}$.

Suppose $f(x)g(x) \neq 0$. Then $f(x)g(x)$ has a unique degree.

If $a_nb_m \neq 0$, then $\deg[f(x)g(x)] = n + m = \deg f(x) + \deg g(x)$.

Also if $a_nb_m = 0$, then $\deg[f(x)g(x)] < n + m$.

Corollary 1: Suppose D is an integral domain and $f(x)$, $g(x)$ are two non-zero elements of $D[x]$. Then

$$\deg[f(x)g(x)] = \deg f(x) + \deg g(x).$$

Proof: Since $a_n \neq 0$, $b_m \neq 0$, therefore $a_nb_m \neq 0$ because in an integral domain the product of two non-zero elements cannot be zero. Hence $\deg[f(x)g(x)] = m + n$.

Corollary 2: If F is a field and $f(x)$, $g(x)$ are two non-zero elements of $F[x]$, then

$$\deg[f(x)g(x)] = \deg f(x) + \deg g(x).$$

Proof: Since a field is also free from zero divisors, therefore $a_nb_m \neq 0$ when $a_n \neq 0$ and $b_m \neq 0$. Hence the result.

2.4 Ring of Polynomials

Theorem: The set $R[x]$ of all polynomials over an arbitrary ring R is a ring with respect to addition and multiplication of polynomials.

Proof: Let $f(x), g(x) \in R[x]$. Then $f(x) + g(x)$ and $f(x)g(x)$ are also polynomials over R . Therefore $R[x]$ is closed with respect to addition and multiplication of polynomials.

Now let

$$f(x) = \sum a_i x^i = a_0 x^0 + a_1 x + a_2 x^2 + \dots,$$

$$g(x) = b_0 x^0 + b_1 x + b_2 x^2 + \dots,$$

$$h(x) = c_0 x^0 + c_1 x + c_2 x^2 + \dots \text{ be any arbitrary elements of } R[x].$$

Commutativity of addition: We have

$$\begin{aligned} f(x) + g(x) &= (a_0 + b_0) x^0 + (a_1 + b_1) x + (a_2 + b_2) x^2 + \dots \\ &= (b_0 + a_0) x^0 + (b_1 + a_1) x + (b_2 + a_2) x^2 + \dots = g(x) + f(x). \end{aligned}$$

Associativity of addition: We have

$$\begin{aligned} [f(x) + g(x)] + h(x) &= \sum (a_i + b_i) x^i + \sum c_i x^i = \sum [(a_i + b_i) + c_i] x^i \\ &= \sum [a_i + (b_i + c_i)] x^i = \sum a_i x^i + \sum (b_i + c_i) x^i \\ &= f(x) + [g(x) + h(x)]. \end{aligned}$$

Existence of additive identity: Let $0(x)$ be the zero polynomial over R i.e.,

$$0(x) = 0x^0 + 0x + 0x^2 + \dots$$

$$\begin{aligned} \text{Then } f(x) + 0(x) &= (a_0 + 0) x^0 + (a_1 + 0) x + (a_2 + 0) x^2 + \dots \\ &= a_0 x^0 + a_1 x + a_2 x^2 + \dots = f(x). \end{aligned}$$

$\therefore$ the zero polynomial $0(x)$ is the additive identity.

Existence of additive inverse: Let $-f(x)$ be the polynomial over R defined as

$$-f(x) = (-a_0) x^0 + (-a_1) x + (-a_2) x^2 + \dots$$

$$\begin{aligned} \text{Then } -f(x) + f(x) &= (-a_0 + a_0) x^0 + (-a_1 + a_1) x + (-a_2 + a_2) x^2 + \dots \\ &= 0x^0 + 0x + 0x^2 + \dots = 0(x) = \text{the additive identity.} \end{aligned}$$

$\therefore$ each member of $R[x]$ possesses additive inverse.

Associativity of Multiplication: We have

$$\begin{aligned} f(x) g(x) &= (a_0 x^0 + a_1 x + a_2 x^2 + \dots) (b_0 x^0 + b_1 x + b_2 x^2 + \dots) \\ &= d_0 x^0 + d_1 x + d_2 x^2 + \dots + d_l x^l + \dots, \end{aligned}$$

$$\text{where } d_l = \sum_{i+j=l} a_i b_j.$$

$$\begin{aligned} \text{Now } [f(x) g(x)] h(x) &= (d_0 x^0 + d_1 x + d_2 x^2 + \dots) (c_0 x^0 + c_1 x + c_2 x^2 + \dots) \\ &= e_0 x^0 + e_1 x + e_2 x^2 + \dots + e_n x^n + \dots, \end{aligned}$$

where e_n = the coeff. of x^n in $[f(x) g(x)] h(x)$

$$= \sum_{l+k=n} d_l c_k = \sum_{l+k=n} \left[\left(\sum_{i+j=l} a_i b_j \right) c_k \right] = \sum_{i+j+k=n} a_i b_j c_k.$$

Similarly we can show that the coeff. of x^n in $f(x) [g(x) h(x)]$

$$= \sum_{i+j+k=n} a_i b_j c_k.$$

Thus $[f(x)g(x)]h(x) = f(x)[g(x)h(x)]$ since corresponding coefficients in these two polynomials are equal.

Distributivity of Multiplication with respect to addition: We have

$$f(x)[g(x) + h(x)] \\ = (a_0x^0 + a_1x + a_2x^2 + \dots)[(b_0 + c_0)x^0 + (b_1 + c_1)x + (b_2 + c_2)x^2 + \dots]$$

If n is any non-negative integer, then the coefficient of x^n in $f(x)[g(x) + h(x)]$

$$= \sum_{i+j=n} a_i(b_j + c_j) = \sum_{i+j=n} (a_ib_j + a_ic_j) = \sum_{i+j=n} a_ib_j + \sum_{i+j=n} a_ic_j \\ = \text{Coeff. of } x^n \text{ in } f(x)g(x) + \text{coeff. of } x^n \text{ in } f(x)h(x) \\ = \text{Coeff. of } x^n \text{ in } [f(x)g(x) + f(x)h(x)].$$

$$\therefore f(x)[g(x) + h(x)] = f(x)g(x) + f(x)h(x).$$

Similarly we can prove the right distributive law.

Hence $R[x]$ is a ring. This is called the ring of all polynomials over R . The zero element of this ring is the zero polynomial

$$0x^0 + 0x + 0x^2 + 0x^3 + \dots$$

2.5 R as a Subset of $R[x]$ or Imbedding of R into $R[x]$

Theorem: If R is an arbitrary ring and R' is the set of constant polynomials in $R[x]$, then R' is isomorphic to R .

Proof: We have $R' = \{ax^0 + 0x + 0x^2 + 0x^3 + \dots \text{ such that } a \in R\}$.

Let $\phi: R \rightarrow R'$ such that

$$\phi(a) = ax^0 + 0x + 0x^2 + 0x^3 + \dots \quad \forall a \in R.$$

ϕ is one-one since

$$\phi(a) = \phi(b) \Rightarrow ax^0 + 0x + 0x^2 + \dots = bx^0 + 0x + 0x^2 + \dots \Rightarrow a = b.$$

Also ϕ is obviously onto R' .

$$\text{Again } \phi(a + b) = (a + b)x^0 + 0x + 0x^2 + 0x^3 + \dots$$

$$= [ax^0 + 0x + 0x^2 + \dots] + [bx^0 + 0x + 0x^2 + \dots] = \phi(a) + \phi(b).$$

$$\text{Also } \phi(ab) = abx^0 + 0x + 0x^2 + 0x^3 + \dots$$

$$= (ax^0 + 0x + 0x^2 + \dots)(bx^0 + 0x + 0x^2 + \dots) = \phi(a)\phi(b).$$

$\therefore \phi$ is an isomorphism of R onto R' . Hence $R \cong R'$.

Since R is isomorphic to R' , therefore in $R[x]$ we can identify R' by R i.e., all the constant polynomials in $R[x]$ can be replaced by the corresponding elements of R . This replacement will not affect the addition and multiplication of polynomials.

Hence in future we shall write 0 in place of the zero polynomial. If $ax^0 + 0x + 0x^2 + \dots$ is any constant polynomial in $R[x]$, then we shall simply write a in place of this polynomial.

Also in place of $a_0 x^0 + a_1 x + a_2 x^2 + \dots$ we shall write $a_0 + a_1 x + a_2 x^2 + \dots$. If we are to multiply $f(x)$ by a constant polynomial $ax^0 + 0x + 0x^2 + \dots$, then we shall write $af(x)$ in place of $(ax^0 + 0x + \dots) f(x)$.

2.6 Polynomials over an Integral Domain

Theorem 1: *If D is an integral domain, then the polynomial ring $D[x]$ is also an integral domain.*

Proof: Let D be a commutative ring without zero divisors and with unity element 1. As proved in article 2.4, $D[x]$ is also a ring. To prove that $D[x]$ is an integral domain we should prove that (i) $D[x]$ is commutative, (ii) is without zero divisors and (iii) possesses the unity element.

$D[x]$ is commutative: Let $f(x) = a_0 + a_1 x + a_2 x^2 + \dots$

and $g(x) = b_0 + b_1 x + b_2 x^2 + \dots$ be any two elements of $D[x]$.

If n is any non-negative integer, then the coefficient of x^n in $f(x)g(x)$ is

$$\begin{aligned} &= \sum_{i+j=n} a_i b_j = \sum_{i+j=n} b_j a_i, \text{ since } D \text{ is commutative} \\ &= \text{coefficient of } x^n \text{ in } g(x)f(x). \end{aligned}$$

$\therefore f(x)g(x) = g(x)f(x)$. Hence $D[x]$ is a commutative ring.

If 1 is the unity element of D , then the constant polynomial

$$1 + 0x + 0x^2 + 0x^3 + \dots$$

is the unity element of $D[x]$. We have

$$\begin{aligned} &[a_0 + a_1 x + a_2 x^2 + \dots][1 + 0x + 0x^2 + 0x^3 + \dots] \\ &= (a_0 \cdot 1) + (a_1 \cdot 1)x + (a_2 \cdot 1)x^2 + \dots = a_0 + a_1 x + a_2 x^2 + \dots \end{aligned}$$

$\therefore$ the polynomial $1 + 0x + 0x^2 + \dots$ or simply 1 is the unity element of $D[x]$.

$D[x]$ is without zero divisors: Let

$$\begin{aligned} f(x) &= a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m, a_m \neq 0 \\ g(x) &= b_0 + b_1 x + b_2 x^2 + \dots + b_n x^n, b_n \neq 0 \end{aligned}$$

be two non-zero elements of $D[x]$.

Then $f(x)g(x)$ cannot be a zero polynomial i.e., the zero element of $D[x]$. The reason is that at least one coefficient of $f(x)g(x)$ namely $a_m b_n$ of x^{m+n} is $\neq 0$ because a_m, b_n are non-zero elements of D and D is without zero divisors.

Hence $D[x]$ is an integral domain.

Theorem 2: *If R is an integral domain with unity element, then any unit in $R[x]$ must already be a unit in R .*

Proof: If R is an integral domain with unity element 1, then $R[x]$ is also an integral domain with unity element. Further the constant polynomial 1 is the unity element of $R[x]$. Let $f(x)$ be a unit in $R[x]$ i.e., let $f(x)$ be an invertible element of $R[x]$. Let $g(x)$ be the inverse of $f(x)$ in $R[x]$. Then

$$f(x)g(x) = 1$$

$$\Rightarrow \deg[f(x)g(x)] = 0 \quad [\because \text{degree of the constant polynomial } 1 \text{ is } 0]$$

$$\Rightarrow \deg f(x) + \deg g(x) = 0$$

$$\Rightarrow \deg f(x) = 0, \deg g(x) = 0$$

$$\Rightarrow \text{both } f(x) \text{ and } g(x) \text{ are constant polynomials in } R[x].$$

Let $f(x) = a \in R$ and $g(x) = b \in R$. Then $ab = 1 \Rightarrow a$ is a unit in R . Thus any unit in $R[x]$ must already be a unit in R .

Note: If $a \in R$ is a unit in R , then $a \in R[x]$ is also a unit in $R[x]$. If b is the inverse of a in R , then the constant polynomial b is the inverse of a in $R[x]$.

2.7 Polynomials over a Field

Theorem 1: If F is a field, then the set $F[x]$ of all polynomials over F is an integral domain.

Proof: Every field is an integral domain. So give the same proof as we have given in article 2.4 and then in article 2.6.

We shall call the set $F[x]$ as the polynomial domain over the field F .

Theorem 2: The polynomial domain $F[x]$ over a field F is not a field.

Proof: In order to show that $F[x]$ is not a field, we should show that there exists a non-zero element of $F[x]$ which has no multiplicative inverse. Let $f(x)$ be any member of $F[x]$ such that $\deg f(x)$ is greater than zero. The inverse of $f(x)$ cannot be the zero polynomial because the product of $f(x)$ and the zero polynomial will be equal to the zero polynomial and not equal to the unity element of $F[x]$ which is the polynomial $1 + 0x + 0x^2 + \dots$. Suppose now $g(x)$ is any non-zero polynomial. Then F being a field, we have

$$\deg[f(x)g(x)] = \deg f(x) + \deg g(x) > 0$$

because $\deg f(x) > 0$ and $\deg g(x) \geq 0$.

The degree of the unity element of $F[x]$ is 0. Hence $f(x)g(x)$ cannot be equal to the unity element of $F[x]$. Thus $f(x)$ does not possess multiplicative inverse.

$\therefore F[x]$ is not a field.

Important: The only invertible elements of $F[x]$ are constant polynomials excluding the zero polynomial. No member of $F[x]$ whose degree is greater than 0 is invertible.

Illustrative Examples

Example 1: Show that if a ring R has no zero divisors, then the ring $R[x]$ has also no zero divisors.

Solution: It is given that a ring R has no zero divisors and we have to show that the ring $R[x]$ has also no zero divisors.

Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_mx^m, a_m \neq 0$

and $g(x) = b_0 + b_1x + b_2x^2 + \dots + b_nx^n, b_n \neq 0$

be two non-zero elements of $R[x]$.

Then $f(x)g(x)$ cannot be the zero polynomial i.e., the zero element of $R[x]$. The reason is that at least one coefficient of $f(x)g(x)$ namely a_mb_n of x^{m+n} is $\neq 0$ because a_m, b_n are non-zero elements of R and R is without zero divisors.

Thus in $R[x]$ the product of no two non-zero elements can be the zero element. Hence the ring $R[x]$ has no zero divisors.

Example 2: Consider the following polynomials over the ring $(\mathbb{I}_8, +_8, \times_8)$:

$$f(x) = 2 + 6x + 4x^2, \quad g(x) = 2x + 4x^2, \quad h(x) = 2 + 4x$$

and find (i) $\deg[f(x) + g(x)]$

(ii) $\deg[f(x)g(x)]$

(iii) $\deg[h(x)h(x)]$.

Solution: (i) We have $f(x) + g(x) = (2 + 6x + 4x^2) + (0 + 2x + 4x^2)$

$$= (2 +_8 0) + (6 +_8 2)x + (4 +_8 4)x^2 = 2 + 0x + 0x^2 = 2.$$

Thus $f(x) + g(x)$ is a non-zero constant polynomial and so

$$\deg[f(x) + g(x)] = 0.$$

(ii) We have $f(x)g(x) = (2 + 6x + 4x^2)(2x + 4x^2)$

$$\begin{aligned} &= (2 \times_8 2)x + [(2 \times_8 4) +_8 (6 \times_8 2)]x^2 \\ &\quad + [(6 \times_8 4) +_8 (4 \times_8 2)]x^3 + (4 \times_8 4)x^4 \\ &= 4x + (0 +_8 4)x^2 + (0 +_8 0)x^3 + 0x^4 \\ &= 4x + 4x^2 + 0x^3 + 0x^4 = 4x + 4x^2 \end{aligned}$$

$$\therefore \deg[f(x)g(x)] = 2.$$

(iii) We have $h(x)h(x) = (2 + 4x)(2 + 4x)$

$$\begin{aligned} &= (2 \times_8 2) + [(2 \times_8 4) +_8 (4 \times_8 2)]x + (4 \times_8 4)x^2 \\ &= 4 + (0 +_8 0)x + 0x^2 = 4 + 0x + 0x^2 = 4. \end{aligned}$$

Thus $h(x)h(x)$ is a non-zero constant polynomial and so $\deg[h(x)h(x)] = 0$.

2.8 Ring of Polynomials in n Variables over an Integral Domain

Definition: Let R be an integral domain. Then the ring of polynomials in the n -variables $x_1, \dots, x_n$ over R is denoted by $R[x_1, \dots, x_n]$ and is defined as follows :

Let $R_1 = R[x_1]$, the polynomial ring in x_1 over R ,

$R_2 = R_1[x_2]$, the polynomial ring in x_2 over R_1 ,

$R_3 = R_2[x_3]$, the polynomial ring in x_3 over R_2 ,

.....

.....

$R_n = R_{n-1}[x_n]$, the polynomial ring in x_n over R_{n-1} .

Then R_n is called the ring of polynomials in $x_1, \dots, x_n$ over R and we write $R_n = R[x_1, \dots, x_n]$.

Theorem 1: If R is an integral domain, then so is $R[x_1, \dots, x_n]$.

Proof: If R is an integral domain then $R_1 = R[x_1]$ is also an integral domain. Now R_1 is an integral domain implies that $R_2 = R_1[x_2] = R[x_1, x_2]$ is also an integral domain. Continuing this process a finite number of times we see that $R[x_1, \dots, x_n]$ is an integral domain.

Theorem 2: If F is a field, then $F[x_1, \dots, x_n]$ is an integral domain.

Proof: If F is a field, then $F_1 = F[x_1]$ is an integral domain.

Now F_1 is an integral domain implies that $F_2 = F_1[x_2] = F[x_1, x_2]$ is also an integral domain. Continuing this process a finite number of times we see that $F[x_1, \dots, x_n]$ is an integral domain.

Note: If F is a field, then $F[x_1, \dots, x_n]$ is an integral domain.

Now we can construct the field of quotients of the integral domain $F[x_1, \dots, x_n]$. This field is called the *field of rational functions in $x_1, \dots, x_n$ over F* and is denoted by $F(x_1, \dots, x_n)$.

2.9 Divisibility of Polynomials over a Field

Suppose F is a field. Then $F[x]$ is an integral domain. If $a(x) \neq 0$ and $f(x)$ are elements of $F[x]$, then $a(x)$ is a *divisor (or factor)* of $f(x)$ if and only if there is a polynomial $b(x)$ in $F[x]$ such that $f(x) = a(x)b(x)$. Symbolically we write $a(x) \mid f(x)$.

A **unit** is an element of $F[x]$ which has multiplicative inverse. All the polynomials of zero degree belonging to $F[x]$ are units of $F[x]$. Thus the non-zero elements of F are the only units of $F[x]$.

If $f(x)$ and $g(x)$ are polynomials in $F[x]$, then we call $f(x)$ and $g(x)$ associates if $f(x) = c g(x)$ for some $0 \neq c \in F$. It can be easily proved that two non-zero polynomials $f(x)$ and $g(x)$ in $F[x]$ are associates if and only if $f(x) \mid g(x)$ and $g(x) \mid f(x)$.

If $f(x)$ is any non-zero polynomial in $F[x]$, then $f(x)$ is always divisible by its associates and by all units of $F[x]$. These divisors of $f(x)$ are called its *improper divisors*. All other divisors of $f(x)$, if there are any, are called its *proper divisors*.

Definition of an irreducible polynomial over a field: Let F be a field and $f(x)$ be a non-zero and non-unit polynomial in $F[x]$ i.e., $f(x)$ be a polynomial of positive degree. Then $f(x)$ is said to be irreducible over F (or prime) if it has no proper divisors in $F[x]$; $f(x)$ is reducible over F if it has a proper divisor in $F[x]$.

Thus a positive degree polynomial $f(x)$ in $F[x]$ is irreducible over F if whenever $f(x) = a(x)b(x)$ with $a(x), b(x) \in F[x]$ then one of $a(x)$ or $b(x)$ is a unit in $F[x]$ i.e., has degree 0. Also $f(x)$ is reducible over F if and only if we can find two polynomials $a(x)$ and $b(x)$ in $F[x]$ such that $f(x) = a(x)b(x)$ and none of $a(x)$ and $b(x)$ is a unit in $F[x]$ i.e., has degree 0.

Irreducibility depends on the field. The polynomial $x^2 - 2$ is irreducible over the field of rational numbers while it is reducible over the field of real numbers, since $x^2 - 2 = (x + \sqrt{2})(x - \sqrt{2})$.

The polynomial $x^2 + 1$ is irreducible over the field of real numbers while it is reducible over the field of complex numbers since $x^2 + 1 = (x + i)(x - i)$.

Monic polynomials: Definition:

Let $f(x) = a_0 + a_1x + \dots + a_nx^n$, with $a_n \neq 0$, be a polynomial in $F[x]$ over an arbitrary field F . If the leading coefficient a_n of $f(x)$ is equal to 1, the unity element of F , then the polynomial $f(x)$ will be called monic.

The polynomial $2x - 3x^3$ over the field of real numbers is not monic since its leading coefficient is -3 . But the polynomial $x^3 - 3x + 4$ over the field of real numbers is monic since its leading coefficient is 1.

Greatest common divisor of two polynomials over a field:

Definition: Suppose F is any field. Let $f(x)$ and $g(x)$ be two elements of $F[x]$. A greatest common divisor of $f(x)$ and $g(x)$ is a non-zero polynomial $d(x)$ such that

- (i) $d(x) \mid f(x)$ and $d(x) \mid g(x)$
- (ii) If $c(x)$ is a polynomial such that $c(x) \mid f(x)$ and $c(x) \mid g(x)$ then $c(x) \mid d(x)$.

Relatively prime polynomials: Two polynomials $f(x)$ and $g(x) \in F[x]$ are said to be relatively prime if their greatest common divisor is 1, the unity element of F .

2.10 Division Algorithm for Polynomials over a Field

Theorem: Let $f(x), g(x) \neq 0$ be any two polynomials of the polynomial domain $F[x]$, over the field F . Then there exist uniquely two polynomials $q(x)$ and $r(x)$ in $F[x]$ such that

$$f(x) = q(x) g(x) + r(x) \text{ where either } r(x) = 0 \text{ or } \deg r(x) < \deg g(x).$$

Proof: Suppose

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m, \quad a_m \neq 0$$

$$\text{and } g(x) = b_0 + b_1 x + b_2 x^2 + \dots + b_n x^n, \quad b_n \neq 0.$$

If degree m of $f(x)$ is smaller than the degree n of $g(x)$ or if $f(x) = 0$, then we have nothing to prove. Because we can always write $f(x) = 0 \cdot g(x) + f(x)$. So in this case $q(x) = 0$, $r(x) = f(x)$ and we have either $f(x) = 0$ or $\deg f(x) < \deg g(x)$.

Now let us assume that $m \geq n$. In this case we shall prove the theorem by induction on m i.e., degree of $f(x)$.

If $m = 0$, then $m \geq n \Rightarrow n = 0$. Therefore $f(x)$ and $g(x)$ are both non-zero constant polynomials, $f(x) = a_0, a_0 \neq 0$, and $g(x) = b_0, b_0 \neq 0$.

We have in this case

$$f(x) = a_0 = (a_0 b_0^{-1}) b_0 + 0 = (a_0 b_0^{-1}) g(x) + 0.$$

Thus the theorem is true when $m = 0$ or when the degree of $f(x)$ is less than 1.

We shall now assume that the theorem is true when $f(x)$ is a polynomial of degree less than m and then we shall show that it is also true if $f(x)$ is of degree m and then the proof will be complete by induction.

$$\text{Let } f_1(x) = f(x) - (a_m b_n^{-1}) x^{m-n} g(x) \quad \dots(1)$$

Obviously $\deg f_1(x) < m$. Therefore by our assumed hypothesis, there exist polynomials $s(x)$ and $r(x)$ such that

$$f_1(x) = s(x) g(x) + r(x),$$

$$\text{where } r(x) = 0 \text{ or } \deg r(x) < \deg g(x).$$

Now putting the value of $f_1(x)$ in (1), we get

$$s(x) g(x) + r(x) = f(x) - (a_m b_n^{-1}) x^{m-n} g(x)$$

$$\text{or } f(x) = [(a_m b_n^{-1}) x^{m-n} + s(x)] g(x) + r(x).$$

If we write $q(x)$ in place of $(a_m b_n^{-1}) x^{m-n} + s(x)$, we get

$$f(x) = q(x) g(x) + r(x)$$

$$\text{where } r(x) = 0 \text{ or } \deg r(x) < \deg g(x).$$

This proves the existence of polynomials $q(x)$ and $r(x)$. Now to show that $q(x)$ and $r(x)$ are unique. Let us assume that

$$f(x) = q_1(x) g(x) + r_1(x) = q_2(x) g(x) + r_2(x).$$

$$\text{Then } q_1(x) g(x) + r_1(x) = q_2(x) g(x) + r_2(x)$$

$$\text{or } [q_1(x) - q_2(x)] g(x) = r_2(x) - r_1(x). \quad \dots(2)$$

If $[q_1(x) - q_2(x)] \neq 0$, then $[q_1(x) - q_2(x)] g(x)$ cannot be equal to the zero polynomial because $g(x) \neq 0$ and $F[x]$ is without zero divisors. Also then the degree of $[q_1(x) - q_2(x)] g(x)$ is at least n , the degree of $g(x)$. But $r_2(x) - r_1(x)$ is either equal to the zero polynomial or else its degree is less than n because the degrees of $r_2(x)$ and $r_1(x)$ are both less than n . Hence the equality (2) among two polynomials holds only if $q_1(x) - q_2(x) = 0$ and $r_2(x) - r_1(x) = 0$ i.e., only if $q_1(x) = q_2(x)$ and $r_1(x) = r_2(x)$.

$\therefore$ the polynomials $q(x)$ and $r(x)$ are unique.

Definition: In the division algorithm, the polynomial $q(x)$ is called the **quotient** on dividing $f(x)$ by $g(x)$ and the polynomial $r(x)$ is called the **remainder**.

Theorem: A polynomial domain $F[x]$ over a field F is a principal ideal ring.

Proof: Obviously $F[x]$ is a commutative ring with unity and without zero divisors. Therefore $F[x]$ is a principal ideal ring if every ideal in $F[x]$ is a principal ideal.

Let S be an arbitrary ideal of $F[x]$. If S is the null ideal, then $S = (0)$ i.e., the ideal of $F[x]$ generated by 0. Therefore S is a principal ideal, so let us suppose that S is not a null ideal. Then there exist non-zero polynomials $f(x)$ in S . Let $g(x)$ be a polynomial of lowest degree m belonging to S . We shall show that S is the principal ideal generated by $g(x)$.

Let $f(x)$ be any arbitrary member of S . By division algorithm there exist two polynomials $q(x) \in F[x]$, $r(x) \in F[x]$, such that

$$f(x) = q(x) g(x) + r(x), \text{ where } r(x) = 0 \text{ or } \deg r(x) < \deg g(x).$$

Since S is an ideal, therefore

$$q(x) \in F[x], g(x) \in S \Rightarrow q(x) g(x) \in S.$$

Also $f(x) \in S, q(x) g(x) \in S \Rightarrow f(x) - q(x) g(x) \in S$.

But $f(x) - q(x) g(x) = r(x)$. Therefore $r(x) \in S$.

Now either $r(x) = 0$ or $\deg r(x) < \deg g(x)$. But we have assumed that $g(x)$ is a polynomial of lowest degree belonging to S . Hence $\deg r(x)$ cannot be less than $\deg g(x)$. Therefore we must have $r(x) = 0$. Then $f(x) = q(x) g(x)$. Thus $g(x) \in S$ is such that $f(x) \in S \Rightarrow f(x) = q(x) g(x)$ for some $q(x) \in F[x]$. Therefore S is a principal ideal of $F[x]$ generated by $g(x)$. Hence $F[x]$ is a principal ideal ring.

Important: A polynomial ring over an arbitrary field is a principal ideal ring. But a polynomial ring over an arbitrary ring is not a principal ideal ring as is obvious from the following example.

Example: Show that the polynomial ring $\mathbf{I}[x]$ over the ring of integers is not a principal ideal ring.

Solution: To prove this statement we shall show that the ideal $(2, x)$ of the ring $\mathbf{I}[x]$ generated by two elements 2 and x of $\mathbf{I}[x]$ is not a principal ideal. Let $(2, x)$ be

a principal ideal in $\mathbf{I}[x]$. Then there will exist a non-zero element $g(x) \in \mathbf{I}[x]$ such that $(2, x) = (g(x))$.

Since $2 \in (g(x))$ and $x \in (g(x))$ therefore there will exist elements $\phi(x)$ and $\psi(x)$ belonging to $\mathbf{I}[x]$ such that

$$2 = \phi(x) g(x), \quad \dots(1)$$

$$x = \psi(x) g(x). \quad \dots(2)$$

From (1), we get $2x = [\phi(x) g(x)] x$ and from (2), we get

$$2x = 2 \psi(x) g(x).$$

$$\therefore 2 \psi(x) g(x) = x \phi(x) g(x). \quad [\because \mathbf{I}[x] \text{ is a commutative ring}]$$

$$\therefore 2 \psi(x) = x \phi(x), \text{ since } g(x) \neq 0, \text{ and } \mathbf{I}[x] \text{ is without zero divisors.}$$

Now $2 \psi(x) = x \phi(x)$ implies that the coefficients of $\phi(x)$ must be even integers. Therefore $\phi(x) = 2h(x)$ where $h(x)$ is some polynomial in $\mathbf{I}[x]$. Putting this value of $\phi(x)$ in (1) we get

$$2 = 2h(x) g(x) \quad \text{or} \quad 1 = h(x) g(x).$$

Now $1 = h(x) g(x) \Rightarrow 1 \in (g(x))$. Therefore each element of $\mathbf{I}[x]$ will belong to $(g(x))$. Thus we have $\mathbf{I}[x] = (g(x)) = (2, x)$. Therefore each element of $\mathbf{I}[x]$ will belong to $(2, x)$. We shall show that $1 \notin (2, x)$ and this contradiction will mean that $(2, x)$ is not a principal ideal in $\mathbf{I}[x]$.

Now $1 \in (2, x) \Rightarrow$ we can write

$$1 = 2p(x) + x q(x)$$

where $p(x)$ and $q(x)$ are some elements in $\mathbf{I}[x]$.

$$\text{Let } p(x) = a_0 + a_1 x + a_2 x^2 + \dots \text{ and } q(x) = b_0 + b_1 x + b_2 x^2 + \dots$$

$$\text{Then } 1 = 2(a_0 + a_1 x + a_2 x^2 + \dots) + x(b_0 + b_1 x + b_2 x^2 + \dots)$$

$$\text{or } 1 = 2a_0 + (2a_1 + b_0)x + (2a_2 + b_1)x^2 + \dots$$

This equality implies $1 = 2a_0$ where $a_0 \in \mathbf{I}$.

But for no integer a_0 we can have $1 = 2a_0$. Hence $1 \notin (2, x)$.

$\therefore (2, x)$ is not a principal ideal in $\mathbf{I}[x]$.

2.11 Euclidean Algorithm for Polynomials over a Field

Theorem: Let F be a field and $f(x)$ and $g(x)$ be any two polynomials in $F[x]$, not both of which are zero. Then $f(x)$ and $g(x)$ have a greatest common divisor $d(x)$ which can be expressed in the form

$$d(x) = m(x) f(x) + n(x) g(x)$$

for polynomials $m(x)$ and $n(x)$ in $F[x]$.

Proof: Consider the set

$$S = \{s(x) f(x) + t(x) g(x) : s(x), t(x) \in F[x]\}. \quad \dots(1)$$

We claim that S is an ideal of $F[x]$. The proof is as follows :

Let $s_1(x)f(x) + t_1(x)g(x)$ and $s_2(x)f(x) + t_2(x)g(x)$ be any two elements of S .

$$\begin{aligned} \text{Then } [s_1(x)f(x) + t_1(x)g(x)] - [s_2(x)f(x) + t_2(x)g(x)] \\ = [s_1(x) - s_2(x)]f(x) + [t_1(x) - t_2(x)]g(x) \in S \end{aligned}$$

since $s_1(x) - s_2(x)$ and $t_1(x) - t_2(x)$ are both members of $F[x]$.

Also if $\alpha(x)$ be any member of $F[x]$, then

$$\begin{aligned} \alpha(x)[s_1(x)f(x) + t_1(x)g(x)] \\ = [\alpha(x)s_1(x)]f(x) + [\alpha(x)t_1(x)]g(x) \in S. \end{aligned}$$

Therefore S is an ideal of $F[x]$. Now every ideal in $F[x]$ is a principal ideal. Therefore there exists an element $d(x)$ in S such that every element in S is a multiple of $d(x)$.

Since $d(x) \in S$, therefore from (1) we see that there exist elements $m(x), n(x) \in F[x]$ such that

$$d(x) = m(x)f(x) + n(x)g(x).$$

Now $F[x]$ is a ring with unity element 1.

$\therefore$ Putting $s(x) = 1, t(x) = 0$ in (1), we see that $f(x) \in S$. Also putting $s(x) = 0, t(x) = 1$ in (1), we see that $g(x) \in S$.

Now $f(x), g(x)$ are elements of S . Therefore they are both multiples of $d(x)$. Hence $d(x) \mid f(x)$ and $d(x) \mid g(x)$.

Now suppose $c(x) \mid f(x)$ and $c(x) \mid g(x)$.

Then $c(x) \mid [m(x)f(x)]$ and $c(x) \mid [n(x)g(x)]$. Therefore $c(x)$ is also a divisor of $m(x)f(x) + n(x)g(x)$ i.e., $c(x)$ is a divisor of $d(x)$.

Thus $d(x)$ is a greatest common divisor of $f(x)$ and $g(x)$.

Note: If $d(x)$ is a greatest common divisor of $f(x)$ and $g(x)$ then any associate of $d(x)$ i.e., $k d(x)$ where $0 \neq k \in F$ will also be a greatest common divisor of $f(x)$ and $g(x)$. In particular if $0 \neq b$ is the leading coefficient of the polynomial $d(x)$, then the monic polynomial $b^{-1}d(x)$ will also be a greatest common divisor of $f(x)$ and $g(x)$.

Often while defining greatest common divisor of two polynomials over a field we include one more condition in our definition that the greatest common divisor should be a monic polynomial. The advantage of this extra condition is that now we shall get a unique greatest common divisor as shown below:

Suppose $d_1(x)$ and $d_2(x)$ are two monic polynomials and each is a greatest common divisor of $f(x)$ and $g(x)$. Then $d_1(x) \mid d_2(x)$ and $d_2(x) \mid d_1(x)$. Therefore $d_1(x)$ and $d_2(x)$ are associates and we have $d_1(x) = u d_2(x)$ for some $0 \neq u \in F$. Since $d_1(x)$ and $d_2(x)$ are both monic, therefore $u = 1$.

2.12 Unique Factorization Domain

Definition: An integral domain, R , with unity element 1 is a unique factorization domain if

- (a) any non-zero element in R is either a unit or can be written as the product of a finite number of irreducible (prime) elements of R ;
- (b) the decomposition in part (a) is unique upto the order and associates of the irreducible elements.

Thus if R is a unique factorization domain and if $a \neq 0$ is a non-unit in R , then a can be expressed as a product of a finite number of prime elements of R . Also if

$$a = p_1 p_2 p_3 \dots p_n = p_1' p_2' p_3' \dots p_m'$$

where the p_i and p_j' are prime elements of R , then $m = n$ and each p_i , $1 \leq i \leq n$ is an associate of some p_j' , $1 \leq j \leq m$ and conversely each p_k' is an associate of some p_l .

2.13 The Unique Factorization Theorem for Polynomials over a Field

We shall now prove that every polynomial over a field can be factored uniquely into irreducible factors. Before stating the main factorization theorem, we shall give two preliminary theorems that are needed for its proof.

Theorem 1: Let $f(x)$, $g(x)$ and $h(x)$ be polynomials in $F[x]$ for a field F . If $f(x) \mid g(x)h(x)$ and the greatest common divisor of $f(x)$ and $g(x)$ is 1, then $f(x) \mid h(x)$.

Proof: If the greatest common divisor of $f(x)$ and $g(x)$ is 1, then by theorem of article 2.11 there exist polynomials $m(x)$ and $n(x) \in F[x]$ such that

$$1 = m(x)f(x) + n(x)g(x).$$

Multiplying both members of this equation by $h(x)$, we get

$$h(x) = m(x)f(x)h(x) + n(x)g(x)h(x) \quad \dots(1)$$

But $f(x) \mid g(x)h(x)$, so there exists a polynomial $q(x) \in F[x]$ such that

$$g(x)h(x) = q(x)f(x).$$

Substituting this value of $g(x)h(x)$ in (1), we get

$$\begin{aligned} h(x) &= m(x)f(x)h(x) + n(x)q(x)f(x) \\ &= f(x)[m(x)h(x) + n(x)q(x)], \end{aligned}$$

which shows that $f(x)$ is a divisor of $h(x)$.

Hence the theorem.

Theorem 2: If $f(x)$ is an irreducible polynomial in $F[x]$ for a field F and $f(x) \mid g(x)h(x)$ where $g(x), h(x) \in F[x]$ then $f(x)$ divides at least one of $g(x)$ or $h(x)$.

Proof: Suppose that $f(x)$ does not divide $g(x)$. Since $f(x)$ is prime therefore $f(x)$ does not divide $g(x)$ implies that $f(x)$ and $g(x)$ are relatively prime. Therefore the greatest common divisor of $f(x)$ and $g(x)$ is 1. Hence by theorem 1, we get that $f(x) \mid h(x)$.

Corollary: If $f(x)$ is an irreducible polynomial in $F[x]$ for a field F , and if $f(x)$ divides the product $g_1(x)g_2(x)\dots g_n(x)$ of polynomials in $F[x]$, then $f(x)$ divides $g_i(x)$ for some i , $1 \leq i \leq n$.

This result follows immediately by repeated application of theorem 2.

The Unique Factorization Theorem for polynomials over a field:

Let $f(x)$ be a non-zero polynomial in $F[x]$, where F is a field. Then either $f(x)$ is a unit in $F[x]$ or $f(x) = a(x) p_1(x) p_2(x) \dots p_m(x)$, where each $p_i(x)$, $1 \leq i \leq m$, is an irreducible monic polynomial in $F[x]$ and $a \in F$ is the leading coefficient of $f(x)$. Further the factors $p_1(x), p_2(x), \dots, p_m(x)$ are unique except for the order in which they appear.

Proof: We shall prove the theorem in two parts. First we shall prove that $f(x)$ can be factored as required, and then we shall show that the factors are unique.

Let $f(x)$ be a non-zero element of $F[x]$. Then either $f(x)$ is a unit in $F[x]$ i.e., $\deg f(x)$ is 0 or $\deg f(x) > 0$. If $\deg f(x) > 0$, and the leading coefficient of $f(x)$ is a we are to prove that $f(x)$ can be expressed as a product of a and a finite number of irreducible monic polynomials in $F[x]$. The proof will be by induction on the degree of $f(x)$.

Suppose $f(x)$ is of degree one. Let $f(x) = b + ax$ for $a, b \in F$ and $a \neq 0$. We can write $f(x) = a(a^{-1}b + x)$. Therefore the theorem holds in the case where $f(x)$ has degree one since $a^{-1}b + x$ is irreducible and monic.

Now assume, as the induction hypothesis, that every polynomial of degree less than n can be factored as stated in the theorem. Consider an arbitrary polynomial $f(x)$ of degree n having a as its leading coefficient. We can write $f(x) = a f_1(x)$, where $f_1(x) = a^{-1}f(x)$ and $f_1(x)$ is monic. If $f(x)$ is irreducible, then $f_1(x)$ is also irreducible and the theorem holds. If $f(x)$ is reducible, then it can be factored as $f(x) = g(x) h(x)$ where neither $g(x)$ nor $h(x)$ is a unit in $F[x]$. Now the degree of $f(x)$ is equal to the sum of the degrees of $g(x)$ and $h(x)$. Also $g(x)$ and $h(x)$ are not units in $F[x]$, so each of them must be of degree one or larger. Hence both $g(x)$ and $h(x)$ have degrees less than n . Therefore by our induction hypothesis we can write

$$g(x) = c\alpha_1(x)\alpha_2(x)\dots\alpha_s(x), \quad h(x) = d\beta_1(x)\beta_2(x)\dots\beta_t(x)$$

where each $\alpha_i(x)$ and each $\beta_j(x)$ is monic and irreducible and where c and d are leading coefficients of $g(x)$ and $h(x)$ respectively.

Thus $f(x) = cd\alpha_1(x)\alpha_2(x)\alpha_3(x)\dots\alpha_s(x)\beta_1(x)\beta_2(x)\dots\beta_t(x)$.

Since the leading coefficient of $f(x)$ is a , therefore we must have $a = cd$ because each $\alpha(x)$ and each $\beta(x)$ is monic. Therefore

$$f(x) = a\alpha_1(x)\alpha_2(x)\dots\alpha_s(x)\beta_1(x)\beta_2(x)\dots\beta_t(x).$$

The factorization of $f(x)$ satisfies the requirements of the theorem. Hence the theorem holds for all polynomials of degree n , and by the principle of induction, for all polynomials of arbitrary degree.

In order to prove that the factors are unique, let us suppose that $f(x) = a p_1(x) p_2(x) \dots p_m(x) = a q_1(x) q_2(x) \dots q_n(x)$ where each $p(x)$ and each $q(x)$ is irreducible and monic. Then we shall prove that $n = m$ and each $p(x)$ is equal to some $q(x)$ and each $q(x)$ is equal to some $p(x)$. From these two decompositions of $f(x)$, we have

$$p_1(x) p_2(x) \dots p_m(x) = q_1(x) q_2(x) \dots q_n(x).$$

Now $p_1(x) \mid p_1(x) p_2(x) \dots p_m(x)$.

Therefore $p_1(x) \mid q_1(x) q_2(x) \dots q_n(x)$.

By corollary to theorem 2 of this article $p_1(x)$ must divide at least one of $q_1(x), q_2(x), \dots, q_n(x)$. Since $F[x]$ is a commutative ring, therefore without loss of generality we may suppose that $p_1(x)$ divides $q_1(x)$. But $p_1(x)$ and $q_1(x)$ are both irreducible polynomials in $F[x]$ and $p_1(x) \mid q_1(x)$. Therefore $p_1(x)$ and $q_1(x)$ must be associates and we have $q_1(x) = u p_1(x)$ where u is a unit in $F[x]$ i.e., u is a non-zero element of F . Since $q_1(x)$ and $p_1(x)$ are monic therefore u must be equal to 1 and we have $p_1(x) = q_1(x)$. Thus we have

$$p_1(x) p_2(x) \dots p_m(x) = p_1(x) q_2(x) \dots q_n(x).$$

Cancelling $0 \neq p_1(x)$ from both sides, we get

$$p_2(x) p_3(x) \dots p_m(x) = q_2(x) q_3(x) \dots q_n(x) \quad \dots(1)$$

Now we can repeat the above argument on the relation (1) with $p_2(x)$. If $n > m$, then after m steps the left hand side becomes 1 while the right hand side reduces to a product of a certain number of $q(x)$ (the excess of n over m). But the $q(x)$ are irreducible polynomials so they are not units of $F[x]$ i.e., they are not polynomials of zero degree.

So their product will be a polynomial of degree ≥ 1 . So it cannot be equal to 1. Therefore n cannot be greater than m . Then $n \leq m$. Similarly interchanging the roles of $p(x)$ and $q(x)$, we get $m \leq n$. Hence $m = n$.

Also in the above process we have shown that every $p(x)$ is equal to some $q(x)$ and conversely every $q(x)$ is equal to some $p(x)$. Hence the theorem has been completely established.

Thus we can say that *the ring of polynomials over a field is a unique factorization domain.*

2.14 Value of a Polynomial at $x = c$

Definition: Let $f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$ be a polynomial in $F[x]$ for an arbitrary field F and let c be an element of F . Then

$$f(c) = a_0 + a_1 c + a_2 c^2 + \dots + a_n c^n,$$

where the indicated addition and multiplication are the operations in F , is called the value of $f(x)$ at $x = c$.

Obviously $f(c)$ is an element of F .

Zeros of a polynomial: Definition: If $f(x)$ is a polynomial in $F[x]$ for an arbitrary field F , and $f(c) = 0$ for an element $c \in F$, then c is called a zero of $f(x)$.

Polynomial equations and their roots:

Definition: Let $f(x)$ be a polynomial of degree n over a field F . We say that $f(x) = 0$ is an equation over the field F and n is the degree of the equation.

If c is a zero of the polynomial $f(x)$, then c is a root of the equation $f(x) = 0$. A root of an equation is also called a **solution** of the equation.

Remainder Theorem: If $f(x) \in F[x]$ and $a \in F$, for any field F , then $f(a)$ is the remainder when $f(x)$ is divided by $(x - a)$.

Proof: By division algorithm there exist polynomials $q(x)$ and $r(x)$ such that $f(x) = q(x)(x - a) + r(x)$, where either $r(x) = 0$ or $\deg r(x)$ is less than the degree of $x - a$. But the degree of $(x - a)$ is 1. Therefore $r(x)$ has degree 0 or no degree. Hence $r(x)$ is a constant polynomial i.e., $r(x)$ is simply an element, say, r in F . Thus $f(x) = q(x)(x - a) + r$. Putting $x = a$ in this relation, we get

$$f(a) = q(a)(a - a) + r \Rightarrow f(a) = r.$$

Corollary: Factor Theorem: If $f(x) \in F[x]$ and $a \in F$, for a field F , then $x - a$ divides $f(x)$ if and only if $f(a) = 0$.

Proof: By remainder theorem, $f(a)$ is the remainder when $f(x)$ is divided by $(x - a)$. Therefore if $f(a) = 0$, then $(x - a)$ divides $f(x)$.

Conversely, if $f(x)$ is divisible by $(x - a)$ we get

$$f(x) = (x - a)q(x).$$

Putting $x = a$, we get $f(a) = (a - a)q(a) = 0 \cdot q(a) = 0$.

Illustration: Show that the polynomial $x^2 + x + 4$ is irreducible over F , the field of integers modulo 11.

Solution: The field F is $(\{0, 1, \dots, 10\}, +_{11}, \times_{11})$.

Let $f(x) = x^2 + x + 4$.

If $a \in F$, then by a^n we shall mean $a \times_{11} a \times_{11} a \times_{11} a \dots$ upto n times.

$$\begin{aligned} \text{Now } f(0) &= 0^2 +_{11} 0 +_{11} 4 = 4, f(1) = 1^2 +_{11} 1 +_{11} 4 = 6, \\ f(2) &= 2^2 +_{11} 2 +_{11} 4 = 10, f(3) = 3^2 +_{11} 3 +_{11} 4 = 5, f(4) = 2, \\ f(5) &= 1, f(6) = 6^2 +_{11} 6 +_{11} 4 = 2, f(7) = 5, f(8) = 10, f(9) = 6, \\ f(10) &= 4. \end{aligned}$$

Since $f(a) \neq 0 \forall a \in F$, therefore by factor theorem $x - a$ does not divide $f(x) \forall a \in F$. Therefore $f(x)$ has no proper divisors in $F[x]$. Hence $f(x)$ is irreducible over F .

Comprehensive Exercise 1

1. Resolve $x^4 + 4$ into factors over the field $(\{0, 1, 2, 3, 4\}, +_5, \times_5)$.
2. Resolve $x^2 + 1$ into factors over the field Z_5 .
3. Find the solution of the equation $3x = 2$ in the field $(Z_7, +_7, \times_7)$.
4. Show that $f(x) = x^2 + 8x - 2$ is irreducible over the field of rational numbers $\mathbf{Q}$. Is it irreducible over reals? Give reasons for your answer.
5. Let $f(x) = 2x^4 + 3x^3 + 2$ and $g(x) = 3x^5 + 4x^3 + 2x^2 + 3$ be two polynomials over the field $Z_5 = (\{0, 1, 2, 3, 4\}, +_5, \times_5)$.

Determine (i) $\frac{d}{dx} f(x)$, (ii) $f(x) \cdot g(x)$.

6. If $f(x) = 3x^7 + 2x + 3$, $g(x) = 5x^3 + 2x + 6$ be two polynomials over the field $Z_7 = (\{0, 1, 2, 3, 4, 5, 6\}, +_7, \times_7)$, determine
(i) $\frac{d}{dx} f(x)$, (ii) $f(x) \cdot g(x)$, and (iii) $f(x) + g(x)$.
7. Let $f(x) = x^6 + 3x^5 + 4x^2 - 3x + 2$ and $g(x) = x^2 + 2x - 3$ be in $Z_7[x]$. Find
(i) Sum and product of $f(x)$ and $g(x)$ in $Z_7[x]$.
(ii) Two polynomials $q(x)$ and $r(x)$ in $Z_7[x]$ such that
 $f(x) = q(x)g(x) + r(x)$ with degree $r(x) < 2$.

Answers 1

1. $x^4 + 4 = (x + 1)(x + 2)(x + 3)(x + 4)$.
2. $x^2 + 1 = (x + 2)(x + 3)$.
3. $x = 3$ because $3 \times_7 3 = 2$.
4. not irreducible over reals.
5. (i) $3x^3 + 4x^2$.
(ii) $1 + 4x^2 + 2x^3 + x^4 + 2x^5 + x^6 + 3x^7 + 4x^8 + x^9$.
6. (i) 2
(ii) $4 + 4x + 4x^2 + x^3 + 3x^4 + 4x^7 + 6x^8 + x^{10}$.
(iii) $3x^7 + 5x^3 + 4x + 2$.
7. (i) $f(x) + g(x) = x^6 + 3x^5 + 5x^2 + 6x + 6$.

$$f(x) + g(x) = x^8 + 5x^7 + 3x^6 + 5x^5 + 4x^4 + 5x^3 + 5x^2 + 6x + 1$$

Note that in Z_7 , we have $-3 = 4$, $-1 = 6$ etc.

$$(ii) \quad q(x) = x^4 + x^3 + x^2 + x + 5, \quad r(x) = 4x + 3.$$

2.15 Maximal Ideal

An ideal $S \neq R$ in a ring R is said to be a *maximal ideal* of R if whenever U is an ideal of R such that $S \subseteq U \subseteq R$, then either $R = U$ or $S = U$.

In other words an ideal S of a ring R is said to be *maximal ideal* if there exists no ideal properly contained in R which itself properly contains S i.e., if it is impossible to find an ideal which lies between S and the full ring R . For example, in the ring of integers $\mathbf{I}$, the ideal (6) is not maximal since it is properly contained in the ideal (3) , which in turn is properly contained in $\mathbf{I}$. On the other hand, (5) is a maximal ideal since the only ideal properly containing (5) is $\mathbf{I}$ itself.

Theorem: An ideal S of the ring of integers $\mathbf{I}$ is maximal if and only if S is generated by some prime integer.

Proof: We know that every ideal of the ring of integers $\mathbf{I}$ is a principal ideal. Suppose S is an ideal of $\mathbf{I}$ generated by p so that $S = (p)$. Since p and $-p$ generate the same ideal, therefore we can take p as positive.

Now we are to prove that

(i) S is maximal if p is prime.

(ii) p is prime if S is maximal.

First we shall prove (i). Let p be a prime integer such that $(p) = S$. Let T be an ideal of $\mathbf{I}$ such that $S \subseteq T \subseteq \mathbf{I}$. Since T is also a principal ideal of $\mathbf{I}$, let $T = (q)$ where q is some positive integer.

$$\text{Now } S \subseteq T \Rightarrow (p) \subseteq T \Rightarrow p \in T$$

$$\Rightarrow p \in \{xq : x \in \mathbf{I}\} \Rightarrow p = rq \quad \text{for some positive integer } r.$$

Since p is prime, therefore $q = 1$ or $q = p$.

If $q = 1$, we have $T = (q) = (1) = \mathbf{I}$

and if $q = p$, we have $T = (q) = (p) = S$.

Thus either $T = \mathbf{I}$ or $T = S$.

Hence (p) is a maximal ideal of $\mathbf{I}$.

Now we shall prove (ii). Let $(p) = S$ be a maximal ideal. We are to show that p is prime. Let us suppose that p is a composite integer.

Let $p = mn$, $m \neq 1$, $n \neq 1$.

It is obvious that $(p) \subseteq (m) \subseteq \mathbf{I}$.

But since (p) is a maximal ideal, therefore we have

either $(m) = (p)$ or $(m) = \mathbf{I}$.

If $(m) = \mathbf{I}$, then $m = 1$, which is a contradiction.

If $(m) = (p)$, then m must be equal to lp for some integer l since each element of (p) is a multiple of p .

Therefore $p = mn = lpn = pln$. But $p \neq 0$, therefore $ln = 1$. This gives $n = 1$ which is again a contradiction.

Hence p must be a prime integer.

2.16 More Results on Ideals

Theorem 1: Let S_1, S_2 be ideals of a ring R and let

$$S_1 + S_2 = \{s_1 + s_2 : s_1 \in S_1, s_2 \in S_2\}.$$

Then $S_1 + S_2$ is an ideal of R generated by $S_1 \cup S_2$.

Proof: Let $a_1 + a_2 \in S_1 + S_2, b_1 + b_2 \in S_1 + S_2$. Then

$$a_1, b_1 \in S_1 \text{ and } a_2, b_2 \in S_2.$$

We have $(a_1 + a_2) - (b_1 + b_2) = (a_1 - b_1) + (a_2 - b_2)$.

Since S_1 is an ideal, therefore $a_1, b_1 \in S_1 \Rightarrow a_1 - b_1 \in S_1$.

Similarly $a_2 - b_2 \in S_2$.

$$\therefore (a_1 - b_1) + (a_2 - b_2) \in S_1 + S_2.$$

$$\therefore (a_1 + a_2) - (b_1 + b_2) \in S_1 + S_2.$$

$$\therefore S_1 + S_2 \text{ is a subgroup of the additive group of } R.$$

Let r be any element of R . Then

$$r(a_1 + a_2) = ra_1 + ra_2 \in S_1 + S_2; \text{ since } r \in R, a_1 \in S_1 \Rightarrow ra_1 \in S_1$$

and similarly $ra_2 \in S_2$.

Similarly $(a_1 + a_2)r = a_1r + a_2r \in S_1 + S_2$ since $a_1r \in S_1, a_2r \in S_2$.

Hence $S_1 + S_2$ is an ideal of R .

Since $0 \in S_1$ and also $0 \in S_2$, therefore obviously

$$S_1 \subseteq S_1 + S_2 \text{ and } S_2 \subseteq S_1 + S_2.$$

$$\therefore S_1 \cup S_2 \subseteq S_1 + S_2.$$

Thus $S_1 + S_2$ is an ideal of R containing $S_1 \cup S_2$.

Also if S is any ideal of R containing $S_1 \cup S_2$, then S must contain $S_1 + S_2$. Thus $S_1 + S_2$ is the smallest ideal of R containing $S_1 \cup S_2$ i.e., $S_1 + S_2 = (S_1 \cup S_2)$.

Theorem 2: If an ideal U of a ring R contains a unit of R then $U = R$.

Proof: Let R be a ring with unity element 1 . Let u be a unit of R . Then u is an invertible element of R i.e., u^{-1} exists. Let $u \in U$.

Since U is an ideal, therefore

$$u \in U, u^{-1} \in R \Rightarrow uu^{-1} \in U \Rightarrow 1 \in U.$$

Now let x be any element of R . Then

$$x \in R, 1 \in U \Rightarrow x 1 \in U \Rightarrow x \in U.$$

$$\therefore R \subseteq U.$$

Also $U \subseteq R$ as U is an ideal of R . Hence $R = U$.

Theorem 3: Let R be a commutative ring with unity and a, b be two non-zero elements of R .

Then $(a) = (b)$ iff $a \mid b$ and $b \mid a$.

Proof: Here $(a) =$ the principal ideal of R generated by a

$$= \{ ax : x \in R \}.$$

Similarly $(b) =$ the principal ideal of R generated by b .

$$\text{Let } (a) = (b).$$

$$\text{Then } (a) \subseteq (b) \Rightarrow a \in (b)$$

$$\Rightarrow a = rb \text{ for some } r \in R$$

$$\Rightarrow b \mid a \text{ i.e., } b \text{ is a divisor of } a.$$

Similarly $(a) = (b)$

$$\Rightarrow (b) \subseteq (a) \Rightarrow b \in (a)$$

$$\Rightarrow b = sa \text{ for some } s \in R \Rightarrow a \mid b.$$

$$\text{Thus } (a) = (b) \Rightarrow a \mid b \text{ and } b \mid a.$$

Conversely let $a \mid b$ and $b \mid a$.

Now $a \mid b \Rightarrow b = pa$ for some $p \in R$. Let y be any element of (b) .

$$\text{Then } y = ub \text{ for some } u \in R$$

$$= u(pa) = (up)a \in (a) \text{ since } up \in R.$$

$$\text{Thus } y \in (b) \Rightarrow y \in (a).$$

$$\therefore (b) \subseteq (a).$$

$$\text{Thus } a \mid b \Rightarrow (b) \subseteq (a). \text{ Similarly } b \mid a \Rightarrow (a) \subseteq (b).$$

$$\text{Consequently } a \mid b, b \mid a \Rightarrow (a) = (b).$$

Corollary: Let R be an integral domain with unity and a, b be two non-zero elements of R .

Then $(a) = (b)$ iff a and b are associates.

Theorem 4: An ideal S of a commutative ring R with unity is maximal if and only if the residue class ring R/S is a field.

Proof: Since R is a commutative ring with unity, therefore R/S is also a commutative ring with unity. The zero element of the ring R/S is S and the unity element is the coset $S + 1$ where 1 is the unity element of R .

Let the ideal S be maximal. Then to prove that R/S is a field.

Let $S + b$ be any non-zero element of R/S . Then $S + b \neq S$ i.e., $b \notin S$. To prove that $S + b$ is invertible.

If (b) is the principal ideal of R generated by b , then $S + (b)$ is also an ideal of R . Since $b \notin S$, therefore the ideal S is properly contained in $S + (b)$. But S is a maximal ideal of R . Hence we must have $S + (b) = R$.

Since $1 \in R$, therefore we must obtain 1 on adding an element of S to an element of (b) . Therefore there exists an element $a \in S$ and $\alpha \in R$ such that

$$a + \alpha b = 1 \quad [\text{Note that } (b) = \{\alpha b : \alpha \in R\}]$$

$$\therefore 1 - \alpha b = a \in S.$$

Consequently $S + 1 = S + \alpha b = (S + \alpha)(S + b)$.

$$\therefore S + \alpha = (S + b)^{-1}. \text{ Thus } S + b \text{ is invertible.}$$

$$\therefore R/S \text{ is a field.}$$

Conversely, let S be an ideal of R such that R/S is a field. We shall prove that S is a maximal ideal of R .

Let S' be an ideal of R properly containing S i.e., $S \subseteq S'$ and $S \neq S'$. Then S will be maximal if $S' = R$. The elements of R contained in S already belong to S' since $S \subseteq S'$. Therefore R will be a subset of S' if every element α of R not contained in S also belongs to S' . If $\alpha \in R$ is such that $\alpha \notin S$, then $S + \alpha \neq S$ i.e., $S + \alpha$ is a non-zero element of R/S . Also S' properly contains S . Therefore there exists an element β of S' not contained in S so that $S + \beta$ is also a non-zero element of R/S . Now the non-zero elements of R/S form a group with respect to multiplication because R/S is a field. Therefore there exists a non-zero element $S + \gamma$ of R/S such that

$$(S + \gamma)(S + \beta) = S + \alpha.$$

[We may take $S + \gamma = (S + \alpha)(S + \beta)^{-1}$].

$$\text{Now } (S + \gamma)(S + \beta) = S + \alpha$$

$$\Rightarrow S + \gamma\beta = S + \alpha \Rightarrow \gamma\beta - \alpha \in S \Rightarrow \gamma\beta - \alpha \in S'. \quad [\because S \subseteq S']$$

Now S' is an ideal. Therefore $\gamma \in R, \beta \in S' \Rightarrow \gamma\beta \in S'$.

Again $\gamma\beta \in S', \gamma\beta - \alpha \in S' \Rightarrow \gamma\beta - (\gamma\beta - \alpha) \in S' \text{ i.e., } \alpha \in S'.$

Thus $R \subseteq S'$.

Also $S' \subseteq R$ as S' is an ideal of R .

$$\therefore S' = R.$$

Hence the theorem.

2.17 Prime Ideals

Prime Ideal: Definition: Let R be a ring and S an ideal in R . Then S is said to be a prime ideal of R if $ab \in S, a, b \in R$ implies that either a or b is in S .

For example, in the ring of integers $\mathbf{I}$, the principal ideal (7) is prime. Obviously if ab is in (7) , then a or b must be a multiple of 7. On the other hand, (6) is not a prime ideal in $\mathbf{I}$ since, in particular, $12 = 3 \times 4$ is in (6) , yet neither 3 nor 4 is an element of (6) .

Theorem 1: Let R be a commutative ring and S an ideal of R . Then the ring of residue classes R/S is an integral domain if and only if S is a prime ideal.

Proof: Let R be a commutative ring and S an ideal of R . Then

$$R/S = \{S + a : a \in R\}.$$

Let $S + a, S + b$ be any two elements of R/S . Then $a, b \in R$.

$$\begin{aligned} \text{We have } (S + a)(S + b) &= S + ab \\ &= S + ba \quad [\because R \text{ is a commutative ring}] \\ &= (S + b)(S + a). \end{aligned}$$

$\therefore R/S$ is a commutative ring.

Now let S be a prime ideal of R . Then we are to prove that R/S is an integral domain. For this we are to show that R/S is without zero divisors. The zero element of the ring R/S is the residue class S itself. Let $S + a, S + b$ be any two elements of R/S .

$$\begin{aligned} \text{Then } (S + a)(S + b) &= S \quad (\text{the zero element of } R/S) \\ \Rightarrow S + ab &= S \Rightarrow ab \in S \\ \Rightarrow \text{either } a \text{ or } b &\text{ is in } S, \text{ since } S \text{ is a prime ideal} \\ \Rightarrow \text{either } S + a = S &\text{ or } S + b = S \quad [\text{Note that } a \in S \Leftrightarrow S + a = S] \\ \Rightarrow \text{either } S + a &\text{ or } S + b \text{ is the zero element of } R/S. \\ \therefore R/S &\text{ is without zero divisors.} \end{aligned}$$

Since R/S is a commutative ring without zero divisors, therefore R/S is an integral domain.

Conversely, let R/S be an integral domain. Then we are to prove that S is a prime ideal of R . Let a, b be any two elements in R such that $ab \in S$. We have

$$ab \in S \Rightarrow S + ab = S \Rightarrow (S + a)(S + b) = S.$$

$$\begin{aligned} \text{Since } R/S &\text{ is an integral domain, therefore it is without zero divisors. Therefore} \\ (S + a)(S + b) &= S \quad (\text{the zero element of } R/S) \\ \Rightarrow \text{either } S + a &\text{ or } S + b \text{ is zero} \Rightarrow \text{either } S + a = S \text{ or } S + b = S \\ \Rightarrow \text{either } a \in S &\text{ or } b \in S \Rightarrow S \text{ is a prime ideal.} \end{aligned}$$

This completes the proof of the theorem.

Note: If R is a ring with unity, then R/S is also a ring with unity. The residue class $S + 1$ is the unity element of R/S . Therefore if we define an integral domain as a commutative ring with unity and without zero divisors, even then the above theorem will be true. But in that case R must be a commutative ring with unity.

Theorem 2: Let R be a commutative ring with unity. Then every maximal ideal of R is a prime ideal.

Proof: R is a commutative ring with unit element. Let S be a maximal ideal of R . Then R/S is a field.

Now every field is an integral domain. Therefore R/S is also an integral domain. Hence by theorem 1, S is a prime ideal of R . This completes the proof of the theorem.

But it should be noted that the converse of the above theorem is not true i.e., every prime ideal is not necessarily a maximal ideal.

Illustrative Examples

Example 3: Let $\mathbf{R}$ be the field of real numbers and S the set of all those polynomials $f(x) \in \mathbf{R}[x]$ such that $f(0) = 0 = f(1)$. Prove that S is an ideal of $\mathbf{R}[x]$. Is the residue class ring $\mathbf{R}[x]/S$ an integral domain? Give reasons for your answer.

Solution: Let $f(x), g(x)$ be any elements of S . Then

$$f(0) = 0 = f(1) \text{ and } g(0) = 0 = g(1).$$

Let
$$h(x) = f(x) - g(x).$$

Then
$$h(0) = f(0) - g(0) = 0 - 0 = 0 \text{ and } h(1) = f(1) - g(1) = 0 - 0 = 0.$$

Thus $h(0) = 0 = h(1)$. Therefore $h(x) \in S$.

Thus $f(x), g(x) \in S \Rightarrow h(x) = f(x) - g(x) \in S$.

Further let $f(x)$ be any element of S and $r(x)$ be any element of $\mathbf{R}[x]$.

Then $f(0) = 0 = f(1)$, by definition of S .

Let
$$t(x) = r(x)f(x) = f(x)r(x). \quad [\because \mathbf{R}[x] \text{ is a commutative ring}]$$

Then
$$t(0) = r(0)f(0) = r(0) \cdot 0 = 0$$

and
$$t(1) = r(1)f(1) = r(1) \cdot 0 = 0.$$

$\therefore t(x) \in S$.

Thus $r(x) \in \mathbf{R}[x], f(x) \in S \Rightarrow r(x)f(x) \in S$.

Hence S is an ideal of $\mathbf{R}[x]$.

Now we claim that S is not a prime ideal of $\mathbf{R}[x]$. Let $f(x) = x(x-1)$. Then $f(0) = 0(0-1) = 0$, and $f(1) = 1(1-1) = 0$.

Thus $f(x) = x(x-1)$ is an element of S .

Now let $p(x) = x, q(x) = x-1$.

We have $p(1) = 1 \neq 0$. Therefore $p(x) \notin S$. Also $q(0) = 0-1 = -1 \neq 0$.

Therefore $q(x) \notin S$. Thus $x(x-1) \in S$ while neither $x \in S$ nor $x-1 \in S$. Hence S is not a prime ideal of $\mathbf{R}[x]$.

Since S is not a prime ideal of $\mathbf{R}[x]$, therefore the residue class ring $\mathbf{R}[x]/S$ is not an integral domain.

Example 4: Let R be the ring of all real valued continuous functions defined on the closed interval $[0, 1]$. Let $M = \{f(x) \in R : f\left(\frac{1}{3}\right) = 0\}$.

Show that M is a maximal ideal of R .

Solution: First of all we observe that M is non-empty because the real valued function $e(x)$ on $[0, 1]$ defined by

$$e(x) = 0 \quad \forall \quad x \in [0, 1]$$

belongs to M .

Now let $f(x), g(x)$ be any two elements of M . Then

$$f\left(\frac{1}{3}\right) = 0, g\left(\frac{1}{3}\right) = 0, \text{ by definition of } M.$$

$$\text{Let} \quad h(x) = f(x) - g(x).$$

$$\text{Then} \quad h\left(\frac{1}{3}\right) = f\left(\frac{1}{3}\right) - g\left(\frac{1}{3}\right) = 0 - 0 = 0.$$

Therefore $h(x) \in M$.

$$\text{Thus } f(x), g(x) \in M \Rightarrow h(x) = f(x) - g(x) \in M.$$

Further let $f(x)$ be any element of M and $r(x)$ be any element of R . Then

$$f\left(\frac{1}{3}\right) = 0, \text{ by definition of } M.$$

$$\text{Let } t(x) = r(x) f(x) = f(x) r(x). \quad [\because R \text{ is a commutative ring.}]$$

$$\text{Then } t\left(\frac{1}{3}\right) = r\left(\frac{1}{3}\right) f\left(\frac{1}{3}\right) = r\left(\frac{1}{3}\right) \cdot 0 = 0. \quad \text{Therefore } t(x) \in M.$$

$$\text{Thus } r(x) \in R, f(x) \in M \Rightarrow r(x) f(x) \in M.$$

Hence M is an ideal of R .

Clearly $M \neq R$ because $i(x) \in R$ given by $i(x) = 1 \quad \forall \quad x \in [0, 1]$ does not belong to M .

The ring R is with unity and the element $i(x)$ is its unity element.

Let N be an ideal of R properly containing M i.e., $M \subseteq N$ and $M \neq N$. Then M will be a maximal ideal of R if $N = R$, which will be so if the unity $i(x)$ of R belongs to N . Since M is a proper subset of N , therefore there exists $\lambda(x) \in N$ such that $\lambda(x) \notin M$. This means $\lambda\left(\frac{1}{3}\right) \neq 0$. Put $\lambda\left(\frac{1}{3}\right) = c$ where $c \neq 0$.

Let us define $\beta(x) \in R$ by $\beta(x) = c \quad \forall \quad x \in [0, 1]$. Now consider $\mu(x) \in R$ given by

$$\mu(x) = \lambda(x) - \beta(x).$$

$$\text{We have } \mu\left(\frac{1}{3}\right) = \lambda\left(\frac{1}{3}\right) - \beta\left(\frac{1}{3}\right) = c - c = 0.$$

Therefore $\mu(x) \in M$ and so $\mu(x)$ also belongs to N because N is a super-set of M . Now N is an ideal of R and $\lambda(x), \mu(x)$ are in N . Therefore $\lambda(x) - \mu(x) = \beta(x)$ is also an element of N .

Now define $\gamma(x) \in R$ by $\gamma(x) = 1/c \quad \forall \quad x \in [0, 1]$. Since N is an ideal of R , therefore $\gamma(x) \in R$ and $\beta(x) \in N \Rightarrow \gamma(x) \beta(x) \in N$. We shall show that $\gamma(x) \beta(x) = i(x)$.

For every $x \in [0, 1]$, we have $\gamma(x) \beta(x) = (1/c) c = 1$.

Therefore $\gamma(x) \beta(x) = i(x)$, by definition of $i(x)$.

Thus the unity element $i(x)$ of R belongs to N and consequently $N = R$.

Hence M is a maximal ideal of R .

Example 5: If R is a finite commutative ring (i.e., has only a finite number of elements) with unit element prove that every prime ideal of R is a maximal ideal of R .

Solution: Let R be a finite commutative ring with unit element. Let S be a prime ideal of R . Then to prove that S is a maximal ideal of R .

Since S is a prime ideal of R , therefore the residue class ring R/S is an integral domain. Now

$$R/S = \{ S + a : a \in R \}.$$

Since R is a finite ring, therefore R/S is a finite integral domain. But every finite integral domain is a field. Therefore R/S is a field. Since R is a commutative ring with unity and R/S is a field, therefore S is a maximal ideal of R .

Example 6: Give an example of a ring in which some prime ideal is not a maximal ideal.

Solution: Let $\mathbf{I}[x]$ be the ring of polynomials over the ring of integers $\mathbf{I}$. Let S be the principal ideal of $\mathbf{I}[x]$ generated by x i.e., let $S = (x)$. We shall show that (x) is prime but not maximal.

We have $S = (x) = \{ x f(x) : f(x) \in \mathbf{I}[x] \}$.

First we shall prove that S is prime.

Let $a(x), b(x) \in \mathbf{I}[x]$ be such that $a(x)b(x) \in S$. Then there exists a polynomial $c(x) \in \mathbf{I}[x]$ such that

$$x c(x) = a(x)b(x). \quad \dots(1)$$

$$\begin{aligned} \text{Let } a(x) &= a_0 + a_1x + a_2x^2 + \dots, b(x) = b_0 + b_1x + b_2x^2 + \dots, \\ c(x) &= c_0 + c_1x + c_2x^2 + \dots \end{aligned}$$

Then (1) becomes

$$x(c_0 + c_1x + \dots) = (a_0 + a_1x + \dots)(b_0 + b_1x + \dots).$$

Equating the constant term on both sides, we get

$$a_0 b_0 = 0 \Rightarrow a_0 = 0 \text{ or } b_0 = 0. \quad [\because \mathbf{I} \text{ is without zero divisors}]$$

$$\text{Now } a_0 = 0 \Rightarrow a(x) = a_1x + a_2x^2 + \dots$$

$$\Rightarrow a(x) = x(a_1 + a_2x + \dots) \Rightarrow a(x) \in (x).$$

$$\text{Similarly } b_0 = 0 \Rightarrow b(x) = b_1x + b_2x^2 + \dots$$

$$\Rightarrow b(x) = x(b_1 + b_2x + \dots) \Rightarrow b(x) \in (x).$$

$$\text{Thus } a(x)b(x) \in (x) \Rightarrow \text{either } a(x) \in (x) \text{ or } b(x) \in (x).$$

Hence (x) is a prime ideal.

Now we shall show that (x) is not a maximal ideal of $\mathbf{I}[x]$. For this we must show an ideal N of $\mathbf{I}[x]$ such that (x) is properly contained in N , while N itself is properly contained in $\mathbf{I}[x]$. The ideal $N = (x, 2)$ serves this purpose.

Obviously $(x) \subseteq (x, 2)$. In order to show that (x) is properly contained in $(x, 2)$ we must show an element of $(x, 2)$ which is not in (x) . Clearly $2 \in (x, 2)$. We shall show that $2 \notin (x)$. Let $2 \in (x)$. Then we can write,

$$2 = x f(x) \text{ for some } f(x) \in \mathbf{I}[x].$$

$$\text{Let } f(x) = a_0 + a_1x + \dots$$

$$\text{Then } 2 = x f(x) \Rightarrow 2 = x(a_0 + a_1x + \dots)$$

$$\Rightarrow 2 = a_0x + a_1x^2 + \dots \Rightarrow 2 = 0 + a_0x + a_1x^2 + \dots$$

$$\Rightarrow 2 = 0 \quad [\text{by equality of two polynomials}]$$

But $2 \neq 0$ in the ring of integers. Hence $2 \notin (x)$. Thus (x) is properly contained in $(x, 2)$.

Now obviously $(x, 2) \subseteq \mathbf{I}[x]$. In order to show that $(x, 2)$ is properly contained in $\mathbf{I}[x]$ we must show an element of $\mathbf{I}[x]$ which is not in $(x, 2)$. Clearly $1 \in \mathbf{I}[x]$. We shall show that $1 \notin (x, 2)$. Let $1 \in (x, 2)$. Then we have a relation of the form

$$1 = x f(x) + 2 g(x), \text{ where } f(x), g(x) \in \mathbf{I}[x].$$

$$\text{Let } f(x) = a_0 + a_1x + \dots, g(x) = b_0 + b_1x + \dots$$

$$\text{Then } 1 = x(a_0 + a_1x + \dots) + 2(b_0 + b_1x + \dots)$$

$$\Rightarrow 1 = 2b_0 \quad [\text{Equating constant term on both sides}]$$

But there is no integer b_0 such that $1 = 2b_0$.

Hence $1 \notin (x, 2)$. Thus $(x, 2)$ is properly contained in $\mathbf{I}[x]$.

Therefore (x) is not a maximal ideal of $\mathbf{I}[x]$.

2.18 Euclidean Rings or Euclidean Domains

Definition: Let R be an integral domain i.e., let R be a commutative ring without zero divisors. Then R is said to be a Euclidean ring if to every non-zero element $a \in R$ we can assign a non-negative integer $d(a)$ such that:

(i) For all $a, b \in R$, both non-zero, $d(ab) \geq d(a)$.

(ii) For any $a, b \in R$ and $b \neq 0$, there exist $q, r \in R$ such that $a = qb + r$ where either $r = 0$ or $d(r) < d(b)$. (Lucknow 2007)

The second part of the above definition is known as **division algorithm**. Also we do not assign a value to $d(0)$. Thus $d(a)$ will remain undefined when $a = 0$. Also $d(a)$ will be called d -value of a and $d(a)$ must be some non-negative integer for every non-zero element $a \in R$.

Illustration 1: The ring of integers is a Euclidean ring.

Solution: Let $(\mathbf{I}, +, \cdot)$ be the ring of integers where

$$\mathbf{I} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

Let the d function on the non-zero elements of $\mathbf{I}$ be defined as

$$d(a) = |a| \quad \forall \quad 0 \neq a \in \mathbf{I}.$$

Now if $0 \neq a \in \mathbf{I}$, then $|a|$ is a non-negative integer. Thus we have assigned a non-negative integer to every non-negative element $a \in \mathbf{I}$.

$$[d(-5) = |-5| = 5, d(-1) = |-1| = 1, d(4) = |4| = 4 \text{ etc.}]$$

Further if $a, b \in \mathbf{I}$ and are both non-zero, then

$$|ab| = |a||b| \Rightarrow |ab| \geq |a| \quad [\because |b| \geq 1 \text{ if } 0 \neq b \in \mathbf{I}]$$

$$\Rightarrow d(ab) \geq d(a).$$

Finally we know that if $a \in \mathbf{I}$ and $0 \neq b \in \mathbf{I}$, then there exist two integers q and r such that

$$a = qb + r \text{ where } 0 \leq r < |b|$$

i.e., where either $r = 0$ or $1 \leq r < |b|$

i.e., where either $r = 0$ or $d(r) < d(b)$.

It should be noted that $d(b) = |b|$ and if r is a positive integer then $r = |r| = d(r)$.

Therefore the ring of integers is a Euclidean ring.

Illustration 2: The ring of polynomials over a field is a Euclidean ring.

Solution: Let $F[x]$ be the ring of polynomials over a field F . Let the d function on the non-zero polynomials in $F[x]$ be defined as

$$d[f(x)] = \deg f(x), \quad \forall 0 \neq f(x) \in F[x].$$

Now if $0 \neq f(x) \in F[x]$, then $\deg f(x)$ is a non-negative integer.

Thus we have assigned a non-negative integer to every non-zero element $f(x)$ in $F[x]$.

Further if $f(x), g(x) \in F[x]$ and are both non-zero polynomials, then

$$\deg[f(x)g(x)] = \deg f(x) + \deg g(x)$$

$$\Rightarrow \deg[f(x)g(x)] \geq \deg f(x) \quad [\because \deg g(x) \geq 0]$$

$$\Rightarrow d[f(x)g(x)] \geq d[f(x)].$$

Finally we know that if $f(x) \in F[x]$ and $0 \neq g(x) \in F[x]$, then there exist two polynomials $q(x)$ and $r(x)$ in $F[x]$ such that

$$f(x) = q(x)g(x) + r(x)$$

where either $r(x) = 0$ or $\deg r(x) < \deg g(x)$

i.e., where either $r(x) = 0$ or $d[r(x)] < d[g(x)]$.

Hence the ring of polynomials over a field is a Euclidean ring.

Illustration 3: Every field is a Euclidean ring.

Solution: Let F be any field. Let the d function on the non-zero elements of F be defined as

$$d(a) = 0 \quad \forall 0 \neq a \in F.$$

Thus we have assigned the integer zero to every non-zero element in F .

If a and b are non-zero elements in F then ab is also a non-zero element in F . We have therefore

$$d(ab) = 0 = d(a).$$

Thus we have $d(ab) \geq d(a)$.

Finally if $a \in F$ and $0 \neq b \in F$, then we can write

$$a = (ab^{-1})b + 0$$

i.e., $a = qb + r$, where $q = ab^{-1}$ and $r = 0$.

Hence every field is a Euclidean ring.

Illustration 4: The ring of Gaussian integers is a Euclidean ring.

Solution: Let $(G, +, \cdot)$ be the ring of Gaussian integers where

$$G = \{x + iy : x, y \in \mathbf{I}\}.$$

Let the d function on the non-zero elements of G be defined as

$$d(x + iy) = x^2 + y^2 \quad \forall \quad 0 + i0 \neq x + iy \in G.$$

Now if $x + iy$ is a non-zero element of G , then $(x^2 + y^2)$ is a non-negative integer.

Thus we have assigned a non-negative integer to every non-zero element of G .

If $x + iy$ and $m + in$ are two non-zero elements of G , then

$$\begin{aligned} d[(x + iy)(m + in)] &= d[(xm - ny) + i(my + xn)] \\ &= (xm - ny)^2 + (my + xn)^2 = x^2m^2 + n^2y^2 + m^2y^2 + x^2n^2 \\ &= (x^2 + y^2)(m^2 + n^2) \\ &\geq x^2 + y^2. \end{aligned} \quad [\because m^2 + n^2 \geq 1]$$

Thus $d[(x + iy)(m + in)] \geq d(x + iy)$.

Now to show the existence of division algorithm in G .

Let $\alpha \in G$ and let β be a non-zero element of G . Let $\alpha = x + iy$ and $\beta = m + in$.

Define a complex number λ by the equation

$$\lambda = \frac{\alpha}{\beta} = \frac{x + iy}{m + in} = \frac{(x + iy)(m - in)}{m^2 + n^2} = p + iq,$$

where p, q are rational numbers.

Here λ is not necessarily a Gaussian integer.

Also division by β is possible since $\beta \neq 0$.

Let p' and q' be the nearest integers to p and q respectively.

Then obviously $|p - p'| \leq \frac{1}{2}$, $|q - q'| \leq \frac{1}{2}$.

Let $\lambda' = p' + iq'$. Then λ' is a Gaussian integer.

$$\text{Now } \lambda = \frac{\alpha}{\beta} \Rightarrow \alpha = \lambda \beta \Rightarrow \alpha = \lambda' \beta + (\lambda - \lambda') \beta.$$

$$\text{Thus } \alpha = \lambda' \beta + (\lambda - \lambda') \beta. \quad \dots(1)$$

Since α, β, λ' are Gaussian integers, therefore from (1) it implies that $(\lambda - \lambda') \beta$ is also a Gaussian integer.

Now if p and q are integers then $p = p', q = q'$.

So $\lambda - \lambda' = (p - p') + i(q - q') = 0 + i0$. Thus $(\lambda - \lambda') \beta = 0 + i0$.

If p and q are not both integers, then $(\lambda - \lambda') \beta$ is a non-zero Gaussian integer and we have

$$\begin{aligned}
d[(\lambda - \lambda')\beta] &= d[\{(p - p') + i(q - q')\}(m + in)] \\
&= [(p - p')^2 + (q - q')^2](m^2 + n^2) = [(p - p')^2 + (q - q')^2] d(\beta) \\
&\leq \left[\frac{1}{4} + \frac{1}{4}\right] d(\beta) \quad \left[\because (p - p')^2 \leq \frac{1}{4}, (q - q')^2 \leq \frac{1}{4}\right] \\
&= \frac{1}{2} d(\beta) < d(\beta).
\end{aligned}$$

Thus $\alpha = \lambda' \beta + (\lambda - \lambda') \beta$, where λ' and $(\lambda - \lambda') \beta$ are Gaussian integers and either $(\lambda - \lambda') \beta = 0$

or $d[(\lambda - \lambda') \beta] < d(\beta)$.

Hence the ring of Gaussian integers is a Euclidean ring.

2.19 Properties of Euclidean Rings

Theorem 1: Every Euclidean ring is a principal ideal ring.

Proof: Let R be a Euclidean ring. Let S be an arbitrary ideal of R . If S is the null ideal, then $S = (0)$ i.e., the ideal of R generated by 0. Therefore S is a principal ideal. So let us suppose that S is not a null ideal. Then there exist elements in S not equal to zero. Let b be a non-zero element in S such that $d(b)$ is minimum i.e., there exists no element c in S such that $d(c) < d(b)$. We shall show that $S = (b)$ i.e., S is nothing but the ideal generated by b .

Let a be any element of S . Then by definition of Euclidean ring there exist elements q and r in R such that

$$a = qb + r \text{ where either } r = 0 \text{ or } d(r) < d(b).$$

Now $q \in R, b \in S \Rightarrow qb \in S$ because S is an ideal.

Further $a \in S, qb \in S \Rightarrow a - qb = r \in S$.

Thus $r \in S$ and we have either $r = 0$ or $d(r) < d(b)$.

If $r \neq 0$, then $d(r) < d(b)$ which contradicts our assumption that no element in S has d -value smaller than $d(b)$. Therefore we must have $r = 0$.

Then $a = qb$.

Thus every element a in S is a multiple of the generating element b . Thus $a \in S \Rightarrow a \in (b)$. Therefore $S \subseteq (b)$.

Again if xb is any element of (b) , then $x \in R$.

Now $x \in R, b \in S \Rightarrow xb \in S$. Therefore $(b) \subseteq S$.

Hence $S = (b)$.

Thus every ideal S in R is a principal ideal. Therefore R is a principal ideal ring.

Theorem 2: Every Euclidean ring possesses unity element.

Proof: Let R be a Euclidean ring. Obviously R is an ideal of R . Therefore there exists an element $u_0 \in R$ such that $R = (u_0)$ i.e., there exists an element $u_0 \in R$ such that

every element in R is a multiple of u_0 . Since, in particular, $u_0 \in R$ therefore there exists an element $c \in R$ such that $u_0 = u_0 c$. We shall show that c is the required unity element. Let now a be any element of R . Since $a \in R$, therefore there exists some $x \in R$ such that $a = u_0 x$.

$$\begin{aligned}
 \text{Now } ac &= (u_0 x) c & [\because a = u_0 x] \\
 &= (u_0 c) x & [\because R \text{ is a commutative ring}] \\
 &= u_0 x & [\because u_0 = u_0 c] \\
 &= a. & [\because a = u_0 x]
 \end{aligned}$$

Thus we have $ac = a = ca \quad \forall a \in R$.

Hence c is the unity element.

Theorem3: Let R be a Euclidean ring and a and b be any two elements in R , not both of which are zero. Then a and b have a greatest common divisor d which can be expressed in the form

$$d = \lambda a + \mu b \text{ for some } \lambda, \mu \in R.$$

Proof: Consider the set

$$S = \{sa + tb : s, t \in R\}. \quad \dots(1)$$

We claim that S is an ideal of R . The proof is as follows :

Let $x = s_1 a + t_1 b$, and $y = s_2 a + t_2 b$ be any two elements of S .

Then $s_1, t_1, s_2, t_2 \in R$. We have

$$x - y = (s_1 a + t_1 b) - (s_2 a + t_2 b) = (s_1 - s_2) a + (t_1 - t_2) b \in S$$

since $s_1 - s_2$ and $t_1 - t_2$ are both elements of R .

Thus S is a subgroup of R with respect to addition.

Also if u be any element of R , then

$$xu = ux = u(s_1 a + t_1 b) = (us_1) a + (ut_1) b \in S \text{ since } us_1, ut_1 \in R.$$

Therefore S is an ideal of R . Now every ideal in R is a principal ideal. Therefore there exists an element d in S such that every element in S is a multiple of d .

Since $d \in S$, therefore from (1), we see that there exist elements $\lambda, \mu \in R$ such that

$$d = \lambda a + \mu b.$$

Now R is a ring with unity element 1.

$\therefore$ Putting $s = 1, t = 0$ in (1), we see that $a \in S$. Also putting $s = 0, t = 1$ in (1), we see that $b \in S$.

Now a, b are elements of S . Therefore they are both multiples of d . Hence $d \mid a$ and $d \mid b$.

Now suppose $c \mid a$ and $c \mid b$.

Then $c \mid \lambda a$ and $c \mid \mu b$. Therefore c is also a divisor of $\lambda a + \mu b$ i.e., c is a divisor of d .

Thus d is a greatest common divisor of a and b .

Theorem 4: Let a, b and c be any elements of a Euclidean ring R . Let $(a, b) = 1$ i.e., let the greatest common divisor of a and b be 1. If $a \mid bc$, then $a \mid c$.

Proof: If the greatest common divisor of a and b is 1, then by theorem 3 there exist elements λ and μ in R such that

$$1 = \lambda a + \mu b. \quad \dots(1)$$

Multiplying both members of (1) by c , we get

$$c = \lambda ac + \mu bc. \quad \dots(2)$$

But $a \mid bc$, so there exists an element $q \in R$ such that $bc = qa$.

Substituting this value of bc in (2), we get

$$c = \lambda ac + \mu qa = (\lambda c + \mu q) a,$$

which shows that a is a divisor of c . Hence the theorem.

Theorem 5: If p is a prime element in the Euclidean ring R and $p \mid ab$ where $a, b \in R$ then p divides at least one of a or b .

Proof: If p divides a , we are nothing to prove. So suppose that p does not divide a . Since p is prime and p does not divide a , therefore p and a are relatively prime i.e., the greatest common divisor of p and a is 1. Hence by theorem 4, we get that $p \mid b$.

Corollary: If p is a prime element in the Euclidean ring R and p divides the product $a_1 a_2 \dots a_n$ of elements in R , then p divides at least one of $a_1, a_2, \dots, a_n$.

The result follows immediately by repeated application of theorem 5.

Theorem 6: Let R be a Euclidean ring. Let a and b be two non-zero elements in R . Then

(i) if b is a unit in R , $d(ab) = d(a)$.

(ii) if b is not a unit in R , $d(ab) > d(a)$.

Proof: (i) By the definition of Euclidean ring, we have

$$d(ab) \geq d(a). \quad \dots(1)$$

Now suppose that b is a unit in R . Then b is invertible and b^{-1} exists. We can write

$$a = (ab) b^{-1}.$$

$$\therefore d(a) = d[(ab) b^{-1}].$$

But by the definition of Euclidean ring, we have

$$d[(ab) b^{-1}] \geq d(ab).$$

$$\therefore d(a) \geq d(ab). \quad \dots(2)$$

From (1) and (2), we conclude that

$$d(ab) = d(a).$$

(ii) Suppose now that b is not a unit in R . Since a and b are non-zero elements of the Euclidean ring R , therefore ab is also a non-zero element of R . Now $a \in R$ and $0 \neq ab \in R$, therefore by definition of Euclidean ring there exist elements q and r in R such that

$$a = q(ab) + r \quad \dots(3)$$

where either $r = 0$ or $d(r) < d(ab)$.

If $r = 0$, then

$$\begin{aligned} a &= qab \Rightarrow a - qab = 0 \Rightarrow a(1 - qb) = 0 \\ \Rightarrow 1 - qb &= 0 & [\because a \neq 0 \text{ and } R \text{ is free of zero divisors}] \\ \Rightarrow qb &= 1 \Rightarrow b \text{ is inversible} \Rightarrow b \text{ is a unit in } R. \end{aligned}$$

Thus we get a contradiction. Hence r cannot be zero. Therefore we must have

$$d(r) < d(ab) \quad \text{i.e.,} \quad d(ab) > d(r). \quad \dots(4)$$

Also from (3), we have $r = a - qab = a(1 - qb)$.

$$\therefore d(r) = d[a(1 - qb)].$$

$$\text{But } d[a(1 - qb)] \geq d(a).$$

$$\therefore d(r) \geq d(a). \quad \dots(5)$$

From (4) and (5), we conclude that $d(ab) > d(a)$.

Theorem 7: *The necessary and sufficient condition that the non-zero element a in the Euclidean ring R is a unit is that*

$$d(a) = d(1). \quad \text{(Lucknow 2011)}$$

Proof: Let a be a unit in R . Then to prove that $d(a) = d(1)$.

By the definition of Euclidean ring

$$d(1a) \geq d(1) \Rightarrow d(a) \geq d(1). \quad \dots(1)$$

Since a is a unit in R , therefore a^{-1} exists and we have

$$1 = aa^{-1} \Rightarrow d(1) = d(aa^{-1}).$$

$$\text{But } d(aa^{-1}) \geq d(a).$$

$$\therefore d(1) \geq d(a). \quad \dots(2)$$

From (1) and (2), we conclude that $d(a) = d(1)$.

Conversely let $d(a) = d(1)$. Then to prove that a is a unit in R . If a is not a unit in R , then by theorem 6, we have

$$d(1a) > d(1) \Rightarrow d(a) > d(1).$$

Thus we get a contradiction. Hence a must be a unit in R .

Theorem 8: *Let R be a Euclidean ring. Then every non-zero element in R is either a unit in R or can be written as a product of a finite number of prime elements of R .*

Proof: Let a be a non-zero element of R . We are to prove that either a is a unit in R or it can be written as a product of a finite number of prime elements of R . We shall prove the result by induction on $d(a)$ i.e., by induction on the d -value of a .

Let us first start the induction. We have $a = 1a$. Therefore $d(a) \geq d(1)$. Thus 1 is an element in R which has the minimal d -value. If $d(a) = d(1)$, then a is a unit in R . [See theorem 7]. Thus the result of the theorem is true if $d(a) = d(1)$ and so we have started the induction.

Now assume as our induction hypothesis that the theorem is true for all non-zero elements $x \in R$ such that $d(x) < d(a)$. Then we shall show that the theorem is

true for a also. If a is a prime element of R , the theorem is obviously true. So suppose that a is not prime. Then we can write $a = bc$ where neither b nor c is a unit in R . Since both b and c are not units in R , therefore $d(bc) > d(b)$ and $d(bc) > d(c)$. But $d(a) = d(bc)$. Therefore we have $d(b) < d(a)$ and $d(c) < d(a)$. So by our induction hypothesis each of b and c can be written as a product of a finite number of prime elements of R .

Let $b = p_1 p_2 \dots p_n$, $c = q_1 q_2$ where the p 's and q 's are prime elements of R .

Then $a = bc = p_1 p_2 \dots p_n q_1 q_2 \dots q_m$.

Thus we have written a as a product of a finite number of prime elements of R . This completes the induction and so the theorem has been proved.

Theorem 9: Unique Factorization theorem: *Let R be a Euclidean ring and a be a non-zero non-unit element in R . Suppose that*

$$a = p_1 p_2 \dots p_m = q_1 q_2 \dots q_n$$

where the p 's and q 's are prime elements of R . Then $m = n$ and each p is an associate of some q and each q is an associate of some p .

Proof: We have $p_1 p_2 \dots p_m = q_1 q_2 \dots q_n$. Now p_1 is a divisor of $p_1 p_2 \dots p_m$. Therefore p_1 is also a divisor of $q_1 q_2 \dots q_n$. By Corollary to Theorem 5, p_1 must divide at least one of $q_1, q_2, \dots, q_n$. Since R is a commutative ring, therefore without loss of generality we may suppose that p_1 divides q_1 . But p_1 and q_1 are both prime elements in R . Therefore p_1 and q_1 must be associates and we have $q_1 = up_1$ where u is a unit in R . Thus we have

$$p_1 p_2 \dots p_m = up_1 q_2 \dots q_n.$$

Cancelling $0 \neq p_1$ from both sides, we get

$$p_2 p_3 \dots p_m = uq_2 q_3 \dots q_n. \quad \dots(1)$$

Now we can repeat the above argument on the relation (1) with p_2 . If $n > m$, then after m steps the left hand side becomes 1 while the right hand side reduces to a product of some units in R and certain number of q 's (the excess of n over m). But the q 's are prime elements of R and so they are not units in R . So the product of some units in R and certain number of q 's cannot be equal to 1. Therefore n cannot be greater than m .

Thus $n \leq m$.

Similarly interchanging the roles of p 's and q 's we get $m \leq n$.

Hence $m = n$.

Also in the above process we have shown that every p is an associate of some q and conversely every q is an associate of some p . Hence the theorem has been completely established.

Note: Combining theorems 8 and 9, we can say that *every non-zero element in a Euclidean ring R can be uniquely written (upto associates) as a product of prime elements or is a unit in R . Therefore a Euclidean ring is a Unique Factorization Domain.*

Theorem 10: *An ideal S of the Euclidean ring R is maximal iff S is generated by some prime element of R .*

Proof: We know that every ideal of a Euclidean ring R is a principal ideal. Suppose S is an ideal of R generated by p so that $S = (p)$. Now we are to prove that

(i) S is maximal if p is a prime element of R .

(ii) p is prime if S is maximal.

First we shall prove (i). Let p be a prime element of R such that $(p) = S$. Let T be an ideal of R such that $S \subseteq T \subseteq R$. Since T is also a principal ideal of R , so let $T = (q)$ where $q \in R$.

$$\text{Now } S \subseteq T \Rightarrow (p) \subseteq (q) \Rightarrow p \in (q)$$

$$\Rightarrow p = xq \text{ for some } x \in R \Rightarrow q \mid p.$$

Since p is prime, therefore either q should be a unit in R or q should be an associate of p .

If q is a unit in R , then $T = (q) = R$.

If q is an associate of p , then $T = (q) = (p) = S$.

Thus either $T = R$ or $T = S$.

Now we shall prove (ii). Let $(p) = S$ be a maximal ideal. We are to show that p is prime. Let us suppose that p is composite i.e., p is not prime.

Let $p = mn$ where neither m nor n is a unit in R .

$$\text{Now } p = mn \Rightarrow m \mid p \Rightarrow (p) \subseteq (m).$$

But $(m) \subseteq R$. Therefore we have $(p) \subseteq (m) \subseteq R$.

But (p) is a maximal ideal, therefore we should have either

$$(m) = (p) \text{ or } (m) = R.$$

If $R = (m)$, then $R \subseteq (m)$.

$$\therefore 1 \in R \Rightarrow 1 \in (m) \Rightarrow 1 = ym \text{ for some } y \in R$$

$$\Rightarrow m \text{ is invertible} \Rightarrow m \text{ is a unit in } R.$$

Thus we get a contradiction.

If $(m) = (p)$, then $m \in (p)$. Therefore $m = lp$ for some $l \in R$.

$$\therefore p = mn = lpn = pln.$$

$$\therefore p(1 - ln) = 0 \Rightarrow 1 - ln = 0 \quad [\because p \neq 0 \text{ and } R \text{ is without zero divisors}]$$

$$\Rightarrow ln = 1 \Rightarrow n \text{ is invertible} \Rightarrow n \text{ is a unit in } R.$$

This is again a contradiction. Hence p must be a prime element of R .

2.20 Polynomial Rings over Unique Factorization Domains

We have already defined a unique factorization domain. For the sake of convenience we repeat the definition here.

Unique Factorization Domain: Definition: An integral domain R , with unity element 1 is a unique factorization domain if

- (a) any non-zero element in R is either a unit or can be written as the product of a finite number of irreducible (prime) elements of R .
- (b) the decomposition in part (a) is unique upto the order and associates of the irreducible elements.

In general commutative rings we have defined the greatest common divisors of elements. But the difficulty is that in an arbitrary commutative ring these might not exist. However, in unique factorization domain their existence is assured. Further we know that in an integral domain with unity in case a greatest common divisor of some elements exists, it is unique apart from the distinction between associates.

Theorem 1: Let R be a unique factorization domain and a and b be any two elements in R , not both of which are zero. Then a and b have a greatest common divisor (a, b) in R . Moreover, if a and b are relatively prime (i.e., (a, b) is a unit in R), then $a \mid bc \Rightarrow a \mid c$.

Proof: Suppose a and b are any two elements, not both of which are zero, of a unique factorization domain R . If one of a and b , say, b is 0, then obviously a is the greatest common divisor of a and b . If any of a and b , say a , is a unit in R , then obviously a is the greatest common divisor of a and b . So let us suppose that neither $a = 0$ nor $b = 0$ and none of these is a unit in R . Then each of a and b can be uniquely expressed as the product of a finite number of irreducible elements of R .

$$\text{Let } a = p_1^{m_1} p_2^{m_2} \dots p_r^{m_r}, \quad \dots(1)$$

$$\text{and } b = p_1^{n_1} p_2^{n_2} \dots p_r^{n_r}, \quad \dots(2)$$

where we have arranged the expressions in such a way that the same irreducible factors $p_1, p_2, \dots, p_r$ appear in both. Note that we can definitely do so because the integer 0 can be used as power in any case, if necessary. The elements $p_1, p_2, \dots, p_r$ are all different primes and $m_1, m_2, \dots, m_r, n_1, n_2, \dots, n_r$ are all integers ≥ 0 .

Let $g_i = \text{minimum}(m_i, n_i)$, where $i = 1, 2, \dots, r$. Then obviously

$$p_1^{g_1} p_2^{g_2} \dots p_r^{g_r}$$

is the greatest common divisor of a and b .

This proves the existence of greatest common divisor.

Now suppose that a and b are relatively prime i.e., the greatest common divisor of a and b is a unit in R . Also suppose that

$$a \mid bc.$$

If a is a unit in R , then obviously a is a divisor of c . So let a be not a unit in R . Then a can be uniquely expressed as the product of a finite number of prime elements of R .

$$\text{Let } a = q_1 q_2 \dots q_s,$$

where $q_1, q_2, \dots, q_s$ are prime elements of R .

We have $a \mid bc \Rightarrow bc = ka$ for some $k \in R$

$$\Rightarrow bc = kq_1 q_2 \dots q_s. \quad \dots(3)$$

Since each element of R can be uniquely expressed as the product of a finite number of prime elements of R , therefore each of the prime elements $q_1, q_2, \dots, q_s$ must occur as a factor of either b or c . But none of $q_1, q_2, \dots, q_s$ can be a factor of b because otherwise a and b will not remain relatively prime. Therefore each of $q_1, q_2, \dots, q_s$ must be a factor of c . Hence

$$q_1 q_2 \dots q_s \text{ is a divisor of } c \Rightarrow a \mid c.$$

Note: In a similar manner we can prove that if $a_1, \dots, a_n$ are any n elements of a unique factorization domain, they possess a greatest common divisor which will be unique apart from the distinction between associates. Thus if g_1, g_2 are two greatest common divisors of these n elements, then by the definition of greatest common divisor, we have

$$g_1 \mid g_2 \text{ and } g_2 \mid g_1$$

$$\Rightarrow g_1 \text{ and } g_2 \text{ are associates}$$

$$\Rightarrow g_1 = u g_2 \text{ where } u \text{ is a unit in } R.$$

Thus the greatest common divisor of some elements is unique within units of R .

Theorem 2: If a is a prime element of a unique factorization domain R and b, c are any elements of R , then

$$a \mid bc \Rightarrow a \mid b \text{ or } a \mid c.$$

Proof: If $a \mid b$, then obviously the theorem is proved. So let a be not a divisor of b . Since a is a prime element of R and a is not a divisor of b , therefore we claim that a and b are relatively prime. Since a is a prime element of R , therefore the only divisors of a are the associates of a or the units of R . Now an associate of a cannot be a divisor of b otherwise a itself will be a divisor of b while we have assumed that a is not a divisor of b . Thus the units of R are the only divisors of a which also divide b . Therefore the greatest common divisor of a and b is a unit of R .

Since a and b are relatively prime, therefore by theorem 1, we have

$$a \mid bc \Rightarrow a \mid c.$$

This completes the proof of the theorem.

Polynomial rings over unique factorization domains: Let R be a unique factorization domain. Since R is an integral domain with unity, therefore $R[x]$ is also an integral domain with unity. Also any unit, (invertible element) in $R[x]$ must already be a unit in R . Thus the only units in $R[x]$ are the units of R . A polynomial $p(x) \in R[x]$ is *irreducible* over R i.e., irreducible as an element of $R[x]$ if whenever $p(x) = a(x)b(x)$ with $a(x), b(x) \in R[x]$, then one of $a(x)$ or $b(x)$ is a unit in $R[x]$ i.e., a unit in R . For example, if $\mathbf{I}$ is the ring of integers, then $\mathbf{I}$ is a unique factorization domain. The polynomial $2x^2 + 4 \in \mathbf{I}[x]$ is a reducible element of $\mathbf{I}[x]$. We have $2x^2 + 4 = 2(x^2 + 2)$. Neither 2 nor $x^2 + 2$ is a unit in

$\mathbf{I}[x]$. On the other hand the polynomial $x^2 + 1 \in \mathbf{I}[x]$ is an irreducible element of $\mathbf{I}[x]$.

Content of a polynomial:

Definition: Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ be a polynomial over a unique factorization domain R . Then the content of $f(x)$ denoted by $c(f)$, is defined as the greatest common divisor of the coefficients $a_0, a_1, \dots, a_n$ of $f(x)$. Obviously the content of $f(x)$ is unique within units of R . Thus if c_1 and c_2 are two contents of $f(x)$, then we must have $c_1 = uc_2$ where u is some unit in R .

Primitive polynomial: Definition: Let R be a unique factorization domain. Then a polynomial $f(x) = a_0 + a_1x + \dots + a_nx^n \in R[x]$ is said to be primitive if the greatest common divisor of its coefficients $a_0, a_1, \dots, a_n$ is a unit in R . Thus a polynomial $f(x)$ is primitive if its content is 1 (that is a unit in R). If

$$f(x) = x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n$$

is a monic polynomial over R , then obviously $f(x)$ is primitive.

If $\mathbf{I}$ is the ring of integers, then $3x^3 - 5x^2 + 7$ is a primitive member of $\mathbf{I}[x]$ while $2x^2 - 4x + 8$ is not a primitive member of $\mathbf{I}[x]$.

Every irreducible polynomial of positive degree belonging to $R[x]$ is necessarily primitive. But an irreducible polynomial of zero degree may not be primitive. For example $3 \in \mathbf{I}[x]$ is irreducible but it is not primitive. Further a primitive polynomial may not be irreducible. For example $x^2 + 5x + 6 \in \mathbf{I}[x]$ is primitive and it is not irreducible. We have $x^2 + 5x + 6 = (x + 2)(x + 3)$.

Theorem 3: Let R be a unique factorization domain. Prove that every non-zero member $f(x)$ of $R[x]$ can be written as $f(x) = gf_1(x)$ where $g = c(f)$ and where $f_1(x)$ is primitive. Also prove that this decomposition of $f(x)$ as an element of R by a primitive polynomial in $R[x]$ is unique apart from the distinction between associates.

Proof: Let R be a unique factorization domain and let

$$f(x) = a_0 + a_1x + \dots + a_nx^n \in R[x].$$

Since R is a unique factorization domain, therefore the elements $a_0, a_1, \dots, a_n \in R$ must possess a greatest common divisor. Let $g \in R$ be the greatest common divisor of these elements. Then $g = c(f)$. Let

$$a_i = gb_i \text{ where } i = 0, 1, \dots, n.$$

Then $f(x) = gb_0 + gb_1x + \dots + gb_nx^n = g[b_0 + b_1x + \dots + b_nx^n]$.

Since g is the greatest common divisor of $a_0, a_1, \dots, a_n$, therefore the elements $b_0, b_1, \dots, b_n$ can have no common factor other than units of R . Consequently the polynomial

$$f_1(x) = b_0 + b_1x + \dots + b_nx^n$$

is a primitive member of $R[x]$. Thus we have $f(x) = gf_1(x)$, where $g \in R$ and $f_1(x) \in R[x]$ is primitive.

Now we come to the uniqueness part of the theorem.

If possible, let

$$f(x) = h f_2(x) \text{ where } h \in R \text{ and } f_2(x) \in R[x] \text{ is primitive.}$$

$$\text{Then } g f_1(x) = h f_2(x) \quad \dots(1)$$

Since $f_1(x)$ and $f_2(x)$ are both primitive, therefore the content of the polynomial on the left hand side of (1) is g and the content of the polynomial on the right hand side of (1) is h . But the content of a polynomial is unique upto associates. Therefore g and h are associates

$$\Rightarrow g = hu \text{ where } u \text{ is some unit in } R$$

$$\Rightarrow hu f_1(x) = h f_2(x)$$

$$\Rightarrow u f_1(x) = f_2(x) \quad [\text{by left cancellation law in the integral domain } R[x], \text{ since } h \neq 0]$$

$$\Rightarrow f_1(x) \text{ and } f_2(x) \text{ are associates.}$$

Hence the theorem.

Theorem 4: *If R is a unique factorization domain, then the product of two primitive polynomials in $R[x]$ is again a primitive polynomial in $R[x]$.*

Proof: Let

$$f(x) = a_0 + a_1 x + \dots + a_n x^n \quad \text{and} \quad g(x) = b_0 + b_1 x + \dots + b_m x^m$$

be two primitive polynomials in $R[x]$.

$$\text{Let } h(x) = f(x) g(x) = c_0 + c_1 x + \dots + c_{m+n} x^{m+n}.$$

Suppose $h(x)$ is not primitive. Then all the coefficients of $h(x)$ must be divisible by some prime element p of R . Since $f(x)$ is primitive, therefore the prime element p must not divide some coefficient of $f(x)$. Let a_i be the first coefficient of $f(x)$ which p does not divide. Similarly let b_j be the first coefficient of $g(x)$ which p does not divide. In $f(x) g(x)$, the coefficient of x^{i+j} is

$$c_{i+j} = a_i b_j + (a_{i-1} b_{j+1} + a_{i-2} b_{j+2} + \dots + a_0 b_{i+j}) \\ + (a_{i+1} b_{j-1} + a_{i+2} b_{j-2} + \dots + a_{i+j} b_0)$$

From this relation, we get

$$a_i b_j = c_{i+j} - (a_{i-1} b_{j+1} + a_{i-2} b_{j+2} + \dots + a_0 b_{i+j}) \\ - (a_{i+1} b_{j-1} + a_{i+2} b_{j-2} + \dots + a_{i+j} b_0) \quad \dots(1)$$

Now by our choice of a_i , p is a divisor of each of the elements $a_0, a_1, \dots, a_{i-1}$.

Therefore $p \mid (a_{i-1} b_{j+1} + a_{i-2} b_{j+2} + \dots + a_0 b_{i+j})$.

Similarly, by our choice of b_j , p is a divisor of each of the elements

$b_0, b_1, \dots, b_{j-1}$. Therefore

$$p \mid (a_{i+1} b_{j-1} + a_{i+2} b_{j-2} + \dots + a_{i+j} b_0).$$

Also by assumption $p \mid c_{i+j}$.

Hence from (1), we get

$$p \mid a_i b_j \Rightarrow p \mid a_i \text{ or } p \mid b_j, \text{ since } p \text{ is a prime element of } R.$$

But this is nonsense because according to our assumption p is not a divisor of a_i and also p is not a divisor of b_j .

Hence $h(x)$ must be primitive. This proves the theorem.

Theorem 5: If R is a unique factorization domain and if $f(x), g(x)$ are in $R[x]$, then

$$c(fg) = c(f) c(g) \text{ (upto units).}$$

Proof: The polynomial $f(x)$ in $R[x]$ can be written as $f(x) = a f_1(x)$, where $a = c(f)$ and $f_1(x)$ is primitive. Similarly the polynomial $g(x)$ can be written as $g(x) = b g_1(x)$, where $b = c(g)$ and $g_1(x)$ is primitive. Then

$$f(x) g(x) = ab f_1(x) g_1(x). \quad \dots(1)$$

Since $f_1(x)$ and $g_1(x)$ are both primitive, therefore $f_1(x) g_1(x)$ is also primitive. [Refer theorem 4].

Therefore from (1), we see that the content of $f(x) g(x)$ is either ab or some associate of ab . Thus the content of $f(x) g(x)$ is ab (upto units). Therefore

$$c(fg) = ab = c(f) c(g).$$

This proves the theorem.

Field of Quotients of a unique factorization domain: If R is a unique factorization domain, then R is necessarily an integral domain. Therefore R has a field of quotients. Throughout this section we shall denote the field of quotients of R by F . We can consider $R[x]$ to be a subring of $F[x]$.

Theorem 6: If R is an integral domain (not necessarily a unique factorization domain) and F is its field of quotients, then any element $f(x)$ in $F[x]$ can be written as

$$f(x) = \frac{f_0(x)}{a},$$

where $f_0(x) \in R[x]$ and where $a \in R$.

Proof: Let F be the field of quotients of an integral domain R . Then

$$F = \left\{ \frac{p}{q} : p \in R, 0 \neq q \in R \right\}.$$

Let $f(x)$ be an element of $F[x]$. Let

$$f(x) = \frac{a_0}{b_0} + \frac{a_1}{b_1} x + \dots + \frac{a_n}{b_n} x^n, \text{ where } a_0, a_1, \dots, a_n \in R$$

and $b_0, b_1, \dots, b_n$ are non-zero elements of R .

Now $b_0, b_1, \dots, b_n$ are also non-zero elements of F . So each of them must be invertible. Further $b_0 b_1 \dots b_n$ is also a non-zero element of F and so it is also invertible. Then we can write

$$\begin{aligned} f(x) &= \frac{b_0 b_1 \dots b_n}{b_0 b_1 \dots b_n} \left(\frac{a_0}{b_0} + \frac{a_1}{b_1} x + \dots + \frac{a_n}{b_n} x^n \right) \\ &= \frac{(a_0 b_1 b_2 \dots b_n) + (b_0 a_1 b_2 \dots b_n) x + \dots + (b_0 b_1 \dots b_{n-1} a_n) x^n}{b_0 b_1 \dots b_n} \\ &= \frac{f_0(x)}{a}, \end{aligned}$$

where obviously $f_0(x) = (a_0 b_1 b_2 \dots b_n) + (b_0 a_1 b_2 \dots b_n) x + \dots$
 $+ (b_0 b_1 \dots b_{n-1} a_n) x^n$ is in $R[x]$ and $a = b_0 b_1 \dots b_n$ is in R .

Theorem 7: (Gauss' Lemma): *Let F be the field of quotients of a unique factorization domain R . If the primitive polynomial $f(x) \in R[x]$ can be factored as the product of two polynomials having coefficients in F , then it can be factored as the product of two polynomials having coefficients in R .*

Proof: Let R be a unique factorization domain and F be its field of quotients. Let $f(x) \in R[x]$ be primitive.

Let $f(x) = g(x) h(x)$ where $g(x)$ and $h(x)$ have coefficients in F .

Since $g(x), h(x) \in F[x]$, therefore we can write

$$g(x) = \frac{g_0(x)}{a}, h(x) = \frac{h_0(x)}{b},$$

where $a, b \in R$ and where $g_0(x), h_0(x) \in R[x]$.

Also $g_0(x) = \alpha g_1(x)$, $h_0(x) = \beta h_1(x)$,

where $\alpha = c(g_0)$, $\beta = c(h_0)$ and where $g_1(x), h_1(x)$ are primitive members of $R[x]$. [See theorem 3].

Thus $f(x) = \frac{\alpha\beta}{ab} g_1(x) h_1(x)$

$$\Rightarrow ab f(x) = \alpha\beta g_1(x) h_1(x). \quad \dots(1)$$

Since $g_1(x)$ and $h_1(x)$ are both primitive members of $R[x]$, therefore $g_1(x) h_1(x)$ is also a primitive member of $R[x]$. Therefore from (1), we conclude that $f(x)$ and $g_1(x) h_1(x)$ are associates in $R[x]$. (See theorem 3). Thus

$$f(x) = u g_1(x) h_1(x) \text{ where } u \text{ is a unit in } R[x] \text{ and so a unit of } R.$$

Now $u \in R$, $g_1(x) \in R[x] \Rightarrow u g_1(x)$ is a member of $R[x]$. Also $h_1(x)$ is a member of $R[x]$. Hence $f(x)$ can be factored as the product of two polynomials with coefficients in R .

Note: Let $\mathbf{I}$ be the ring of integers. Then $\mathbf{I}$ is a Euclidean ring and so a unique factorization domain. The field of quotients of $\mathbf{I}$ is the field of rational numbers. If in the above theorem we take $\mathbf{I}$ in place of R , then the statement of the theorem is as follows :

If the primitive polynomial $f(x) \in \mathbf{I}[x]$ can be factored as the product of two polynomials having rational coefficients it can be factored as the product of two polynomials having integer coefficients.

For its proof simply replace R by $\mathbf{I}$ or say, 'let R be the ring of integers'.

Theorem 8: *Let F be the field of quotients of a unique factorization domain R . If $f(x) \in R[x]$ is both primitive and irreducible as an element of $R[x]$, then it is irreducible as an element of $F[x]$. Conversely, if the primitive element $f(x)$ in $R[x]$ is irreducible as an element of $F[x]$, it is also irreducible as an element of $R[x]$.*

Proof: Let $f(x)$ be a primitive member of $R[x]$. Suppose $f(x)$ is irreducible in $R[x]$ but is reducible in $F[x]$. Since F is a field and $f(x)$ is reducible in $F[x]$, therefore we must have $f(x) = g(x)h(x)$, where $g(x), h(x)$ are in $F[x]$ and are of positive degree. Now we can write

$$g(x) = \frac{g_0(x)}{a}, h(x) = \frac{h_0(x)}{b},$$

where $a, b \in R$, and where $g_0(x), h_0(x) \in R[x]$.

Also $g_0(x) = \alpha g_1(x), h_0(x) = \beta h_1(x)$

where $\alpha = c(g_0), \beta = c(h_0)$, and where $g_1(x), h_1(x)$ are primitive members of $R[x]$. [See theorem 3].

$$\text{Thus } f(x) = \frac{\alpha\beta}{ab} g_1(x) h_1(x) \Rightarrow ab f(x) = \alpha\beta g_1(x) h_1(x). \quad \dots(1)$$

Since $g_1(x)$ and $h_1(x)$ are both primitive members of $R[x]$, therefore $g_1(x)h_1(x)$ is also a primitive member of $R[x]$. Therefore from (1), we conclude that $f(x)$ and $g_1(x)h_1(x)$ are associates in $R[x]$. [See theorem 3].

Thus $f(x) = ug_1(x)h_1(x)$ where u is a unit in $R[x]$ and so a unit in R .

Let $ug_1(x) = g_2(x)$. Then

$$f(x) = g_2(x)h_1(x), \text{ where } g_2(x), h_1(x) \in R[x].$$

We have $\deg g_2(x) = \deg g(x)$, and $\deg h_1(x) = \deg h(x)$.

Thus $\deg g_2(x) > 0, \deg h_1(x) > 0$.

Therefore neither $g_2(x)$ nor $h_1(x)$ is a unit in $R[x]$.

Thus $f(x) = g_2(x)h_1(x)$ is a proper factorization of $f(x)$ in $R[x]$. This contradicts the given statement that $f(x)$ is irreducible in $R[x]$. Hence $f(x)$ must be irreducible in $F[x]$.

Converse: Suppose $f(x)$ is a primitive member of $R[x]$ and is irreducible as an element of $F[x]$. Then to prove that $f(x)$ is also irreducible as an element of $R[x]$.

Let $f(x) = g(x)h(x)$, where $g(x), h(x) \in R[x]$.

Then $f(x)$ will be irreducible in $R[x]$ if one of $g(x)$ or $h(x)$ is a unit in $R[x]$ i.e., a unit in R .

Now $g(x), h(x) \in R[x]$ can also be treated as $g(x), h(x) \in F[x]$.

Since $f(x)$ is irreducible as an element of $F[x]$, therefore one of $g(x)$ or $h(x)$ must be of degree 0. Suppose $\deg g(x) = 0$. Then $g(x)$ is a constant polynomial.

Let $g(x) = k \in R$. Then

$$f(x) = kh(x).$$

Now $f(x)$ is a primitive member of $R[x]$. Therefore $c(f)$ is a unit in R . If k is not a unit in R , then content of $kh(x)$ cannot be a unit in R and so it cannot be equal to $c(f)$. Hence k must be a unit in R . Consequently $f(x)$ is irreducible as an element of $R[x]$.

This completes the proof of the theorem.

Theorem 9: Let F be the field of quotients of a unique factorization domain R . If $f_1(x), f_2(x)$ are two primitive members of $R[x]$ and are associates in $F[x]$, then they are also associates in $R[x]$.

Proof: Since $f_1(x), f_2(x)$ are associates in $F[x]$, therefore we have

$$f_1(x) = k f_2(x) \text{ where } 0 \neq k \in F$$

i.e., k is a unit in $F[x]$.

Note that the only units of $F[x]$ are the non-zero elements of F .

Now F is the field of quotients of R . Therefore

$$0 \neq k \in F \Rightarrow k = \frac{g}{h} \text{ where } g, 0 \neq h \in R.$$

$$\therefore f_1(x) = \frac{g}{h} f_2(x) \Rightarrow h f_1(x) = g f_2(x).$$

Since $h, g \in R$ and $f_1(x), f_2(x)$ are primitive members of $R[x]$, therefore by theorem 3, $f_1(x)$ and $f_2(x)$ are associates in $R[x]$.

Theorem 10: If R is a unique factorization domain and if $p(x)$ is a primitive polynomial in $R[x]$, then it can be factored in a unique way as the product of irreducible elements in $R[x]$. Hence show that the polynomial ring $R[x]$ over a unique factorization domain R is itself a unique factorization domain.

Proof: Let F be the field of quotients of a unique factorization domain R . Let $p(x)$ be a primitive member of $R[x]$. We can regard $p(x)$ as a member of $F[x]$. Since F is a field, therefore $F[x]$ is a unique factorization domain. Recall that the ring of polynomials over a field is a unique factorization domain. Therefore $p(x) \in F[x]$ can be factored as

$$p(x) = p_1(x) p_2(x) \dots p_k(x), \text{ where}$$

$p_1(x), p_2(x), \dots, p_k(x)$ are irreducible polynomials in $F[x]$. Now each $p_i(x), 1 \leq i \leq k$, can be written as $p_i(x) = \frac{f_i(x)}{a_i}$, where $a_i \in R$ and $f_i(x) \in R[x]$. Further $f_i(x)$ can be

written as

$$f_i(x) = c_i q_i(x),$$

where $c_i \in R$ and $q_i(x)$ is a primitive member of $R[x]$. Thus each $p_i(x) = \frac{c_i}{a_i} q_i(x)$,

where $c_i, a_i \in R$ and $q_i(x)$ is a primitive member of $R[x]$. Since $p_i(x)$ is irreducible in $F[x]$, therefore $q_i(x)$ must also be irreducible in $F[x]$. Now $q_i(x)$ is a primitive member of $R[x]$ and $q_i(x)$ is irreducible in $F[x]$. Therefore, by converse part of theorem 8, $q_i(x)$ is irreducible in $R[x]$.

Now $p(x) = p_1(x) p_2(x) \dots p_k(x)$

$$= \frac{c_1 c_2 \dots c_k}{a_1 a_2 \dots a_k} q_1(x) q_2(x) \dots q_k(x).$$

$$\therefore a_1 a_2 \dots a_k p(x) = c_1 c_2 \dots c_k q_1(x) q_2(x) \dots q_k(x). \quad \dots(1)$$

Since $q_1(x), \dots, q_k(x)$ are all primitive members of $R[x]$, therefore

$$q_1(x) q_2(x) \dots q_k(x)$$

is also a primitive member of $R[x]$. Further $p(x)$ is primitive.

Therefore from the relation (1), we conclude with the help of theorem 3 that $p(x)$ and $q_1(x) q_2(x) \dots q_k(x)$ are associates in $R[x]$. Therefore

$$p(x) = u q_1(x) q_2(x) \dots q_k(x)$$

where u is some unit in $R[x]$ and hence in R .

If $q_1(x)$ is irreducible in $R[x]$ then $u q_1(x)$ is also irreducible in $R[x]$. If we simply replace $u q_1(x)$ by $q_1(x)$, then we get

$$p(x) = q_1(x) q_2(x) \dots q_k(x).$$

Thus we have factored $p(x)$ in $R[x]$ as a product of irreducible elements.

Now to show that the above factorization of $p(x)$ is unique upto the order and associates of irreducible elements.

Let $p(x) = r_1(x) r_2(x) \dots r_m(x)$, where the $r_i(x)$ are irreducible in $R[x]$. Since $p(x)$ is primitive, therefore each $r_i(x)$ must be primitive. Consequently by theorem 8, each $r_i(x)$ must be irreducible in $F[x]$. But $F[x]$ is a unique factorization domain. Therefore $p(x) \in F[x]$ can be uniquely expressed as the product of irreducible elements of $F[x]$. Hence the $r_i(x)$ and the $q_i(x)$ regarded as the elements of $F[x]$ are equal (upto associates) in some order.

Since $r_i(x)$ and $q_i(x)$ are primitive members of $R[x]$ and are associates in $F[x]$, therefore by theorem 9, they are also associates in $R[x]$. Thus $p(x)$ has a unique factorization as a product of irreducible elements of $R[x]$.

Now we are in a position to prove that if R is a unique factorization domain, then so is $R[x]$.

Let $f(x) \in R[x]$ be arbitrary. Then we can write $f(x)$ in a unique way as $f(x) = c g(x)$ where $c \in R$ and $g(x)$ is a primitive member of $R[x]$.

Now by the above discussion $g(x)$ can be uniquely expressed as the product of irreducible elements of $R[x]$. What about c ?

Let $c = h_1(x) h_2(x) \dots h_s(x)$, where $h_1(x), \dots, h_s(x) \in R[x]$.

We have $0 = \deg c = \deg h_1(x) + \deg h_2(x) + \dots + \deg h_s(x)$

$\Rightarrow$ each $h_i(x)$ must be of degree 0

$\Rightarrow$ each $h_i(x)$ is an element of R .

Thus the only factorization of c as an element of $R[x]$ are those it had as an element of R . In particular if $\alpha \in R$ is irreducible, then $\alpha \in R[x]$ is also irreducible. But R is a unique factorization domain. Therefore $c \in R$ can be uniquely expressed as the product of irreducible elements of R and hence of $R[x]$.

Finally, we conclude that $f(x) = c g(x)$ can be uniquely expressed as the product of irreducible elements of $R[x]$. Hence $R[x]$ is a unique factorization domain.

Corollary 1: If R is a unique factorization domain then so is $R[x_1, \dots, x_n]$.

Proof: If R is a unique factorization domain, then we know that $R_1 = R[x_1]$ is a unique factorization domain. Now R_1 is a unique factorization domain implies that

$$R_2 = R_1[x_2] = R[x_1, x_2]$$

is a unique factorization domain. Continuing this process a finite number of times we conclude that $R[x_1, \dots, x_n]$ is a unique factorization domain.

Corollary 2: *If F is a field, then $F[x_1, x_2, \dots, x_n]$ is a unique factorization domain.*

Proof: If F is a field, then we know that $F_1 = F[x_1]$ is a unique factorization domain. Now F_1 is a unique factorization domain implies that $F_2 = F_1[x_2] = F[x_1, x_2]$ is a unique factorization domain. Continuing this process a finite number of times, we conclude that $F[x_1, \dots, x_n]$ is a unique factorization domain.

Eisenstein's Criterion of Irreducibility:

Theorem 11: *Let F be the field of quotients of a unique factorization domain R . If*

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \in R[x]$$

and p is a prime element of R such that

$$p \mid a_0, p \mid a_1, p \mid a_2, \dots, p \mid a_{n-1}$$

whereas p is not a divisor of a_n and p^2 is not a divisor of a_0 , then $f(x)$ is irreducible in $F[x]$.

Proof: Without loss of generality we may take $f(x)$ to be primitive, for taking out the greatest common factor of its coefficients does not disturb the hypothesis, since p is not a divisor of a_n . Now suppose that $f(x)$ is reducible in $F[x]$. Then $f(x)$ can be factored as the product of two polynomials of positive degree in $F[x]$. Therefore by Gauss lemma $f(x)$ can be factored as the product of two polynomials of positive degree in $R[x]$. Thus if we assume that $f(x)$ is reducible in $F[x]$, then

$$\begin{aligned} f(x) &= a_0 + a_1x + \dots + a_nx^n \\ &= (b_0 + b_1x + \dots + b_rx^r)(c_0 + c_1x + \dots + c_sx^s) \end{aligned} \quad \dots(1)$$

where the b 's and c 's are elements of R and where $r > 0$ and $s > 0$.

We have from (1), $a_0 = b_0c_0$.

Since p is a prime element of R , therefore

$$p \mid a_0 \Rightarrow p \mid b_0 \quad \text{or} \quad p \mid c_0.$$

Since p^2 is not a divisor of a_0 , therefore p cannot divide both b_0 and c_0 . Suppose that

$$p \mid b_0 \text{ and } p \text{ is not a divisor of } c_0.$$

If p is a divisor of all the coefficients $b_0, b_1, \dots, b_r$, then from (1) we see that p is a divisor of all the coefficients of $f(x)$. But p is not a divisor of a_n . Therefore not all the coefficients $b_0, b_1, \dots, b_r$ can be divisible by p . Let b_k , where $k \leq r$, be the first b which is not divisible by p . Then each of $b_0, b_1, \dots, b_{k-1}$ is divisible by p and b_k is not divisible by p .

Also $k < n$, since $r < n$.

Now from (1), we have

$$\begin{aligned} a_k &= b_k c_0 + b_{k-1} c_1 + b_{k-2} c_2 + \dots + b_0 c_k \\ \Rightarrow b_k c_0 &= a_k - b_{k-1} c_1 - b_{k-2} c_2 - \dots - b_0 c_k. \end{aligned} \quad \dots(2)$$

Now $k < n$. Therefore $p \mid a_k$. Also $p \mid b_{k-1}, b_{k-2}, \dots, b_0$.

Therefore from (2), we have

$$p \mid b_k c_0 \Rightarrow p \mid b_k \text{ or } p \mid c_0, \text{ since } p \text{ is a prime element of } R.$$

But this is nonsense because according to our initial assumptions p is neither a divisor of b_k nor a divisor of c_0 . Hence $f(x)$ must be irreducible in $F[x]$. This completes the proof of the theorem.

Note: In the above theorem if we take the ring of integers $\mathbf{I}$ in place of the unique factorization domain R , then the field of quotients of $\mathbf{I}$ is the field of rational numbers. The statement of the theorem will be as follows :

Let $f(x) = a_0 + a_1 x + \dots + a_n x^n$ be a polynomial with integer coefficients. If p is a prime number such that

$$p \mid a_0, p \mid a_1, \dots, p \mid a_{n-1}$$

whereas p is not a divisor of a_n and p^2 is not a divisor of a_0 , then $f(x)$ is irreducible over the field of rational numbers.

There will be no difference in proof.

Comprehensive Exercise 2

1. If p is a prime number, prove that the polynomial $x^n - p$ is irreducible over the field of rational numbers.
2. Show that the polynomial $x^2 - 3$ is irreducible over the field of rational numbers.
3. Prove that the polynomial $1 + x + \dots + x^{p-1}$, where p is a prime number, is irreducible over the field of rational numbers.
4. Show that the polynomial $x^4 + x^3 + x^2 + x + 1$ is irreducible over the field of rational numbers.
5. Let R be a unique factorization domain. Then show that every prime element in R generates a prime ideal.

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The polynomial $x^2 + 1$ is reducible over the field of
 - rational numbers
 - real numbers
 - complex numbers
 - none of these.
- If $f(x) \in F[x]$ and $a \in F$, for a field F , then $(x - a)$ divides $f(x)$ if and only if
 - $f(a) = 0$
 - $f(a) \neq 0$
 - $f'(a) = 0$
 - $f'(a) \neq 0$.
- Let $f(x) = 2 + 6x + 4x^2$, $g(x) = 2x + 4x^2$ be two polynomials over the ring $(\mathbb{I}_8, +_8, \times_8)$. Then $\deg[f(x) + g(x)]$ is
 - 0
 - 1
 - 2
 - 4.
- Let R be a Euclidean ring. Let a and b be two non-zero elements in R . Then if b is a unit in R ,
 - $d(ab) > d(a)$
 - $d(ab) < d(a)$
 - $d(ab) = d(a)$
 - none of these.
- The necessary and sufficient condition that the non-zero element a in the Euclidean ring R is a unit is that
 - $d(a) > d(1)$
 - $d(a) < d(1)$
 - $d(a) = d(1)$
 - none of these.

Fill in the Blank(s)

Fill in the blanks ".....", so that the following statements are complete and correct.

- If R is an arbitrary ring and R' is the set of constant polynomials in $R[x]$, then R' is to R .
- Two polynomials $f(x)$ and $g(x) \in F[x]$ are said to be if their greatest common divisor is 1, the unity element of the field F .
- If $f(x) \in F[x]$ and $a \in F$, for any field F , then is the remainder when $f(x)$ is divided by $(x - a)$.
- An ideal S of the ring of integers $\mathbb{I}$ is maximal if and only if S is generated by some integer.
- Every Euclidean ring is a ring.
- Every Euclidean ring possesses element.

True or False

Write 'T' for true and 'F' for false statement.

1. Every field is a principal ideal ring.
2. The polynomial domain $F[x]$ over a field F is a field.
3. The polynomial ring $\mathbf{I}[x]$ over the ring of integers is not a principal ideal ring.
4. The field of real numbers is a prime field.
5. Every prime integer is a prime Gaussian integer.
6. A Euclidean ring is a unique Factorization Domain.
7. The ring of Gaussian integers is not a Euclidean ring.
8. Every field is a Euclidean ring.

Answers

Multiple Choice Questions

1. (c). 2. (a). 3. (a). 4. (c). 5. (c).

Fill in the Blank(s)

1. isomorphic. 2. relatively prime. 3. $f(a)$.
 4. prime. 5. principal ideal. 6. unity.

True or False

1. T. 2. F. 3. T. 4. F. 5. F.
 6. T. 7. F. 8. T.



Chapter

3



Linear Transformations

3.1 Quotient Space

Right and Left Cosets: Let W be any subspace of a vector space V (F). Let α be any element of V . Then the set

$$W + \alpha = \{\gamma + \alpha : \gamma \in W\}$$

is called a **right coset** of W in V generated by α . Similarly the set

$$\alpha + W = \{\alpha + \gamma : \gamma \in W\}$$

is called a **left coset** of W in V generated by α .

Obviously $W + \alpha$ and $\alpha + W$ are both subsets of V . Since addition in V is commutative, therefore we have $W + \alpha = \alpha + W$. Hence we shall call $W + \alpha$ as simply a **coset** of W in V generated by α .

The following results about cosets are both to be remembered :

- (i) We have $\mathbf{0} \in V$ and $W + \mathbf{0} = W$. Therefore W itself is a coset of W in V .
 - (ii) $\alpha \in W \Rightarrow W + \alpha = W$.
-

Proof:(i) First we shall prove that $W + \alpha \subseteq W$.

Let $\gamma + \alpha$ be any arbitrary element of $W + \alpha$. Then $\gamma \in W$.

Now W is a subspace of V . Therefore

$$\gamma \in W, \alpha \in W \Rightarrow \gamma + \alpha \in W.$$

Thus every element of $W + \alpha$ is also an element of W . Hence

$$W + \alpha \subseteq W.$$

Now we shall prove that $W \subseteq W + \alpha$.

Let $\beta \in W$. Since W is a subspace, therefore

$$\alpha \in W \Rightarrow -\alpha \in W.$$

Consequently $\beta \in W, -\alpha \in W \Rightarrow \beta - \alpha \in W$. Now we can write

$$\beta = (\beta - \alpha) + \alpha \in W + \alpha \text{ since } \beta - \alpha \in W.$$

Thus $\beta \in W \Rightarrow \beta \in W + \alpha$. Therefore $W \subseteq W + \alpha$.

Hence $W = W + \alpha$.

(ii) If $W + \alpha$ and $W + \beta$ are two cosets of W in V , then

$$W + \alpha = W + \beta \Leftrightarrow \alpha - \beta \in W.$$

Proof: Since $\mathbf{0} \in W$, therefore $\mathbf{0} + \alpha \in W + \alpha$. Thus

$$\alpha \in W + \alpha.$$

Now $W + \alpha = W + \beta \Rightarrow \alpha \in W + \beta$

$$\Rightarrow \alpha - \beta \in W + (\beta - \beta)$$

$$\Rightarrow \alpha - \beta \in W + \mathbf{0} \Rightarrow \alpha - \beta \in W.$$

Conversely,

$$\alpha - \beta \in W \Rightarrow W + (\alpha - \beta) = W$$

$$\Rightarrow W + [(\alpha - \beta) + \beta] = W + \beta$$

$$\Rightarrow W + \alpha = W + \beta.$$

Let V/W denote the set of all cosets of W in V i.e., let

$$V/W = \{W + \alpha : \alpha \in V\}.$$

We have just seen that if $\alpha - \beta \in W$, then $W + \alpha = W + \beta$. Thus a coset of W in V can have more than one representation.

Now if $V(F)$ is a vector space, then we shall give a vector space structure to the set V/W over the same field F . For this we shall have to define addition in V/W i.e., addition of cosets of W in V and multiplication of a coset by an element of F i.e., scalar multiplication.

Theorem: If W is any subspace of a vector space $V(F)$, then the set V/W of all cosets $W + \alpha$ where α is any arbitrary element of V , is a vector space over F for the addition and scalar multiplication compositions defined as follows :

$$(W + \alpha) + (W + \beta) = W + (\alpha + \beta) \quad \forall \quad \alpha, \beta \in V$$

$$\text{and} \quad a(W + \alpha) = W + a\alpha; \quad a \in F, \alpha \in V.$$

Proof: We have $\alpha, \beta \in V \Rightarrow \alpha + \beta \in V$.

Also $a \in F, \alpha \in V \Rightarrow a\alpha \in V$.

Therefore $W + (\alpha + \beta) \in V / W$ and also $W + a\alpha \in V / W$. Thus V / W is closed with respect to addition of cosets and scalar multiplication as defined above. Now first of all we shall show that these two compositions are well defined *i.e.*, are independent of the particular representation chosen to denote a coset.

Let $W + \alpha = W + \alpha', \alpha, \alpha' \in V$

and $W + \beta = W + \beta', \beta, \beta' \in V$.

We have $W + \alpha = W + \alpha' \Rightarrow \alpha - \alpha' \in W$

and $W + \beta = W + \beta' \Rightarrow \beta - \beta' \in W$.

Now W is a subspace, therefore

$$\begin{aligned} & \alpha - \alpha' \in W, \beta - \beta' \in W \\ \Rightarrow & (\alpha - \alpha') + (\beta - \beta') \in W \\ \Rightarrow & (\alpha + \beta) - (\alpha' + \beta') \in W \\ \Rightarrow & W + (\alpha + \beta) = W + (\alpha' + \beta') \\ \Rightarrow & (W + \alpha) + (W + \beta) = (W + \alpha') + (W + \beta'). \end{aligned}$$

Therefore addition in V/W is well defined.

Again $a \in F, \alpha - \alpha' \in W \Rightarrow a(\alpha - \alpha') \in W$

$\Rightarrow a\alpha - a\alpha' \in W \Rightarrow W + a\alpha = W + a\alpha'$.

$\therefore$ scalar multiplication in V/W is also well defined.

Commutativity of addition: Let $W + \alpha, W + \beta$ be any two elements of V/W .

Then

$$\begin{aligned} (W + \alpha) + (W + \beta) &= W + (\alpha + \beta) = W + (\beta + \alpha) \\ &= (W + \beta) + (W + \alpha). \end{aligned}$$

Associativity of addition: Let $W + \alpha, W + \beta, W + \gamma$ be any three elements of V/W . Then

$$\begin{aligned} (W + \alpha) + [(W + \beta) + (W + \gamma)] &= (W + \alpha) + [W + (\beta + \gamma)] \\ &= W + [\alpha + (\beta + \gamma)] \\ &= W + [(\alpha + \beta) + \gamma] \\ &= [W + (\alpha + \beta)] + (W + \gamma) \\ &= [(W + \alpha) + (W + \beta)] + (W + \gamma). \end{aligned}$$

Existence of additive identity: If $\mathbf{0}$ is the zero vector of V , then

$$W + \mathbf{0} = W \in V/W.$$

If $W + \alpha$ is any element of V/W , then

$$(W + \mathbf{0}) + (W + \alpha) = W + (\mathbf{0} + \alpha) = W + \alpha.$$

$\therefore W + \mathbf{0} = W$ is the additive identity.

Existence of additive inverse: If $W + \alpha$ is any element of V / W , then

$$W + (-\alpha) = W - \alpha \in V/W.$$

Also we have

$$(W + \alpha) + (W - \alpha) = W + (\alpha - \alpha) = W + \mathbf{0} = W.$$

$\therefore W - \alpha$ is the additive inverse of $W + \alpha$.

Thus V/W is an abelian group with respect to addition composition. Further we observe that if

$a, b \in F$ and $W + \alpha, W + \beta \in V / W$, then

1. $a [(W + \alpha) + (W + \beta)] = a [W + (\alpha + \beta)]$

$$= W + a (\alpha + \beta) = W + (a\alpha + a\beta)$$

$$= (W + a\alpha) + (W + a\beta)$$

$$= a (W + \alpha) + a (W + \beta).$$
2. $(a + b) (W + \alpha) = W + (a + b) \alpha$

$$= W + (a\alpha + b\alpha)$$

$$= (W + a\alpha) + (W + b\alpha)$$

$$= a (W + \alpha) + b (W + \alpha).$$
3. $(ab) (W + \alpha) = W + (ab) \alpha = W + a (b\alpha)$

$$= a (W + b\alpha) = a [b (W + \alpha)].$$
4. $1 (W + \alpha) = W + 1\alpha = W + \alpha.$

$\therefore V/W$ is a vector space over F for these two compositions. The vector space V/W is called the **Quotient Space** of V relative to W . The coset W is the zero vector of this vector space.

3.2 Dimension of a Quotient Space

Theorem: If W be a subspace of a finite dimensional vector space $V (F)$, then

$$\dim V/W = \dim V - \dim W.$$

Proof: Let m be the dimension of the subspace W of the vector space $V (F)$.

Let $S = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$

be a basis of W . Since S is a linearly independent subset of V , therefore it can be extended to form a basis of V . Let

$$S' = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l\}$$

be a basis of V . Then $\dim V = m + l$.

$$\therefore \dim V - \dim W = (m + l) - m = l.$$

So we should prove that $\dim V / W = l$.

We claim that the set of l cosets

$$S_1 = \{W + \beta_1, W + \beta_2, \dots, W + \beta_l\}$$

is a basis of V/W .

First we show that S_1 is linearly independent. The zero vector of V / W is W .

$$\begin{aligned}
\text{Let } & a_1 (W + \beta_1) + a_2 (W + \beta_2) + \dots + a_l (W + \beta_l) = W \\
\Rightarrow & (W + a_1 \beta_1) + (W + a_2 \beta_2) + \dots + (W + a_l \beta_l) = W \\
\Rightarrow & W + (a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l) = W + \mathbf{0} \\
\Rightarrow & a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l \in W \\
\Rightarrow & a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l = b_1 \alpha_1 + b_2 \alpha_2 + \dots + b_m \alpha_m \\
& \quad [\because \text{any vector in } W \text{ can be expressed as a linear} \\
& \quad \text{combination of its basis vectors}] \\
\Rightarrow & a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l - b_1 \alpha_1 - b_2 \alpha_2 - \dots - b_m \alpha_m = \mathbf{0} \\
\Rightarrow & a_1 = 0, a_2 = 0, \dots, a_l = 0 \text{ since the vectors} \\
& \quad \beta_1, \beta_2, \dots, \beta_l, \alpha_1, \alpha_2, \dots, \alpha_m \text{ are linearly independent.}
\end{aligned}$$

$\therefore$ The set S_1 is linearly independent.

Now to show that $L(S_1) = V/W$. Let $W + \alpha$ be any element of V/W . The vector $\alpha \in V$ can be expressed as

$$\begin{aligned}
\alpha &= c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_m \alpha_m + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l \\
&= \gamma + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l \quad \text{where} \\
& \quad \gamma = c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_m \alpha_m \in W.
\end{aligned}$$

$$\begin{aligned}
\text{So } W + \alpha &= W + (\gamma + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l) \\
&= (W + \gamma) + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l \\
&= W + (d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l) \quad [\because \gamma \in W \Rightarrow W + \gamma = W] \\
&= (W + d_1 \beta_1) + (W + d_2 \beta_2) + \dots + (W + d_l \beta_l) \\
&= d_1 (W + \beta_1) + d_2 (W + \beta_2) + \dots + d_l (W + \beta_l).
\end{aligned}$$

Thus any element $W + \alpha$ of V/W can be expressed as a linear combination of elements of S_1 .

$\therefore V/W = L(S_1)$. $\therefore S_1$ is a basis of V/W .

$\therefore \dim V/W = l$. Hence the theorem.

3.3 Direct Sum of Spaces

Vector space as a direct sum of subspaces.

Definition: Let $V(F)$ be a vector space and let $W_1, W_2, \dots, W_m$ be subspaces of V . Then V is said to be the direct sum of $W_1, W_2, \dots, W_m$ if every element $\alpha \in V$ can be written in one and only one way as $\alpha = \alpha_1 + \alpha_2 + \dots + \alpha_m$ where

$$\alpha_1 \in W_1, \alpha_2 \in W_2, \dots, \alpha_m \in W_m.$$

If a vector space $V(F)$ is a direct sum of its two subspaces W_1 and W_2 then we should have not only $V = W_1 + W_2$ but also that each vector of V can be uniquely expressed as sum of an element of W_1 and an element of W_2 . Symbolically the direct sum is represented by the notation $V = W_1 \oplus W_2$.

Illustration 1: Let $V_2(F)$ be the vector space of all ordered pairs of F . Then $W_1 = \{(a, 0) : a \in F\}$ and $W_2 = \{(0, b) : b \in F\}$ are two subspaces of $V_2(F)$. Obviously

any element $(x, y) \in V_2$ (F) can be uniquely expressed as the sum of an element of W_1 and an element of W_2 . The unique expression is $(x, y) = (x, 0) + (0, y)$. Thus V_2 (F) is the direct sum of W_1 and W_2 . Also we observe that the only element common to both W_1 and W_2 is the zero vector $(0, 0)$.

Illustration 2: Let $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$ and $W_2 = \{(0, 0, c) : c \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$.

Now any vector $(a, b, c) \in \mathbf{R}^3$ can be uniquely expressed as the sum of a vector in W_1 and a vector in W_2 : $(a, b, c) = (a, b, 0) + (0, 0, c)$.

Thus $\mathbf{R}^3$ is the direct sum of W_1 and W_2 .

Also we observe that the only element common to both W_1 and W_2 is the zero vector $(0, 0, 0)$.

Illustration 3: Let $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$ and $W_2 = \{(0, b, c) : b, c \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$. Then $\mathbf{R}^3 = W_1 + W_2$ since every vector in $\mathbf{R}^3$ is the sum of a vector in W_1 and a vector in W_2 . However, $\mathbf{R}^3$ is not the direct sum of W_1 and W_2 since such sums are not unique.

For example, $(4, 6, 8) = (4, 5, 0) + (0, 1, 8)$.

Also $(4, 6, 8) = (4, -1, 0) + (0, 7, 8)$.

Here we observe that $W_1 \cap W_2 \neq \{0\}$.

3.4 Disjoint Subspaces

Definition. Two subspaces W_1 and W_2 of the vector space V (F) are said to be disjoint if their intersection is the zero subspace i.e., if $W_1 \cap W_2 = \{0\}$.

Theorem. The necessary and sufficient conditions for a vector space V (F) to be a direct sum of its two subspaces W_1 and W_2 are that

- (i) $V = W_1 + W_2$
- (ii) $W_1 \cap W_2 = \{0\}$ i.e., W_1 and W_2 are disjoint.

Proof: The conditions are necessary.

Let V be direct sum of its two subspaces W_1 and W_2 . Then each element of V is expressible uniquely as sum of an element of W_1 and an element of W_2 . Therefore we have

$$V = W_1 + W_2.$$

Let, if possible $0 \neq \alpha \in W_1 \cap W_2$. Then $\alpha \in W_1, \alpha \in W_2$. Also $\alpha \in V$ and we can write

$$\alpha = 0 + \alpha \text{ where } 0 \in W_1, \alpha \in W_2$$

$$\text{and } \alpha = \alpha + 0 \text{ where } \alpha \in W_1, 0 \in W_2.$$

Thus $\alpha \in V$ can be expressed in at least two different ways as sum of an element of W_1 and an element of W_2 . This contradicts the fact that V is direct sum of W_1 and W_2 . Hence $\mathbf{0}$ is the only vector common to both W_1 and W_2 i.e., $W_1 \cap W_2 = \{\mathbf{0}\}$. Thus the conditions are necessary.

The conditions are sufficient:

Let $V = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$. Then to show that V is direct sum of W_1 and W_2 .

$V = W_1 + W_2 \Rightarrow$ that each element of V can be expressed as sum of an element of W_1 and an element of W_2 . Now to show that this expression is unique.

Let, if possible,

$$\alpha = \alpha_1 + \alpha_2, \alpha \in V, \alpha_1 \in W_1, \alpha_2 \in W_2,$$

$$\text{and} \quad \alpha = \beta_1 + \beta_2, \beta_1 \in W_1, \beta_2 \in W_2.$$

Then to show that $\alpha_1 = \beta_1$ and $\alpha_2 = \beta_2$.

$$\text{We have} \quad \alpha_1 + \alpha_2 = \beta_1 + \beta_2$$

$$\Rightarrow \quad \alpha_1 - \beta_1 = \beta_2 - \alpha_2.$$

Since W_1 is a subspace, therefore

$$\alpha_1 \in W_1, \beta_1 \in W_1 \Rightarrow \alpha_1 - \beta_1 \in W_1.$$

$$\text{Similarly,} \quad \beta_2 - \alpha_2 \in W_2.$$

$$\therefore \quad \alpha_1 - \beta_1 = \beta_2 - \alpha_2 \in W_1 \cap W_2.$$

But $\mathbf{0}$ is the only vector which belongs to $W_1 \cap W_2$. Therefore

$$\alpha_1 - \beta_1 = \mathbf{0} \Rightarrow \alpha_1 = \beta_1.$$

$$\text{Also} \quad \beta_2 - \alpha_2 = \mathbf{0} \Rightarrow \alpha_2 = \beta_2.$$

Thus each vector $\alpha \in V$ is uniquely expressible as sum of an element of W_1 and an element of W_2 . Hence $V = W_1 \oplus W_2$.

3.5 Dimension of a Direct Sum

Theorem: If a finite dimensional vector space V (F) is a direct sum of two subspaces W_1 and W_2 , then

$$\dim V = \dim W_1 + \dim W_2.$$

Proof: Let $\dim W_1 = m$ and $\dim W_2 = l$. Also let the sets of vectors

$$S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_m\} \quad \text{and} \quad S_2 = \{\beta_1, \beta_2, \dots, \beta_l\}$$

be the bases of W_1 and W_2 respectively.

$$\text{We have} \quad \dim W_1 + \dim W_2 = m + l.$$

In order to prove that $\dim V = \dim W_1 + \dim W_2$, we should therefore prove that $\dim V = m + l$. We claim that the set

$$S = S_1 \cup S_2 = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l\}$$

is a basis of V .

First we show that the set S is linearly independent. Let

$$a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m + b_1 \beta_1 + b_2 \beta_2 + \dots + b_l \beta_l = \mathbf{0}.$$

$$\Rightarrow a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m = -(b_1 \beta_1 + b_2 \beta_2 + \dots + b_l \beta_l).$$

$$\text{Now } a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m \in W_1$$

$$\text{and } -(b_1 \beta_1 + b_2 \beta_2 + \dots + b_l \beta_l) \in W_2.$$

$$\text{Therefore } a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m \in W_1 \cap W_2$$

$$\text{and } -(b_1 \beta_1 + b_2 \beta_2 + \dots + b_l \beta_l) \in W_1 \cap W_2.$$

But V is the direct sum of W_1 and W_2 . Therefore $\mathbf{0}$ is the only vector belonging to $W_1 \cap W_2$. Then we have

$$a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m = \mathbf{0}, b_1 \beta_1 + b_2 \beta_2 + \dots + b_l \beta_l = \mathbf{0}.$$

Since both the sets $\{\alpha_1, \alpha_2, \dots, \alpha_m\}$ and $\{\beta_1, \beta_2, \dots, \beta_l\}$ are linearly independent, therefore we have

$$a_1 = 0, a_2 = 0, \dots, a_m = 0, b_1 = 0, b_2 = 0, \dots, b_l = 0.$$

Therefore S is linearly independent.

Now we shall show that $L(S) = V$. Let α be any element of V . Then

$$\alpha = \text{an element of } W_1 + \text{an element of } W_2.$$

$$= \text{a linear combination of } S_1 + \text{a linear combination of } S_2$$

$$= \text{a linear combination of elements of } S.$$

$$\therefore L(S) = V$$

$$\therefore S \text{ is a basis of } V. \text{ Therefore } \dim V = m + l.$$

Hence the theorem.

A sort of converse of this theorem is true. It has been proved in the following theorem.

Theorem: Let V be a finite dimensional vector space and let W_1, W_2 be subspaces of V such that $V = W_1 + W_2$ and $\dim V = \dim W_1 + \dim W_2$.

$$\text{Then } V = W_1 \oplus W_2.$$

Proof: Let $\dim W_1 = l$ and $\dim W_2 = m$. Then

$$\dim V = l + m.$$

$$\text{Let } S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_l\}$$

be a basis of W_1 and

$$S_2 = \{\beta_1, \beta_2, \dots, \beta_m\}$$

be a basis of W_2 . We shall show that $S_1 \cup S_2$ is a basis of V .

$$\text{Let } \alpha \in V. \text{ Since } V = W_1 + W_2, \text{ therefore we can write}$$

$$\alpha = \gamma + \delta \text{ where } \gamma \in W_1, \delta \in W_2.$$

Now $\gamma \in W_1$ can be expressed as a linear combination of the elements of S_1 and $\delta \in W_2$ can be expressed as a linear combination of the elements of S_2 . Therefore $\alpha \in V$ can be expressed as a linear combination of the elements of $S_1 \cup S_2$. Therefore $V = L(S_1 \cup S_2)$. Since $\dim V = l + m$ and $L(S_1 \cup S_2) = V$, therefore

the number of distinct elements in $S_1 \cup S_2$ cannot be less than $l + m$. Thus $S_1 \cup S_2$ has $l + m$ distinct elements and therefore $S_1 \cup S_2$ is a basis of V . Therefore the set

$$\{\alpha_1, \alpha_2, \dots, \alpha_l, \beta_1, \beta_2, \dots, \beta_m\}$$

is linearly independent.

Now we shall show that

$$W_1 \cap W_2 = \{\mathbf{0}\}.$$

Let $\alpha \in W_1 \cap W_2$.

Then $\alpha \in W_1, \alpha \in W_2$.

Therefore $\alpha = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_l \alpha_l$

and $\alpha = b_1 \beta_1 + b_2 \beta_2 + \dots + b_m \beta_m$

for some a 's and b 's $\in F$.

$$\therefore a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_l \alpha_l = b_1 \beta_1 + b_2 \beta_2 + \dots + b_m \beta_m$$

$$\Rightarrow a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_l \alpha_l - b_1 \beta_1 - b_2 \beta_2 - \dots - b_m \beta_m = \mathbf{0}$$

$$\Rightarrow a_1 = 0, a_2 = 0, \dots, a_l = 0, b_1 = 0, b_2 = 0, \dots, b_m = 0$$

$$\Rightarrow \alpha = \mathbf{0}.$$

$$\therefore W_1 \cap W_2 = \{\mathbf{0}\}.$$

3.6 Complementary Subspaces

Definition: Let $V (F)$ be a vector space and W_1, W_2 be two subspaces of V . Then the subspace W_2 is called the complement of W_1 in V if V is the direct sum of W_1 and W_2 .

Existence of complementary subspaces:

Theorem: Corresponding to each subspace W_1 of a finite dimensional vector space $V (F)$, there exists a subspace W_2 such that V is the direct sum of W_1 and W_2 .

Proof: Let $\dim W_1 = m$. Let the set

$$S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$$

be a basis of W_1 .

Since S_1 is a linearly independent subset of V , therefore S_1 can be extended to form a basis of V . Let the set

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l\}$$

be a basis of V .

Let W_2 be the subspace of V generated by the set

$$S_2 = \{\beta_1, \beta_2, \dots, \beta_l\}.$$

We shall prove that V is the direct sum of W_1 and W_2 . For this we shall prove that $V = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$.

Let α be any element of V . Then we can express

$$\alpha = \text{a linear combination of elements of } S$$

= a linear combination of S_1 + a linear combination of S_2
 = an element of W_1 + an element of W_2 .

$\therefore V = W_1 + W_2$.

Again let $\beta \in W_1 \cap W_2$. Then β can be expressed as a linear combination of S_1 and also as a linear combination of S_2 . So we have

$$\beta = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m = b_1 \beta_1 + b_2 \beta_2 + \dots + b_l \beta_l.$$

$$\Rightarrow a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m - b_1 \beta_1 - b_2 \beta_2 - \dots - b_l \beta_l = \mathbf{0}$$

$$\Rightarrow a_1 = 0, a_2 = 0, \dots, a_m = 0, b_1 = 0, b_2 = 0, \dots, b_l = 0$$

since $\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l$ are linearly independent.

$\therefore \beta = \mathbf{0}$ (zero vector).

Thus $W_1 \cap W_2 = \{\mathbf{0}\}$.

Hence V is the direct sum of W_1 and W_2 .

Dimension of a Quotient space: Alternative method:

Theorem: If W_1 and W_2 are complementary subspaces of a vector space V , then the mapping f which assigns to each vector β in W_2 the coset $W_1 + \beta$ is an isomorphism between W_2 and V/W_1 .

Proof: It is given that

$$V = W_1 \oplus W_2$$

and $f : W_2 \rightarrow V/W_1$ such that

$$f(\beta) = W_1 + \beta \quad \forall \beta \in W_2.$$

We shall show that f is an isomorphism of W_2 onto V/W_1 .

(i) **f is one-one:**

If $\beta_1, \beta_2 \in W_2$, then

$$f(\beta_1) = f(\beta_2) \Rightarrow W_1 + \beta_1 = W_1 + \beta_2 \quad [\text{by def. of } f]$$

$$\Rightarrow \beta_1 - \beta_2 \in W_1$$

$$\Rightarrow \beta_1 - \beta_2 \in W_1 \cap W_2 \quad [\because \beta_1 - \beta_2 \in W_2 \text{ because } W_2 \text{ is a subspace}]$$

$$\Rightarrow \beta_1 - \beta_2 = \mathbf{0} \quad [\because W_1 \cap W_2 = \{\mathbf{0}\}]$$

$$\Rightarrow \beta_1 = \beta_2.$$

$\therefore f$ is one-one.

(ii) **f is onto:**

Let $W_1 + \alpha$ be any coset in V/W_1 , where $\alpha \in V$. Since V is direct sum of W_1 and W_2 , therefore we can write

$$\alpha = \gamma + \beta \text{ where } \gamma \in W_1, \beta \in W_2.$$

This gives $\gamma = \alpha - \beta \in W_1$.

Since $\alpha - \beta \in W_1$, therefore $W_1 + \alpha = W_1 + \beta$.

$$\begin{aligned} \text{Now } f(\beta) &= W_1 + \beta && [\text{by def. of } f] \\ &= W_1 + \alpha. \end{aligned}$$

Thus $W_1 + \alpha \in V/W_1 \Rightarrow \exists \beta \in W_2$ such that

$$f(\beta) = W_1 + \alpha.$$

Therefore f is onto.

(iii) f is a linear transformation:

Let $a, b \in F$ and $\beta_1, \beta_2 \in W_2$.

$$\begin{aligned} \text{Then } f(a\beta_1 + b\beta_2) &= W_1 + (a\beta_1 + b\beta_2) \\ &= (W_1 + a\beta_1) + (W_1 + b\beta_2) \\ &= a(W_1 + \beta_1) + b(W_1 + \beta_2) \\ &= af(\beta_1) + bf(\beta_2). \end{aligned}$$

Therefore f is a linear transformation.

Hence f is an isomorphism between W_2 and V/W_1 .

Corollary: Dimension of a quotient space: *If W is an m -dimensional subspace of an n -dimensional vector space V , then the dimension of the quotient space V/W is $n - m$.*

Proof: Since W is a subspace of a finite dimensional vector space V , therefore there exists a subspace W_1 of V such that

$$V = W \oplus W_1.$$

$$\text{Also } \dim V = \dim W + \dim W_1$$

$$\text{or } \dim W_1 = \dim V - \dim W = n - m.$$

Now by the above theorem, we have

$$V/W \cong W_1.$$

$$\therefore \dim V/W = \dim W_1 = n - m.$$

3.7 Direct Sum of Several Subspaces

Now we shall have some discussion on the direct sum of several subspaces. For this purpose we shall first define the concept of independence of subspaces analogous to the disjointness condition on two subspaces.

Definition: Suppose $W_1, W_2, \dots, W_k$ are subspaces of the vector space V . We shall say that $W_1, W_2, \dots, W_k$ are **independent**, if

$$\alpha_1 + \alpha_2 + \dots + \alpha_k = \mathbf{0}, \alpha_i \in W_i \text{ implies that each } \alpha_i = \mathbf{0}.$$

Theorem 1: Suppose $V(F)$ is a vector space. Let $W_1, W_2, \dots, W_k$ be subspaces of V and let $W = W_1 + W_2 + \dots + W_k$. The following are equivalent :

(i) $W_1, W_2, \dots, W_k$ are independent.

(ii) Each vector α in W can be uniquely expressed in the form

$$\alpha = \alpha_1 + \alpha_2 + \dots + \alpha_k \text{ with } \alpha_i \in W_i, i = 1, \dots, k.$$

(iii) For each $j, 2 \leq j \leq k$, the subspace W_j is disjoint from the sum $(W_1 + \dots + W_{j-1})$.

Proof: In order to prove the equivalence of the three statements we shall prove that (i) $\Rightarrow$ (ii), (ii) $\Rightarrow$ (iii) and (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii). Suppose $W_1, \dots, W_k$ are independent. Let $\alpha \in W$.

Since $W = W_1 + \dots + W_k$, therefore we can write

$$\alpha = \alpha_1 + \dots + \alpha_k \text{ with } \alpha_i \text{ in } W_i.$$

Suppose that also $\alpha = \beta_1 + \dots + \beta_k$ with β_i in W_i .

Then $\alpha_1 + \dots + \alpha_k = \beta_1 + \dots + \beta_k$

$$\Rightarrow (\alpha_1 - \beta_1) + \dots + (\alpha_k - \beta_k) = \mathbf{0} \text{ with } \alpha_i - \beta_i \text{ in } W_i \text{ as } W_i \text{ is a subspace}$$

$$\Rightarrow \alpha_i - \beta_i = \mathbf{0} \quad [\because W_1, \dots, W_k \text{ are independent}]$$

$$\Rightarrow \alpha_i = \beta_i, i = 1, \dots, k.$$

Therefore the α_i 's are uniquely determined by α .

(ii) $\Rightarrow$ (iii). Let $\alpha \in W_j \cap (W_1 + \dots + W_{j-1})$.

Then $\alpha \in W_j$ and $\alpha \in W_1 + \dots + W_{j-1}$.

Now $\alpha \in W_1 + \dots + W_{j-1}$ implies that there exist vectors $\alpha_1, \dots, \alpha_{j-1}$ with α_i in W_i such that

$$\alpha = \alpha_1 + \dots + \alpha_{j-1}.$$

Also $\alpha \in W_j$.

Therefore we get two expressions for α as a sum of vectors, one in each W_i . These are

$$\alpha = \alpha_1 + \dots + \alpha_{j-1} + \mathbf{0} + \dots + \mathbf{0}$$

in which the vector belonging to W_j is $\mathbf{0}$

$$\text{and } \alpha = \mathbf{0} + \dots + \mathbf{0} + \alpha + \dots + \mathbf{0}$$

in which the vector belonging to W_j is α .

Since the expression for α is given to be unique, therefore we must have

$$\alpha_1 = \dots = \alpha_{j-1} = \mathbf{0} = \alpha.$$

Thus $W_j \cap (W_1 + \dots + W_{j-1}) = \{\mathbf{0}\}$.

(iii) $\Rightarrow$ (i).

Let $\alpha_1 + \dots + \alpha_k = \mathbf{0}$ where $\alpha_i \in W_i, i = 1, \dots, k$(1)

Then we are to prove that each $\alpha_i = \mathbf{0}$.

Suppose that for some i we have $\alpha_i \neq \mathbf{0}$.

Let j be the largest integer i between 1 and k such that $\alpha_i \neq \mathbf{0}$. Obviously j must be ≥ 2 and at the most j can be equal to k . Then (1) reduces to

$$\alpha_1 + \dots + \alpha_j = \mathbf{0}, \alpha_j \neq \mathbf{0}$$

$$\Rightarrow \alpha_j = -\alpha_1 - \dots - \alpha_{j-1}$$

$$\Rightarrow \alpha_j \in W_1 + \dots + W_{j-1} \quad [\because -\alpha_1 - \dots - \alpha_{j-1} \in W_1 + \dots + W_{j-1}]$$

$$\Rightarrow \alpha_j \in W_j \cap (W_1 + \dots + W_{j-1}) \quad [\Rightarrow \alpha_j \in W_j]$$

$$\Rightarrow \alpha_j = \mathbf{0}.$$

Thus we get a contradiction. Hence each $\alpha_i = \mathbf{0}$.

Note: If any (and hence all) of the three conditions of theorem 1 hold for $W_1, \dots, W_k$, then we shall say that W is the **direct sum** of $W_1, \dots, W_k$ and we write

$$W = W_1 \oplus \dots \oplus W_k .$$

Theorem 2: Let V (F) be a vector space. Let $W_1, \dots, W_n$ be subspaces of V . Suppose that

$$V = W_1 + \dots + W_n$$

and that $W_i \cap (W_1 + \dots + W_{i-1} + W_{i+1} + \dots + W_n) = \{0\}$

for every $i = 1, 2, \dots, n$. Prove that V is the direct sum of $W_1, \dots, W_n$.

Proof: In order to prove that V is the direct sum of $W_1, \dots, W_n$, we should prove that each vector $\alpha \in V$ can be uniquely expressed as

$$\alpha = \alpha_1 + \dots + \alpha_n \text{ where } \alpha_i \in W_i, i = 1, \dots, n.$$

Since $V = W_1 + \dots + W_n$, therefore any vector α in V can be written as

$$\alpha = \alpha_1 + \dots + \alpha_n \text{ where } \alpha_i \in W_i . \quad \dots(1)$$

To show that $\alpha_1, \dots, \alpha_n$ are unique.

$$\text{Let } \alpha = \beta_1 + \dots + \beta_n \text{ where } \beta_i \in W_i . \quad \dots(2)$$

From (1) and (2), we get

$$\begin{aligned} \alpha_1 + \dots + \alpha_n &= \beta_1 + \dots + \beta_n \\ (\alpha_1 - \beta_1) + \dots + (\alpha_{i-1} - \beta_{i-1}) + (\alpha_i - \beta_i) \\ &\quad + (\alpha_{i+1} - \beta_{i+1}) + \dots + (\alpha_n - \beta_n) = 0. \quad \dots(3) \end{aligned}$$

Now each W_i is a subspace of V . Therefore $\alpha_i - \beta_i$ and also its additive inverse

$$\beta_i - \alpha_i \in W_i, i = 1, \dots, n.$$

From (3), we get

$$\begin{aligned} (\alpha_i - \beta_i) &= (\beta_1 - \alpha_1) + \dots + (\beta_{i-1} - \alpha_{i-1}) \\ &\quad + (\beta_{i+1} - \alpha_{i+1}) + \dots + (\beta_n - \alpha_n). \quad \dots(4) \end{aligned}$$

Now the vector on the right hand side of (4) and consequently the vector $\alpha_i - \beta_i$ is in $W_1 + \dots + W_{i-1} + W_{i+1} + \dots + W_n$.

Also $\alpha_i - \beta_i \in W_i$.

$$\therefore \alpha_i - \beta_i \in W_i \cap (W_1 + \dots + W_{i-1} + W_{i+1} + \dots + W_n).$$

But for every $i = 1, \dots, n$, it is given that

$$W_i \cap (W_1 + \dots + W_{i-1} + W_{i+1} + \dots + W_n) = \{0\}.$$

Therefore $\alpha_i - \beta_i = 0, i = 1, \dots, n$

$$\Rightarrow \alpha_i = \beta_i, i = 1, \dots, n$$

$\Rightarrow$ the expression (1) for α is unique.

Hence V is the direct sum of $W_1, \dots, W_n$.

Theorem 3: Let V (F) be a finite dimensional vector space and let $W_1, \dots, W_k$ be subspaces of V . Then the following two statements are equivalent.

(i) V is the direct sum of $W_1, \dots, W_k$.

(ii) If B_i is a basis of $W_i, i = 1, \dots, k$, then the union

$$B = \bigcup_{i=1}^k B_i \text{ is also a basis for } V.$$

Proof: Let $B_i = \{\alpha_1^i, \alpha_2^i, \dots, \alpha_{n_i}^i\}$ be a basis for W_i .

Here $n_i = \dim W_i$ = number of vectors in B_i . Also let B be the union of the bases B_i .

(i) $\Rightarrow$ (ii). It is given that V is the direct sum of $W_1, \dots, W_k$, therefore for any $\alpha \in V$, we can write

$$\alpha = \alpha_1 + \dots + \alpha_k \text{ for } \alpha_i \in W_i, i = 1, \dots, k.$$

Now α_i can be expressed as a linear combination of the vectors in B_i which is a basis of W_i . Therefore α can be expressed as a linear combination of the elements of $B = \bigcup_{i=1}^k B_i$. Therefore $L(B) = V$ i.e., B spans V .

Now to show that B is linearly independent. Let

$$\sum_{i=1}^k (a_1^i \alpha_1^i + a_2^i \alpha_2^i + \dots + a_{n_i}^i \alpha_{n_i}^i) = \mathbf{0}. \quad \dots(1)$$

Since V is the direct sum of $W_1, \dots, W_k$, therefore $\mathbf{0} \in V$ can be uniquely expressed as a sum of vectors one in each W_i . This unique expression is

$$\mathbf{0} = \mathbf{0} + \dots + \mathbf{0} \text{ where } \mathbf{0} \in W_i, i = 1, \dots, k.$$

Now $a_1^i \alpha_1^i + \dots + a_{n_i}^i \alpha_{n_i}^i \in W_i$. Therefore from (1) which is an expression for $\mathbf{0} \in V$ as a sum of vectors one in each W_i , we get

$$\begin{aligned} a_1^i \alpha_1^i + \dots + a_{n_i}^i \alpha_{n_i}^i &= \mathbf{0}, i = 1, \dots, k \\ a_1^i &= \dots = a_{n_i}^i = 0 \end{aligned}$$

since $\{\alpha_1^i, \dots, \alpha_{n_i}^i\}$ is linearly independent being a basis for W_i .

Therefore $B = \bigcup_{i=1}^k B_i$ is linearly independent. Therefore B is a basis of V .

(ii) $\Rightarrow$ (i). It is given that $B = \bigcup_{i=1}^k B_i$ is a basis of V .

Therefore for any $\alpha \in V$, we can write

$$\begin{aligned} \alpha &= \sum_{i=1}^k (a_1^i \alpha_1^i + a_2^i \alpha_2^i + \dots + a_{n_i}^i \alpha_{n_i}^i) \\ &= \alpha_1 + \alpha_2 + \dots + \alpha_k \end{aligned} \quad \dots(2)$$

where $\alpha_i = a_1^i \alpha_1^i + \dots + a_{n_i}^i \alpha_{n_i}^i \in W_i$.

Thus each vector in V can be expressed as a sum of vectors one in each W_i .

Now V will be the direct sum of $W_1, \dots, W_k$ if the expression (2) for α is unique. Let

$$\alpha = \beta_1 + \beta_2 + \dots + \beta_k \quad \dots(3)$$

where $\beta_i = b_1^i \alpha_1^i + \dots + b_{n_i}^i \alpha_{n_i}^i \in W_i$.

From (2) and (3), we get

$$\begin{aligned} \alpha_1 + \dots + \alpha_k &= \beta_1 + \dots + \beta_k \\ \Rightarrow (\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i) + \dots + (\alpha_k - \beta_k) &= \mathbf{0} \end{aligned}$$

$$\begin{aligned}
 \Rightarrow & \sum_{i=1}^k (\alpha_i - \beta_i) = \mathbf{0} \\
 \Rightarrow & \sum_{i=1}^k [(a_1^i - b_1^i) \alpha_1^i + \dots + (a_{n_i}^i - b_{n_i}^i) \alpha_{n_i}^i] = \mathbf{0} \\
 \Rightarrow & a_1^i - b_1^i = \dots = a_{n_i}^i - b_{n_i}^i = 0, i = 1, \dots, k \\
 & [\because \bigcup_{i=1}^k B_i \text{ is linearly independent being a basis of } V] \\
 \Rightarrow & a_1^i = b_1^i, \dots, a_{n_i}^i = b_{n_i}^i, i = 1, \dots, k \\
 \Rightarrow & \alpha_i = \beta_i, i = 1, \dots, k \\
 \Rightarrow & \text{the expression (2) for } \alpha \text{ is unique.}
 \end{aligned}$$

Hence V is the direct sum of $W_1, \dots, W_k$.

Note: While proving this theorem we have proved that if a finite dimensional vector space V (F) is the direct sum of its subspaces $W_1, \dots, W_k$, then

$$\dim V = \dim W_1 + \dots + \dim W_k.$$

Illustrative Examples

Example 1: Find dimension of V / W where $V = \mathbf{R}^3$, $W = \{(a, 0, 0) : a \in \mathbf{R}\}$.

Solution: Here $V = \mathbf{R}^3$. $\dim V = 3$.

$W = \{(a, 0, 0) : a \in \mathbf{R}\}$. $\{(1, 0, 0)\}$ is a basis of W so $\dim W = 1$.

Now $\dim V / W = \dim V - \dim W = 3 - 1 = 2$.

Example 2: If $W_1 = \{(x, 0, z) : x, z \in \mathbf{R}\}$, $W_2 = \{(0, y, z) : y, z \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$ then show that $\mathbf{R}^3 = W_1 + W_2$ but $\mathbf{R}^3 \neq W_1 \oplus W_2$.

Solution: Any $(x, y, z) \in \mathbf{R}^3$ can be written as a sum of a vector of W_1 and a vector of W_2 as shown below.

We have $(x, y, z) = \left(x, 0, \frac{1}{2}z\right) + \left(0, y, \frac{1}{2}z\right)$, where $\left(x, 0, \frac{1}{2}z\right) \in W_1$

and $\left(0, y, \frac{1}{2}z\right) \in W_2$.

Thus $\mathbf{R}^3 = W_1 + W_2$.

Also we have $(x, y, z) = \left(x, 0, \frac{1}{4}z\right) + \left(0, y, \frac{3}{4}z\right)$ where $\left(x, 0, \frac{1}{4}z\right) \in W_1$ and

$\left(0, y, \frac{3}{4}z\right) \in W_2$.

It follows that representation of (x, y, z) as a sum of a vector of W_1 and a vector of W_2 is not unique.

Hence $\mathbf{R}^3 \neq W_1 \oplus W_2$.

Example 3: If $W_1 = \{(a, b, c) : a = b = c\}$, $W_2 = \{(a, b, c) : a = 0\}$ are two subspaces of $\mathbf{R}^3$, show that $\mathbf{R}^3 = W_1 \oplus W_2$.

Solution: We claim that $\mathbf{R}^3 = W_1 + W_2$.

For if $\alpha = (a, b, c) \in \mathbf{R}^3$, then $\alpha = (a, a, a) + (0, b - a, c - a)$ where $(a, a, a) \in W_1$ and $(0, b - a, c - a) \in W_2$.

Also we have $W_1 \cap W_2 = \{\mathbf{0}\}$.

For $\alpha = (a, b, c) \in W_1 \cap W_2 \Rightarrow a = b = c$ and $a = 0$

$\Rightarrow a = 0, b = 0, c = 0$ i.e., $\alpha = \mathbf{0}$.

Thus $\mathbf{R}^3 = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$.

Hence $\mathbf{R}^3 = W_1 \oplus W_2$.

Example 4: Construct three subspaces W_1, W_2, W_3 of a vector space V so that $V = W_1 \oplus W_2 = W_1 \oplus W_3$ but $W_2 \neq W_3$.

Solution: Take the vector space $V = \mathbf{R}^2$.

Obviously $W_1 = \{(a, 0) : a \in \mathbf{R}\}$,

$W_2 = \{(0, a) : a \in \mathbf{R}\}$, and $W_3 = \{(a, a) : a \in \mathbf{R}\}$,

are three subspaces of $\mathbf{R}^2$.

We have $V = W_1 + W_2$ and $W_1 \cap W_2 = \{(0, 0)\}$. $\therefore V = W_1 \oplus W_2$.

Also it can be easily shown that

$V = W_1 + W_3$ and $W_1 \cap W_3 = \{(0, 0)\}$. $\therefore V = W_1 \oplus W_3$.

Thus $V = W_1 \oplus W_2 = W_1 \oplus W_3$ but $W_2 \neq W_3$.

Comprehensive Exercise 1

1. Let V be the vector space of square matrices of order n over the field $\mathbf{R}$. Let W_1 and W_2 be the subspaces of symmetric and antisymmetric matrices respectively. Show that $V = W_1 \oplus W_2$.
2. Let V be the vector space of all functions from the real field $\mathbf{R}$ into $\mathbf{R}$. Let U be the subspace of even functions and W the subspace of odd functions. Show that $V = U \oplus W$.
3. Let W_1, W_2 and W_3 be the following subspaces of $\mathbf{R}^3$:

$$W_1 = \{(a, b, c) : a + b + c = 0\}, W_2 = \{(a, b, c) : a = c\},$$

$$W_3 = \{(0, 0, c) : c \in \mathbf{R}\}.$$

Show that (i) $\mathbf{R}^3 = W_1 + W_2$; (ii) $\mathbf{R}^3 = W_1 + W_3$; (iii) $\mathbf{R}^3 = W_2 + W_3$.

When is the sum direct?

4. Let W be a subspace of a vector space V over a field F . Show that $\alpha \in \beta + W$ iff $\alpha - \beta \in W$.
5. Let W_1 and W_2 be two subspaces of a finite dimensional vector space V . If $\dim V = \dim W_1 + \dim W_2$ and $W_1 \cap W_2 = \{0\}$, prove that $V = W_1 \oplus W_2$.

Answers 1

3. The sum is direct in (ii) and (iii)

3.8 Linear Transformation or Vector Space Homomorphism

Definition: Let $U(F)$ and $V(F)$ be two vector spaces over the same field F . A linear transformation from U into V is a function T from U into V such that

$$T(a\alpha + b\beta) = aT(\alpha) + bT(\beta) \quad \dots(1)$$

for all α, β in U and for all a, b in F .

The condition (1) is also called **linearity property**. It can be easily seen that the condition (1) is equivalent to the condition

$$T(a\alpha + \beta) = aT(\alpha) + T(\beta)$$

for all α, β in U and all scalars a in F .

Linear operator: Definition: Let $V(F)$ be a vector space. A linear operator on V is a function T from V into V such that

$$T(a\alpha + b\beta) = aT(\alpha) + bT(\beta)$$

for all α, β in V and for all a, b in F .

Thus T is a linear operator on V if T is a linear transformation from V into V itself.

Illustration 1: The function

$$T : V_3(\mathbf{R}) \rightarrow V_2(\mathbf{R})$$

defined by $T(a, b, c) = (a, b) \forall a, b, c \in \mathbf{R}$ is a linear transformation from $V_3(\mathbf{R})$ into $V_2(\mathbf{R})$.

Let $\alpha = (a_1, b_1, c_1), \beta = (a_2, b_2, c_2) \in V_3(\mathbf{R})$.

If $a, b \in \mathbf{R}$, then

$$\begin{aligned} T(a\alpha + b\beta) &= T[a(a_1, b_1, c_1) + b(a_2, b_2, c_2)] \\ &= T(aa_1 + ba_2, ab_1 + bb_2, ac_1 + bc_2) \\ &= (aa_1 + ba_2, ab_1 + bb_2) \quad [\text{by def. of } T] \\ &= (aa_1, ab_1) + (ba_2, bb_2) \\ &= a(a_1, b_1) + b(a_2, b_2) \end{aligned}$$

$$\begin{aligned}
 &= aT(a_1, b_1, c_1) + bT(a_2, b_2, c_2) \\
 &= aT(\alpha) + bT(\beta).
 \end{aligned}$$

$\therefore T$ is a linear transformation from $V_3(\mathbf{R})$ into $V_2(\mathbf{R})$.

Illustration 2: Let $V(F)$ be the vector space of all $m \times n$ matrices over the field F . Let P be a fixed $m \times m$ matrix over F , and let Q be a fixed $n \times n$ matrix over F . The correspondence T from V into V defined by

$$T(A) = PAQ \quad \forall A \in V$$

is a linear operator on V .

If A is an $m \times n$ matrix over the field F , then PAQ is also an $m \times n$ matrix over the field F . Therefore T is a function from V into V . Now let $A, B \in V$ and $a, b \in F$. Then

$$\begin{aligned}
 T(aA + bB) &= P(aA + bB)Q && [\text{by def. of } T] \\
 &= (aPA + bPB)Q = aPAQ + bPBQ = aT(A) + bT(B).
 \end{aligned}$$

$\therefore T$ is a linear transformation from V into V . Thus T is a linear operator on V .

Illustration 3: Let $V(F)$ be the vector space of all polynomials over the field F . Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \in V$ be a polynomial of degree n in the indeterminate x . Let us define

$$Df(x) = a_1 + 2a_2x + \dots + na_nx^{n-1} \quad \text{if } n > 1$$

and $Df(x) = 0$ if $f(x)$ is a constant polynomial.

Then the correspondence D from V into V is a linear operator on V .

If $f(x)$ is a polynomial over the field F , then $Df(x)$ as defined above is also a polynomial over the field F . Thus if $f(x) \in V$, then $Df(x) \in V$. Therefore D is a function from V into V .

Also if $f(x), g(x) \in V$ and $a, b \in F$, then

$$D[af(x) + bg(x)] = aDf(x) + bDg(x).$$

$\therefore D$ is a linear transformation from V into V .

The operator D on V is called the differentiation operator. It should be noted that for polynomials the definition of differentiation can be given purely algebraically, and does not require the usual theory of limiting processes.

Illustration 4: Let $V(\mathbf{R})$ be the vector space of all continuous functions from $\mathbf{R}$ into $\mathbf{R}$. If $f \in V$ and we define T by

$$(Tf)(x) = \int_0^x f(t) dt \quad \forall x \in \mathbf{R},$$

then T is a linear transformation from V into V .

If f is real valued continuous function, then Tf , as defined above, is also a real valued continuous function. Thus

$$f \in V \Rightarrow Tf \in V.$$

Also the operation of integration satisfies the linearity property. Therefore T is a linear transformation from V into V .

3.9 Some Particular Transformations

1. Zero Transformation: Let $U (F)$ and $V (F)$ be two vector spaces. The function T , from U into V defined by

$$T (\alpha) = \mathbf{0} \text{ (zero vector of } V) \quad \forall \alpha \in U$$

is a linear transformation from U into V .

Let $\alpha, \beta \in U$ and $a, b \in F$. Then $a\alpha + b\beta \in U$.

We have $T (a\alpha + b\beta) = \mathbf{0}$ [by def. of T]

$$= a\mathbf{0} + b\mathbf{0} = aT (\alpha) + bT (\beta).$$

$\therefore T$ is a linear transformation from U into V . It is called **zero transformation** and we shall in future denote it by $\hat{\mathbf{0}}$.

2. Identity operator: Let $V (F)$ be a vector space. The function I from V into V defined by $I (\alpha) = \alpha \quad \forall \alpha \in V$ is a linear transformation from V into V .

If $\alpha, \beta \in V$ and $a, b \in F$, then $a\alpha + b\beta \in V$ and we have

$$I (a\alpha + b\beta) = a\alpha + b\beta \quad [\text{by def. of } I]$$

$$= aI (\alpha) + bI (\beta).$$

$\therefore I$ is a linear transformation from V into V . The transformation I is called **identity operator** on V and we shall always denote it by I .

3. Negative of a linear transformation: Let $U (F)$ and $V (F)$ be two vector spaces. Let T be a linear transformation from U into V . The correspondence $-T$ defined by

$$(-T) (\alpha) = -[T (\alpha)] \quad \forall \alpha \in U$$

is a linear transformation from U into V .

Since $T (\alpha) \in V \Rightarrow -T (\alpha) \in V$, therefore $-T$ is a function from U into V .

Let $\alpha, \beta \in U$ and $a, b \in F$. Then $a\alpha + b\beta \in U$ and we have

$$(-T) (a\alpha + b\beta) = -[T (a\alpha + b\beta)] \quad [\text{by def. of } -T]$$

$$= -[aT (\alpha) + bT (\beta)] \quad [\because T \text{ is a linear transformation}]$$

$$= a[-T (\alpha)] + b[-T (\beta)] = a[(-T) \alpha] + b[(-T) \beta].$$

$\therefore -T$ is a linear transformation from U into V . The linear transformation $-T$ is called the **negative** of the linear transformation T .

3.10 Properties of Linear Transformations.

Theorem: Let T be a linear transformation from a vector space $U (F)$ into a vector space $V (F)$. Then

(i) $T (\mathbf{0}) = \mathbf{0}$ where $\mathbf{0}$ on the left hand side is zero vector of U and $\mathbf{0}$ on the right hand side is zero vector of V .

(ii) $T (-\alpha) = -T (\alpha) \quad \forall \alpha \in U$.

$$(iii) \quad T(\alpha - \beta) = T(\alpha) - T(\beta) \quad \forall \quad \alpha, \beta \in U.$$

$$(iv) \quad T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n) = a_1T(\alpha_1) + a_2T(\alpha_2) + \dots + a_nT(\alpha_n)$$

where $\alpha_1, \alpha_2, \dots, \alpha_n \in U$ and $a_1, a_2, \dots, a_n \in F$.

Proof: (i) Let $\alpha \in U$. Then $T(\alpha) \in V$. We have

$$\begin{aligned} T(\alpha) + \mathbf{0} &= T(\alpha) & [\because \mathbf{0} \text{ is zero vector of } V \text{ and } T(\alpha) \in V] \\ &= T(\alpha + \mathbf{0}) & [\because \mathbf{0} \text{ is zero vector of } U] \\ &= T(\alpha) + T(\mathbf{0}) & [\because T \text{ is a linear transformation}] \end{aligned}$$

Now in the vector space V , we have

$$T(\alpha) + \mathbf{0} = T(\alpha) + T(\mathbf{0})$$

$$\Rightarrow \quad \mathbf{0} = T(\mathbf{0}), \text{ by left cancellation law for addition in } V.$$

Note: When we write $T(\mathbf{0}) = \mathbf{0}$, there should be no confusion about the vector $\mathbf{0}$.

Here T is a function from U into V . Therefore if $\mathbf{0} \in U$, then its image under T i.e., $T(\mathbf{0}) \in V$. Thus in $T(\mathbf{0}) = \mathbf{0}$, the zero on the right hand side is zero vector of V .

$$(ii) \quad \text{We have } T[\alpha + (-\alpha)] = T(\alpha) + T(-\alpha) \quad [\because T \text{ is a linear transformation}]$$

$$\text{But } T[\alpha + (-\alpha)] = T(\mathbf{0}) = \mathbf{0} \in V. \quad [\text{by (i)}]$$

Thus in V , we have

$$T(\alpha) + T(-\alpha) = \mathbf{0}$$

$$\Rightarrow \quad T(-\alpha) = -T(\alpha).$$

$$\begin{aligned} (iii) \quad T(\alpha - \beta) &= T[\alpha + (-\beta)] \\ &= T(\alpha) + T(-\beta) & [\because T \text{ is linear}] \\ &= T(\alpha) + [-T(\beta)] & [\text{by (ii)}] \\ &= T(\alpha) - T(\beta). \end{aligned}$$

(iv) We shall prove the result by induction on n , the number of vectors in the linear combination $a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$. Suppose

$$\begin{aligned} T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_{n-1}\alpha_{n-1}) &= a_1T(\alpha_1) + a_2T(\alpha_2) \\ &\quad + \dots + a_{n-1}T(\alpha_{n-1}). \quad \dots(1) \end{aligned}$$

$$\begin{aligned} \text{Then } T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n) &= [T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_{n-1}\alpha_{n-1}) + a_n\alpha_n] \\ &= T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_{n-1}\alpha_{n-1}) + a_nT(\alpha_n) \\ &= [a_1T(\alpha_1) + a_2T(\alpha_2) + \dots + a_{n-1}T(\alpha_{n-1})] + a_nT(\alpha_n) \quad [\text{by (1)}] \\ &= a_1T(\alpha_1) + a_2T(\alpha_2) + \dots + a_{n-1}T(\alpha_{n-1}) + a_nT(\alpha_n). \end{aligned}$$

Now the proof is complete by induction since the result is true when the number of vectors in the linear combination is 1.

Note: On account of this property sometimes we say that a linear transformation **preserves linear combinations**.

3.11 Range and Null Space of a Linear Transformation

Range of a linear transformation:

Definition: Let $U(F)$ and $V(F)$ be two vector spaces and let T be a linear transformation from U into V . Then the range of T written as $R(T)$ is the set of all vectors β in V such that $\beta = T(\alpha)$ for some α in U .

Thus the range of T is the image set of U under T i.e.,

$$\text{Range}(T) = \{T(\alpha) \in V : \alpha \in U\}.$$

Theorem 1: If $U(F)$ and $V(F)$ are two vector spaces and T is a linear transformation from U into V , then range of T is a subspace of V .

Proof: Obviously $R(T)$ is a non-empty subset of V .

Let $\beta_1, \beta_2 \in R(T)$. Then there exist vectors α_1, α_2 in U such that

$$T(\alpha_1) = \beta_1, T(\alpha_2) = \beta_2.$$

Let a, b be any elements of the field F . We have

$$\begin{aligned} a\beta_1 + b\beta_2 &= aT(\alpha_1) + bT(\alpha_2) \\ &= T(a\alpha_1 + b\alpha_2) \quad [\because T \text{ is a linear transformation}] \end{aligned}$$

Now U is a vector space. Therefore $\alpha_1, \alpha_2 \in U$ and

$$a, b \in F \Rightarrow a\alpha_1 + b\alpha_2 \in U.$$

Consequently $T(a\alpha_1 + b\alpha_2) = a\beta_1 + b\beta_2 \in R(T)$.

Thus $a, b \in F$ and $\beta_1, \beta_2 \in R(T) \Rightarrow a\beta_1 + b\beta_2 \in R(T)$.

Therefore $R(T)$ is a subspace of V .

Null space of linear transformation:

Definition: Let $U(F)$ and $V(F)$ be two vector spaces and let T be a linear transformation from U into V . Then the null space of T written as $N(T)$ is the set of all vectors α in U such that $T(\alpha) = \mathbf{0}$ (zero vector of V). Thus

$$N(T) = \{\alpha \in U : T(\alpha) = \mathbf{0} \in V\}.$$

If we regard the linear transformation T from U into V as a vector space homomorphism of U into V , then the null space of T is also called the **kernel of T** .

Theorem 2: If $U(F)$ and $V(F)$ are two vector spaces and T is a linear transformation from U into V , then the kernel of T or the null space of T is a subspace of U .

Proof: Let $N(T) = \{\alpha \in U : T(\alpha) = \mathbf{0} \in V\}$.

Since $T(\mathbf{0}) = \mathbf{0} \in V$, therefore at least $\mathbf{0} \in N(T)$.

Thus $N(T)$ is a non-empty subset of U .

Let $\alpha_1, \alpha_2 \in N(T)$. Then $T(\alpha_1) = \mathbf{0}$ and $T(\alpha_2) = \mathbf{0}$.

Let $a, b \in F$. Then $a\alpha_1 + b\alpha_2 \in U$ and

$$T(a\alpha_1 + b\alpha_2) = aT(\alpha_1) + bT(\alpha_2) \quad [\because T \text{ is a linear transformation}]$$

$$= a\mathbf{0} + b\mathbf{0} = \mathbf{0} + \mathbf{0} = \mathbf{0} \in V.$$

$$\therefore a\alpha_1 + b\alpha_2 \in N(T).$$

Thus $a, b \in F$ and $\alpha_1, \alpha_2 \in N(T) \Rightarrow a\alpha_1 + b\alpha_2 \in N(T)$. Therefore $N(T)$ is a subspace of U .

3.12 Rank and Nullity of a Linear Transformation

Theorem 1: Let T be a linear transformation from a vector space $U(F)$ into a vector space $V(F)$. If U is finite dimensional, then the range of T is a finite dimensional subspace of V .

Proof: Since U is finite dimensional, therefore there exists a finite subset of U , say $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ which spans U .

Let $\beta \in \text{range of } T$. Then there exists α in U such that $T(\alpha) = \beta$.

Now $\alpha \in U \Rightarrow \exists a_1, a_2, \dots, a_n \in F$ such that

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$$

$$\Rightarrow T(\alpha) = T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n)$$

$$\Rightarrow \beta = a_1 T(\alpha_1) + a_2 T(\alpha_2) + \dots + a_n T(\alpha_n). \quad \dots(1)$$

Now the vectors $T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)$ are in the range of T . If β is any vector in the range of T , then from (1), we see that β can be expressed as a linear combination of $T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)$.

Therefore range of T is spanned by the vectors

$$T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n).$$

Hence range of T is finite dimensional.

Now we are in a position to define **rank** and **nullity** of a linear transformation.

Rank and nullity of a linear transformation:

Definition: Let T be a linear transformation from a vector space $U(F)$ into a vector space $V(F)$ with U as finite dimensional. The rank of T denoted by $\rho(T)$ is the dimension of the range of T i.e.,

$$\rho(T) = \dim R(T).$$

The nullity of T denoted by $\nu(T)$ is the dimension of the null space of T i.e.,

$$\nu(T) = \dim N(T).$$

Theorem 2: Let U and V be vector spaces over the field F and let T be a linear transformation from U into V . Suppose that U is finite dimensional. Then

$$\text{rank}(T) + \text{nullity}(T) = \dim U.$$

Proof: Let N be the null space of T . Then N is a subspace of U . Since U is finite dimensional, therefore N is finite dimensional. Let $\dim N = \text{nullity}(T) = k$ and let $\{\alpha_1, \alpha_2, \dots, \alpha_k\}$ be a basis for N .

Since $\{\alpha_1, \alpha_2, \dots, \alpha_k\}$ is a linearly independent subset of U , therefore we can extend it to form a basis of U . Let $\dim U = n$ and let

$$\{\alpha_1, \alpha_2, \dots, \alpha_k, \alpha_{k+1}, \dots, \alpha_n\}$$

be a basis for U .

The vectors $T(\alpha_1), T(\alpha_2), \dots, T(\alpha_k), T(\alpha_{k+1}), \dots, T(\alpha_n)$ are in the range of T . We claim that $\{T(\alpha_{k+1}), T(\alpha_{k+2}), \dots, T(\alpha_n)\}$ is a basis for the range of T .

(i) First we shall prove that the vectors

$$T(\alpha_{k+1}), T(\alpha_{k+2}), \dots, T(\alpha_n) \text{ span the range of } T.$$

Let $\beta \in \text{range of } T$. Then there exists $\alpha \in U$ such that

$$T(\alpha) = \beta.$$

Now $\alpha \in U \Rightarrow \exists a_1, a_2, \dots, a_n \in F$ such that

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$$

$$\Rightarrow T(\alpha) = T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n)$$

$$\Rightarrow \beta = a_1T(\alpha_1) + \dots + a_kT(\alpha_k) + a_{k+1}T(\alpha_{k+1}) + \dots + a_nT(\alpha_n)$$

$$\Rightarrow \beta = a_{k+1}T(\alpha_{k+1}) + a_{k+2}T(\alpha_{k+2}) + \dots + a_nT(\alpha_n)$$

$$[\because \alpha_1, \alpha_2, \dots, \alpha_k \in N \Rightarrow T(\alpha_1) = \mathbf{0}, \dots, T(\alpha_k) = \mathbf{0}]$$

$\therefore$ the vectors $T(\alpha_{k+1}), \dots, T(\alpha_n)$ span the range of T .

(ii) Now we shall show that the vectors

$$T(\alpha_{k+1}), \dots, T(\alpha_n)$$

are linearly independent.

Let $c_{k+1}, \dots, c_n \in F$ be such that

$$c_{k+1}T(\alpha_{k+1}) + \dots + c_nT(\alpha_n) = \mathbf{0}$$

$$\Rightarrow T(c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n) = \mathbf{0}$$

$$\Rightarrow c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n \in \text{null space of } T \text{ i.e., } N$$

$$\Rightarrow c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n = b_1\alpha_1 + b_2\alpha_2 + \dots + b_k\alpha_k$$

$$\text{for some } b_1, b_2, \dots, b_k \in F$$

$$[\because \text{each vector in } N \text{ can be expressed as a linear combination of the vectors } \alpha_1, \dots, \alpha_k \text{ forming a basis of } N]$$

$$\Rightarrow b_1\alpha_1 + \dots + b_k\alpha_k - c_{k+1}\alpha_{k+1} - \dots - c_n\alpha_n = \mathbf{0}$$

$$\Rightarrow b_1 = \dots = b_k = c_{k+1} = \dots = c_n = 0$$

$$[\because \alpha_1, \alpha_2, \dots, \alpha_k, \alpha_{k+1}, \dots, \alpha_n \text{ are linearly independent, being a basis for } U]$$

$$\Rightarrow \text{the vectors } T(\alpha_{k+1}), \dots, T(\alpha_n) \text{ are linearly independent.}$$

$$\therefore \text{the vectors } T(\alpha_{k+1}), \dots, T(\alpha_n) \text{ form a basis of range of } T.$$

$$\therefore \text{rank } T = \dim \text{ of range of } T = n - k.$$

$$\therefore \text{rank } (T) + \text{nullity } (T) = (n - k) + k = n = \dim U.$$

Note: If in place of the vector space V , we take the vector space U i.e., if T is a linear transformation on an n -dimensional vector space U , even then as a special case of the above theorem,

$$\rho(T) + \nu(T) = n.$$

Illustrative Examples

Example 5: Show that the mapping $T : V_3(\mathbf{R}) \rightarrow V_2(\mathbf{R})$ defined as

$$T(a_1, a_2, a_3) = (3a_1 - 2a_2 + a_3, a_1 - 3a_2 - 2a_3)$$

is a linear transformation from $V_3(\mathbf{R})$ into $V_2(\mathbf{R})$.

Solution: Let $\alpha = (a_1, a_2, a_3), \beta = (b_1, b_2, b_3) \in V_3(\mathbf{R})$.

Then $T(\alpha) = T(a_1, a_2, a_3) = (3a_1 - 2a_2 + a_3, a_1 - 3a_2 - 2a_3)$

and $T(\beta) = (3b_1 - 2b_2 + b_3, b_1 - 3b_2 - 2b_3)$.

Also let $a, b \in \mathbf{R}$. Then $a\alpha + b\beta \in V_3(\mathbf{R})$. We have

$$\begin{aligned} T(a\alpha + b\beta) &= T[a(a_1, a_2, a_3) + b(b_1, b_2, b_3)] \\ &= T(aa_1 + bb_1, aa_2 + bb_2, aa_3 + bb_3) \\ &= (3(aa_1 + bb_1) - 2(aa_2 + bb_2) + aa_3 + bb_3, \\ &\quad aa_1 + bb_1 - 3(aa_2 + bb_2) - 2(aa_3 + bb_3)) \\ &= (a(3a_1 - 2a_2 + a_3) + b(3b_1 - 2b_2 + b_3), \\ &\quad a(a_1 - 3a_2 - 2a_3) + b(b_1 - 3b_2 - 2b_3)) \\ &= a(3a_1 - 2a_2 + a_3, a_1 - 3a_2 - 2a_3) \\ &\quad + b(3b_1 - 2b_2 + b_3, b_1 - 3b_2 - 2b_3) \\ &= aT(\alpha) + bT(\beta). \end{aligned}$$

Hence T is a linear transformation from $V_3(\mathbf{R})$ into $V_2(\mathbf{R})$.

Example 6: Show that the mapping $T : V_2(\mathbf{R}) \rightarrow V_3(\mathbf{R})$ defined as

$$T(a, b) = (a + b, a - b, b)$$

is a linear transformation from $V_2(\mathbf{R})$ into $V_3(\mathbf{R})$. Find the range, rank, null-space and nullity of T .

Solution: Let $\alpha = (a_1, b_1), \beta = (a_2, b_2) \in V_2(\mathbf{R})$.

Then $T(\alpha) = T(a_1, b_1) = (a_1 + b_1, a_1 - b_1, b_1)$

and $T(\beta) = (a_2 + b_2, a_2 - b_2, b_2)$.

Also let $a, b \in \mathbf{R}$.

$$\begin{aligned} \text{Then } a\alpha + b\beta &\in V_2(\mathbf{R}) \text{ and } T(a\alpha + b\beta) = T[a(a_1, b_1) + b(a_2, b_2)] \\ &= T(aa_1 + ba_2, ab_1 + bb_2) \\ &= (aa_1 + ba_2 + ab_1 + bb_2, aa_1 + ba_2 - ab_1 - bb_2, ab_1 + bb_2) \\ &= (a[a_1 + b_1] + b[a_2 + b_2], a[a_1 - b_1] + b[a_2 - b_2], ab_1 + bb_2) \end{aligned}$$

$$= a (a_1 + b_1, a_1 - b_1, b_1) + b (a_2 + b_2, a_2 - b_2, b_2) \\ = aT(\alpha) + bT(\beta).$$

$\therefore T$ is a linear transformation from $V_2(\mathbf{R})$ into $V_3(\mathbf{R})$.

Now $\{(1, 0), (0, 1)\}$ is a basis for $V_2(\mathbf{R})$.

We have $T(1, 0) = (1 + 0, 1 - 0, 0) = (1, 1, 0)$

and $T(0, 1) = (0 + 1, 0 - 1, 1) = (1, -1, 1)$.

The vectors $T(1, 0), T(0, 1)$ span the range of T .

Thus the range of T is the subspace of $V_3(\mathbf{R})$ spanned by the vectors $(1, 1, 0), (1, -1, 1)$.

Now the vectors $(1, 1, 0), (1, -1, 1) \in V_3(\mathbf{R})$ are linearly independent because if $x, y \in \mathbf{R}$, then

$$\begin{aligned} & x(1, 1, 0) + y(1, -1, 1) = (0, 0, 0) \\ \Rightarrow & (x + y, x - y, y) = (0, 0, 0) \\ \Rightarrow & x + y = 0, x - y = 0, y = 0 \Rightarrow x = 0, y = 0. \end{aligned}$$

$\therefore$ the vectors $(1, 1, 0), (1, -1, 1)$ form a basis for the range of T .

Hence $\text{rank } T = \dim \text{ of range of } T = 2$.

Nullity of $T = \dim \text{ of } V_2(\mathbf{R}) - \text{rank } T = 2 - 2 = 0$.

$\therefore$ null space of T must be the zero subspace of $V_2(\mathbf{R})$.

Otherwise, $(a, b) \in \text{null space of } T$

$$\begin{aligned} \Rightarrow & T(a, b) = (0, 0, 0) \\ \Rightarrow & (a + b, a - b, b) = (0, 0, 0) \\ \Rightarrow & a + b = 0, a - b = 0, b = 0 \\ \Rightarrow & a = 0, b = 0. \end{aligned}$$

$\therefore (0, 0)$ is the only element of $V_2(\mathbf{R})$ which belongs to the null space of T .

$\therefore$ null space of T is the zero subspace of $V_2(\mathbf{R})$.

Example 7: Let V be the vector space of all $n \times n$ matrices over the field F and let B be a fixed $n \times n$ matrix. If

$$T(A) = AB - BA \quad \forall A \in V,$$

verify that T is a linear transformation from V into V .

Solution: If $A \in V$, then $T(A) = AB - BA \in V$ because $AB - BA$ is also an $n \times n$ matrix over the field F . Thus T is a function from V into V .

Let $A_1, A_2 \in V$ and $a, b \in F$.

Then $aA_1 + bA_2 \in V$

$$\begin{aligned} \text{and } T(aA_1 + bA_2) &= (aA_1 + bA_2)B - B(aA_1 + bA_2) \\ &= aA_1B + bA_2B - aBA_1 - bBA_2 \\ &= a(A_1B - BA_1) + b(A_2B - BA_2) \\ &= aT(A_1) + bT(A_2). \end{aligned}$$

$\therefore T$ is a linear transformation from V into V .

Example 8: Let V be an n -dimensional vector space over the field F and let T be a linear transformation from V into V such that the range and null space of T are identical. Prove that n is even. Give an example of such a linear transformation.

Solution: Let N be the null space of T . Then N is also the range of T .

Now $\rho(T) + \nu(T) = \dim V$

i.e., $\dim$ of range of T + $\dim$ of null space of $T = \dim V = n$

i.e., $2 \dim N = n$ [$\because$ range of $T =$ null space of $T = N$]

i.e., n is even.

Example of such a transformation:

Let $T : V_2(\mathbf{R}) \rightarrow V_2(\mathbf{R})$ be defined by

$$T(a, b) = (b, 0) \quad \forall a, b \in \mathbf{R}.$$

Let $\alpha = (a_1, b_1), \beta = (a_2, b_2) \in V_2(\mathbf{R})$ and let $x, y \in \mathbf{R}$.

$$\begin{aligned} \text{Then } T(x\alpha + y\beta) &= T(x(a_1, b_1) + y(a_2, b_2)) = T(xa_1 + ya_2, xb_1 + yb_2) \\ &= (xb_1 + yb_2, 0) = (xb_1, 0) + (yb_2, 0) = x(b_1, 0) + y(b_2, 0) \\ &= xT(a_1, b_1) + yT(a_2, b_2) = xT(\alpha) + yT(\beta). \end{aligned}$$

$\therefore T$ is a linear transformation from $V_2(\mathbf{R})$ into $V_2(\mathbf{R})$.

Now $\{(1, 0), (0, 1)\}$ is a basis of $V_2(\mathbf{R})$.

We have $T(1, 0) = (0, 0)$ and $T(0, 1) = (1, 0)$.

Thus the range of T is the subspace of $V_2(\mathbf{R})$ spanned by the vectors $(0, 0)$ and $(1, 0)$.

The vector $(0, 0)$ can be omitted from this spanning set because it is zero vector.

Therefore the range of T is the subspace of $V_2(\mathbf{R})$ spanned by the vector $(1, 0)$. Thus the range of $T = \{a(1, 0) : a \in \mathbf{R}\} = \{(a, 0) : a \in \mathbf{R}\}$.

Now let $(a, b) \in N$ (the null space of T).

$$\text{Then } (a, b) \in N \Rightarrow T(a, b) = (0, 0) \Rightarrow (b, 0) = (0, 0) \Rightarrow b = 0.$$

$$\therefore \text{null space of } T = \{(a, 0) : a \in \mathbf{R}\}.$$

Thus range of $T =$ null space of T .

Also we observe that $\dim V_2(\mathbf{R}) = 2$ which is even.

Example 9: Let V be a vector space and T a linear transformation from V into V . Prove that the following two statements about T are equivalent :

(i) The intersection of the range of T and the null space of T is the zero subspace of V i.e., $R(T) \cap N(T) = \{0\}$.

(ii) $T[T(\alpha)] = 0 \Rightarrow T(\alpha) = 0$.

Solution: First we shall show that (i) $\Rightarrow$ (ii).

We have $T[T(\alpha)] = 0 \Rightarrow T(\alpha) \in N(T)$

$$\Rightarrow T(\alpha) \in R(T) \cap N(T) \quad [\because \alpha \in V \Rightarrow T(\alpha) \in R(T)]$$

$$\Rightarrow T(\alpha) = 0 \text{ because } R(T) \cap N(T) = \{0\}.$$

Now we shall show that (ii) $\Rightarrow$ (i).

Let $\alpha \neq \mathbf{0}$ and $\alpha \in R(T) \cap N(T)$.

Then $\alpha \in R(T)$ and $\alpha \in N(T)$.

Since $\alpha \in N(T)$, therefore $T(\alpha) = \mathbf{0}$(1)

Also $\alpha \in R(T) \Rightarrow \exists \beta \in V$ such that $T(\beta) = \alpha$.

Now $T(\beta) = \alpha$

$\Rightarrow T[T(\beta)] = T(\alpha) = \mathbf{0}$ [From (1)]

Thus $\exists \beta \in V$ such that $T[T(\beta)] = \mathbf{0}$ but $T(\beta) = \alpha \neq \mathbf{0}$.

This contradicts the given hypothesis (ii).

Therefore there exists no $\alpha \in R(T) \cap N(T)$ such that $\alpha \neq \mathbf{0}$.

Hence $R(T) \cap N(T) = \{\mathbf{0}\}$.

Example 10: Consider the basis $S = \{\alpha_1, \alpha_2, \alpha_3\}$ of $\mathbf{R}^3$ where

$$\alpha_1 = (1, 1, 1), \alpha_2 = (1, 1, 0), \alpha_3 = (1, 0, 0).$$

Express $(2, -3, 5)$ in terms of the basis $\alpha_1, \alpha_2, \alpha_3$.

Let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be defined as

$$T(\alpha_1) = (1, 0), T(\alpha_2) = (2, -1), T(\alpha_3) = (4, 3).$$

Find $T(2, -3, 5)$.

Solution: Let $(2, -3, 5) = a\alpha_1 + b\alpha_2 + c\alpha_3$

$$= a(1, 1, 1) + b(1, 1, 0) + c(1, 0, 0).$$

Then $a + b + c = 2, a + b = -3, a = 5$.

Solving these equations, we get

$$a = 5, b = -8, c = 5.$$

$$\therefore (2, -3, 5) = 5\alpha_1 - 8\alpha_2 + 5\alpha_3.$$

$$\text{Now } T(2, -3, 5) = T(5\alpha_1 - 8\alpha_2 + 5\alpha_3)$$

$$= 5T(\alpha_1) - 8T(\alpha_2) + 5T(\alpha_3)$$

[$\because T$ is a linear transformation]

$$= 5(1, 0) - 8(2, -1) + 5(4, 3)$$

$$= (5, 0) - (16, -8) + (20, 15)$$

$$= (9, 23).$$

Comprehensive Exercise 2

1. Show that the mapping $T : V_3(\mathbf{R}) \rightarrow V_2(\mathbf{R})$ defined as

$$T(a_1, a_2, a_3) = (a_1 - a_2, a_1 - a_3)$$

is a linear transformation.

2. Show that the mapping $T : \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined as $T(x, y, z) = (z, x + y)$ is linear.

3. Show that the following functions are linear :

- (i) $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $T(a, b) = (b, a)$
- (ii) $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $T(a, b) = (a + b, a)$
- (iii) $T : \mathbf{R}^3 \rightarrow \mathbf{R}$ defined by $T(a, b, c) = 2a - 3b + 4c$.

4. Show that the following mappings T are not linear :

- (i) $T : \mathbf{R}^2 \rightarrow \mathbf{R}$ defined by $T(x, y) = xy$;
- (ii) $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $T(x, y) = (1 + x, y)$;
- (iii) $T : \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $T(x, y, z) = (|x|, 0)$;
- (iv) $T : \mathbf{R}^2 \rightarrow \mathbf{R}$ defined by $T(x, y) = |x - y|$.

5. Show that the mapping $T : \mathbf{R}^2 \rightarrow \mathbf{R}^3$ defined as

$$T(a, b) = (a - b, b - a, -a)$$

is a linear transformation from $\mathbf{R}^2$ into $\mathbf{R}^3$. Find the range, rank, null space and nullity of T .

6. Let F be the field of complex numbers and let T be the function from F^3 into F^3 defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2 + 2x_3, 2x_1 + x_2 - x_3, -x_1 - 2x_2).$$

Verify that T is a linear transformation. Describe the null space of T .

7. Let F be the field of complex numbers and let T be the function from F^3 into F^3 defined by

$$T(a, b, c) = (a - b + 2c, 2a + b, -a - 2b + 2c).$$

Show that T is a linear transformation. Find also the rank and the nullity of T .

8. Let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear transformation defined by :

$$T(x, y, z) = (x + 2y - z, y + z, x + y - 2z).$$

Find a basis and the dimension of (i) the range of T (ii) the null space of T .

9. Let $T : V_4(\mathbf{R}) \rightarrow V_3(\mathbf{R})$ be a linear transformation defined by

$$T(a, b, c, d) = (a - b + c + d, a + 2c - d, a + b + 3c - 3d).$$

Then obtain the basis and dimension of the range space of T and null space of T .

10. Let V be the vector space of polynomials in x over $\mathbf{R}$. Let $D : V \rightarrow V$ be the differential operator: $D(f) = \frac{df}{dx}$. Find the image (i.e. range) and kernel of D .

11. Let V be the vector space of $n \times n$ matrices over the field F and let E be an arbitrary matrix in V . Let $T : V \rightarrow V$ be defined as $T(A) = AE + EA$, $A \in V$. Show that T is linear.

12. Let V be the vector space of 2×2 matrices over $\mathbf{R}$ and let $M = \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}$. Let $T : V \rightarrow V$ be the linear function defined by $T(A) = MA$ for $A \in V$. Find a basis and the dimension of (i) the kernel of T and (ii) the range of T .
13. Let V be the vector space of 2×2 matrices over $\mathbf{R}$ and let $M = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Let $T : V \rightarrow V$ be the linear transformation defined by $T(A) = AM - MA$. Find a basis and the dimension of the kernel of T .
14. Let V be the space of $n \times 1$ matrices over a field F and let W be the space of $m \times 1$ matrices over F . Let A be a fixed $m \times n$ matrix over F and let T be the linear transformation from V into W defined by $T(X) = AX$. Prove that T is the zero transformation if and only if A is the zero matrix.
15. Let $U(F)$ and $V(F)$ be two vector spaces and let T_1, T_2 be two linear transformations from U to V . Let x, y be two given elements of F . Then the mapping T defined as $T(\alpha) = x T_1(\alpha) + y T_2(\alpha) \forall \alpha \in U$ is a linear transformation from U into V .

Answers 2

5. Null space of $T = \{0\}$; nullity of $T = 0$, rank $T = 2$
The set $\{(1, -1, -1), (-1, 1, 0)\}$ is a basis set for $R(T)$
6. Null space of $T = \{0\}$ 7. Rank $T = 2$; nullity of $T = 1$
8. (i) $\{(1, 0, 1), (2, 1, 1)\}$ is a basis of $R(T)$ and dim. $R(T) = 2$
(ii) $\{(3, -1, 1)\}$ is a basis of $N(T)$ and dim $N(T) = 1$
9. $\{(1, 1, 1), (0, 1, 2)\}$ is a basis of $R(T)$ and dim. $R(T) = 2$
 $\{(1, 2, 0, 1), (2, 1, -1, 0)\}$ is a basis of $N(T)$ and dim. $N(T) = 2$
10. The image of D is the whole space V
The kernel of D is the set of constant polynomials
12. (i) $\left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}\right\}$ is a basis of the kernel T and dim (kernel T) = 2
(ii) $\left\{\begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}\right\}$ is a basis for $R(T)$ and dim $R(T) = 2$
13. $\left\{\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ is a basis of the kernel of T and dim (kernel T) = 2

3.13 Linear Transformations as Vectors

Let $L(U, V)$ be the set of all linear transformations from a vector space $U(F)$ into a vector space $V(F)$. Sometimes we denote this set by $\text{Hom}(U, V)$. Now we want to impose a vector space structure on the set $L(U, V)$ over the same field F . For this purpose we shall have to suitably define addition in $L(U, V)$ and scalar multiplication in $L(U, V)$ over F .

Theorem 1: Let U and V be vector spaces over the field F . Let T_1 and T_2 be linear transformations from U into V . The function $T_1 + T_2$ defined by

$$(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha) \quad \forall \alpha \in U$$

is a linear transformation from U into V . If c is any element of F , the function (cT) defined by

$$(cT)(\alpha) = cT(\alpha) \quad \forall \alpha \in U$$

is a linear transformation from U into V . The set $L(U, V)$ of all linear transformations from U into V , together with the addition and scalar multiplication defined above is a vector space over the field F .

Proof: Suppose T_1 and T_2 are linear transformations from U into V and we define $T_1 + T_2$ as follows :

$$(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha) \quad \forall \alpha \in U. \quad \dots(1)$$

Since $T_1(\alpha) + T_2(\alpha) \in V$, therefore $T_1 + T_2$ is a function from U into V .

Let $a, b \in F$ and $\alpha, \beta \in U$. Then

$$\begin{aligned} (T_1 + T_2)(a\alpha + b\beta) &= T_1(a\alpha + b\beta) + T_2(a\alpha + b\beta) && [\text{by (1)}] \\ &= [aT_1(\alpha) + bT_1(\beta)] + [aT_2(\alpha) + bT_2(\beta)] \\ & \quad [\because T_1 \text{ and } T_2 \text{ are linear transformations}] \\ &= a[T_1(\alpha) + T_2(\alpha)] + b[T_1(\beta) + T_2(\beta)] && [\because V \text{ is a vector space}] \\ &= a(T_1 + T_2)(\alpha) + b(T_1 + T_2)(\beta) && [\text{by (1)}] \end{aligned}$$

$\therefore T_1 + T_2$ is a linear transformation from U into V . Thus

$$T_1, T_2 \in L(U, V) \Rightarrow T_1 + T_2 \in L(U, V).$$

Therefore $L(U, V)$ is closed with respect to addition defined in it.

Again let $T \in L(U, V)$ and $c \in F$. Let us define cT as follows :

$$(cT)(\alpha) = cT(\alpha) \quad \forall \alpha \in U. \quad \dots(2)$$

Since $cT(\alpha) \in V$, therefore cT is a function from U into V .

Let $a, b \in F$ and $\alpha, \beta \in U$. Then

$$\begin{aligned} (cT)(a\alpha + b\beta) &= cT(a\alpha + b\beta) && [\text{by (2)}] \\ &= c[aT(\alpha) + bT(\beta)] && [\because T \text{ is a linear transformation}] \\ &= c[aT(\alpha)] + c[bT(\beta)] = (ca)T(\alpha) + (cb)T(\beta) \\ &= (ac)T(\alpha) + (bc)T(\beta) = a[cT(\alpha)] + b[cT(\beta)] \\ &= a[(cT)(\alpha)] + b[(cT)(\beta)]. \end{aligned}$$

$\therefore cT$ is a linear transformation from U into V . Thus

$$T \in L(U, V) \text{ and } c \in F \Rightarrow cT \in L(U, V).$$

Therefore $L(U, V)$ is closed with respect to scalar multiplication defined in it.

Associativity of addition in $L(U, V)$:

Let $T_1, T_2, T_3 \in L(U, V)$. If $\alpha \in U$, then

$$\begin{aligned} [T_1 + (T_2 + T_3)](\alpha) &= T_1(\alpha) + (T_2 + T_3)(\alpha) \\ &\quad [\text{by (1) i.e., by def. of addition in } L(U, V)] \\ &= T_1(\alpha) + [T_2(\alpha) + T_3(\alpha)] \quad [\text{by (1)}] \\ &= [T_1(\alpha) + T_2(\alpha)] + T_3(\alpha) \quad [\because \text{addition in } V \text{ is associative}] \\ &= (T_1 + T_2)(\alpha) + T_3(\alpha) \quad [\text{by (1)}] \\ &= [(T_1 + T_2) + T_3](\alpha) \quad [\text{by (1)}] \end{aligned}$$

$$\begin{aligned} \therefore T_1 + (T_2 + T_3) &= (T_1 + T_2) + T_3 \\ &\quad [\text{by def. of equality of two functions}] \end{aligned}$$

Commutativity of addition in $L(U, V)$: Let $T_1, T_2 \in L(U, V)$. If α is any element of U , then

$$\begin{aligned} (T_1 + T_2)(\alpha) &= T_1(\alpha) + T_2(\alpha) \quad [\text{by (1)}] \\ &= T_2(\alpha) + T_1(\alpha) \quad [\because \text{addition in } V \text{ is commutative}] \\ &= (T_2 + T_1)(\alpha) \quad [\text{by (1)}] \\ \therefore T_1 + T_2 &= T_2 + T_1 \quad [\text{by def. of equality of two functions}] \end{aligned}$$

Existence of additive identity in $L(U, V)$: Let $\hat{0}$ be the zero function from U into V i.e., $\hat{0}(\alpha) = \mathbf{0} \in V \forall \alpha \in U$.

Then $\hat{0} \in L(U, V)$. If $T \in L(U, V)$ and $\alpha \in U$, we have

$$\begin{aligned} (\hat{0} + T)(\alpha) &= \hat{0}(\alpha) + T(\alpha) \quad [\text{by (1)}] \\ &= \mathbf{0} + T(\alpha) \quad [\text{by def. of } \hat{0}] \\ &= T(\alpha) \quad [\mathbf{0} \text{ being additive identity in } V] \end{aligned}$$

$$\therefore \hat{0} + T = T \quad \forall T \in L(U, V).$$

$$\therefore \hat{0} \text{ is the additive identity in } L(U, V).$$

Existence of additive inverse of each element in $L(U, V)$:

Let $T \in L(U, V)$. Let us define $-T$ as follows :

$$(-T)(\alpha) = -T(\alpha) \quad \forall \alpha \in U.$$

Then $-T \in L(U, V)$. If $\alpha \in U$, we have

$$\begin{aligned} (-T + T)(\alpha) &= (-T)(\alpha) + T(\alpha) \\ &\quad [\text{by def. of addition in } L(U, V)] \\ &= -T(\alpha) + T(\alpha) \quad [\text{by def. of } -T] \\ &= \mathbf{0} \in V \\ &= \hat{0}(\alpha) \quad [\text{by def. of } \hat{0}] \end{aligned}$$

$\therefore -T + T = \hat{0}$ for every $T \in L(U, V)$.

Thus each element in $L(U, V)$ possesses additive inverse.

Therefore $L(U, V)$ is an **abelian group** with respect to addition defined in it.

Further we make the following observations :

(i) Let $c \in F$ and $T_1, T_2 \in L(U, V)$. If α is any element in U , we have

$$\begin{aligned}
 [c(T_1 + T_2)](\alpha) &= c[(T_1 + T_2)(\alpha)] && [\text{by (2) i.e., by def. of scalar multiplication in } L(U, V)] \\
 &= c[T_1(\alpha) + T_2(\alpha)] && [\text{by (1)}] \\
 &= cT_1(\alpha) + cT_2(\alpha) \\
 & \quad [\because c \in F \text{ and } T_1(\alpha), T_2(\alpha) \in V \text{ which is a vector space}] \\
 &= (cT_1)(\alpha) + (cT_2)(\alpha) && [\text{by (2)}] \\
 &= (cT_1 + cT_2)(\alpha) && [\text{by (1)}]
 \end{aligned}$$

$\therefore c(T_1 + T_2) = cT_1 + cT_2$.

(ii) Let $a, b \in F$ and $T \in L(U, V)$. If $\alpha \in U$, we have

$$\begin{aligned}
 [(a + b)T](\alpha) &= (a + b)T(\alpha) && [\text{by (2)}] \\
 &= aT(\alpha) + bT(\alpha) && [\because V \text{ is a vector space}] \\
 &= (aT)(\alpha) + (bT)(\alpha) && [\text{by (2)}] \\
 &= (aT + bT)(\alpha) && [\text{by (1)}]
 \end{aligned}$$

$\therefore (a + b)T = aT + bT$.

(iii) Let $a, b \in F$ and $T \in L(U, V)$. If $\alpha \in U$, we have

$$\begin{aligned}
 [(ab)T](\alpha) &= (ab)T(\alpha) && [\text{by (2)}] \\
 &= a[bT(\alpha)] && [\because V \text{ is a vector space}] \\
 &= a[(bT)(\alpha)] && [\text{by (2)}] \\
 &= [a(bT)](\alpha) && [\text{by (2)}]
 \end{aligned}$$

$\therefore (ab)T = a(bT)$.

(iv) Let $1 \in F$ and $T \in L(U, V)$. If $\alpha \in U$, we have

$$\begin{aligned}
 (1T)(\alpha) &= 1T(\alpha) && [\text{by (2)}] \\
 &= T(\alpha) && [\because V \text{ is a vector space}]
 \end{aligned}$$

$\therefore 1T = T$.

Hence $L(U, V)$ is a vector space over the field F .

Note: If in place of the vector space V , we take U , then we observe that the set of all linear operators on U forms a vector space with respect to addition and scalar multiplication defined as above.

Dimension of $L(U, V)$: Now we shall prove that if $U(F)$ and $V(F)$ are finite dimensional, then the vector space of linear transformations from U into V is also finite dimensional. For this purpose we shall require an **important result** which we prove in the following theorem :

Theorem 2: Let U be a finite dimensional vector space over the field F and let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be an ordered basis for U . Let V be a vector space over the same field F and let $\beta_1, \dots, \beta_n$ be any vectors in V . Then there exists a unique linear transformation T from U into V such that

$$T(\alpha_i) = \beta_i, i = 1, 2, \dots, n.$$

Proof: Existence of T :

Let $\alpha \in U$.

Since $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a basis for U , therefore there exist unique scalars $x_1, x_2, \dots, x_n$ such that

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \dots + x_n\alpha_n.$$

For this vector α , let us define

$$T(\alpha) = x_1\beta_1 + x_2\beta_2 + \dots + x_n\beta_n.$$

Obviously $T(\alpha)$ as defined above is a unique element of V . Therefore T is a well-defined rule for associating with each vector α in U a unique vector $T(\alpha)$ in V . Thus T is a function from U into V .

The unique representation of $\alpha_i \in U$ as a linear combination of the vectors belonging to the basis B is

$$\alpha_i = 0\alpha_1 + 0\alpha_2 + \dots + 1\alpha_i + 0\alpha_{i+1} + \dots + 0\alpha_n.$$

Therefore according to our definition of T , we have

$$T(\alpha_i) = 0\beta_1 + 0\beta_2 + \dots + 1\beta_i + 0\beta_{i+1} + \dots + 0\beta_n$$

i.e., $T(\alpha_i) = \beta_i, i = 1, 2, \dots, n.$

Now to show that T is a linear transformation.

Let $a, b \in F$ and $\alpha, \beta \in U$. Let

$$\alpha = x_1\alpha_1 + \dots + x_n\alpha_n \quad \text{and} \quad \beta = y_1\alpha_1 + \dots + y_n\alpha_n.$$

Then $T(a\alpha + b\beta) = T[a(x_1\alpha_1 + \dots + x_n\alpha_n) + b(y_1\alpha_1 + \dots + y_n\alpha_n)]$

$$= T[(ax_1 + by_1)\alpha_1 + \dots + (ax_n + by_n)\alpha_n]$$

$$= (ax_1 + by_1)\beta_1 + \dots + (ax_n + by_n)\beta_n \quad [\text{by def. of } T]$$

$$= a(x_1\beta_1 + \dots + x_n\beta_n) + b(y_1\beta_1 + \dots + y_n\beta_n)$$

$$= aT(\alpha) + bT(\beta) \quad [\text{by def. of } T]$$

$\therefore T$ is a linear transformation from U into V . Thus there exists a linear transformation T from U into V such that

$$T(\alpha_i) = \beta_i, i = 1, 2, \dots, n.$$

Uniqueness of T : Let T' be a linear transformation from U into V such that

$$T'(\alpha_i) = \beta_i, i = 1, 2, \dots, n.$$

For the vector $\alpha = x_1\alpha_1 + \dots + x_n\alpha_n \in U$, we have

$$T'(\alpha) = T'(x_1\alpha_1 + \dots + x_n\alpha_n)$$

$$= x_1T'(\alpha_1) + \dots + x_nT'(\alpha_n)$$

[$\because T'$ is a linear transformation]

$$\begin{aligned}
 &= x_1 \beta_1 + \dots + x_n \beta_n && [\text{by def. of } T'] \\
 &= T(\alpha). && [\text{by def. of } T]
 \end{aligned}$$

Thus $T'(\alpha) = T(\alpha) \forall \alpha \in U$.

$\therefore T' = T$.

This shows the uniqueness of T .

Note: From this theorem we conclude that if T is a linear transformation from a finite dimensional vector space $U(F)$ into a vector space $V(F)$, then T is completely defined if we mention under T the images of the elements of a basis set of U . If S and T are two linear transformations from U into V such that

$$\begin{aligned}
 S(\alpha_i) &= T(\alpha_i) \forall \alpha_i \text{ belonging to a basis of } U, \text{ then} \\
 S(\alpha) &= T(\alpha) \forall \alpha \in U, \text{ i.e., } S = T.
 \end{aligned}$$

Thus two linear transformations from U into V are equal if they agree on a basis of U .

Theorem 3: Let U be an n -dimensional vector space over the field F , and let V be an m -dimensional vector space over F . Then the vector space $L(U, V)$ of all linear transformations from U into V is finite dimensional and is of dimension mn .

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_m\}$ be ordered bases for U and V respectively. By theorem 2, there exists a unique linear transformation T_{11} from U into V such that

$$T_{11}(\alpha_1) = \beta_1, T_{11}(\alpha_2) = \mathbf{0}, \dots, T_{11}(\alpha_n) = \mathbf{0}$$

where $\beta_1, \mathbf{0}, \dots, \mathbf{0}$ are vectors in V .

In fact, for each pair of integers (p, q) with $1 \leq p \leq m$ and $1 \leq q \leq n$, there exists a unique linear transformation T_{pq} from U into V such that

$$T_{pq}(\alpha_i) = \begin{cases} \mathbf{0}, & \text{if } i \neq q \\ \beta_p, & \text{if } i = q \end{cases}$$

$$\text{i.e., } T_{pq}(\alpha_i) = \delta_{iq} \beta_p, \quad \dots(1)$$

where $\delta_{iq} \in F$ is Kronecker delta i.e., $\delta_{iq} = 1$ if $i = q$ and $\delta_{iq} = 0$ if $i \neq q$.

Since p can be any of $1, 2, \dots, m$ and q any of $1, 2, \dots, n$, there are mn such T_{pq} 's. Let B_1 denote the set of these mn transformations T_{pq} 's. We shall show that B_1 is a basis for $L(U, V)$.

(i) First we shall show that $L(U, V)$ is a linear span of B_1 .

Let $T \in L(U, V)$. Since $T(\alpha_1) \in V$ and any element in V is a linear combination of $\beta_1, \beta_2, \dots, \beta_m$, therefore

$$T(\alpha_1) = a_{11} \beta_1 + a_{21} \beta_2 + \dots + a_{m1} \beta_m,$$

for some $a_{11}, a_{21}, \dots, a_{m1} \in F$. In fact for each $i, 1 \leq i \leq n$,

$$T(\alpha_i) = a_{1i} \beta_1 + a_{2i} \beta_2 + \dots + a_{mi} \beta_m = \sum_{p=1}^m a_{pi} \beta_p \quad \dots(2)$$

$$\text{Now consider } S = \sum_{p=1}^m \sum_{q=1}^n a_{pq} T_{pq}.$$

Obviously S is a linear combination of elements of B_1 which is a subset of $L(U, V)$.

Since $L(U, V)$ is a vector space, therefore $S \in L(U, V)$ i.e., S is also a linear transformation from U into V . We shall show that $S = T$.

Let us compute $S(\alpha_i)$ where α_i is any vector in the basis B of U . We have

$$\begin{aligned} S(\alpha_i) &= \left[\sum_{p=1}^m \sum_{q=1}^n a_{pq} T_{pq} \right] (\alpha_i) = \sum_{p=1}^m \sum_{q=1}^n a_{pq} T_{pq} (\alpha_i) \\ &= \sum_{p=1}^m \sum_{q=1}^n a_{pq} \delta_{iq} \beta_p \quad [\text{From (1)}] \\ &= \sum_{p=1}^m a_{pi} \beta_p \quad [\text{On summing with respect to } q. \text{ Remember} \\ &\quad \text{that } \delta_{iq} = 1 \text{ when } q = i \text{ and } \delta_{iq} = 0 \text{ when } q \neq i] \\ &= T(\alpha_i). \quad [\text{From (2)}] \end{aligned}$$

Thus $S(\alpha_i) = T(\alpha_i) \quad \forall \alpha_i \in B$. Therefore S and T agree on a basis of U . So we must have $S = T$. Thus T is also a linear combination of the elements of B_1 . Therefore $L(U, V)$ is a linear span of B_1 .

(ii) Now we shall show that B_1 is linearly independent. For b_{pq} 's $\in F$, let

$$\begin{aligned} &\sum_{p=1}^m \sum_{q=1}^n b_{pq} T_{pq} = \hat{0} \text{ i.e., zero vector of } L(U, V) \\ \Rightarrow &\left[\sum_{p=1}^m \sum_{q=1}^n b_{pq} T_{pq} \right] (\alpha_i) = \hat{0}(\alpha_i) \quad \forall \alpha_i \in B \\ \Rightarrow &\sum_{p=1}^m \sum_{q=1}^n b_{pq} T_{pq} (\alpha_i) = \mathbf{0} \in V \quad [\because \hat{0} \text{ is zero transformation}] \\ \Rightarrow &\sum_{p=1}^m \sum_{q=1}^n b_{pq} \delta_{iq} \beta_p = \mathbf{0} \quad \Rightarrow \quad \sum_{p=1}^m b_{pi} \beta_p = \mathbf{0} \\ \Rightarrow &b_{1i} \beta_1 + b_{2i} \beta_2 + \dots + b_{mi} \beta_m = \mathbf{0}, 1 \leq i \leq n \\ \Rightarrow &b_{1i} = 0, b_{2i} = 0, \dots, b_{mi} = 0, 1 \leq i \leq n \\ &\quad [\because \beta_1, \beta_2, \dots, \beta_m \text{ are linearly independent}] \\ \Rightarrow &b_{pq} = 0 \text{ where } 1 \leq p \leq m \text{ and } 1 \leq q \leq n \\ \Rightarrow &B_1 \text{ is linearly independent. Therefore } B_1 \text{ is a basis of } L(U, V). \\ \therefore &\dim L(U, V) = \text{number of elements in } B_1 = mn. \end{aligned}$$

Corollary: The vector space $L(U, U)$ of all linear operators on an n -dimensional vector space U is of dimension n^2 .

Note: Suppose $U(F)$ is an n -dimensional vector space and $V(F)$ is an m -dimensional vector space. If $U \neq \{\mathbf{0}\}$ and $V \neq \{\mathbf{0}\}$, then $n \geq 1$ and $m \geq 1$. Therefore $L(U, V)$ does not just consist of the element $\hat{0}$, because dimension of $L(U, V)$ is $mn \geq 1$.

3.14 Product of Linear Transformations

Theorem 1: Let U, V and W be vector spaces over the field F . Let T be a linear transformation from U into V and S a linear transformation from V into W . Then the composite function ST (called product of linear transformations) defined by

$$(ST)(\alpha) = S[T(\alpha)] \quad \forall \alpha \in U$$

is a linear transformation from U into W .

Proof: T is a function from U into V and S is a function from V into W .

So $\alpha \in U \Rightarrow T(\alpha) \in V$. Further

$$T(\alpha) \in V \Rightarrow S[T(\alpha)] \in W. \text{ Thus } (ST)(\alpha) \in W.$$

Therefore ST is a function from U into W . Now to show that ST is a linear transformation from U into W .

Let $a, b \in F$ and $\alpha, \beta \in U$. Then

$$\begin{aligned} (ST)(a\alpha + b\beta) &= S[T(a\alpha + b\beta)] \quad [\text{by def. of product of two functions}] \\ &= S[aT(\alpha) + bT(\beta)] \quad [\because T \text{ is a linear transformation}] \\ &= aS[T(\alpha)] + bS[T(\beta)] \quad [\because S \text{ is a linear transformation}] \\ &= a(ST)(\alpha) + b(ST)(\beta). \end{aligned}$$

Hence ST is a linear transformation from U into W .

Note: If T and S are linear operators on a vector space $V(F)$, then both the products ST as well as TS exist and each is a linear operator on V . However, in general $TS \neq ST$ as is obvious from the following examples.

Illustrative Examples

Example 11: Let T_1 and T_2 be linear operators on $\mathbf{R}^2$ defined as follows :

$$T_1(x_1, x_2) = (x_2, x_1) \quad \text{and} \quad T_2(x_1, x_2) = (x_1, 0).$$

Show that $T_1T_2 \neq T_2T_1$.

Solution: We have

$$\begin{aligned} (T_1T_2)(x_1, x_2) &= T_1[T_2(x_1, x_2)] \\ &= T_1(x_1, 0), & [\text{by def. of } T_2] \\ &= (0, x_1), & [\text{by def. of } T_1]. \end{aligned}$$

$$\begin{aligned} \text{Also } (T_2T_1)(x_1, x_2) &= T_2[T_1(x_1, x_2)], & [\text{by def. of } T_2T_1] \\ &= T_2(x_2, x_1), & [\text{by def. of } T_1] \\ &= (x_2, 0), & [\text{by def. of } T_2]. \end{aligned}$$

Thus we see that $(T_1T_2)(x_1, x_2) \neq (T_2T_1)(x_1, x_2) \quad \forall (x_1, x_2) \in \mathbf{R}^2$.

Hence by the definition of equality of two mappings, we have

$$T_1T_2 \neq T_2T_1.$$

Example 12: Let $S(\mathbf{R})$ be the vector space of all polynomial functions in x with coefficients as elements of the field $\mathbf{R}$ of real numbers. Let D and T be two linear operators on V defined by

$$D(f(x)) = \frac{d}{dx} f(x) \quad \dots(1)$$

and $T(f(x)) = \int_0^x f(x) dx \quad \dots(2)$

for every $f(x) \in V$.

Then show that $DT = I$ (identity operator) and $TD \neq I$.

Solution: Let $f(x) = a_0 + a_1x + a_2x^2 + \dots \in V$.

We have $(DT)(f(x)) = D[T(f(x))]$

$$= D\left[\int_0^x f(x) dx\right] = D\left[\int_0^x (a_0 + a_1x + a_2x^2 + \dots) dx\right]$$

$$= D\left[a_0x + \frac{a_1}{2}x^2 + \frac{a_2}{3}x^3 + \dots\right]_0^x = \frac{d}{dx}\left[a_0x + \frac{a_1}{2}x^2 + \dots\right]$$

$$= a_0 + a_1x + a_2x^2 + \dots = f(x) = I[f(x)].$$

Thus we have $(DT)[f(x)] = I[f(x)] \quad \forall f(x) \in V$. Therefore $DT = I$.

Now $(TD)f(x) = T[Df(x)]$

$$= T\left[\frac{d}{dx}(a_0 + a_1x + a_2x^2 + \dots)\right] = T(a_1 + 2a_2x + \dots)$$

$$= \int_0^x (a_1 + 2a_2x + \dots) dx = [a_1x + a_2x^2 + \dots]_0^x$$

$$= a_1x + a_2x^2 + \dots$$

$$\neq f(x) \text{ unless } a_0 = 0.$$

Thus $\exists f(x) \in V$ such that $(TD)[f(x)] \neq I[f(x)]$.

$\therefore TD \neq I$.

Hence $TD \neq DT$,

showing that product of linear operators is not in general commutative.

Example 13: Let $V(\mathbf{R})$ be the vector space of all polynomials in x with coefficients in the field $\mathbf{R}$. Let D and T be two linear transformations on V defined as

$$D[f(x)] = \frac{d}{dx} f(x) \quad \forall f(x) \in V \quad \text{and} \quad T[f(x)] = xf(x) \quad \forall f(x) \in V.$$

Then show that $DT \neq TD$.

Solution: We have

$$(DT)[f(x)] = D[T(f(x))] = D[xf(x)] = \frac{d}{dx}[xf(x)]$$

$$= f(x) + x \frac{d}{dx} f(x). \quad \dots(1)$$

Also $(TD)[f(x)] = T[D(f(x))] = T\left[\frac{d}{dx}(f(x))\right]$

$$= x \frac{d}{dx} f(x). \quad \dots(2)$$

From (1) and (2), we see that $\exists f(x) \in V$ such that

$$(DT)(f(x)) \neq (TD)(f(x)) \Rightarrow DT \neq TD.$$

Also we see that

$$\begin{aligned} (DT - TD)(f(x)) &= (DT)(f(x)) - (TD)(f(x)) \\ &= f(x) = I(f(x)). \end{aligned}$$

$$\therefore DT - TD = I.$$

Theorem 2: Let $V(F)$ be a vector space and A, B, C be linear transformations on V .

Then

- (i) $A\hat{0} = \hat{0} = \hat{0}A$ (ii) $AI = A = IA$
- (iii) $A(BC) = (AB)C$ (iv) $A(B + C) = AB + AC$
- (v) $(A + B)C = AC + BC$
- (vi) $c(AB) = (cA)B = A(cB)$ where c is any element of F .

Proof: Just for the sake of convenience we first mention here our definitions of addition, scalar multiplication and product of linear transformations :

$$(A + B)(\alpha) = A(\alpha) + B(\alpha) \quad \dots(1)$$

$$(cA)(\alpha) = cA(\alpha) \quad \dots(2)$$

$$(AB)(\alpha) = A[B(\alpha)] \quad \dots(3)$$

$\forall \alpha \in V$ and $\forall c \in F$.

Now we shall prove the above results.

- (i) We have $\forall \alpha \in V, (A\hat{0})(\alpha) = A[\hat{0}(\alpha)]$ [by (3)]
 $= A(\hat{0})$ [$\because \hat{0}$ is zero transformation]
 $= \hat{0} = \hat{0}(\alpha).$

$$\therefore A\hat{0} = \hat{0}. \quad \text{[by def. of equality of two functions]}$$

Similarly we can show that $\hat{0}A = \hat{0}$.

- (ii) We have $\forall \alpha \in V,$
 $(AI)(\alpha) = A[I(\alpha)]$
 $= A(\alpha)$ [$\because I$ is identity transformation]

$$\therefore AI = A.$$

Similarly we can show that $IA = A$.

- (iii) We have $\forall \alpha \in V$
 $[A(BC)](\alpha) = A[(BC)(\alpha)]$ [by (3)]
 $= A[B(C(\alpha))]$ [by (3)]
 $= (AB)[C(\alpha)]$ [by (3)]
 $= [(AB)C](\alpha).$ [by (3)]

$$\therefore A(BC) = (AB)C.$$

(iv) We have $\forall \alpha \in V$,

$$\begin{aligned} [A(B + C)](\alpha) &= A[(B + C)(\alpha)] && [\text{by (3)}] \\ &= A[B(\alpha) + C(\alpha)] && [\text{by (1)}] \\ &= A[B(\alpha)] + A[C(\alpha)] && [\because A \text{ is a linear} \\ &\quad \text{transformation and } B(\alpha), C(\alpha) \in V] \\ &= (AB)(\alpha) + (AC)(\alpha) && [\text{by (3)}] \\ &= (AB + AC)(\alpha) && [\text{by (1)}] \end{aligned}$$

$$\therefore A(B + C) = AB + AC.$$

(v) We have $\forall \alpha \in V$,

$$\begin{aligned} [(A + B)C](\alpha) &= (A + B)[C(\alpha)] && [\text{by (3)}] \\ &= A[C(\alpha)] + B[C(\alpha)] && [\text{by (1) since } C(\alpha) \in V] \\ &= (AC)(\alpha) + (BC)(\alpha) && [\text{by (3)}] \\ &= (AC + BC)(\alpha) && [\text{by (1)}] \end{aligned}$$

$$\therefore (A + B)C = AC + BC.$$

(vi) We have $\forall \alpha \in V$,

$$\begin{aligned} [c(AB)](\alpha) &= c[(AB)(\alpha)] && [\text{by (2)}] \\ &= c[A(B(\alpha))] && [\text{by (3)}] \\ &= (cA)[B(\alpha)] && [\text{by (2) since } B(\alpha) \in V] \\ &= [(cA)B](\alpha) && [\text{by (3)}] \end{aligned}$$

$$\therefore c(AB) = (cA)B.$$

$$\begin{aligned} \text{Again } [c(AB)](\alpha) &= c[(AB)(\alpha)] && [\text{by (2)}] \\ &= c[A(B(\alpha))] && [\text{by (3)}] \\ &= A[cB(\alpha)] && [\because A \text{ is a linear transformation and } B(\alpha) \in V] \\ &= A[(cB)(\alpha)] && [\text{by (2)}] \\ &= [A(cB)](\alpha). && [\text{by (3)}] \end{aligned}$$

$$\therefore c(AB) = A(cB).$$

3.15 Ring of Linear Operators on a Vector Space

Ring: Definition: A non-empty set R with two binary operations, to be denoted additively and multiplicatively, is called a ring if the following postulates are satisfied :

R_1 . R is closed with respect to addition i.e.,

$$a + b \in R \quad \forall a, b \in R.$$

R_2 . $(a + b) + c = a + (b + c) \quad \forall a, b, c \in R.$

R_3 . $a + b = b + a \quad \forall a, b \in R.$

R_4 . $\exists$ an element 0 (called zero element) in R such that

$$0 + a = a \quad \forall a \in R.$$

$R_5. a \in R \Rightarrow \exists -a \in R$ such that

$$(-a) + a = 0.$$

$R_6. R$ is closed with respect to multiplication i.e.,

$$ab \in R, \forall a, b \in R$$

$R_7. (ab)c = a(bc) \forall a, b, c \in R.$

$R_8. Multiplication is distributive with respect to addition, i.e.,$

$$a(b+c) = ab+ac \quad \text{and} \quad (a+b)c = ac+bc \quad \forall a, b, c \in R.$$

Ring with unity element: Definition:

If in a ring R there exists an element $1 \in R$ such that

$$1a = a = a1 \quad \forall a \in R,$$

then R is called a ring with unity element. The element 1 is called the unity element of the ring.

Theorem: The set $L(V, V)$ of all linear transformations from a vector space $V(F)$ into itself is a ring with unity element with respect to addition and multiplication of linear transformations defined as below :

$$(S+T)(\alpha) = S(\alpha) + T(\alpha)$$

and $(ST)(\alpha) = S[T(\alpha)] \quad \forall S, T \in L(V, V) \text{ and } \forall \alpha \in V.$

Proof: The students should themselves write the complete proof of this theorem. We have proved all the steps here and there. They should show here that all the ring postulates are satisfied in the set $L(V, V)$. The transformation $\hat{0}$ will act as the zero element and the identity transformation I will act as the unity element of this ring.

3.16 Algebra or Linear Algebra

Definition: Let F be a field. A vector space V over F is called a linear algebra over F if there is defined an additional operation in V called **multiplication of vectors** and satisfying the following postulates :

1. $\alpha\beta \in V \quad \forall \alpha, \beta \in V$
2. $\alpha(\beta\gamma) = (\alpha\beta)\gamma \quad \forall \alpha, \beta, \gamma \in V$
3. $\alpha(\beta+\gamma) = \alpha\beta + \alpha\gamma$ and $(\alpha+\beta)\gamma = \alpha\gamma + \beta\gamma \quad \forall \alpha, \beta, \gamma \in V.$
4. $c(\alpha\beta) = (c\alpha)\beta = \alpha(c\beta) \quad \forall \alpha, \beta \in V \text{ and } c \in F.$

If there is an element 1 in V such that

$$1\alpha = \alpha = \alpha 1 \quad \forall \alpha \in V,$$

then we call V a **linear algebra with identity over F** . Also 1 is then called the identity of V . The algebra V is **Commutative** if

$$\alpha\beta = \beta\alpha \quad \forall \alpha, \beta \in V.$$

Theorem: Let $V(F)$ be a vector space. The vector space $L(V, V)$ over F of all linear transformations from V into V is a linear algebra with identity with respect to the product of linear transformations as the multiplication composition in $L(V, V)$.

Proof: The students should write the complete proof here. All the necessary steps have been proved here and there.

3.17 Polynomials

Let T be a linear transformation on a vector space V (F). Then T is also a linear transformation on V . We shall write $T^1 = T$ and $T^2 = T T$. Since the product of linear transformations is an associative operation, therefore if m is a positive integer, we shall define

$$T^m = T T T \dots \text{upto } m \text{ times.}$$

Obviously T^m is a linear transformation on V .

Also we define $T^0 = I$ (identity transformation).

If m and n are non-negative integers, it can be easily seen that

$$T^m T^n = T^{m+n} \text{ and } (T^m)^n = T^{mn}.$$

The set $L(V, V)$ of all linear transformations on V is a vector space over the field F . If $a_0, a_1, \dots, a_n \in F$, then

$$p(T) = a_0 I + a_1 T + a_2 T^2 + \dots + a_n T^n \in L(V, V)$$

i.e., $p(T)$ is also a linear transformation on V because it is a linear combination over F of elements of $L(V, V)$. We call $p(T)$ as a polynomial in linear transformation T . The polynomials in a linear transformation behave like ordinary polynomials.

3.18 Invertible Linear Transformations

Definition: Let U and V be vector spaces over the field F . Let T be a linear transformation from U into V such that T is one-one onto. Then T is called invertible.

If T is a function from U into V , then T is said to be 1-1 if

$$\alpha_1, \alpha_2 \in U \text{ and } \alpha_1 \neq \alpha_2 \Rightarrow T(\alpha_1) \neq T(\alpha_2).$$

In other words T is said to be 1-1 if

$$\alpha_1, \alpha_2 \in U \text{ and } T(\alpha_1) = T(\alpha_2) \Rightarrow \alpha_1 = \alpha_2.$$

Further T is said to be onto if

$$\beta \in V \Rightarrow \exists \alpha \in U \text{ such that } T(\alpha) = \beta.$$

If T is one-one and onto, then we define a function from V into U , called the inverse of T and denoted by T^{-1} as follows :

Let β be any vector in V . Since T is onto, therefore

$$\beta \in V \Rightarrow \exists \alpha \in U \text{ such that } T(\alpha) = \beta.$$

Also α determined in this way is a unique element of U because T is one-one and therefore

$$\alpha_0, \alpha \in U \text{ and } \alpha_0 \neq \alpha \Rightarrow \beta = T(\alpha) \neq T(\alpha_0).$$

We define $T^{-1}(\beta)$ to be α . Thus

$$T^{-1} : V \Rightarrow U \text{ such that}$$

$$T^{-1}(\beta) = \alpha \Leftrightarrow T(\alpha) = \beta.$$

The function T^{-1} is itself one-one and onto. In the following theorem, we shall prove that T^{-1} is a linear transformation from V into U .

Theorem 1: Let U and V be vector spaces over the field F and let T be a linear transformation from U into V . If T is one-one and onto, then the inverse function T^{-1} is a linear transformation from V into U .

Proof: Let $\beta_1, \beta_2 \in V$ and $a, b \in F$.

Since T is one-one and onto, therefore there exist unique vectors $\alpha_1, \alpha_2 \in U$ such that $T(\alpha_1) = \beta_1, T(\alpha_2) = \beta_2$.

By definition of T^{-1} , we have

$$T^{-1}(\beta_1) = \alpha_1, T^{-1}(\beta_2) = \alpha_2.$$

Now $a\alpha_1 + b\alpha_2 \in U$ and we have by linearity of T ,

$$\begin{aligned} T(a\alpha_1 + b\alpha_2) &= aT(\alpha_1) + bT(\alpha_2) \\ &= a\beta_1 + b\beta_2 \in V. \end{aligned}$$

$\therefore$ by def. of T^{-1} , we have

$$\begin{aligned} T^{-1}(a\beta_1 + b\beta_2) &= a\alpha_1 + b\alpha_2 \\ &= aT^{-1}(\beta_1) + bT^{-1}(\beta_2). \end{aligned}$$

$\therefore T^{-1}$ is a linear transformation from V into U .

Theorem 2: Let T be an invertible linear transformation on a vector space $V(F)$. Then

$$T^{-1}T = I = TT^{-1}.$$

Proof: Let α be any element of V and let $T(\alpha) = \beta$. Then

$$T^{-1}(\beta) = \alpha.$$

We have $T(\alpha) = \beta$

$$\Rightarrow T^{-1}[T(\alpha)] = T^{-1}(\beta) \Rightarrow (T^{-1}T)(\alpha) = \alpha$$

$$\Rightarrow (T^{-1}T)(\alpha) = I(\alpha) \Rightarrow T^{-1}T = I.$$

Let β be any element of V . Since T is onto, therefore $\beta \in V \Rightarrow \exists \alpha \in V$ such that $T(\alpha) = \beta$. Then $T^{-1}(\beta) = \alpha$.

Now $T^{-1}(\beta) = \alpha$

$$\Rightarrow T[T^{-1}(\beta)] = T(\alpha)$$

$$\Rightarrow (TT^{-1})(\beta) = \beta$$

$$\Rightarrow (TT^{-1})(\beta) = \beta = I(\beta)$$

$$\Rightarrow TT^{-1} = I.$$

Theorem 3: If A, B and C are linear transformations on a vector space V (F) such that

$$AB = CA = I,$$

then A is invertible and $A^{-1} = B = C$.

Proof: In order to show that A is invertible, we are to show that A is one-one and onto.

(i) **A is one-one:**

Let $\alpha_1, \alpha_2 \in V$. Then

$$A(\alpha_1) = A(\alpha_2)$$

$$\Rightarrow C[A(\alpha_1)] = C[A(\alpha_2)]$$

$$\Rightarrow (CA)(\alpha_1) = (CA)(\alpha_2)$$

$$\Rightarrow I(\alpha_1) = I(\alpha_2)$$

$$\Rightarrow \alpha_1 = \alpha_2.$$

$\therefore A$ is **one-one**.

(ii) **A is onto:**

Let β be any element of V . Since B is a linear transformation on V , therefore $B(\beta) \in V$. Let $B(\beta) = \alpha$. Then

$$B(\beta) = \alpha$$

$$\Rightarrow A[B(\beta)] = A(\alpha)$$

$$\Rightarrow (AB)(\beta) = A(\alpha)$$

$$\Rightarrow I(\beta) = A(\alpha) \quad [\because AB = I]$$

$$\Rightarrow \beta = A(\alpha).$$

Thus $\beta \in V \Rightarrow \exists \alpha \in V$ such that $A(\alpha) = \beta$.

$\therefore A$ is onto.

Since A is one-one and onto therefore A is invertible i.e., A^{-1} exists.

(iii) Now we shall show that $A^{-1} = B = C$.

We have $AB = I$

$$\Rightarrow A^{-1}(AB) = A^{-1}I \Rightarrow (A^{-1}A)B = A^{-1}$$

$$\Rightarrow IB = A^{-1} \Rightarrow B = A^{-1}.$$

Again $CA = I$

$$\Rightarrow (CA)A^{-1} = IA^{-1} \Rightarrow C(AA^{-1}) = A^{-1}$$

$$\Rightarrow CI = A^{-1} \Rightarrow C = A^{-1}.$$

Hence the theorem.

Theorem 4: The necessary and sufficient condition for a linear transformation A on a vector space V (F) to be invertible is that there exists a linear transformation B on V such that

$$AB = I = BA.$$

Proof: The condition is necessary. For proof see theorem 2.

The condition is sufficient: For proof see theorem 3. Take B in place of C . Also we note that $B = A^{-1}$ and $A = B^{-1}$.

Theorem 5: Uniqueness of inverse: Let A be an invertible linear transformation on a vector space V (F). Then A possesses unique inverse.

Proof: Let B and C be two inverses of A . Then

$$AB = I = BA \quad \text{and} \quad AC = I = CA.$$

$$\text{We have} \quad C(AB) = CI = C. \quad \dots(1)$$

$$\text{Also} \quad (CA)B = IB = B. \quad \dots(2)$$

Since product of linear transformations is associative, therefore from (1) and (2), we get

$$C(AB) = (CA)B$$

$$\Rightarrow \quad C = B.$$

Hence the inverse of A is unique.

Theorem 6: Let V (F) be a vector space and let A, B be linear transformations on V . Then show that

(i) If A and B are invertible, then AB is invertible and

$$(AB)^{-1} = B^{-1} A^{-1}.$$

(ii) If A is invertible and $a \neq 0 \in F$, then aA is invertible and

$$(aA)^{-1} = \frac{1}{a} A^{-1}.$$

(iii) If A is invertible, then A^{-1} is invertible and $(A^{-1})^{-1} = A$.

Proof: (i) We have

$$\begin{aligned} (B^{-1} A^{-1})(AB) &= B^{-1} [A^{-1}(AB)] = B^{-1} [(A^{-1} A)B] \\ &= B^{-1} (IB) = B^{-1} B = I. \end{aligned}$$

$$\begin{aligned} \text{Also} \quad (AB)(B^{-1} A^{-1}) &= A[B(B^{-1} A^{-1})] = A[(BB^{-1})A^{-1}] \\ &= A(IA^{-1}) = AA^{-1} = I. \end{aligned}$$

$$\text{Thus} \quad (AB)(B^{-1} A^{-1}) = I = (B^{-1} A^{-1})(AB).$$

$\therefore$ By theorem 3, AB is invertible and $(AB)^{-1} = B^{-1} A^{-1}$.

$$(ii) \quad \text{We have} \quad (aA)\left(\frac{1}{a} A^{-1}\right) = a\left[A\left(\frac{1}{a} A^{-1}\right)\right]$$

$$= a\left[\frac{1}{a} (AA^{-1})\right] = \left(a \frac{1}{a}\right) (AA^{-1}) = I I = I.$$

$$\begin{aligned} \text{Also} \quad \left(\frac{1}{a} A^{-1}\right)(aA) &= \frac{1}{a} [A^{-1}(aA)] = \frac{1}{a} [a(A^{-1} A)] \\ &= \left(\frac{1}{a} a\right) (A^{-1} A) = I I = I. \end{aligned}$$

Thus $(aA) \left(\frac{1}{a} A^{-1} \right) = I = \left(\frac{1}{a} A^{-1} \right) (aA).$

$\therefore$ by theorem 3, aA is invertible and

$$(aA)^{-1} = \frac{1}{a} A^{-1}.$$

(iii) Since A is invertible, therefore

$$AA^{-1} = I = A^{-1} A.$$

$\therefore$ by theorem 3, A^{-1} is invertible and

$$A = (A^{-1})^{-1}.$$

Singular and Non-singular transformations:

Definition: Let T be a linear transformation from a vector space $U (F)$ into a vector space $V (F)$. Then T is said to be **non-singular** if the null space of T (i.e., $\ker T$) consists of the zero vector alone i.e., if

$$\alpha \in U \text{ and } T(\alpha) = \mathbf{0} \Rightarrow \alpha = \mathbf{0}.$$

If there exists a vector $\mathbf{0} \neq \alpha \in U$ such that $T(\alpha) = \mathbf{0}$, then T is said to be singular.

Theorem 7: Let T be a linear transformation from a vector space $U (F)$ into a vector space $V (F)$. Then T is non-singular if and only if T is one-one.

Proof: Given that T is non-singular. Then to prove that T is one-one.

Let $\alpha_1, \alpha_2 \in U$. Then

$$T(\alpha_1) = T(\alpha_2)$$

$$\Rightarrow T(\alpha_1) - T(\alpha_2) = \mathbf{0} \Rightarrow T(\alpha_1 - \alpha_2) = \mathbf{0}$$

$$\Rightarrow \alpha_1 - \alpha_2 = \mathbf{0} \quad [\because T \text{ is non-singular}]$$

$$\Rightarrow \alpha_1 = \alpha_2.$$

$\therefore T$ is one-one.

Conversely let T be one-one. We know that $T(\mathbf{0}) = \mathbf{0}$. Since T is one-one, therefore $\alpha \in U$ and $T(\alpha) = \mathbf{0} = T(\mathbf{0}) \Rightarrow \alpha = \mathbf{0}$. Thus the null space of T consists of zero vector alone. Therefore T is non-singular.

Theorem 8: Let T be a linear transformation from U into V . Then T is non-singular if and only if T carries each linearly independent subset of U onto a linearly independent subset of V .

Proof: First suppose that T is non-singular.

Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$

be a linearly independent subset of U . Then image of B under T is the subset B' of V given by

$$B' = \{T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)\}.$$

To prove that B' is linearly independent.

Let $a_1, a_2, \dots, a_n \in F$ and let

$$\begin{aligned}
& a_1 T(\alpha_1) + \dots + a_n T(\alpha_n) = \mathbf{0} \\
\Rightarrow & T(a_1 \alpha_1 + \dots + a_n \alpha_n) = \mathbf{0} & [\because T \text{ is linear}] \\
\Rightarrow & a_1 \alpha_1 + \dots + a_n \alpha_n = \mathbf{0} & [\because T \text{ is non-singular}] \\
\Rightarrow & a_i = 0, i = 1, 2, \dots, n & [\because \alpha_1, \dots, \alpha_n \text{ are linearly independent}]
\end{aligned}$$

Thus the image of B under T is linearly independent.

Conversely suppose that T carries independent subsets onto independent subsets. Then to prove that T is non-singular.

Let $\alpha \neq \mathbf{0} \in U$. Then the set $S = \{\alpha\}$ consisting of the one non-zero vector α is linearly independent. The image of S under T is the set

$$S' = \{T(\alpha)\}.$$

It is given that S' is also linearly independent. Therefore $T(\alpha) \neq \mathbf{0}$ because the set consisting of zero vector alone is linearly dependent. Thus

$$\mathbf{0} \neq \alpha \in U \Rightarrow T(\alpha) \neq \mathbf{0}.$$

This shows that the null space of T consists of the zero vector alone. Therefore T is non-singular.

Theorem 9: Let U and V be finite dimensional vector spaces over the field F such that $\dim U = \dim V$. If T is a linear transformation from U into V , the following are equivalent.

- (i) T is invertible.
- (ii) T is non-singular.
- (iii) The range of T is V .
- (iv) If $\{\alpha_1, \dots, \alpha_n\}$ is any basis for U , then $\{T(\alpha_1), \dots, T(\alpha_n)\}$ is a basis for V .
- (v) There is some basis $\{\alpha_1, \dots, \alpha_n\}$ for U such that $\{T(\alpha_1), \dots, T(\alpha_n)\}$ is a basis for V .

Proof: (i) $\Rightarrow$ (ii).

If T is invertible, then T is one-one. Therefore T is non-singular.

(ii) $\Rightarrow$ (iii).

Let T be non-singular. Let $\{\alpha_1, \dots, \alpha_n\}$ be a basis for U . Then $\{\alpha_1, \dots, \alpha_n\}$ is a linearly independent subset of U . Since T is non-singular therefore $\{T(\alpha_1), \dots, T(\alpha_n)\}$ is a linearly independent subset of V and it contains n vectors. Since $\dim V$ is also n , therefore this set of vectors is a basis for V . Now let β be any vector in V . Then there exist scalars $a_1, \dots, a_n \in F$ such that

$$\beta = a_1 T(\alpha_1) + \dots + a_n T(\alpha_n) = T(a_1 \alpha_1 + \dots + a_n \alpha_n)$$

which shows that β is in the range of T because

$$a_1 \alpha_1 + \dots + a_n \alpha_n \in U.$$

Thus every vector in V is in the range of T . Hence range of T is V .

(iii) $\Rightarrow$ (iv).

Now suppose that range of T is V i.e., T is onto. If $\{\alpha_1, \dots, \alpha_n\}$ is any basis for U , then the vectors $T(\alpha_1), \dots, T(\alpha_n)$ span the range of T which is equal to V . Thus the vectors $T(\alpha_1), \dots, T(\alpha_n)$ which are n in number span V whose dimension is also n . Therefore $\{T(\alpha_1), \dots, T(\alpha_n)\}$ must be a basis set for V .

$$(iv) \Rightarrow (v).$$

Since U is finite dimensional, therefore there exists a basis for U . Let $\{\alpha_1, \dots, \alpha_n\}$ be a basis for U . Then $\{T(\alpha_1), \dots, T(\alpha_n)\}$ is a basis for V as it is given in (iv).

$$(v) \Rightarrow (i).$$

Suppose there is some basis $\{\alpha_1, \dots, \alpha_n\}$ for U such that $\{T(\alpha_1), \dots, T(\alpha_n)\}$ is a basis for V . The vectors $\{T(\alpha_1), \dots, T(\alpha_n)\}$ span the range of T . Also they span V . Therefore the range of T must be all of V i.e., T is onto.

If $\alpha = c_1 \alpha_1 + \dots + c_n \alpha_n$ is in the null space of T , then

$$T(c_1 \alpha_1 + \dots + c_n \alpha_n) = \mathbf{0}$$

$$\Rightarrow c_1 T(\alpha_1) + \dots + c_n T(\alpha_n) = \mathbf{0}$$

$$\Rightarrow c_i = 0, 1 \leq i \leq n \text{ because } T(\alpha_1), \dots, T(\alpha_n) \text{ are linearly independent}$$

$$\Rightarrow \alpha = \mathbf{0}.$$

$\therefore T$ is non-singular and consequently T is one-one. Hence T is invertible.

Illustrative Examples

Example 14: Describe explicitly the linear transformation $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ such that $T(2, 3) = (4, 5)$ and $T(1, 0) = (0, 0)$.

Solution: First we shall show that the set $\{(2, 3), (1, 0)\}$ is a basis of $\mathbf{R}^2$. For linear independence of this set let

$$a(2, 3) + b(1, 0) = (0, 0), \text{ where } a, b \in \mathbf{R}.$$

$$\text{Then } (2a + b, 3a) = (0, 0)$$

$$\Rightarrow 2a + b = 0, 3a = 0$$

$$\Rightarrow a = 0, b = 0.$$

Hence the set $\{(2, 3), (1, 0)\}$ is linearly independent.

Now we shall show that the set $\{(2, 3), (1, 0)\}$ spans $\mathbf{R}^2$. Let $(x_1, x_2) \in \mathbf{R}^2$ and let $(x_1, x_2) = a(2, 3) + b(1, 0) = (2a + b, 3a)$.

Then $2a + b = x_1, 3a = x_2$. Therefore

$$a = \frac{x_2}{3}, b = \frac{3x_1 - 2x_2}{3}.$$

$$\therefore (x_1, x_2) = \frac{x_2}{3}(2, 3) + \frac{3x_1 - 2x_2}{3}(1, 0). \quad \dots(1)$$

From the relation (1) we see that the set $\{(2, 3), (1, 0)\}$ spans $\mathbf{R}^2$. Hence this set is a basis for $\mathbf{R}^2$.

Now let (x_1, x_2) be any member of $\mathbf{R}^2$. Then we are to find a formula for $T(x_1, x_2)$ under the conditions that $T(2, 3) = (4, 5)$, $T(1, 0) = (0, 0)$.

$$\begin{aligned} \text{We have } T(x_1, x_2) &= T\left[\frac{x_2}{3}(2, 3) + \frac{3x_1 - 2x_2}{3}(1, 0)\right], \text{ by (1)} \\ &= \frac{x_2}{3}T(2, 3) + \frac{3x_1 - 2x_2}{3}T(1, 0), \quad \text{by linearity of } T \\ &= \frac{x_2}{3}(4, 5) + \frac{3x_1 - 2x_2}{3}(0, 0) = \left(\frac{4x_2}{3}, \frac{5x_2}{3}\right). \end{aligned}$$

Example 15: Describe explicitly a linear transformation from $V_3(\mathbf{R})$ into $V_3(\mathbf{R})$ which has its range the subspace spanned by $(1, 0, -1)$ and $(1, 2, 2)$.

Solution: The set $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a basis for $V_3(\mathbf{R})$.

Also $\{(1, 0, -1), (1, 2, 2), (0, 0, 0)\}$ is a subset of $V_3(\mathbf{R})$. It should be noted that in this subset the number of vectors has been taken the same as is the number of vectors in the set B .

There exists a unique linear transformation T from $V_3(\mathbf{R})$ into $V_3(\mathbf{R})$ such that

$$\text{and } \left. \begin{aligned} T(1, 0, 0) &= (1, 0, -1), \\ T(0, 1, 0) &= (1, 2, 2), \\ T(0, 0, 1) &= (0, 0, 0). \end{aligned} \right\} \quad \dots(1)$$

Now the vectors $T(1, 0, 0)$, $T(0, 1, 0)$, $T(0, 0, 1)$ span the range of T . In other words the vectors

$$(1, 0, -1), (1, 2, 2), (0, 0, 0)$$

span the range of T . Thus the range of T is the subspace of $V_3(\mathbf{R})$ spanned by the set $\{(1, 0, -1), (1, 2, 2)\}$ because the zero vector $(0, 0, 0)$ can be omitted from the spanning set. Therefore T defined in (1) is the required transformation.

Now let us find an explicit expression for T . Let (a, b, c) be any element of $V_3(\mathbf{R})$. Then we can write

$$\begin{aligned} (a, b, c) &= a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1). \\ \therefore T(a, b, c) &= aT(1, 0, 0) + bT(0, 1, 0) + cT(0, 0, 1) \\ &= a(1, 0, -1) + b(1, 2, 2) + c(0, 0, 0) \quad [\text{from (1)}] \\ &= (a + b, 2b, 2b - a). \end{aligned}$$

Example 16: Let T be a linear operator on $V_3(\mathbf{R})$ defined by

$$T(a, b, c) = (3a, a - b, 2a + b + c) \quad \forall (a, b, c) \in V_3(\mathbf{R}).$$

Is T invertible? If so, find a rule for T^{-1} like the one which defines T .

Solution: Let us see that T is one-one or not.

Let $\alpha = (a_1, b_1, c_1)$, $\beta = (a_2, b_2, c_2) \in V_3(\mathbf{R})$.

Then $T(\alpha) = T(\beta)$

$$\Rightarrow T(a_1, b_1, c_1) = T(a_2, b_2, c_2)$$

$$\begin{aligned} \Rightarrow & (3a_1, a_1 - b_1, 2a_1 + b_1 + c_1) = (3a_2, a_2 - b_2, 2a_2 + b_2 + c_2) \\ \Rightarrow & 3a_1 = 3a_2, a_1 - b_1 = a_2 - b_2, 2a_1 + b_1 + c_1 = 2a_2 + b_2 + c_2 \\ \Rightarrow & a_1 = a_2, b_1 = b_2, c_1 = c_2. \end{aligned}$$

$\therefore T$ is one-one.

Now T is a linear transformation on a finite dimensional vector space $V_3(\mathbf{R})$ whose dimension is 3. Since T is one-one, therefore T must be onto also and thus T is invertible.

If $T(a, b, c) = (p, q, r)$, then $T^{-1}(p, q, r) = (a, b, c)$.

Now $T(a, b, c) = (p, q, r)$

$$\Rightarrow (3a, a - b, 2a + b + c) = (p, q, r)$$

$$\Rightarrow p = 3a, q = a - b, r = 2a + b + c$$

$$\Rightarrow a = \frac{p}{3}, b = \frac{p}{3} - q, c = r - 2a - b = r - \frac{2p}{3} - \frac{p}{3} + q = r - p + q.$$

$$\therefore T^{-1}(p, q, r) = \left(\frac{p}{3}, \frac{p}{3} - q, r - p + q \right) \quad \forall (p, q, r) \in V_3(\mathbf{R})$$

is the rule which defines T^{-1} .

Example 17: A linear transformation T is defined on $V_2(\mathbf{C})$ by

$$T(a, b) = (\alpha a + \beta b, \gamma a + \delta b),$$

where $\alpha, \beta, \gamma, \delta$ are fixed elements of $\mathbf{C}$. Prove that T is invertible if and only if $\alpha\delta - \beta\gamma \neq 0$.

Solution: The vector space $V_2(\mathbf{C})$ is of dimension 2. Therefore T is a linear transformation on a finite-dimensional vector space. T will be invertible if and only if the null space of T consists of zero vector alone. The zero vector of the space $V_2(\mathbf{C})$ is the ordered pair $(0, 0)$. Thus T is invertible

$$\text{iff } T(x, y) = (0, 0) \Rightarrow x = 0, y = 0$$

$$\text{i.e.,} \quad \text{iff } (\alpha x + \beta y, \gamma x + \delta y) = (0, 0) \Rightarrow x = 0, y = 0$$

$$\text{i.e.,} \quad \text{iff } \alpha x + \beta y = 0, \gamma x + \delta y = 0 \Rightarrow x = 0, y = 0.$$

Now the necessary and sufficient condition for the equations $\alpha x + \beta y = 0$, $\gamma x + \delta y = 0$ to have the only solution $x = 0, y = 0$ is that

$$\begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix} \neq 0.$$

Hence T is invertible iff $\alpha\delta - \beta\gamma \neq 0$.

Example 18: Find two linear operators T and S on $V_2(\mathbf{R})$ such that

$$TS = \hat{0} \quad \text{but} \quad ST \neq \hat{0}.$$

Solution: Consider the linear transformations T and S on $V_2(\mathbf{R})$ defined by

$$T(a, b) = (a, 0) \quad \forall (a, b) \in V_2(\mathbf{R})$$

and $S(a, b) = (0, a) \quad \forall (a, b) \in V_2(\mathbf{R})$.

$$\begin{aligned} \text{We have } (TS)(a, b) &= T[S(a, b)] = T(0, a) = (0, 0) \\ &= \hat{0}(a, b) \quad \forall (a, b) \in V_2(\mathbf{R}). \end{aligned}$$

$$\therefore TS = \hat{0}.$$

$$\begin{aligned} \text{Again } (ST)(a, b) &= S[T(a, b)] = S(a, 0) = (0, a) \\ &\neq \hat{0}(a, b) \quad \forall (a, b) \in V_2(\mathbf{R}). \end{aligned}$$

$$\text{Thus } ST \neq \hat{0}.$$

Example 19: Let V be a vector space over the field F and T a linear operator on V . If $T^2 = \hat{0}$, what can you say about the relation of the range of T to the null space of T ? Give an example of a linear operator T on $V_2(\mathbf{R})$ such that $T^2 = \hat{0}$ but $T \neq \hat{0}$.

Solution: We have $T^2 = \hat{0}$

$$\Rightarrow T^2(\alpha) = \hat{0}(\alpha) \quad \forall \alpha \in V$$

$$\Rightarrow T[T(\alpha)] = \hat{0} \quad \forall \alpha \in V$$

$$\Rightarrow T(\alpha) \in \text{null space of } T \quad \forall \alpha \in V.$$

But $T(\alpha) \in \text{range of } T \quad \forall \alpha \in V$.

$$\therefore T^2 = \hat{0} \Rightarrow \text{range of } T \subseteq \text{null space of } T.$$

For the second part of the question, consider the linear transformation T on $V_2(\mathbf{R})$ defined by

$$T(a, b) = (0, a) \quad \forall (a, b) \in V_2(\mathbf{R}).$$

Then obviously $T \neq \hat{0}$.

$$\begin{aligned} \text{We have } T^2(a, b) &= T[T(a, b)] = T(0, a) = (0, 0) \\ &= \hat{0}(a, b) \quad \forall (a, b) \in V_2(\mathbf{R}). \end{aligned}$$

$$\therefore T^2 = \hat{0}.$$

Example 20: If $T : U \rightarrow V$ is a linear transformation and U is finite dimensional, show that U and range of T have the same dimension iff T is non-singular. Determine all non-singular linear transformations

$$T : V_4(\mathbf{R}) \rightarrow V_3(\mathbf{R}).$$

Solution: We know that

$$\begin{aligned} \dim U &= \text{rank}(T) + \text{nullity}(T) \\ &= \dim \text{of range of } T + \dim \text{of null space of } T. \end{aligned}$$

$$\therefore \dim U = \dim \text{of range of } T$$

iff $\dim \text{of null space of } T$ is zero

i.e., iff null space of T consists of zero vector alone

i.e., iff T is non-singular.

Let T be a linear transformation from $V_4(\mathbf{R})$ into $V_3(\mathbf{R})$. Then T will be non-singular iff

$$\dim \text{ of } V_4(\mathbf{R}) = \dim \text{ of range of } T.$$

Now $\dim V_4(\mathbf{R}) = 4$ and $\dim \text{ of range of } T \leq 3$ because $\text{range of } T \subseteq V_3(\mathbf{R})$.

$\therefore \dim V_4(\mathbf{R})$ cannot be equal to $\dim \text{ of range of } T$.

Hence T cannot be non-singular. Thus there can be no non-singular linear transformation from $V_4(\mathbf{R})$ into $V_3(\mathbf{R})$.

Example 21: If A and B are linear transformations (on the same vector space), then a necessary and sufficient condition that both A and B be invertible is that both AB and BA be invertible.

Solution: Let A and B be two invertible linear transformations on a vector space V .

We have $(AB)(B^{-1}A^{-1}) = I = (B^{-1}A^{-1})(AB)$.

$\therefore AB$ is invertible.

Also we have $(BA)(A^{-1}B^{-1}) = I = (A^{-1}B^{-1})(BA)$.

$\therefore BA$ is also invertible.

Thus the condition is necessary.

Conversely, let AB and BA be both invertible. Then AB and BA are both one-one and onto.

First we shall show that A is invertible.

A is one-one: Let $\alpha_1, \alpha_2 \in V$. Then

$$\begin{aligned} A(\alpha_1) = A(\alpha_2) &\Rightarrow B[A(\alpha_1)] = B[A(\alpha_2)] \\ \Rightarrow (BA)(\alpha_1) = (BA)(\alpha_2) &\Rightarrow \alpha_1 = \alpha_2 \quad [\because BA \text{ is one-one}] \end{aligned}$$

$\therefore A$ is one-one.

A is onto:

Let $\beta \in V$. Since AB is onto, therefore there exists $\alpha \in V$ such that

$$(AB)(\alpha) = \beta \Rightarrow A[B(\alpha)] = \beta.$$

Thus $\beta \in V \Rightarrow \exists B(\alpha) \in V$ such that $A[B(\alpha)] = \beta$.

$\therefore A$ is onto.

$\therefore A$ is invertible.

Interchanging the roles played by AB and BA in the above proof, we can prove that B is invertible.

Example 22: If A is a linear transformation on a vector space V such that

$$A^2 - A + I = \hat{0},$$

then A is invertible.

Solution: If $A^2 - A + I = \hat{0}$, then $A^2 - A = -I$.

First we shall prove that A is one-one. Let $\alpha_1, \alpha_2 \in V$.

$$\text{Then } A(\alpha_1) = A(\alpha_2) \quad \dots(1)$$

$$\Rightarrow A[A(\alpha_1)] = A[A(\alpha_2)]$$

$$\Rightarrow A^2(\alpha_1) = A^2(\alpha_2) \quad \dots(2)$$

$$\Rightarrow A^2(\alpha_1) - A(\alpha_1) = A^2(\alpha_2) - A(\alpha_2) \quad [\text{From (2) and (1)}]$$

$$\Rightarrow (A^2 - A)(\alpha_1) = (A^2 - A)(\alpha_2)$$

$$\Rightarrow (-I)(\alpha_1) = (-I)(\alpha_2) \Rightarrow -[I(\alpha_1)] = -[I(\alpha_2)]$$

$$\Rightarrow -\alpha_1 = -\alpha_2 \Rightarrow \alpha_1 = \alpha_2.$$

$\therefore A$ is one-one.

Now to prove that A is onto.

Let $\alpha \in V$. Then $\alpha - A(\alpha) \in V$.

$$\begin{aligned} \text{We have } A[\alpha - A(\alpha)] &= A(\alpha) - A^2(\alpha) = (A - A^2)(\alpha) \\ &= I(\alpha) \quad [\because A^2 - A = -I \Rightarrow A - A^2 = I] \\ &= \alpha. \end{aligned}$$

Thus $\alpha \in V \Rightarrow \exists \alpha - A(\alpha) \in V$ such that $A[\alpha - A(\alpha)] = \alpha$.

$\therefore A$ is onto.

Hence A is invertible.

Example 23: Let V be a finite dimensional vector space and T be a linear operator on V . Suppose that $\text{rank}(T^2) = \text{rank}(T)$. Prove that the range and null space of T are disjoint i.e., have only the zero vector in common.

Solution: We have

$$\dim V = \text{rank}(T) + \text{nullity}(T) \text{ and } \dim V = \text{rank}(T^2) + \text{nullity}(T^2).$$

Since $\text{rank}(T) = \text{rank}(T^2)$, therefore we get $\text{nullity}(T) = \text{nullity}(T^2)$

i.e., $\dim$ of null space of $T = \dim$ of null space of T^2 .

Now $T(\alpha) = \mathbf{0}$

$$\Rightarrow T[T(\alpha)] = T(\mathbf{0}) \Rightarrow T^2(\alpha) = \mathbf{0}.$$

$$\therefore \alpha \in \text{null space of } T \Rightarrow \alpha \in \text{null space of } T^2.$$

$$\therefore \text{null space of } T \subseteq \text{null space of } T^2.$$

But null space of T and null space of T^2 are both subspaces of V and have the same dimension.

$$\therefore \text{null space of } T = \text{null space of } T^2.$$

$$\therefore \text{null space of } T^2 \subseteq \text{null space of } T$$

$$\text{i.e., } T^2(\alpha) = \mathbf{0} \Rightarrow T(\alpha) = \mathbf{0}.$$

$$\therefore \text{range and null space of } T \text{ are disjoint.} \quad [\text{See Example 5 after 3.12}]$$

Example 24: Let V be a finite dimensional vector space over the field F .

Let $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $\{\beta_1, \beta_2, \dots, \beta_n\}$ be two ordered bases for V . Show that there exists a unique invertible linear transformation T on V such that

$$T(\alpha_i) = \beta_i, 1 \leq i \leq n.$$

Solution: We have proved in one of the previous theorems that there exists a unique linear transformation T on V such that $T(\alpha_i) = \beta_i, 1 \leq i \leq n$.

Here we are to show that T is invertible. Since V is finite dimensional therefore in order to prove that T is invertible, it is sufficient to prove that T is non-singular.

Let $\alpha \in V$ and $T(\alpha) = \mathbf{0}$.

Let $\alpha = a_1 \alpha_1 + \dots + a_n \alpha_n$ where $a_1, \dots, a_n \in F$.

We have $T(\alpha) = \mathbf{0}$

$$\Rightarrow T(a_1 \alpha_1 + \dots + a_n \alpha_n) = \mathbf{0} \Rightarrow a_1 T(\alpha_1) + \dots + a_n T(\alpha_n) = \mathbf{0}$$

$$\Rightarrow a_1 \beta_1 + \dots + a_n \beta_n = \mathbf{0}$$

$$\Rightarrow a_i = 0 \text{ for each } 1 \leq i \leq n \quad [\because \beta_1, \dots, \beta_n \text{ are linearly independent}]$$

$$\Rightarrow \alpha = \mathbf{0}.$$

$\therefore T$ is non-singular because null space of T consists of zero vector alone.

Hence T is invertible.

Comprehensive Exercise 3

1. Describe explicitly the linear transformation T from F^2 to F^2 such that $T(e_1) = (a, b)$, $T(e_2) = (c, d)$ where $e_1 = (1, 0)$, $e_2 = (0, 1)$.
2. Find a linear transformation $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ such that $T(1, 0) = (1, 1)$ and $T(0, 1) = (-1, 2)$. Prove that T maps the square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$ into a parallelogram.
3. Let $T : \mathbf{R}^2 \rightarrow \mathbf{R}$ be the linear transformation for which $T(1, 1) = 3$ and $T(0, 1) = -2$. Find $T(a, b)$.
4. Describe explicitly a linear transformation from $V_3(\mathbf{R})$ into $V_4(\mathbf{R})$ which has its range the subspace spanned by the vectors $(1, 2, 0, -4)$, $(2, 0, -1, -3)$.
5. Find a linear mapping $T : \mathbf{R}^3 \rightarrow \mathbf{R}^4$ whose image is generated by $(1, -1, 2, 3)$ and $(2, 3, -1, 0)$.
6. Let F be any field and let T be a linear operator on F^2 defined by $T(a, b) = (a + b, a)$. Show that T is invertible and find a rule for T^{-1} like the one which defines T .
7. Show that the operator T on $\mathbf{R}^3$ defined by $T(x, y, z) = (x + z, x - z, y)$ is invertible and find similar rule defining T^{-1} .

8. For the linear operator T of Solved Example 16, after article 3.18, prove that
- $$(T^2 - I)(T - 3I) = \hat{0}.$$
9. Let T and U be the linear operators on $\mathbf{R}^2$ defined by $T(a, b) = (b, a)$ and $U(a, b) = (a, 0)$. Give rules like the one defining T and U for each of the linear transformation $(U + T), UT, TU, T^2, U^2$.
10. Let T be the (unique) linear operator on $\mathbf{C}^3$ for which
- $$T(1, 0, 0) = (1, 0, i), \quad T(0, 1, 0) = (0, 1, 1), \quad T(0, 0, 1) = (i, 1, 0).$$
- Show that T is not invertible.
11. Show that if two linear transformations of a finite dimensional vector space coincide on a basis of that vector space, then they are identical.
12. If T is a linear transformation on a finite dimensional vector space V such that $\text{range}(T)$ is a proper subset of V , show that there exists a non-zero element α in V with $T(\alpha) = \hat{0}$.
13. Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be defined as $T(a, b, c) = (0, a, b)$. Show that
- $$T \neq \hat{0}, \quad T^2 \neq \hat{0} \quad \text{but} \quad T^3 = \hat{0}.$$
14. Let T be a linear transformation from a vector space U into a vector space V with $\text{Ker } T \neq \hat{0}$. Show that there exist vectors α_1 and α_2 in U such that $\alpha_1 \neq \alpha_2$ and $T\alpha_1 = T\alpha_2$.
15. Let T be a linear transformation from $V_3(\mathbf{R})$ into $V_2(\mathbf{R})$, and let S be a linear transformation from $V_2(\mathbf{R})$ into $V_3(\mathbf{R})$. Prove that the transformation ST is not invertible.
16. Let A and B be linear transformations on a finite dimensional vector space V and let $AB = I$. Then A and B are both invertible and $A^{-1} = B$. Give an example to show that this is false when V is not finite dimensional.
17. If A and B are linear transformations (on the same vector space) and if $AB = I$, then A is called a left inverse of B and B is called a right inverse of A . Prove that if A has exactly one right inverse, say B , then A is invertible.
18. Prove that the set of invertible linear operators on a vector space V with the operation of composition forms a group. Check if this group is commutative.
19. Let V and W be vector spaces over the field F and let U be an isomorphism of V onto W . Prove that $T \rightarrow UTU^{-1}$ is an isomorphism of $L(V, V)$ onto $L(W, W)$.
20. If $\{\alpha_1, \dots, \alpha_k\}$ and $\{\beta_1, \dots, \beta_k\}$ are linearly independent sets of vectors in a finite dimensional vector space V , then there exists an invertible linear transformation T on V such that
- $$T(\alpha_i) = \beta_i, \quad i = 1, \dots, k.$$

Answers 3

1. $T(x_1, x_2) = (x_1 a + x_2 c, x_1 b + x_2 d)$
2. $T(x_1, x_2) = (x_1 - x_2, x_1 + 2x_2)$
3. $T(a, b) = 5a - 2b$
4. $T(a, b, c) = (a + 2b, 2a - b, -4a - 3b)$
5. $T(a, b, c) = (a + 2b, -a + 3b, 2a - b, 3a)$
6. $T^{-1}(p, q) = (q, p - q)$
7. $T^{-1}(x, y, z) = \left(\frac{1}{2}x + \frac{1}{2}y, z, \frac{1}{2}x - \frac{1}{2}y\right)$
9. $(U + T)(a, b) = (a + b, a); (UT)(a, b) = (b, 0); (TU)(a, b) = (0, \alpha)$
 $T^2(a, b) = (a, b); U^2(a, b) = (a, 0)$
18. Not commutative

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. If W be a subspace of a finite dimensional vector space $V(F)$, then $\dim V/W =$
 - (a) $\dim V - \dim W$
 - (b) $\dim V + \dim W$
 - (c) $(\dim V) / (\dim W)$
 - (d) None of these.
2. Two finite dimensional vector spaces U and V over a field F are isomorphic if
 - (a) $\dim U = \dim V$
 - (b) $\dim U \neq \dim V$
 - (c) $\dim U < \dim V$
 - (d) $\dim U > \dim V$
3. The dimension of $\frac{V}{W}$, where $V = R^3$, $W = \{(a, 0, 0) : a \in R\}$ is
 - (a) 1
 - (b) 3
 - (c) 2
 - (d) 4
4. If $U(F)$ and $V(F)$ are two vector spaces and T is a linear transformation from U into V , then range of T is a subspace of
 - (a) U
 - (b) V
 - (c) $U \cup V$
 - (d) None of these.

5. Let T be a linear transformation from a vector space U (F) into a vector space V (F) with U as finite dimensional. The rank of T is the dimension of the
 - (a) range of T
 - (b) null space of T
 - (c) vector space U
 - (d) vector space V .
6. Let U be an n -dimensional vector space over the field F , and let V be an m -dimensional vector space over F . Then the vector space $L(U, V)$ of all linear transformations from U into V is finite dimensional and is of dimension
 - (a) m
 - (b) n
 - (c) mn
 - (d) none of these.
7. If $T : V_2(\mathbf{R}) \rightarrow V_3(\mathbf{R})$ defined as $T(a, b) = (a + b, a - b, b)$ is a linear transformation, then nullity of T is
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) none of these.
8. If T is a linear transformation from a vector space V into a vector space W , then the condition for T^{-1} to be a linear transformation from W to V is
 - (a) T should be one-one
 - (b) T should be onto
 - (c) T should be one-one and onto
 - (d) none of these.
9. Let F be any field and let T be a linear operator on F^2 defined by $T(a, b) = (a + b, a)$. Then $T^{-1}(a, b) =$
 - (a) $(b, a - b)$
 - (b) $(a - b, b)$
 - (c) $(a, a + b)$
 - (d) none of these.
10. If T is a linear transformation $T(a, b) = (\alpha a + \beta b, \gamma a + \delta b)$, the T is invertible if
 - (a) $\alpha\beta - \gamma\delta = 0$
 - (b) $\alpha\beta - \gamma\delta \neq 0$
 - (c) $\alpha\delta - \beta\gamma = 0$
 - (d) $\alpha\delta - \beta\gamma \neq 0$
11. Let $V(F)$ be a vector space and let T_1, T_2 be linear Transformations on V . If T_1 and T_2 are invertible then
 - (a) $(T_1 T_2)^{-1} = T_2 T_1^{-1}$
 - (b) $(T_1 T_2)^{-1} = T_1 T_2^{-1}$
 - (c) $(T_1 T_2)^{-1} = T_2^{-1} T_1^{-1}$
 - (d) $(T_1 T_2)^{-1} = T_1^{-1} T_2^{-1}$

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. If W is an m -dimensional subspace of an n -dimensional vector space V , then the dimension of the quotient space $\frac{V}{W}$ is
2. Two subspaces W_1 and W_2 of the vector space $V(F)$ are said to be disjoint if $W_1 \cap W_2 = \dots\dots\dots$

3. Let $V(F)$ be a vector space. A linear operator on V is a function T from V into V such that $T(a\alpha + b\beta) = \dots$ for all α, β in V and for all a, b in F .
4. If $U(F)$ and $V(F)$ are two vector spaces and T is a linear transformation from U into V , then the kernel of T is a subspace of U .
5. Let U and V be vector spaces over the field F and let T be a linear transformation from U into V . Suppose that U is finite dimensional. Then $\text{rank}(T) + \text{nullity}(T) = \dots$.
6. Let V be an n -dimensional vector space over the field F and let T be a linear transformation from V into V such that the range and null space of T are identical. Then n is $\dots$.
7. The vector space of all linear operators on an n -dimensional vector space U is of dimension $\dots$.
8. Let $V(F)$ be a vector space and let A, B be linear transformations on V . If A and B are invertible then AB is invertible and $(AB)^{-1} = \dots$.
9. A linear operator T on $\mathbf{R}^2$ defined by $T(x, y) = (ax + by, cx + dy)$ will be invertible iff $\dots$.

True or False

Write 'T' for true and 'F' for false statement.

1. If a finite dimensional vector space $V(F)$ is a direct sum of two subspaces W_1 and W_2 , then $\dim V = \dim W_1 + \dim W_2$.
2. Let $U(F)$ and $V(F)$ be two vector spaces and let T be a linear transformation from U into V . Then the null space of T is the set of all vectors α in U such that $T(\alpha) = \alpha$.
3. The function $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $T(a, b) = (1 + a, b)$ is a linear transformation.
4. Two linear transformations from U into V are equal if they agree on a basis of U .
5. For two linear operators T and S on $\mathbf{R}^2$, $TS = \hat{0} \rightarrow ST = \hat{0}$.
6. If S and T are linear operators on a vector space U , then $(S + T)^2 = S^2 + 2ST + T^2$.
7. The identity operator on a vector space is always invertible.

Answers

Multiple Choice Questions

- | | | | |
|--------|---------|---------|--------|
| 1. (a) | 2. (a) | 3. (c) | 4. (b) |
| 5. (a) | 6. (c) | 7. (a) | 8. (c) |
| 9. (a) | 10. (d) | 11. (c) | |

Fill in the Blank(s)

- | | | |
|------------|--------------------|-----------------------------|
| 1. $n - m$ | 2. $\{0\}$ | 3. $aT(\alpha) + bT(\beta)$ |
| 4. U | 5. $\dim U$ | 6. even |
| 7. n^2 | 8. $B^{-1} A^{-1}$ | 9. $ad - bc \neq 0$ |

True or False

- | | | | |
|--------|--------|--------|--------|
| 1. T | 2. F | 3. F | 4. T |
| 5. F | 6. F | 7. T | |



Chapter

4



Matrices and Linear Transformations

4.1 Matrix

Definition: Let F be any field. A set of mn elements of F arranged in the form of a rectangular array having m rows and n columns is called an $m \times n$ matrix over the field F .

An $m \times n$ matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}.$$

In a compact form the above matrix is represented by $A = [a_{ij}]_{m \times n}$. The element a_{ij} is called the $(i, j)^{\text{th}}$ element of the matrix A . In this element the first suffix i will always denote the number of row in which this element occurs.

If in a matrix A the number of rows is equal to the number of columns and is equal to n , then A is called a **square matrix** of order n and the elements a_{ij} for which $i = j$ constitute its **principal diagonal**.

Unit matrix: A square matrix each of whose diagonal elements is equal to 1 and each of whose non-diagonal elements is equal to zero is called a unit matrix or an identity matrix. We shall denote it by I . Thus if I is unit matrix of order n , then $I = [\delta_{ij}]_{n \times n}$ where δ_{ij} is Kronecker delta.

Diagonal matrix: A square matrix is said to be a diagonal matrix if all the elements lying above and below the principal diagonal are equal to 0. For example,

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 2+i & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

is a diagonal matrix of order 4 over the field of complex numbers.

Null matrix: The $m \times n$ matrix whose elements are all zero is called the null matrix or (zero matrix) of the type $m \times n$.

Equality of two matrices: Definition:

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{m \times n}$. Then
 $A = B$ if $a_{ij} = b_{ij}$ for each pair of subscripts i and j .

Addition of two matrices: Definition:

Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$. Then we define
 $A + B = [a_{ij} + b_{ij}]_{m \times n}$.

Multiplication of a matrix by a scalar: Definition:

Let $A = [a_{ij}]_{m \times n}$ and $a \in F$ i.e., a be a scalar. Then we define
 $aA = [aa_{ij}]_{m \times n}$.

Multiplication of two matrices: Definition:

Let $A = [a_{ij}]_{m \times n}$, $B = [b_{jk}]_{n \times p}$ i.e., the number of columns in the matrix A is equal to the number of rows in the matrix B . Then we define

$$AB = \left[\sum_{j=1}^n a_{ij} b_{jk} \right]_{m \times p} \quad \text{i.e., } AB \text{ is an } m \times p \text{ matrix whose } (i, k)^{\text{th}}$$

element is equal to $\sum_{j=1}^n a_{ij} b_{jk}$.

If A and B are both square matrices of order n , then both the products AB and BA exist but in general $AB \neq BA$.

Transpose of a matrix: Definition:

Let $A = [a_{ij}]_{m \times n}$. The $n \times m$ matrix A^T obtained by interchanging the rows and columns of A is called the transpose of A . Thus $A^T = [b_{ij}]_{n \times m}$, where $b_{ij} = a_{ji}$, i.e., the $(i, j)^{\text{th}}$ element of A^T is the $(j, i)^{\text{th}}$ element of A . If A is an $m \times n$ matrix and B is an $n \times p$ matrix, it can be shown that $(AB)^T = B^T A^T$. The transpose of a matrix A is also denoted by A^t or by A' .

Determinant of a square matrix: Let P_n denote the group of all permutations of degree n on the set $\{1, 2, \dots, n\}$. If $\theta \in P_n$, then $\theta(i)$ will denote the image of i under θ . The symbol $(-1)^\theta$ for $\theta \in P_n$ will mean $+1$ if θ is an even permutation and -1 if θ is an odd permutation.

Definition: Let $A = [a_{ij}]_{n \times n}$. Then the *determinant* of A , written as $\det A$ or $|A|$ or $|a_{ij}|_{n \times n}$ is the element

$$\sum_{\theta \in P_n} (-1)^\theta a_{1\theta(1)} a_{2\theta(2)} \dots a_{n\theta(n)} \text{ in } F.$$

The number of terms in this summation is $n!$ because there are $n!$ permutations in the set P_n .

We shall often use the notation

$$\begin{vmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \dots & \dots & \dots \\ \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} \end{vmatrix}$$

for the determinant of the matrix $[a_{ij}]_{n \times n}$.

The following properties of determinants are worth to be noted :

- (i) The determinant of a unit matrix is always equal to 1.
- (ii) The determinant of a null matrix is always equal to 0.
- (iii) If $A = [a_{ij}]_{n \times n}$, $B = [b_{ij}]_{n \times n}$, then $\det(AB) = (\det A)(\det B)$.

Cofactors: Definition: Let $A = [a_{ij}]_{n \times n}$. We define

$$\begin{aligned} A_{ij} &= \text{cofactor of } a_{ij} \text{ in } A \\ &= (-1)^{i+j} \cdot [\text{determinant of the matrix of order } n-1 \text{ obtained} \\ &\quad \text{by deleting the row and column of } A \text{ passing through } a_{ij}]. \end{aligned}$$

It should be noted that

$$\sum_{i=1}^n a_{ik} A_{ij} = 0 \quad \text{if } k \neq j$$

$$\text{or} \quad = \det A \quad \text{if } k = j.$$

Adjoint of a square matrix: Definition: Let $A = [a_{ij}]_{n \times n}$.

The $n \times n$ matrix which is the transpose of the matrix of cofactors of A is called the adjoint of A and is denoted by $\text{adj } A$.

It should be remembered that

$$A(\text{adj } A) = (\text{adj } A)A = (\det A)I$$

where I is the unit matrix of order n .

Inverse of a square matrix: Definition: Let A be a square matrix of order n . If there exists a square matrix B of order n such that

$$AB = I = BA$$

then A is said to be invertible and B is called the inverse of A .

Also we write $B = A^{-1}$.

The following results should be remembered :

- (i) The necessary and sufficient condition for a square matrix A to be invertible is that $\det A \neq 0$.

(ii) If A is invertible, then A^{-1} is unique and

$$A^{-1} = \frac{1}{\det A} (\text{adj. } A).$$

(iii) If A and B are invertible square matrices of order n , then AB is also invertible and $(AB)^{-1} = B^{-1} A^{-1}$.

(iv) If A is invertible, so is A^{-1} and $(A^{-1})^{-1} = A$.

Elementary row operations on a matrix:

Definition: Let A be an $m \times n$ matrix over the field F . The following three operations are called elementary row operations :

- (1) multiplication of any row of A by a non-zero element c of F .
- (2) addition to the elements of any row of A the corresponding elements of any other row of A multiplied by any element a in F .
- (3) interchange of two rows of A .

Row equivalent matrices: Definition: If A and B are $m \times n$ matrices over the field F , then B is said to be row equivalent to A if B can be obtained from A by a finite sequence of elementary row operations. It can be easily seen that the relation of being row equivalent is an equivalence relation in the set of all $m \times n$ matrices over F .

Row reduced Echelon matrix: Definition:

An $m \times n$ matrix R is called a row reduced echelon matrix if :

- (1) Every row of R which has all its entries 0 occurs below every row which has a non-zero entry.
- (2) The first non-zero entry in each non-zero row is equal to 1.
- (3) If the first non-zero entry in row i appears in column k_i , then all other entries in column k_i are zero.
- (4) If r is the number of non-zero rows, then

$$k_1 < k_2 < \dots < k_r$$

(i.e., the first non-zero entry in row i is to the left of the first non-zero entry in row $i + 1$).

Row and column rank of a matrix:

Definition: Let $A = [a_{ij}]_{m \times n}$ be an $m \times n$ matrix over the field F . The row vectors of A are the vectors $\alpha_1, \dots, \alpha_m \in V_n(F)$ defined by

$$\alpha_i = (a_{i1}, a_{i2}, \dots, a_{in}), 1 \leq i \leq m.$$

The row space of A is the subspace of $V_n(F)$ spanned by these vectors. The row rank of A is the dimension of the row space of A .

The column vectors of A are the vectors $\beta_1, \dots, \beta_n \in V_m(F)$ defined by

$$\beta_j = (a_{1j}, a_{2j}, \dots, a_{mj}), 1 \leq j \leq n.$$

The **column space** of A is the subspace of $V_m(F)$ spanned by these vectors. The **column rank** of A is the dimension of the column space of A .

The following two results are to be remembered :

- (1) Row equivalent matrices have the same row space.
- (2) If R is a non-zero row reduced Echelon matrix, then the non-zero row vectors of R are linearly independent and therefore they form a basis for the row space of R .

In order to find the row rank of a matrix A , we should reduce it to row reduced Echelon matrix R by elementary row operations. The number of non-zero rows in R will give us the row rank of A .

4.2 Representation of Transformations by Matrices

Matrix of a linear transformation: Let U be an n -dimensional vector space over the field F and let V be an m -dimensional vector space over F . Let

$$B = \{\alpha_1, \dots, \alpha_n\} \text{ and } B' = \{\beta_1, \dots, \beta_m\}$$

be ordered bases for U and V respectively. Suppose T is a linear transformation from U into V . We know that T is completely determined by its action on the vectors α_j belonging to a basis for U . Each of the n vectors $T(\alpha_j)$ is uniquely expressible as a linear combination of $\beta_1, \dots, \beta_m$ because $T(\alpha_j) \in V$ and these m vectors form a basis for V . Let for $j = 1, 2, \dots, n$,

$$T(\alpha_j) = a_{1j} \beta_1 + a_{2j} \beta_2 + \dots + a_{mj} \beta_m = \sum_{i=1}^m a_{ij} \beta_i.$$

The scalars $a_{1j}, a_{2j}, \dots, a_{mj}$ are the coordinates of $T(\alpha_j)$ in the ordered basis B' . The $m \times n$ matrix whose j^{th} column ($j = 1, 2, \dots, n$) consists of these coordinates is called the **matrix of the linear transformation T relative to the pair of ordered bases B and B'** . We shall denote it by the symbol $[T; B; B']$ or simply by $[T]$ if the bases are understood. Thus

$$\begin{aligned} [T] &= [T; B; B'] \\ &= \text{matrix of } T \text{ relative to the ordered bases } B \text{ and } B' \\ &= [a_{ij}]_{m \times n} \end{aligned}$$

$$\text{where } T(\alpha_j) = \sum_{i=1}^m a_{ij} \beta_i, \text{ for each } j = 1, 2, \dots, n. \quad \dots(1)$$

The coordinates of $T(\alpha_1)$ in the ordered basis B' form the first column of this matrix, the coordinates of $T(\alpha_2)$ in the ordered basis B' form the second column of this matrix and so on.

The $m \times n$ matrix $[a_{ij}]_{m \times n}$ completely determines the linear transformation T through the formulae given in (1). Therefore the matrix $[a_{ij}]_{m \times n}$ represents the transformation T .

Note: Let T be a linear transformation from an n -dimensional vector space $V(F)$ into itself. Then in order to represent T by a matrix, it is most convenient to use the same ordered basis in each case, i.e., to take $B = B'$. The representing matrix will then be called the **matrix of T relative to the ordered basis B** and will be denoted by $[T; B]$ or sometimes also by $[T]_B$.

Thus if $B = \{\alpha_1, \dots, \alpha_n\}$ is an ordered basis for V , then

$$[T]_B \text{ or } [T; B] = \text{matrix of } T \text{ relative to the ordered basis } B \\ = [a_{ij}]_{n \times n},$$

where
$$T(\alpha_j) = \sum_{i=1}^n a_{ij} \alpha_i, \text{ for each } j = 1, 2, \dots, n.$$

Illustrative Examples

Example 1: Let T be a linear transformation on the vector space $V_2(F)$ defined by $T(a, b) = (a, 0)$.

Write the matrix of T relative to the standard ordered basis of $V_2(F)$.

Solution: Let $B = \{\alpha_1, \alpha_2\}$ be the standard ordered basis for $V_2(F)$. Then

$$\alpha_1 = (1, 0), \alpha_2 = (0, 1).$$

We have $T(\alpha_1) = T(1, 0) = (1, 0)$.

Now let us express $T(\alpha_1)$ as a linear combination of vectors in B . We have

$$T(\alpha_1) = (1, 0) = 1(1, 0) + 0(0, 1) = 1\alpha_1 + 0\alpha_2.$$

Thus 1, 0 are the coordinates of $T(\alpha_1)$ with respect to the ordered basis B . These coordinates will form the first column of matrix of T relative to ordered basis B .

Again $T(\alpha_2) = T(0, 1) = (0, 0) = 0(1, 0) + 0(0, 1)$.

Thus 0, 0 are the coordinates of $T(\alpha_2)$ and will form second column of matrix of T relative to ordered basis B .

Thus matrix of T relative to ordered basis B

$$= [T]_B \text{ or } [T; B] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Example 2: Let $V(\mathbf{R})$ be the vector space of all polynomials in x with coefficients in $\mathbf{R}$ of the form

$$f(x) = a_0 x^0 + a_1 x + a_2 x^2 + a_3 x^3$$

i.e., the space of polynomials of degree three or less. The differentiation operator D is a linear transformation on V . The set

$$B = \{\alpha_1, \dots, \alpha_4\} \text{ where } \alpha_1 = x^0, \alpha_2 = x^1, \alpha_3 = x^2, \alpha_4 = x^3$$

is an ordered basis for V . Write the matrix of D relative to the ordered basis B .

Solution: We have

$$\begin{aligned}
D(\alpha_1) &= D(x^0) = 0 = 0x^0 + 0x^1 + 0x^2 + 0x^3 \\
&= 0\alpha_1 + 0\alpha_2 + 0\alpha_3 + 0\alpha_4 \\
D(\alpha_2) &= D(x^1) = x^0 = 1x^0 + 0x^1 + 0x^2 + 0x^3 \\
&= 1\alpha_1 + 0\alpha_2 + 0\alpha_3 + 0\alpha_4 \\
D(\alpha_3) &= D(x^2) = 2x^1 = 0x^0 + 2x^1 + 0x^2 + 0x^3 \\
&= 0\alpha_1 + 2\alpha_2 + 0\alpha_3 + 0\alpha_4 \\
D(\alpha_4) &= D(x^3) = 3x^2 = 0x^0 + 0x^1 + 3x^2 + 0x^3 \\
&= 0\alpha_1 + 0\alpha_2 + 3\alpha_3 + 0\alpha_4.
\end{aligned}$$

$\therefore$ the matrix of D relative to the ordered basis B

$$= [D; B] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{4 \times 4}.$$

Theorem 1: Let U be an n -dimensional vector space over the field F and let V be an m -dimensional vector space over F . Let B and B' be ordered bases for U and V respectively. Then corresponding to every matrix $[a_{ij}]_{m \times n}$ of mn scalars belonging to F there corresponds a unique linear transformation T from U into V such that

$$[T; B; B'] = [a_{ij}]_{m \times n}.$$

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_m\}$.

Now $\sum_{i=1}^m a_{i1} \beta_i, \sum_{i=1}^m a_{i2} \beta_i, \dots, \sum_{i=1}^m a_{in} \beta_i$

are vectors belonging to V because each of them is a linear combination of the vectors belonging to a basis for V . It should be noted that the vector $\sum_{i=1}^m a_{ij} \beta_i$ has been obtained with the help of the j^{th} column of the matrix $[a_{ij}]_{m \times n}$.

Since B is a basis for U , therefore by the theorem 2 of article 3.13 of chapter 3 there exists a unique linear transformation T from U into V such that

$$T(\alpha_j) = \sum_{i=1}^m a_{ij} \beta_i \text{ where } j = 1, 2, \dots, n. \quad \dots(1)$$

By our definition of matrix of a linear transformation, we have from (1)

$$[T; B; B'] = [a_{ij}]_{m \times n}.$$

Note: If we take $V = U$, then in place of B' , we also take B . In that case the above theorem will run as :

Let V be an n -dimensional vector space over the field F and B be an ordered basis or co-ordinate system for V . Then corresponding to every matrix $[a_{ij}]_{n \times n}$ of n^2 scalars belonging to F there corresponds a unique linear transformation T from V into V such that

$$[T; B] \text{ or } [T]_B = [a_{ij}]_{n \times n}.$$

Explicit expression for a linear transformation in terms of its matrix: Now our aim is to establish a formula which will give us the image of any vector under a linear transformation T in terms of its matrix.

Theorem 2: Let T be a linear transformation from an n -dimensional vector space U into an m -dimensional vector space V and let B and B' be ordered bases for U and V respectively. If A is the matrix of T relative to B and B' then $\forall \alpha \in U$, we have

$$[T(\alpha)]_{B'} = A [\alpha]_B$$

where $[\alpha]_B$ is the co-ordinate matrix of α with respect to ordered basis B and $[T(\alpha)]_{B'}$ is co-ordinate matrix of $T(\alpha) \in V$ with respect to B' .

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$

and $B' = \{\beta_1, \beta_2, \dots, \beta_m\}$.

Then $A = [T; B; B'] = [a_{ij}]_{m \times n}$,

where $T(\alpha_j) = \sum_{i=1}^m a_{ij} \beta_i, j = 1, 2, \dots, n.$... (1)

If $\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n$ is a vector in U , then

$$\begin{aligned} T(\alpha) &= T\left(\sum_{j=1}^n x_j \alpha_j\right) \\ &= \sum_{j=1}^n x_j T(\alpha_j) \quad [\because T \text{ is a linear transformation}] \\ &= \sum_{j=1}^n x_j \sum_{i=1}^m a_{ij} \beta_i \quad [\text{From (1)}] \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j\right) \beta_i. \quad \dots (2) \end{aligned}$$

The co-ordinate matrix of $T(\alpha)$ with respect to ordered basis B' is an $m \times 1$ matrix. From (2), we see that the i^{th} entry of this column matrix $[T(\alpha)]_{B'}$

$$= \sum_{j=1}^n a_{ij} x_j$$

i.e., the coefficient of β_i in the linear combination (2) for $T(\alpha)$.

If X is the co-ordinate matrix $[\alpha]_B$ of α with respect to ordered basis B , then X is an $n \times 1$ matrix. The product AX will be an $m \times 1$ matrix. The i^{th} entry of this column matrix AX will be

$$= \sum_{j=1}^n a_{ij} x_j.$$

$\therefore [T(\alpha)]_{B'} = AX = A [\alpha]_B = [T; B; B'] [\alpha]_B.$

Note: If we take $U = V$, then the above result will be

$$[T(\alpha)]_B = [T]_B [\alpha]_B.$$

Matrices of Identity and Zero transformations.

Theorem 3: Let V (F) be an n -dimensional vector space and B be any ordered basis for V .

If I be the identity transformation and $\hat{0}$ be the zero transformation on V , then

(i) $[I; B] = I$ (unit matrix of order n) and

(ii) $[\hat{0}; B] = \text{null matrix of the type } n \times n.$

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$.

(i) We have $I(\alpha_j) = \alpha_j, j = 1, 2, \dots, n$

$$= 0\alpha_1 + \dots + 1\alpha_j + 0\alpha_{j+1} + \dots + 0\alpha_n$$

$$= \sum_{i=1}^n \delta_{ij} \alpha_i, \text{ where } \delta_{ij} \text{ is Kronecker delta.}$$

$\therefore$ By def. of matrix of a linear transformation, we have

$$[I; B] = [\delta_{ij}]_{n \times n} = I \text{ i.e., unit matrix of order } n.$$

(ii) We have $\hat{0}(\alpha_j) = 0, j = 1, 2, \dots, n$

$$= 0\alpha_1 + 0\alpha_2 + \dots + 0\alpha_n$$

$$= \sum_{i=1}^n o_{ij} \alpha_i, \text{ where each } o_{ij} = 0.$$

$\therefore$ By def. of matrix of a linear transformation, we have

$$[\hat{0}; B] = [o_{ij}]_{n \times n} = \text{null matrix of the type } n \times n.$$

Theorem 4: Let T and S be linear transformations from an n -dimensional vector space U into an m -dimensional vector space V and let B and B' be ordered bases for U and V respectively. Then

(i) $[T + S; B; B'] = [T; B; B'] + [S; B; B']$

(ii) $[cT; B; B'] = c[T; B; B']$ where c is any scalar.

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_m\}$.

Let $[a_{ij}]_{m \times n}$ be the matrix of T relative to B, B' . Then

$$T(\alpha_j) = \sum_{i=1}^m a_{ij} \beta_i, j = 1, 2, \dots, n.$$

Also let $[b_{ij}]_{m \times n}$ be the matrix of S relative to B, B' . Then

$$S(\alpha_j) = \sum_{i=1}^m b_{ij} \beta_i, j = 1, 2, \dots, n.$$

(i) We have

$$(T + S)(\alpha_j) = T(\alpha_j) + S(\alpha_j), j = 1, 2, \dots, n$$

$$= \sum_{i=1}^m a_{ij} \beta_i + \sum_{i=1}^m b_{ij} \beta_i = \sum_{i=1}^m (a_{ij} + b_{ij}) \beta_i.$$

$\therefore$ matrix of $T + S$ relative to $B, B' = [a_{ij} + b_{ij}]_{m \times n}$

$$= [a_{ij}]_{m \times n} + [b_{ij}]_{m \times n},$$

$$\therefore [T + S ; B ; B'] = [T ; B ; B'] + [S ; B ; B'].$$

$$\begin{aligned} \text{(ii) We have } (cT)(\alpha_j) &= cT(\alpha_j), j = 1, 2, \dots, n \\ &= c \sum_{i=1}^m a_{ij} \beta_i = \sum_{i=1}^m (ca_{ij}) \beta_i. \end{aligned}$$

$$\begin{aligned} \therefore [cT ; B ; B'] &= \text{matrix of } cT \text{ relative to } B, B' \\ &= [ca_{ij}]_{m \times n} = c [a_{ij}]_{m \times n} = c [T ; B ; B']. \end{aligned}$$

Theorem 5: Let U, V and W be finite dimensional vector spaces over the field F ; let T be a linear transformation from U into V and S a linear transformation from V into W . Further let B, B' and B'' be ordered bases for spaces U, V and W respectively. If A is the matrix of T relative to the pair B, B' and D is the matrix of S relative to the pair B', B'' then the matrix of the composite transformation ST relative to the pair B, B'' is the product matrix $C = DA$.

Proof: Let $\dim U = n, \dim V = m$ and $\dim W = p$. Further let

$$B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}, B' = \{\beta_1, \beta_2, \dots, \beta_m\}$$

$$\text{and } B'' = \{\gamma_1, \gamma_2, \dots, \gamma_p\}.$$

Let $A = [a_{ij}]_{m \times n}, D = [d_{ki}]_{p \times m}$ and $C = [c_{kj}]_{p \times n}$. Then

$$T(\alpha_j) = \sum_{i=1}^m a_{ij} \beta_i, j = 1, 2, \dots, n, \quad \dots(1)$$

$$S(\beta_i) = \sum_{k=1}^p d_{ki} \gamma_k, i = 1, 2, \dots, m. \quad \dots(2)$$

$$\text{and } (ST)(\alpha_j) = \sum_{k=1}^p c_{kj} \gamma_k, j = 1, 2, \dots, n. \quad \dots(3)$$

$$\text{We have } (ST)(\alpha_j) = S[T(\alpha_j)], j = 1, 2, \dots, n$$

$$= S\left(\sum_{i=1}^m a_{ij} \beta_i\right) \quad [\text{From (1)}]$$

$$= \sum_{i=1}^m a_{ij} S(\beta_i) \quad [\because S \text{ is linear}]$$

$$= \sum_{i=1}^m a_{ij} \sum_{k=1}^p d_{ki} \gamma_k \quad [\text{From (2)}]$$

$$= \sum_{k=1}^p \left(\sum_{i=1}^m d_{ki} a_{ij} \right) \gamma_k. \quad \dots(4)$$

Therefore from (3) and (4), we have

$$c_{kj} = \sum_{i=1}^m d_{ki} a_{ij}, j = 1, 2, \dots, n; k = 1, 2, \dots, p.$$

$$\therefore [c_{kj}]_{p \times n} = \left[\sum_{i=1}^m d_{ki} a_{ij} \right]_{p \times n}$$

$$= [d_{ki}]_{p \times m} [a_{ij}]_{m \times n}, \text{ by def. of product of two matrices.}$$

Thus $C = DA$.

Note: If $U = V = W$, then the statement and proof of the above theorem will be as follows :

Let V be an n -dimensional vector space over the field F ;let T and S be linear transformations of V . Further let B be an ordered basis for V . If A is the matrix of T relative to B , and D is the matrix of S relative to B , then the matrix of the composite transformation ST relative to B is the product matrix

$$C = DA \text{ i.e., } [ST]_B = [S]_B [T]_B .$$

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$.

Let $A = [a_{ij}]_{n \times n}$, $D = [d_{ki}]_{n \times n}$ and $C = [c_{kj}]_{n \times n}$. Then

$$T(\alpha_j) = \sum_{i=1}^n a_{ij} \alpha_i, \quad j = 1, 2, \dots, n, \quad \dots(1)$$

$$S(\alpha_i) = \sum_{k=1}^n d_{ki} \alpha_k, \quad i = 1, 2, \dots, n, \quad \dots(2)$$

$$\text{and} \quad (ST)(\alpha_j) = \sum_{k=1}^n c_{kj} \alpha_k, \quad j = 1, 2, \dots, n. \quad \dots(3)$$

We have $(ST)(\alpha_j) = S[T(\alpha_j)]$

$$\begin{aligned} &= S\left(\sum_{i=1}^n a_{ij} \alpha_i\right) = \sum_{i=1}^n a_{ij} S(\alpha_i) = \sum_{i=1}^n a_{ij} \sum_{k=1}^n d_{ki} \alpha_k \\ &= \sum_{k=1}^n \left(\sum_{i=1}^n d_{ki} a_{ij}\right) \alpha_k. \end{aligned} \quad \dots(4)$$

$\therefore$ from (3) and (4), we have

$$c_{kj} = \sum_{i=1}^n d_{ki} a_{ij} .$$

$$\therefore [c_{kj}]_{n \times n} = \left[\sum_{i=1}^n d_{ki} a_{ij} \right]_{n \times n} = [d_{ki}]_{n \times n} [a_{ij}]_{n \times n} .$$

$$\therefore C = DA.$$

Theorem 6: Let U be an n -dimensional vector space over the field F and let V be an m -dimensional vector space over F . For each pair of ordered bases B, B' for U and V respectively, the function which assigns to a linear transformation T its matrix relative to B, B' is an isomorphism between the space $L(U, V)$ and the space of all $m \times n$ matrices over the field F .

Proof. Let $B = \{\alpha_1, \dots, \alpha_n\}$ and $B' = \{\beta_1, \dots, \beta_m\}$.

Let M be the vector space of all $m \times n$ matrices over the field F . Let

$$\psi : L(U, V) \rightarrow M \text{ such that}$$

$$\psi(T) = [T; B; B'] \quad \forall T \in L(U, V).$$

Let $T_1, T_2 \in L(U, V)$; and let

$$[T_1; B; B'] = [a_{ij}]_{m \times n} \text{ and } [T_2; B; B'] = [b_{ij}]_{m \times n} .$$

Then $T_1 (\alpha_j) = \sum_{i=1}^m a_{ij} \beta_i, j = 1, 2, \dots, n$

and $T_2 (\alpha_j) = \sum_{i=1}^m b_{ij} \beta_i, j = 1, 2, \dots, n.$

To prove that ψ is one-one:

We have $\psi (T_1) = \psi (T_2)$

$\Rightarrow [T_1 ; B ; B'] = [T_2 ; B ; B']$ [by def. of ψ]

$\Rightarrow [a_{ij}]_{m \times n} = [b_{ij}]_{m \times n}$

$\Rightarrow a_{ij} = b_{ij}$ for $i = 1, \dots, m$ and $j = 1, \dots, n$

$\Rightarrow \sum_{i=1}^m a_{ij} \beta_i = \sum_{i=1}^m b_{ij} \beta_i$ for $j = 1, \dots, n$

$\Rightarrow T_1 (\alpha_j) = T_2 (\alpha_j)$ for $j = 1, \dots, n$

$\Rightarrow T_1 = T_2$ [$\because T_1$ and T_2 agree on a basis for U]

$\therefore \psi$ is one-one.

ψ is onto:

Let $[c_{ij}]_{m \times n} \in M$. Then there exists a linear transformation T from U into V such that

$$T (\alpha_j) = \sum_{i=1}^m c_{ij} \beta_i, j = 1, 2, \dots, n.$$

We have $[T ; B ; B'] = [c_{ij}]_{m \times n} \Rightarrow \psi (T) = [c_{ij}]_{m \times n}.$

$\therefore \psi$ is onto.

ψ is a linear transformation:

If $a, b \in F$, then

$$\psi (aT_1 + bT_2) = [aT_1 + bT_2 ; B ; B'] \quad \text{[by def. of } \psi]$$

$$= [aT_1 ; B ; B'] + [bT_2 ; B ; B'] \quad \text{[by theorem 4]}$$

$$= a [T_1 ; B ; B'] + b [T_2 ; B ; B'] \quad \text{[by theorem 4]}$$

$$= a\psi (T_1) + b\psi (T_2), \text{ by def. of } \psi.$$

$\therefore \psi$ is a linear transformation.

Hence ψ is an isomorphism from $L(U, V)$ onto M .

Note: It should be noted that in the above theorem if $U \rightarrow V$, then ψ also preserves products and I i.e.,

$$\psi (T_1 T_2) = \psi (T_1) \psi (T_2)$$

and $\psi (I) = I$ i.e., unit matrix.

Theorem 7: Let T be a linear operator on an n -dimensional vector space V and let B be an ordered basis for V . Prove that T is invertible iff $[T]_B$ is an invertible matrix. Also if T is invertible, then

$$[T^{-1}]_B = [T]_B^{-1},$$

i.e., the matrix of T^{-1} relative to B is the inverse of the matrix of T relative to B .

Proof: Let T be invertible. Then T^{-1} exists and we have

$$\begin{aligned} T^{-1} T &= I = T T^{-1} \Rightarrow [T^{-1} T]_B = [I]_B = [T T^{-1}]_B \\ \Rightarrow [T^{-1}]_B [T]_B &= I = [T]_B [T^{-1}]_B \\ \Rightarrow [T]_B &\text{ is invertible and } ([T]_B)^{-1} = [T^{-1}]_B. \end{aligned}$$

Conversely, let $[T]_B$ be an invertible matrix. Let $[T]_B = A$. Let $C = A^{-1}$ and let S be the linear transformation of V such that

$$[S]_B = C.$$

We have $C A = I = A C$

$$\begin{aligned} \Rightarrow [S]_B [T]_B &= I = [T]_B [S]_B \Rightarrow [ST]_B = [I]_B = [TS]_B \\ \Rightarrow ST &= I = TS \Rightarrow T \text{ is invertible.} \end{aligned}$$

4.3 Change of Basis

Suppose V is an n -dimensional vector space over the field F . Let B and B' be two ordered bases for V . If α is any vector in V , then we are now interested to know what is the relation between its coordinates with respect to B and its coordinates with respect to B' .

Theorem 8: Let $V (F)$ be an n -dimensional vector space and let B and B' be two ordered bases for V . Then there is a unique necessarily invertible, $n \times n$ matrix A with entries in F such that

- (1) $[\alpha]_B = A [\alpha]_{B'}$
- (2) $[\alpha]_{B'} = A^{-1} [\alpha]_B$ for every vector α in V .

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_n\}$.

Then there exists a unique linear transformation T from V into V such that

$$T(\alpha_j) = \beta_j, \quad j = 1, 2, \dots, n. \quad \dots(1)$$

Since T maps a basis B onto a basis B' , therefore T is necessarily invertible. The matrix of T relative to B i.e., $[T]_B$ will be a unique $n \times n$ matrix with elements in F . Also this matrix will be invertible because T is invertible.

Let $[T]_B = A = [a_{ij}]_{n \times n}$. Then

$$T(\alpha_j) = \sum_{i=1}^n a_{ij} \alpha_i, \quad j = 1, 2, \dots, n. \quad \dots(2)$$

Let $x_1, x_2, \dots, x_n$ be the coordinates of α with respect to B and $y_1, y_2, \dots, y_n$ be the coordinates of α with respect to B' . Then

$$\begin{aligned} \alpha &= y_1 \beta_1 + y_2 \beta_2 + \dots + y_n \beta_n = \sum_{j=1}^n y_j \beta_j \\ &= \sum_{j=1}^n y_j T(\alpha_j) && \text{[From (1)]} \\ &= \sum_{j=1}^n y_j \sum_{i=1}^n a_{ij} \alpha_i && \text{[From (2)]} \end{aligned}$$

$$= \sum_{i=1}^n \left(\sum_{j=1}^n a_{ij} y_j \right) \alpha_i .$$

Also
$$\alpha = \sum_{i=1}^n x_i \alpha_i .$$

$$\therefore x_i = \sum_{j=1}^n a_{ij} y_j$$

because the expression for α as a linear combination of elements of B is unique. Now $[\alpha]_B$ is a column matrix of the type $n \times 1$. Also $[\alpha]_{B'}$ is a column matrix of the type $n \times 1$. The product matrix $A [\alpha]_{B'}$ will also be of the type $n \times 1$.

The i^{th} entry of $[\alpha]_B = x_i = \sum_{j=1}^n a_{ij} y_j = i^{\text{th}}$ entry of $A [\alpha]_{B'}$.

$$\therefore [\alpha]_B = A [\alpha]_{B'} \quad \Rightarrow \quad A^{-1} [\alpha]_B = A^{-1} A [\alpha]_{B'}$$

$$\Rightarrow A^{-1} [\alpha]_B = I [\alpha]_B \quad \Rightarrow \quad A^{-1} [\alpha]_B = [\alpha]_{B'} .$$

Note: The matrix $A = [T]_B$ is called the transition matrix from B to B' . It expresses the coordinates of each vector in V relative to B in terms of its coordinates relative to B' .

How to write the transition matrix from one basis to another ?

Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_n\}$ be two ordered bases for the n -dimensional vector space $V (F)$. Let A be the transition matrix from the basis B to the basis B' . Let T be the linear transformation from V into V which maps the basis B onto the basis B' . Then A is the matrix of T relative to B i.e., $A = [T]_B$. So in order to find the matrix A , we should first express each vector in the basis B' as a linear combination over F of the vectors in B . Thus we write the relations

$$\beta_1 = a_{11} \alpha_1 + a_{21} \alpha_2 + \dots + a_{n1} \alpha_n$$

$$\beta_2 = a_{12} \alpha_1 + a_{22} \alpha_2 + \dots + a_{n2} \alpha_n$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$\beta_n = a_{1n} \alpha_1 + a_{2n} \alpha_2 + \dots + a_{nn} \alpha_n .$$

Then the matrix $A = [a_{ij}]_{n \times n}$ i.e., A is the transpose of the matrix of coefficients in the above relations. Thus

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} .$$

Now suppose α is any vector in V . If $[\alpha]_B$ is the coordinate matrix of α relative to the basis B and $[\alpha]_{B'}$ its coordinate matrix relative to the basis B' then

$$[\alpha]_B = A [\alpha]_{B'} \text{ and } [\alpha]_{B'} = A^{-1} [\alpha]_B .$$

Theorem 9: Let $B = \{\alpha_1, \dots, \alpha_n\}$ and $B' = \{\beta_1, \dots, \beta_n\}$ be two ordered bases for an n -dimensional vector space $V(F)$. If $(x_1, \dots, x_n)$ is an ordered set of n scalars, let

$$\alpha = \sum_{i=1}^n x_i \alpha_i \text{ and } \beta = \sum_{i=1}^n x_i \beta_i.$$

Then show that $T(\alpha) = \beta$, where T is the linear operator on V defined by

$$T(\alpha_i) = \beta_i, \quad i = 1, 2, \dots, n.$$

Proof: We have $T(\alpha) = T\left(\sum_{i=1}^n x_i \alpha_i\right) = \sum_{i=1}^n x_i T(\alpha_i)$ [$\because T$ is linear]

$$= \sum_{i=1}^n x_i \beta_i = \beta.$$

Similarity:

Similarity of matrices. Definition: Let A and B be square matrices of order n over the field F . Then B is said to be similar to A if there exists an $n \times n$ invertible square matrix C with elements in F such that

$$B = C^{-1} A C.$$

Theorem 10: The relation of similarity is an equivalence relation in the set of all $n \times n$ matrices over the field F .

Proof: If A and B are two $n \times n$ matrices over the field F , then B is said to be similar to A if there exists an $n \times n$ invertible matrix C over F such that

$$B = C^{-1} A C.$$

Reflexive: Let A be any $n \times n$ matrix over F . We can write $A = I^{-1} A I$, where I is $n \times n$ unit matrix over F .

$\therefore A$ is similar to A because I is definitely invertible.

Symmetric: Let A be similar to B . Then there exists an $n \times n$ invertible matrix P over F such that

$$A = P^{-1} B P$$

$$\Rightarrow P A P^{-1} = P (P^{-1} B P) P^{-1}$$

$$\Rightarrow P A P^{-1} = B \quad \Rightarrow \quad B = P A P^{-1}$$

$$\Rightarrow B = (P^{-1})^{-1} A P^{-1}$$

$$[\because P \text{ is invertible means } P^{-1} \text{ is invertible and } (P^{-1})^{-1} = P]$$

$$\Rightarrow B \text{ is similar to } A.$$

Transitive: Let A be similar to B and B be similar to C . Then

$$A = P^{-1} B P \quad \text{and} \quad B = Q^{-1} C Q,$$

where P and Q are invertible $n \times n$ matrices over F .

$$\text{We have } A = P^{-1} B P = P^{-1} (Q^{-1} C Q) P$$

$$\begin{aligned}
&= (P^{-1} Q^{-1}) C (QP) \\
&= (QP)^{-1} C (QP) \quad [\because P \text{ and } Q \text{ are invertible means } QP \\
&\quad \text{is invertible and } (QP)^{-1} = P^{-1} Q^{-1}]
\end{aligned}$$

$\therefore$ A is similar to C .

Hence similarity is an equivalence relation on the set of $n \times n$ matrices over the field F .

Theorem 11: *Similar matrices have the same determinant.*

Proof: Let B be similar to A . Then there exists an invertible matrix C such that

$$B = C^{-1} A C$$

$$\Rightarrow \det B = \det (C^{-1} A C) \Rightarrow \det B = (\det C^{-1}) (\det A) (\det C)$$

$$\Rightarrow \det B = (\det C^{-1}) (\det C) (\det A) \Rightarrow \det B = (\det C^{-1} C) (\det A)$$

$$\Rightarrow \det B = (\det I) (\det A) \Rightarrow \det B = 1 (\det A) \Rightarrow \det B = \det A.$$

Similarity of linear transformations: Definition: Let A and B be linear transformations on a vector space $V (F)$. Then B is said to be similar to A if there exists an invertible linear transformation C on V such that

$$B = C A C^{-1}.$$

Theorem 12: *The relation of similarity is an equivalence relation in the set of all linear transformations on a vector space $V (F)$.*

Proof: If A and B are two linear transformations on the vector space $V (F)$, then B is said to be similar to A if there exists an invertible linear transformation C on V such that

$$B = C A C^{-1}.$$

Reflexive: Let A be any linear transformation on V . We can write

$$A = I A I^{-1},$$

where I is identity transformation on V .

$\therefore$ A is similar to A because I is definitely invertible.

Symmetric: Let A be similar to B . Then there exists an invertible linear transformation P on V such that

$$A = P B P^{-1}$$

$$\Rightarrow P^{-1} A P = P^{-1} (P B P^{-1}) P$$

$$\Rightarrow P^{-1} A P = B \Rightarrow B = P^{-1} A P$$

$$\Rightarrow B = P^{-1} A (P^{-1})^{-1}$$

$$\Rightarrow B \text{ is similar to } A.$$

Transitive: Let A be similar to B and B be similar to C .

Then $A = P B P^{-1},$

and $B = Q C Q^{-1},$

where P and Q are invertible linear transformations on V .

$$\begin{aligned}\text{We have } A &= PBP^{-1} = P(QCQ^{-1})P^{-1} \\ &= (PQ)C(Q^{-1}P^{-1}) = (PQ)C(PQ)^{-1}.\end{aligned}$$

$\therefore$ A is similar to C .

Hence similarity is an equivalence relation on the set of all linear transformations on $V(F)$.

Theorem 13: Let T be a linear operator on an n -dimensional vector space $V(F)$ and let B and B' be two ordered bases for V . Then the matrix of T relative to B' is similar to the matrix of T relative to B .

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \dots, \beta_n\}$.

$$\begin{aligned}\text{Let } A &= [a_{ij}]_{n \times n} \text{ be the matrix of } T \text{ relative to } B \\ \text{and } C &= [c_{ij}]_{n \times n} \text{ be the matrix of } T \text{ relative to } B'. \text{ Then} \\ T(\alpha_j) &= \sum_{i=1}^n a_{ij} \alpha_i, j = 1, 2, \dots, n\end{aligned} \quad \dots(1)$$

$$\text{and } T(\beta_j) = \sum_{i=1}^n c_{ij} \beta_i, j = 1, 2, \dots, n. \quad \dots(2)$$

Let S be the linear operator on V defined by

$$S(\alpha_j) = \beta_j, j = 1, 2, \dots, n. \quad \dots(3)$$

Since S maps a basis B onto a basis B' , therefore S is necessarily invertible. Let P be the matrix of S relative to B . Then P is also an invertible matrix.

$$\begin{aligned}\text{If } P &= [p_{ij}]_{n \times n}, \text{ then} \\ S(\alpha_j) &= \sum_{i=1}^n p_{ij} \alpha_i, j = 1, 2, \dots, n\end{aligned} \quad \dots(4)$$

$$\text{We have } T(\beta_j) = T[S(\alpha_j)] \quad [\text{From (3)}]$$

$$\begin{aligned}&= T\left(\sum_{k=1}^n p_{kj} \alpha_k\right) \\ &\quad [\text{From (4), on replacing } i \text{ by } k \text{ which is immaterial}] \\ &= \sum_{k=1}^n p_{kj} T(\alpha_k) \\ &\quad [\because T \text{ is linear}]\end{aligned}$$

$$\begin{aligned}&= \sum_{k=1}^n p_{kj} \sum_{i=1}^n a_{ik} \alpha_i \\ &\quad [\text{From (1), on replacing } j \text{ by } k] \\ &= \sum_{i=1}^n \left(\sum_{k=1}^n a_{ik} p_{kj}\right) \alpha_i.\end{aligned} \quad \dots(5)$$

$$\text{Also } T(\beta_j) = \sum_{k=1}^n c_{kj} \beta_k \quad [\text{From (2), on replacing } i \text{ by } k]$$

$$= \sum_{k=1}^n c_{kj} S(\alpha_k) \quad [\text{From (3)}]$$

$$= \sum_{k=1}^n c_{kj} \sum_{i=1}^n p_{ik} \alpha_i \quad [\text{From (4), on replacing } j \text{ by } k]$$

$$= \sum_{i=1}^n \left(\sum_{k=1}^n p_{ik} c_{kj} \right) \alpha_i . \quad \dots(6)$$

From (5) and (6), we have

$$\begin{aligned} & \sum_{i=1}^n \left(\sum_{k=1}^n a_{ik} p_{kj} \right) \alpha_i = \sum_{i=1}^n \left(\sum_{k=1}^n p_{ik} c_{kj} \right) \alpha_i \\ \Rightarrow & \sum_{k=1}^n a_{ik} p_{kj} = \sum_{k=1}^n p_{ik} c_{kj} \\ \Rightarrow & [a_{ik}]_{n \times n} [p_{kj}]_{n \times n} = [p_{ik}]_{n \times n} [c_{kj}]_{n \times n} \\ & \hspace{15em} [\text{by def. of matrix multiplication}] \\ \Rightarrow & AP = PC \\ \Rightarrow & P^{-1} AP = P^{-1} PC \quad [\because P^{-1} \text{ exists}] \\ \Rightarrow & P^{-1} AP = IC \Rightarrow P^{-1} AP = C \\ \Rightarrow & C \text{ is similar to } A. \end{aligned}$$

Note: Suppose B and B' are two ordered bases for an n -dimensional vector space $V(F)$. Let T be a linear operator on V . Suppose A is the matrix of T relative to B and C is the matrix of T relative to B' . If P is the transition matrix from the basis B to the basis B' , then $C = P^{-1}AP$.

This result will enable us to find the matrix of T relative to the basis B' when we already knew the matrix of T relative to the basis B .

Theorem 14: Let V be an n -dimensional vector space over the field F and T_1 and T_2 be two linear operators on V . If there exist two ordered bases B and B' for V such that $[T_1]_B = [T_2]_{B'}$, then show that T_2 is similar to T_1 .

Proof: Let $B = \{\alpha_1, \dots, \alpha_n\}$ and $B' = \{\beta_1, \dots, \beta_n\}$.

Let $[T_1]_B = [T_2]_{B'} = A = [a_{ij}]_{n \times n}$. Then

$$T_1(\alpha_j) = \sum_{i=1}^n a_{ij} \alpha_i, \quad j = 1, 2, \dots, n, \quad \dots(1)$$

$$\text{and} \quad T_2(\beta_j) = \sum_{i=1}^n a_{ij} \beta_i, \quad j = 1, 2, \dots, n. \quad \dots(2)$$

Let S be the linear operator on V defined by

$$S(\alpha_j) = \beta_j, \quad j = 1, 2, \dots, n. \quad \dots(3)$$

Since S maps a basis of V onto a basis of V , therefore S is invertible.

We have $T_2(\beta_j) = T_2[S(\alpha_j)]$ [From (3)]

$$= (T_2 S)(\alpha_j). \quad \dots(4)$$

$$\text{Also} \quad T_2(\beta_j) = \sum_{i=1}^n a_{ij} \beta_i \quad [\text{From (2)}]$$

$$= \sum_{i=1}^n a_{ij} S(\alpha_i) \quad [\text{From (3)}]$$

$$\begin{aligned}
 &= S \left(\sum_{i=1}^n a_{ij} \alpha_i \right) && [\because S \text{ is linear}] \\
 &= S [T_1 (\alpha_j)] && [\text{From (1)}] \\
 &= (ST_1) (\alpha_j). && \dots(5)
 \end{aligned}$$

From (4) and (5), we have

$$(T_2 S) (\alpha_j) = (ST_1) (\alpha_j), j = 1, 2, \dots, n.$$

Since $T_2 S$ and ST_1 agree on a basis for V , therefore we have $T_2 S = ST_1$

$$\Rightarrow T_2 S S^{-1} = ST_1 S^{-1} \Rightarrow T_2 I = ST_1 S^{-1}$$

$$\Rightarrow T_2 = ST_1 S^{-1} \Rightarrow T_2 \text{ is similar to } T_1.$$

Determinant of a linear transformation on a finite dimensional vector space: Let T be a linear operator on an n -dimensional vector space $V (F)$. If B and

B' are two ordered bases for V , then $[T]_B$ and $[T]_{B'}$

are similar matrices. Also similar matrices have the same determinant. This enables us to make the following definition :

Definition: Let T be a linear operator on an n -dimensional vector space $V (F)$. Then the determinant of T is the determinant of the matrix of T relative to any ordered basis for V . By the above discussion the determinant of T as defined by us will be a unique element of F and thus our definition is sensible.

Scalar Transformation: Definition: Let $V (F)$ be a vector space. A linear transformation T on V is said to be a scalar transformation of V if

$$T (\alpha) = c\alpha \quad \forall \alpha \in V,$$

where c is a fixed scalar in F .

Also then we write $T = c$ and we say that the linear transformation T is equal to the scalar c .

Also obviously if the linear transformation T is equal to the scalar c , then we have $T = cI$, where I is the identity transformation on V .

Trace of a Matrix: Definition: Let A be a square matrix of order n over a field F . The sum of the elements of A lying along the principal diagonal is called the trace of A . We shall write the trace of A as **trace A**. Thus if $A = [a_{ij}]_{n \times n}$, then

$$\text{tr } A = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}.$$

In the following two theorems we have given some fundamental properties of the trace function.

Theorem 15: Let A and B be two square matrices of order n over a field F and $\lambda \in F$. Then

- (i) $\text{tr } (\lambda A) = \lambda \text{tr } A$;
- (ii) $\text{tr } (A + B) = \text{tr } A + \text{tr } B$;
- (iii) $\text{tr } (AB) = \text{tr } (BA)$.

Proof: Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$.

(i) We have $\lambda A = [\lambda a_{ij}]_{n \times n}$, by def. of multiplication of a matrix by a scalar.

$$\therefore \quad \text{tr}(\lambda A) = \sum_{i=1}^n \lambda a_{ii} = \lambda \sum_{i=1}^n a_{ii} = \lambda \text{tr } A.$$

(ii) We have $A + B = [a_{ij} + b_{ij}]_{n \times n}$.

$$\therefore \quad \text{tr}(A + B) = \sum_{i=1}^n (a_{ii} + b_{ii}) = \sum_{i=1}^n a_{ii} + \sum_{i=1}^n b_{ii} = \text{tr } A + \text{tr } B.$$

(iii) We have $AB = [c_{ij}]_{n \times n}$ where $c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$.

Also $BA = [d_{ij}]_{n \times n}$ where $d_{ij} = \sum_{k=1}^n b_{ik} a_{kj}$.

$$\text{Now } \text{tr}(AB) = \sum_{i=1}^n c_{ii} = \sum_{i=1}^n \left(\sum_{k=1}^n a_{ik} b_{ki} \right)$$

$$= \sum_{k=1}^n \sum_{i=1}^n a_{ik} b_{ki},$$

interchanging the order of summation in the last sum

$$= \sum_{k=1}^n \left(\sum_{i=1}^n b_{ki} a_{ik} \right) = \sum_{k=1}^n d_{kk}$$

$$= d_{11} + d_{22} + \dots + d_{nn} = \text{tr}(BA).$$

Theorem 16: *Similar matrices have the same trace.*

Proof: Suppose A and B are two similar matrices. Then there exists an invertible matrix C such that $B = C^{-1} A C$.

$$\text{Let } C^{-1} A = D.$$

$$\text{Then } \text{tr } B = \text{tr}(DC)$$

$$= \text{tr}(CD)$$

[by theorem 15]

$$= \text{tr}(CC^{-1} A) = \text{tr}(IA) = \text{tr } A.$$

Trace of a linear transformation on a finite dimensional vector space: Let T be a linear operator on an n -dimensional vector space $V(F)$. If B and B' are two ordered bases for V , then $[T]_B$ and $[T]_{B'}$

are similar matrices. Also similar matrices have the same trace. This enables us to make the following definition.

Definition of trace of a linear transformation. Let T be a linear operator on an n -dimensional vector space $V(F)$. Then the trace of T is the trace of the matrix of T relative to any ordered basis for V .

By the above discussion the trace of T as defined by us will be a unique element of F and thus our definition is sensible.

Illustrative Examples

Example 3: Find the matrix of the linear transformation T on $V_3(\mathbf{R})$ defined as $T(a, b, c) = (2b + c, a - 4b, 3a)$, with respect to the ordered basis B and also with respect to the ordered basis B' where

(i) $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

(ii) $B' = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$.

Solution: (i) We have

$$T(1, 0, 0) = (0, 1, 3) = 0(1, 0, 0) + 1(0, 1, 0) + 3(0, 0, 1)$$

$$T(0, 1, 0) = (2, -4, 0) = 2(1, 0, 0) - 4(0, 1, 0) + 0(0, 0, 1)$$

and $T(0, 0, 1) = (1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1)$.

$\therefore$ by def. of matrix of T with respect to B , we have

$$[T]_B = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -4 & 0 \\ 3 & 0 & 0 \end{bmatrix}.$$

Note: In order to find the matrix of T relative to the standard ordered basis B , it is sufficient to compute $T(1, 0, 0)$, $T(0, 1, 0)$ and $T(0, 0, 1)$. There is no need of further expressing these vectors as linear combinations of $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$. Obviously the co-ordinates of the vectors $T(1, 0, 0)$, $T(0, 1, 0)$ and $T(0, 0, 1)$ respectively constitute the first, second and third columns of the matrix $[T]_B$.

(ii) We have $T(1, 1, 1) = (3, -3, 3)$.

Now our aim is to express $(3, -3, 3)$ as a linear combination of vectors in B' .

Let
$$(a, b, c) = x(1, 1, 1) + y(1, 1, 0) + z(1, 0, 0)$$

$$= (x + y + z, x + y, x).$$

Then $x + y + z = a$, $x + y = b$, $x = c$

i.e., $x = c$, $y = b - c$, $z = a - b$(1)

Putting $a = 3$, $b = -3$, and $c = 3$ in (1), we get

$$x = 3, y = -6 \text{ and } z = 6.$$

$\therefore T(1, 1, 1) = (3, -3, 3) = 3(1, 1, 1) - 6(1, 1, 0) + 6(1, 0, 0)$.

Also $T(1, 1, 0) = (2, -3, 3)$.

Putting $a = 2$, $b = -3$ and $c = 3$ in (1), we get

$$T(1, 1, 0) = (2, -3, 3) = 3(1, 1, 1) - 6(1, 1, 0) + 5(1, 0, 0).$$

Finally, $T(1, 0, 0) = (0, 1, 3)$.

Putting $a = 0$, $b = 1$ and $c = 3$ in (1), we get

$$T(1, 0, 0) = (0, 1, 3) = 3(1, 1, 1) - 2(1, 0, 0) - 1(1, 0, 0).$$

$$\therefore [T]_{B'} = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 6 & 5 & -1 \end{bmatrix}.$$

Example 4: Let T be the linear operator on $\mathbf{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

What is the matrix of T in the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$ where

$$\alpha_1 = (1, 0, 1), \alpha_2 = (-1, 2, 1) \text{ and } \alpha_3 = (2, 1, 1)?$$

Solution: By def. of T , we have

$$T(\alpha_1) = T(1, 0, 1) = (4, -2, 3).$$

Now our aim is to express $(4, -2, 3)$ as a linear combination of the vectors in the basis $B = \{\alpha_1, \alpha_2, \alpha_3\}$. Let

$$\begin{aligned} (a, b, c) &= x\alpha_1 + y\alpha_2 + z\alpha_3 \\ &= x(1, 0, 1) + y(-1, 2, 1) + z(2, 1, 1) \\ &= (x - y + 2z, 2y + z, x + y + z). \end{aligned}$$

Then $x - y + 2z = a$, $2y + z = b$, $x + y + z = c$.

Solving these equations, we have

$$x = \frac{-a - 3b + 5c}{4}, \quad y = \frac{b + c - a}{4}, \quad z = \frac{b - c + a}{2}. \quad \dots(1)$$

Putting $a = 4, b = -2, c = 3$ in (1), we get

$$x = \frac{17}{4}, \quad y = -\frac{3}{4}, \quad z = -\frac{1}{2}.$$

$$\therefore T(\alpha_1) = \frac{17}{4}\alpha_1 - \frac{3}{4}\alpha_2 - \frac{1}{2}\alpha_3.$$

Also $T(\alpha_2) = T(-1, 2, 1) = (-2, 4, 9).$

Putting $a = -2, b = 4, c = 9$ in (1), we get $x = \frac{35}{4}, y = \frac{15}{4}, z = \frac{-7}{2}.$

$$\therefore T(\alpha_2) = \frac{35}{4}\alpha_1 + \frac{15}{4}\alpha_2 - \frac{7}{2}\alpha_3.$$

Finally $T(\alpha_3) = T(2, 1, 1) = (7, -3, 4).$

Putting $a = 7, b = -3, c = 4$ in (1), we get $x = \frac{11}{2}, y = -\frac{3}{2}, z = 0.$

$$\therefore T(\alpha_3) = \frac{11}{2}\alpha_1 - \frac{3}{2}\alpha_2 + 0\alpha_3.$$

$$\therefore [T]_B = \begin{bmatrix} \frac{17}{4} & \frac{35}{4} & \frac{11}{2} \\ -\frac{3}{4} & \frac{15}{4} & -\frac{3}{2} \\ -\frac{1}{2} & -\frac{7}{2} & 0 \end{bmatrix}.$$

Example 5: Let T be a linear operator on $\mathbf{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

Prove that T is invertible and find a formula for T^{-1} .

Solution: Suppose B is the standard ordered basis for $\mathbf{R}^3$. Then

$$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

Let $A = [T]_B$ i.e., let A be the matrix of T with respect to B . First we shall compute A . We have

$$T(1, 0, 0) = (3, -2, -1), T(0, 1, 0) = (0, 1, 2) \text{ and}$$

$$T(0, 0, 1) = (1, 0, 4).$$

$$\therefore A = [T]_B = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix}.$$

Now T will be invertible if the matrix $[T]_B$ is invertible. [See theorem 7 of article 4.2].

$$\text{We have } \det A = |A| = \begin{vmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{vmatrix} = 3(4 - 0) + 1(-4 + 1) = 9.$$

Since $\det A \neq 0$, therefore the matrix A is invertible and consequently T is invertible.

Now we shall compute the matrix A^{-1} . For this let us first find $\text{adj. } A$.

The cofactors of the elements of the first row of A are

$$\begin{vmatrix} 1 & 0 \\ 2 & 4 \end{vmatrix}, -\begin{vmatrix} -2 & 0 \\ -1 & 4 \end{vmatrix}, \begin{vmatrix} -2 & 1 \\ -1 & 2 \end{vmatrix} \text{ i.e., } 4, 8, -3.$$

The cofactors of the elements of the second row of A are

$$-\begin{vmatrix} 0 & 1 \\ 2 & 4 \end{vmatrix}, \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix}, -\begin{vmatrix} 3 & 0 \\ -1 & 2 \end{vmatrix} \text{ i.e., } 2, 13, -6.$$

The cofactors of the elements of the third row of A are

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}, -\begin{vmatrix} 3 & 1 \\ -2 & 0 \end{vmatrix}, \begin{vmatrix} 3 & 0 \\ -2 & 1 \end{vmatrix} \text{ i.e., } -1, -2, 3.$$

$$\therefore \text{Adj. } A = \text{transpose of the matrix } \begin{bmatrix} 4 & 8 & -3 \\ 2 & 13 & -6 \\ -1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 2 & -1 \\ 8 & 13 & -2 \\ -3 & -6 & 3 \end{bmatrix}.$$

$$\therefore A^{-1} = \frac{1}{\det A} \text{Adj. } A = \frac{1}{9} \begin{bmatrix} 4 & 2 & -1 \\ 8 & 13 & -2 \\ -3 & -6 & 3 \end{bmatrix}.$$

$$\text{Now } [T^{-1}]_B = ([T]_B)^{-1} = A^{-1}.$$

[See theorem 7 of article 4.2]

We shall now find a formula for T^{-1} . Let $\alpha = (a, b, c)$ be any vector belonging to $\mathbf{R}^3$. Then

$$\begin{aligned} [T^{-1}(\alpha)]_B &= [T^{-1}]_B [\alpha]_B \text{ [See Note of Theorem 2, article 4.2]} \\ &= \frac{1}{9} \begin{bmatrix} 4 & 2 & -1 \\ 8 & 13 & -2 \\ -3 & -6 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 4a + 2b - c \\ 8a + 13b - 2c \\ -3a - 6b + 3c \end{bmatrix} \end{aligned}$$

Since B is the standard ordered basis for $\mathbf{R}^3$,

$$\therefore T^{-1}(\alpha) = T^{-1}(a, b, c) = \frac{1}{9}(4a + 2b - c, 8a + 13b - 2c, -3a - 6b + 3c).$$

Example 6: Let T be the linear operator on $\mathbf{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

(i) What is the matrix of T in the standard ordered basis B for $\mathbf{R}^3$?

(ii) Find the transition matrix P from the ordered basis B to the ordered basis $B' = \{\alpha_1, \alpha_2, \alpha_3\}$ where $\alpha_1 = (1, 0, 1)$, $\alpha_2 = (-1, 2, 1)$, and $\alpha_3 = (2, 1, 1)$. Hence find the matrix of T relative to the ordered basis B' .

Solution: (i) Let $A = [T]_B$. Then

$$A = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix}. \quad \text{[For calculation work see Example 5]}$$

(ii) Since B is the standard ordered basis, therefore the transition matrix P from B to B' can be immediately written as

$$P = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Now $[T]_{B'} = P^{-1} [T]_B P$. [See note of theorem 13, article 4.2]

In order to compute the matrix P^{-1} , we find that $\det P = -4$.

$$\text{Therefore } P^{-1} = \frac{1}{\det P} \text{Adj. } P = -\frac{1}{4} \begin{bmatrix} 1 & 3 & -5 \\ 1 & -1 & -1 \\ -2 & -2 & 2 \end{bmatrix}.$$

$$\begin{aligned} \therefore [T]_{B'} &= -\frac{1}{4} \begin{bmatrix} 1 & 3 & -5 \\ 1 & -1 & -1 \\ -2 & -2 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} 2 & -7 & -19 \\ 6 & -3 & -3 \\ -4 & 2 & 6 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \end{aligned}$$

$$= -\frac{1}{4} \begin{bmatrix} -17 & -35 & -22 \\ 3 & -15 & 6 \\ 2 & 14 & 0 \end{bmatrix} = \begin{bmatrix} \frac{17}{4} & \frac{35}{4} & \frac{11}{2} \\ -\frac{3}{4} & \frac{15}{4} & -\frac{3}{2} \\ -\frac{1}{2} & -\frac{7}{2} & 0 \end{bmatrix}.$$

[Note that this result tallies with that of Example 4].

Example 7: Let T be a linear operator on $\mathbf{R}^2$ defined by :

$$T(x, y) = (2y, 3x - y).$$

Find the matrix representation of T relative to the basis $\{(1, 3), (2, 5)\}$.

Solution: Let $\alpha_1 = (1, 3)$ and $\alpha_2 = (2, 5)$. By def. of T , we have

$$T(\alpha_1) = T(1, 3) = (2 \cdot 3, 3 \cdot 1 - 3) = (6, 0)$$

and

$$T(\alpha_2) = T(2, 5) = (2 \cdot 5, 3 \cdot 2 - 5) = (10, 1).$$

Now our aim is to express the vectors $T(\alpha_1)$ and $T(\alpha_2)$ as linear combinations of the vectors in the basis $\{\alpha_1, \alpha_2\}$.

$$\text{Let } (a, b) = p\alpha_1 + q\alpha_2 = p(1, 3) + q(2, 5) = (p + 2q, 3p + 5q).$$

$$\text{Then } p + 2q = a, 3p + 5q = b.$$

Solving these equations, we get

$$p = -5a + 2b, q = 3a - b. \quad \dots(1)$$

Putting $a = 6, b = 0$ in (1), we get $p = -30, q = 18$.

$$\therefore T(\alpha_1) = (6, 0) = -30\alpha_1 + 18\alpha_2. \quad \dots(2)$$

Again putting $a = 10, b = 1$ in (1), we get

$$p = -48, q = 29.$$

$$\therefore T(\alpha_2) = (10, 1) = -48\alpha_1 + 29\alpha_2. \quad \dots(3)$$

From the relations (2) and (3), we see that the matrix of T relative to the basis

$$\{\alpha_1, \alpha_2\} \text{ is } \begin{bmatrix} -30 & -48 \\ 18 & 29 \end{bmatrix}.$$

Example 8: Consider the vector space $V(\mathbf{R})$ of all 2×2 matrices over the field $\mathbf{R}$ of real numbers. Let T be the linear transformation on V that sends each matrix X onto AX , where

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}. \text{ Find the matrix of } T \text{ with respect to the ordered basis } B = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} \text{ for}$$

V where

$$\alpha_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \alpha_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Solution: We have

$$T(\alpha_1) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$= 1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

$$T(\alpha_2) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T(\alpha_3) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$= 1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

and

$$T(\alpha_4) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\therefore [T]_B = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Example 9: If the matrix of a linear transformation T on $V_2(\mathbb{C})$, with respect to the ordered basis $B = \{(1, 0), (0, 1)\}$ is $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, what is the matrix of T with respect to the ordered basis $B' = \{(1, 1), (1, -1)\}$?

Solution: Let us first define T explicitly. It is given that

$$[T]_B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

$\therefore T(1, 0) = 1(1, 0) + 1(0, 1) = (1, 1)$, and $T(0, 1) = 1(1, 0) + 1(0, 1) = (1, 1)$.

If $(a, b) \in V_2(\mathbb{C})$, then we can write $(a, b) = a(1, 0) + b(0, 1)$.

$$\begin{aligned} \therefore T(a, b) &= aT(1, 0) + bT(0, 1) \\ &= a(1, 1) + b(1, 1) = (a + b, a + b). \end{aligned}$$

This is the explicit expression for T .

Now let us find the matrix of T with respect to B' .

We have $T(1, 1) = (2, 2)$.

Let $(2, 2) = x(1, 1) + y(1, -1) = (x + y, x - y)$.

Then $x + y = 2, x - y = 2$

$$\Rightarrow x = 2, y = 0.$$

$\therefore (2, 2) = 2(1, 1) + 0(1, -1)$.

Also $T(1, -1) = (0, 0) = 0(1, 1) + 0(1, -1)$.

$$\therefore [T]_{B'} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

Note: If P is the transition matrix from the basis B to the basis B' , then

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

We can compute $[T]_{B'}$ by using the formula

$$[T]_{B'} = P^{-1} [T]_B P.$$

Example 10: Show that the vectors $\alpha_1 = (1, 0, -1)$, $\alpha_2 = (1, 2, 1)$, $\alpha_3 = (0, -3, 2)$ form a basis for $\mathbf{R}^3$. Express each of the standard basis vectors as a linear combination of $\alpha_1, \alpha_2, \alpha_3$.

Solution: Let a, b, c be scalars i.e., real numbers such that

$$a\alpha_1 + b\alpha_2 + c\alpha_3 = \mathbf{0}$$

$$\text{i.e., } a(1, 0, -1) + b(1, 2, 1) + c(0, -3, 2) = (0, 0, 0)$$

$$\text{i.e., } (a + b + 0c, 0a + 2b - 3c, -a + b + 2c) = (0, 0, 0)$$

$$\text{i.e., } \begin{cases} a + b + 0c = 0, \\ 0a + 2b - 3c = 0, \\ -a + b + 2c = 0. \end{cases} \quad \dots(1)$$

The coefficient matrix A of these equations is

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & -3 \\ -1 & 1 & 2 \end{bmatrix}.$$

$$\text{We have } \det A = |A| = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & -3 \\ -1 & 1 & 2 \end{vmatrix}.$$

$$= 1(4 + 3) - 1(0 - 3) = 7 + 3 = 10.$$

Since $\det A \neq 0$, therefore the matrix A is non-singular and $\text{rank } A = 3$ i.e., equal to the number of unknowns a, b, c . Hence $a = 0, b = 0, c = 0$ is the only solution of the equations (1). Therefore the vectors $\alpha_1, \alpha_2, \alpha_3$ are linearly independent over $\mathbf{R}$. Since $\dim \mathbf{R}^3 = 3$, therefore the set $\{\alpha_1, \alpha_2, \alpha_3\}$ containing three linearly independent vectors forms a basis for $\mathbf{R}^3$.

Now let $B = \{e_1, e_2, e_3\}$ be the standard ordered basis for $\mathbf{R}^3$.

$$\text{Then } e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1).$$

$$\text{Let } B' = \{\alpha_1, \alpha_2, \alpha_3\}.$$

$$\text{We have } \alpha_1 = (1, 0, -1) = 1e_1 + 0e_2 - 1e_3$$

$$\alpha_2 = (1, 2, 1) = 1e_1 + 2e_2 + 1e_3$$

$$\alpha_3 = (0, -3, 2) = 0e_1 - 3e_2 + 2e_3.$$

If P is the transition matrix from the basis B to the basis B' , then

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & -3 \\ -1 & 1 & 2 \end{bmatrix}.$$

Let us find the matrix P^{-1} . For this let us first find $\text{Adj. } P$.

The cofactors of the elements of the first row of P are

$$\begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}, -\begin{bmatrix} 0 & -3 \\ -1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix}, \text{ i.e., } 7, 3, 2.$$

The cofactors of the elements of the second row of P are

$$-\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, \begin{vmatrix} 1 & 0 \\ -1 & 2 \end{vmatrix}, -\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} \text{ i.e., } -2, 2, -2.$$

The cofactors of the elements of the third row of P are

$$\begin{vmatrix} 1 & 0 \\ 2 & -3 \end{vmatrix}, -\begin{vmatrix} 1 & 0 \\ 0 & -3 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix}, \text{ i.e., } -3, 3, 2$$

$$\therefore \text{Adj } P = \text{transpose of the matrix } \begin{bmatrix} 7 & 3 & 2 \\ -2 & 2 & -2 \\ -3 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 7 & -2 & -3 \\ 3 & 2 & 3 \\ 2 & -2 & 2 \end{bmatrix}.$$

$$\therefore P^{-1} = \frac{1}{\det P} \text{Adj } P = \frac{1}{10} \begin{bmatrix} 7 & -2 & -3 \\ 3 & 2 & 3 \\ 2 & -2 & 2 \end{bmatrix}.$$

Now $e_1 = 1e_1 + 0e_2 + 0e_3$.

$$\therefore \text{Coordinate matrix of } e_1 \text{ relative to the basis } B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

$\therefore$ Co-ordinate matrix of e_1 relative to the basis B'

$$\begin{aligned} &= [e_1]_{B'} = P^{-1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 7 & -2 & -3 \\ 3 & 2 & 3 \\ 2 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 7 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 7/10 \\ 3/10 \\ 2/10 \end{bmatrix}. \end{aligned}$$

$$\therefore e_1 = \frac{7}{10} \alpha_1 + \frac{3}{10} \alpha_2 + \frac{2}{10} \alpha_3.$$

Also $[e_2]_B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $[e_3]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

$\therefore [e_2]_{B'} = P^{-1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $[e_3]_{B'} = P^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

Thus $[e_2]_{B'} = \frac{1}{10} \begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix}$, $[e_3]_{B'} = \frac{1}{10} \begin{bmatrix} -3 \\ 3 \\ 2 \end{bmatrix}$.

$\therefore e_2 = -\frac{2}{10} \alpha_1 + \frac{2}{10} \alpha_2 - \frac{2}{10} \alpha_3$

and $e_3 = -\frac{3}{10} \alpha_1 + \frac{3}{10} \alpha_2 + \frac{2}{10} \alpha_3$.

Example 11: Let A be an $m \times n$ matrix with real entries. Prove that $A = 0$ (null matrix) if and only if $\text{trace}(A^t A) = 0$.

Solution: Let $A = [a_{ij}]_{m \times n}$.

Then $A^t = [b_{ij}]_{n \times m}$, where $b_{ij} = a_{ji}$.

Now $A^t A$ is a matrix of the type $n \times n$.

Let $A^t A = [c_{ij}]_{n \times n}$. Then

c_{ii} = the sum of the products of the corresponding elements of the i^{th} row of A^t and the i^{th} column of A

$$= b_{i1} a_{1i} + b_{i2} a_{2i} + \dots + b_{im} a_{mi}$$

$$= a_{1i} a_{1i} + a_{2i} a_{2i} + \dots + a_{mi} a_{mi} \quad [\because b_{ij} = a_{ji}]$$

$$= a_{1i}^2 + a_{2i}^2 + \dots + a_{mi}^2.$$

$$\text{Now trace}(A^t A) = \sum_{i=1}^n c_{ii} = \sum_{i=1}^n \{a_{1i}^2 + a_{2i}^2 + \dots + a_{mi}^2\}$$

= the sum of the squares of all the elements of A .

Now the elements of A are all real numbers.

Therefore $\text{trace}(A^t A) = 0$

$\Rightarrow$ the sum of the squares of all the elements of A is zero

$\Rightarrow$ each element of A is zero

$\Rightarrow A$ is a null matrix.

Conversely if A is a null matrix, then $A^t A$ is also a null matrix and so $\text{trace}(A^t A) = 0$.

Hence $\text{trace}(A^t A) = 0$ iff $A = 0$.

Example 12: Show that the only matrix similar to the identity matrix I is I itself.

Solution: The identity matrix I is invertible and we can write $I = I^{-1} I I$. Therefore I is similar to I . Further let B be a matrix similar to I . Then there exists an invertible matrix P such that

$$B = P^{-1} I P$$

$$\Rightarrow B = P^{-1} P \quad [\because P^{-1} I = P^{-1}]$$

$$\Rightarrow B = I.$$

Hence the only matrix similar to I is I itself.

Example 13: If T and S are similar linear transformations on a finite dimensional vector space V (F), then $\det T = \det S$.

Solution: Since T and S are similar, therefore there exists an invertible linear transformation P on V such that $T = PSP^{-1}$.

$$\begin{aligned} \text{Therefore } \det T &= \det (PSP^{-1}) = (\det P) (\det S) (\det P^{-1}) \\ &= (\det P) (\det P^{-1}) (\det S) = [\det (PP^{-1})] (\det S) \\ &= (\det I) (\det S) = 1 (\det S) = \det S. \end{aligned}$$

Comprehensive Exercise 1

1. Let $V = \mathbf{R}^3$ and $T : V \rightarrow V$ be a linear mapping defined by

$$T(x, y, z) = (x + z, -2x + y, -x + 2y + z).$$

What is matrix of T relative to the basis $B = \{(1, 0, 1), (-1, 1, 1), (0, 1, 1)\}$?

2. Find matrix representation of linear mapping $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ given by

$$T(x, y, z) = (z, y + z, x + y + z) \text{ relative to the basis}$$

$$B = \{(1, 0, 1), (-1, 2, 1), (2, 1, 1)\}.$$

3. Find the coordinates of the vector $(2, 1, 3, 4)$ of $\mathbf{R}^4$ relative to the basis vectors

$$\alpha_1 = (1, 1, 0, 0), \alpha_2 = (1, 0, 1, 1), \alpha_3 = (2, 0, 0, 2), \alpha_4 = (0, 0, 2, 2).$$

4. Let T be the linear operator on $\mathbf{R}^2$ defined by

$$T(x, y) = (4x - 2y, 2x + y).$$

Compute the matrix of T relative to the basis $\{\alpha_1, \alpha_2\}$ where $\alpha_1 = (1, 1)$, $\alpha_2 = (-1, 0)$.

5. Let T be the linear operator on $\mathbf{R}^2$ defined by

$$T(x, y) = (4x - 2y, 2x + y).$$

(i) What is the matrix of T in the standard ordered basis B for $\mathbf{R}^2$?

- (ii) Find the transition matrix P from the ordered basis B to the ordered basis $B' = \{\alpha_1, \alpha_2\}$ where $\alpha_1 = (1, 1), \alpha_2 = (-1, 0)$. Hence find the matrix of T relative to the ordered basis B' .
6. Let T be the linear operator on $\mathbf{R}^2$ defined by $T(a, b) = (a, 0)$. Write the matrix of T in the standard ordered basis $B = \{(1, 0), (0, 1)\}$. If $B' = \{(1, 1), (2, 1)\}$ is another ordered basis for $\mathbf{R}^2$, find the transition matrix P from the basis B to the basis B' . Hence find the matrix of T relative to the basis B' .
7. The matrix of a linear transformation T on $V_3(\mathbf{C})$ relative to the basis

$$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \text{ is } \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}.$$

What is the matrix of T relative to the basis

$$B' = \{(0, 1, -1), (1, -1, 1), (-1, 0, 1)\}?$$

8. Find the matrix relative to the basis

$$\alpha_1 = \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right), \alpha_2 = \left(\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3}\right), \alpha_3 = \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right)$$

of $\mathbf{R}^3$, of the linear transformation $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ whose matrix relative to

$$\text{the standard ordered basis is } \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

9. Find the matrix representation of the linear mappings relative to the usual bases for $\mathbf{R}^n$.
- (i) $T: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $T(x, y, z) = (2x - 4y + 9z, 5x + 3y - 2z)$.
- (ii) $T: \mathbf{R} \rightarrow \mathbf{R}^2$ defined by $T(x) = (3x, 5x)$.
- (iii) $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by $T(x, y, z) = (x, y, 0)$.
- (iv) $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by $T(x, y, z) = (z, y + z, x + y + z)$.
10. Let $B = \{(1, 0), (0, 1)\}$ and $B' = \{(1, 2), (2, 3)\}$ be any two bases of $\mathbf{R}^2$ and $T(x, y) = (2x - 3y, x + y)$.
- (i) Find the transition matrices P and Q from B to B' and from B' to B respectively.
- (ii) Verify that $[\alpha]_B = P[\alpha]_{B'} \quad \forall \alpha \in \mathbf{R}^2$
- (iii) Verify that $P^{-1}[T]_B P = [T]_{B'}$.
11. Let V be the space of all 2×2 matrices over the field F and let P be a fixed 2×2 matrix over F . Let T be the linear operator on V defined by $T(A) = PA, \forall A \in V$. Prove that $\text{trace}(T) = 2 \text{trace}(P)$.

12. Let V be the space of 2×2 matrices over $\mathbf{R}$ and let $M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Let T be the linear operator on V defined by $T(A) = MA$. Find the trace of T .

13. Find the trace of the operator T on $\mathbf{R}^3$ defined by

$$T(x, y, z) = (a_1 x + a_2 y + a_3 z, b_1 x + b_2 y + b_3 z, c_1 x + c_2 y + c_3 z).$$

14. Show that the only matrix similar to the zero matrix is the zero matrix itself.

Answers 1

1. $\begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ -2 & 2 & 0 \end{bmatrix}$

2. $\begin{bmatrix} \frac{3}{2} & 0 & \frac{13}{4} \\ \frac{1}{2} & 1 & \frac{5}{4} \\ 0 & 1 & -\frac{1}{2} \end{bmatrix}$

3. $(1, 0, \frac{1}{2}, \frac{3}{2})$

4. $\begin{bmatrix} 3 & -2 \\ 1 & 2 \end{bmatrix}$

5. (i) $\begin{bmatrix} 4 & -2 \\ 2 & 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}; \begin{bmatrix} 3 & -2 \\ 1 & 2 \end{bmatrix}$

6. $[T]_B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; P = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}; [T]_{B'} = \begin{bmatrix} -1 & -2 \\ 1 & 2 \end{bmatrix}$

7. $[T]_{B'} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 0 & -2 \\ 0 & 0 & -3 \end{bmatrix}$

8. $\begin{bmatrix} 3 & -2/3 & -2/3 \\ -2/3 & 10/3 & 0 \\ -2/3 & 0 & 8/3 \end{bmatrix}$

9. (i) $\begin{bmatrix} 2 & -4 & 9 \\ 5 & 3 & -2 \end{bmatrix}$

(ii) $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(iv) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

10. (i) $P = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}; Q = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$

12. $\text{trace}(T) = 10$

13. $\text{trace}(T) = a_1 + b_2 + c_3$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. Let T be a linear transformation on the vector space $V_2(F)$ defined by

$$T(a, b) = (a, 0).$$

The matrix of T relative to the ordered basis $\{(1, 0), (0, 1)\}$ of $V_2(F)$ is

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$

2. The transition matrix P from the standard ordered basis to the ordered basis $\{(1, 1), (-1, 0)\}$ is

(a) $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$

3. Let V be the vector space of 2×2 matrices over $\mathbf{R}$ and let

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Let T be the linear operator on V defined by $T(A) = MA$. Then the trace of T is

(a) 5

(b) 10

(c) 0

(d) None of these.

Fill in the Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

1. If T is a linear operator on $\mathbf{R}^2$ defined by

$$T(x, y) = (x - y, y), \text{ then } T^2(x, y) = \dots\dots\dots$$

2. Let A be a square matrix of order n over a field F . The sum of the elements of A lying along the principal diagonal is called the of A .
3. Let T and S be similar linear operators on the finite dimensional vector space $V(F)$, then $\det(T) \dots\dots \det(S)$.

4. Let $V(F)$ be an n -dimensional vector space and B be any ordered basis for V . If I be the identity transformation on V , then $[I; B] = \dots\dots$
5. Let T be a linear operator on an n -dimensional vector space $V(F)$ and let B and B' be two ordered bases for V . Then the matrix of T relative to B' is $\dots\dots$ to the matrix of T relative to B .

True or False

Write 'T' for true and 'F' for false statement.

1. The relation of similarity is an equivalence relation in the set of all linear transformations on a vector space $V(F)$.
2. Similar matrices have the same trace.

Answers

Multiple Choice Questions

1. (a) 2. (b) 3. (b)

Fill in the Blank(s)

1. $(x - 2y, y)$ 2. trace
3. = 4. unit matrix of order n 5. similar

True or False

1. T 2. T



Chapter

5



Linear Functionals

5.1 Linear Functionals

Let $V(F)$ be a vector space. We know that the field F can be regarded as a vector space over F . This is the vector space $F(F)$ or F^1 . We shall simply denote it by F . A linear transformation from V into F is called a **linear functional** on V . We shall now give independent definition of a linear functional.

Linear Functionals: Definition: Let $V(F)$ be a vector space. A function f from V into F is said to be a linear functional on V if

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta) \quad \forall a, b \in F \text{ and } \forall \alpha, \beta \in V.$$

If f is a linear functional on $V(F)$, then $f(\alpha)$ is in F for each α belonging to V . Since $f(\alpha)$ is a scalar, therefore a linear functional on V is a **scalar valued function**.

Illustration 1: Let $V_n(F)$ be the vector space of ordered n -tuples of the elements of the field F .

Let $x_1, x_2, \dots, x_n$ be n field elements of F . If

$$\alpha = (a_1, a_2, \dots, a_n) \in V_n(F),$$

let f be a function from $V_n(F)$ into F defined by

$$f(\alpha) = x_1 a_1 + x_2 a_2 + \dots + x_n a_n.$$

Let $\beta = (b_1, b_2, \dots, b_n) \in V_n(F)$. If $a, b \in F$, we have

$$\begin{aligned} f(a\alpha + b\beta) &= f[a(a_1, \dots, a_n) + b(b_1, \dots, b_n)] \\ &= f(aa_1 + bb_1, \dots, aa_n + bb_n) \\ &= x_1(aa_1 + bb_1) + \dots + x_n(aa_n + bb_n) \\ &= a(x_1 a_1 + \dots + x_n a_n) + b(x_1 b_1 + \dots + x_n b_n) \\ &= af(a_1, \dots, a_n) + bf(b_1, \dots, b_n) = af(\alpha) + bf(\beta). \end{aligned}$$

$\therefore f$ is a linear functional on $V_n(F)$.

Illustration 2: Now we shall give a very important example of a linear functional.

We shall prove that *the trace function is a linear functional on the space of all $n \times n$ matrices over a field F .*

Let n be a positive integer and F a field. Let $V(F)$ be the vector space of all $n \times n$ matrices over F . If $A = [a_{ij}]_{n \times n} \in V$, then the trace of A is the scalar

$$\text{tr } A = a_{11} + a_{22} + \dots + a_{nn} = \sum_{i=1}^n a_{ii}.$$

Thus the trace of A is the scalar obtained by adding the elements of A lying along the principal diagonal.

The trace function is a linear functional on V because if

$$\begin{aligned} a, b \in F \text{ and } A = [a_{ij}]_{n \times n}, B = [b_{ij}]_{n \times n} \in V, \text{ then} \\ \text{tr}(aA + bB) &= \text{tr}(a[a_{ij}]_{n \times n} + b[b_{ij}]_{n \times n}) = \text{tr}([aa_{ij} + bb_{ij}]_{n \times n}) \\ &= \sum_{i=1}^n (aa_{ii} + bb_{ii}) = a \sum_{i=1}^n a_{ii} + b \sum_{i=1}^n b_{ii} = a(\text{tr } A) + b(\text{tr } B). \end{aligned}$$

Illustration 3: Now we shall give another important example of a linear functional.

Let V be a finite-dimensional vector space over the field F and let B be an ordered basis for V . The function f_i which assigns to each vector α in V the i^{th} coordinate of α relative to the ordered basis B is a linear functional on V .

Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$.

If $\alpha = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n \in V$, then by definition of f_i , we have

$$f_i(\alpha) = a_i.$$

Similarly if $\beta = b_1 \alpha_1 + \dots + b_n \alpha_n \in V$, then $f_i(\beta) = b_i$.

If $a, b \in F$, we have

$$\begin{aligned} f_i(a\alpha + b\beta) &= f_i[a(a_1 \alpha_1 + \dots + a_n \alpha_n) + b(b_1 \alpha_1 + \dots + b_n \alpha_n)] \\ &= f_i[(aa_1 + bb_1) \alpha_1 + \dots + (aa_n + bb_n) \alpha_n] \\ &= aa_i + bb_i = af_i(\alpha) + bf_i(\beta). \end{aligned}$$

Hence f_i is a linear functional on V .

Illustration 4: Let V be the vector space of polynomials in x over $\mathbf{R}$. Let $T : V \rightarrow \mathbf{R}$ be the integral operator defined by $T(f(x)) = \int_0^1 f(x) dx$. Then T is linear and hence it is a linear functional on V .

5.2 Some Particular Linear Functionals

1. Zero functional: Let V be a vector space over the field F . The function f from V into F defined by

$$f(\alpha) = 0 \text{ (zero of } F) \forall \alpha \in V$$

is a linear functional on V .

Proof: Let $\alpha, \beta \in V$ and $a, b \in F$. We have

$$\begin{aligned} f(a\alpha + b\beta) &= 0 && \text{(by def. of } f) \\ &= a0 + b0 = af(\alpha) + bf(\beta). \end{aligned}$$

$\therefore f$ is a linear functional on V . It is called the zero functional and we shall in future denote it by $\hat{0}$.

2. Negative of a linear functional: Let V be a vector space over the field F . Let f be a linear functional on V . The correspondence $-f$ defined by

$$(-f)(\alpha) = -[f(\alpha)] \quad \forall \alpha \in V$$

is a linear functional on V .

Proof: Since $f(\alpha) \in F \Rightarrow -f(\alpha) \in F$, therefore $-f$ is a function from V into F .

Let $a, b \in F$ and $\alpha, \beta \in V$. Then

$$\begin{aligned} (-f)(a\alpha + b\beta) &= -[f(a\alpha + b\beta)] && \text{[by def. of } -f] \\ &= -[af(\alpha) + bf(\beta)] && [\because f \text{ is a linear functional}] \\ &= a[-f(\alpha)] + b[-f(\beta)] \\ &= a[(-f)(\alpha)] + b[(-f)(\beta)]. \end{aligned}$$

$\therefore -f$ is a linear functional on V .

Properties of a linear functional:

Theorem 1: Let f be a linear functional on a vector space $V(F)$. Then

(i) $f(\mathbf{0}) = 0$ where $\mathbf{0}$ on the left hand side is zero vector of V , and 0 on the right hand side is zero element of F .

(ii) $f(-\alpha) = -f(\alpha) \quad \forall \alpha \in V$.

Proof. (i) Let $\alpha \in V$. Then $f(\alpha) \in F$.

$$\begin{aligned} \text{We have } f(\alpha) + 0 &= f(\alpha) && [\because 0 \text{ is zero element of } F] \\ &= f(\alpha + \mathbf{0}) && [\because \mathbf{0} \text{ is zero element of } V] \\ &= f(\alpha) + f(\mathbf{0}) && [\because f \text{ is a linear functional}] \end{aligned}$$

Now F is a field. Therefore

$$f(\alpha) + 0 = f(\alpha) + f(0)$$

$\Rightarrow f(0) = 0$, by left cancellation law for addition in F .

(ii) We have $f[\alpha + (-\alpha)] = f(\alpha) + f(-\alpha)$ [$\because f$ is a linear functional]

But $f[\alpha + (-\alpha)] = f(0) = 0$ [by (i)]

Thus in F , we have

$$f(\alpha) + f(-\alpha) = 0$$

$$\Rightarrow f(-\alpha) = -f(\alpha).$$

5.3 Dual Spaces

Let V' be the set of all linear functionals on a vector space V (F). Sometimes we denote this set by V^* . Now our aim is to impose a vector space structure on the set V' over the same field F . For this purpose we shall have to suitably define addition in V' and scalar multiplication in V' over F .

Theorem: Let V be a vector space over the field F . Let f_1 and f_2 be linear functionals on V . The function $f_1 + f_2$ defined by

$$(f_1 + f_2)(\alpha) = f_1(\alpha) + f_2(\alpha) \quad \forall \alpha \in V$$

is a linear functional on V . If c is any element of F , the function cf defined by

$$(cf)(\alpha) = cf(\alpha) \quad \forall \alpha \in V$$

is a linear functional on V . The set V' of all linear functionals on V , together with the addition and scalar multiplication defined as above is a vector space over the field F .

Proof: Suppose f_1 and f_2 are linear functionals on V and we define $f_1 + f_2$ as follows :

$$(f_1 + f_2)(\alpha) = f_1(\alpha) + f_2(\alpha) \quad \forall \alpha \in V. \quad \dots(1)$$

Since $f_1(\alpha) + f_2(\alpha) \in F$, therefore $f_1 + f_2$ is a function from V into F .

Let $a, b \in F$ and $\alpha, \beta \in V$. Then

$$\begin{aligned} (f_1 + f_2)(a\alpha + b\beta) &= f_1(a\alpha + b\beta) + f_2(a\alpha + b\beta) && [\text{by (1)}] \\ &= [af_1(\alpha) + bf_1(\beta)] + [af_2(\alpha) + bf_2(\beta)] \\ & \quad [\because f_1 \text{ and } f_2 \text{ are linear functionals}] \\ &= a[f_1(\alpha) + f_2(\alpha)] + b[f_1(\beta) + f_2(\beta)] \\ &= a[(f_1 + f_2)(\alpha)] + b[(f_1 + f_2)(\beta)] && [\text{by (1)}] \end{aligned}$$

$\therefore (f_1 + f_2)$ is a linear functional on V . Thus

$$f_1, f_2 \in V' \Rightarrow f_1 + f_2 \in V'.$$

Therefore V' is closed with respect to addition defined in it.

Again let $f \in V'$ and $c \in F$. Let us define cf as follows :

$$(cf)(\alpha) = cf(\alpha) \quad \forall \alpha \in V. \quad \dots(2)$$

Since $cf(\alpha) \in F$, therefore cf is a function from V into F .

Let $a, b \in F$ and $\alpha, \beta \in V$. Then

$$\begin{aligned}
 (cf)(a\alpha + b\beta) &= cf(a\alpha + b\beta) && [\text{by (2)}] \\
 &= c[af(\alpha) + bf(\beta)] && [\because f \text{ is linear functional}] \\
 &= c[af(\alpha)] + c[bf(\beta)] && [\because F \text{ is a field}] \\
 &= (ca)f(\alpha) + (cb)f(\beta) \\
 &= (ac)f(\alpha) + (bc)f(\beta) \\
 &= a[cf(\alpha)] + b[cf(\beta)] \\
 &= a[(cf)(\alpha)] + b[(cf)(\beta)].
 \end{aligned}$$

$\therefore cf$ is a linear functional on V . Thus

$$f \in V' \text{ and } c \in F \Rightarrow cf \in V'.$$

Therefore V' is closed with respect to scalar multiplication defined in it.

Associativity of addition in V' .

Let $f_1, f_2, f_3 \in V'$. If $\alpha \in V$, then

$$\begin{aligned}
 [f_1 + (f_2 + f_3)](\alpha) &= f_1(\alpha) + (f_2 + f_3)(\alpha) && [\text{by (1)}] \\
 &= f_1(\alpha) + [f_2(\alpha) + f_3(\alpha)] && [\text{by (1)}] \\
 &= [f_1(\alpha) + f_2(\alpha)] + f_3(\alpha) && [\because \text{addition in } F \text{ is associative}] \\
 &= (f_1 + f_2)(\alpha) + f_3(\alpha) && [\text{by (1)}] \\
 &= [(f_1 + f_2) + f_3](\alpha) && [\text{by (1)}]
 \end{aligned}$$

$\therefore f_1 + (f_2 + f_3) = (f_1 + f_2) + f_3$
[by def. of equality of two functions]

Commutativity of addition in V' . Let $f_1, f_2 \in V'$. If α is any element of V , then

$$\begin{aligned}
 (f_1 + f_2)(\alpha) &= f_1(\alpha) + f_2(\alpha) && [\text{by (1)}] \\
 &= f_2(\alpha) + f_1(\alpha) && [\because \text{addition in } F \text{ is commutative}] \\
 &= (f_2 + f_1)(\alpha) && [\text{by (1)}]
 \end{aligned}$$

$\therefore f_1 + f_2 = f_2 + f_1$.

Existence of additive identity in V' . Let $\hat{0}$ be the zero linear functional in V i.e.,

$$\hat{0}(\alpha) = 0 \quad \forall \alpha \in V.$$

Then $\hat{0} \in V'$. If $f \in V'$ and $\alpha \in V$, we have

$$\begin{aligned}
 (\hat{0} + f)(\alpha) &= \hat{0}(\alpha) + f(\alpha) && [\text{by (1)}] \\
 &= 0 + f(\alpha) && [\text{by def. of } \hat{0}] \\
 &= f(\alpha) && [0 \text{ being additive identity in } F]
 \end{aligned}$$

$$\therefore \quad \hat{0} + f = f \quad \forall f \in V'.$$

$$\therefore \quad \hat{0} \text{ is the additive identity in } V'.$$

Existence of additive inverse of each element in V' .

Let $f \in V'$. Let us define $-f$ as follows :

$$(-f)(\alpha) = -f(\alpha) \quad \forall \alpha \in V.$$

Then $-f \in V'$. If $\alpha \in V$, we have

$$\begin{aligned} (-f + f)(\alpha) &= (-f)(\alpha) + f(\alpha) && [\text{by (1)}] \\ &= -f(\alpha) + f(\alpha) && [\text{by def. of } -f] \\ &= 0 \\ &= \hat{0}(\alpha) && [\text{by def. of } \hat{0}] \end{aligned}$$

$$\therefore \quad -f + f = \hat{0} \text{ for every } f \in V'.$$

Thus each element in V' possesses additive inverse. Therefore V' is an **abelian group** with respect to addition defined in it.

Further we make the following observations :

(i) Let $c \in F$ and $f_1, f_2 \in V'$. If α is any element in V , we have

$$\begin{aligned} [c(f_1 + f_2)](\alpha) &= c[(f_1 + f_2)(\alpha)] && [\text{by (2)}] \\ &= c[f_1(\alpha) + f_2(\alpha)] && [\text{by (1)}] \\ &= cf_1(\alpha) + cf_2(\alpha) \\ &= (cf_1)(\alpha) + (cf_2)(\alpha) && [\text{by (2)}] \\ &= (cf_1 + cf_2)(\alpha) && [\text{by (1)}] \end{aligned}$$

$$\therefore \quad c(f_1 + f_2) = cf_1 + cf_2.$$

(ii) Let $a, b \in F$ and $f \in V'$. If $\alpha \in V$, we have

$$\begin{aligned} [(a + b)f](\alpha) &= (a + b)f(\alpha) && [\text{by (2)}] \\ &= af(\alpha) + bf(\alpha) && [\because F \text{ is a field}] \\ &= (af)(\alpha) + (bf)(\alpha) && [\text{by (2)}] \\ &= (af + bf)(\alpha) && [\text{by (1)}] \end{aligned}$$

$$\therefore \quad (a + b)f = af + bf.$$

(iii) Let $a, b \in F$ and $f \in V'$. If $\alpha \in V$, we have

$$\begin{aligned} [(ab)f](\alpha) &= (ab)f(\alpha) && [\text{by (2)}] \\ &= a[bf(\alpha)] && [\because \text{multiplication in } F \text{ is associative}] \\ &= a[(bf)(\alpha)] && [\text{by (2)}] \\ &= [a(bf)](\alpha) && [\text{by (2)}] \end{aligned}$$

$$\therefore \quad (ab)f = a(bf).$$

(iv) Let 1 be the multiplicative identity of F and $f \in V'$. If $\alpha \in V$, we have

$$\begin{aligned} (1f)(\alpha) &= 1f(\alpha) && [\text{by (2)}] \\ &= f(\alpha) && [\because F \text{ is a field}] \end{aligned}$$

$$\therefore 1f = f.$$

Hence V' is a vector space over the field F .

Dual Space:

Definition: Let V be a vector space over the field F . Then the set V' of all linear functionals on V is also a vector space over the field F . The vector space V' is called the dual space of V .

Sometimes V^* and $\hat{V}$ are also used to denote the dual space of V . The dual space of V is also called the **conjugate space** of V .

5.4 Dual Bases

Theorem 1: Let V be an n -dimensional vector space over the field F and let $B = \{\alpha_1, \dots, \alpha_n\}$ be an ordered basis for V . If $\{x_1, \dots, x_n\}$ is any ordered set of n scalars, then there exists a unique linear functional f on V such that

$$f(\alpha_i) = x_i, \quad i = 1, 2, \dots, n.$$

Proof: **Existence of f .** Let $\alpha \in V$.

Since $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a basis for V , therefore there exist unique scalars $a_1, a_2, \dots, a_n$ such that

$$\alpha = a_1\alpha_1 + \dots + a_n\alpha_n.$$

For this vector α , let us define

$$f(\alpha) = a_1x_1 + \dots + a_nx_n.$$

Obviously $f(\alpha)$ as defined above is a unique element of F . Therefore f is a well-defined rule for associating with each vector α in V a unique scalar $f(\alpha)$ in F . Thus f is a function from V into F .

The unique representation of $\alpha_i \in V$ as a linear combination of the vectors belonging to the basis B is

$$\alpha_i = 0\alpha_1 + 0\alpha_2 + \dots + 1\alpha_i + 0\alpha_{i+1} + \dots + 0\alpha_n.$$

Therefore according to our definition of f , we have

$$f(\alpha_i) = 0x_1 + 0x_2 + \dots + 1x_i + 0x_{i+1} + \dots + 0x_n$$

$$\text{i.e.,} \quad f(\alpha_i) = x_i, \quad i = 1, 2, \dots, n.$$

Now to show that f is a linear functional.

Let $a, b \in F$ and $\alpha, \beta \in V$. Let

$$\alpha = a_1\alpha_1 + \dots + a_n\alpha_n,$$

$$\text{and} \quad \beta = b_1\alpha_1 + \dots + b_n\alpha_n.$$

$$\begin{aligned}
\text{Then } f(a\alpha + b\beta) &= f[a(a_1\alpha_1 + \dots + a_n\alpha_n) + b(b_1\alpha_1 + \dots + b_n\alpha_n)] \\
&= f[(aa_1 + bb_1)\alpha_1 + \dots + (aa_n + bb_n)\alpha_n] \\
&= (aa_1 + bb_1)x_1 + \dots + (aa_n + bb_n)x_n \quad [\text{by def. of } f] \\
&= a(a_1x_1 + \dots + a_nx_n) + b(b_1x_1 + \dots + b_nx_n) = af(\alpha) + bf(\beta).
\end{aligned}$$

$\therefore f$ is a linear functional on V . Thus there exists a linear functional f on V such that $f(\alpha_i) = x_i, i = 1, 2, \dots, n$.

Uniqueness of f : Let g be a linear functional on V such that

$$g(\alpha_i) = x_i, \quad i = 1, 2, \dots, n.$$

For any vector $\alpha = a_1\alpha_1 + \dots + a_n\alpha_n \in V$, we have

$$\begin{aligned}
g(\alpha) &= g(a_1\alpha_1 + \dots + a_n\alpha_n) \\
&= a_1g(\alpha_1) + \dots + a_ng(\alpha_n) \quad [\because g \text{ is linear}] \\
&= a_1x_1 + \dots + a_nx_n \quad [\text{by def. of } g] \\
&= f(\alpha). \quad [\text{by def. of } f]
\end{aligned}$$

Thus $g(\alpha) = f(\alpha) \quad \forall \alpha \in V$.

$$\therefore g = f.$$

This shows the uniqueness of f .

Remark: From this theorem we conclude that if f is a linear functional on a finite dimensional vector space V , then f is completely determined if we mention under f the images of the elements of a basis set of V . If f and g are two linear functionals on V such that $f(\alpha_i) = g(\alpha_i)$ for all α_i belonging to a basis of V , then

$$f(\alpha) = g(\alpha) \quad \forall \alpha \in V \quad \text{i.e., } f = g.$$

Thus two linear functionals of V are equal if they agree on a basis of V .

Theorem 2: Let V be an n -dimensional vector space over the field F and let $B = \{\alpha_1, \dots, \alpha_n\}$ be a basis for V . Then there is a uniquely determined basis $B' = \{f_1, \dots, f_n\}$ for V' such that $f_i(\alpha_j) = \delta_{ij}$. Consequently the dual space of an n -dimensional space is n -dimensional.

The basis B' is called the **dual basis** of B .

Proof: $B = \{\alpha_1, \dots, \alpha_n\}$ is an ordered basis for V . Therefore by theorem 1, there exists a unique linear functional f_1 on V such that

$$f_1(\alpha_1) = 1, f_1(\alpha_2) = 0, \dots, f_1(\alpha_n) = 0$$

where $\{1, 0, \dots, 0\}$ is an ordered set of n scalars.

In fact, for each $i = 1, 2, \dots, n$ there exists a unique linear functional f_i on V such that

$$f_i(\alpha_j) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$$\text{i.e., } f_i(\alpha_j) = \delta_{ij}, \quad \dots(1)$$

where $\delta_{ij} \in F$ is Kronecker delta i.e., $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

Let $B' = \{f_1, \dots, f_n\}$. Then B' is a subset of V' containing n distinct elements of V' . We shall show that B' is a basis for V' .

First we shall show that B' is linearly independent.

$$\text{Let } c_1 f_1 + c_2 f_2 + \dots + c_n f_n = \hat{0}$$

$$\Rightarrow (c_1 f_1 + \dots + c_n f_n)(\alpha) = \hat{0}(\alpha) \quad \forall \alpha \in V$$

$$\Rightarrow c_1 f_1(\alpha) + \dots + c_n f_n(\alpha) = 0 \quad \forall \alpha \in V \quad [\because \hat{0}(\alpha) = 0]$$

$$\Rightarrow \sum_{i=1}^n c_i f_i(\alpha) = 0 \quad \forall \alpha \in V$$

$$\Rightarrow \sum_{i=1}^n c_i f_i(\alpha_j) = 0, \quad j = 1, 2, \dots, n \quad [\text{Putting } \alpha = \alpha_j \text{ where } j = 1, 2, \dots, n]$$

$$\Rightarrow \sum_{i=1}^n c_i \delta_{ij} = 0, \quad j = 1, 2, \dots, n$$

$$\Rightarrow c_j = 0, \quad j = 1, 2, \dots, n$$

$$\Rightarrow f_1, f_2, \dots, f_n \text{ are linearly independent.}$$

In the second place, we shall show that the linear span of B' is equal to V' .

Let f be any element of V' . The linear functional f will be completely determined if we define it on a basis for V . So let

$$f(\alpha_i) = a_i, \quad i = 1, 2, \dots, n. \quad \dots(2)$$

We shall show that

$$f = a_1 f_1 + \dots + a_n f_n = \sum_{i=1}^n a_i f_i.$$

We know that two linear functionals on V are equal if they agree on a basis of V . So let $\alpha_j \in B$ where $j = 1, \dots, n$. Then

$$\begin{aligned} \left[\sum_{i=1}^n a_i f_i \right](\alpha_j) &= \sum_{i=1}^n a_i f_i(\alpha_j) \\ &= \sum_{i=1}^n a_i \delta_{ij} \end{aligned} \quad [\text{from (1)}]$$

$$\begin{aligned} &= a_j, \quad \text{on summing with respect to } i \text{ and} \\ &\quad \text{remembering that } \delta_{ij} = 1 \text{ when } i = j \\ &\quad \text{and } \delta_{ij} = 0 \text{ when } i \neq j \\ &= f(\alpha_j) \end{aligned} \quad [\text{from (2)}]$$

$$\text{Thus } \left[\sum_{i=1}^n a_i f_i \right](\alpha_j) = f(\alpha_j) \quad \forall \alpha_j \in B.$$

Therefore $f = \sum_{i=1}^n a_i f_i$. Thus every element f in V' can be expressed as a linear combination of $f_1, \dots, f_n$.

$\therefore V' = \text{linear span of } B'$. Hence B' is a basis for V' .

Now $\dim V' = \text{number of distinct elements in } B' = n$.

Corollary: If V is an n -dimensional vector space over the field F , then V is isomorphic to its dual space V' .

Proof: We have $\dim V' = \dim V = n$.

$\therefore$ V is isomorphic to V' .

Theorem 3: Let V be an n -dimensional vector space over the field F and let $B = \{\alpha_1, \dots, \alpha_n\}$ be a basis for V . Let $B' = \{f_1, \dots, f_n\}$ be the dual basis of B . Then for each linear functional f on V , we have

$$f = \sum_{i=1}^n f(\alpha_i) f_i$$

and for each vector α in V we have

$$\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i.$$

Proof: Since B' is dual basis of B , therefore

$$f_i(\alpha_j) = \delta_{ij}. \quad \dots(1)$$

If f is a linear functional on V , then $f \in V'$ for which B' is basis. Therefore f can be expressed as a linear combination of $f_1, \dots, f_n$. Let $f = \sum_{i=1}^n c_i f_i$.

$$\text{Then } f(\alpha_j) = \left(\sum_{i=1}^n c_i f_i \right)(\alpha_j) = \sum_{i=1}^n c_i f_i(\alpha_j)$$

$$= \sum_{i=1}^n c_i \delta_{ij} \quad [\text{From (1)}]$$

$$= c_j, j = 1, 2, \dots, n.$$

$$\therefore f = \sum_{i=1}^n f(\alpha_i) f_i.$$

Now let α be any vector in V . Let

$$\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n. \quad \dots(2)$$

$$\text{Then } f_i(\alpha) = f_i \left(\sum_{j=1}^n x_j \alpha_j \right) \quad \left[\text{From (2), } \alpha = \sum_{j=1}^n x_j \alpha_j \right]$$

$$= \sum_{j=1}^n x_j f_i(\alpha_j) \quad [\because f_i \text{ is linear functional}]$$

$$= \sum_{j=1}^n x_j \delta_{ij} \quad [\text{From (1)}]$$

$$= x_i.$$

$$\therefore \alpha = f_1(\alpha) \alpha_1 + \dots + f_n(\alpha) \alpha_n = \sum_{i=1}^n f_i(\alpha) \alpha_i.$$

Important: It should be noted that if $B = \{\alpha_1, \dots, \alpha_n\}$ is an ordered basis for V and $B' = \{f_1, \dots, f_n\}$ is the dual basis, then f_i is precisely the function which assigns to each vector α in V the i^{th} coordinate of α relative to the ordered basis B .

Theorem 4: Let V be an n -dimensional vector space over the field F . If α is a non-zero vector in V , there exists a linear functional f on V such that $f(\alpha) \neq 0$.

Proof: Since $\alpha \neq 0$, therefore $\{\alpha\}$ is a linearly independent subset of V . So it can be extended to form a basis for V . Thus there exists a basis $B = \{\alpha_1, \dots, \alpha_n\}$ for V such that $\alpha_1 = \alpha$.

If $B' = \{f_1, \dots, f_n\}$ is the dual basis, then

$$f_1(\alpha) = f_1(\alpha_1) = 1 \neq 0.$$

Thus there exists linear functional f_1 such that

$$f_1(\alpha) \neq 0.$$

Corollary: Let V be an n -dimensional vector space over the field F . If

$$f(\alpha) = 0 \quad \forall f \in V', \text{ then } \alpha = 0.$$

Proof: Suppose $\alpha \neq 0$. Then there is a linear functional f on V such that $f(\alpha) \neq 0$.

This contradicts the hypothesis that

$$f(\alpha) = 0 \quad \forall f \in V'.$$

Hence we must have $\alpha = 0$.

Theorem 5: Let V be an n -dimensional vector space over the field F . If α, β are any two different vectors in V , then there exists a linear functional f on V such that $f(\alpha) \neq f(\beta)$.

Proof: We have $\alpha \neq \beta \Rightarrow \alpha - \beta \neq 0$.

Now $\alpha - \beta$ is a non-zero vector in V . Therefore by theorem 4, there exists a linear functional f on V such that

$$f(\alpha - \beta) \neq 0 \quad \Rightarrow \quad f(\alpha) - f(\beta) \neq 0 \quad \Rightarrow \quad f(\alpha) \neq f(\beta)$$

Hence the result.

5.5 Reflexivity

Second dual space (or Bi-dual space): We know that every vector space V possesses a dual space V' consisting of all linear functionals on V . Now V' is also a vector space. Therefore it will also possess a dual space $(V')'$ consisting of all linear functionals on V' . This dual space of V' is called the **Second dual space** or **Bi-dual space** of V and for the sake of simplicity we shall denote it by V'' .

If V is finite-dimensional, then

$$\dim V = \dim V' = \dim V''$$

showing that they are isomorphic to each other.

Theorem 1: Let V be a finite dimensional vector space over the field F . If α is any vector in V , the function L_α on V' defined by

$$L_\alpha(f) = f(\alpha) \quad \forall f \in V'$$

is a linear functional on V' i.e., $L_\alpha \in V''$.

Also the mapping $\alpha \Rightarrow L_\alpha$ is an isomorphism of V onto V'' .

Proof: If $\alpha \in V$ and $f \in V'$, then $f(\alpha)$ is a unique element of F . Therefore the correspondence L_α defined by

$$L_\alpha(f) = f(\alpha) \quad \forall f \in V' \quad \dots(1)$$

is a function from V' into F .

Let $a, b \in F$ and $f, g \in V'$, then

$$L_\alpha(af + bg) = (af + bg)(\alpha) \quad [\text{From (1)}]$$

$$= (af)(\alpha) + (bg)(\alpha)$$

$$= af(\alpha) + bg(\alpha)$$

[by scalar multiplication of linear functionals]

$$= a[L_\alpha(f)] + b[L_\alpha(g)]. \quad [\text{From (1)}]$$

Therefore L_α is a linear functional on V' and thus $L_\alpha \in V''$.

Now let ψ be the function from V into V'' defined by

$$\psi(\alpha) = L_\alpha \quad \forall \alpha \in V.$$

ψ is one-one: If $\alpha, \beta \in V$, then

$$\psi(\alpha) = \psi(\beta)$$

$$\Rightarrow L_\alpha = L_\beta \Rightarrow L_\alpha(f) = L_\beta(f) \quad \forall f \in V'$$

$$\Rightarrow f(\alpha) = f(\beta) \quad \forall f \in V' \quad [\text{From (1)}]$$

$$\Rightarrow f(\alpha) - f(\beta) = 0 \quad \forall f \in V'$$

$$\Rightarrow f(\alpha - \beta) = 0 \quad \forall f \in V'$$

$$\Rightarrow \alpha - \beta = \mathbf{0}$$

[$\because$ by theorem 4 of 2.16, if $\alpha - \beta \neq \mathbf{0}$, then $\exists$ a linear

functional f on V such that $f(\alpha - \beta) \neq 0$. Here we have

$$f(\alpha - \beta) = 0 \quad \forall f \in V' \text{ and so } \alpha - \beta \text{ must be } \mathbf{0}]$$

$$\Rightarrow \alpha = \beta.$$

$$\therefore \psi \text{ is one-one.}$$

ψ is a linear transformation:

Let $a, b \in F$ and $\alpha, \beta \in V$. Then

$$\psi(a\alpha + b\beta) = L_{a\alpha + b\beta} \quad [\text{by def. of } \psi]$$

For every $f \in V'$, we have

$$\begin{aligned} L_{a\alpha + b\beta}(f) &= f(a\alpha + b\beta) \\ &= af(\alpha) + bf(\beta) \end{aligned} \quad [\text{From (1)}]$$

$$= aL_\alpha(f) + bL_\beta(f) \quad [\text{From (1)}]$$

$$= (aL_\alpha)(f) + (bL_\beta)(f) = (aL_\alpha + bL_\beta)(f).$$

$$\therefore L_{a\alpha + b\beta} = aL_\alpha + bL_\beta = a\psi(\alpha) + b\psi(\beta).$$

Thus $\psi(a\alpha + b\beta) = a\psi(\alpha) + b\psi(\beta)$.

$\therefore \psi$ is a linear transformation from V into V'' . We have $\dim V = \dim V''$.

Therefore ψ is one-one implies that ψ must also be onto.

Hence ψ is an isomorphism of V onto V'' .

Note: The correspondence $\alpha \rightarrow L_\alpha$ as defined in the above theorem is called the **natural correspondence** between V and V'' . It is important to note that the above theorem shows not only that V and V'' are isomorphic—this much is obvious from the fact that they have the same dimension — but that the natural correspondence is an isomorphism. This property of vector spaces is called **reflexivity**. Thus in the above theorem we have proved that *every finite-dimensional vector space is reflexive*.

In future we shall identify V'' with V through the natural isomorphism $\alpha \leftrightarrow L_\alpha$. We shall say that the element L of V'' is the same as the element α of V iff $L = L_\alpha$ i.e., iff

$$L(f) = f(\alpha) \quad \forall f \in V'.$$

It will be in this sense that we shall regard $V'' = V$.

Theorem 2: Let V be a finite dimensional vector space over the field F . If L is a linear functional on the dual space V' of V , then there is a unique vector α in V such that

$$L(f) = f(\alpha) \quad \forall f \in V'.$$

Proof: This theorem is an immediate corollary of theorem 1. We should first prove theorem 1. Then we should conclude like this :

The correspondence $\alpha \rightarrow L_\alpha$ is a one-to-one correspondence between V and V'' . Therefore if $L \in V''$, there exists a unique vector α in V such that $L = L_\alpha$ i.e., such that

$$L(f) = f(\alpha) \quad \forall f \in V'.$$

Theorem 3: Let V be a finite dimensional vector space over the field F . Each basis for V' is the dual of some basis for V .

Proof: Let $B' = \{f_1, f_2, \dots, f_n\}$ be a basis for V' . Then there exists a dual basis $(B')' = \{L_1, L_2, \dots, L_n\}$ for V'' such that

$$L_i(f_j) = \delta_{ij}. \quad \dots(1)$$

By previous theorem, for each i there is a vector α_i in V such that

$$L_i = L_{\alpha_i} \text{ where } L_{\alpha_i}(f) = f(\alpha_i) \quad \forall f \in V'. \quad \dots(2)$$

The correspondence $\alpha \leftrightarrow L_\alpha$ is an isomorphism of V onto V'' . Under an isomorphism a basis is mapped onto a basis. Therefore $B = \{\alpha_1, \dots, \alpha_n\}$ is a basis for V because it is the image set of a basis for V'' under the above isomorphism.

Putting $f = f_j$ in (2), we get

$$\begin{aligned} f_j(\alpha_i) &= L_{\alpha_i}(f_j) = L_i(f_j) \\ &= \delta_{ij}. \end{aligned} \quad [\text{From (1)}]$$

$\therefore B' = \{f_1, \dots, f_n\}$ is the dual of the basis B .

Hence the result.

Theorem 4: Let V be a finite dimensional vector space over the field F . Let B be a basis for V and B' be the dual basis of B . Then show that

$$B'' = (B')' = B.$$

Proof: Let $B = \{\alpha_1, \dots, \alpha_n\}$ be a basis for V ,

$B' = \{f_1, \dots, f_n\}$ be the dual basis of B in V' and

$B'' = (B')' = \{L_1, \dots, L_n\}$ be the dual basis of B in V'' . Then

$$f_i(\alpha_j) = \delta_{ij},$$

and

$$L_i(f_j) = \delta_{ij}, i = 1, \dots, n; j = 1, \dots, n.$$

If

$\alpha \in V$, then there exists $L_\alpha \in V''$ such that

$$L_\alpha(f) = f(\alpha) \quad \forall f \in V'.$$

Taking α_i in place of α , we see that for each $j = 1, \dots, n$,

$$L_{\alpha_i}(f_j) = f_j(\alpha_i) = \delta_{ij} = L_i(f_j).$$

Thus L_{α_i} and L_i agree on a basis for V' . Therefore $L_{\alpha_i} = L_i$.

If we identify V'' with V through the natural isomorphism $\alpha \leftrightarrow L_\alpha$, then we consider L_α as the same element as α .

So $L_i = L_{\alpha_i} = \alpha_i$ where $i = 1, 2, \dots, n$.

Thus $B'' = B$.

Illustrative Examples

Example 1: Find the dual basis of the basis set

$$B = \{(1, -1, 3), (0, 1, -1), (0, 3, -2)\} \text{ for } V_3(\mathbf{R}).$$

Solution: Let $\alpha_1 = (1, -1, 3), \alpha_2 = (0, 1, -1), \alpha_3 = (0, 3, -2)$.

Then $B = \{\alpha_1, \alpha_2, \alpha_3\}$.

If $B' = \{f_1, f_2, f_3\}$ is dual basis of B , then

$$f_1(\alpha_1) = 1, f_1(\alpha_2) = 0, f_1(\alpha_3) = 0,$$

$$f_2(\alpha_1) = 0, f_2(\alpha_2) = 1, f_2(\alpha_3) = 0,$$

and $f_3(\alpha_1) = 0, f_3(\alpha_2) = 0, f_3(\alpha_3) = 1$.

Now to find explicit expressions for f_1, f_2, f_3 .

Let $(a, b, c) \in V_3(\mathbf{R})$.

$$\begin{aligned} \text{Let } (a, b, c) &= x(1, -1, 3) + y(0, 1, -1) + z(0, 3, -2) \\ &= x\alpha_1 + y\alpha_2 + z\alpha_3. \end{aligned} \quad \dots(1)$$

Then $f_1(a, b, c) = x, f_2(a, b, c) = y$, and $f_3(a, b, c) = z$.

Now to find the values of x, y, z .

From (1), we have

$$x = a, -x + y + 3z = b, 3x - y - 2z = c.$$

Solving these equations, we have

$$x = a, y = 7a - 2b - 3c, z = b + c - 2a.$$

Hence $f_1(a, b, c) = a$, $f_2(a, b, c) = 7a - 2b - 3c$,

and $f_3(a, b, c) = -2a + b + c$.

Therefore $B' = \{f_1, f_2, f_3\}$ is a dual basis of B where f_1, f_2, f_3 are as defined above.

Example 2: The vectors $\alpha_1 = (1, 1, 1)$, $\alpha_2 = (1, 1, -1)$, and $\alpha_3 = (1, -1, -1)$ form a basis of $V_3(\mathbb{C})$. If $\{f_1, f_2, f_3\}$ is the dual basis and if $\alpha = (0, 1, 0)$, find $f_1(\alpha)$, $f_2(\alpha)$ and $f_3(\alpha)$.

Solution: Let $\alpha = a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$. Then

$$f_1(\alpha) = a_1, f_2(\alpha) = a_2, f_3(\alpha) = a_3.$$

Now

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$$

$$\Rightarrow (0, 1, 0) = a_1(1, 1, 1) + a_2(1, 1, -1) + a_3(1, -1, -1)$$

$$\Rightarrow (0, 1, 0) = (a_1 + a_2 + a_3, a_1 + a_2 - a_3, a_1 - a_2 - a_3)$$

$$\Rightarrow a_1 + a_2 + a_3 = 0, a_1 + a_2 - a_3 = 1, a_1 - a_2 - a_3 = 0$$

$$\Rightarrow a_1 = 0, a_2 = \frac{1}{2}, a_3 = -\frac{1}{2}.$$

$$\therefore f_1(\alpha) = 0, f_2(\alpha) = \frac{1}{2}, f_3(\alpha) = -\frac{1}{2}.$$

Example 3: If f is a non-zero linear functional on a vector space V and if x is an arbitrary scalar, does there necessarily exist a vector α in V such that $f(\alpha) = x$?

Solution: f is a non-zero linear functional on V . Therefore there must be some non-zero vector β in V such that

$$f(\beta) = y \text{ where } y \text{ is a non-zero element of } F.$$

If x is any element of F , then

$$x = (xy^{-1})y = (xy^{-1})f(\beta)$$

$$= f[(xy^{-1})\beta] \quad [\because f \text{ is linear functional}]$$

Thus there exists $\alpha = (xy^{-1})\beta \in V$ such that $f(\alpha) = x$.

Note: If f is a non-zero linear functional on $V(F)$, then f is necessarily a function from V onto F .

Important Note: In some books $f(\alpha)$ is written as $[\alpha, f]$.

Example 4. Prove that if f is a linear functional on an n -dimensional vector space $V(F)$, then the set of all those vectors α for which $f(\alpha) = 0$ is a subspace of V , what is the dimension of that subspace?

Solution: Let $N = \{\alpha \in V : f(\alpha) = 0\}$.

N is not empty because at least $0 \in N$. Remember that

$$f(0) = 0.$$

Let $\alpha, \beta \in N$. Then $f(\alpha) = 0, f(\beta) = 0$.

If $a, b \in F$, we have

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta) = a0 + b0 = 0.$$

$\therefore a\alpha + b\beta \in N$.

Thus $a, b \in F$ and $\alpha, \beta \in N$

$\Rightarrow a\alpha + b\beta \in N$.

$\therefore N$ is a subspace of V . This subspace N is the null space of f .

We know that $\dim V = \dim N + \dim(\text{range of } f)$.

(i) If f is zero linear functional, then range of f consists of zero element of F alone. Therefore $\dim(\text{range of } f) = 0$ in this case.

$\therefore$ In this case, we have

$$\dim V = \dim N + 0$$

$\Rightarrow n = \dim N$.

(ii) If f is a non-zero linear functional on V , then f is onto F . So range of f consists of all F in this case. The dimension of the vector space F is 1.

$\therefore$ In this case we have

$$\dim V = \dim N + 1$$

$\Rightarrow \dim N = n - 1$.

Example 5: Let V be a vector space over the field F . Let f be a non-zero linear functional on V and let N be the null space of f . Fix a vector α_0 in V which is not in N . Prove that for each α in V there is a scalar c and a vector β in N such that $\alpha = c\alpha_0 + \beta$. Prove that c and β are unique.

Solution: Since f is a non-zero linear functional on V , therefore there exists a non-zero vector α_0 in V such that $f(\alpha_0) \neq 0$. Consequently $\alpha_0 \notin N$. Let $f(\alpha_0) = y \neq 0$.

Let α be any element of V and let $f(\alpha) = x$.

We have $f(\alpha) = x$

$\Rightarrow f(\alpha) = (x y^{-1}) y \quad [\because 0 \neq y \in F \Rightarrow y^{-1} \text{ exists}]$

$\Rightarrow f(\alpha) = cy$ where $c = x y^{-1} \in F$

$\Rightarrow f(\alpha) = cf(\alpha_0)$

$\Rightarrow f(\alpha) = f(c\alpha_0) \quad [\because f \text{ is a linear functional}]$

$\Rightarrow f(\alpha) - f(c\alpha_0) = 0$

$\Rightarrow f(\alpha - c\alpha_0) = 0$

$\Rightarrow \alpha - c\alpha_0 \in N$

$\Rightarrow \alpha - c\alpha_0 = \beta$ for some $\beta \in N$.

$\Rightarrow \alpha = c\alpha_0 + \beta$.

If possible, let

$$\alpha = c' \alpha_0 + \beta' \text{ where } c' \in F \text{ and } \beta' \in N.$$

$$\text{Then } c \alpha_0 + \beta = c' \alpha_0 + \beta' \quad \dots(1)$$

$$\Rightarrow (c - c') \alpha_0 + (\beta - \beta') = \mathbf{0}$$

$$\Rightarrow f [(c - c') \alpha_0 + (\beta - \beta')] = f (\mathbf{0})$$

$$\Rightarrow (c - c') f (\alpha_0) + f (\beta - \beta') = 0$$

$$\Rightarrow (c - c') f (\alpha_0) = 0$$

$$[\because \beta, \beta' \in N \Rightarrow \beta - \beta' \in N \text{ and thus } f (\beta - \beta') = 0]$$

$$\Rightarrow (c - c') = 0 \quad [\because f (\alpha_0) \text{ is a non-zero element of } F]$$

$$c = c'.$$

Putting $c = c'$ in (1), we get $c \alpha_0 + \beta = c \alpha_0 + \beta' \Rightarrow \beta = \beta'$.

Hence c and β are unique.

Example 6: If f and g are in V' such that $f (\alpha) = 0 \Rightarrow g (\alpha) = 0$, prove that $g = kf$ for some $k \in F$.

Solution: It is given that $f (\alpha) = 0 \Rightarrow g (\alpha) = 0$. Therefore if α belongs to the null space of f , then α also belongs to the null space of g . Thus null space of f is a subset of the null space of g .

(i) If f is zero linear functional, then null space of f is equal to V . Therefore in this case V is a subset of null space of g . Hence null space of g is equal to V . So g is also zero linear functional. Hence we have

$$g = k f \quad \forall k \in F.$$

(ii) Let f be non-zero linear functional on V . Then there exists a non-zero vector $\alpha_0 \in V$ such that $f (\alpha_0) = y$ where y is a non-zero element of F .

$$\text{Let } k = \frac{g (\alpha_0)}{f (\alpha_0)}.$$

If $\alpha \in V$, then we can write

$$\alpha = c \alpha_0 + \beta \text{ where } c \in F \text{ and } \beta \in \text{null space of } f.$$

$$\text{We have } g (\alpha) = g (c \alpha_0 + \beta) = cg (\alpha_0) + g (\beta)$$

$$= cg (\alpha_0)$$

$$[\because \beta \in \text{null space of } f \Rightarrow f (\beta) = 0 \text{ and so } g (\beta) = 0]$$

$$\text{Also } (k f) (\alpha) = k f (\alpha) = k f (c \alpha_0 + \beta)$$

$$= k [cf (\alpha_0) + f (\beta)]$$

$$= kc f (\alpha_0)$$

$$[\because f (\beta) = 0]$$

$$= \frac{g (\alpha_0)}{f (\alpha_0)} cf (\alpha_0) = cg (\alpha_0).$$

$$\text{Thus } g (\alpha) = (k f) (\alpha) \quad \forall \alpha \in V.$$

$$\therefore g = k f.$$

Comprehensive Exercise 1

1. Let $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ and $g : \mathbf{R}^2 \rightarrow \mathbf{R}$ be the linear functionals defined by $f(x, y) = x + 2y$ and $g(x, y) = 3x - y$. Find (i) $f + g$ (ii) $4f$ (iii) $2f - 5g$.
2. Let $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ and $g : \mathbf{R}^2 \rightarrow \mathbf{R}$ be the linear functionals defined by $f(x, y, z) = 2x - 3y + z$ and $g(x, y, z) = 4x - 2y + 3z$. Find (i) $f + g$ (ii) $3f$ (iii) $2f - 5g$.
3. Find the dual basis of the basis set $\{(2, 1), (3, 1)\}$ for $\mathbf{R}^2$.

4. Find the dual basis of the basis set

$$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \text{ for } V_3(\mathbf{R}).$$

5. Find the dual basis of the basis set

$$B = \{(1, -2, 3), (1, -1, 1), (2, -4, 7)\} \text{ of } V_3(\mathbf{R}).$$

6. If $B = \{(-1, 1, 1), (1, -1, 1), (1, 1, -1)\}$ is a basis of $V_3(\mathbf{R})$, then find the dual basis of B .

7. Let V be the vector space of polynomials over $\mathbf{R}$ of degree ≤ 1 i.e.,

$$V = \{a + bx : a, b \in \mathbf{R}\}.$$

Let ϕ_1 and ϕ_2 be linear functionals on V defined by

$$\phi_1[f(x)] = \int_0^1 f(x) dx \text{ and } \phi_2[f(x)] = \int_0^2 f(x) dx.$$

Find the basis $\{f_1, f_2\}$ of V which is dual to $\{\phi_1, \phi_2\}$.

8. Let V be the vector space of polynomials over $\mathbf{R}$ of degree ≤ 2 . Let ϕ_1, ϕ_2 and ϕ_3 be the linear functionals on V defined by

$$\phi_1[f(x)] = \int_0^1 f(x) dx, \quad \phi_2[f(x)] = f'(1), \quad \phi_3[f(x)] = f(0).$$

Here $f(x) = a + bx + cx^2 \in V$ and $f'(x)$ denotes the derivative of $f(x)$. Find the basis $\{f_1(x), f_2(x), f_3(x)\}$ of V which is dual to $\{\phi_1, \phi_2, \phi_3\}$.

9. Prove that every finite dimensional vector space V is isomorphic to its second conjugate space V^{**} under an isomorphism which is independent of the choice of a basis in V .
10. Define a non-zero linear functional f on $\mathbf{C}^3$ such that $f(\alpha) = 0 = f(\beta)$ where $\alpha = (1, 1, 1)$ and $\beta = (1, 1, -1)$.

Answers 1

1. (i) $4x + y$ (ii) $4x + 8y$ (iii) $-13x + 9y$
2. (i) $6x - 5y + 4z$ (ii) $6x - 9y + 3z$ (iii) $-16x + 4y - 13z$
3. $B' = \{f_1, f_2\}$ where $f_1(a, b) = 3b - a$, $f_2(a, b) = a - 2b$
4. $B' = \{f_1, f_2, f_3\}$ where $f_1(a, b, c) = a$, $f_2(a, b, c) = b$, $f_3(a, b, c) = c$
5. $B' = \{f_1, f_2, f_3\}$ where $f_1(a, b, c) = -3a - 5b - 2c$, $f_2(a, b, c) = 2a + b$,
 $f_3(a, b, c) = a + 2b + c$
6. $B' = \{f_1, f_2, f_3\}$, where $f_1(a, b, c) = \frac{1}{2}(b + c)$, $f_2(a, b, c) = \frac{1}{2}(a + c)$,
 $f_3(a, b, c) = \frac{1}{2}(a + b)$
7. $f_1(x) = 2 - 2x$, $f_2(x) = -\frac{1}{2} + x$
8. $f_1(x) = 3x - \frac{3}{2}x^2$, $f_2(x) = -\frac{1}{2}x + \frac{3}{4}x^2$, $f_3(x) = 1 - 3x + \frac{3}{2}x^2$
10. $f(a, b, c) = x(a - b)$, where x is any non-zero scalar.

5.6 Annihilators

Definition: If V is a vector space over the field F and S is a subset of V , the annihilator of S is the set S^0 of all linear functionals f on V such that

$$f(\alpha) = 0 \quad \forall \alpha \in S.$$

Sometimes $A(S)$ is also used to denote the annihilator of S .

Thus $S^0 = \{f \in V' : f(\alpha) = 0 \quad \forall \alpha \in S\}$.

It should be noted that we have defined the annihilator of S which is simply a subset of V . S should not necessarily be a subspace of V .

If $S = \text{zero subspace of } V$, then $S^0 = V'$.

If $S = V$, then $S^0 = V^0 = \text{zero subspace of } V'$.

If V is finite dimensional and S contains a non-zero vector, then $S^0 \neq V'$. If $0 \neq \alpha \in S$, then there is a linear functional f on V such that $f(\alpha) \neq 0$. Thus there is $f \in V'$ such that $f \notin S^0$. Therefore $S^0 \neq V'$.

Theorem 1: If S is any subset of a vector space $V(F)$, then S^0 is a subspace of V' .

Proof: First we see that S^0 is a non-empty subset of V' because at least $\hat{0} \in S^0$.

We have $\hat{0}(\alpha) = 0 \forall \alpha \in S$.

Let $f, g \in S^0$. Then $f(\alpha) = 0 \forall \alpha \in S$, and $g(\alpha) = 0 \forall \alpha \in S$.

If $a, b \in F$, then

$$(af + bg)(\alpha) = (af)(\alpha) + (bg)(\alpha) = af(\alpha) + bg(\alpha) = a0 + b0 = 0.$$

$$\therefore af + bg \in S^0.$$

Thus $a, b \in F$ and $f, g \in S^0 \Rightarrow af + bg \in S^0$.

$\therefore S^0$ is a subspace of V' .

Dimension of annihilator:

Theorem 2: Let V be a finite dimensional vector space over the field F , and let W be a subspace of V . Then

$$\dim W + \dim W^0 = \dim V.$$

Proof: If W is zero subspace of V , then $W^0 = V'$.

$$\therefore \dim W^0 = \dim V' = \dim V.$$

Also in this case $\dim W = 0$. Hence the result.

Similarly the result is obvious when $W = V$.

Let us now suppose that W is a proper subspace of V .

Let $\dim V = n$, and $\dim W = m$ where $0 < m < n$.

Let $B_1 = \{\alpha_1, \dots, \alpha_m\}$ be a basis for W . Since B_1 is a linearly independent subset of V also, therefore it can be extended to form a basis for V . Let $B = \{\alpha_1, \dots, \alpha_m, \alpha_{m+1}, \dots, \alpha_n\}$ be a basis for V .

Let $B' = \{f_1, \dots, f_m, f_{m+1}, \dots, f_n\}$ be the dual basis of B . Then B' is a basis for V' such that $f_i(\alpha_j) = \delta_{ij}$.

We claim that $S = \{f_{m+1}, \dots, f_n\}$ is a basis for W^0 .

Since $S \subset B'$, therefore S is linearly independent because B' is linearly independent. So S will be a basis for W^0 , if W^0 is equal to the subspace of V' spanned by S i.e., if $W^0 = L(S)$.

First we shall show that $W^0 \subseteq L(S)$. Let $f \in W^0$. Then $f \in V'$. So let

$$f = \sum_{i=1}^n x_i f_i. \quad \dots(1)$$

Now $f \in W^0 \Rightarrow f(\alpha) = 0 \forall \alpha \in W$

$$\Rightarrow f(\alpha_j) = 0 \text{ for each } j = 1, \dots, m \quad [\because \alpha_1, \dots, \alpha_m \text{ are in } W]$$

$$\Rightarrow \left(\sum_{i=1}^n x_i f_i \right)(\alpha_j) = 0 \quad [\text{From (1)}]$$

$$\Rightarrow \sum_{i=1}^n x_i f_i(\alpha_j) = 0 \Rightarrow \sum_{i=1}^n x_i \delta_{ij} = 0$$

$\Rightarrow x_j = 0$ for each $j = 1, \dots, m$.

Putting $x_1 = 0, x_2 = 0, \dots, x_m = 0$ in (1), we get

$$f = x_{m+1} f_{m+1} + \dots + x_n f_n$$

= a linear combination of the elements of S .

$\therefore f \in L(S)$.

Thus $f \in W^0 \Rightarrow f \in L(S)$.

$\therefore W^0 \subseteq L(S)$.

Now we shall show that $L(S) \subseteq W^0$.

Let $g \in L(S)$. Then g is a linear combination of

$$f_{m+1}, \dots, f_n.$$

Let $g = \sum_{k=m+1}^n y_k f_k$ (2)

Let $\alpha \in W$. Then α is a linear combination of $\alpha_1, \dots, \alpha_m$. Let

$$\alpha = \sum_{j=1}^m c_j \alpha_j. \quad \dots (3)$$

We have $g(\alpha) = g\left(\sum_{j=1}^m c_j \alpha_j\right)$ [From (3)]

$$= \sum_{j=1}^m c_j g(\alpha_j) \quad [\because g \text{ is linear functional}]$$

$$= \sum_{j=1}^m c_j \left(\sum_{k=m+1}^n y_k f_k \right) (\alpha_j) \quad [\text{From (2)}]$$

$$= \sum_{j=1}^m c_j \sum_{k=m+1}^n y_k f_k(\alpha_j) = \sum_{j=1}^m c_j \sum_{k=m+1}^n y_k \delta_{kj}$$

$$= \sum_{j=1}^m c_j \cdot 0 \quad [\because \delta_{kj} = 0 \text{ if } k \neq j \text{ which is so for each}$$

$$k = m+1, \dots, n \text{ and for each } j = 1, \dots, m]$$

$$= 0.$$

Thus $g(\alpha) = 0 \forall \alpha \in W$. Therefore $g \in W^0$.

Thus $g \in L(S) \Rightarrow g \in W^0$.

$\therefore L(S) \subseteq W^0$.

Hence $W^0 = L(S)$ and S is a basis for W^0 .

$\therefore \dim W^0 = n - m = \dim V - \dim W$

or $\dim V = \dim W + \dim W^0$.

Corollary: If V is finite-dimensional and W is a subspace of V , then W' is isomorphic to V'/W^0 .

Proof: Let $\dim V = n$ and $\dim W = m$. W' is dual space of W ,

so $\dim W' = \dim W = m$.

Now $\dim V'/W^0 = \dim V' - \dim W^0$

$$= \dim V - (\dim V - \dim W) = \dim W = m.$$

Since $\dim W' = \dim V'/W^0$, therefore $W' \cong V'/W^0$.

Annihilator of an annihilator: Let V be a vector space over the field F . If S is any subset of V , then S^0 is a subspace of V' . By definition of an annihilator, we have

$$(S^0)^0 = S^{00} = \{L \in V'' : L(f) = 0 \quad \forall f \in S^0\}.$$

Obviously S^{00} is a subspace of V'' . But if V is finite dimensional, then we have identified V'' with V through the natural isomorphism $\alpha \leftrightarrow L_\alpha$. Therefore we may regard S^{00} as a subspace of V . Thus

$$S^{00} = \{\alpha \in V : f(\alpha) = 0 \quad \forall f \in S^0\}.$$

Theorem 3: Let V be a finite dimensional vector space over the field F and let W be a subspace of V . Then $W^{00} = W$.

Proof: We have

$$W^0 = \{f \in V' : f(\alpha) = 0 \quad \forall \alpha \in W\} \quad \dots(1)$$

$$\text{and} \quad W^{00} = \{\alpha \in V : f(\alpha) = 0 \quad \forall f \in W^0\}. \quad \dots(2)$$

Let $\alpha \in W$. Then from (1), $f(\alpha) = 0 \quad \forall f \in W^0$ and so from (2), $\alpha \in W^{00}$.

$$\therefore \quad \alpha \in W \Rightarrow \alpha \in W^{00}.$$

Thus $W \subseteq W^{00}$. Now W is a subspace of V and W^{00} is also a subspace of V . Since $W \subseteq W^{00}$, therefore W is a subspace of W^{00} .

$$\text{Now } \dim W + \dim W^0 = \dim V. \quad [\text{by theorem (2)}]$$

Applying the same theorem for vector space V' and its subspace W^0 , we get

$$\dim W^0 + \dim W^{00} = \dim V' = \dim V.$$

$$\therefore \quad \dim W = \dim V - \dim W^0 = \dim V - [\dim V - \dim W^{00}] \\ = \dim W^{00}.$$

Since W is a subspace of W^{00} and $\dim W = \dim W^{00}$, therefore $W = W^{00}$.

Illustrative Examples

Example 7: If S_1 and S_2 are two subsets of a vector space V such that $S_1 \subseteq S_2$, then show that $S_2^0 \subseteq S_1^0$.

Solution: Let $f \in S_2^0$. Then $f(\alpha) = 0 \quad \forall \alpha \in S_2$

$$\begin{aligned} \Rightarrow f(\alpha) &= 0 \quad \forall \alpha \in S_1 & [\because S_1 \subseteq S_2] \\ \Rightarrow f &\in S_1^0. \\ \therefore S_2^0 &\subseteq S_1^0. \end{aligned}$$

Example 8: Let V be a vector space over the field F . If S is any subset of V , then show that $S^0 = [L(S)]^0$.

Solution: We know that $S \subseteq L(S)$.

$$\therefore [L(S)]^0 \subseteq S^0. \quad \dots(1)$$

Now let $f \in S^0$. Then $f(\alpha) = 0 \quad \forall \alpha \in S$.

If β is any element of $L(S)$, then

$$\beta = \sum_{i=1}^n x_i \alpha_i \text{ where each } \alpha_i \in S.$$

We have $f(\beta) = \sum_{i=1}^n x_i f(\alpha_i) = 0$, since each $f(\alpha_i) = 0$.

Thus $f(\beta) = 0 \quad \forall \beta \in L(S)$.

$$\therefore f \in (L(S))^0.$$

$$\text{Therefore } S^0 \subseteq (L(S))^0 \quad \dots(2)$$

From (1) and (2), we conclude that $S^0 = (L(S))^0$.

Example 9: Let V be a finite-dimensional vector space over the field F . If S is any subset of V , then $S^{00} = L(S)$.

Solution: We have $S^0 = (L(S))^0$. [See Example 8]

$$\therefore S^{00} = (L(S))^{00}. \quad \dots(1)$$

But V is finite-dimensional and $L(S)$ is a subspace of V . Therefore by theorem 3,

$$(L(S))^{00} = L(S).$$

$\therefore$ from (1), we have

$$S^{00} = L(S).$$

Example 10: Let W_1 and W_2 be subspaces of a finite dimensional vector space V .

(a) Prove that $(W_1 + W_2)^0 = W_1^0 \cap W_2^0$.

(b) Prove that $(W_1 \cap W_2)^0 = W_1^0 + W_2^0$.

Solution: (a) First we shall prove that

$$W_1^0 \cap W_2^0 \subseteq (W_1 + W_2)^0.$$

Let $f \in W_1^0 \cap W_2^0$. Then $f \in W_1^0$, $f \in W_2^0$.

Suppose α is any vector in $W_1 + W_2$. Then

$$\alpha = \alpha_1 + \alpha_2 \text{ where } \alpha_1 \in W_1, \alpha_2 \in W_2.$$

$$\begin{aligned} \text{We have } f(\alpha) &= f(\alpha_1 + \alpha_2) = f(\alpha_1) + f(\alpha_2) \\ &= 0 + 0 \quad [\because \alpha_1 \in W_1 \text{ and } f \in W_1^0 \Rightarrow f(\alpha_1) = 0 \\ &\quad \text{and similarly } f(\alpha_2) = 0] \\ &= 0. \end{aligned}$$

Thus $f(\alpha) = 0 \quad \forall \alpha \in W_1 + W_2$.

$$\therefore f \in (W_1 + W_2)^0.$$

$$\therefore W_1^0 \cap W_2^0 \subseteq (W_1 + W_2)^0. \quad \dots(1)$$

Now we shall prove that

$$(W_1 + W_2)^0 \subseteq W_1^0 \cap W_2^0.$$

We have $W_1 \subseteq W_1 + W_2$.

$$\therefore (W_1 + W_2)^0 \subseteq W_1^0. \quad \dots(2)$$

Similarly, $W_2 \subseteq W_1 + W_2$.

$$\therefore (W_1 + W_2)^0 \subseteq W_2^0. \quad \dots(3)$$

From (2) and (3), we have

$$(W_1 + W_2)^0 \subseteq W_1^0 \cap W_2^0. \quad \dots(4)$$

From (1) and (4), we have

$$(W_1 + W_2)^0 = W_1^0 \cap W_2^0.$$

(b) Let us use the result (a) for the vector space V' in place of the vector space V . Thus replacing W_1 by W_1^0 and W_2 by W_2^0 in (a), we get

$$\begin{aligned} (W_1^0 + W_2^0)^0 &= W_1^{00} \cap W_2^{00} \\ \Rightarrow (W_1^0 + W_2^0)^0 &= W_1 \cap W_2 \quad [\because W_1^{00} = W_1 \text{ etc.}] \\ \Rightarrow (W_1^0 + W_2^0)^{00} &= (W_1 \cap W_2)^0 \\ \Rightarrow W_1^0 + W_2^0 &= (W_1 \cap W_2)^0. \end{aligned}$$

Example 11: If W_1 and W_2 are subspaces of a vector space V , and if $V = W_1 \oplus W_2$, then $V' = W_1^0 \oplus W_2^0$.

Solution: To prove that $V' = W_1^0 \oplus W_2^0$, we are to prove that

$$(i) \quad W_1^0 \cap W_2^0 = \{\hat{0}\}$$

and (ii) $V' = W_1^0 + W_2^0$ i.e., each $f \in V'$ can be written as $f_1 + f_2$

where $f_1 \in W_1^0, f_2 \in W_2^0$.

(i) First to prove that $W_1^0 \cap W_2^0 = \{\hat{0}\}$.

Let $f \in W_1^0 \cap W_2^0$. Then $f \in W_1^0$ and $f \in W_2^0$.

If α is any vector in V , then, V being the direct sum of W_1 and W_2 , we can write

$$\alpha = \alpha_1 + \alpha_2 \text{ where } \alpha_1 \in W_1, \alpha_2 \in W_2.$$

We have $f(\alpha) = f(\alpha_1 + \alpha_2)$

$$= f(\alpha_1) + f(\alpha_2) \quad [\because f \text{ is linear functional}]$$

$$= 0 + 0 \quad [\because f \in W_1^0 \text{ and } \alpha_1 \in W_1 \Rightarrow f(\alpha_1) = 0 \text{ and similarly } f(\alpha_2) = 0]$$

$$= 0.$$

Thus $f(\alpha) = 0 \quad \forall \alpha \in V$.

$$\therefore f = \hat{0}.$$

$$\therefore W_1^0 \cap W_2^0 = \{\hat{0}\}.$$

(ii) Now to prove that $V' = W_1^0 + W_2^0$.

Let $f \in V'$.

If $\alpha \in V$, then α can be uniquely written as

$$\alpha = \alpha_1 + \alpha_2 \text{ where } \alpha_1 \in W_1, \alpha_2 \in W_2.$$

For each f , let us define two functions f_1 and f_2 from V into F such that

$$f_1(\alpha) = f_1(\alpha_1 + \alpha_2) = f(\alpha_2) \quad \dots(1)$$

$$\text{and} \quad f_2(\alpha) = f_2(\alpha_1 + \alpha_2) = f(\alpha_1). \quad \dots(2)$$

First we shall show that f_1 is a linear functional on V . Let $a, b \in F$ and $\alpha = \alpha_1 + \alpha_2, \beta = \beta_1 + \beta_2 \in V$ where $\alpha_1, \beta_1 \in W_1$ and $\alpha_2, \beta_2 \in W_2$. Then

$$\begin{aligned} f_1(a\alpha + b\beta) &= f_1[a(\alpha_1 + \alpha_2) + b(\beta_1 + \beta_2)] \\ &= f_1[(a\alpha_1 + b\beta_1) + (a\alpha_2 + b\beta_2)] \\ &= f(a\alpha_2 + b\beta_2) \\ &= f(a\alpha_2 + b\beta_2) \quad [\because a\alpha_1 + b\beta_1 \in W_1, a\alpha_2 + b\beta_2 \in W_2] \\ &= af(\alpha_2) + bf(\beta_2) \quad [\because f \text{ is linear functional}] \\ &= af_1(\alpha) + bf_1(\beta) \quad [\text{From (1)}] \end{aligned}$$

$\therefore f_1$ is linear functional on V i.e., $f_1 \in V'$.

Now we shall show that $f_1 \in W_1^0$.

Let α_1 be any vector in W_1 . Then α_1 is also in V . We can write

$$\alpha_1 = \alpha_1 + \mathbf{0}, \text{ where } \alpha_1 \in W_1, \mathbf{0} \in W_2.$$

$\therefore$ from (1), we have

$$f_1(\alpha_1) = f_1(\alpha_1 + \mathbf{0}) = f(\mathbf{0}) = 0.$$

Thus $f_1(\alpha_1) = 0 \quad \forall \alpha_1 \in W_1$.

$$\therefore f_1 \in W_1^0.$$

Similarly we can show that f_2 is a linear functional on V and $f_2 \in W_2^0$.

Now we claim that $f = f_1 + f_2$.

Let α be any element in V . Let

$$\alpha = \alpha_1 + \alpha_2, \text{ where } \alpha_1 \in W_1, \alpha_2 \in W_2.$$

Then $(f_1 + f_2)(\alpha) = f_1(\alpha) + f_2(\alpha) = f(\alpha_2) + f(\alpha_1)$ [From (1) and (2)]

$$= f(\alpha_1) + f(\alpha_2) = f(\alpha_1 + \alpha_2)$$

[$\because f$ is linear functional]

$$= f(\alpha).$$

Thus $(f_1 + f_2)(\alpha) = f(\alpha) \forall \alpha \in V$.

$$\therefore f = f_1 + f_2.$$

Thus $f \in V' \Rightarrow f = f_1 + f_2$ where $f_1 \in W_1^0, f_2 \in W_2^0$.

$$\therefore V' = W_1^0 + W_2^0.$$

$$\text{Hence } V' = W_1^0 \oplus W_2^0$$

5.7 Invariant Direct-Sum Decompositions

Let T be a linear operator on a vector space V (F). If S is a non-empty subset of V , then by $T(S)$ we mean the set of those elements of V which are images under T of the elements in S . Thus

$$T(S) = \{T(\alpha) \in V : \alpha \in S\}.$$

Obviously $T(S) \subseteq V$. We call it the **image** of S under T .

Let V be a vector space and T a linear operator on V . If W is a subspace of V , we say that W is invariant under T if

$$\alpha \in W \Rightarrow T(\alpha) \in W.$$

If T is any linear operator on V , then V is invariant under T . If $\alpha \in V$, then $T(\alpha) \in V$ because T is a linear operator on V . Thus V is invariant under T .

The zero subspace of V is also invariant under T . The zero subspace contains only one vector i.e., $\mathbf{0}$ and we know that $T(\mathbf{0}) = \mathbf{0}$ which is in zero subspace.

Let $V(F)$ be the vector space of all polynomials in x over the field F and let D be the differentiation operator on V . Let W be the subspace of V consisting of all polynomials of degree not greater than n .

If $f(x) \in W$, then $D[f(x)] \in W$ because differentiation operator D is degree decreasing. Therefore W is invariant under D .

Let W be a subspace of the vector space V and let W be invariant under the linear operator T on V i.e., let

$$\alpha \in W \Rightarrow T(\alpha) \in W.$$

We know that W itself is a vector space. If we ignore the fact that T is defined outside W , then we may regard T as a linear operator on W . Thus the linear operator T induces a linear operator T_W on the vector space W defined by

$$T_W(\alpha) = T(\alpha) \quad \forall \alpha \in W.$$

It should be noted that T_W is quite a different object from T because the domain of T_W is W while the domain of T is V .

Invariance can be considered for several linear transformations also. Thus W is invariant under a set of linear transformations if it is invariant under each member of the set.

Matrix interpretation of invariance: Let V be a finite dimensional vector space over the field F and let T be a linear operator on V . Suppose V has a subspace W which is invariant under T . Then we can choose suitable ordered basis B for V so that the matrix of T with respect to B takes some particular simple form.

Let $B_1 = \{\alpha_1, \dots, \alpha_m\}$ be an ordered basis for W where $\dim W = m$. We can extend B_1 to form a basis for V . Let

$$B = \{\alpha_1, \dots, \alpha_m, \alpha_{m+1}, \dots, \alpha_n\}$$

be an ordered basis for V where $\dim V = n$.

Let $A = [a_{ij}]_{n \times n}$ be the matrix of T with respect to the ordered basis B . Then

$$T(\alpha_j) = \sum_{i=1}^n a_{ij} \alpha_i, \quad j = 1, 2, \dots, n. \quad \dots(1)$$

If $1 \leq j \leq m$, then α_j is in W . But W is invariant under T . Therefore if $1 \leq j \leq m$, then $T(\alpha_j)$ is in W and so it can be expressed as a linear combination of the vectors $\alpha_1, \dots, \alpha_m$, which form a basis for W . This means that

$$T(\alpha_j) = \sum_{i=1}^m a_{ij} \alpha_i, \quad 1 \leq j \leq m. \quad \dots(2)$$

In other words in the relation (1), the scalars a_{ij} are all zero if $1 \leq j \leq m$ and $m+1 \leq i \leq n$.

Therefore the matrix A takes the simple form

$$A = \begin{bmatrix} M & C \\ O & D \end{bmatrix}$$

where M is an $m \times m$ matrix, C is an $m \times (n-m)$ matrix, O is the null matrix of the type $(n-m) \times m$ and D is an $(n-m) \times (n-m)$ matrix.

From the relation (2) it is obvious that the matrix M is nothing but the matrix of the induced operator T_W on W relative to the ordered basis B_1 for W .

Reducibility: Definition: Let W_1 and W_2 be two subspaces of a vector space V and let T be a linear operator on V . Then T is said to be reduced by the pair (W_1, W_2) if

- (i) $V = W_1 \oplus W_2$,
- (ii) both W_1 and W_2 are invariant under T .

It should be noted that if a subspace W_1 of V is invariant under T , then there are many ways of finding a subspace W_2 of V such that $V = W_1 \oplus W_2$, but it is not necessary that some W_2 will also be invariant under T . In other words among the collection of all subspaces invariant under T we may not be able to select any two other than V and the zero subspace with the property that V is their direct sum.

The definition of reducibility can be extended to more than two subspaces. Thus let $W_1, \dots, W_k$ be k subspaces of a vector space V and let T be a linear operator on V . Then T is said to be reduced by $(W_1, \dots, W_k)$ if

(i) V is the direct sum of the subspaces $W_1, \dots, W_k$,

and (ii) Each of the subspaces W_i is invariant under T .

Direct sum of linear operators: Definition:

Suppose T is a linear operator on the vector space V . Let

$$V = W_1 \oplus \dots \oplus W_k$$

be a direct sum decomposition of V in which each subspace W_i is invariant under T . Then T induces a linear operator T_i on each W_i by restricting its domain from V to W_i . If $\alpha \in V$, then there exist unique vectors $\alpha_1, \dots, \alpha_k$ with α_i in W_i such that

$$\alpha = \alpha_1 + \dots + \alpha_k$$

$$\Rightarrow T(\alpha) = T(\alpha_1 + \dots + \alpha_k)$$

$$\Rightarrow T(\alpha) = T(\alpha_1) + \dots + T(\alpha_k) \quad [\because T \text{ is linear}]$$

$$\Rightarrow T(\alpha) = T_1(\alpha_1) + \dots + T_k(\alpha_k) \quad [\because \text{if } \alpha_i \in W_i, \text{ then by def. of } T_i, \text{ we have } T(\alpha_i) = T_i(\alpha_i)]$$

Thus we can find the action of T on V with the help of independent action of the operators T_i on the subspaces W_i . In such situation we say that the operator T is the **direct sum of the operators $T_1, \dots, T_k$** . It should be noted carefully that T is a linear operator on V , while the T_i are linear operators on the various subspaces W_i .

Matrix representation of reducibility: If T is a linear operator on a finite dimensional vector space V and is reduced by the pair (W_1, W_2) , then by choosing a suitable basis B for V , we can give a particularly simple form to the matrix of T with respect to B .

Let $\dim V = n$ and $\dim W_1 = m$. Then $\dim W_2 = n - m$ since V is the direct sum of W_1 and W_2 .

Let $B_1 = \{\alpha_1, \dots, \alpha_m\}$ be a basis for W_1 and $B_2 = \{\alpha_{m+1}, \dots, \alpha_n\}$ be a basis for W_2 . Then $B = B_1 \cup B_2 = \{\alpha_1, \dots, \alpha_m, \alpha_{m+1}, \dots, \alpha_n\}$ is a basis for V .

It can be easily seen, as in the case of invariance, that

$$[T]_B = \begin{bmatrix} M & O \\ O & N \end{bmatrix}$$

where M is an $m \times m$ matrix, N is an $(n - m) \times (n - m)$ matrix and O are null matrices of suitable sizes.

Also if T_1 and T_2 are linear operators induced by T on W_1 and W_2 respectively, then

$$M = [T_1]_{B_1}, \text{ and } N = [T_2]_{B_2}.$$

Illustrative Examples

Example 12: If T is a linear operator on a vector space V and if W is any subspace of V , then $T(W)$ is a subspace of V . Also W is invariant under T iff $T(W) \subseteq W$.

Solution: We have, by definition

$$T(W) = \{T(\alpha) : \alpha \in W\}.$$

Since $\mathbf{0} \in W$ and $T(\mathbf{0}) = \mathbf{0}$, therefore $T(W)$ is not empty because at least $\mathbf{0} \in T(W)$.

Now let $T(\alpha_1), T(\alpha_2)$ be any two elements of $T(W)$ where α_1, α_2 are any two elements of W .

If $a, b \in F$, then

$$aT(\alpha_1) + bT(\alpha_2) = T(a\alpha_1 + b\alpha_2), \text{ because } T \text{ is linear.}$$

But W is a subspace of V .

Therefore $\alpha_1, \alpha_2 \in W$ and $a, b \in F \Rightarrow a\alpha_1 + b\alpha_2 \in W$. Consequently

$$T(a\alpha_1 + b\alpha_2) \in T(W).$$

Thus $a, b \in F$ and $T(\alpha_1), T(\alpha_2) \in T(W)$

$$\Rightarrow aT(\alpha_1) + bT(\alpha_2) \in T(W).$$

$\therefore T(W)$ is a subspace of V .

Second Part: Suppose W is invariant under T .

Let $T(\alpha)$ be any element of $T(W)$ where $\alpha \in W$.

Since $\alpha \in W$ and W is invariant under T , therefore

$$T(\alpha) \in W. \text{ Thus } T(\alpha) \in T(W) \Rightarrow T(\alpha) \in W.$$

Therefore $T(W) \subseteq W$.

Conversely suppose that $T(W) \subseteq W$.

Then $T(\alpha) \in W \forall \alpha \in W$. Therefore W is invariant under T .

Example 13: If T is any linear operator on a vector space V , then the range of T and the null space of T are both invariant under T .

Solution: Let $N(T)$ be the null space of T . Then

$$N(T) = \{\alpha \in V : T(\alpha) = \mathbf{0}\}.$$

If $\beta \in N(T)$, then $T(\beta) = \mathbf{0} \in N(T)$ because $N(T)$ is a subspace.

$\therefore N(T)$ is invariant under T .

Again let $R(T)$ be the range of T . Then

$$R(T) = \{T(\alpha) \in V : \alpha \in V\}.$$

Since $R(T)$ is a subset of V , therefore $\beta \in R(T) \Rightarrow \beta \in V$.

Now $\beta \in V \Rightarrow T(\beta) \in R(T)$.

Thus $\beta \in R(T) \Rightarrow T(\beta) \in R(T)$. Therefore $R(T)$ is invariant under T .

Example 14: Give an example of a linear transformation T on a finite-dimensional vector space V such that V and the zero subspace are the only subspaces invariant under T .

Solution: Let T be the linear operator on $V_2(\mathbf{R})$ which is represented in the standard ordered basis by the matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Let W be a proper subspace of $V_2(\mathbf{R})$ which is invariant under T . Then W must be of dimension 1. Let W be the subspace spanned by some non-zero vector α . Now $\alpha \in W$ and W is invariant under T . Therefore $T(\alpha) \in W$.

$\therefore T(\alpha) = c\alpha$ for some $c \in \mathbf{R}$

$\Rightarrow T(\alpha) = cI(\alpha)$ where I is identity operator on V

$\Rightarrow [T - cI](\alpha) = \mathbf{0}$

$\Rightarrow T - cI$ is singular

$[\because \alpha \neq \mathbf{0}]$

$\Rightarrow T - cI$ is not invertible.

If B denotes the standard ordered basis for $V_2(\mathbf{R})$, then

$$\begin{aligned} [T - cI]_B &= [T]_B - c[I]_B \\ &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} - c \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -c & -1 \\ 1 & -c \end{bmatrix}. \end{aligned}$$

Now $\det \begin{bmatrix} -c & -1 \\ 1 & -c \end{bmatrix} = \begin{vmatrix} -c & -1 \\ 1 & -c \end{vmatrix} = c^2 + 1 \neq 0$ for any real number c .

$\therefore \begin{bmatrix} -c & -1 \\ 1 & -c \end{bmatrix}$ i.e., $[T - cI]_B$ is invertible.

Consequently $T - cI$ is invertible which is contradictory to the result that $T - cI$ is not invertible.

Hence no proper subspace W of $V_2(\mathbf{R})$ can be invariant under T .

5.8 The Adjoint or the Transpose of a Linear Transformation

In order to bring some simplicity in our work we shall introduce a few changes in our notation of writing the image of an element of a vector space under a linear transformation and that under a linear functional. If T is a linear transformation on a vector space V and $\alpha \in V$, then in place of writing $T(\alpha)$ we shall simply write $T\alpha$ i.e., we shall omit the brackets. Thus $T\alpha$ will mean the image of α under T . If T_1 and T_2 are two linear transformations of V , then in our new notation $T_1 T_2 \alpha$ will stand for $T_1 [T_2(\alpha)]$.

Let f be a linear functional on V . If $\alpha \in V$, then in place of writing $f(\alpha)$ we shall write $[\alpha, f]$. This is the square brackets notation to write the image of a vector

under a linear functional. Thus $[\alpha, f]$ will stand for $f(\alpha)$. If $a, b \in F$ and $\alpha, \beta \in V$, then in this new notation the linearity property of f

$$i.e., \quad f(a\alpha + b\beta) = af(\alpha) + bf(\beta)$$

will be written as

$$[a\alpha + b\beta, f] = a[\alpha, f] + b[\beta, f].$$

Also if f and g are two linear functionals on V and $a, b \in F$, then the property defining addition and scalar multiplication of linear functionals *i.e.*, the property

$$(af + bg)(\alpha) = af(\alpha) + bg(\alpha)$$

will be written as

$$[\alpha, af + bg] = a[\alpha, f] + b[\alpha, g].$$

Note that in this new notation, we get

$$[\alpha, f] = f(\alpha), [\alpha, g] = g(\alpha).$$

Theorem 1: Let U and V be vector spaces over the field F . For each linear transformation T from U into V , there is a unique linear transformation T' from V' into U' such that

$$[T'(g)](\alpha) = g[T(\alpha)] \quad (in \text{ old notation})$$

$$or \quad [\alpha, T'g] = [T\alpha, g] \quad (in \text{ new notation})$$

for every g in V' and α in U .

The linear transformation T' is called the **adjoint** or the **transpose** or the **dual** of T . In some books it is denoted by T^t or by T^* .

Proof: T is a linear transformation from U to V . U' is the dual space of U and V' is the dual space of V . Suppose $g \in V'$ *i.e.*, g is a linear functional on V . Let us define

$$f(\alpha) = g[T(\alpha)] \quad \forall \alpha \in U \quad \dots(1)$$

Then f is a function from U into F . We see that f is nothing but the product or composite of the two functions T and g where $T: U \rightarrow V$ and $g: V \rightarrow F$. Since both T and g are linear therefore f is also linear. Thus f is a linear functional on U *i.e.*, $f \in U'$. In this way T provides us with a rule T' which associates with each functional g on V a linear functional

$$f = T'(g) \text{ on } U, \text{ defined by (1).}$$

Thus $T': V' \rightarrow U'$ such that

$$T'(g) = f \quad \forall g \in V' \text{ where } f(\alpha) = g[T(\alpha)] \quad \forall \alpha \in U.$$

Putting $f = T'(g)$ in (1), we see that T' is a function from V' into U' such that

$$[T'(g)](\alpha) = g[T(\alpha)]$$

or in square brackets notation

$$[\alpha, T'g] = [T\alpha, g] \quad \dots(2)$$

$$\forall g \in V' \text{ and } \forall \alpha \in U.$$

Now we shall show that T' is a linear transformation from V' into U' . Let $g_1, g_2 \in V'$ and $a, b \in F$.

Then we are to prove that

$$T' (ag_1 + bg_2) = aT' g_1 + bT' g_2 \quad \dots(3)$$

where $T' g_1$ stands for $T' (g_1)$ and $T' g_2$ stands for $T' (g_2)$.

We see that both the sides of (3) are elements of U' i.e., both are linear functionals on U . So if α is any element of U , we have

$$\begin{aligned} [\alpha, T' (ag_1 + bg_2)] &= [T\alpha, ag_1 + bg_2] \quad [\text{From (2) because } ag_1 + bg_2 \in V'] \\ &= [T\alpha, ag_1] + [T\alpha, bg_2] \quad [\text{by def. of addition in } V'] \\ &= a [T\alpha, g_1] + b [Ta, g_2] \\ &\quad [\text{by def. of scalar multiplication in } V'] \\ &= a [\alpha, T' g_1] + b [\alpha, T' g_2] \quad [\text{From (2)}] \\ &= [\alpha, aT' g_1] + [\alpha, bT' g_2] \\ &\quad [\text{by def. of scalar multiplication in } U'] \\ &\quad \text{Note that } T' g_1, T' g_2 \in U' \\ &= [\alpha, aT' g_1 + bT' g_2] \quad [\text{By addition in } U'] \end{aligned}$$

Thus $\forall \alpha \in U$, we have

$$[\alpha, T' (ag_1 + bg_2)] = [\alpha, aT' g_1 + bT' g_2].$$

$$\therefore T' (ag_1 + bg_2) = aT' g_1 + bT' g_2 \quad [\text{by def. of equality of two functions}]$$

Hence T' is a linear transformation from V' into U' .

Now let us show that T' is uniquely determined for a given T . If possible, let T_1 be a linear transformation from V' into U' such that

$$[\alpha, T_1 g] = [T\alpha, g] \quad \forall g \in V' \text{ and } \alpha \in U. \quad \dots(4)$$

Then from (2) and (4), we get

$$[\alpha, T_1 g] = [\alpha, T' g] \quad \forall \alpha \in U, \forall g \in V'$$

$$\Rightarrow T_1 g = T' g \quad \forall g \in V'$$

$$\Rightarrow T_1 = T'.$$

$\therefore T'$ is uniquely determined for each T . Hence the theorem.

Note: If T is a linear transformation on the vector space V , then in the proof of the above theorem we should simply replace U by V .

Theorem 2: If T is a linear transformation from a vector space U into a vector space V , then

(i) the annihilator of the range of T is equal to the null space of T' i.e.,

$$[R(T)]^0 = N(T').$$

If in addition U and V are finite dimensional, then

$$(ii) \quad \rho(T') = \rho(T)$$

and (iii) the range of T' is the annihilator of the null space of T i.e.,

$$R(T') = [N(T)]^0.$$

Proof : (i) If $g \in V'$, then by definition of T' , we have

$$[\alpha, T'g] = [T\alpha, g] \quad \forall \alpha \in U. \quad \dots(1)$$

Let $g \in N(T')$ which is a subspace of V' . Then

$T'g = \hat{0}$ where $\hat{0}$ is zero element of U' i.e., $\hat{0}$ is zero functional on U . Therefore from (1), we get

$$[T\alpha, g] = [\alpha, \hat{0}] \quad \forall \alpha \in U$$

$$\Rightarrow [T\alpha, g] = 0 \quad \forall \alpha \in U \quad [\because \hat{0}(\alpha) = 0 \quad \forall \alpha \in U]$$

$$\Rightarrow g(\beta) = 0 \quad \forall \beta \in R(T) \quad [\because R(T) = \{\beta \in V : \beta = T(\alpha) \text{ for some } \alpha \in U\}]$$

$$\Rightarrow g \in [R(T)]^0.$$

$$\therefore N(T') \subseteq [R(T)]^0.$$

Now let $g \in [R(T)]^0$ which is a subspace of V' . Then

$$g(\beta) = 0 \quad \forall \beta \in R(T)$$

$$\Rightarrow [T\alpha, g] = 0 \quad \forall \alpha \in U \quad [\because \forall \alpha \in U, T\alpha \in R(T)]$$

$$\Rightarrow [\alpha, T'g] = 0 \quad \forall \alpha \in U \quad [\text{From (1)}]$$

$$\Rightarrow T'g = \hat{0} \quad (\text{zero functional on } U) \Rightarrow g \in N(T').$$

$$\therefore [R(T)]^0 \subseteq N(T').$$

$$\text{Hence } [R(T)]^0 = N(T').$$

(ii) Suppose U and V are finite dimensional. Let $\dim U = n$, $\dim V = m$. Let $r = \rho(T)$ = the dimension of $R(T)$.

Now $R(T)$ is a subspace of V . Therefore

$$\dim R(T) + \dim [R(T)]^0 = \dim V. \quad [\text{See Th. 2 of article 5.6}]$$

$$\begin{aligned} \therefore \dim [R(T)]^0 &= \dim V - \dim R(T) \\ &= \dim V - r = m - r. \end{aligned}$$

By part (i) of this theorem $[R(T)]^0 = N(T')$.

$$\therefore \dim N(T') = m - r \Rightarrow \text{nullity of } T' = m - r.$$

But T' is a linear transformation from V' into U' .

$$\therefore \rho(T') + \nu(T') = \dim V'$$

$$\begin{aligned} \text{or } \rho(T') &= \dim V' - \nu(T') \\ &= \dim V - \text{nullity of } T' = m - (m - r) = r. \end{aligned}$$

$$\therefore \rho(T) = \rho(T') = r.$$

(iii) T' is a linear transformation from V' into U' . Therefore $R(T')$ is a subspace of U' . Also $[N(T)]^0$ is a subspace of U' because $N(T)$ is a subspace of U . First we shall show that $R(T') \subseteq [N(T)]^0$.

Let $f \in R(T')$.

Then $f = T'g$ for some $g \in V'$.

If α is any vector in $N(T)$, then $T\alpha = \mathbf{0}$.

We have

$$[\alpha, f] = [\alpha, T'g] = [T\alpha, g] = [\mathbf{0}, g] = 0.$$

Thus $f(\alpha) = 0 \quad \forall \alpha \in N(T)$.

Therefore $f \in [N(T)]^0$.

$\therefore R(T') \subseteq [N(T)]^0$

$\Rightarrow R(T')$ is a subspace of $[N(T)]^0$.

Now $\dim N(T) + \dim [N(T)]^0 = \dim U$. [Theorem 2 of article 5.6]

$\therefore \dim [N(T)]^0 = \dim U - \dim N(T)$
 $= \dim R(T) \quad [\because \dim U = \dim R(T) + \dim N(T)]$
 $= \rho(T) = \rho(T') = \dim R(T').$

Thus $\dim R(T') = \dim [N(T)]^0$

and $R(T') \subseteq [N(T)]^0$.

$\therefore R(T') = [N(T)]^0$.

Note: If T is a linear transformation on a vector space V , then in the proof of the above theorem we should replace U by V and m by n .

Theorem 3: Let U and V be finite-dimensional vector spaces over the field F . Let B be an ordered basis for U with dual basis B' , and let B_1 be an ordered basis for V with dual basis B_1' . Let T be a linear transformation from U into V . Let A be the matrix of T relative to B, B_1 and let C be the matrix of T' relative to B_1', B' . Then $C = A'$ i.e., the matrix C is the transpose of the matrix A .

Proof: Let $\dim U = n, \dim V = m$.

Let $B = \{\alpha_1, \dots, \alpha_n\}, B' = \{f_1, \dots, f_n\},$

$B_1 = \{\beta_1, \dots, \beta_m\}, B_1' = \{g_1, \dots, g_m\}.$

Now T is a linear transformation from U into V and T' is that from V' into U' . The matrix A of T relative to B, B_1 will be of the type $m \times n$. If $A = [a_{ij}]_{m \times n}$, then by definition

$$T(\alpha_j) \text{ or simply } T\alpha_j = \sum_{i=1}^m a_{ij} \beta_i, j = 1, 2, \dots, n. \quad \dots(1)$$

The matrix C of T' relative to B_1', B' will be of the type $n \times m$. If $C = [c_{ji}]_{n \times m}$, then by definition

$$T'(g_i) \text{ or simply } T'g_i = \sum_{j=1}^n c_{ji} f_j, i = 1, 2, \dots, m \quad \dots(2)$$

Now $T'g_i$ is an element of U' i.e., $T'g_i$ is a linear functional on U . If f is any linear functional on U , then we know that

$$f = \sum_{j=1}^n f(\alpha_j) f_j . \quad [\text{See theorem 3 of Chapter 5, article 5.4}]$$

Applying this formula for $T' g_i$ in place of f , we get

$$T' g_i = \sum_{j=1}^n \{(T' g_i)(\alpha_j)\} f_j . \quad \dots(3)$$

Now let us find $(T' g_i)(\alpha_j)$.

We have

$$\begin{aligned} (T' g_i)(\alpha_j) &= g_i(T(\alpha_j)) && [\text{by def. of } T'] \\ &= g_i\left(\sum_{k=1}^m a_{kj} \beta_k\right) && [\text{From (1), replacing the suffix } i \\ &&& \text{by } k \text{ which is immaterial}] \\ &= \sum_{k=1}^m a_{kj} g_i(\beta_k) && [\because g_i \text{ is linear}] \\ &= \sum_{k=1}^m a_{kj} \delta_{ik} && [\because g_i \in B_1' \text{ which is dual basis of } B_1] \\ &= a_{ij} && [\text{On summing with respect to } k \text{ and remembering} \\ &&& \text{that } \delta_{ik} = 1 \text{ when } k = i \text{ and } \delta_{ik} = 0 \text{ when } k \neq i] \end{aligned}$$

Putting this value of $(T' g_i)(\alpha_j)$ in (3), we get

$$T' g_i = \sum_{j=1}^m a_{ij} f_j . \quad \dots(4)$$

Since $f_1, \dots, f_n$ are linearly independent, therefore from (2) and (4), we get

$$c_{ji} = a_{ij} .$$

Hence by definition of transpose of a matrix, we have

$$C = A' .$$

Note: If T is a linear transformation on a finite-dimensional vector space V , then in the above theorem we put $U = V$ and $m = n$. Also according to our convention we take $B_1 = B$. The students should write the complete proof themselves.

Theorem 4: Let A be any $m \times n$ matrix over the field F . Then the row rank of A is equal to the column rank of A .

Proof : Let $A = [a_{ij}]_{m \times n}$. Let

$$B = \{\alpha_1, \dots, \alpha_n\} \text{ and } B_1 = \{\beta_1, \dots, \beta_m\}$$

be the standard ordered bases for $V_n(F)$ and $V_m(F)$ respectively. Let T be the linear transformation from $V_n(F)$ into $V_m(F)$ whose matrix is A relative to ordered bases B and B_1 . Then obviously the vectors $T(\alpha_1), \dots, T(\alpha_n)$ are nothing but the column vectors of the matrix A . Also these vectors span the range of T because $\alpha_1, \dots, \alpha_n$ form a basis for the domain of T i.e., $V_n(F)$.

$\therefore$ the range of T = the column space of A

$\Rightarrow$ the dimension of the range of T = the dimension of the column space of A
 $\Rightarrow \rho(T) = \text{the column rank of } A. \quad \dots(1)$

If T' is the adjoint of the linear transformation T , then the matrix of T' relative to the dual bases B_1' and B' is the matrix A' which is the transpose of the matrix A . The columns of the matrix A' are nothing but the rows of the matrix A . By the same reasoning as given in proving the result (1), we have

$$\begin{aligned} \rho(T') &= \text{the column rank of } A' \\ &= \text{the row rank of } A \end{aligned} \quad \dots(2)$$

Since $\rho(T) = \rho(T')$, therefore from (1) and (2), we get the result that the column rank of A = the row rank of A .

Theorem 5 : Prove the following properties of adjoints of linear operators on a vector space $V(F)$:

- (i) $\hat{0}' = \hat{0}$;
- (ii) $I' = I$;
- (iii) $(T_1 + T_2)' = T_1' + T_2'$;
- (iv) $(T_1 T_2)' = T_2' T_1'$;
- (v) $(aT)' = aT'$ where $a \in F$;
- (vi) $(T^{-1})' = (T')^{-1}$ if T is invertible ;
- (vii) $(T')' = T'' = T$ if V is finite-dimensional.

Proof : (i) If $\hat{0}$ is the zero transformation on V , then by the definition of the adjoint of a linear transformation, we have

$$\begin{aligned} [\alpha, \hat{0}' g] &= [\hat{0} \alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V \\ &= \text{for every } g \text{ in } V' \quad [\because \hat{0}(\alpha) = 0 \quad \forall \alpha \in V] \\ &= 0 \quad [\because g(0) = 0] \\ &= [\alpha, \hat{0}] \quad \forall \alpha \in V \quad [\text{Here } \hat{0} \in V' \text{ and } \hat{0}(\alpha) = 0] \\ &= [\alpha, \hat{0} g] \quad \forall g \in V' \text{ and } \alpha \in V \\ &\quad [\text{Here } \hat{0} \text{ is the zero transformation on } V'] \end{aligned}$$

Thus we have

$$[\alpha, \hat{0}' g] = [\alpha, \hat{0} g] \text{ for all } g \text{ in } V' \text{ and } \alpha \text{ in } V.$$

$$\therefore \hat{0}' = \hat{0}.$$

(ii) If I is the identity transformation on V , then by the definition of the adjoint of a linear transformation, we have

$$\begin{aligned} [\alpha, I' g] &= [I\alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V \\ &= [\alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V \quad [\because I(\alpha) = \alpha \quad \forall \alpha \in V] \end{aligned}$$

$$= [\alpha, Ig] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

[Here I is the identity operator on V']

$$\therefore I' = I.$$

(iii) If T_1, T_2 are linear operators on V , then $T_1 + T_2$ is also a linear operator on V .

By the definition of adjoint, we have

$$\begin{aligned} [\alpha, (T_1 + T_2)' g] &= [(T_1 + T_2) \alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V \\ &= [T_1 \alpha + T_2 \alpha, g] \\ &\quad \text{[by def. of addition of linear transformations]} \\ &= [T_1 \alpha, g] + [T_2 \alpha, g] \quad \text{[by linearity property of } g] \\ &= [\alpha, T_1' g] + [\alpha, T_2' g] \quad \text{[by def. of adjoint]} \\ &= [\alpha, T_1' g + T_2' g] \quad \text{[by def. of addition of linear} \\ &\quad \text{functionals. Note that } T_1' g, T_2' g \text{ are} \\ &\quad \text{elements of } V'] \\ &= [\alpha, (T_1' + T_2') g]. \end{aligned}$$

Thus we have

$$[\alpha, (T_1 + T_2)' g] = [\alpha, (T_1' + T_2') g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V.$$

$$\therefore (T_1 + T_2)' g = (T_1' + T_2') g \quad \forall g \in V'.$$

$$\therefore (T_1 + T_2)' = T_1' + T_2'.$$

(iv) If T_1, T_2 are linear operators on V , then $T_1 T_2$ is also a linear operator on V . By the definition of adjoint, we have

$$\begin{aligned} [\alpha, (T_1 T_2)' g] &= [(T_1 T_2) \alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V \\ &= [(T_1) T_2 \alpha, g] \\ &\quad \text{[by def. of product of linear transformations]} \\ &= [T_2 \alpha, T_1' g] \quad \text{[by def. of adjoint]} \\ &= [\alpha, T_2' T_1' g] \quad \text{[by def. of adjoint]} \end{aligned}$$

Thus we have

$$[\alpha, (T_1 T_2)' g] = [\alpha, T_2' T_1' g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V.$$

$$\therefore (T_1 T_2)' = T_2' T_1'.$$

Note: This is called the reversal law for the adjoint of the product of two linear transformations.

(v) If T is a linear operator on V and $a \in F$, then aT is also a linear operator on V . By the definition of the adjoint, we have

$$\begin{aligned} [\alpha, (aT)' g] &= [(aT) \alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V \\ &= [a (T\alpha), g] \quad \text{[by def. of scalar multiplication of a} \\ &\quad \text{linear transformation]} \\ &= a [T\alpha, g] \quad \text{[}\because g \text{ is linear]} \\ &= a [\alpha, T' g] \quad \text{[by def. of adjoint]} \\ &= [\alpha, a (T' g)] \text{ [by def. of scalar multiplication in } V'. \end{aligned}$$

Note that $T'g \in V'$

$$= [\alpha, (aT')g]$$

[by def. of scalar multiplication of T' by a]

$$\therefore (aT)' = aT'.$$

(vi) Suppose T is an invertible linear operator on V . If T^{-1} is the inverse of T , we have

$$T^{-1}T = I = TT^{-1}$$

$$\Rightarrow (T^{-1}T)' = I' = (TT^{-1})'$$

$$\Rightarrow T'(T^{-1}) = I = (T^{-1})'T' \quad [\text{Using results (ii) and (iv)}]$$

$$\therefore T' \text{ is invertible and } (T')^{-1} = (T^{-1})'.$$

(vii) V is a finite dimensional vector space. T is a linear operator on V , T' is a linear operator on V' and $(T')'$ or T'' is a linear operator on V'' . We have identified V'' with V through natural isomorphism $\alpha \leftrightarrow L_\alpha$ where $\alpha \in V$ and $L_\alpha \in V''$. Here L_α is a linear functional on V' and is such that

$$L_\alpha(g) = g(\alpha) \quad \forall g \in V'. \quad \dots(1)$$

Through this natural isomorphism we shall take $\alpha = L_\alpha$ and thus T'' will be regarded as a linear operator on V .

Now T' is a linear operator on V' . Therefore by the definition of adjoint, we have

$$[g, T''L_\alpha] = [g, (T')'L_\alpha] = [T'g, L_\alpha] \text{ for every } g \in V' \text{ and } \alpha \in V.$$

Now $T'g$ is an element of V' . Therefore from (1), we have

$$[T'g, L_\alpha] = [\alpha, T'g]$$

$$[\text{Note that from (1), } L_\alpha(T'g) = (T'g)\alpha]$$

$$= [T\alpha, g]. \quad [\text{by def. of adjoint}]$$

Again $T''L_\alpha$ is an element of V'' . Therefore from (1), we have

$$[g, T''L_\alpha] = [\beta, g] \text{ where } \beta \in V \text{ and } \beta \leftrightarrow T''L_\alpha$$

under natural isomorphism

$$= [T''\alpha, g] [\because \beta = T''L_\alpha = T''\alpha \text{ when we regard}$$

T'' as linear operator on V in place of V'']

Thus, we have

$$[T\alpha, g] = [T''\alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$\Rightarrow g(T\alpha) = g(T''\alpha) \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$\Rightarrow g(T\alpha - T''\alpha) = 0 \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$\Rightarrow T\alpha - T''\alpha = \mathbf{0} \text{ for every } \alpha \text{ in } V$$

$$\Rightarrow (T - T'')\alpha = \mathbf{0} \text{ for every } \alpha \text{ in } V$$

$$\Rightarrow T - T'' = \hat{\mathbf{0}}$$

$$\Rightarrow T = T''.$$

Illustrative Examples

Example 15 : Let f be the linear functional on $\mathbf{R}^2$ defined by $f(x, y) = 2x - 5y$. For each linear mapping

$$T : \mathbf{R}^3 \rightarrow \mathbf{R}^2 \text{ find } [T' (f)](x, y, z), \text{ where}$$

$$(i) \quad T(x, y, z) = (x - y, y + z),$$

$$(ii) \quad T(x, y, z) = (x + y + 2z, 2x + y),$$

$$(iii) \quad T(x, y, z) = (x + y, 0).$$

Solution : By definition of the transpose mapping,

$$T' (f) = f \circ T \text{ i.e., } [T' (f)] \alpha = f [T (\alpha)] \text{ for every } \alpha \in \mathbf{R}^3.$$

$$(i) \quad [T' (f)](x, y, z) = f [T (x, y, z)] \\ = f (x - y, y + z) = 2(x - y) - 5(y + z) = 2x - 7y - 5z.$$

$$(ii) \quad [T' (f)](x, y, z) = f [T (x, y, z)] \\ = f (x + y + 2z, 2x + y) \\ = 2(x + y + 2z) - 5(2x + y) = -8x - 3y + 4z.$$

$$(iii) \quad [T' (f)](x, y, z) = f [T (x, y, z)] \\ = f (x + y, 0) = 2(x + y) - 5 \cdot 0 = 2x + 2y.$$

Example 16 : If A and B are similar linear transformations on a vector space V , then so also are A' and B' .

Solution : A is similar to B means that there exists an invertible linear transformation C on V such that

$$A = CBC^{-1}$$

$$\Rightarrow A = (CBC^{-1})'$$

$$\Rightarrow A' = (C^{-1})' B' C'.$$

Now C is invertible implies that C' is also invertible and

$$(C')^{-1} = (C^{-1})'.$$

$$\therefore A' = (C')^{-1} B' C' \Rightarrow C' A' (C')^{-1} = B'$$

[Multiplying on right by $(C')^{-1}$ and on left by C']

$$\Rightarrow B' \text{ is similar to } A'$$

$$\Rightarrow A' \text{ and } B' \text{ are similar.}$$

Example 17 : Let V be a finite dimensional vector space over the field F . Show that $T \rightarrow T'$ is an isomorphism of $L(V, V)$ onto $L(V', V')$.

Solution : Let $\dim V = n$.

Then $\dim V' = n$.

Also $\dim L(V, V) = n^2, \dim L(V', V') = n^2$.

Let $\psi : L(V, V) \rightarrow L(V', V')$ such that
 $\psi(T) = T' \quad \forall T \in L(V, V)$.

(i) ψ is linear transformation:

Let $a, b \in F$ and $T_1, T_2 \in L(V, V)$. Then

$$\begin{aligned} \psi(aT_1 + bT_2) &= (aT_1 + bT_2)' && [\text{by def. of } \psi] \\ &= (aT_1)' + (bT_2)' && [\because (A + B)' = A' + B'] \\ &= aT_1' + bT_2' && [\because (aA)' = aA'] \\ &= a\psi(T_1) + b\psi(T_2) && [\text{by def. of } \psi] \end{aligned}$$

$\therefore \psi$ is a linear transformation from $L(V, V)$ into $L(V', V')$.

(ii) ψ is one-one:

Let $T_1, T_2 \in L(V, V)$.
 Then $\psi(T_1) = \psi(T_2)$
 $\Rightarrow T_1' = T_2'$
 $\Rightarrow T_1'' = T_2''$
 $\Rightarrow T_1 = T_2 \quad [\because V \text{ is finite-dimensional}]$
 $\therefore \psi$ is one-one.

(iii) ψ is onto:

We have $\dim L(V, V) = \dim L(V', V') = n^2$.

Since ψ is a linear transformation from $L(V, V)$ into $L(V', V')$ therefore ψ is one-one implies that ψ must be onto.

Hence ψ is an isomorphism of $L(V, V)$ onto $L(V', V')$.

Example 18 : If A and B are linear transformations on an n -dimensional vector space V , then prove that

(i) $\rho(AB) \geq \rho(A) + \rho(B) - n$.

(ii) $v(AB) \leq v(A) + v(B)$. (Sylvester's law of nullity)

Solution : (i) First we shall prove, that if T is a linear transformation on V and W_1 is an h -dimensional subspace of V , then the dimension of $T(W_1)$ is $\geq h - v(T)$.

Since V is finite-dimensional, therefore the subspace W_1 will possess complement.

Let $V = W_1 \oplus W_2$. Then

$$\dim W_2 = n - h = k \quad (\text{say}).$$

Since $V = W_1 + W_2$, therefore

$$T(V) = T(W_1) + T(W_2), \text{ as can be easily seen.}$$

$\therefore \dim T(V) = \dim [T(W_1) + T(W_2)]$
 $\leq \dim T(W_1) + \dim T(W_2)$
 $[\because \text{the dimension of a sum is } \leq \text{the sum of the dimensions}]$

But $T(V) = \text{the range of } T$.

$\therefore \dim T(V) = \rho(T)$.

Thus $\dim T(W_1) + \dim T(W_2) \geq \rho(T)$ (1)

Now $T(W_2)$ is a subspace of W_2 . Therefore

$$\dim W_2 \geq \dim T(W_2). \quad \dots (2)$$

From (1) and (2), we get

$$\begin{aligned} & \dim T(W_1) + \dim W_2 \geq \rho(T) \\ \Rightarrow & \dim T(W_1) \geq \rho(T) - \dim W_2 \\ \Rightarrow & \dim T(W_1) \geq n - v(T) - k \quad [\because \rho(T) + v(T) = n] \\ \Rightarrow & \dim T(W_1) \geq n - k - v(T) \\ \Rightarrow & \dim T(W_1) \geq h - v(T). \quad \dots (3) \end{aligned}$$

Now taking $T = A$ and $W_1 = B(V)$ in (3), we get

$$\begin{aligned} & \dim A[B(V)] \geq \dim B(V) - v(A) \\ \Rightarrow & \dim (AB)(V) \geq \rho(B) - v(A) \quad [\because B(V) = \text{the range of } B] \\ \Rightarrow & \rho(AB) \geq \rho(B) - [n - \rho(A)] \\ \Rightarrow & \rho(AB) \geq \rho(A) + \rho(B) - n. \end{aligned}$$

(ii) We have $\rho(AB) + v(AB) = n$.

$\therefore \rho(AB) = n - v(AB)$.

But $\rho(AB) \geq \rho(A) + \rho(B) - n$.

$\therefore n - v(AB) \geq \rho(A) + \rho(B) - n$

$\Rightarrow v(AB) \leq [n - \rho(A)] + [n - \rho(B)]$

$\Rightarrow v(AB) \leq v(A) + v(B) \quad [\because \rho(A) + v(A) = n]$

Comprehensive Exercise 2

- Let V be a finite dimensional vector space over the field F . If W_1 and W_2 are subspaces of V , then $W_1^0 = W_2^0$ iff $W_1 = W_2$.
- If W_1 and W_2 are subspaces of a finite-dimensional vector space V and if $V = W_1 \oplus W_2$, then
 - W_1' is isomorphic to W_2^0 .
 - W_2' is isomorphic to W_1^0 .
- Let W be the subspace of $\mathbf{R}^3$ spanned by $(1, 1, 0)$ and $(0, 1, 1)$. Find a basis of the annihilator of W .
- Let W be the subspace of $\mathbf{R}^4$ spanned by $(1, 2, -3, 4)$, $(1, 3, -2, 6)$ and $(1, 4, -1, 8)$. Find a basis of the annihilator of W .

5. If the set $S = \{W_i\}$ is the collection of subspaces of a vector space V which are invariant under T , then show that $W = \bigcap_i W_i$ is also invariant under T .
6. Prove that the subspace spanned by two subspaces each of which is invariant under some linear operator T , is itself invariant under T .
7. Let V be a vector space over the field F , and let T be a linear operator on V and let $f(t)$ be a polynomial in the indeterminate t over the field F . If W is the null space of the operator $f(T)$, then W is invariant under T .
8. Let T be the linear operator on $\mathbf{R}^2$, the matrix of which in the standard ordered basis is

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}.$$

- (a) Prove that the only subspaces of $\mathbf{R}^2$ invariant under T are $\mathbf{R}^2$ and the zero subspace.
- (b) If U is the linear operator on $\mathbf{C}^2$, the matrix of which in the standard ordered basis is A , show that U has one dimensional invariant subspaces.
9. Let T be the linear operator on $\mathbf{R}^2$, the matrix of which in the standard ordered basis is

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

If W_1 is the subspace of $\mathbf{R}^2$, spanned by the vector $(1, 0)$, prove that W_1 is invariant under T .

10. Show that the space generated by $(1, 1, 1)$ and $(1, 2, 1)$ is an invariant subspace of $\mathbf{R}^3$ under T , where $T(x, y, z) = (x + y - z, x + y, x + y - z)$.
11. If A and B are linear transformations on a finite-dimensional vector space V , then prove that
 - (i) $\rho(A + B) \leq \rho(A) + \rho(B)$
 - (ii) $\rho(AB) \leq \min\{\rho(A), \rho(B)\}$.
 - (iii) If B is invertible, then $\rho(AB) = \rho(BA) = \rho(A)$.

Answers 2

3. Basis is $\{f(x, y, z) = x - y + z\}$
4. Basis is $\{f_1, f_2\}$ where $f_1(x, y, z, w) = 5x - y + z$, $f_2(x, y, z, w) = 2y - w$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- Let V be an n -dimensional vector space over the field F . The dimension of the dual space of V is
 - n
 - n^2
 - $\frac{1}{2}n$
 - none of these
- If the dual basis of the basis set $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ for $V_3(\mathbf{R})$ is $B' = \{f_1, f_2, f_3\}$, then
 - $f_1(a, b, c) = a, f_2(a, b, c) = b, f_3(a, b, c) = c$
 - $f_1(a, b, c) = b, f_2(a, b, c) = c, f_3(a, b, c) = a$
 - $f_1(a, b, c) = c, f_2(a, b, c) = a, f_3(a, b, c) = b$
 - None of these.
- If $V(F)$ is a vector space and f be a linear functional from $V \rightarrow F$, then
 - $f(\mathbf{0}) = 0$
 - $f(\mathbf{0}) \neq 0$
 - $f(0) = \mathbf{0}$
 - $f(0) \neq \mathbf{0}$
- If $V(F)$ is a vector space and f is a linear functional from $V \rightarrow F$, then
 - $f(-x) = f(x)$
 - $f(-x) = -f(x)$
 - $f(-x) \neq -f(x)$
 - $f(-x) \neq f(x)$
- Let V be a vector space over a field F . A linear functional on V is a linear mapping from
 - F into V
 - V into F
 - V into itself
 - none of these
- If S is any subset of a vector space $V(F)$, then S^0 is a subspace of
 - V
 - V'
 - V''
 - None of these.
- Let V be a finite dimensional vector space over the field F . If S is any subset of V , then $S^{00} =$
 - S
 - $L(S)$
 - $[L(S)]^0$
 - None of these.

Fill in the Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

1. Let V be a vector space over the field F . The dual space of V is also called the space of V .
2. If V is an n -dimensional vector space over the field F , then V is to its dual space V' .
3. Let V be a finite dimensional vector space over the field F , and let W be a subspace of V . Then

$$\dim W + \dim W^0 = \dots\dots\dots$$
4. Let V be a finite dimensional vector space over the field F and let W be a subspace of V . Then $W^{00} = \dots\dots\dots$

True or False

Write ‘T’ for true and ‘F’ for false statement.

1. Let $V(F)$ be a vector space. A linear functional on V is a vector valued function.
2. Every finite dimensional vector space is not reflexive.
3. If $V(F)$ is a vector space, then mapping $f : V \rightarrow F : f(\alpha) = 0$, then f is a linear functional.
4. If V is finite-dimensional and W is a subspace of V then W' is isomorphic to V'/W^0 .
5. If T is a linear transformation from a vector space U into a vector space V then $\rho(T') \neq \rho(T)$.

Answers

Multiple Choice Questions

1. (a)
2. (a)
3. (a)
4. (b)
5. (b)
6. (b)
7. (b)

Fill in the Blank(s)

1. conjugate
2. isomorphic
3. $\dim V$
4. W

True or False

1. F
2. F
3. T
4. T
5. F



Chapter

6

Characteristic Values and Annihilating Polynomials

6.1 Characteristic Values and Characteristic Vectors (or Eigen Values and Eigen Vectors)

Throughout this discussion T will be regarded as a linear operator on a finite dimensional vector space.

Definition: Let T be a linear operator on an n -dimensional vector space V over the field F . Then a scalar $c \in F$ is called a **characteristic value** of T if there is a **non-zero vector** α in V such that $T\alpha = c\alpha$. Also if c is a characteristic value of T , then any non-zero vector α in V such that $T\alpha = c\alpha$ is called a **characteristic vector** of T belonging to the characteristic value c .

Characteristic values are sometimes also called **proper values**, **eigen values**, or **spectral values**. Similarly characteristic vectors are called **proper vectors**, **eigen vectors**, or **spectral vectors**.

The set of all characteristic values of T is called the **spectrum** of T .

Theorem 1: If α is a characteristic vector of T corresponding to the characteristic value c , then $k\alpha$ is also a characteristic vector of T corresponding to the same characteristic value c . Here k is any non-zero scalar.

Proof: Since α is a characteristic vector of T corresponding to the characteristic value c , therefore $\alpha \neq \mathbf{0}$ and

$$T(\alpha) = c\alpha. \quad \dots(1)$$

If k is any non-zero scalar, then $k\alpha \neq \mathbf{0}$.

$$\text{Also } T(k\alpha) = kT(\alpha) = k(c\alpha) = (kc)\alpha = (ck)\alpha = c(k\alpha).$$

$\therefore k\alpha$ is a characteristic vector of T corresponding to the characteristic value c . Thus corresponding to a characteristic value c , there may correspond more than one characteristic vectors.

Theorem 2: If α is a characteristic vector of T , then α cannot correspond to more than one characteristic values of T .

Proof: Let α be a characteristic vector of T corresponding to two distinct characteristic values c_1 and c_2 of T . Then

$$T\alpha = c_1\alpha \quad \text{and} \quad T\alpha = c_2\alpha$$

$$\therefore c_1\alpha = c_2\alpha \Rightarrow (c_1 - c_2)\alpha = \mathbf{0} \Rightarrow c_1 - c_2 = 0 \quad [\because \alpha \neq \mathbf{0}]$$

$$\Rightarrow c_1 = c_2.$$

Theorem 3: Let T be a linear operator on a finite dimensional vector space V and let c be a characteristic value of T . Then the set $W_c = \{\alpha \in V : T\alpha = c\alpha\}$ is a subspace of V .

Proof: Let $\alpha, \beta \in W_c$. Then $T\alpha = c\alpha$, and $T\beta = c\beta$.

If $a, b \in F$, then

$$T(a\alpha + b\beta) = aT\alpha + bT\beta = a(c\alpha) + b(c\beta) = c(a\alpha + b\beta).$$

$$\therefore a\alpha + b\beta \in W_c.$$

Therefore W_c is a subspace of V .

Note: The set W_c is nothing but the set of all characteristic vectors of T corresponding to the characteristic value c provided we include the zero vector in this set. In other words W_c is the null space of the linear operator $T - cI$. The subspace W_c of V is called the **characteristic space of the characteristic value c** of the linear operator T . It is also called the space of characteristic vectors of T associated with the characteristic value c .

Theorem 4: Distinct characteristic vectors of T corresponding to distinct characteristic values of T are linearly independent.

Proof: Let $c_1, c_2, \dots, c_m$ be m distinct characteristic values of T and let $\alpha_1, \alpha_2, \dots, \alpha_m$ be the characteristic vectors of T corresponding to these characteristic values respectively. Then

$$T\alpha_i = c_i\alpha_i \quad \text{where } 1 \leq i \leq m.$$

Let $S = \{\alpha_1, \dots, \alpha_m\}$.

Then to prove that the set S is linearly independent. We shall prove the theorem by induction on m , the number of vectors in S .

If $m = 1$, then S is linearly independent because S contains only one non-zero vector. Note that a characteristic vector cannot be $\mathbf{0}$ by our definition.

Now suppose that the set

$$S_1 = \{\alpha_1, \dots, \alpha_k\}, \text{ where } k < m,$$

is linearly independent.

Consider the set $S_2 = \{\alpha_1, \dots, \alpha_k, \alpha_{k+1}\}$.

We shall show that S_2 is linearly independent.

Let $a_1, \dots, a_{k+1} \in F$ and let

$$a_1 \alpha_1 + \dots + a_{k+1} \alpha_{k+1} = \mathbf{0} \quad \dots(1)$$

$$\Rightarrow T(a_1 \alpha_1 + \dots + a_{k+1} \alpha_{k+1}) = T(\mathbf{0})$$

$$\Rightarrow a_1 T(\alpha_1) + \dots + a_{k+1} T(\alpha_{k+1}) = \mathbf{0}$$

$$\Rightarrow a_1 (c_1 \alpha_1) + \dots + a_{k+1} (c_{k+1} \alpha_{k+1}) = \mathbf{0} \quad \dots(2)$$

Multiplying (1) by the scalar c_{k+1} and subtracting from (2), we get

$$a_1 (c_1 - c_{k+1}) \alpha_1 + \dots + a_k (c_k - c_{k+1}) \alpha_k = \mathbf{0}.$$

$\therefore a_1 = 0, \dots, a_k = 0$ since $\alpha_1, \dots, \alpha_k$ are linearly independent according to our assumption and $c_1, \dots, c_{k+1}$ are all distinct.

Putting each of $a_1, \dots, a_k$ equal to 0 in (1), we get

$$a_{k+1} \alpha_{k+1} = \mathbf{0} \quad \Rightarrow \quad a_{k+1} = 0 \text{ since } \alpha_{k+1} \neq \mathbf{0}.$$

Thus the relation (1) implies that

$$a_1 = 0, \dots, a_k = 0, a_{k+1} = 0.$$

$\therefore$ the set S_2 is linearly independent.

Now the proof is complete by induction.

Corollary: If T is a linear operator on an n -dimensional vector space V , then T cannot have more than n distinct characteristic values.

Proof: Suppose T has more than n distinct characteristic values. Then the corresponding set of distinct characteristic vectors of T will be linearly independent. Thus we shall have a linearly independent subset of V containing more than n vectors which is not possible because V is of dimension n . Hence T cannot have more than n distinct characteristic values.

Theorem 5: Let T be a linear operator on a finite-dimensional vector space V . Then the following are equivalent.

- (i) c is a characteristic value of T .
- (ii) the operator $T - cI$ is singular (not invertible).
- (iii) $\det(T - cI) = 0$.

Proof: (i) $\Rightarrow$ (ii).

c is a characteristic value of T implies that there exists a non-zero vector α in V such that

$$T\alpha = c\alpha$$

or $T\alpha = cI\alpha$ where I is the identity operator on V

or $T\alpha = (cI)\alpha$ or $(T - cI)\alpha = \mathbf{0}$.

Thus $(T - cI)\alpha = \mathbf{0}$ while $\alpha \neq \mathbf{0}$. Therefore the operator $T - cI$ is singular and thus $T - cI$ is not invertible.

(ii) $\Rightarrow$ (iii).

If the operator $T - cI$ is singular, then it is not invertible.

Therefore $\det(T - cI) = 0$.

(iii) $\Rightarrow$ (i).

If $\det(T - cI) = 0$, then $T - cI$ is not invertible.

If $T - cI$ is not invertible, then $T - cI$ is singular because every non-singular operator on a finite-dimensional vector space is invertible. Now $T - cI$ is singular means that there is a non-zero vector α in V such that

$$(T - cI)\alpha = \mathbf{0} \quad \text{or} \quad T\alpha - cI\alpha = \mathbf{0} \quad \text{or} \quad T\alpha = c\alpha.$$

$\therefore c$ is a characteristic value of T .

This completes the proof of the theorem.

Let T be a linear operator on an n -dimensional vector space V . Let B be an ordered basis for V and let A be the matrix of T with respect to B i.e., let $A = [T]_B$. If c is any scalar, we have

$$\begin{aligned} [T - cI]_B &= [T]_B - c[I]_B \\ &= A - cI \quad \text{where } I \text{ is the unit matrix of order } n. \end{aligned}$$

[Note that $[I]_B = I$].

We have $\det(T - cI) = \det[T - cI]_B = \det(A - cI)$.

Therefore c is a characteristic value of T iff $\det(A - cI) = 0$. This enables us to make the following definition.

6.2 Characteristic Values of a Matrix

Definition: Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n over the field F . An element c in F is called a characteristic value of A if

$$\det(A - cI) = 0 \quad \text{where } I \text{ is the unit matrix of order } n.$$

Now suppose T is a linear operator on an n -dimensional vector space V and A is the matrix of T with respect to any ordered basis B . Then c is a characteristic value of T iff c is a characteristic value of the matrix A . Therefore our definition of characteristic values of a matrix is sensible.

6.3 Characteristic Equation of a Matrix

Definition: Let A be a square matrix of order n over the field F . Consider the matrix $A - xI$. The elements of this matrix are polynomials in x of degree at most 1. If we evaluate $\det(A - xI)$, then it will be a polynomial in x of degree n . The coefficients of x in this polynomial will be $(-1)^n$. Let us denote this polynomial by $f(x)$.

Then $f(x) = \det(A - xI)$ is called the **characteristic polynomial** of the matrix A . The equation $f(x) = 0$ is called the **characteristic equation** of the matrix A . Now c is a characteristic value of the matrix A iff $\det(A - cI) = 0$ i.e., iff $f(c) = 0$ i.e., iff c is a root of the characteristic equation of A . Thus in order to find the characteristic values of a matrix we should first obtain its characteristic equation and then we should find the roots of this equation.

6.4 Characteristic Vector of a Matrix

Definition: If c is a characteristic value of an $n \times n$ matrix A , then a non-zero matrix X of the type $n \times 1$ such that $AX = cX$ is called a **characteristic vector** of A corresponding to the characteristic value c .

Theorem 6: Let T be a linear operator on an n -dimensional vector space V and A be the matrix of T relative to any ordered basis B . Then a vector α in V is an **eigenvector** of T corresponding to its eigenvalue c if and only if its coordinate vector X relative to the basis B is an **eigen-vector** of A corresponding to its eigenvalue c .

Proof: We have

$$[T - cI]_B = [T]_B - c[I]_B = A - cI.$$

If $\alpha \neq 0$, then the coordinate vector X of α is also non-zero.

$$\begin{aligned} \text{Now } [(T - cI)(\alpha)]_B &= [T - cI]_B [\alpha]_B \quad [\text{See theorem 2 of chap. 4, article 4.2}] \\ &= (A - cI)X. \end{aligned}$$

$$\therefore (T - cI)(\alpha) = 0 \quad \text{iff } (A - cI)X = 0$$

$$\text{or } T(\alpha) = c\alpha \quad \text{iff } AX = cX$$

$$\text{or } \alpha \text{ is an eigenvector of } T \quad \text{iff } X \text{ is an eigenvector of } A.$$

Thus with the help of this theorem we see that our definition of characteristic vector of a matrix is sensible. Now we shall define the characteristic polynomial of a linear operator. Before doing so we shall prove the following theorem.

Theorem 7: Similar matrices A and B have the same characteristic polynomial and hence the same eigenvalues. If X is an eigenvector of A corresponding to the eigenvalue c , then $P^{-1}X$ is an eigenvector of B corresponding to the eigenvalue c where

$$B = P^{-1}AP.$$

Proof: Suppose A and B are similar matrices. Then there exists an invertible matrix P such that

$$B = P^{-1}AP.$$

$$\begin{aligned} \text{We have } B - xI &= P^{-1}AP - xI = P^{-1}AP - P^{-1}(xI)P \\ &= P^{-1}(A - xI)P. \end{aligned}$$

$$\begin{aligned} \therefore \det(B - xI) &= \det P^{-1} \det(A - xI) \det P \\ &= \det P^{-1} \cdot \det P \cdot \det(A - xI) = \det(P^{-1}P) \cdot \det(A - xI) \\ &= \det I \cdot \det(A - xI) = 1 \cdot \det(A - xI) = \det(A - xI). \end{aligned}$$

Thus the matrices A and B have the same characteristic polynomial and consequently they will have the same characteristic values.

If c is an eigenvalue of A and X is a corresponding eigenvector, then $AX = cX$, and hence

$$B(P^{-1}X) = (P^{-1}AP)P^{-1}X = P^{-1}AX = P^{-1}(cX) = c(P^{-1}X).$$

$\therefore P^{-1}X$ is an eigenvector of B corresponding to c . This completes the proof of the theorem.

Now suppose that T is a linear operator on an n -dimensional vector space V . If B_1, B_2 are any two ordered bases for V , then we know that the matrices $[T]_{B_1}$ and $[T]_{B_2}$ are similar. Also similar matrices have the same characteristic polynomial. This enables us to define sensibly the characteristic polynomial of T .

6.5 Characteristic Polynomial of a Linear Operator

Definition: Let T be a linear operator on an n -dimensional vector space V . The characteristic polynomial of T is the characteristic polynomial of any $n \times n$ matrix which represents T in some ordered basis for V . On account of the above discussion the polynomial of T as defined by us will be unique.

If B is any ordered basis for V and A is the matrix of T with respect to B then

$$\begin{aligned} \det(T - xI) &= \det[T - xI]_B = \det([T]_B - x[I]_B) = \det(A - xI) \\ &= \text{the characteristic polynomial of } A \text{ and so also that of } T. \end{aligned}$$

$$\therefore \text{the characteristic polynomial of } T = \det(T - xI).$$

The equation $\det(T - xI) = 0$ is called the characteristic equation of T .

6.6 Existence of Characteristic Values

Let T be a linear operator on an n -dimensional vector space V over the field F . Then c belonging to F will be a characteristic value of T iff c is a root of the characteristic equation of T i.e., iff c is a root of the equation

$$\det(T - xI) = 0. \quad \dots(1)$$

The equation (1) is of degree n in x . If the field F is *algebraically closed* i.e., if every polynomial equation in F possesses a root then T will definitely have at least one characteristic value. If the field F is not algebraically closed, then T may or may not have a characteristic value according as the equation (1) has or has not a root in F . Since the equation (1) is of degree n in x , therefore if T has a characteristic value then it cannot have more than n distinct characteristic values. The field of complex numbers is algebraically closed. By fundamental theorem of algebra we know that every polynomial equation over the field of complex numbers is solvable. Therefore if F is the field of complex numbers then T will definitely have at least one characteristic value. The field of real numbers is not algebraically closed. If F is the field of real numbers, then T may or may not have a characteristic value.

Illustration: Consider the linear operator T on $V_2(\mathbf{R})$ which is represented in the standard ordered basis by the matrix

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The characteristic polynomial for T (or for A) is

$$\det(A - xI) = \det \begin{bmatrix} 0 - x & -1 \\ 1 & 0 - x \end{bmatrix} = \begin{vmatrix} -x & -1 \\ 1 & -x \end{vmatrix} = x^2 + 1.$$

The polynomial equation $x^2 + 1 = 0$ has no roots in $\mathbf{R}$. Therefore T has no characteristic values.

However if T is a linear operator on $V_2(\mathbf{C})$, then the characteristic equation of T has two distinct roots i and $-i$ in $\mathbf{C}$. In this case T has two characteristic values i and $-i$.

6.7 Algebraic and Geometric Multiplicity of a Characteristic Value

Definition: Let T be a linear operator on an n -dimensional vector space V and let c be a characteristic value of T . By *geometric multiplicity* of c we mean the dimension of the characteristic space W_c of c . By *algebraic multiplicity* of c we mean the multiplicity of c as root of the characteristic equation of T .

Method of finding the characteristic values and the corresponding characteristic vectors of a linear operator T : Let T be a linear operator on an n -dimensional vector space V over the field F . Let B be any ordered basis for V and let $A = [T]_B$. The roots of the equation $\det(A - xI) = 0$ will give the characteristic values of A or also of T . Let c be a characteristic value of T . Then $\mathbf{0} \neq \alpha$ will be a characteristic vector corresponding to this characteristic value if

$$(T - cI)\alpha = \mathbf{0} \quad \text{i.e.,} \quad \text{if} \quad [T - cI]_B [\alpha]_B = [\mathbf{0}]_B$$

$$\text{i.e., if } (A - cI)X = O, \quad \dots(1)$$

where $X = [\alpha]_B$ = a column matrix of the type $n \times 1$ and O is the null matrix of the type $n \times 1$. Thus to find the coordinate matrix of α with respect to B , we should solve the matrix equation (1) for X .

6.8 Matric Polynomials

Definition: An expression of the form

$$f(x) = A_0 + A_1x + A_2x^2 + \dots + A_mx^m,$$

where $A_0, A_1, A_2, \dots, A_m$ are all square matrices of order n , is called a *Matric polynomial of degree m* provided A_m is not a null matrix. The symbol x is called **indeterminate**.

Equality of Matric Polynomials: Two matric polynomials are equal iff the coefficients of the like powers of x are the same.

Lemma: Every square matrix over the field F whose elements are ordinary polynomials in x over F , can essentially be expressed as a matric polynomial in x of degree m , where m is the highest power of x occurring in any element of the matrix.

We shall explain this theorem by the following illustration:

Consider the matrix

$$A = \begin{bmatrix} 1+2x+3x^2 & x^2 & 4-6x \\ 1+x^3 & 3+4x^2 & 1-2x+4x^3 \\ 2-3x+2x^3 & 5 & 6 \end{bmatrix}$$

in which the highest power of x occurring in any element is 3. Rewriting each element as a cubic in x , supplying missing coefficients with zeros, we get

$$A = \begin{bmatrix} 1+2.x+3.x^2+0.x^3 & 0+0.x+1.x^2+0.x^3 & 4-6.x+0.x^2+0.x^3 \\ 1+0.x+0.x^2+1.x^3 & 3+0.x+4.x^2+0.x^3 & 1-2.x+0.x^2+4.x^3 \\ 2-3.x+0.x^2+2.x^3 & 5+0.x+0.x^2+0.x^3 & 6+0.x+0.x^2+0.x^3 \end{bmatrix}.$$

Obviously A can be written as the matrix polynomial

$$A = \begin{bmatrix} 1 & 0 & 4 \\ 1 & 3 & 1 \\ 2 & 5 & 6 \end{bmatrix} + x \begin{bmatrix} 2 & 0 & -6 \\ 0 & 0 & -2 \\ -3 & 0 & 0 \end{bmatrix} + x^2 \begin{bmatrix} 3 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} + x^3 \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 4 \\ 2 & 0 & 0 \end{bmatrix}.$$

Theorem 8: The Cayley-Hamilton Theorem: Let T be a linear operator on an n -dimensional vector space V (F). Then T satisfies its characteristic equation i.e., if $f(x)$ be the characteristic polynomial of T , then $f(T) = \hat{0}$.

Or

Every square matrix satisfies its characteristic equation .

चववर्षि Let T be a linear operator on an n -dimensional vector space V over the field F . Let B be any ordered basis for V and A be the matrix of T relative to B i.e., let $A = [T]_B$. The characteristic polynomial of T is the same as the characteristic polynomial of A . If $A = [a_{ij}]_{n \times n}$, then the characteristic polynomial $f(x)$ of A is given by

$$f(x) = \det(A - xI) = \begin{vmatrix} a_{11} - x & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - x & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} - x \end{vmatrix}$$

$$= a_0 + a_1x + a_2x^2 + \dots + a_nx^n \quad (\text{say}), \quad \dots(1)$$

where the a_i 's are in F .

The characteristic equation of A is $f(x) = 0$

i.e., $a_0 + a_1x + a_2x^2 + \dots + a_nx^n = 0$.

Since the elements of the matrix $A - xI$ are polynomials at most of the first degree in x , therefore the elements of the matrix $\text{adj}(A - xI)$ are ordinary polynomials in x of degree $n - 1$ or less. Note that the elements of the matrix $\text{adj}(A - xI)$ are the cofactors of the elements of the matrix $A - xI$. Therefore $\text{adj}(A - xI)$ can be written as a matrix polynomial in x in the form

$$\text{adj}(A - xI) = B_0 + B_1x + B_2x^2 + \dots + B_{n-1}x^{n-1}, \quad \dots(2)$$

where the B_i 's are square matrices of order n over F with elements independent of x .

Now by the property of adjoints, we know that

$$(A - xI) \cdot \text{adj}(A - xI) = \{\det(A - xI)\} I.$$

$$\therefore (A - xI) \{B_0 + xB_1 + x^2B_2 + \dots + x^{n-1}B_{n-1}\}$$

$$= \{a_0 + a_1x + \dots + a_nx^n\} I$$

[from (1) and (2)]

Equating the coefficients of like powers of x on both sides, we get

$$AB_0 = a_0 I$$

$$AB_1 - IB_0 = a_1 I$$

$$AB_2 - IB_1 = a_2 I$$

.....

.....

$$\begin{aligned} AB_{n-1} - IB_{n-2} &= a_{n-1} I \\ -IB_{n-1} &= a_n I. \end{aligned}$$

Premultiplying these equations successively by $I, A, A^2, \dots, A^n$ and adding, we get

$$a_0 I + a_1 A + a_2 A^2 + \dots + a_n A^n = O,$$

where O is the null matrix of order n .

Thus $f(A) = O$.

Now $f(T) = a_0 I + a_1 T + a_2 T^2 + \dots + a_n T^n$.

$$\begin{aligned} \therefore [f(T)]_B &= [a_0 I + a_1 T + a_2 T^2 + \dots + a_n T^n]_B \\ &= a_0 [I]_B + a_1 [T]_B + a_2 [T^2]_B + \dots + a_n [T^n]_B \\ &= f(A). \end{aligned}$$

$$\begin{aligned} \therefore f(A) = O &\Rightarrow [f(T)]_B = O = [\hat{0}]_B \Rightarrow f(T) = \hat{0} \\ \Rightarrow a_0 I + a_1 T + a_2 T^2 + \dots + a_n T^n &= \hat{0}. \end{aligned} \quad \dots(3)$$

Corollary: We have

$$f(x) = a_0 + a_1 x + \dots + a_n x^n = \det. (A - xI).$$

$\therefore f(0) = a_0 = \det. A = \det. T$.

If T is non-singular, then T is invertible and

$$\det. T \neq 0 \text{ i.e., } a_0 \neq 0.$$

Then from (3), we get

$$\begin{aligned} a_0 I &= -(a_1 T + a_2 T^2 + \dots + a_n T^n) \\ \Rightarrow I &= -\left(\frac{a_1}{a_0} I + \frac{a_2}{a_0} T + \dots + \frac{a_n}{a_0} T^{n-1}\right) T \\ \Rightarrow T^{-1} &= -\left(\frac{a_1}{a_0} I + \frac{a_2}{a_0} T + \dots + \frac{a_n}{a_0} T^{n-1}\right). \end{aligned}$$

6.9 Diagonalizable Operators

Definition: Suppose T is a linear operator on the finite-dimensional vector space V . Then T is said to be diagonalizable if there is a basis B for V each vector of which is a characteristic vector of T .

Matrix of a Diagonalizable Operator: Let T be a diagonalizable operator on an n -dimensional vector space V . Let $B = \{\alpha_1, \dots, \alpha_n\}$ be an ordered basis for V such that each α_j is a characteristic vector of T . Let $T\alpha_i = c_i \alpha_i$. Then

$$\begin{aligned} T\alpha_1 &= c_1 \alpha_1 = c_1 \alpha_1 + 0\alpha_2 + \dots + 0\alpha_n \\ T\alpha_2 &= c_2 \alpha_2 = 0\alpha_1 + c_2 \alpha_2 + \dots + 0\alpha_n \\ &\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ &\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ T\alpha_n &= c_n \alpha_n = 0\alpha_1 + 0\alpha_2 + \dots + 0\alpha_{n-1} + c_n \alpha_n. \end{aligned}$$

Therefore the matrix of T relative to B is

$$[T]_B = \begin{bmatrix} c_1 & 0 & \dots & 0 \\ 0 & c_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & c_n \end{bmatrix}.$$

This matrix is a diagonal matrix. Note that a square matrix of order n is said to be a diagonal matrix if all the elements lying above and below the principal diagonal are equal to zero. The scalars $c_1, \dots, c_n$ need not all be distinct. If V is n dimensional, then T is diagonalizable iff T has n linearly independent characteristic vectors.

Diagonalizable Matrix: Definition: A matrix A over a field F is said to be diagonalizable if it is similar to a diagonal matrix over the field F . Thus a matrix A is diagonalizable if there exists an invertible matrix P such that $P^{-1}AP = D$ where D is a diagonal matrix. Also the matrix P is then said to diagonalize A or transform A to diagonal form.

Theorem 9: A necessary and sufficient condition that an $n \times n$ matrix A over a field F be diagonalizable is that A has n linearly independent characteristic vectors in $V_n(F)$.

Proof: If A is diagonalizable, then A is similar to a diagonal matrix D . Therefore there exists an invertible matrix P such that

$$P^{-1}AP = D \quad \text{or} \quad AP = PD. \quad \dots(1)$$

If $c_1, c_2, \dots, c_n$ are the diagonal elements of D , then $c_1, c_2, \dots, c_n$ are the characteristic values of D as can be easily seen. But similar matrices have the same characteristic values. Therefore $c_1, c_2, \dots, c_n$ are the characteristic values of A .

Now suppose $P_1, P_2, \dots, P_n$ are the column vectors of the matrix P . Then equating corresponding columns on each side of (1), we get

$$AP_i = c_i P_i \quad (i = 1, 2, \dots, n). \quad \dots(2)$$

But (2) shows that P_i is a characteristic vector of A corresponding to the characteristic value c_i . Since the matrix P is invertible, therefore its column vectors $P_1, P_2, \dots, P_n$ are n linearly independent vectors belonging to $V_n(F)$. Thus A has n linearly independent characteristic vectors $P_1, \dots, P_n$.

Conversely, if $P_1, \dots, P_n$ are n linearly independent characteristic vectors of A corresponding to characteristic values $c_1, \dots, c_n$, then equations (2) holds. Therefore equation (1) holds where P is the matrix with columns $P_1, \dots, P_n$. Since the columns of P are linearly independent, therefore P is invertible and hence (1) implies $P^{-1}AP = D$. Thus A is similar to a diagonal matrix and so is diagonalizable.

This completes the proof of the theorem.

Remark: In the proof of the above theorem we have shown that if A is diagonalizable and P diagonalizes A , then

$$P^{-1}AP = \begin{bmatrix} c_1 & 0 & \dots & 0 \\ 0 & c_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & c_n \end{bmatrix}.$$

if and only if the j^{th} column of P is a characteristic vector of A corresponding to the characteristic value c_j of A , ($j = 1, 2, \dots, n$).

Theorem 10: A linear operator T on an n -dimensional vector space $V(F)$ is diagonalizable if and only if its matrix A relative to any ordered basis B of V is diagonalizable.

Proof: Suppose T is diagonalizable. Then T has n linearly independent characteristic vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ in V . Suppose $X_1, X_2, \dots, X_n$ are the co-ordinate vectors of $\alpha_1, \alpha_2, \dots, \alpha_n$ relative to the basis B . Then $X_1, \dots, X_n$ are also linearly independent since V is isomorphic to $V_n(F)$ by isomorphism which takes a vector in V to its co-ordinate vector in $V_n(F)$. Under an isomorphism a linearly independent set is mapped onto a linearly independent set. Further $X_1, \dots, X_n$ are the characteristic vectors of the matrix A [see theorem 6]. Therefore the matrix A is diagonalizable. [See theorem 9].

Conversely suppose the matrix A is diagonalizable. Then A has n linearly independent characteristic vectors $X_1, \dots, X_n$ in $V_n(F)$. If $\alpha_1, \dots, \alpha_n$ are the vectors in V having $X_1, \dots, X_n$ as their coordinate vectors, then $\alpha_1, \dots, \alpha_n$ will be n linearly independent characteristic vectors of T . So T is diagonalizable.

Theorem 11: Let T be any linear operator on a finite dimensional vector space V , let $c_1, c_2, \dots, c_k$ be the distinct characteristic values of T , and let W_i be the null space of $(T - c_i I)$. Then the subspaces $W_1, \dots, W_k$ are independent.

Further show that if in addition T is diagonalizable, then V is the direct sum of the subspaces $W_1, \dots, W_k$.

Proof: By definition of W_i , we have

$$W_i = \{\alpha : \alpha \in V \text{ and } (T - c_i I)\alpha = \mathbf{0} \text{ i.e., } T\alpha = c_i\alpha\}.$$

Now let α_i be in W_i , $i = 1, \dots, k$, and suppose that

$$\alpha_1 + \alpha_2 + \dots + \alpha_k = \mathbf{0}. \quad \dots(1)$$

Let j be any integer between 1 and k and let

$$U_j = \prod_{\substack{1 \leq i \leq k \\ i \neq j}} (T - c_i I).$$

Note that U_j is the product of the operators $(T - c_i I)$ for $i \neq j$. In other words $U_j = (T - c_1 I)(T - c_2 I) \dots (T - c_k I)$ where in the product the factor $T - c_j I$ is missing.

Let us find $U_j \alpha_i$, $i = 1, \dots, k$. By the definition of W_i , we have

$(T - c_i I) \alpha_i = \mathbf{0}$. Since the operators $(T - c_i I)$ all commute, being polynomials in T , therefore $U_j \alpha_i = \mathbf{0}$ for $i \neq j$. Note that for each $i \neq j$, U_j contains a factor $(T - c_i I)$ and $(T - c_i I) \alpha_i = \mathbf{0}$.

$$\begin{aligned}
 \text{Also } U_j \alpha_j &= [(T - c_1 I) \dots (T - c_k I)] \alpha_j \\
 &= [(T - c_1 I) \dots (T - c_{k-1} I)] (T \alpha_j - c_k I \alpha_j) \\
 &= [(T - c_1 I) \dots (T - c_{k-1} I)] (c_j \alpha_j - c_k \alpha_j) \\
 &\quad [\because T \alpha_j = c_j \alpha_j \text{ and } I \alpha_j = \alpha_j] \\
 &= [(T - c_1 I) \dots (T - c_{k-1} I)] (c_j - c_k) \alpha_j \\
 &= (c_j - c_k) [(T - c_1 I) \dots (T - c_{k-1} I)] \alpha_j \\
 &= (c_j - c_k) (c_j - c_{k-1}) \dots (c_j - c_1) \alpha_j, \\
 &\quad \text{the factor } c_j - c_j \text{ will be missing.}
 \end{aligned}$$

$$\text{Thus } U_j \alpha_j = \left[\prod_{\substack{1 \leq i \leq k \\ i \neq j}} (c_j - c_i) \right] \alpha_j. \quad \dots(2)$$

Now applying U_j to both sides of (1), we get

$$\begin{aligned}
 &U_j \alpha_1 + U_j \alpha_2 + \dots + U_j \alpha_k = \mathbf{0} \\
 \Rightarrow &U_j \alpha_j = \mathbf{0} \quad [\because U_j \alpha_i = \mathbf{0} \text{ if } i \neq j] \\
 \Rightarrow &\left[\prod_{i \neq j} (c_j - c_i) \right] \alpha_j = \mathbf{0} \quad [\text{by (2)}]
 \end{aligned}$$

Since the scalars c_i are all distinct, therefore the product

$$\prod_{i \neq j} (c_j - c_i), \text{ is a non-zero scalar.}$$

$$\text{Hence } \left[\prod_{i \neq j} (c_j - c_i) \right] \alpha_j = \mathbf{0} \Rightarrow \alpha_j = \mathbf{0}.$$

Thus $\alpha_j = \mathbf{0}$ for every integer j between 1 and k .

$$\text{In this way } \alpha_1 + \dots + \alpha_k = \mathbf{0}$$

$$\Rightarrow \alpha_i = \mathbf{0} \text{ for each } i.$$

Hence the subspaces $W_1, \dots, W_k$ are independent.

Second Part: Now suppose that T is diagonalizable. Then we shall show that $V = W_1 + \dots + W_k$. Since T is diagonalizable, therefore there exists a basis of V each vector of which is a characteristic vector of T . Thus there exists a basis of V consisting of vectors belonging to the characteristic subspaces $W_1, \dots, W_k$. If $\alpha \in V$, then α can be expressed as a linear combination of these basis vectors. Thus α can be written as $\alpha = \alpha_1 + \dots + \alpha_k$ where $\alpha_i \in W_i, i = 1, \dots, k$. In this way $\alpha \in W_1 + \dots + W_k$. Therefore $V = W_1 + \dots + W_k$. But in the first part, we have proved that the subspaces $W_1, \dots, W_k$ are independent. Hence

$$V = W_1 \oplus \dots \oplus W_k.$$

Illustrative Examples

Example 1: Let V be an n -dimensional vector space over F . What is the characteristic polynomial of (i) the identity operator on V , (ii) the zero operator on V ?

Solution: Let B be any ordered basis for V .

(i) If I is the identity operator on V , then

$$[I]_B = I.$$

The characteristic polynomial of $I = \det(I - xI)$

$$\begin{vmatrix} 1-x & 0 & \dots & 0 \\ 0 & 1-x & \dots & 0 \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ 0 & 0 & \dots & 1-x \end{vmatrix} = (1-x)^n.$$

(ii) If $\hat{0}$ is the zero operator on V , then $[\hat{0}]_B = O$ i.e., the null matrix of order n .

The characteristic polynomial of $\hat{0} = \det(O - xI)$

$$= \begin{vmatrix} -x & 0 & \dots & 0 \\ 0 & -x & \dots & 0 \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ 0 & 0 & \dots & -x \end{vmatrix} = (-1)^n x^n.$$

Example 2: Let T be a linear operator on a finite dimensional vector space V and let c be a characteristic value of T . Show that the characteristic space of c i.e., W_c is invariant under T .

Solution: We have by definition, $W_c = \{\alpha \in V : T\alpha = c\alpha\}$.

Let $\alpha \in W_c$. Then $T\alpha = c\alpha$.

Since W_c is a subspace, therefore

$$c \in F \text{ and } \alpha \in W_c \Rightarrow c\alpha \in W_c.$$

Thus $\alpha \in W_c \Rightarrow T\alpha \in W_c$.

Hence W_c is invariant under T .

Example 3: If A and B are similar linear operators on a finite dimensional vector space V , then A and B have the same characteristic polynomial.

Solution: Suppose A and B are similar linear transformations on a finite dimensional vector space V . Then there exists an invertible linear operator C on V such that $A = CBC^{-1}$.

We have $A - xI = CBC^{-1} - xI = CBC^{-1} - C(xI)C^{-1} = C(B - xI)C^{-1}$.

$$\begin{aligned} \therefore \det(A - xI) &= \det\{C(B - xI)C^{-1}\} \\ &= \det C \cdot \det(B - xI) \cdot \det C^{-1} = \det C \cdot \det C^{-1} \cdot \det(B - xI) \\ &= \det(CC^{-1}) \det(B - xI) = \det I \cdot \det(B - xI) = 1 \cdot \det(B - xI) \\ &= \det(B - xI). \end{aligned}$$

Now the characteristic polynomial of $A = \det(A - xI)$.

$\therefore$ the characteristic polynomial of A = the characteristic polynomial of B .

Example 4: Find all (complex) characteristic values and characteristic vectors of the following matrices

$$(a) \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution: (a) Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

We have $A - xI = \begin{bmatrix} 1-x & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{bmatrix}$.

$\therefore$ the characteristic polynomial of A is

$$= \det(A - xI) = \begin{vmatrix} 1-x & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix}$$

$$= \begin{vmatrix} 3-x & 1 & 1 \\ 3-x & 1-x & 1 \\ 3-x & 1 & 1-x \end{vmatrix} C_1 + C_2 + C_3$$

$$= (3-x) \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix}$$

$$= (3-x) \begin{vmatrix} 1 & 1 & 1 \\ 0 & -x & 0 \\ 0 & 0 & -x \end{vmatrix} R_2 - R_1, R_3 - R_1$$

$$= (3-x)x^2.$$

$\therefore$ the characteristic equation of A is $(3-x)x^2 = 0$.

The only roots of this equation are $x = 3, 0$.

$\therefore$ 0 and 3 are the only characteristic values of A .

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the coordinate matrix of a characteristic vector corresponding to

the characteristic value $x = 0$. Then X will be given by a non-zero solution of the equation

$$(A - 0I) X = O$$

$$\text{i.e., } \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} x_1 + x_2 + x_3 \\ x_1 + x_2 + x_3 \\ x_1 + x_2 + x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

or $x_1 + x_2 + x_3 = 0$.

This equation has two linearly independent solutions i.e.,

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \text{ and } X_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

Every non-zero multiple of these column matrices X_1 and X_2 is a characteristic vector of A corresponding to the characteristic value 0.

The characteristic space of this characteristic value will be the subspace W spanned by these two vectors X_1 and X_2 . Any non-zero vector in W will be a characteristic vector corresponding to this characteristic value.

To find the characteristic vectors corresponding to the characteristic value 3 we consider the equation

$$(A - 3I) X = O$$

$$\text{i.e., } \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{i.e., } \begin{bmatrix} -2x_1 + x_2 + x_3 \\ x_1 - 2x_2 + x_3 \\ x_1 + x_2 - 2x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{i.e., } -2x_1 + x_2 + x_3 = 0, x_1 - 2x_2 + x_3 = 0, x_1 + x_2 - 2x_3 = 0.$$

Solving these equations, we get $x_1 = x_2 = x_3 = k$.

$$\therefore X = \begin{bmatrix} k \\ k \\ k \end{bmatrix}, \text{ where } k \neq 0.$$

$$(b) \text{ Let } A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

The characteristic equation of A is $(1-x)^3 = 0$.

$\therefore x = 1$ is the only characteristic value of A .

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the coordinate matrix of a characteristic vector corresponding to

the characteristic value 1. Then X will be given by a non-zero solution of the equation

$$(A - I) X = O$$

$$\text{i.e., } \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{i.e., } \begin{bmatrix} x_2 + x_3 \\ x_3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{i.e., } x_2 + x_3 = 0, x_3 = 0.$$

$$\therefore x_1 = k, x_2 = 0, x_3 = 0.$$

$$\text{Thus } X = \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix}, \text{ where } k \neq 0.$$

Example 5: Let T be the linear operator on $\mathbf{R}^3$ which is represented in the standard basis by the matrix

$$\begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}.$$

Prove that T is diagonalizable.

Solution: Let $A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}.$

The characteristic equation of A is

$$\begin{vmatrix} -9-x & 4 & 4 \\ -8 & 3-x & 4 \\ -16 & 8 & 7-x \end{vmatrix} = 0$$

$$\text{or } \begin{vmatrix} -1-x & 4 & 4 \\ -1-x & 3-x & 4 \\ -1-x & 8 & 7-x \end{vmatrix} = 0, \text{ applying } C_1 + C_2 + C_3$$

$$\text{or } -(1+x) \begin{vmatrix} 1 & 4 & 4 \\ 1 & 3-x & 4 \\ 1 & 8 & 7-x \end{vmatrix} = 0$$

$$\text{or} \quad (1+x) \begin{vmatrix} 1 & 4 & 4 \\ 0 & -1-x & 0 \\ 0 & 4 & 3-x \end{vmatrix} = 0, \text{ applying } R_2 - R_1, R_3 - R_1$$

$$\text{or} \quad (1+x)(1+x)(3-x) = 0.$$

The roots of this equation are $-1, -1, 3$.

$\therefore$ The eigenvalues of the matrix A are $-1, -1, 3$.

The characteristic vectors X of A corresponding to the eigenvalue -1 are given by the equation

$$(A - (-1)I)X = O \quad \text{or} \quad (A + I)X = O$$

$$\text{or} \quad \begin{bmatrix} -8 & 4 & 4 \\ -8 & 4 & 4 \\ -16 & 8 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

These equations are equivalent to the equations

$$\begin{bmatrix} -8 & 4 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ applying } R_2 - R_1, R_3 - 2R_1.$$

The matrix of coefficients of these equations has rank 1. Therefore these equations have two linearly independent solutions. We see that these equations reduce to the single equation

$$-2x_1 + x_2 + x_3 = 0.$$

Obviously

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

are two linearly independent solutions of this equation. Therefore X_1 and X_2 are two linearly independent eigenvectors of A corresponding to the eigenvalue -1 .

Now the eigenvectors of A corresponding to the eigenvalue 3 are given by

$$(A - 3I)X = O$$

$$\text{i.e.,} \quad \begin{bmatrix} -12 & 4 & 4 \\ -8 & 0 & 4 \\ -16 & 8 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

These equations are equivalent to the equations

$$\begin{bmatrix} -12 & 4 & 4 \\ 4 & -4 & 0 \\ -4 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ applying } R_2 - R_1, R_3 - R_1.$$

The matrix of coefficients of these equations has rank 2. Therefore these equations

will have a non-zero solution. Also these equations will have $3 - 2 = 1$ linearly independent solution. These equations can be written as

$$-12x_1 + 4x_2 + 4x_3 = 0$$

$$4x_1 - 4x_2 = 0$$

$$-4x_1 + 4x_2 = 0.$$

From these, we get $x_1 = x_2 = 1$, say.

Then $x_3 = 2$.

$$\therefore X_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

is an eigenvector of A corresponding to the eigenvalue 3.

$$\text{Now let } P = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix}.$$

We have $\det P = 1 \neq 0$. Therefore the matrix P is invertible. Therefore the columns of P are linearly independent vectors belonging to $\mathbf{R}^3$. Since the matrix A has three linearly independent eigenvectors in $\mathbf{R}^3$, therefore it is diagonalizable. Consequently the linear operator T is diagonalizable. Also the diagonal form D of A is given by

$$P^{-1}AP = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix} = D.$$

Example 6: Prove that the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ is not diagonalizable over the field $\mathbf{C}$.

Solution: The characteristic equation of A is

$$\begin{vmatrix} 1-x & 2 \\ 0 & 1-x \end{vmatrix} = 0 \quad \text{or} \quad (1-x)^2 = 0.$$

The roots of this equation are 1, 1. Therefore the only distinct eigenvalue of A is 1.

The eigenvectors of A corresponding to this eigenvalue are given by

$$\begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{or} \quad 0x_1 + 2x_2 = 0.$$

This equation has only one linearly independent solution. We see that

$$X = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

is the only linearly independent eigenvector of A . Since A has not two linearly independent eigenvectors, therefore it is not diagonalizable.

Comprehensive Exercise 1

1. Show that the characteristic values of a diagonal matrix are precisely the elements in the diagonal. Hence show that if a matrix B is similar to a diagonal matrix D , then the diagonal elements of D are the characteristic values of B .
2. Let T be a linear operator on a finite dimensional vector space V . Then show that 0 is a characteristic value of T iff T is not invertible.
3. Suppose S and T are two linear operators on a finite dimensional vector space V . If S and T have the same characteristic polynomial, then

$$\det S = \det T.$$

4. Find all (complex) proper values and proper vectors of the following matrices:

(a) $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 \\ 0 & i \end{bmatrix}.$

5. Find the characteristic equation of the matrix

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

and verify that it is satisfied by A and hence obtain A^{-1} .

6. Show that the characteristic equation of the complex matrix

$$A = \begin{bmatrix} 0 & 0 & c \\ 1 & 0 & b \\ 0 & 1 & a \end{bmatrix}$$

is $x^3 - ax^2 - bx - c = 0$.

7. Find the eigenvalues and the corresponding eigen space for the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

8. Find the characteristic roots and the characteristic spaces of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}.$$

9. Let T be a linear operator on the n -dimensional vector space V , and suppose that T has n distinct characteristic values. Prove that T is diagonalizable.

10. Show that distinct eigenvectors of a matrix A corresponding to distinct eigenvalues of A are linearly independent.
11. Let T be the linear operator on $\mathbf{R}^3$ which is represented in the standard ordered basis by the matrix

$$A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}.$$

Find the characteristic values of A and prove that T is diagonalizable.

12. Is the matrix $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

similar over the field $\mathbf{R}$ to a diagonal matrix? Is A similar over the field $\mathbf{C}$ to a diagonal matrix?

13. Show that the matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is not diagonalizable.

14. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$. Is A similar to a diagonal matrix?

15. Is the matrix $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{bmatrix}$

similar over the field $\mathbf{R}$ to a diagonal matrix? Is A similar over the field $\mathbf{C}$ to a diagonal matrix?

16. For each of the following matrices over the field $\mathbf{C}$, find the diagonal form and a diagonalizing matrix P .

(a) $\begin{bmatrix} 20 & 18 \\ -27 & -25 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} -17 & 18 & -6 \\ -18 & 19 & -6 \\ -9 & 9 & 2 \end{bmatrix}.$

Answers 1

4. (a) 0 is the only characteristic value of A . The corresponding characteristic vectors are given by $[k \ 0]'$ where k is any non-zero scalar.
- (b) Characteristic values are 1, i . Characteristic vectors corresponding to the value 1 are given by $[k \ 0]'$ and corresponding to the value i are given by $[0, \ c]'$, where k and c are any non-zero scalars.

- (c) 1, i are characteristic values. Characteristic values corresponding to 1 are given by $[k \ 0]'$ and corresponding to i are given by $[c \ (i-1)c]'$, where k and c are any non-zero scalars.

5. $x^3 - 6x^2 + 9x - 4 = 0$; $A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$.

7. 0, 3, 15. Corresponding eigenspaces are spanned by

$$\begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ -2 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} \text{ respectively.}$$

8. 1, 2, 2. Corresponding eigenspaces are spanned by $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

respectively.

11. 1, 2, 2.

12. Not similar over $\mathbf{R}$ to a diagonal matrix; similar over $\mathbf{C}$ to a diagonal matrix.

14. Yes.

15. Not similar over the field $\mathbf{R}$ as well as $\mathbf{C}$ to a diagonal matrix.

16. (a) $D = \begin{bmatrix} 2 & 0 \\ 0 & -7 \end{bmatrix}, P = \begin{bmatrix} 1 & -3 \\ -1 & 2 \end{bmatrix},$

(b) $D = \begin{bmatrix} 3+4i & 0 \\ 0 & 3-4i \end{bmatrix}, P = \begin{bmatrix} 1 & -1 \\ i & i \end{bmatrix},$

(c) $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}, P = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{bmatrix},$

(d) $D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P = \begin{bmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & 0 & 3 \end{bmatrix}.$

6.10 Minimal Polynomial and Minimal Equation of a Linear Operator or of a Matrix

Annihilating Polynomials: Suppose T is a linear operator on a finite dimensional vector space over the field F and $f(x)$ is a polynomial over F . If $f(T) = \hat{0}$, then we say that polynomial $f(x)$ *annihilates* the linear operator T .

Similarly suppose A is a square matrix of order n over the field F and $f(x)$ is a polynomial over F . If $f(A) = O$, then we say that the polynomial $f(x)$ *annihilates* the matrix A . We know that every linear operator T on an n -dimensional vector space $V(F)$ satisfies its characteristic equation. Also the characteristic polynomial of T is a non-zero polynomial *i.e.*, a polynomial in which the coefficients of various terms are not all zero. Note that if A is the matrix of T in some ordered basis, then the characteristic polynomial of T is $|A - xI|$ in which the coefficient of x^n is $(-1)^n$ which is not zero. Thus we see that at least the characteristic polynomial of T is a non-zero polynomial which annihilates T . Therefore the set of those non-zero polynomials which annihilate T is not empty.

Monic polynomial: **Definition:** A polynomial in x over a field F is called a monic polynomial if the coefficient of the highest power of x in it is unity. Thus $x^3 - 2x^2 + \frac{5}{7}x + 5$ is a monic polynomial of degree 3 over the field of rational numbers.

Among these non-zero polynomials which annihilate a linear operator T , the polynomial which is monic and which is of the lowest degree is of special interest. It is called the *minimal polynomial* of the linear operator T .

Minimal polynomial of a linear operator: **Definition:** Suppose T is a linear operator on an n -dimensional vector space $V(F)$. The monic polynomial of lowest degree over the field F that annihilates T is called the *minimal polynomial* of T . Also if $f(x)$ is the minimal polynomial of T , the equation $f(x) = 0$ is called the *minimal equation* of the linear operator T .

Similarly we can define the minimal polynomial of a matrix. Suppose A is a square matrix of order n over the field F . The monic polynomial of lowest degree over the field F that annihilates A is called the *minimal polynomial* of A .

Now suppose T is a linear operator on an n -dimensional vector space $V(F)$ and A is the matrix of T in some ordered basis B . If $f(x)$ is any polynomial over F , then $[f(T)]_B = f(A)$. Therefore $f(T) = \hat{0}$ if and only if $f(A) = O$. Thus $f(x)$ annihilates T iff it annihilates A . Therefore if $f(x)$ is the polynomial of lowest degree that annihilates T , then it is also the polynomial of lowest degree that annihilates A and conversely. Hence T and A have the same minimal polynomial. Further the characteristic polynomial of the matrix A is of degree n . Since the characteristic polynomial of A annihilates A , therefore the minimal polynomial of A cannot be of degree greater than n . Its degree must be less than or equal to n .

Theorem 1: *The minimal polynomial of a matrix or of a linear operator is unique.*

Proof: Suppose the minimal polynomial of a matrix A is of degree r . Then no non-zero polynomial of degree less than r can annihilate A . Let

$$f(x) = x^r + a_1x^{r-1} + a_2x^{r-2} + \dots + a_{r-1}x + a_r$$

and $g(x) = x^r + b_1 x^{r-1} + b_2 x^{r-2} + \dots + b_{r-1} x + b_r$

be two minimal polynomials of A . Then both $f(x)$ and $g(x)$ annihilate A . Therefore we have $f(A) = O$ and $g(A) = O$. These give

$$A^r + a_1 A^{r-1} + \dots + a_{r-1} A + a_r I = O, \quad \dots(1)$$

$$\text{and} \quad A^r + b_1 A^{r-1} + \dots + b_{r-1} A + b_r I = O. \quad \dots(2)$$

Subtracting (1) from (2), we get

$$(b_1 - a_1) A^{r-1} + \dots + (b_r - a_r) I = O. \quad \dots(3)$$

From (3), we see that the polynomial $(b_1 - a_1) x^{r-1} + \dots + (b_r - a_r)$ also annihilates A . Since its degree is less than r , therefore it must be a zero polynomial. This gives $b_1 - a_1 = 0, b_2 - a_2 = 0, \dots, b_r - a_r = 0$. Thus $a_1 = b_1, \dots, a_r = b_r$.

Therefore $f(x) = g(x)$ and thus the minimal polynomial of A is unique.

Theorem 2: *The minimal polynomial of a matrix (linear operator) is a divisor of every polynomial that annihilates the matrix (linear operator).*

Proof: Suppose $m(x)$ is the minimal polynomial of a matrix A . Let $h(x)$ be any polynomial that annihilates A . Since $m(x)$ and $h(x)$ are two polynomials, therefore by the division algorithm there exist two polynomials $q(x)$ and $r(x)$ such that

$$h(x) = m(x) q(x) + r(x), \quad \dots(1)$$

where either $r(x)$ is a zero polynomial or its degree is less than the degree of $m(x)$. Putting $x = A$ on both sides of (1), we get

$$h(A) = m(A) q(A) + r(A)$$

$$\Rightarrow O = O q(A) + r(A) \quad [\because \text{both } m(x) \text{ and } h(x) \text{ annihilate } A]$$

$$\Rightarrow r(A) = O.$$

Thus $r(x)$ is a polynomial which also annihilates A . If $r(x) \neq 0$, then it is a non-zero polynomial of degree smaller than the degree of the minimal polynomial $m(x)$ and thus we arrive at a contradiction that $m(x)$ is the minimal polynomial of A . Therefore $r(x)$ must be a zero polynomial. Then (1) gives

$$h(x) = m(x) q(x) \Rightarrow m(x) \text{ is a divisor of } h(x).$$

Corollary: *The minimal polynomial of a matrix is a divisor of the characteristic polynomial of that matrix.*

Proof: Suppose $f(x)$ is the characteristic polynomial of a matrix A . Then $f(A) = O$ by Cayley-Hamilton theorem. Thus $f(x)$ annihilates A . If $m(x)$ is the minimal polynomial of A , then by the above theorem we see that $m(x)$ must be a divisor of $f(x)$.

Theorem 3: *Let T be a linear operator on an n -dimensional vector space V [or, let A be an $n \times n$ matrix]. The characteristic and minimal polynomials for T [for A] have the same roots, except for multiplicities.*

Proof: Suppose $f(x)$ is the characteristic polynomial of a linear operator T and $m(x)$ is its minimal polynomial. First we shall prove that every root of the equation $m(x) = 0$ is also a root of the equation $f(x) = 0$. We know that the minimal polynomial is a divisor of the characteristic polynomial. Therefore $m(x)$ is a divisor of $f(x)$. Then there exists a polynomial $q(x)$ such that

$$f(x) = m(x) q(x). \quad \dots(1)$$

Suppose c is a root of the equation $m(x) = 0$. Then $m(c) = 0$. Putting $x = c$ on both sides of (1) we get $f(c) = m(c) q(c) = 0$ $q(c) = 0$. Therefore c is also a root of $f(x) = 0$. Thus c is also a characteristic root of the linear operator T .

Conversely suppose that c is a characteristic value of T . Then there exists a non-zero vector α such that $T\alpha = c\alpha$. Since $m(x)$ is a polynomial, therefore we have

$$[m(T)](\alpha) = m(c) \alpha.$$

But $m(x)$ is the minimal polynomial for T . So $m(x)$ annihilates

$$T \text{ i.e., } m(T) = \hat{0}.$$

$$\begin{aligned} \therefore \quad \hat{0}(\alpha) = m(c) \alpha &\Rightarrow \quad \mathbf{0} = m(c) \alpha & [\because \hat{0}(\alpha) = \mathbf{0}] \\ \Rightarrow \quad m(c) = 0. & & [\because \alpha \neq \mathbf{0}] \end{aligned}$$

Thus c is a root of the minimal equation of T .

Hence every root of the minimal equation of T is also a root of its characteristic equation and every root of the characteristic equation of T is also a root of its minimal equation.

Theorem 4: Let T be a diagonalizable linear operator and let $c_1, \dots, c_k$ be the distinct characteristic values of T . Then the minimal polynomial for T is the polynomial

$$p(x) = (x - c_1) \dots (x - c_k).$$

Proof: We know that each characteristic value of T is a root of the minimal polynomial for T . Therefore each of the scalars $c_1, \dots, c_k$ is a root of the minimal polynomial for T and so each of the polynomials $x - c_1, \dots, x - c_k$ is a factor of the minimal polynomial for T . Therefore the polynomial $p(x) = (x - c_1) \dots (x - c_k)$ will be the minimal polynomial for T provided it annihilates T i.e., provided $p(T) = \hat{0}$.

Let α be a characteristic vector of T . Then one of the operators $T - c_1 I, \dots, T - c_k I$ sends α into $\mathbf{0}$. Therefore

$$(T - c_1 I) \dots (T - c_k I) \alpha = \mathbf{0} \text{ i.e., } p(T) \alpha = \mathbf{0}$$

for every characteristic vector α .

Now T is a diagonalizable operator. Let V be the underlying vector space. Then there exists a basis B for V which consists of characteristic vectors. If β is any vector in V , then β can be expressed as a linear combination of the vectors in the basis B . But we have just shown that $p(T) \alpha = \mathbf{0}$ for every characteristic vector α . Therefore we have

$$p(T)\beta = \mathbf{0}, \forall \beta \in V \Rightarrow p(T) = \hat{\mathbf{0}}.$$

$\therefore$ $p(x)$ annihilates T and so $p(x)$ is the minimal polynomial for T .

Thus we have proved that if T is a diagonalizable linear operator, the minimal polynomial for T is a product of distinct linear factors.

Corollary: If the roots of the characteristic equation of a linear operator T are all distinct say $c_1, c_2, \dots, c_n$, then the minimal polynomial for T is the polynomial

$$p(x) = (x - c_1) \dots (x - c_n).$$

Proof: Since the roots of the characteristic equation of T are all distinct, therefore T is diagonalizable. Hence by the above theorem, the minimal polynomial for T is the polynomial

$$(x - c_1)(x - c_2) \dots (x - c_n).$$

Illustrative Examples

Example 7: Let V be a finite-dimensional vector space. What is the minimal polynomial for the identity operator on V ? What is the minimal polynomial for the zero operator?

Solution: We have $I - 1I = I - I = \hat{\mathbf{0}}$. Therefore the monic polynomial $x - 1$ annihilates the identity operator I and it is the polynomial of lowest degree that annihilates I . Hence $x - 1$ is the minimal polynomial for I .

Again we see that the monic polynomial x annihilates the zero operator $\hat{\mathbf{0}}$ and it is the polynomial of lowest degree that annihilates $\hat{\mathbf{0}}$. Hence x is the minimal polynomial for $\hat{\mathbf{0}}$.

Example 8: Let V be an n -dimensional vector space and let T be a linear operator on V . Suppose that there exists some positive integer k so that $T^k = \hat{\mathbf{0}}$. Prove that $T^n = \hat{\mathbf{0}}$.

Solution: Since $T^k = \hat{\mathbf{0}}$, therefore the polynomial x^k annihilates T . So the minimal polynomial for T is a divisor of x^k . Let x^r be the minimal polynomial for T where $r \leq k$. Then $T^r = \hat{\mathbf{0}}$.

Now $T^n = T^{n-r} T^r = T^{n-r} \hat{\mathbf{0}} = \hat{\mathbf{0}}$.

Example 9: Find the minimal polynomial for the real matrix

$$A = \begin{bmatrix} 7 & 4 & -1 \\ 4 & 7 & -1 \\ -4 & -4 & 4 \end{bmatrix}.$$

Solution: We have $|A - xI| = \begin{vmatrix} 7-x & 4 & -1 \\ 4 & 7-x & -1 \\ -4 & -4 & 4-x \end{vmatrix}$

$$= \begin{vmatrix} 7-x & 4 & -1 \\ 4 & 7-x & -1 \\ 0 & 3-x & 3-x \end{vmatrix}, \text{ by } R_3 + R_2$$

$$= (3-x) \begin{vmatrix} 7-x & 4 & -1 \\ 4 & 7-x & -1 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= (3-x) \begin{vmatrix} 7-x & 4 & -5 \\ 4 & 7-x & x-8 \\ 0 & 1 & 0 \end{vmatrix}, \text{ by } C_3 - C_2$$

$$= -(3-x) \begin{vmatrix} 7-x & -5 \\ 4 & x-8 \end{vmatrix}, \text{ expanding along third row}$$

$$= -(3-x) \begin{vmatrix} 3-x & 3-x \\ 4 & x-8 \end{vmatrix}, \text{ by } R_1 - R_2$$

$$= -(3-x)^2 \begin{vmatrix} 1 & 1 \\ 4 & x-8 \end{vmatrix} = -(3-x)^2 (x-12).$$

Therefore the roots of the equation $|A - xI| = 0$ are $x = 3, 3, 12$. These are the characteristic roots of A .

Let us now find the minimal polynomial of A . We know that each characteristic root of A is also a root of its minimal polynomial. So if $m(x)$ is the minimal polynomial for A , then both $x - 3$ and $x - 12$ are factors of $m(x)$. Let us try whether the polynomial $h(x) = (x - 3)(x - 12) = x^2 - 15x + 36$ annihilates A or not.

We have $A^2 = \begin{bmatrix} 69 & 60 & -15 \\ 60 & 69 & -15 \\ -60 & -60 & 24 \end{bmatrix}$.

$$\therefore A^2 - 15A + 36I = \begin{bmatrix} 69 & 60 & -15 \\ 60 & 69 & -15 \\ -60 & -60 & 24 \end{bmatrix} - 15 \begin{bmatrix} 7 & 4 & -1 \\ 4 & 7 & -1 \\ -4 & -4 & 4 \end{bmatrix} + \begin{bmatrix} 36 & 0 & 0 \\ 0 & 36 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

$$= \begin{bmatrix} 105 & 60 & -15 \\ 60 & 105 & -15 \\ -60 & -60 & 60 \end{bmatrix} - \begin{bmatrix} 105 & 60 & -15 \\ 60 & 105 & -15 \\ -60 & -60 & 60 \end{bmatrix} = O.$$

$\therefore h(x)$ annihilates A . Thus $h(x)$ is the monic polynomial of lowest degree which annihilates A . Hence $h(x)$ is the minimal polynomial for A .

Note: In order to find the minimal polynomial of a matrix A , we should not forget that each characteristic root of A must also be a root of the minimal polynomial. We should try to find the monic polynomial of lowest degree which annihilates A and which has also the characteristic roots of A as its roots.

Comprehensive Exercise 2

1. Show that the minimal polynomial of the real matrix

$$A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix} \text{ is } (x-1)(x-2).$$

2. Write the characteristic polynomial and the minimal polynomial of the

$$\text{matrix } A = \begin{bmatrix} 4 & 3 & 0 \\ 2 & 1 & 0 \\ 5 & 7 & 9 \end{bmatrix}.$$

3. Show that the minimal polynomial of the real matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

is $x^2 - 1$.

4. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix}$.

Show that A and B have different characteristic polynomials but have the same minimal polynomial.

5. Show that similar matrices have the same minimal polynomial.

Answers 2

2. $(9-x)(x^2 - 5x - 2); (x-9)(x^2 - 5x - 2).$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. If V is an n -dimensional vector space over the field F , then the characteristic polynomial of the identity operator on V is
(a) x^n (b) $(1+x)^n$
(c) $(1+x)^{2n}$ (d) none of these.
2. The characteristic values of the matrix $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$ are
(a) $1, i$ (b) $1, -i$
(c) $-1, i$ (d) $-1, -i$.
3. Let V be a finite-dimensional vector space. The minimal polynomial for the zero operator is
(a) $-x$ (b) x
(c) $x-1$ (d) none of these.
4. The minimal polynomial of the real matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is
(a) $x^2 + 1$ (b) $x^2 - 1$
(c) $x + 1$ (d) $x - 1$.
5. The eigen values of the triangular matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$ are
(a) $1, 2, 3$ (b) $2, 3, 4$
(c) $1, 3, 4$ (d) none of these.
6. Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. The eigen values of A are
(a) $1, 5$ (b) $-1, 5$
(c) $1, -5$ (d) $-1, -5$.
7. The only eigen value of the matrix $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is
(a) 1 (b) 2
(c) 3 (d) 4 .

Fill in the Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

1. Distinct characteristic vectors of T corresponding to distinct characteristic values of T are linearly
2. Let T be a linear operator on a finite dimensional vector space V . Then c is a characteristic value of T iff $\det(A - cI) = \dots\dots$
3. If c is a characteristic value of an invertible transformation T , then c^{-1} is a characteristic value of
4. Let V be a finite-dimensional vector space. The minimal polynomial for the identity operator on V is
5. If the characteristic equation of a linear operator T has n distinct roots, say $c_1, c_2, \dots, c_n$, then the minimal polynomial for T is the polynomial $p(x) = \dots\dots$

True or False

Write ‘T’ for true and ‘F’ for false statement.

1. The minimal polynomial of a matrix or of a linear operator is not unique.
2. The minimal polynomial of a matrix is a divisor of the characteristic polynomial of that matrix.
3. Cayley Hamilton theorem states that ‘Every square matrix satisfies its characteristic equation’.
4. The characteristic roots of a diagonal matrix are just the diagonal elements of the matrix.
5. Similar matrices do not have the same minimal polynomial.
6. 0 is the only eigen value of the matrix $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

Answers

Multiple Choice Questions

- | | | | | |
|--------|--------|--------|--------|---------|
| 1. (d) | 2. (a) | 3. (b) | 4. (b) | 5. (a). |
| 6. (b) | 7. (a) | | | |

Fill in the Blank(s)

- | | | | |
|---|------|-------------|------------|
| 1. independent | 2. 0 | 3. T^{-1} | 4. $x - 1$ |
| 5. $(x - c_1)(x - c_2) \dots (x - c_n)$ | | | |

True or False

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. F. | 2. T. | 3. T. | 4. T. | 5. F. |
| 6. T. | | | | |



Chapter

7



Inner Product Spaces

7.1 Introduction

Throughout this chapter we shall deal only with *real* or *complex* vector spaces. Thus if V is the vector space over the field F , then F will not be an arbitrary field. In this chapter F will be either the field $\mathbf{R}$ of real numbers or the field $\mathbf{C}$ of complex numbers.

Before defining inner product and inner product spaces, we shall just give some important properties of complex numbers.

Let $z \in \mathbf{C}$ i.e., let z be a complex number. Then $z = x + iy$ where $x, y \in \mathbf{R}$ and $i = \sqrt{-1}$. Here x is called the real part of z and y is called the imaginary part of z . We write $x = \operatorname{Re} z$, and $y = \operatorname{Im} z$. The *modulus* of the complex number $z = x + iy$ is the non-negative real number $\sqrt{(x^2 + y^2)}$ and is denoted by $|z|$. Also if $z = x + iy$ is a complex number, then the complex number $\bar{z} = x - iy$ is called the *conjugate complex* of z . If $z = \bar{z}$, then $x + iy = x - iy$ and therefore $y = 0$. Thus $z = \bar{z}$ implies that z is real. Obviously we have

$$(i) \quad z + \bar{z} = 2x = 2 \operatorname{Re} z,$$

$$(ii) \quad z - \bar{z} = 2iy = 2i \operatorname{Im} z,$$

$$(iii) \quad z\bar{z} = x^2 + y^2 = |z|^2,$$

$$(iv) \quad |z| = 0 \Leftrightarrow x = 0, y = 0$$

$$\text{i.e., } |z| = 0 \Leftrightarrow z = 0,$$

- (v) $\overline{\overline{z}} = z$, (vi) $|\overline{z}| = |z|$, and
 (vii) $|z| = \sqrt{x^2 + y^2} \geq x$ i.e., $|z| \geq \operatorname{Re} z$

If z_1 and z_2 are two complex numbers, then

- (i) $|z_1 + z_2| \leq |z_1| + |z_2|$ (ii) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
 (iii) $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$, and (iv) $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$.

7.2 Inner Product Spaces

(Lucknow 2011)

Definition: Let $V(F)$ be a vector space where F is either the field of real numbers or the field of complex numbers. An **inner product** on V is a function from $V \times V$ into F which assigns to each ordered pair of vectors α, β in V a scalar (α, β) in such a way that

- (1) $(\alpha, \beta) = \overline{(\beta, \alpha)}$ [Here $\overline{(\beta, \alpha)}$ denotes the conjugate complex of the number (β, α)].
 (2) $(a\alpha + b\beta, \gamma) = a(\alpha, \gamma) + b(\beta, \gamma)$
 (3) $(\alpha, \alpha) \geq 0$ and $(\alpha, \alpha) = 0 \Rightarrow \alpha = \mathbf{0}$
 for any $\alpha, \beta, \gamma \in V$ and $a, b \in F$.

Also the vector space V is then said to be an **inner product space** with respect to the specified inner product defined on it.

It should be noted that in the above definition (α, β) does not denote the ordered pair of the vectors α and β . But it denotes the inner product of the vectors α and β . It is an element of F which has been assigned by the function (named as inner product) to the vectors α and β . Sometimes the inner product of the ordered pair of vectors α, β is also written as $(\alpha | \beta)$. If $F = \mathbf{R}$, then (α, β) is a real number and if $F = \mathbf{C}$, then (α, β) is a complex number.

If F is the field of real numbers, then the complex conjugate appearing in (1) is superfluous and (1) should be read as $(\alpha, \beta) = (\beta, \alpha)$. If F is the field of complex numbers, then from (1), we have $(\alpha, \alpha) = \overline{(\alpha, \alpha)}$ and therefore (α, α) is real. Thus (α, α) is always real whether $F = \mathbf{R}$ or $F = \mathbf{C}$. Therefore the inequality given in (3) makes sense.

If $V(F)$ is an inner product space, then it is called a **Euclidean space** if F is the field of real numbers. Also it is called a **Unitary space** if F is the field of complex numbers.

Note 1: The property (3) in the definition of inner product is called *non-negativity*. The property (2) is called the *linearity property*. If $F = \mathbf{R}$, then the property (1) is called *symmetry* and if $F = \mathbf{C}$, then it is called *conjugate symmetry*.

Note 2: If in an inner product space $V(F)$, the vector α is $\mathbf{0}$, then $(\alpha, \alpha) = 0$.

We have $(\mathbf{0}, \mathbf{0}) = (0 \mathbf{0}, \mathbf{0})$ [$\because 0 \mathbf{0} = \mathbf{0}$ in V]

$= 0 (\mathbf{0}, \mathbf{0})$ [by linearity property of inner product]

$= 0$ [$\because (\mathbf{0}, \mathbf{0}) \in F$ and therefore $0 (\mathbf{0}, \mathbf{0}) = 0$]

Illustration 1: On $V_n(\mathbf{C})$ there is an inner product which we call the **standard inner product**.

If $\alpha = (a_1, a_2, \dots, a_n), \beta = (b_1, b_2, \dots, b_n) \in V_n(\mathbf{C})$,

then we define

$$(\alpha, \beta) = a_1 \bar{b}_1 + a_2 \bar{b}_2 + \dots + a_n \bar{b}_n = \sum_{i=1}^n a_i \bar{b}_i. \quad \dots(1)$$

Let us see that all the postulates of an inner product hold in (1).

(i) Conjugate symmetry : From the definition of inner product given in (1), we have $(\beta, \alpha) = b_1 \bar{a}_1 + \dots + b_n \bar{a}_n$.

$$\begin{aligned} \therefore (\beta, \alpha) &= \overline{(a_1 \bar{b}_1 + \dots + a_n \bar{b}_n)} = \overline{(b_1 \bar{a}_1)} + \dots + \overline{(b_n \bar{a}_n)} \\ &= \bar{b}_1 \overline{(\bar{a}_1)} + \dots + \bar{b}_n \overline{(\bar{a}_n)} = \bar{b}_1 a_1 + \dots + \bar{b}_n a_n \\ &= a_1 \bar{b}_1 + \dots + a_n \bar{b}_n \quad [\because \text{multiplication in } \mathbf{C} \text{ is commutative}] \\ &= (\alpha, \beta). \end{aligned}$$

Thus $(\alpha, \beta) = \overline{(\beta, \alpha)}$.

(ii) Linearity : Let $\gamma = (c_1, \dots, c_n) \in V_n(\mathbf{C})$ and let $a, b \in \mathbf{C}$.

We have $a\alpha + b\beta = a(a_1, \dots, a_n) + b(b_1, \dots, b_n)$

$$= (aa_1 + bb_1, \dots, aa_n + bb_n).$$

$$\begin{aligned} \therefore (a\alpha + b\beta, \gamma) &= (aa_1 + bb_1) \bar{c}_1 + \dots + (aa_n + bb_n) \bar{c}_n \quad [\text{by (1)}] \\ &= (aa_1 \bar{c}_1 + \dots + aa_n \bar{c}_n) + (bb_1 \bar{c}_1 + \dots + bb_n \bar{c}_n) \\ &= a(a_1 \bar{c}_1 + \dots + a_n \bar{c}_n) + b(b_1 \bar{c}_1 + \dots + b_n \bar{c}_n) \\ &= a(\alpha, \gamma) + b(\beta, \gamma) \quad [\text{by (1)}] \end{aligned}$$

(iii) Non-negativity :

$$\begin{aligned} (\alpha, \alpha) &= a_1 \bar{a}_1 + \dots + a_n \bar{a}_n \quad [\text{by (1)}] \\ &= |a_1|^2 + \dots + |a_n|^2. \quad \dots(2) \end{aligned}$$

Now a_i is a complex number. Therefore $|a_i|^2 \geq 0$. Thus (2) is a sum of n non-negative real numbers and therefore it is ≥ 0 . Thus $(\alpha, \alpha) \geq 0$. Also $(\alpha, \alpha) = 0$

$$\Rightarrow |a_1|^2 + \dots + |a_n|^2 = 0$$

$$\Rightarrow \text{each } |a_i|^2 = 0 \text{ and so each } a_i = 0 \Rightarrow \alpha = \mathbf{0}.$$

Hence the product defined in (1) is an inner product on $V_n(\mathbf{C})$ and with respect to this inner product $V_n(\mathbf{C})$ is an inner product space.

If α, β are two vectors in $V_n(\mathbf{C})$, then the standard inner product of α and β is also called the **dot product** of α and β and is denoted by $\alpha \cdot \beta$. Thus if $\alpha = (a_1, \dots, a_n), \beta = (b_1, \dots, b_n)$ then $\alpha \cdot \beta = a_1 \bar{b}_1 + a_2 \bar{b}_2 + \dots + a_n \bar{b}_n$.

Illustration 2: If $\alpha = (a_1, a_2), \beta = (b_1, b_2) \in V_2(\mathbf{R})$, let us define

$$(\alpha, \beta) = a_1 b_1 - a_2 b_1 - a_1 b_2 + 4a_2 b_2. \quad \dots(1)$$

We shall show that all the postulates of an inner product hold good in (1).

(i) **Symmetry** : We have

$$\begin{aligned}(\beta, \alpha) &= b_1 a_1 - b_2 a_1 - b_1 a_2 + 4b_2 a_2 && [\text{from (1)}] \\&= a_1 b_1 - a_2 b_1 - a_1 b_2 + 4a_2 b_2 && [\because a_1, a_2, b_1, b_2 \in \mathbf{R}] \\&= (\alpha, \beta).\end{aligned}$$

(ii) **Linearity** : If $a, b \in \mathbf{R}$, we have

$$a\alpha + b\beta = a(a_1, a_2) + b(b_1, b_2) = (aa_1 + bb_1, aa_2 + bb_2).$$

Let $\gamma = (c_1, c_2) \in V_2(\mathbf{R})$. Then

$$\begin{aligned}(a\alpha + b\beta, \gamma) &= (aa_1 + bb_1)c_1 - (aa_2 + bb_2)c_1 - (aa_1 + bb_1)c_2 \\&\quad + 4(aa_2 + bb_2)c_2 \quad [\text{from (1)}] \\&= (aa_1c_1 - aa_2c_1 - aa_1c_2 + 4aa_2c_2) + (bb_1c_1 - bb_2c_1 - bb_1c_2 + 4bb_2c_2) \\&= a(a_1c_1 - a_2c_1 - a_1c_2 + 4a_2c_2) + b(b_1c_1 - b_2c_1 - b_1c_2 + 4b_2c_2) \\&= a(\alpha, \gamma) + b(\beta, \gamma).\end{aligned}$$

[from (1)]

(iii) **Non-negativity**: We have

$$\begin{aligned}(\alpha, \alpha) &= a_1a_1 - a_2a_1 - a_1a_2 + 4a_2a_2 = a_1^2 - 2a_1a_2 + 4a_2^2 \\&= (a_1 - a_2)^2 + 3a_2^2.\end{aligned} \quad \dots(2)$$

Now (2) is a sum of two non-negative real numbers. Therefore it is ≥ 0 . Thus $(\alpha, \alpha) \geq 0$.

$$\begin{aligned}\text{Also } (\alpha, \alpha) &= 0 && \Rightarrow (a_1 - a_2)^2 + 3a_2^2 = 0 \\ \Rightarrow (a_1 - a_2)^2 &= 0, 3a_2^2 = 0 && \Rightarrow a_1 - a_2 = 0, a_2 = 0 \\ \Rightarrow a_1 &= 0, a_2 = 0 && \Rightarrow \alpha = \mathbf{0}.\end{aligned}$$

Hence the product defined in (1) is an inner product on $V_2(\mathbf{R})$. Also with respect to this inner product $V_2(\mathbf{R})$ is an inner product space.

Note: There can be defined more than one inner product on a vector space. For example, we have the standard inner product on $V_2(\mathbf{R})$ and the inner product defined in example 3.

Illustration 3: Let $V(\mathbf{C})$ be the vector space of all continuous complex-valued functions on the unit interval, $0 \leq t \leq 1$. If $f(t), g(t) \in V$, let us define

$$(f(t), g(t)) = \int_0^1 f(t) \overline{g(t)} dt. \quad \dots(1)$$

We shall show that all the postulates of an inner product hold in (1).

(i) **Conjugate Symmetry**: We have

$$(g(t), f(t)) = \int_0^1 g(t) \overline{f(t)} dt \quad [\text{from (1)}]$$

$$\begin{aligned}\therefore \overline{(g(t), f(t))} &= \overline{\int_0^1 g(t) \overline{f(t)} dt} = \int_0^1 \overline{[g(t) \overline{f(t)}]} dt \\&= \int_0^1 \overline{g(t)} f(t) dt = \int_0^1 f(t) \overline{g(t)} dt = (f(t), g(t))\end{aligned}$$

(ii) **Linearity**: Let $a, b \in \mathbf{C}$ and $h(t) \in V$. Then

$$\begin{aligned}
 (af(t) + bg(t), h(t)) &= \int_0^1 [af(t) + bg(t)] \overline{h(t)} dt \\
 &= a \int_0^1 f(t) \overline{h(t)} dt + b \int_0^1 g(t) \overline{h(t)} dt \\
 &= a(f(t), h(t)) + b(g(t), h(t)).
 \end{aligned}$$

(iii) **Non-negativity:** We have

$$(f(t), f(t)) = \int_0^1 f(t) \overline{f(t)} dt = \int_0^1 |f(t)|^2 dt \quad \dots(2)$$

Since $|f(t)|^2 \geq 0$ for every t lying in the closed interval $[0, 1]$, therefore $(2) \geq 0$.

Thus $(f(t), f(t)) \geq 0$

Also $(f(t), f(t)) = 0$

$$\Rightarrow \int_0^1 |f(t)|^2 dt = 0$$

$$\Rightarrow |f(t)|^2 = 0 \text{ for every } t \text{ lying in } [0, 1]$$

$$\Rightarrow f(t) = 0 \text{ for every } t \text{ lying in } [0, 1] \Rightarrow f(t) = 0.$$

Hence the product defined in (1) is an inner product on $V(\mathbb{C})$.

7.3 Norm or Length of a Vector in an Inner Product Space

Consider the vector space $V_3(\mathbb{R})$ with standard inner product defined on it. If $\alpha = (a_1, a_2, a_3) \in V_3(\mathbb{R})$, we have $(\alpha, \alpha) = a_1^2 + a_2^2 + a_3^2$.

Now we know that in the three dimensional Euclidean space $\sqrt{a_1^2 + a_2^2 + a_3^2}$ is the length of the vector $\alpha = (a_1, a_2, a_3)$. Taking motivation from this fact, we make the following definition.

Definition: Let V be an inner product space. If $\alpha \in V$, then the norm or the length of the vector α , written as $\|\alpha\|$, is defined as the positive square root of (α, α) i.e.,

$$\|\alpha\| = \sqrt{(\alpha, \alpha)}.$$

Unit vector. Definition: Let V be an inner product space. If $\alpha \in V$ is such that $\|\alpha\| = 1$, then α is called a unit vector or is said to be normalized. Thus in an inner product space a vector is called a unit vector if its length is 1.

Theorem 1: In an inner product space $V(F)$, prove that

$$(i) (a\alpha - b\beta, \gamma) = a(\alpha, \gamma) - b(\beta, \gamma) \quad (ii) (\alpha, a\beta + b\gamma) = \overline{a}(\alpha, \beta) + \overline{b}(\alpha, \gamma).$$

Proof: (i) We have

$$\begin{aligned}
 (a\alpha - b\beta, \gamma) &= (a\alpha + (-b)\beta, \gamma) \\
 &= a(\alpha, \gamma) + (-b)(\beta, \gamma) && \text{[by linearity property]} \\
 &= a(\alpha, \gamma) - b(\beta, \gamma).
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad (\alpha, a\beta + b\gamma) &= \overline{(a\beta + b\gamma, \alpha)} && \text{[by conjugate symmetry]} \\
 &= \overline{a(\beta, \alpha) + b(\gamma, \alpha)} && \text{[by linearity property]}
 \end{aligned}$$

$$\begin{aligned}
 &= \overline{a}(\beta, \alpha) + \overline{b}(\gamma, \alpha) = \overline{a}(\beta, \alpha) + \overline{b}(\gamma, \alpha) \\
 &= \overline{a}(\alpha, \beta) + \overline{b}(\alpha, \gamma).
 \end{aligned}$$

Note 1: If $F = \mathbf{R}$, then the result (ii) can be simply read as

$$(\alpha, a\beta + b\gamma) = a(\alpha, \beta) + b(\alpha, \gamma).$$

Note 2: Similarly it can be proved that $(\alpha, a\beta - b\gamma) = \overline{a}(\alpha, \beta) - \overline{b}(\alpha, \gamma)$.

Also $(\alpha, \beta + \gamma) = (\alpha, 1\beta + 1\gamma) = \overline{1}(\alpha, \beta) + \overline{1}(\alpha, \gamma) = (\alpha, \beta) + (\alpha, \gamma)$.

Theorem 2: In an inner product space $V(F)$, prove that

(i) $\|\alpha\| \geq 0$; and $\|\alpha\| = 0$; if and only if $\alpha = \mathbf{0}$.

(ii) $\|a\alpha\| = |a| \cdot \|\alpha\|$.

Proof: (i) We have $\|\alpha\| = \sqrt{(\alpha, \alpha)}$ [by def. of norm]

$$\Rightarrow \|\alpha\|^2 = (\alpha, \alpha)$$

$$\Rightarrow \|\alpha\|^2 \geq 0 \quad [\because (\alpha, \alpha) \geq 0]$$

$$\Rightarrow \|\alpha\|^2 \geq 0$$

Also $(\alpha, \alpha) = 0$ iff $\alpha = \mathbf{0}$

$\therefore \|\alpha\|^2 = 0$ iff $\alpha = \mathbf{0}$ i.e., $\|\alpha\| = 0$ iff $\alpha = \mathbf{0}$.

Thus in an inner product space, $\|\alpha\| > 0$ iff $\alpha \neq \mathbf{0}$.

(ii) We have $\|a\alpha\|^2 = (a\alpha, a\alpha)$ [by def. of norm]

$$= a(\alpha, a\alpha) \quad [\text{by linearity property}]$$

$$= a\overline{a}(\alpha, \alpha) \quad [\text{by theorem 1}]$$

$$= |a|^2 \cdot \|\alpha\|^2.$$

$$\text{Thus } \|a\alpha\|^2 = |a|^2 \cdot \|\alpha\|^2$$

Taking square root, we get $\|a\alpha\| = |a| \cdot \|\alpha\|$.

Note: If α is any non-zero vector of an inner product space V , then $\frac{1}{\|\alpha\|}\alpha$ is a unit

vector in V . We have $\|\alpha\| \neq 0$ because $\alpha \neq \mathbf{0}$.

Therefore $\frac{1}{\|\alpha\|}\alpha \in V$.

$$\text{Now } \left(\frac{1}{\|\alpha\|}\alpha, \frac{1}{\|\alpha\|}\alpha \right) = \frac{1}{\|\alpha\|} \left(\alpha, \frac{1}{\|\alpha\|}\alpha \right) = \frac{1}{\|\alpha\|^2}(\alpha, \alpha) = \frac{1}{\|\alpha\|^2} \|\alpha\|^2 = 1.$$

Therefore $\left\| \frac{\alpha}{\|\alpha\|} \right\| = 1$ and thus $\frac{\alpha}{\|\alpha\|}$ is unit vector.

For example if $\alpha = (2, 1, 2)$ is a vector in $V_3(\mathbf{R})$ with standard inner product, then $\|\alpha\| = \sqrt{(\alpha, \alpha)} = \sqrt{4 + 1 + 4} = 3$.

Therefore $\frac{1}{3}(2, 1, 2)$ i.e., $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$ is a unit vector.

Theorem 3: Cauchy-Schwarz's Inequality. *In an inner product space $V(F)$, prove that*

$$|\langle \alpha, \beta \rangle| \leq \|\alpha\| \cdot \|\beta\|. \quad (\text{Lucknow 2011})$$

Proof: If $\alpha = \mathbf{0}$, then $\|\alpha\| = 0$. Also in that case

$$\langle \alpha, \beta \rangle = \langle \mathbf{0}, \beta \rangle = \langle \mathbf{0}, \mathbf{0} \rangle = 0 \quad \langle \mathbf{0}, \beta \rangle = 0.$$

$$\therefore |\langle \alpha, \beta \rangle| = 0.$$

Thus if $\alpha = \mathbf{0}$, then $|\langle \alpha, \beta \rangle| = 0$ and $\|\alpha\| \|\beta\| = 0$.

$\therefore$ the inequality $|\langle \alpha, \beta \rangle| \leq \|\alpha\| \cdot \|\beta\|$ is valid.

Now let $\alpha \neq \mathbf{0}$. Then $\|\alpha\| > 0$.

Therefore $\frac{1}{\|\alpha\|^2}$ is a positive real number. Consider the vector $\gamma = \beta - \frac{\langle \beta, \alpha \rangle}{\|\alpha\|^2} \alpha$.

We have

$$\begin{aligned} \langle \gamma, \gamma \rangle &= \left\langle \beta - \frac{\langle \beta, \alpha \rangle}{\|\alpha\|^2} \alpha, \beta - \frac{\langle \beta, \alpha \rangle}{\|\alpha\|^2} \alpha \right\rangle \\ &= \left\langle \beta, \beta - \frac{\langle \beta, \alpha \rangle}{\|\alpha\|^2} \alpha \right\rangle - \frac{\langle \beta, \alpha \rangle}{\|\alpha\|^2} \left\langle \alpha, \beta - \frac{\langle \beta, \alpha \rangle}{\|\alpha\|^2} \alpha \right\rangle \quad [\text{by linearity property}] \\ &= \langle \beta, \beta \rangle - \frac{\overline{\langle \beta, \alpha \rangle}}{\|\alpha\|^2} \langle \beta, \alpha \rangle - \frac{\langle \beta, \alpha \rangle}{\|\alpha\|^2} \langle \alpha, \beta \rangle + \frac{\langle \beta, \alpha \rangle \overline{\langle \beta, \alpha \rangle}}{\|\alpha\|^2 \cdot \|\alpha\|^2} \langle \alpha, \alpha \rangle \\ &= \|\beta\|^2 - \frac{\overline{\langle \beta, \alpha \rangle} \langle \beta, \alpha \rangle}{\|\alpha\|^2} - \frac{\langle \alpha, \beta \rangle \langle \alpha, \beta \rangle}{\|\alpha\|^2} + \frac{\overline{\langle \beta, \alpha \rangle} \langle \beta, \alpha \rangle}{\|\alpha\|^2 \|\alpha\|^2} \cdot \|\alpha\|^2 \\ &= \|\beta\|^2 - \frac{\langle \alpha, \beta \rangle \overline{\langle \alpha, \beta \rangle}}{\|\alpha\|^2}, \text{ the second and the fourth terms cancel} \\ &= \|\beta\|^2 - \frac{|\langle \alpha, \beta \rangle|^2}{\|\alpha\|^2}. \end{aligned}$$

$[\because z\bar{z} = |z|^2 \text{ if } z \in \mathbf{C}]$

But $\langle \gamma, \gamma \rangle = \|\gamma\|^2 \geq 0$

$$\therefore \|\beta\|^2 - \frac{|\langle \alpha, \beta \rangle|^2}{\|\alpha\|^2} \geq 0 \quad \text{or} \quad \|\beta\|^2 \cdot \|\alpha\|^2 \geq |\langle \alpha, \beta \rangle|^2$$

or $|\langle \alpha, \beta \rangle| \leq \|\alpha\| \cdot \|\beta\|$, taking square root of both sides.

Schwarz's inequality has very important applications in mathematics.

Theorem 4: Triangle inequality. *If α, β are vectors in an inner product space V , prove that*

$$\|\alpha + \beta\| \leq \|\alpha\| + \|\beta\|. \quad (\text{Lucknow 2007})$$

Proof: We have $\|\alpha + \beta\|^2 = \langle \alpha + \beta, \alpha + \beta \rangle$ [by def. of norm]

$$= \langle \alpha, \alpha + \beta \rangle + \langle \beta, \alpha + \beta \rangle \quad [\text{by linearity property}]$$

$$= \langle \alpha, \alpha \rangle + \langle \alpha, \beta \rangle + \langle \beta, \alpha \rangle + \langle \beta, \beta \rangle \quad [\text{by theorem 1}]$$

$$= \|\alpha\|^2 + \langle \alpha, \beta \rangle + \overline{\langle \alpha, \beta \rangle} + \|\beta\|^2 \quad [\because \langle \beta, \alpha \rangle = \overline{\langle \alpha, \beta \rangle}]$$

$$= \|\alpha\|^2 + 2\operatorname{Re} \langle \alpha, \beta \rangle + \|\beta\|^2 \quad [\because z + \bar{z} = 2\operatorname{Re} z]$$

$$\begin{aligned}
&\leq ||\alpha||^2 + 2|(\alpha, \beta)| + ||\beta||^2 && [\because \operatorname{Re} z \leq |z|] \\
&\leq ||\alpha||^2 + 2||\alpha|| \cdot ||\beta|| + ||\beta||^2 \\
&\quad [\because \text{by Schwarz inequality } |(\alpha, \beta)| \leq ||\alpha|| \cdot ||\beta||] \\
&= (||\alpha|| + ||\beta||)^2.
\end{aligned}$$

Thus $||\alpha + \beta||^2 \leq (||\alpha|| + ||\beta||)^2$.

Taking square root of both sides, we get

$$||\alpha + \beta|| \leq ||\alpha|| + ||\beta||.$$

Geometrical interpretation: Let α, β be the vectors in the inner product space $V_3(\mathbf{R})$ with standard inner product defined on it. Suppose the vectors α, β represent the sides AB and BC respectively of a triangle ABC in the three dimensional Euclidean space. Then $||\alpha|| = AB, ||\beta|| = BC$. Also the vector $\alpha + \beta$ represents the side AC of the triangle ABC and $||\alpha + \beta|| = AC$. Then from the above inequality, we have $AC \leq AB + BC$.

Special Cases Of Cauchy-Schwarz's Inequality.

Case I. Consider the vector space $V_n(\mathbf{C})$ with standard inner product defined on it.

Let $\alpha = (a_1, a_2, \dots, a_n)$ and $\beta = (b_1, b_2, \dots, b_n) \in V_n(\mathbf{C})$

Then $a_1, \dots, a_n, b_1, \dots, b_n$ are all complex numbers. (Lucknow 2006)

We have $(\alpha, \beta) = a_1 \bar{b}_1 + a_2 \bar{b}_2 + \dots + a_n \bar{b}_n$.

$$|(\alpha, \beta)|^2 = |a_1 \bar{b}_1 + a_2 \bar{b}_2 + \dots + a_n \bar{b}_n|^2.$$

Also $||\alpha||^2 = (\alpha, \alpha) = a_1 \bar{a}_1 + \dots + a_n \bar{a}_n = |a_1|^2 + \dots + |a_n|^2$.

Similarly $||\beta||^2 = |b_1|^2 + \dots + |b_n|^2$.

By Schwarz's inequality, we have $|(\alpha, \beta)|^2 \leq ||\alpha||^2 \cdot ||\beta||^2$

$\therefore$ If $a_1, \dots, a_n, b_1, \dots, b_n$ are complex numbers, then

$$\begin{aligned}
&|a_1 \bar{b}_1 + a_2 \bar{b}_2 + \dots + a_n \bar{b}_n|^2 \\
&\leq (|a_1|^2 + \dots + |a_n|^2)(|b_1|^2 + \dots + |b_n|^2).
\end{aligned}$$

This inequality is known as **Cauchy's inequality**.

If $a_1, \dots, a_n, b_1, \dots, b_n$ are all real numbers, then this inequality gives that

$$(a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2 \leq (a_1^2 + \dots + a_n^2)(b_1^2 + \dots + b_n^2).$$

Case II. Consider the vector space $V(\mathbf{C})$ of all continuous, complex-valued functions on the unit interval $0 \leq t \leq 1$, with inner product defined by

$$(f(t), g(t)) = \int_0^1 f(t) \overline{g(t)} dt.$$

We have $||f(t)||^2 = (f(t), f(t)) = \int_0^1 f(t) \overline{f(t)} dt = \int_0^1 |f(t)|^2 dt$.

Similarly $||g(t)||^2 = \int_0^1 |g(t)|^2 dt$.

Also $|(f(t), g(t))|^2 = \left| \int_0^1 f(t) \overline{g(t)} dt \right|^2$.

By Schwarz's inequality, we have

$$|(f(t), g(t))|^2 \leq \|f(t)\|^2 \cdot \|g(t)\|^2.$$

Therefore if $f(t), g(t)$ are continuous complex valued functions on the unit interval $[0, 1]$, then

$$\left| \int_0^1 f(t) \overline{g(t)} dt \right|^2 \leq \left(\int_0^1 |f(t)|^2 dt \right) \left(\int_0^1 |g(t)|^2 dt \right).$$

Case III. Consider the vector space $V_3(\mathbf{R})$ with standard inner product defined on it i.e., if

$$\alpha = (a_1, a_2, a_3), \beta = (b_1, b_2, b_3) \in V_3(\mathbf{R}),$$

$$\text{then } (\alpha, \beta) = a_1 b_1 + a_2 b_2 + a_3 b_3. \quad \dots(1)$$

We see that (1) is nothing but the dot product of two vectors α and β in three dimensional Euclidean space. If θ is the angle between the non-zero vectors α and β , then we know that

$$\begin{aligned} \cos^2 \theta &= \frac{(a_1 b_1 + a_2 b_2 + a_3 b_3)^2}{(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)} = \frac{\{(\alpha, \beta)\}^2}{(\alpha, \alpha)(\beta, \beta)} \\ &= \frac{|(\alpha, \beta)|^2}{\|\alpha\|^2 \|\beta\|^2} \quad [\because \text{if } (\alpha, \beta) \text{ is real then } \{(\alpha, \beta)\}^2 = |(\alpha, \beta)|^2] \end{aligned}$$

But by Schwarz's inequality, we have

$$|(\alpha, \beta)|^2 \leq \|\alpha\|^2 \cdot \|\beta\|^2$$

$$\therefore \cos^2 \theta \leq \frac{\|\alpha\|^2 \|\beta\|^2}{\|\alpha\|^2 \|\beta\|^2} \quad \text{i.e., } \cos^2 \theta \leq 1.$$

Thus the absolute value of the cosine of a real angle cannot be greater than 1.

Normed vector space. Definition: Let $V(F)$ be a vector space where F is either the field of real numbers or the field of complex numbers. Then V is said to be a normed vector space if to each vector α there corresponds a real number denoted by $\|\alpha\|$ called the norm of α in such a manner that

$$(1) \quad \|\alpha\| \geq 0 \text{ and } \|\alpha\| = 0 \Rightarrow \alpha = \mathbf{0}.$$

$$(2) \quad \|a\alpha\| = |a| \cdot \|\alpha\|, \forall a \in F.$$

$$(3) \quad \|\alpha + \beta\| \leq \|\alpha\| + \|\beta\|, \forall \alpha, \beta \in V.$$

We have shown in theorems 2 and 4 that the norm of an inner product space satisfies all the three conditions of the norm of a normed vector space. Hence every inner product space is a normed vector space.

But the converse is not true i.e., every normed vector space is not an inner product space. For example, consider a normed vector space $\mathbf{R}^2(\mathbf{R})$ with a norm defined by

$$\|\alpha\| = \max(|a|, |b|), \text{ where } \alpha = (a, b).$$

It is impossible to define an inner product, denoted by $(\cdot)$ on $\mathbf{R}^2(\mathbf{R})$ such that $(\alpha, \alpha) = \|\alpha\|^2$.

Distance in an inner product space:

Definition: Let $V(F)$ be an inner product space. Then we define the distance (α, β) between two vectors α and β by

$$d(\alpha, \beta) = \|\alpha - \beta\| = \sqrt{[(\alpha - \beta, \alpha - \beta)]}.$$

Theorem 5: In an inner product space $V(F)$ we define the distance $d(\alpha, \beta)$ from α to β by $d(\alpha, \beta) = \|\alpha - \beta\|$. Prove that

$$(1) \quad d(\alpha, \beta) \geq 0 \text{ and } d(\alpha, \beta) = 0 \text{ iff } \alpha = \beta.$$

$$(2) \quad d(\alpha, \beta) = d(\beta, \alpha).$$

$$(3) \quad d(\alpha, \beta) \leq d(\alpha, \gamma) + d(\gamma, \beta).$$

[Triangle inequality]

$$(4) \quad d(\alpha, \beta) = d(\alpha + \gamma, \beta + \gamma).$$

Proof: (1) We have $d(\alpha, \beta) = \|\alpha - \beta\|$.

[by definition]

Now $\|\alpha - \beta\| \geq 0$ and $\|\alpha - \beta\| = 0$ if and only if $\alpha - \beta = \mathbf{0}$.

$$d(\alpha, \beta) \geq 0 \text{ and } d(\alpha, \beta) = 0 \text{ if and only if } \alpha = \beta.$$

$$(2) \quad \text{We have } d(\alpha, \beta) = \|\alpha - \beta\|$$

(by def.)

$$= \|(-1)(\beta - \alpha)\|$$

$$= |-1| \|\beta - \alpha\|$$

$$[\because \|a\alpha\| = |a| \|\alpha\|]$$

$$= \|\beta - \alpha\| = d(\beta, \alpha)$$

$$(3) \quad \text{We have } d(\alpha, \beta) = \|\alpha - \beta\| = \|(\alpha - \gamma) + (\gamma - \beta)\|$$

$$\leq \|\alpha - \gamma\| + \|\gamma - \beta\|$$

[by theorem 4]

$$= d(\alpha, \gamma) + d(\gamma, \beta).$$

$$\therefore d(\alpha, \beta) \leq d(\alpha, \gamma) + d(\gamma, \beta)$$

$$(4) \quad \text{We have } d(\alpha, \beta) = \|(\alpha - \beta)\| = \|(a + \gamma) - (\beta + \gamma)\| = d(\alpha + \gamma, \beta + \gamma).$$

Matrix of an Inner Product:

Before defining the matrix of an inner product, we shall define some special types of matrices over the complex field $\mathbf{C}$.

Conjugate transpose of a Matrix: Definition: Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n over the field $\mathbf{C}$ of complex numbers. Then the matrix $[\bar{a}_{ji}]_{n \times n}$ is called the conjugate transpose of A and we shall denote it by A^* .

Thus in order to obtain A^* from A , we should first replace each element of A by its conjugate complex and then we should write transpose of the new matrix.

If in place of the field $\mathbf{C}$ we take the field $\mathbf{R}$ of the real numbers, then $\bar{a}_{ji} = a_{ji}$. So in this case A^* will simply be the transpose of the matrix A .

If $A = A^*$, then the matrix A is said to be a **self-adjoint** matrix or a **Hermitian matrix**.

Symmetric matrix: Definition: A square matrix A over a field F is said to be a symmetric matrix if it is equal to its transpose i.e., if

$$A = A^T.$$

Obviously a Hermitian matrix over the field of real numbers is a symmetric matrix.

Theorem: If $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is an ordered basis of a finite dimensional vector space V , then an inner product on V is completely determined by the values which it takes on pairs of vectors in B .

Proof: Suppose we are given a particular inner product on V . We shall show that this inner product on V is completely determined by the values

$$g_{ij} = (\alpha_j, \alpha_i) \text{ where } i = 1, 2, \dots, n; j = 1, 2, \dots, n.$$

Let $\alpha = \sum_{j=1}^n x_j \alpha_j$ and $\beta = \sum_{i=1}^n y_i \alpha_i$ be any two vectors in V . Then

$$(\alpha, \beta) = \left(\sum_{j=1}^n x_j \alpha_j, \beta \right) = \sum_{j=1}^n x_j (\alpha_j, \beta),$$

by linearity property of the inner product

$$= \sum_{j=1}^n x_j \left(\alpha_j, \sum_{i=1}^n y_i \alpha_i \right) = \sum_{j=1}^n x_j \sum_{i=1}^n \bar{y}_i (\alpha_j, \alpha_i)$$

[See theorem 1, part (ii) of 4.3]

$$= \sum_{j=1}^n \sum_{i=1}^n \bar{y}_i g_{ij} x_j$$

$$= Y^* GX, \quad \dots(1)$$

where X, Y are the coordinate matrices of α, β in the ordered basis B and G is the matrix $[g_{ij}]_{n \times n}$.

From (1) we observe that the inner product (α, β) is completely determined by the matrix G i.e., by the scalars g_{ij} . Hence the result of the theorem.

Definition: Let $B = \{\alpha_1, \dots, \alpha_n\}$ be an ordered basis for an n -dimensional inner product space V . The matrix $G = [g_{ij}]_{n \times n}$, where $g_{ij} = (\alpha_j, \alpha_i)$, is called the matrix of the underlined inner product in the ordered basis B .

We observe that $\overline{g_{ij}} = \overline{(\alpha_j, \alpha_i)} = (\alpha_i, \alpha_j) = g_{ji}$. Therefore the matrix G is such that $G^* = G$. Thus G is a Hermitian matrix.

Further we know that in an inner product space, $(\alpha, \alpha) > 0$ if $\alpha \neq 0$. Therefore from the relation (1) given above we observe that the matrix G is such that

$$X^* GX > 0, X \neq 0. \quad \dots(2)$$

From the relation (2) we conclude that the matrix G is invertible. For if G is not invertible then there exists an $X \neq 0$ such that $GX = 0$. For any such X the relation (2) is impossible. Hence G must be invertible.

In the last, from the relation (2) we observe that if $x_1, x_2, \dots, x_n$ are scalars not all of which are zero, then

$$\sum_{i=1}^n \sum_{j=1}^n \bar{x}_i g_{ij} x_j > 0. \quad \dots(3)$$

Now suppose that out of the n scalars $x_1, \dots, x_n$ we take $x_i = 1$ and each of the remaining $n - 1$ scalars is taken as 0. Then from (3) we conclude that $g_{ii} > 0$. Thus $g_{ii} > 0$ for each $i = 1, \dots, n$. Hence each entry along the principal diagonal of the matrix G is positive.

Illustrative Examples

Example 1: Show that we can always define an inner product on a finite dimensional vector space real or complex.

Solution: Let V be a finite dimensional vector space over the field F real or complex.

Let $B = \{\alpha_1, \dots, \alpha_n\}$ be a basis for V .

Let $\alpha, \beta \in V$. Then we can write $\alpha = a_1\alpha_1 + \dots + a_n\alpha_n$,

and $\beta = b_1\alpha_1 + \dots + b_n\alpha_n$

where $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are uniquely determined elements of F . Let us define

$$(\alpha, \beta) = a_1 \bar{b}_1 + \dots + a_n \bar{b}_n. \quad \dots(1)$$

We shall show that (1) satisfies all the conditions for an inner product.

(i) **Conjugate Symmetry:** We have

$$(\beta, \alpha) = b_1 \bar{a}_1 + \dots + b_n \bar{a}_n.$$

$$\begin{aligned} \therefore \quad \overline{(\beta, \alpha)} &= \overline{(b_1 \bar{a}_1 + \dots + b_n \bar{a}_n)} = \bar{b}_1 a_1 + \dots + \bar{b}_n a_n \\ &= a_1 \bar{b}_1 + \dots + a_n \bar{b}_n = (\alpha, \beta). \end{aligned}$$

(ii) **Linearity:** Let $\gamma = c_1\alpha_1 + \dots + c_n\alpha_n \in V$ and $a, b \in F$. We have

$$\begin{aligned} a\alpha + b\beta &= a(a_1\alpha_1 + \dots + a_n\alpha_n) + b(b_1\alpha_1 + \dots + b_n\alpha_n) \\ &= (aa_1 + bb_1)\alpha_1 + \dots + (aa_n + bb_n)\alpha_n. \end{aligned}$$

$$\begin{aligned} \therefore \quad (a\alpha + b\beta, \gamma) &= (aa_1 + bb_1)\bar{c}_1 + \dots + (aa_n + bb_n)\bar{c}_n \\ &= a(a_1\bar{c}_1 + \dots + a_n\bar{c}_n) + b(b_1\bar{c}_1 + \dots + b_n\bar{c}_n) \\ &= a(\alpha, \gamma) + b(\beta, \gamma). \end{aligned}$$

(iii) **Non-negativity:** We have

$$(\alpha, \alpha) = a_1 \bar{a}_1 + \dots + a_n \bar{a}_n = |a_1|^2 + \dots + |a_n|^2 \geq 0.$$

$$\text{Also } (\alpha, \alpha) = 0 \quad \Rightarrow \quad |a_1|^2 + \dots + |a_n|^2 = 0$$

$$\Rightarrow \quad |a_1|^2 = 0, \dots, |a_n|^2 = 0$$

$$\Rightarrow \quad a_1 = 0, \dots, a_n = 0 \quad \Rightarrow \quad \alpha = \mathbf{0}.$$

Hence (1) is an inner product on V .

Example 2: In $V_2(F)$ define for $\alpha = (a_1, a_2)$ and $\beta = (b_1, b_2)$,

$$(\alpha, \beta) = 2a_1 \bar{b}_1 + a_1 \bar{b}_2 + a_2 \bar{b}_1 + a_2 \bar{b}_2.$$

Show that this defines an inner product on $V_2(F)$.

Solution: (1) **Conjugate Symmetry:** We have

$$\begin{aligned} (\beta, \alpha) &= 2b_1 \bar{a}_1 + b_1 \bar{a}_2 + b_2 \bar{a}_1 + b_2 \bar{a}_2. \\ \therefore \overline{(\beta, \alpha)} &= \overline{(2b_1 \bar{a}_1 + b_1 \bar{a}_2 + b_2 \bar{a}_1 + b_2 \bar{a}_2)} = 2\bar{b}_1 a_1 + \bar{b}_1 a_2 + \bar{b}_2 a_1 + \bar{b}_2 a_2 \\ &= 2a_1 \bar{b}_1 + a_1 \bar{b}_2 + a_2 \bar{b}_1 + a_2 \bar{b}_2 = (\alpha, \beta). \end{aligned}$$

(2) **Linearity:** Let $a, b \in F$ and $\gamma = (c_1, c_2) \in V_2(F)$. Then

$$\begin{aligned} a\alpha + b\beta &= a(a_1, a_2) + b(b_1, b_2) = (aa_1 + bb_1, aa_2 + bb_2). \\ \therefore (a\alpha + b\beta, \gamma) &= 2(aa_1 + bb_1)\bar{c}_1 + (aa_1 + bb_1)\bar{c}_2 \\ &\quad + (aa_2 + bb_2)\bar{c}_1 + (aa_2 + bb_2)\bar{c}_2 \\ &= a(2a_1\bar{c}_1 + a_1\bar{c}_2 + a_2\bar{c}_1 + a_2\bar{c}_2) + b(2b_1\bar{c}_1 + b_1\bar{c}_2 + b_2\bar{c}_1 + b_2\bar{c}_2) \\ &= a(\alpha, \gamma) + b(\beta, \gamma). \end{aligned}$$

(3) **Non-negativity:** We have

$$\begin{aligned} (\alpha, \alpha) &= 2a_1\bar{a}_1 + a_1\bar{a}_2 + a_2\bar{a}_1 + a_2\bar{a}_2 = a_1\bar{a}_1 + (a_1 + a_2)(\bar{a}_1 + \bar{a}_2) \\ &= |a_1|^2 + (a_1 + a_2)\overline{(a_1 + a_2)} = |a_1|^2 + |a_1 + a_2|^2 \geq 0. \end{aligned}$$

$$\text{Also } (\alpha, \alpha) = 0 \Rightarrow |a_1|^2 + |a_1 + a_2|^2 = 0$$

$$a_1 = 0, a_1 + a_2 = 0 \Rightarrow a_1 = 0, a_2 = 0 \Rightarrow \alpha = \mathbf{0}.$$

Hence the result.

Example 3: If α and β are vectors in an inner product space then show that

$$\|\alpha + \beta\|^2 + \|\alpha - \beta\|^2 = 2\|\alpha\|^2 + 2\|\beta\|^2. \quad \text{(Lucknow 2010)} \\ \text{(Parallelogram law)}$$

$$\begin{aligned} \text{Solution: We have } \|\alpha + \beta\|^2 &= (\alpha + \beta, \alpha + \beta) \quad [\text{by def. of norm}] \\ &= (\alpha, \alpha + \beta) + (\beta, \alpha + \beta) \quad [\text{by linearity property}] \\ &= (\alpha, \alpha) + (\alpha, \beta) + (\beta, \alpha) + (\beta, \beta) \\ &= \|\alpha\|^2 + (\alpha, \beta) + (\beta, \alpha) + \|\beta\|^2 \quad \dots(1) \end{aligned}$$

$$\begin{aligned} \text{Also } \|\alpha - \beta\|^2 &= (\alpha - \beta, \alpha - \beta) = (\alpha, \alpha - \beta) - (\beta, \alpha - \beta) \\ &= (\alpha, \alpha) - (\alpha, \beta) - (\beta, \alpha) + (\beta, \beta) \\ &= \|\alpha\|^2 - (\alpha, \beta) - (\beta, \alpha) + \|\beta\|^2 \quad \dots(2) \end{aligned}$$

Adding (1) and (2), we get

$$\|\alpha + \beta\|^2 + \|\alpha - \beta\|^2 = 2\|\alpha\|^2 + 2\|\beta\|^2.$$

Geometrical interpretation: Let α and β be vectors in the vector space $V_2(\mathbf{R})$ with standard inner product defined on it. Suppose the vector α is represented by the side AB and the vector β by the side BC of a parallelogram $ABCD$. Then the vectors $\alpha + \beta$ and $\alpha - \beta$ represent the diagonals AC and DB of the parallelogram.

$$\therefore AC^2 + DB^2 = 2AB^2 + 2BC^2,$$

i.e., the sum of the squares of the sides of a parallelogram is equal to the sum of the squares of its diagonals.

Example 4: If α, β are vectors in an inner product space $V(F)$ and $a, b \in F$, then prove that

$$(i) \quad \|a\alpha + b\beta\|^2 = |a|^2 \|\alpha\|^2 + a\bar{b}(\alpha, \beta) + \bar{a}b(\beta, \alpha) + |b|^2 \|\beta\|^2,$$

$$(ii) \quad \operatorname{Re}(\alpha, \beta) = \frac{1}{4} \|\alpha + \beta\|^2 - \frac{1}{4} \|\alpha - \beta\|^2.$$

Solution: (i) We have $\|a\alpha + b\beta\|^2 = (a\alpha + b\beta, a\alpha + b\beta)$

$$\begin{aligned} &= a(\alpha, a\alpha + b\beta) + b(\beta, a\alpha + b\beta) \\ &= a\{\bar{a}(\alpha, \alpha) + \bar{b}(\alpha, \beta)\} + b\{\bar{a}(\beta, \alpha) + \bar{b}(\beta, \beta)\} \\ &= a\bar{a}(\alpha, \alpha) + a\bar{b}(\alpha, \beta) + b\bar{a}(\beta, \alpha) + b\bar{b}(\beta, \beta) \\ &= |a|^2 \|\alpha\|^2 + a\bar{b}(\alpha, \beta) + \bar{a}b(\beta, \alpha) + |b|^2 \|\beta\|^2 \end{aligned}$$

$$\begin{aligned} (ii) \quad \text{We have } \|\alpha + \beta\|^2 &= (\alpha + \beta, \alpha + \beta) = (\alpha, \alpha + \beta) + (\beta, \alpha + \beta) \\ &= (\alpha, \alpha) + (\alpha, \beta) + (\beta, \alpha) + (\beta, \beta) = \|\alpha\|^2 + (\alpha, \beta) + \overline{(\alpha, \beta)} + \|\beta\|^2 \\ &= \|\alpha\|^2 + 2\operatorname{Re}(\alpha, \beta) + \|\beta\|^2 \end{aligned} \quad \dots(1)$$

$$\begin{aligned} \text{Also } \|\alpha - \beta\|^2 &= (\alpha - \beta, \alpha - \beta) = (\alpha, \alpha - \beta) - (\beta, \alpha - \beta) \\ &= (\alpha, \alpha) - (\alpha, \beta) - (\beta, \alpha) + (\beta, \beta) = \|\alpha\|^2 - \{(\alpha, \beta) + (\beta, \alpha)\} + \|\beta\|^2 \\ &= \|\alpha\|^2 - \{(\alpha, \beta) + \overline{(\alpha, \beta)}\} + \|\beta\|^2 \\ &= \|\alpha\|^2 - 2\operatorname{Re}(\alpha, \beta) + \|\beta\|^2 \end{aligned} \quad \dots(2)$$

Subtracting (2) from (1), we get

$$\|\alpha + \beta\|^2 - \|\alpha - \beta\|^2 = 4\operatorname{Re}(\alpha, \beta).$$

$$\therefore \operatorname{Re}(\alpha, \beta) = \frac{1}{4} \|\alpha + \beta\|^2 - \frac{1}{4} \|\alpha - \beta\|^2.$$

Note: If $F = \mathbf{R}$, then $\operatorname{Re}(\alpha, \beta) = (\alpha, \beta)$.

Example 5: Suppose that α and β are vectors in an inner product space V . If $|(\alpha, \beta)| = \|\alpha\| \|\beta\|$ (that is, if the Schwarz inequality reduces to an equality), then α and β are linearly dependent.

Solution: It is given that $|(\alpha, \beta)| = \|\alpha\| \cdot \|\beta\|$(1)

If $\alpha = \mathbf{0}$, then (1) is satisfied. Therefore if α and β satisfy (1), then α can be $\mathbf{0}$ also. If $\alpha \neq \mathbf{0}$, the vectors α and β are linearly dependent because any set of vectors containing zero vector is linearly dependent.

Let us now suppose that α and β satisfy (1) and $\alpha \neq \mathbf{0}$.

If $\alpha \neq \mathbf{0}$, then $\|\alpha\| > 0$. Consider the vector

$$\gamma = \beta - \frac{(\alpha, \beta)}{\|\alpha\|^2} \alpha.$$

$$\text{We have } (\gamma, \gamma) = \left(\beta - \frac{(\beta, \alpha)}{\|\alpha\|^2} \alpha, \beta - \frac{(\beta, \alpha)}{\|\alpha\|^2} \alpha \right)$$

$$\begin{aligned}
&= \left(\beta, \beta - \frac{(\beta, \alpha)}{\|\alpha\|^2} \alpha \right) - \frac{(\alpha, \beta)}{\|\alpha\|^2} \left(\alpha, \beta - \frac{(\beta, \alpha)}{\|\alpha\|^2} \alpha \right) \\
&= (\beta, \beta) - \frac{\overline{(\beta, \alpha)}}{\|\alpha\|^2} (\beta, \alpha) - \frac{(\beta, \alpha)}{\|\alpha\|^2} (\alpha, \beta) + \frac{(\beta, \alpha) \overline{(\beta, \alpha)}}{\|\alpha\|^2 \|\alpha\|^2} (\alpha, \alpha) \\
&= \|\beta\|^2 - \frac{\overline{(\beta, \alpha)} (\beta, \alpha)}{\|\alpha\|^2} - \frac{\overline{(\alpha, \beta)} (\alpha, \beta)}{\|\alpha\|^2} + \frac{(\beta, \alpha) \overline{(\beta, \alpha)}}{\|\alpha\|^2 \|\alpha\|^2} \|\alpha\|^2 \\
&= \|\beta\|^2 - \frac{|(\alpha, \beta)|^2}{\|\alpha\|^2} = \frac{\|\beta\|^2 \|\alpha\|^2 - |(\alpha, \beta)|^2}{\|\alpha\|^2} = 0. \quad [\text{from (1)}]
\end{aligned}$$

Now $(\gamma, \gamma) = 0 \Rightarrow \gamma = \mathbf{0} \Rightarrow \beta - \frac{(\beta, \alpha)}{\|\alpha\|^2} \alpha = \mathbf{0}$

$\Rightarrow \alpha$ and β are linearly dependent.

Example 6: If in an inner product space the vectors α and β are linearly dependent, then

$$|(\alpha, \beta)| = \|\alpha\| \cdot \|\beta\|.$$

Solution: If $\alpha = \mathbf{0}$, then $|(\mathbf{0}, \beta)| = 0$ and $\|\alpha\| = 0$. Therefore the given result is true.

Also if $\beta = \mathbf{0}$, then $(\alpha, \mathbf{0}) = \overline{(\mathbf{0}, \alpha)} = \overline{0} = 0$ and $\|\beta\| = 0$.

So let us suppose that both α and β are non-zero vectors. Since they are linearly dependent, therefore $\alpha = c\beta$ where c is some scalar. We have

$$(\alpha, \beta) = (c\beta, \beta) = c (\beta, \beta) = c \|\beta\|^2.$$

$$\therefore |(\alpha, \beta)| = |c| \|\beta\|^2.$$

$$\text{Also } \|\alpha\| = \|c\beta\| = |c| \cdot \|\beta\|.$$

$$\therefore \|\alpha\| \|\beta\| = |c| \cdot \|\beta\|^2.$$

$$\text{Hence } |(\alpha, \beta)| = \|\alpha\| \cdot \|\beta\|.$$

Comprehensive Exercise 1

1. If $\alpha = (a_1, a_2, \dots, a_n), \beta = (b_1, b_2, \dots, b_n) \in V_n(\mathbf{R})$, then prove that

$$(\alpha, \beta) = a_1 b_1 + a_2 b_2 + \dots + a_n b_n \quad \dots(1)$$

defines an inner product on $V_n(\mathbf{R})$.

2. Show that for the vectors $\alpha = (x_1, x_2)$ and $\beta = (y_1, y_2)$ from $\mathbf{R}^2$ the following defines an inner product on $\mathbf{R}^2$:

$$(\alpha, \beta) = x_1 y_1 - x_2 y_1 - x_1 y_2 + 2x_2 y_2. \quad \dots(1)$$

3. Which of the following define inner products in $V_2(\mathbf{R})$? Give reasons.
[Assume $\alpha = (x_1, x_2), \beta = (y_1, y_2)$].

- (a) $(\alpha, \beta) = x_1 y_1 + 2x_1 y_2 + 2x_2 y_1 + 5x_2 y_2$
 (b) $(\alpha, \beta) = x_1^2 - 2x_1 y_2 - 2x_2 y_1 + y_1^2$.
 (c) $(\alpha, \beta) = 2x_1 y_1 + 5x_2 y_2$.
 (d) $(\alpha, \beta) = x_1 y_1 - 2x_1 y_2 - 2x_2 y_1 + 4x_2 y_2$.
4. Let $\alpha = (a_1, a_2)$ and $\beta = (b_1, b_2)$ be any two vectors $\in V_2(\mathbf{C})$. Prove that $(\alpha, \beta) = a_1 \bar{b}_1 + (a_1 + a_2)(\bar{b}_1 + \bar{b}_2)$ defines an inner product in $V_2(\mathbf{C})$. Show that the norm of the vector $(3, 4)$ in this inner product space is $\sqrt{58}$.
5. Let V be an inner product space.
 (a) Show that $(0, \beta) = 0$ for all β in V .
 (b) Show that if $(\alpha, \beta) = 0$ for all β in V , then $\alpha = 0$.
6. Let V be an inner product space, and α, β be vectors in V . Show that $\alpha = \beta$ if and only if $(\alpha, \gamma) = (\beta, \gamma)$ for every γ in V .
7. Normalize each of the following vectors in the Euclidean space $\mathbf{R}^3$:
 (i) $(2, 1, -1)$, (ii) $\left(\frac{1}{2}, \frac{2}{3}, -\frac{1}{4}\right)$.
8. Let $V(\mathbf{R})$ be a vector space of polynomials with inner product defined by

$$(f(t), g(t)) = \int_0^1 f(t) g(t) dt.$$

 If $f(t) = t^2 + t - 4$, $g(t) = t - 1$, then find (f, g) and $\|g\|$.
9. If α, β be vectors in a real inner product space such that $\|\alpha\| = \|\beta\|$, then prove that $(\alpha + \beta, \alpha - \beta) = 0$.
10. Prove that if α and β are vectors in a unitary space, then
 (i) $4(\alpha, \beta) = \|\alpha + \beta\|^2 - \|\alpha - \beta\|^2 + i\|\alpha + i\beta\|^2 - i\|\alpha - i\beta\|^2$.
 (ii) $(\alpha, \beta) = \operatorname{Re}(\alpha, \beta) + i \operatorname{Re}(\alpha, i\beta)$.
11. Show that any two vectors α, β of an inner product space are linearly dependent if and only if $|(\alpha, \beta)| = \|\alpha\| \cdot \|\beta\|$.
12. If in an inner product space $\|\alpha + \beta\| = \|\alpha\| + \|\beta\|$, then prove that the vectors α and β are linearly dependent. Give an example to show that the converse of this statement is false.

Answers 1

3. (a) and (c) are inner products, (b) and (d) are not
 7. (i) $\left(\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$ (ii) $\frac{1}{\sqrt{109}}(6, 8, -3)$
 8. $\frac{7}{4}; \frac{1}{\sqrt{3}}$

7.4 Orthogonality

(Lucknow 2008)

Definition: Let α and β be vectors in an inner product space V . Then α is said to be **orthogonal** to β if $(\alpha, \beta) = 0$.

The relation of orthogonality in an inner product space is **symmetric**. We have α is orthogonal to $\beta \Rightarrow (\alpha, \beta) = 0 \Rightarrow \overline{(\alpha, \beta)} = \overline{0} \Rightarrow (\beta, \alpha) = 0 \Rightarrow \beta$ is orthogonal to α .

So we can say that *two vectors α and β in an inner product space are **orthogonal** if $(\alpha, \beta) = 0$.*

Note 1: If α is orthogonal to β , then every scalar multiple of α is orthogonal to β . Let k be any scalar. Then

$$(k\alpha, \beta) = k(\alpha, \beta) = k \cdot 0 = 0. \quad [\because (\alpha, \beta) = 0]$$

Therefore $k\alpha$ is orthogonal to β .

Note 2: The zero vector is orthogonal to every vector. For every vector α in V , we have $(0, \alpha) = 0$.

Note 3: The zero vector is the only vector which is orthogonal to itself.

We have α is orthogonal to $\alpha \Rightarrow (\alpha, \alpha) = 0$

$\Rightarrow \alpha = 0$, by def. of an inner product space.

Definition: A vector α is said to be orthogonal to a set S if it is orthogonal to each vector in S . Similarly two subspaces are called orthogonal if every vector in each is orthogonal to every vector in the other.

Orthogonal set. Definition: Let S be a set of vectors in an inner product space V . Then S is said to be an **orthogonal set** provided that any two distinct vectors in S are orthogonal.

Theorem 1: Let $S = \{\alpha_1, \dots, \alpha_m\}$ be an orthogonal set of non-zero vectors in an inner product space V . If a vector β in V is in the linear span of S , then

$$\beta = \sum_{k=1}^m \frac{(\beta, \alpha_k)}{\|\alpha_k\|^2} \alpha_k.$$

Proof: Since $\beta \in L(S)$, therefore β can be expressed as a linear combination of the vectors in S . Let

$$\beta = c_1 \alpha_1 + \dots + c_m \alpha_m = \sum_{j=1}^m c_j \alpha_j.$$

We have for each k where $1 \leq k \leq m$,

$$(\beta, \alpha_k) = \left(\sum_{j=1}^m c_j \alpha_j, \alpha_k \right) = \sum_{j=1}^m c_j (\alpha_j, \alpha_k)$$

[by linearity property of inner product]

$$= c_k (\alpha_k, \alpha_k).$$

[On summing with respect to j . Note that S is an orthogonal set of non-zero vectors and so $(\alpha_j, \alpha_k) = 0$ if $j \neq k$]

Now $\alpha_k \neq \mathbf{0}$. Therefore $(\alpha_k, \alpha_k) \neq 0$. Thus $\|\alpha_k\|^2 \neq 0$.

$$\therefore c_k = \frac{(\beta, \alpha_k)}{\|\alpha_k\|^2}, 1 \leq k \leq m.$$

Putting these values of $c_1, \dots, c_m$ in (1), we get

$$\beta = \sum_{k=1}^m \frac{(\beta, \alpha_k)}{\|\alpha_k\|^2} \alpha_k.$$

Theorem 2: Any orthogonal set of non-zero vectors in an inner product space V is linearly independent.

Proof: Let S be an orthogonal set of non-zero vectors in an inner product space V . Let $S_1 = \{\alpha_1, \dots, \alpha_m\}$ be a finite subset of S containing m distinct vectors. Let

$$\sum_{j=1}^m c_j \alpha_j = c_1 \alpha_1 + \dots + c_m \alpha_m = \mathbf{0}. \quad \dots(1)$$

We have, for each k where $1 \leq k \leq m$,

$$\begin{aligned} \left(\sum_{j=1}^m c_j \alpha_j, \alpha_k \right) &= \sum_{j=1}^m c_j (\alpha_j, \alpha_k) = c_k (\alpha_k, \alpha_k) \quad [\because (\alpha_j, \alpha_k) = 0 \text{ if } j \neq k] \\ &= c_k \|\alpha_k\|^2. \end{aligned}$$

But from (1), $\sum_{j=1}^m c_j \alpha_j = \mathbf{0}$. Therefore $\left(\sum_{j=1}^m c_j \alpha_j, \alpha_k \right) = (\mathbf{0}, \alpha_k) = 0$.

$\therefore$ (1) implies that $c_k \|\alpha_k\|^2 = 0, 1 \leq k \leq m$

$$\Rightarrow c_k = 0 \quad [\because \alpha_k \neq \mathbf{0} \Rightarrow \|\alpha_k\|^2 \neq 0]$$

$\therefore$ The set S_1 is linearly independent. Thus every finite subset of S is linearly independent. Therefore S is linearly independent.

Orthonormal set. Definition: Let S be a set of vectors in an inner product space V . Then S is said to be an **orthonormal set** if

$$(i) \quad \alpha \in S \Rightarrow \|\alpha\| = 1 \text{ i.e., } (\alpha, \alpha) = 1,$$

and (ii) $\alpha, \beta \in S$ and $\alpha \neq \beta \Rightarrow (\alpha, \beta) = 0$.

Thus an orthonormal set is an orthogonal set with the additional property that each vector in it is of length 1. In other words a set S consisting of mutually orthogonal unit vectors is called an orthonormal set. Obviously an orthonormal set cannot contain zero vector because $\|\mathbf{0}\| = 0$.

A finite set $S = \{\alpha_1, \dots, \alpha_m\}$ is orthonormal if

$$(\alpha_i, \alpha_j) = \delta_{ij} \text{ where } \delta_{ij} = 1 \text{ if } i = j \text{ and } \delta_{ij} = 0 \text{ if } i \neq j.$$

Existence of an orthonormal set: Every inner product space V which is not equal to zero space possesses an orthonormal set.

Let $\mathbf{0} \neq \alpha \in V$. Then $\|\alpha\| \neq 0$. The set $\left\{ \frac{\alpha}{\|\alpha\|} \right\}$ containing only one vector is necessarily an orthonormal set.

$$\begin{aligned} \text{We have } \left(\frac{\alpha}{\|\alpha\|}, \frac{\alpha}{\|\alpha\|} \right) &= \frac{1}{\|\alpha\|} \left(\alpha, \frac{\alpha}{\|\alpha\|} \right) = \frac{1}{\|\alpha\|}, \frac{1}{\|\alpha\|} (\alpha, \alpha) \\ &= \frac{1}{\|\alpha\|^2} \|\alpha\|^2 = 1. \end{aligned}$$

Theorem 3: Let $S = \{\alpha_1, \dots, \alpha_m\}$ be an orthonormal set of vectors in an inner product space V . If a vector β is in the linear span of S , then $\beta = \sum_{k=1}^m (\beta, \alpha_k) \alpha_k$.

Proof: Since $\beta \in L(S)$, therefore β can be expressed as a linear combination of the vectors in S . Let

$$\beta = c_1 \alpha_1 + \dots + c_m \alpha_m = \sum_{j=1}^m c_j \alpha_j. \quad \dots(1)$$

We have for each k where $1 \leq k \leq m$,

$$\begin{aligned} (\beta, \alpha_k) &= \left(\sum_{j=1}^m c_j \alpha_j, \alpha_k \right) = \sum_{j=1}^m c_j (\alpha_j, \alpha_k) \\ &\quad \text{[by linearity of inner product]} \\ &= \sum_{j=1}^m c_j \delta_{jk} \quad [\because S \text{ is an orthonormal set}] \\ &= c_k. \quad \text{[On summing with respect to } j \text{ and remembering} \\ &\quad \text{that } \delta_{jk} = 1 \text{ if } j = k \text{ and } \delta_{jk} = 0 \text{ if } j \neq k] \end{aligned}$$

Putting the values of $c_1, \dots, c_m$ in (1), we get

$$\beta = \sum_{k=1}^m (\beta, \alpha_k) \alpha_k.$$

Theorem 4: If $S = \{\alpha_1, \dots, \alpha_m\}$ is an orthonormal set in V and if $\beta \in V$ then $\gamma = \beta - \sum_{i=1}^m (\beta, \alpha_i) \alpha_i$ is orthogonal to each of $\alpha_1, \dots, \alpha_m$ and, consequently, to the subspace spanned by S .

Proof: We have for each k where $1 \leq k \leq m$,

$$\begin{aligned} (\gamma, \alpha_k) &= \left(\beta - \sum_{i=1}^m (\beta, \alpha_i) \alpha_i, \alpha_k \right) \\ &= (\beta, \alpha_k) - \left(\sum_{i=1}^m (\beta, \alpha_i) \alpha_i, \alpha_k \right) \quad \text{[by linearity of inner product]} \\ &= (\beta, \alpha_k) - \sum_{i=1}^m (\beta, \alpha_i) (\alpha_i, \alpha_k) \quad \text{[by linearity of inner product]} \\ &= (\beta, \alpha_k) - \sum_{i=1}^m (\beta, \alpha_i) \delta_{ik} \quad [\because \alpha_i, \alpha_k \text{ belong to an orthonormal set}] \\ &= (\beta, \alpha_k) - (\beta, \alpha_k) \quad [\because \delta_{ik} = 1 \text{ if } i = k \text{ and } \delta_{ik} = 0 \text{ if } i \neq k] \\ &= 0. \end{aligned}$$

Hence the first part of the theorem.

Now let δ be any vector in the subspace spanned by S i.e., let $\delta \in L(S)$. Then $\delta = \sum_{i=1}^m a_i \alpha_i$ where each a_i is some scalar.

$$\text{We have } (\gamma, \delta) = \left(\gamma, \sum_{i=1}^m a_i \alpha_i \right) = \sum_{i=1}^m \overline{a_i} (\gamma, \alpha_i) = \sum_{i=1}^m \overline{a_i} 0 = 0.$$

Thus γ is orthogonal to every vector δ in $L(S)$. Therefore γ is orthogonal to $L(S)$.

Theorem 5: Any orthonormal set of vectors in an inner product space is linearly independent.

Proof: Let S be any orthonormal set of vectors in an inner product space V . Let $S_1 = \{\alpha_1, \dots, \alpha_m\}$ be a finite subset of S containing m distinct vectors. Let

$$\sum_{j=1}^m c_j \alpha_j = c_1 \alpha_1 + \dots + c_m \alpha_m = \mathbf{0}. \quad \dots(1)$$

We have, for each k where $1 \leq k \leq m$,

$$\begin{aligned} \left(\sum_{j=1}^m c_j \alpha_j, \alpha_k \right) &= \sum_{j=1}^m c_j (\alpha_j, \alpha_k) \quad [\text{by linearity of inner product}] \\ &= \sum_{j=1}^m c_j \delta_{jk} \quad [\because (\alpha_j, \alpha_k) = \delta_{jk}] \\ &= c_k. \quad [\text{On summing with respect to } j] \end{aligned}$$

But from (1), $\sum_{j=1}^m c_j \alpha_j = \mathbf{0}$. Therefore $\left(\sum_{j=1}^m c_j \alpha_j, \alpha_k \right) = (\mathbf{0}, \alpha_k) = 0$.

$\therefore$ (1) implies that $c_k = 0$ for each $1 \leq k \leq m$.

$\therefore$ the set S_1 is linearly independent. Thus every finite subset of S is linearly independent. Therefore S is linearly independent.

Complete orthonormal set: Definition: An orthonormal set is said to be complete if it is not contained in any larger orthonormal set.

Orthonormal dimension of a finite-dimensional vector space: Definition: Let V be a finite-dimensional inner product space of dimension n . If S is any orthonormal set in V then S is linearly independent. Therefore S cannot contain more than n distinct vectors because in an n -dimensional vector space a linearly independent set cannot contain more than n vectors.

The **orthogonal dimension** of V is defined as the largest number of vectors an orthonormal set in V can contain.

Obviously the orthogonal dimension of V will be $\leq n$ where n is the linear dimension of V .

The following theorem gives us a characterization of completeness i.e., it gives us equivalent definitions of completeness.

Theorem 6: If $S = \{\alpha_1, \dots, \alpha_n\}$ is any finite orthonormal set in an inner product space V , then the following six conditions on S are equivalent:

(i) The orthonormal set S is complete.

- (ii) If $(\beta, \alpha_i) = 0$ for $i = 1, \dots, n$, then $\beta = \mathbf{0}$.
- (iii) The linear span of S is equal to V i.e., $L(S) = V$.
- (iv) If $\beta \in V$, then $\beta = \sum_{i=1}^n (\beta, \alpha_i) \alpha_i$.
- (v) If β and γ are in V , then $(\beta, \gamma) = \sum_{i=1}^n (\beta, \alpha_i) (\alpha_i, \gamma)$.
- (vi) If β is in V , then $\sum_{i=1}^n |(\beta, \alpha_i)|^2 = \|\beta\|^2$.

Proof: (i) $\Rightarrow$ (ii).

It is given that S is a complete orthonormal set. Let $\beta \in V$ and $(\beta, \alpha_i) = 0$ for each $i = 1, \dots, n$.

Then β is orthogonal to each of the vectors $\alpha_1, \dots, \alpha_n$.

If $\beta \neq \mathbf{0}$, then adjoining the vector $\frac{\beta}{\|\beta\|}$ to the set S we obtain an orthonormal set larger than S . This contradicts the given statement that S is a complete orthonormal set. Hence $\beta = \mathbf{0}$.

(ii) $\Rightarrow$ (iii).

It is given that if $(\beta, \alpha_i) = 0$ for $i = 1, \dots, n$, then $\beta = \mathbf{0}$. To prove that $L(S) = V$.

Let γ be any vector in V . Consider the vector $\delta = \gamma - \sum_{i=1}^n (\gamma, \alpha_i) \alpha_i$.

We know that δ is orthogonal to each of the vectors $\alpha_1, \dots, \alpha_n$ i.e., $(\delta, \alpha_i) = 0$ for each $i = 1, \dots, n$. Therefore according to the given statement $\delta = \mathbf{0}$. This gives $\gamma = \sum_{i=1}^n (\gamma, \alpha_i) \alpha_i$. Thus every vector γ in V can be expressed as a linear combination of $\alpha_1, \dots, \alpha_n$. Therefore $L(S) = V$.

(iii) $\Rightarrow$ (iv).

It is given that $L(S) = V$. Therefore if $\beta \in V$, then β can be expressed as a linear combination of $\alpha_1, \dots, \alpha_n$. From theorem 3, we know that this expression for β will be

$$\beta = \sum_{i=1}^n (\beta, \alpha_i) \alpha_i.$$

(iv) $\Rightarrow$ (v).

It is given that if β is in V , then $\beta = \sum_{i=1}^n (\beta, \alpha_i) \alpha_i$. If γ is another vector in V , then

$$\gamma = \sum_{i=1}^n (\gamma, \alpha_i) \alpha_i.$$

$$\begin{aligned} \text{We have } (\beta, \gamma) &= \left(\sum_{i=1}^n (\beta, \alpha_i) \alpha_i, \sum_{j=1}^n (\gamma, \alpha_j) \alpha_j \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n (\beta, \alpha_i) \overline{(\gamma, \alpha_j)} (\alpha_i, \alpha_j) = \sum_{i=1}^n (\beta, \alpha_i) \overline{(\gamma, \alpha_i)} \end{aligned}$$

[On summing with respect to j]

$$= \sum_{i=1}^n (\beta, \alpha_i) (\alpha_i, \gamma).$$

(v) $\Rightarrow$ (vi).

It is given that if β and γ are in V , then $(\beta, \gamma) = \sum_{i=1}^n (\beta, \alpha_i) (\alpha_i, \gamma)$.

If β is in V then taking $\gamma = \beta$ in the given result, we get

$$(\beta, \beta) = \sum_{i=1}^n (\beta, \alpha_i) (\alpha_i, \beta) = \sum_{i=1}^n (\beta, \alpha_i) \overline{(\beta, \alpha_i)}$$

$$\Rightarrow \|\beta\|^2 = \sum_{i=1}^n |(\beta, \alpha_i)|^2.$$

(vi) $\Rightarrow$ (i).

It is given that if β is in V , then $\|\beta\|^2 = \sum_{i=1}^n |(\beta, \alpha_i)|^2$.. To prove that S is a complete orthonormal set.

Let S be not a complete orthonormal set i.e., let S be contained in a larger orthonormal set S_1 .

Then there exists a vector α_0 in S_1 such that $\|\alpha_0\| = 1$ and α_0 is orthogonal to each of the vectors $\alpha_1, \dots, \alpha_n$. Since α_0 is in V , therefore from the given condition,

$$\text{we have } \|\alpha_0\|^2 = \sum_{i=1}^n |(\alpha_0, \alpha_i)|^2 = 0.$$

This contradicts the fact that $\|\alpha_0\| = 1$. Hence S must be complete.

Corollary: Every complete orthonormal set in a finite-dimensional inner product space V forms a basis for V .

Proof: Let S be a complete orthonormal set in a finite-dimensional inner product space V . Then S is linearly independent. Also by the above theorem $L(S) = V$. Hence S must be a basis for V .

In the next theorem we shall prove that the orthogonal dimension of a finite dimensional inner product space is equal to its linear dimension.

Theorem 7: If V is an n -dimensional inner product space, then there exist complete orthonormal sets in V , and every complete orthonormal set in V contains exactly n elements. The orthogonal dimension of V is the same as its linear dimension.

Proof: Let $0 \neq \alpha \in V$. Then $\left\{ \frac{\alpha}{\|\alpha\|} \right\}$ is an orthonormal set in V . If it is not complete,

then we can enlarge it by adding one more vector to it so that the resulting set is also an orthonormal set. If this resulting orthonormal set is still not complete, then we enlarge it again. Thus we proceed by induction. Ultimately we must reach a complete orthonormal set because an orthonormal set is linearly independent and so it can contain at most n elements. Thus there exist complete orthonormal sets in V .

Now suppose $S = \{\alpha_1, \dots, \alpha_m\}$ is a complete orthonormal set in V . Then S is linearly independent. Also the linear span of S is V . Therefore S is a basis for V . Hence the number of vectors in S must be n . Thus we must have $m = n$.

Thus we have proved that there exist complete orthonormal sets in V and each of them will have n elements. Thus n is the largest number of vectors that an orthonormal set in V will contain. Therefore the orthogonal dimension of V is equal to n which is also the linear dimension of V .

Now we shall give an alternative proof of theorem 7. This proof will be a constructive proof *i.e.*, it will also give us a process to construct an orthonormal basis for a finite dimensional inner product space.

Orthonormal basis: Definition: A basis of an inner product space that consists of mutually orthogonal unit vectors is called an orthonormal basis.

Gram-Schmidt orthogonalization process:

Theorem 8: Every finite-dimensional inner product space has an orthonormal basis.

(Lucknow 2007)

Proof: Let V be an n -dimensional inner product space and let $B = \{\beta_1, \beta_2, \dots, \beta_n\}$ be a basis for V . From this set we shall construct an orthonormal set $B_1 = \{\alpha_1, \dots, \alpha_n\}$ of n distinct vectors by means of a construction known as Gram-Schmidt orthogonalization process. The main idea behind this construction is that each $\alpha_j, 1 \leq j \leq n$ will be in the linear span of $\beta_1, \dots, \beta_j$.

We have $\beta_1 \neq \mathbf{0}$ because the set B is linearly independent. Let $\alpha_1 = \frac{\beta_1}{\|\beta_1\|}$.

We have $(\alpha_1, \alpha_1) = \left(\frac{\beta_1}{\|\beta_1\|}, \frac{\beta_1}{\|\beta_1\|} \right) = \frac{1}{\|\beta_1\|^2} (\beta_1, \beta_1) = \frac{1}{\|\beta_1\|^2} \cdot \|\beta_1\|^2 = 1$.

Thus we have constructed an orthonormal set $\{\alpha_1\}$ containing one vector. Also α_1 is in the linear span of β_1 .

Now let $\gamma_2 = \beta_2 - (\beta_2, \alpha_1) \alpha_1$. By theorem 4, γ_2 is orthogonal to α_1 . Also $\gamma_2 \neq \mathbf{0}$ because if $\gamma_2 = \mathbf{0}$, then β_2 is a scalar multiple of α_1 and therefore of β_1 . But this is not possible because the vectors β_1 and β_2 are linearly independent. Hence $\gamma_2 \neq \mathbf{0}$. Let us now put $\alpha_2 = \frac{\gamma_2}{\|\gamma_2\|}$. Then $\|\alpha_2\| = 1$. Also α_2 is orthogonal to α_1 because α_2 is

simply a scalar multiple of γ_2 which is orthogonal to α_1 . Further $\alpha_2 \neq \alpha_1$. For otherwise β_2 will become a scalar multiple of β_1 . Thus $\{\alpha_1, \alpha_2\}$ is an orthonormal set containing two distinct vectors such that α_1 is in the linear span of β_1 and α_2 is in the linear span of β_1, β_2 .

The way ahead is now clear. Suppose that we have constructed an orthonormal set $\{\alpha_1, \dots, \alpha_k\}$ of k (where $k < n$) distinct vectors such that each α_j ($j = 1, \dots, k$) is a linear combination of $\beta_1, \dots, \beta_j$. Consider the vector

$$\gamma_{k+1} = \beta_{k+1} - (\beta_{k+1}, \alpha_1) \alpha_1 - (\beta_{k+1}, \alpha_2) \alpha_2 - \dots - (\beta_{k+1}, \alpha_k) \alpha_k \dots (1)$$

By theorem 4, γ_{k+1} is orthogonal to each of the vectors $\alpha_1, \dots, \alpha_k$. Suppose $\gamma_{k+1} = \mathbf{0}$. Then β_{k+1} is a linear combination of $\alpha_1, \dots, \alpha_k$. But according to our assumption each α_j ($j = 1, \dots, k$) is a linear combination of $\beta_1, \beta_2, \dots, \beta_j$. Therefore β_{k+1} is a linear combination of $\beta_1, \dots, \beta_k$. This is not possible because $\beta_1, \dots, \beta_k, \beta_{k+1}$ are linearly independent.

Therefore we must have $\gamma_{k+1} \neq \mathbf{0}$.

$$\text{Let us now put } \alpha_{k+1} = \frac{\gamma_{k+1}}{\|\gamma_{k+1}\|}. \quad \dots(2)$$

We have $\|\alpha_{k+1}\| = 1$. Also α_{k+1} is orthogonal to each of the vectors $\alpha_1, \dots, \alpha_k$, because α_{k+1} is simply a scalar multiple of γ_{k+1} which is orthogonal to each of the vectors $\alpha_1, \dots, \alpha_k$. Further obviously $\alpha_{k+1} \neq \alpha_j$, $j = 1, \dots, k$. For otherwise from (1) and (2), we see that β_{k+1} will become a linear combination of $\beta_1, \dots, \beta_k$. Also from (1) and (2), we see that α_{k+1} is in the linear span of $\beta_1, \dots, \beta_{k+1}$.

Thus we have been able to construct an orthonormal set

$$\{\alpha_1, \dots, \alpha_k, \alpha_{k+1}\}$$

containing $k+1$ distinct vectors such that α_j ($j = 1, 2, \dots, k+1$) is in the linear span of $\beta_1, \dots, \beta_j$. Our aim is now complete by induction. Thus continuing in this way we shall ultimately obtain an orthonormal set $B_1 = \{\alpha_1, \dots, \alpha_n\}$ containing n distinct vectors. The set B_1 is linearly independent because it is an orthonormal set. Therefore B_1 is a basis for V because the number of vectors in B_1 is equal to the dimension of V . Also the set B_1 is a complete orthonormal set because the maximum number of vectors in an orthonormal set in V can be n . Thus there exist complete orthonormal sets in V . Also the orthogonal dimension of V is equal to n i.e., equal to the linear dimension of V .

Note: In the above construction the vector α_2 will be $\frac{\gamma_2}{\|\gamma_2\|}$ where

$$\gamma_2 = \beta_2 - (\beta_2, \alpha_1) \alpha_1. \text{ Similarly the vector } \alpha_3 \text{ will be } \frac{\gamma_3}{\|\gamma_3\|} \text{ where}$$

$$\gamma_3 = \beta_3 - (\beta_3, \alpha_1) \alpha_1 - (\beta_3, \alpha_2) \alpha_2. \text{ Similarly the other vectors can be found.}$$

How to apply Gram-Schmidt orthogonalization process to numerical problems ?

Suppose $B = \{\beta_1, \beta_2, \dots, \beta_n\}$ is a given basis of a finite dimensional inner product space V . Let $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be an orthonormal basis for V which we are required to construct from the basis B . The vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ will be obtained in the following way.

$$\text{Take } \alpha_1 = \frac{\beta_1}{\|\beta_1\|},$$

$$\alpha_2 = \frac{\gamma_2}{\|\gamma_2\|} \text{ where } \gamma_2 = \beta_2 - (\beta_2, \alpha_1) \alpha_1,$$

$$\alpha_3 = \frac{\gamma_3}{\|\gamma_3\|} \text{ where } \gamma_3 = \beta_3 - (\beta_3, \alpha_1) \alpha_1 - (\beta_3, \alpha_2) \alpha_2,$$

.....

$$\alpha_n = \frac{\gamma_n}{\|\gamma_n\|} \text{ where } \gamma_n = \beta_n - (\beta_n, \alpha_1) \alpha_1 - (\beta_n, \alpha_2) \alpha_2 - \dots - (\beta_n, \alpha_{n-1}) \alpha_{n-1}.$$

Now we shall give an example to illustrate the Gram-Schmidt process.

Example 7: Apply the Gram-Schmidt process to the vectors $\beta_1 = (1, 0, 1), \beta_2 = (1, 0, -1), \beta_3 = (0, 3, 4)$, to obtain an orthonormal basis for $V_3(\mathbf{R})$ with the standard inner product. (Lucknow 2009)

Solution: We have $\|\beta_1\|^2 = (\beta_1, \beta_1) = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = (1)^2 + (0)^2 + (1)^2 = 2$.

$$\text{Let } \alpha_1 = \frac{\beta_1}{\|\beta_1\|} = \frac{1}{\sqrt{2}}(1, 0, 1) = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right).$$

$$\text{Now let } \gamma_2 = \beta_2 - (\beta_2, \alpha_1) \alpha_1.$$

$$\text{We have } (\beta_2, \alpha_1) = 1 \cdot \frac{1}{\sqrt{2}} + 0 \cdot 0 + (-1) \cdot \frac{1}{\sqrt{2}} = 0.$$

$$\therefore \gamma_2 = (1, 0, -1) - 0 \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) = (1, 0, -1).$$

$$\text{Now } \|\gamma_2\|^2 = (\gamma_2, \gamma_2) = (1)^2 + (0)^2 + (-1)^2 = 2.$$

$$\text{Let } \alpha_2 = \frac{\gamma_2}{\|\gamma_2\|} = \frac{1}{\sqrt{2}}(1, 0, -1) = \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right).$$

$$\text{Now let } \gamma_3 = \beta_3 - (\beta_3, \alpha_1) \alpha_1 - (\beta_3, \alpha_2) \alpha_2.$$

$$\text{We have } (\beta_3, \alpha_1) = \left((0, 3, 4), \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)\right) = 0 \cdot \frac{1}{\sqrt{2}} + 3 \cdot 0 + 4 \cdot \frac{1}{\sqrt{2}} = 2\sqrt{2}.$$

$$(\beta_3, \alpha_2) = \left((0, 3, 4), \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)\right) = 0 \cdot \frac{1}{\sqrt{2}} + 3 \cdot 0 - 4 \cdot \frac{1}{\sqrt{2}} = 2\sqrt{2}.$$

$$\therefore \gamma_3 = (0, 3, 4) - 2\sqrt{2} \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) + 2\sqrt{2} \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$$

$$\therefore = (0, 3, 4) - (2, 0, 2) + (2, 0, -2) = (0, 3, 0).$$

$$\text{Now } \|\gamma_3\|^2 = (\gamma_3, \gamma_3) = (0)^2 + (3)^2 + (0)^2 = 9.$$

$$\text{Put } \alpha_3 = \frac{\gamma_3}{\|\gamma_3\|} = \frac{1}{3}(0, 3, 0) = (0, 1, 0).$$

$$\text{Now } \{\alpha_1, \alpha_2, \alpha_3\} \text{ i.e., } \left\{\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right), (0, 1, 0)\right\}$$

is the required orthonormal basis for $V_3(\mathbf{R})$.

Theorem 9: Bessel's inequality:

(Lucknow 2007)

If $B = \{\alpha_1, \dots, \alpha_m\}$ is any finite orthonormal set in an inner product space V and if β is any vector in V , then $\sum_{i=1}^m |(\beta, \alpha_i)|^2 \leq \|\beta\|^2$.

Furthermore, equality holds if and only if β is in the subspace spanned by $\alpha_1, \dots, \alpha_m$.

Proof: Consider the vector $\gamma = \beta - \sum_{i=1}^m (\beta, \alpha_i) \alpha_i$.

$$\begin{aligned} \text{We have } \|\gamma\|^2 &= (\gamma, \gamma) = \left(\beta - \sum_{i=1}^m (\beta, \alpha_i) \alpha_i, \beta - \sum_{j=1}^m (\beta, \alpha_j) \alpha_j \right) \\ &= (\beta, \beta) - \sum_{i=1}^m (\beta, \alpha_i) (\alpha_i, \beta) - \sum_{j=1}^m \overline{(\beta, \alpha_j)} (\beta, \alpha_j) \\ &\quad + \sum_{i=1}^m \sum_{j=1}^m (\beta, \alpha_i) \overline{(\beta, \alpha_j)} (\alpha_i, \alpha_j) \\ &= (\beta, \beta) - \sum_{i=1}^m (\beta, \alpha_i) \overline{(\beta, \alpha_i)} - \sum_{j=1}^m \overline{(\beta, \alpha_j)} (\beta, \alpha_j) + \sum_{i=1}^m (\beta, \alpha_i) \overline{(\beta, \alpha_i)} \\ &\quad [\text{On summing with respect to } j \text{ and remembering that} \\ &\quad (\alpha_i, \alpha_j) = 1 \text{ when } j = i \text{ and } (\alpha_i, \alpha_j) = 0 \text{ when } j \neq i] \\ &= \|\beta\|^2 - \sum_{i=1}^m |(\beta, \alpha_i)|^2 - \sum_{i=1}^m |(\beta, \alpha_i)|^2 + \sum_{i=1}^m |(\beta, \alpha_i)|^2. \end{aligned}$$

$$\therefore \|\gamma\|^2 = \|\beta\|^2 - \sum_{i=1}^m |(\beta, \alpha_i)|^2 \quad \dots(1)$$

Now $\|\gamma\|^2 \geq 0$.

$$\therefore \|\beta\|^2 - \sum_{i=1}^m |(\beta, \alpha_i)|^2 \geq 0 \quad \text{or} \quad \sum_{i=1}^m |(\beta, \alpha_i)|^2 \leq \|\beta\|^2,$$

If the equality holds i.e., if $\sum_{i=1}^m |(\beta, \alpha_i)|^2 = \|\beta\|^2$ then from (1) we have $\|\gamma\|^2 = 0$.

This implies that $\gamma = \mathbf{0}$ i.e., $\beta = \sum_{i=1}^m (\beta, \alpha_i) \alpha_i$.

Thus if the equality holds, then β is a linear combination of $\alpha_1, \dots, \alpha_m$.

If β is a linear combination of $\alpha_1, \dots, \alpha_m$, then from theorem 3, we know that $\beta = \sum_{i=1}^m (\beta, \alpha_i) \alpha_i$. This implies that $\gamma = \mathbf{0}$ which in itself implies that $\|\gamma\|^2 = 0$.

Then from (1), we get $\sum_{i=1}^m |(\beta, \alpha_i)|^2 = \|\beta\|^2$

and thus the equality holds.

Note: Another statement of Bessel's inequality:

Let $\{\alpha_1, \dots, \alpha_m\}$ be an orthogonal set of non-zero vectors in an inner product space V . If β is any vector in V , then

$$\sum_{i=1}^m \frac{|(\beta, \alpha_i)|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2.$$

(Lucknow 2010)

Proof: Let $B = \{\delta_1, \dots, \delta_m\}$,

where $\delta_i = \frac{\alpha_i}{\|\alpha_i\|}$, $1 \leq i \leq m$.

Then $\|\delta_i\| = 1$. Thus the set B is an orthonormal set. Now proceeding as in the previous theorem, we get

$$\sum_{i=1}^m |(\beta, \delta_i)|^2 \leq \|\beta\|^2. \quad \dots(1)$$

Also $(\beta, \delta_i) = \left(\beta, \frac{\alpha_i}{\|\alpha_i\|} \right) = \frac{1}{\|\alpha_i\|} (\beta, \alpha_i)$.

$$|(\beta, \delta_i)|^2 = \frac{|(\beta, \alpha_i)|^2}{\|\alpha_i\|^2}. \quad \dots(2)$$

From (1) and (2), we get the required result.

Corollary: If V is finite dimensional and if $\{\alpha_1, \dots, \alpha_m\}$ is an orthonormal set in V such that $\sum_{i=1}^m |(\beta, \alpha_i)|^2 = \|\beta\|^2$ for every $\beta \in V$, prove that $\{\alpha_1, \dots, \alpha_m\}$ must be a basis of V .

Proof: Let β be any vector in V . Consider the vector

$$\gamma = \beta - \sum_{i=1}^m (\beta, \alpha_i) \alpha_i. \quad \dots(1)$$

As in the proof of Bessel's inequality, we have

$$\begin{aligned} \|\gamma\|^2 &= (\gamma, \gamma) = \|\beta\|^2 - \sum_{i=1}^m |(\beta, \alpha_i)|^2 \\ &= 0, \text{ by the given condition.} \end{aligned} \quad [\text{prove it here}]$$

$$\therefore \gamma = \mathbf{0} \text{ i.e., } \beta = \sum_{i=1}^m (\beta, \alpha_i) \alpha_i. \quad [\text{from (1)}]$$

Thus every vector β in V can be expressed as a linear combination of the vectors in the set $S = \{\alpha_1, \dots, \alpha_m\}$ i.e., $L(S) = V$. Also S is linearly independent because it is an orthonormal set. Hence S must be a basis for V .

Orthogonal complement. Definition: Let V be an inner product space, and let S be any set of vectors in V . The orthogonal complement of S , written as $S^\perp$ and read as S perpendicular, is defined by

$$S^\perp = \{\alpha \in V : (\alpha, \beta) = 0 \forall \beta \in S\}.$$

Thus $S^\perp$ is the set of all those vectors in V which are orthogonal to every vector in S .

Theorem 10: Let S be any set of vectors in an inner product space V . Then $S^\perp$ is a subspace of V .

Proof: We have, by definition,

$$S^\perp = \{\alpha \in V : (\alpha, \beta) = 0 \forall \beta \in S\}.$$

Since $(0, \beta) = 0 \quad \forall \quad \beta \in S$, therefore at least $0 \in S^\perp$ and thus $S^\perp$ is not empty.

Let $a, b \in F$ and $\gamma, \delta \in S^\perp$. Then $(\gamma, \beta) = 0 \quad \forall \quad \beta \in S$ and $(\delta, \beta) = 0 \quad \forall \quad \beta \in S$.

For every $\beta \in S$, we have

$$(a\gamma + b\delta, \beta) = a(\gamma, \beta) + b(\delta, \beta) = a0 + b0 = 0.$$

Therefore $a\gamma + b\delta \in S^\perp$. Hence $S^\perp$ is a subspace of V .

Note: The orthogonal complement of V is the zero subspace and the orthogonal complement of the zero subspace is V itself.

Orthogonal complement of an orthogonal complement:

Definition: Let S be any subset of an inner product space V . Then $S^\perp$ is a subset of V . We define $(S^\perp)^\perp$, written as $S^{\perp\perp}$, by

$$S^{\perp\perp} = \{\alpha \in V : (\alpha, \beta) = 0 \quad \forall \quad \beta \in S^\perp\}.$$

Obviously $S^{\perp\perp}$ is a subspace of V . Also it can be easily seen that $S \subseteq S^{\perp\perp}$.

Let $\alpha \in S$. Then $(\alpha, \beta) = 0 \quad \forall \quad \beta \in S^\perp$. Therefore by definition of $(S^\perp)^\perp$,

$$\alpha \in (S^\perp)^\perp. \text{ Thus } \alpha \in S \Rightarrow \alpha \in S^{\perp\perp}. \text{ Therefore } S \subseteq S^{\perp\perp}.$$

Theorem 11: Projection Theorem: The following theorem is known as the projection theorem and is very important.

Let W be any subspace of a finite dimensional inner product space V . Then

$$(i) \quad V = W \oplus W^\perp, \quad \text{and} \quad (ii) \quad W^{\perp\perp} = W.$$

Proof: (i) First we shall prove that $V = W + W^\perp$.

Since W is a subspace of a finite dimensional vector space V therefore W itself is also finite-dimensional. Let $\dim V = n$ and $\dim W = m$.

Now every finite-dimensional vector space possesses an orthonormal basis. Let $B_1 = \{\alpha_1, \dots, \alpha_m\}$ be an orthonormal basis for W .

Let β be any vector in V . Consider the vector

$$\gamma = \beta - \sum_{i=1}^m (\beta, \alpha_i) \alpha_i. \quad \dots(1)$$

By theorem 4, the vector γ is orthogonal to each of the vectors $\alpha_1, \dots, \alpha_m$ and consequently γ is orthogonal to the subspace W spanned by these vectors. Thus γ is orthogonal to every vector in W . Therefore $\gamma \in W^\perp$. Also the vector $\sum_{i=1}^m (\beta, \alpha_i) \alpha_i$ is

in W because it is a linear combination of vectors belonging to a basis for W .

Now from (1), we have

$$\beta = \left[\sum_{i=1}^m (\beta, \alpha_i) \alpha_i \right] + \gamma \text{ where } \sum_{i=1}^m (\beta, \alpha_i) \alpha_i \text{ is in } W \text{ and } \gamma \text{ is in } W^\perp.$$

Therefore $V = W + W^\perp$.

Now we shall prove that the subspaces W and $W^\perp$ are disjoint. Let $\alpha \in W \cap W^\perp$.

Then $\alpha \in W$ and $\alpha \in W^\perp$. Since $\alpha \in W^\perp$, therefore α is orthogonal to every vector in W . In particular α is orthogonal to α because $\alpha \in W$. Now $(\alpha, \alpha) = 0 \Rightarrow \alpha = \mathbf{0}$. Thus $\mathbf{0}$ is the only vector which belongs to both W and $W^\perp$. Hence W and $W^\perp$ are disjoint.

$$\therefore V = W \oplus W^\perp.$$

$$(ii) \quad \text{We have } V = W \oplus W^\perp. \quad \dots(2)$$

Now $W^\perp$ is also a subspace of V . Therefore taking $W^\perp$ in place of W and using the result (2), we get $V = W^\perp \oplus W^{\perp\perp}$(3)

Since V is the direct sum of W and $W^\perp$ and V is finite-dimensional, therefore
$$\dim V = \dim W + \dim W^\perp. \quad \dots(4)$$

Similarly from (3), we get $\dim V = \dim W^\perp + \dim W^{\perp\perp}. \quad \dots(5)$

From (4) and (5), we get $\dim W = \dim W^{\perp\perp}. \quad \dots(6)$

Now we shall prove that $W \subseteq W^{\perp\perp}$.

Let $\alpha \in W$. Then $(\alpha, \beta) = 0 \quad \forall \beta \in W^\perp$. Therefore by definition of $(W^\perp)^\perp$, $\alpha \in (W^\perp)^\perp$. Thus $\alpha \in W \Rightarrow \alpha \in W^{\perp\perp}$. Therefore $W \subseteq W^{\perp\perp}$.

Since $W \subseteq W^{\perp\perp}$, therefore W is a subspace of $W^{\perp\perp}$. Also $\dim W = \dim W^{\perp\perp}$. Hence $W = W^{\perp\perp}$.

Corollary: Let W be any subspace of a finite-dimensional inner product space V . Then
$$\dim W^\perp = \dim V - \dim W.$$

Proof: Since V is finite dimensional and $V = W \oplus W^\perp$,

therefore,
$$\dim V = \dim W + \dim W^\perp$$

$$\Rightarrow \dim W^\perp = \dim V - \dim W.$$

Definition: If W is a subspace of a finite dimensional inner product space V , then $V = W \oplus W^\perp$. Therefore every vector α in V can be uniquely expressed as $\alpha = \alpha_1 + \alpha_2$ where $\alpha_1 \in W$ and $\alpha_2 \in W^\perp$.

The vectors α_1 and α_2 are then called the **orthogonal projections** of α on the subspaces W and $W^\perp$.

Illustrative Examples

Example 8: State whether the following statement is true or false. Give reasons to support your answer.

α is an element of an n -dimensional unitary space V and α is perpendicular to n linearly independent vectors from V , then $\alpha = \mathbf{0}$.

Solution: True. Suppose α is perpendicular to n linearly independent vectors $\alpha_1, \dots, \alpha_n$.

Since V is of dimension n , therefore the n linearly independent vectors $\alpha_1, \dots, \alpha_n$ constitute a basis for V . So we can write $\alpha = a_1\alpha_1 + \dots + a_n\alpha_n$.

$$\begin{aligned} \text{Now } (\alpha, \alpha) &= (a_1\alpha_1 + \dots + a_n\alpha_n, \alpha) = a_1(\alpha_1, \alpha) + \dots + a_n(\alpha_n, \alpha) \\ &= a_1 \times 0 + \dots + a_n \times 0 \quad [\because \alpha \text{ is } \perp \text{ to each of the vectors } \alpha_1, \dots, \alpha_n] \\ &= 0. \end{aligned}$$

$$\therefore \alpha = 0.$$

Example 9: If α and β are orthogonal unit vectors (that is, $\{\alpha, \beta\}$ is an orthonormal set), what is the distance between α and β ?

Solution: If $d(\alpha, \beta)$ denotes the distance between α and β ,

$$\text{Then } d(\alpha, \beta) = \|\alpha - \beta\|.$$

$$\begin{aligned} \text{We have } \|\alpha - \beta\|^2 &= (\alpha - \beta, \alpha - \beta) = (\alpha, \alpha - \beta) - (\beta, \alpha - \beta) \\ &= (\alpha, \alpha) - (\alpha, \beta) - (\beta, \alpha) + (\beta, \beta) \\ &= \|\alpha\|^2 - 0 - 0 + \|\beta\|^2 \quad [\because \alpha \text{ is orthogonal to } \beta] \\ &= 1 + 1 \quad [\alpha \text{ and } \beta \text{ are unit vectors}] \\ &= 2. \end{aligned}$$

$$\therefore d(\alpha, \beta) = \|\alpha - \beta\| = \sqrt{2}.$$

Example 10: Prove that two vectors α and β in a real inner product space are orthogonal if and only if $\|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$.

Solution: Let α, β be two vectors in a real inner product space V . We have

$$\begin{aligned} \|\alpha + \beta\|^2 &= (\alpha + \beta, \alpha + \beta) = (\alpha, \alpha) + (\alpha, \beta) + (\beta, \alpha) + (\beta, \beta) \\ &= \|\alpha\|^2 + 2(\alpha, \beta) + \|\beta\|^2 \quad [\because (\beta, \alpha) = (\alpha, \beta)] \end{aligned}$$

Thus in a real inner product space V , we have

$$\|\alpha + \beta\|^2 = \|\alpha\|^2 + 2(\alpha, \beta) + \|\beta\|^2 \quad \dots(1)$$

If α and β are orthogonal, $(\alpha, \beta) = 0$.

Therefore from (1), we get $\|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$

Conversely, suppose that $\|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$.

Then from (1), we get $2(\alpha, \beta) = 0$ i.e., $(\alpha, \beta) = 0$.

Therefore α and β are orthogonal.

Note 1: The above result is known as the **Pythagorean theorem**. Its geometrical interpretation is that if ABC is a triangle in three dimensional Euclidean space, then the angle B is a right angle if and only if $AB^2 + BC^2 = AC^2$.

Note 2: If V is a complex inner product space, then the above result becomes false.

$$\text{In this case } \|\alpha + \beta\|^2 = \|\alpha\|^2 + (\alpha, \beta) + \overline{(\alpha, \beta)} + \|\beta\|^2$$

$$= \|\alpha\|^2 + 2\operatorname{Re}(\alpha, \beta) + \|\beta\|^2.$$

If α and β are orthogonal, then $(\alpha, \beta) = 0$.

So $\operatorname{Re}(\alpha, \beta) = 0$ and we get $\|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$

But if $\|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$, then we get $2\operatorname{Re}(\alpha, \beta) = 0$. This implies that $\operatorname{Re}(\alpha, \beta) = 0$.

This does not necessarily imply that $(\alpha, \beta) = 0$ i.e., α and β are orthogonal. Thus in a complex inner product space if α and β are orthogonal, then we have $\|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$. But if we have $\|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$, then it is not necessary that α and β are orthogonal.

Example 11: If α and β are vectors in a real inner product space, and if $\alpha + \beta$ is orthogonal to $\alpha - \beta$, then prove that $\|\alpha\| = \|\beta\|$. Interpret the result geometrically.

Solution: We have $\alpha + \beta$ is orthogonal to $\alpha - \beta$

$$\Rightarrow (\alpha - \beta, \alpha + \beta) = 0 \Rightarrow (\alpha, \alpha + \beta) - (\beta, \alpha + \beta) = 0$$

$$\Rightarrow (\alpha, \alpha) + (\alpha, \beta) - (\beta, \alpha) - (\beta, \beta) = 0$$

$$\Rightarrow \|\alpha\|^2 + (\alpha, \beta) - (\alpha, \beta) - \|\beta\|^2 = 0 \Rightarrow \|\alpha\|^2 = \|\beta\|^2 \Rightarrow \|\alpha\| = \|\beta\|.$$

Geometrical Interpretation: Let V be the three dimensional Euclidean space i.e., let V be the inner product space $V_3(\mathbf{R})$ with standard inner product defined on it. Let vectors α and β represent the sides AB and BC of a parallelogram $ABCD$. Then the vectors $\alpha + \beta$ and $\alpha - \beta$ are along the diagonals AC and DB of a parallelogram. If these diagonals are at right angles, then the length of α is equal to the length of β . So $AB = BC$ and the parallelogram is a rhombus.

Example 12: If S is a subset of an inner product space V , then prove that $S^\perp = S^{\perp\perp\perp}$.

Solution: We know that $S \subseteq S^{\perp\perp}$. Taking $S^\perp$ in place of S , we see that

$$S^\perp \subseteq (S^\perp)^{\perp\perp}$$

$$\text{i.e., } S^\perp \subseteq (S^\perp)^{\perp\perp\perp}. \quad \dots(1)$$

$$\text{Also } S \subseteq S^{\perp\perp} \Rightarrow (S^{\perp\perp})^\perp \subseteq S^\perp \quad [\because S_1 \subseteq S_2 \Rightarrow S_2^\perp \subseteq S_1^\perp]$$

$$\Rightarrow S^{\perp\perp\perp} \subseteq S^\perp. \quad \dots(2)$$

From (1) and (2), we get $S^\perp = S^{\perp\perp\perp}$.

Example 13: If $A = \{\alpha_1, \dots, \alpha_m\}$ is an orthonormal basis for subspace W of a finite dimensional inner product space V and $B = \{\beta_1, \dots, \beta_t\}$ is an orthonormal basis for $W^\perp$, then prove that $S = \{\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_t\}$ is an orthonormal basis for V .

Solution: First we shall prove that the set S is an orthonormal set. Obviously each vector in S is a unit vector. So it remains to prove that two distinct vectors in S are orthogonal.

Now $(\alpha_i, \alpha_j) = 0 \quad \forall \quad i = 1, \dots, m, j = 1, \dots, m, i \neq j. \quad [\because A \text{ is orthonormal}]$

Similarly $(\beta_i, \beta_j) = 0 \quad \forall \quad i = 1, \dots, t, j = 1, \dots, t, i \neq j$.

Lastly we are to verify that $(\alpha_i, \beta_j) = 0 \quad \forall i = 1, \dots, m$ and $j = 1, \dots, t$. But this is true since $\alpha_i \in W$ and $\beta_j \in W^\perp$.

Hence the set S is an orthogonal set. Therefore it is a linearly independent set. So S will be a basis for V if $L(S) = V$.

Let $\alpha \in V$. Since $V = W \oplus W^\perp$, therefore we can write $\alpha = \gamma + \delta$ where $\gamma \in W$ and $\delta \in W^\perp$. Now $\gamma \in W$ can be expressed as a linear combination of the vectors belonging to the basis A of W . Similarly $\delta \in W^\perp$ can be expressed as a linear combination of the vectors belonging to the basis B of $W^\perp$. Therefore α can be expressed as a linear combination of the vectors belonging to $A \cup B$ i.e., belonging to S . Therefore $L(S) = V$.

Hence S is a basis for V .

Example 14: If W_1 and W_2 are subspaces of a finite-dimensional inner product space, then

$$(i) \quad (W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp \qquad (ii) \quad (W_1 \cap W_2)^\perp = W_1^\perp + W_2^\perp$$

Solution: (i) We have $W_1 \subseteq W_1 + W_2$ (1)

$$\therefore (W_1 + W_2)^\perp \subseteq W_1^\perp.$$

Also $W_2 \subseteq W_1 + W_2$.

$$\therefore (W_1 + W_2)^\perp \subseteq W_2^\perp. \qquad \dots (2)$$

From (1) and (2), we conclude that $(W_1 + W_2)^\perp \subseteq W_1^\perp \cap W_2^\perp$.

Now we shall show that $W_1^\perp \cap W_2^\perp \subseteq (W_1 + W_2)^\perp$.

Let $\alpha \in W_1^\perp \cap W_2^\perp$. Then $\alpha \in W_1^\perp$ and $\alpha \in W_2^\perp$. Therefore α is orthogonal to every vector in W_1 and also to every vector in W_2 .

Let β be any vector in $W_1 + W_2$.

Then we can write $\beta = \gamma_1 + \gamma_2$ where $\gamma_1 \in W_1, \gamma_2 \in W_2$.

We have $(\alpha, \beta) = (\alpha, \gamma_1 + \gamma_2) = (\alpha, \gamma_1) + (\alpha, \gamma_2) = 0 + 0 = 0$.

Therefore α is orthogonal to every vector β in $W_1 + W_2$.

$$\text{So} \quad \alpha \in (W_1 + W_2)^\perp. \qquad \dots (3)$$

$$\text{Thus} \quad \alpha \in W_1^\perp \cap W_2^\perp \Rightarrow \alpha \in (W_1 + W_2)^\perp.$$

$$\therefore W_1^\perp \cap W_2^\perp \subseteq (W_1 + W_2)^\perp \qquad \dots (4)$$

From (3) and (4), we have $(W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp$.

(ii) $W_1^\perp$ and $W_2^\perp$ are also subspaces of V . Taking $W_1^\perp$ in place of W_1 and $W_2^\perp$ in place of W_2 in the result (i), we get $(W_1^\perp + W_2^\perp)^\perp = W_1^{\perp\perp} \cap W_2^{\perp\perp}$

$$\Rightarrow (W_1^\perp + W_2^\perp)^\perp = W_1 \cap W_2$$

[$\because V$ is finite dimensional and so $W_1^{\perp\perp} = W_1$ etc.]

$$\Rightarrow (W_1^\perp + W_2^\perp)^{\perp\perp} = (W_1 \cap W_2)^\perp$$

$$\Rightarrow W_1^\perp + W_2^\perp = (W_1 \cap W_2)^\perp$$

$$\Rightarrow (W_1 \cap W_2)^\perp = W_1^\perp + W_2^\perp.$$

Example 15: Find two mutually orthogonal vectors each of which is orthogonal to the vector $\alpha = (4, 2, 3)$ of $V_3(\mathbf{R})$ with respect to standard inner product.

Solution: Let $\beta = (x_1, x_2, x_3)$ be any vector orthogonal to the vector $(4, 2, 3)$. Then $4x_1 + 2x_2 + 3x_3 = 0$.

Obviously $\beta = (3, -3, -2)$ is a solution of this equation. We now require a third vector $\gamma = (y_1, y_2, y_3)$ orthogonal to both α and β . This means γ must be a solution vector of the system of equations

$$4y_1 + 2y_2 + 3y_3 = 0, 3y_1 - 3y_2 - 2y_3 = 0.$$

Obviously $\gamma = (5, 17, -18)$ is a solution of these equations. Thus, β and γ are orthogonal to each other and to α . The solution is, of course, by no means unique.

Comprehensive Exercise 2

- Let $V_3(\mathbf{R})$ be the inner product space relative to the standard inner product. Then find
 - two linearly independent vectors each of which is orthogonal to the vector $(1, 1, 2)$.
 - two mutually orthogonal vectors, each of which is orthogonal to $(5, 2, -1)$.
 - two mutually orthogonal unit vectors, each of which is orthogonal to $(2, -1, 3)$.
 - the projections of the vector $(3, 4, 1)$ onto the space spanned by $(1, 1, 1)$ and on its orthogonal complement.
- Let $V_3(\mathbf{R})$ be the inner product space with respect to the standard inner product and let W be the subspace of $V_3(\mathbf{R})$ spanned by the vector $\alpha = (2, -1, 6)$. Find the projections of the vector $\beta = (4, 1, 2)$ on W and $W^\perp$.
- Verify that the vectors $(\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3})$, $(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3})$ and $(\frac{2}{2}, \frac{2}{3}, -\frac{1}{3})$ form an orthonormal basis for $V_3(\mathbf{R})$ relative to the standard inner product.
- Given the basis $(1, 0, 0)$, $(1, 1, 0)$, $(1, 1, 1)$ for $V_3(\mathbf{R})$, construct from it by the Gram-Schmidt process an orthonormal basis relative to the standard inner product.
- Given the basis $(2, 0, 1)$, $(3, -1, 5)$, and $(0, 4, 2)$ for $V_3(\mathbf{R})$, construct from it by the Gram-Schmidt process an orthonormal basis relative to the standard inner product.
- Find an orthonormal basis of the vector space V of all real polynomials of degree not greater than two, in which the inner product is defined as

$$(\phi(x), \psi(x)) = \int_{-1}^1 \phi(x) \Psi(x) dx, \text{ where } \phi(x), \psi(x) \in V.$$

7. If α and β are vectors in a real inner product space, and if $\|\alpha\| = \|\beta\|$, then $\alpha - \beta$ and $\alpha + \beta$ are orthogonal. Interpret the result geometrically.
8. If V is an inner product space, then prove that
 - (i) $\{0\}^\perp = V$, (ii) $V^\perp = \{0\}$.
9. If V is an inner product space and S, S_1, S_2 are subsets of V , then
 - (i) $S_1 \subseteq S_2 \Rightarrow S_2^\perp \subseteq S_1^\perp$ (ii) $S^\perp = [L(S)]^\perp$
 - (iii) $L(S) \subseteq S^{\perp\perp}$
 - (iv) $L(S) = S^{\perp\perp}$ if V is finite dimensional.
10. Let W be a subspace of an inner product space V . If $\{\alpha_1, \dots, \alpha_n\}$ is a basis for W , then $\beta \in W^\perp$ if and only if $(\beta, \alpha_i) = 0 \quad \forall \quad i = 1, 2, \dots, n$.
11. Let V be a finite-dimensional inner product space, and let $\{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis for V . Show that for any vectors α, β in V

$$(\alpha, \beta) = \sum_{k=1}^n (\alpha, \alpha_k) \overline{(\beta, \alpha_k)}.$$
12. If $W_1, \dots, W_k$ are pairwise orthogonal subspaces in an inner product space V , and if $\alpha = \alpha_1 + \dots + \alpha_k$ with α_i in W_i for $i = 1, \dots, k$, then

$$\|\alpha\|^2 = \|\alpha_1\|^2 + \dots + \|\alpha_k\|^2$$
13. Find a vector of unit length which is orthogonal to the vector $\alpha = (2, -1, 6)$ of $V_3(\mathbf{R})$ with respect to standard inner product.
14. Two vectors α and β in a complex inner product space are orthogonal if and only if $\|a\alpha + b\beta\|^2 = \|a\alpha\|^2 + \|b\beta\|^2$ for all pairs of scalars a and b .
15. Let W be a finite dimensional proper subspace of an inner product space V . Let $\alpha \in V$ and $\alpha \notin W$. Show that there is a vector $\beta \in W$ such that $\alpha - \beta \perp W$.

Answers 2

1. (a), (b), (c). Check yourself
(d) $\left(\frac{8}{3}, \frac{8}{3}, \frac{8}{3}\right)$ and $\left(\frac{1}{3}, \frac{4}{3}, -\frac{5}{3}\right)$
2. $\left(\frac{38}{41}, -\frac{19}{41}, \frac{114}{41}\right), \left(\frac{126}{41}, \frac{60}{41}, -\frac{32}{41}\right)$
4. $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$
5. $\frac{1}{\sqrt{5}}(2, 0, 1), \frac{1}{\sqrt{(270)}}(-7, -5, 14), \frac{1}{3\sqrt{6}}(-1, 7, 2)$

6. $\left\{ \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{\sqrt{2}}x, \frac{\sqrt{5}}{\sqrt{8}}(3x^2 - 1) \right\}$

13. $\left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \right)$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- If in an inner product space $V(F)$, the vector α is $\mathbf{0}$ then $(\alpha, \alpha) =$
 - 0
 - 1
 - $\mathbf{0}$
 - None of these.
- In an inner product space $V(F)$, $\|\alpha + \beta\| \leq$
 - $\|\alpha\| \cdot \|\beta\|$
 - $\|\alpha\| / \|\beta\|$
 - $\|\alpha\| + \|\beta\|$
 - $\|\alpha\| - \|\beta\|$.
- The norm of $\alpha = (3, 4) \in \mathbf{R}^2$ with respect to the usual inner product is :
 - 5
 - 3
 - 1
 - None of these.
- Any two vectors α, β of an inner product space are linearly dependent if and only if :
 - $|(\alpha, \beta)| = 0$
 - $|(\alpha, \beta)| = \|\alpha\| \cdot \|\beta\|$
 - $|(\alpha, \beta)| = 1$
 - None of these.
- If α and β are orthogonal unit vectors, the distance between α and β is :
 - 2
 - $\sqrt{2}$
 - 0
 - 1.
- If S is a subset of an inner product space V , then $S^{\perp\perp} =$
 - S
 - $S^\perp$
 - $S^{\perp\perp}$
 - None of these.
- If W_1 and W_2 are subspaces of a finite-dimensional inner product space, then $(W_1 + W_2)^\perp =$
 - $W_1^\perp \cap W_2^\perp$
 - $W_1^\perp \cup W_2^\perp$
 - $W_1^\perp + W_2^\perp$
 - None of these.

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- In an inner product space $V(F)$, $(\alpha, \alpha) = 0 \Rightarrow \alpha = \dots\dots$

2. In an inner product space $V(F)$, if $\alpha \in V$ then $\|\alpha\| = \dots\dots\dots$
3. In an inner product space $V(F)$, $|\langle \alpha, \beta \rangle| \leq \dots\dots\dots$
4. Two vectors α and β in an inner product space are orthogonal if $\langle \alpha, \beta \rangle = \dots\dots\dots$
5. A set S consisting of mutually orthogonal unit vectors is called an $\dots\dots\dots$ set.
6. If $B = \{\alpha_1, \dots, \alpha_m\}$ is any finite orthonormal set in an inner product space V and if β is any vector in V , then $\sum_{i=1}^m |\langle \beta, \alpha_i \rangle|^2 \leq \dots\dots\dots$

True or False

Write 'T' for true and 'F' for false statement.

1. If $\alpha = (a_1, \dots, a_n)$ and $\beta = (b_1, \dots, b_n)$ are two vectors in $V_n(\mathbb{C})$, then the standard inner product of α and β is

$$\alpha \cdot \beta = a_1 b_1 + a_2 b_2 + \dots + a_n b_n.$$
2. In an inner product space $V(F)$, $\langle \alpha, a\beta + b\gamma \rangle = \bar{a} \langle \alpha, \beta \rangle + \bar{b} \langle \alpha, \gamma \rangle$.
3. In an inner product space $V(F)$, the distance $d(\alpha, \beta)$ from α to β is given by

$$d(\alpha, \beta) = \|\alpha - \beta\|.$$
4. If α and β are vectors in an inner product space, then

$$\|\alpha + \beta\|^2 + \|\alpha - \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$$
5. Any orthogonal set of non-zero vectors in an inner product space V is linearly dependent.
6. Every finite-dimensional inner product space has an orthonormal basis.
7. If V is an inner product space, then $V^\perp = \{0\}$.

Answers

Multiple Choice Questions

- | | | | |
|--------|--------|--------|--------|
| 1. (a) | 2. (c) | 3. (a) | 4. (b) |
| 5. (b) | 6. (b) | 7. (a) | |

Fill in the Blank(s)

- | | | |
|------|--|---------------------------------|
| 1. 0 | 2. $\sqrt{\langle \alpha, \alpha \rangle}$ | 3. $\ \alpha\ \cdot \ \beta\ $ |
| 4. 0 | 5. orthonormal | 6. $\ \beta\ ^2$ |

True or False

- | | | | |
|------|------|------|------|
| 1. F | 2. T | 3. T | 4. F |
| 5. F | 6. T | 7. T | |



Chapter

8

Bilinear, Quadratic and Hermitian Forms

8.1 External Direct Product of Two Vector Spaces

Suppose U and V are two vector spaces over the same field F . Let

$$W = U \times V \quad \text{i.e.,} \quad W = \{(\alpha, \beta) : \alpha \in U, \beta \in V\}.$$

If (α_1, β_1) and (α_2, β_2) are two elements in W , then we define their equality as follows :

$$(\alpha_1, \beta_1) = (\alpha_2, \beta_2) \text{ if } \alpha_1 = \alpha_2 \text{ and } \beta_1 = \beta_2 .$$

Also we define the sum of (α_1, β_1) and (α_2, β_2) as follows :

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1 + \beta_2).$$

If c is any element in F and (α, β) is any element in W , then we define scalar multiplication in W as follows :

$$c (\alpha, \beta) = (c\alpha, c\beta).$$

It can be easily shown that with respect to addition and scalar multiplication as defined above, W is a vector space over the field F . We call W as the external direct product of the vector spaces U and V and we shall write $W = U \oplus V$.

Now we shall consider some special type of scalar-valued functions on W known as *bilinear forms*.

8.2 Bilinear Forms

Definition: Let U and V be two vector spaces over the same field F . A bilinear form on $W = U \oplus V$ is a function f from W into F , which assigns to each element (α, β) in W a scalar $f(\alpha, \beta)$ in such a way that

$$f(a\alpha_1 + b\alpha_2, \beta) = af(\alpha_1, \beta) + bf(\alpha_2, \beta)$$

and $f(\alpha, a\beta_1 + b\beta_2) = af(\alpha, \beta_1) + bf(\alpha, \beta_2)$.

Here $f(\alpha, \beta)$ is an element of F . It denotes the image of (α, β) under the function f . Thus a bilinear form on W is a function from W into F which is linear as a function of either of its arguments when the other is fixed.

If $U = V$, then in place of saying that f is a bilinear form on $W = U \oplus V$, we shall simply say that f is a bilinear form on V .

Thus if V is a vector space over the field F , then a bilinear form on V is a function f , which assigns to each ordered pair of vectors α, β in V a scalar $f(\alpha, \beta)$ in F , and which satisfies

$$f(a\alpha_1 + b\alpha_2, \beta) = af(\alpha_1, \beta) + bf(\alpha_2, \beta)$$

and $f(\alpha, a\beta_1 + b\beta_2) = af(\alpha, \beta_1) + bf(\alpha, \beta_2)$.

Illustrative Examples

Example 1: Suppose V is a vector space over the field F . Let L_1, L_2 be linear functionals on V . Let f be a function from $V \times V$ into F defined as

$$f(\alpha, \beta) = L_1(\alpha) L_2(\beta).$$

Then f is a bilinear form on V .

Solution: If $\alpha, \beta \in V$, then $L_1(\alpha), L_2(\beta)$ are scalars. We have

$$\begin{aligned} f(a\alpha_1 + b\alpha_2, \beta) &= L_1(a\alpha_1 + b\alpha_2) L_2(\beta) \\ &= [aL_1(\alpha_1) + bL_1(\alpha_2)] L_2(\beta) \\ &= aL_1(\alpha_1) L_2(\beta) + bL_1(\alpha_2) L_2(\beta) \\ &= af(\alpha_1, \beta) + bf(\alpha_2, \beta). \end{aligned}$$

$$\begin{aligned} \text{Also } f(\alpha, a\beta_1 + b\beta_2) &= L_1(\alpha) L_2(a\beta_1 + b\beta_2) \\ &= L_1(\alpha) [aL_2(\beta_1) + bL_2(\beta_2)] \\ &= aL_1(\alpha) L_2(\beta_1) + bL_1(\alpha) L_2(\beta_2) \\ &= af(\alpha, \beta_1) + bf(\alpha, \beta_2). \end{aligned}$$

Hence f is a bilinear form on V .

Example 2: Suppose V is a vector space over the field F . Let T be a linear operator on V and f a bilinear form on V . Suppose g is a function from $V \times V$ into F defined as

$$g(\alpha, \beta) = f(T\alpha, T\beta).$$

Then g is a bilinear form on V .

Solution: We have $g(a\alpha_1 + b\alpha_2, \beta) = f(T(a\alpha_1 + b\alpha_2), T\beta)$
 $= f(aT\alpha_1 + bT\alpha_2, T\beta)$
 $= af(T\alpha_1, T\beta) + bf(T\alpha_2, T\beta)$
 $= ag(\alpha_1, \beta) + bg(\alpha_2, \beta).$

Also $g(\alpha, a\beta_1 + b\beta_2) = f(T\alpha, T(a\beta_1 + b\beta_2))$
 $= f(T\alpha, aT\beta_1 + bT\beta_2)$
 $= af(T\alpha, T\beta_1) + bf(T\alpha, T\beta_2)$
 $= ag(\alpha, \beta_1) + bg(\alpha, \beta_2).$

Hence g is a bilinear form on V .

8.3 Matrix of a Bilinear Form

Theorem 1: If U is an n -dimensional vector space with basis $\{\alpha_1, \dots, \alpha_n\}$, if V is an m -dimensional vector space with basis $\{\beta_1, \dots, \beta_m\}$, and if $\{a_{ij}\}$ is any set of nm scalars ($i = 1, \dots, n$; $j = 1, \dots, m$), then there is one and only one bilinear form f on $U \oplus V$ such that $f(\alpha_i, \beta_j) = a_{ij}$ for all i and j .

Proof: Let $\alpha = \sum_{i=1}^n x_i \alpha_i \in U$ and $\beta = \sum_{j=1}^m y_j \beta_j \in V$.

Let us define a function f from $U \times V$ into F such that

$$f(\alpha, \beta) = \sum_{i=1}^n \sum_{j=1}^m x_i y_j a_{ij}. \quad \dots(1)$$

We shall show that f is a bilinear form on $U \times V$.

Let $a, b \in F$ and let $\alpha_1, \alpha_2 \in U$.

$$\text{Let } \alpha_1 = \sum_{i=1}^n a_i \alpha_i, \alpha_2 = \sum_{i=1}^n b_i \alpha_i.$$

$$\text{Then } f(\alpha_1, \beta) = \sum_{i=1}^n \sum_{j=1}^m a_i y_j a_{ij}$$

$$\text{and } f(\alpha_2, \beta) = \sum_{i=1}^n \sum_{j=1}^m b_i y_j a_{ij}.$$

$$\text{Also } a\alpha_1 + b\alpha_2 = a \sum_{i=1}^n a_i \alpha_i + b \sum_{i=1}^n b_i \alpha_i = \sum_{i=1}^n (aa_i + bb_i) \alpha_i.$$

$$\begin{aligned} \therefore f(a\alpha_1 + b\alpha_2, \beta) &= \sum_{i=1}^n \sum_{j=1}^m (aa_i + bb_i) y_j a_{ij} \\ &= \sum_{i=1}^n \sum_{j=1}^m aa_i y_j a_{ij} + \sum_{i=1}^n \sum_{j=1}^m bb_i y_j a_{ij} \\ &= a \sum_{i=1}^n \sum_{j=1}^m a_i y_j a_{ij} + b \sum_{i=1}^n \sum_{j=1}^m b_i y_j a_{ij} \\ &= af(\alpha_1, \beta) + bf(\alpha_2, \beta). \end{aligned}$$

Similarly, we can prove that if $a, b \in F$, and $\beta_1, \beta_2 \in V$, then

$$f(\alpha, a\beta_1 + b\beta_2) = af(\alpha, \beta_1) + bf(\alpha, \beta_2).$$

Therefore f is a bilinear form on $U \times V$.

Now $\alpha_i = 0\alpha_1 + \dots + 0\alpha_{i-1} + 1\alpha_i + 0\alpha_{i+1} + \dots + 0\alpha_n$

and $\beta_j = 0\beta_1 + \dots + 0\beta_{j-1} + 1\beta_j + 0\beta_{j+1} + \dots + 0\beta_m$.

Therefore from (1), we have

$$f(\alpha_i, \beta_j) = a_{ij}.$$

Thus there exists a bilinear form f on $U \times V$ such that

$$f(\alpha_i, \beta_j) = a_{ij}.$$

Now to show that f is unique.

Let g be a bilinear form on $U \times V$ such that $g(\alpha_i, \beta_j) = a_{ij}$ (2)

If $\alpha = \sum_{i=1}^n x_i \alpha_i$ be in U and $\beta = \sum_{j=1}^m y_j \beta_j$ be in V , then

$$\begin{aligned} g(\alpha, \beta) &= g\left(\sum_{i=1}^n x_i \alpha_i, \sum_{j=1}^m y_j \beta_j\right) \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j g(\alpha_i, \beta_j) \quad [\because g \text{ is a bilinear form}] \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j a_{ij} \quad [\text{from (2)}] \\ &= f(\alpha, \beta). \quad [\text{from (1)}] \end{aligned}$$

$\therefore$ By the equality of two functions, we have

$$g = f.$$

Thus f is unique.

Matrix of a bilinear form: Definition: Let V be a finite-dimensional vector space and let $B = \{\alpha_1, \dots, \alpha_n\}$ be an ordered basis for V . If f is a bilinear form on V , the **matrix of f in the ordered basis B** is the $n \times n$ matrix $\mathbf{A} = [a_{ij}]_{n \times n}$ such that

$$f(\alpha_i, \alpha_j) = a_{ij}, \quad i = 1, \dots, n; \quad j = 1, \dots, n.$$

We shall denote this matrix $\mathbf{A}$ by $[f]_B$.

Rank of a bilinear form: Definition: The rank of a bilinear form is defined as the rank of the matrix of the form in any ordered basis.

Let us describe all bilinear forms on a finite-dimensional vector space V of dimension n .

If $\alpha = \sum_{i=1}^n x_i \alpha_i$, and $\beta = \sum_{j=1}^n y_j \alpha_j$ are vectors in V , then

$$\begin{aligned} f(\alpha, \beta) &= f\left(\sum_{i=1}^n x_i \alpha_i, \sum_{j=1}^n y_j \alpha_j\right) = \sum_{i=1}^n \sum_{j=1}^n x_i y_j f(\alpha_i, \alpha_j) \\ &= \sum_{i=1}^n \sum_{j=1}^n x_i y_j a_{ij} = \mathbf{X}' \mathbf{A} \mathbf{Y}, \end{aligned}$$

where $\mathbf{X}$ and $\mathbf{Y}$ are coordinate matrices of α and β in the ordered basis $\mathbf{B}$ and $\mathbf{X}'$ is the transpose of the matrix $\mathbf{X}$. Thus $f(\alpha, \beta) = [\alpha]_B' \mathbf{A} [\beta]_B$.

From the definition of the matrix of a bilinear form, we note that if f is a bilinear form on an n -dimensional vector space V over the field F and B is an ordered basis of V , then there exists a unique $n \times n$ matrix $\mathbf{A} = [a_{ij}]_{n \times n}$ over the field F such that

$$\mathbf{A} = [f]_B.$$

Conversely, if $\mathbf{A} = [a_{ij}]_{n \times n}$ be an $n \times n$ matrix over the field F , then from theorem 1, we see that there exists a unique bilinear form f on V such that

$$[f]_B = [a_{ij}]_{n \times n}.$$

If $\alpha = \sum_{i=1}^n x_i \alpha_i$, $\beta = \sum_{j=1}^n y_j \alpha_j$ are vectors in V , then the bilinear form f is de-

$$\text{fined as } f(\alpha, \beta) = \sum_{i=1}^n \sum_{j=1}^n x_i y_j a_{ij} = \mathbf{X}' \mathbf{A} \mathbf{Y}, \quad \dots(1)$$

where $\mathbf{X}$, $\mathbf{Y}$ are the coordinate matrices of α, β in the ordered basis B . Hence the bilinear forms on V are precisely those obtained from an $n \times n$ matrix as in (1).

Illustrative Examples

Example 3: Let f be the bilinear form on $V_2(\mathbf{R})$ defined by

$$f((x_1, y_1), (x_2, y_2)) = x_1 y_1 + x_2 y_2. \quad (\text{Lucknow 2011})$$

Find the matrix of f in the ordered basis

$$B = \{(1, -1), (1, 1)\} \text{ of } V_2(\mathbf{R}).$$

Solution: Let $B = \{\alpha_1, \alpha_2\}$

where $\alpha_1 = (1, -1), \alpha_2 = (1, 1)$.

We have $f(\alpha_1, \alpha_1) = f((1, -1), (1, -1)) = -1 - 1 = -2$,

$$f(\alpha_1, \alpha_2) = f((1, -1), (1, 1)) = -1 + 1 = 0,$$

$$f(\alpha_2, \alpha_1) = f((1, 1), (1, -1)) = 1 - 1 = 0,$$

$$f(\alpha_2, \alpha_2) = f((1, 1), (1, 1)) = 1 + 1 = 2.$$

$$\therefore [f]_B = \begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix}.$$

8.4 Bilinear Form Corresponding to a Given Matrix

Bilinear Form: Definition: Let V_m and V_n be two vector spaces over the same field F and let $\mathbf{A} = [a_{ij}]_{m \times n}$ be an $m \times n$ matrix over the field F .

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_m \end{bmatrix}$ and $\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix}$ be any two elements of V_m and V_n respectively so that

$\mathbf{X}^T$ = the transpose of the column matrix $\mathbf{X}$

$$= [x_1 \quad x_2 \quad \dots \quad x_m]$$

and $\mathbf{Y}^T = [y_1 \quad y_2 \quad \dots \quad y_n]$.

Then an expression of the form

$$b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} x_i y_j \quad \dots(1)$$

is called a bilinear form over the field F corresponding to the matrix $\mathbf{A}$.

It should be noted that $b(\mathbf{X}, \mathbf{Y})$ is an element of the field F and $b(\mathbf{X}, \mathbf{Y})$ is a mapping from

$$V_m \times V_n \rightarrow F.$$

The matrix $\mathbf{A}$ is called the **matrix** of the bilinear form (1) and the rank of the matrix $\mathbf{A}$ is called the rank of the bilinear form (1).

It should be noted that the coefficient of the product $x_i y_j$ in (1) is the element a_{ij} of the matrix $\mathbf{A}$ which occurs in the i th row and the j th column.

Symmetric bilinear form: The bilinear form (1) is said to be **symmetric** if its matrix $\mathbf{A}$ is a symmetric matrix.

If the field F is the real field $\mathbf{R}$, then the bilinear form (1) is said to be a **real bilinear form**. Thus in a real bilinear form $b(\mathbf{X}, \mathbf{Y})$ assumes real values.

If the vectors $\mathbf{X}$ and $\mathbf{Y}$ belong to the same vector space V_n over a field F , then $\mathbf{A}$ is a square matrix and $\mathbf{X}^T \mathbf{A} \mathbf{Y}$ is called a bilinear form on the vector space V_n over the field F .

In order to show that $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y}$ given by (1) is a bilinear form, first we show that the mapping $b(\mathbf{X}, \mathbf{Y})$ is a linear mapping from $V_m \rightarrow F$.

Let $\mathbf{X}_1, \mathbf{X}_2$ be any two elements of V_m and α, β be any two elements of the field F and let the vector $\mathbf{Y}$ be fixed. Then

$$\begin{aligned} b(\alpha \mathbf{X}_1 + \beta \mathbf{X}_2, \mathbf{Y}) &= (\alpha \mathbf{X}_1 + \beta \mathbf{X}_2)^T \mathbf{A} \mathbf{Y} \\ &= (\alpha \mathbf{X}_1^T + \beta \mathbf{X}_2^T) \mathbf{A} \mathbf{Y} \\ &= \alpha \mathbf{X}_1^T \mathbf{A} \mathbf{Y} + \beta \mathbf{X}_2^T \mathbf{A} \mathbf{Y} \\ &= \alpha b(\mathbf{X}_1, \mathbf{Y}) + \beta b(\mathbf{X}_2, \mathbf{Y}). \end{aligned}$$

$\therefore$ The mapping $b(\mathbf{X}, \mathbf{Y})$ is a linear mapping from $V_m \rightarrow F$.

Now we show that the mapping $b(\mathbf{X}, \mathbf{Y})$ is a linear mapping from $V_n \rightarrow F$.

Let $\mathbf{Y}_1, \mathbf{Y}_2$ be any two elements of V_n and α, β be any two elements of the field F and let the vector $\mathbf{X}$ be fixed. Then

$$\begin{aligned}
 b(\mathbf{X}, \alpha \mathbf{Y}_1 + \beta \mathbf{Y}_2) &= \mathbf{X}^T \mathbf{A} (\alpha \mathbf{Y}_1 + \beta \mathbf{Y}_2) \\
 &= \alpha \mathbf{X}^T \mathbf{A} \mathbf{Y}_1 + \beta \mathbf{X}^T \mathbf{A} \mathbf{Y}_2 \\
 &= \alpha b(\mathbf{X}, \mathbf{Y}_1) + \beta b(\mathbf{X}, \mathbf{Y}_2).
 \end{aligned}$$

$\therefore$ The mapping $b(\mathbf{X}, \mathbf{Y})$ is a linear mapping from $V_n \rightarrow F$.

Hence, $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} x_i y_j$ is a bilinear form.

Example of a real bilinear form:

$$\begin{aligned}
 \text{Let } b(\mathbf{X}, \mathbf{Y}) &= [x_1 \ x_2] \begin{bmatrix} 5 & 3 & 1 \\ 7 & 4 & 9 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\
 &= 5x_1 y_1 + 3x_1 y_2 + x_1 y_3 + 7x_2 y_1 + 4x_2 y_2 + 9x_2 y_3.
 \end{aligned}$$

Then $b(\mathbf{X}, \mathbf{Y})$ is an example of a real bilinear form.

The matrix $\begin{bmatrix} 5 & 3 & 1 \\ 7 & 4 & 9 \end{bmatrix}$ is the matrix of this bilinear form. The rank of this matrix is 2 and so the rank of this bilinear form is 2.

Equivalent Matrices: Definition: Let $\mathbf{A}$ and $\mathbf{B}$ be two $m \times n$ matrices over the same field F . The matrix $\mathbf{A}$ is said to be equivalent to the matrix $\mathbf{B}$ if there exist non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$ of orders m and n respectively such that

$$\mathbf{B} = \mathbf{P}^T \mathbf{A} \mathbf{Q}.$$

Equivalent Bilinear Forms: Definition: Let V_m and V_n be two vector spaces over the same field F and let $\mathbf{A}$ and $\mathbf{D}$ be two $m \times n$ matrices over the field F .

The bilinear form $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y}$ is said to be equivalent to the bilinear form $b(\mathbf{U}, \mathbf{V}) = \mathbf{U}^T \mathbf{D} \mathbf{V}$, where $\mathbf{X}$ and $\mathbf{U}$ are in V_m , $\mathbf{Y}$ and $\mathbf{V}$ are in V_n if there exist non-singular matrices $\mathbf{B}$ and $\mathbf{C}$ of orders m and n respectively such that

$$\mathbf{D} = \mathbf{B}^T \mathbf{A} \mathbf{C}$$

i.e., the matrices $\mathbf{A}$ and $\mathbf{D}$ are equivalent.

Thus, the bilinear form $b(\mathbf{U}, \mathbf{V})$ equivalent to the bilinear $b(\mathbf{X}, \mathbf{Y})$ is given by

$$b(\mathbf{U}, \mathbf{V}) = \mathbf{U}^T (\mathbf{B}^T \mathbf{A} \mathbf{C}) \mathbf{V} = \mathbf{U}^T \mathbf{D} \mathbf{V},$$

where $\mathbf{D} = \mathbf{B}^T \mathbf{A} \mathbf{C}$ and $\mathbf{B}$ and $\mathbf{C}$ are non-singular matrices of orders m and n respectively. The transformations of vectors yielding these equivalent bilinear forms are $\mathbf{X} = \mathbf{B} \mathbf{U}$, $\mathbf{Y} = \mathbf{C} \mathbf{V}$.

The matrices $\mathbf{B}$ and $\mathbf{C}$ are called the transformation matrices yielding these equivalent bilinear forms.

Equivalent Canonical form of a given bilinear form.

Let V_m and V_n be two vector spaces over the same field F and let A be an $m \times n$ matrix over the field F .

Let $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y}$,
 where $\mathbf{X}^T = [x_1 \quad x_2 \quad \dots \quad x_m]$,
 $\mathbf{Y}^T = [y_1 \quad y_2 \quad \dots \quad y_n]$

be a given bilinear form over the field F .

If the matrix $\mathbf{A}$ is of rank r , then there exist non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$ of orders m and n respectively such that

$$\mathbf{P}^T \mathbf{A} \mathbf{Q} = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$$

is in the normal form.

If we transform the vectors $\mathbf{X}$ and $\mathbf{Y}$ to $\mathbf{U}$ and $\mathbf{V}$ by the transformations

$$\mathbf{X} = \mathbf{P} \mathbf{U}, \quad \mathbf{Y} = \mathbf{Q} \mathbf{V},$$

then the bilinear form $b(\mathbf{U}, \mathbf{V})$ equivalent to the bilinear form $b(\mathbf{X}, \mathbf{Y})$ is given by

$$\begin{aligned} b(\mathbf{U}, \mathbf{V}) &= \mathbf{U}^T (\mathbf{P}^T \mathbf{A} \mathbf{Q}) \mathbf{V} = \mathbf{U}^T \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \mathbf{V} \\ &= u_1 v_1 + u_2 v_2 + \dots + u_r v_r, \end{aligned}$$

where $\mathbf{U} = [u_1 \quad u_2 \quad \dots \quad u_m]^T$ and $\mathbf{V} = [v_1 \quad v_2 \quad \dots \quad v_n]^T$.

The bilinear form $b(\mathbf{U}, \mathbf{V}) = u_1 v_1 + \dots + u_r v_r$ is called the *equivalent canonical form* or the *equivalent normal form* of the bilinear form

$$b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y}.$$

Congruent Matrices. Definition: A square matrix $\mathbf{B}$ of order n over a field F is said to be congruent to another square matrix $\mathbf{A}$ of order n over F , if there exists a non-singular matrix $\mathbf{P}$ over F such that

$$\mathbf{B} = \mathbf{P}^T \mathbf{A} \mathbf{P}.$$

Cogradient Transformations: Definition : Let $\mathbf{X}$ and $\mathbf{Y}$ be vectors belonging to the same vector space V_n over a field F and let $\mathbf{A}$ be a square matrix of order n over F . Let

$$b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y}$$

be a bilinear form on the vector space V_n over F .

Let $\mathbf{B}$ be a non-singular matrix of order n and let the vectors $\mathbf{X}$ and $\mathbf{Y}$ be transformed to the vectors $\mathbf{U}$ and $\mathbf{V}$ by the transformations

$$\mathbf{X} = \mathbf{B} \mathbf{U}, \quad \mathbf{Y} = \mathbf{B} \mathbf{V}.$$

Then the bilinear form $b(\mathbf{X}, \mathbf{Y})$ transforms to the equivalent bilinear form

$$b(\mathbf{U}, \mathbf{V}) = \mathbf{U}^T (\mathbf{B}^T \mathbf{A} \mathbf{B}) \mathbf{V} = \mathbf{U}^T \mathbf{D} \mathbf{V},$$

where $\mathbf{D} = \mathbf{B}^T \mathbf{A} \mathbf{B}$ and $\mathbf{U}$ and $\mathbf{V}$ are both n -vectors.

The bilinear form $\mathbf{U}^T \mathbf{D} \mathbf{V}$ is said to be congruent to the bilinear form $\mathbf{X}^T \mathbf{A} \mathbf{Y}$. Under such circumstances when $\mathbf{X}$ and $\mathbf{Y}$ are subjected to the same transformation $\mathbf{X} = \mathbf{B} \mathbf{U}$ and $\mathbf{Y} = \mathbf{B} \mathbf{V}$, we say that $\mathbf{X}$ and $\mathbf{Y}$ are transformed cogradiently.

Here, the matrix $\mathbf{A}$ is congruent to $\mathbf{B}$ because $\mathbf{D} = \mathbf{B}^T \mathbf{A} \mathbf{B}$, where $\mathbf{B}$ is non-singular.

Illustrative Examples

Example 4: Find the matrix $\mathbf{A}$ of each of the following bilinear forms $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y}$.

- (i) $3x_1 y_1 + x_1 y_2 - 2x_2 y_1 + 3x_2 y_2 - 3x_1 y_3$
(ii) $2x_1 y_1 + x_1 y_2 + x_1 y_3 + 3x_2 y_1 - 2x_2 y_3 + x_3 y_2 - 5x_3 y_3$

Which of the above forms is symmetric?

Solution: (i) The element a_{ij} of the matrix $\mathbf{A}$ is the coefficient of $x_i y_j$ in the given bilinear form.

$$\therefore \mathbf{A} = \begin{bmatrix} 3 & 1 & -3 \\ -2 & 3 & 0 \end{bmatrix}.$$

The matrix $\mathbf{A}$ is not a symmetric matrix. So the given bilinear form is not symmetric.

(ii) The element a_{ij} of the matrix $\mathbf{A}$ which occurs in the i th row and the j th column is the coefficient of $x_i y_j$ in the given bilinear form.

$$\therefore \mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 0 & -2 \\ 0 & 1 & -5 \end{bmatrix}.$$

The matrix $\mathbf{A}$ is not symmetric. So the given bilinear form is not symmetric.

Example 5: Transform the bilinear form $\mathbf{X}^T \mathbf{A} \mathbf{Y}$ to the equivalent canonical form where

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}.$$

Solution: We write $\mathbf{A} = \mathbf{I}_3 \mathbf{A} \mathbf{I}_3$ i.e.,

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_1 \leftrightarrow R_3$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 4 & 2 & 2 \\ 2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_2 \rightarrow R_2 - 4R_1$, $R_3 \rightarrow R_3 - 2R_1$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -6 & -6 \\ 0 & -3 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -4 \\ 1 & 0 & -2 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $C_2 \rightarrow C_2 - 2C_1, C_3 \rightarrow C_3 - 2C_1$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -6 & -6 \\ 0 & -3 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -4 \\ 1 & 0 & -2 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_2 \rightarrow -\frac{1}{6} R_2, R_3 \rightarrow -\frac{1}{3} R_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{6} & \frac{2}{3} \\ -\frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix} A \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_3 \rightarrow R_3 - R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{6} & \frac{2}{3} \\ -\frac{1}{3} & \frac{1}{6} & 0 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $C_3 \rightarrow C_3 - C_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{6} & \frac{2}{3} \\ -\frac{1}{3} & \frac{1}{6} & 0 \end{bmatrix} A \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\therefore \mathbf{P}^T \mathbf{A} \mathbf{Q} = \begin{bmatrix} \mathbf{I}_2 & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}, \text{ where } \mathbf{P} = \begin{bmatrix} 0 & 0 & -\frac{1}{3} \\ 0 & -\frac{1}{6} & \frac{1}{6} \\ 1 & \frac{2}{3} & 0 \end{bmatrix}$$

and $\mathbf{Q} = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$

Hence if the vectors $\mathbf{X} = [x_1 \ x_2 \ x_3]^T$ and $\mathbf{Y} = [y_1 \ y_2 \ y_3]^T$ are transformed to the vectors $\mathbf{U} = [u_1 \ u_2 \ u_3]^T$ and $\mathbf{V} = [v_1 \ v_2 \ v_3]^T$ respectively by the transformations $\mathbf{X} = \mathbf{P}\mathbf{U}$, $\mathbf{Y} = \mathbf{Q}\mathbf{V}$,

then the bilinear form

$$[x_1 \ x_2 \ x_3] \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

transforms to the equivalent canonical form

$$[u_1 \ u_2 \ u_3] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = u_1 v_1 + u_2 v_2.$$

8.5 Quadratic Forms

Definition: An expression of the form $\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$, where a_{ij} 's are elements of a field F , is called a quadratic form in the n variables $x_1, x_2, \dots, x_n$ over the field F .

Real Quadratic Form: Definition : An expression of the form

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j,$$

where a_{ij} 's are all real numbers, is called a real quadratic form in the n variables $x_1, x_2, \dots, x_n$.

For example,

- (i) $2x^2 + 7xy + 5y^2$ is a real quadratic form in the two variables x and y .
- (ii) $2x^2 - y^2 + 2z^2 - 2yz - 4zx + 6xy$ is a real quadratic form in the three variables x, y and z .
- (iii) $x_1^2 - 2x_2^2 + 4x_3^2 - 4x_4^2 - 2x_1 x_2 + 3x_1 x_4 + 4x_2 x_3 - 5x_3 x_4$ is a real quadratic form in the four variables x_1, x_2, x_3 and x_4 .

Theorem: Every quadratic form over a field F in n variables $x_1, x_2, \dots, x_n$ can be expressed in the form $\mathbf{X}'\mathbf{B}\mathbf{X}$ where $\mathbf{X} = [x_1, x_2, \dots, x_n]^T$ is a column vector and $\mathbf{B}$ is a symmetric matrix of order n over the field F .

Proof: Let $\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$, ... (1)

be a quadratic form over the field F in the n variables $x_1, x_2, \dots, x_n$.

In (1) it is assumed that $x_i x_j = x_j x_i$. Then the total coefficient of $x_i x_j$ in (1) is $a_{ij} + a_{ji}$. Let us assign half of this coefficient to x_{ij} and half to x_{ji} . Thus we define another set of scalars b_{ij} , such that $b_{ii} = a_{ii}$ and $b_{ij} = b_{ji} = \frac{1}{2}(a_{ij} + a_{ji})$, $i \neq j$. Then

$$\text{we have } \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j = \sum_{i=1}^n \sum_{j=1}^n b_{ij} x_i x_j.$$

Let $\mathbf{B} = [b_{ij}]_{n \times n}$. Then $\mathbf{B}$ is a symmetric matrix of order n over the field F since $b_{ij} = b_{ji}$.

$$\text{Let } \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ \dots \\ x_n \end{bmatrix}. \text{ Then } \mathbf{X}^T \text{ or } \mathbf{X}' = [x_1 \quad x_2 \quad \dots \quad x_n].$$

Now $\mathbf{X}^T \mathbf{B} \mathbf{X}$ is a matrix of the type 1×1 . It can be easily seen that the single element of this matrix is $\sum_{i=1}^n \sum_{j=1}^n b_{ij} x_i x_j$.

If we identify a 1×1 matrix with its single element *i.e.*, if we regard a 1×1 matrix equal to its single element, then we have

$$\mathbf{X}^T \mathbf{B} \mathbf{X} = \sum_{i=1}^n \sum_{j=1}^n b_{ij} x_i x_j = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j.$$

Hence the result.

8.6 Matrix of a Quadratic Form

Definition: If $\phi = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$ is a quadratic form in n variables $x_1, x_2, \dots, x_n$, then there exists a unique symmetric matrix $\mathbf{B}$ of order n such that $\phi = \mathbf{X}^T \mathbf{B} \mathbf{X}$ where $\mathbf{X} = [x_1 \ x_2 \ \dots \ x_n]^T$. The symmetric matrix $\mathbf{B}$ is called the matrix of the quadratic form $\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$.

Since every quadratic form can always be so written that matrix of its coefficients is a symmetric matrix, therefore we shall be considering quadratic forms which are so adjusted that the coefficient matrix is symmetric.

8.7 Quadratic form Corresponding to a Symmetric Matrix

Let $\mathbf{A} = [a_{ij}]_{n \times n}$ be a symmetric matrix over the field F and let

$$\mathbf{X} = [x_1 \ x_2 \ \dots \ x_n]^T$$

be a column vector. Then $\mathbf{X}^T \mathbf{A} \mathbf{X}$ determines a unique quadratic form $\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$ in n variables $x_1, x_2, \dots, x_n$ over the field F .

Thus we have seen that there exists a one-to-one correspondence between the set of all quadratic forms in n variables over a field F and the set of all n -rowed symmetric matrices over F .

8.8 Quadratic Form Corresponding to any Given Square Matrix

Definition: Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ \dots \\ x_n \end{bmatrix}$ be any n -vector in the vector space V_n over a field F so

that $\mathbf{X}^T = [x_1 \ x_2 \ \dots \ x_n]$.

Let $\mathbf{A} = [a_{ij}]_{n \times n}$ be any given square matrix of order n over the field F . Then any polynomial of the form

$$q(x_1, x_2, \dots, x_n) = \mathbf{X}^T \mathbf{A} \mathbf{X} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

is called a quadratic form of order n over F in the n variables $x_1, x_2, \dots, x_n$.

We can always find a unique symmetric matrix $B = [b_{ij}]_{n \times n}$ of order n such that $\mathbf{X}^T \mathbf{A} \mathbf{X} = \mathbf{X}^T \mathbf{B} \mathbf{X}$. We have $b_{ij} = b_{ji} = \frac{1}{2}(a_{ij} + a_{ji})$.

Discriminant of a quadratic form: Singular and Non-singular Quadratic forms: By the *discriminant* of a quadratic form $\mathbf{X}^T \mathbf{A} \mathbf{X}$, we mean $\det \mathbf{A}$. The quadratic form $\mathbf{X}^T \mathbf{A} \mathbf{X}$ is said to be *non-singular* if $\det \mathbf{A} \neq 0$, and it is said to be *singular* if $\det \mathbf{A} = 0$.

Illustrative Examples

Example 6: Write down the matrix of each of the following quadratic forms and verify that they can be written as matrix products $\mathbf{X}^T \mathbf{A} \mathbf{X}$:

(i) $x_1^2 - 18x_1 x_2 + 5x_2^2$.

(ii) $x_1^2 + 2x_2^2 - 5x_3^2 - x_1 x_2 + 4x_2 x_3 - 3x_3 x_1$.

Solution: (i) The given quadratic form can be written as

$$x_1 x_1 - 9x_1 x_2 - 9x_2 x_1 + 5x_2 x_2.$$

Let $\mathbf{A}$ be the matrix of this quadratic form. Then $\mathbf{A} = \begin{bmatrix} 1 & -9 \\ -9 & 5 \end{bmatrix}$.

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Then $\mathbf{X}' = [x_1 \ x_2]$.

We have $\mathbf{X}' \mathbf{A} = [x_1 \ x_2] \begin{bmatrix} 1 & -9 \\ -9 & 5 \end{bmatrix} = [x_1 - 9x_2 \ -9x_1 + 5x_2]$.

$$\begin{aligned} \therefore \mathbf{X}' \mathbf{A} \mathbf{X} &= [x_1 - 9x_2 \ -9x_1 + 5x_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= x_1(x_1 - 9x_2) + x_2(-9x_1 + 5x_2) \\ &= x_1^2 - 9x_1 x_2 - 9x_2 x_1 + 5x_2^2 \\ &= x_1^2 - 18x_1 x_2 + 5x_2^2. \end{aligned}$$

(ii) The given quadratic form can be written as

$$\begin{aligned} x_1 x_1 - \frac{1}{2} x_1 x_2 - \frac{3}{2} x_1 x_3 - \frac{1}{2} x_2 x_1 + 2x_2 x_2 + 2x_2 x_3 - \frac{3}{2} x_3 x_1 \\ + 2x_3 x_2 - 5x_3 x_3. \end{aligned}$$

Let $\mathbf{A}$ be the matrix of this quadratic form. Then

$$\mathbf{A} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} \\ -\frac{1}{2} & 2 & 2 \\ -\frac{3}{2} & 2 & -5 \end{bmatrix}.$$

Obviously $\mathbf{A}$ is a symmetric matrix.

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Then $\mathbf{X}' = [x_1 \quad x_2 \quad x_3]$.

We have $\mathbf{X}'\mathbf{A} = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} \\ -\frac{1}{2} & 2 & 2 \\ -\frac{3}{2} & 2 & -5 \end{bmatrix}$

$$= [x_1 - \frac{1}{2}x_2 - \frac{3}{2}x_3 \quad -\frac{1}{2}x_1 + 2x_2 + 2x_3 \quad -\frac{3}{2}x_1 + 2x_2 - 5x_3].$$

$$\begin{aligned} \therefore \mathbf{X}'\mathbf{A}\mathbf{X} &= [x_1 - \frac{1}{2}x_2 - \frac{3}{2}x_3 \quad -\frac{1}{2}x_1 + 2x_2 + 2x_3 \quad -\frac{3}{2}x_1 + 2x_2 - 5x_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= x_1(x_1 - \frac{1}{2}x_2 - \frac{3}{2}x_3) + x_2(-\frac{1}{2}x_1 + 2x_2 + 2x_3) + x_3(-\frac{3}{2}x_1 + 2x_2 - 5x_3) \\ &= x_1^2 - \frac{1}{2}x_1x_2 - \frac{3}{2}x_1x_3 - \frac{1}{2}x_2x_1 + 2x_2^2 + 2x_2x_3 - \frac{3}{2}x_3x_1 \\ &\quad + 2x_3x_2 - 5x_3^2 \\ &= x_1^2 + 2x_2^2 - 5x_3^2 - x_1x_2 + 4x_2x_3 - 3x_3x_1. \end{aligned}$$

Example 7: Obtain the matrices corresponding to the following quadratic forms :

(i) $x^2 + 2y^2 + 3z^2 + 4xy + 5yz + 6zx$.

(ii) $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$.

Solution: (i) The given quadratic form can be written as

$$x^2 + 2xy + 3xz + 2yx + 2y^2 + \frac{5}{2}yz + 3zx + \frac{5}{2}zy + 3z^2.$$

$\therefore$ if $\mathbf{A}$ is the matrix of this quadratic form, then

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & \frac{5}{2} \\ 3 & \frac{5}{2} & 3 \end{bmatrix}, \text{ which is a symmetric matrix of order 3.}$$

(ii) The given quadratic form can be written as

$$ax^2 + hxy + gxz + hyx + by^2 + fyz + gzx + fzy + cz^2.$$

$\therefore$ if $\mathbf{A}$ is the matrix of this quadratic form, then

$$\mathbf{A} = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}.$$

Example 8: Write down the quadratic forms corresponding to the following matrices :

$$(i) \begin{bmatrix} 0 & 5 & -1 \\ 5 & 1 & 6 \\ -1 & 6 & 2 \end{bmatrix} \quad (ii) \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}.$$

Solution: (i) Let $\mathbf{X} = [x_1 \ x_2 \ x_3]^T$ and $\mathbf{A}$ denote the given symmetric matrix. Then $\mathbf{X}^T \mathbf{A} \mathbf{X}$ is the quadratic form corresponding to this matrix. We have

$$\mathbf{X}^T \mathbf{A} = [x_1 \ x_2 \ x_3] \begin{bmatrix} 0 & 5 & -1 \\ 5 & 1 & 6 \\ -1 & 6 & 2 \end{bmatrix}$$

$$= [5x_2 - x_3 \quad 5x_1 + x_2 + 6x_3 \quad -x_1 + 6x_2 + 2x_3].$$

$$\therefore \mathbf{X}^T \mathbf{A} \mathbf{X} = x_1 (5x_2 - x_3) + x_2 (5x_1 + x_2 + 6x_3) + x_3 (-x_1 + 6x_2 + 2x_3) \\ = x_2^2 + 2x_3^2 + 10x_1 x_2 - 2x_1 x_3 + 12x_2 x_3.$$

(ii) Let $\mathbf{X} = [x_1 \ x_2 \ x_3 \ x_4]^T$ and $\mathbf{A}$ denote the given symmetric matrix. Then $\mathbf{X}^T \mathbf{A} \mathbf{X}$ is the quadratic form corresponding to this matrix. We have

$$\mathbf{X}^T \mathbf{A} = [x_1 \ x_2 \ x_3 \ x_4] \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$

$$= [x_2 + 2x_3 + 3x_4 \quad x_1 + 2x_2 + 3x_3 + 4x_4 \quad 2x_1 + 3x_2 + 4x_3 + 5x_4 \\ 3x_1 + 4x_2 + 5x_3 + 6x_4]$$

$$\therefore \mathbf{X}^T \mathbf{A} \mathbf{X} = x_1 (x_2 + 2x_3 + 3x_4) + x_2 (x_1 + 2x_2 + 3x_3 + 4x_4) \\ + x_3 (2x_1 + 3x_2 + 4x_3 + 5x_4) + x_4 (3x_1 + 4x_2 + 5x_3 + 6x_4) \\ = 2x_2^2 + 4x_3^2 + 6x_4^2 + 2x_1 x_2 + 4x_1 x_3 + 6x_1 x_4 \\ + 6x_2 x_3 + 8x_2 x_4 + 10x_3 x_4.$$

Comprehensive Exercise 1

1. Which of the following functions f , defined on vectors $\alpha = (x_1, x_2)$ and $\beta = (y_1, y_2)$ in $\mathbf{R}^2$, are bilinear forms ?

(i) $f(\alpha, \beta) = x_1 y_2 - x_2 y_1$

(ii) $f(\alpha, \beta) = (x_1 - y_1)^2 + x_2 y_2.$

2. Find the matrix $\mathbf{A}$ of each of the following bilinear forms $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y}$.

(i) $-5x_1y_1 - x_1y_2 + 2x_2y_1 - x_3y_1 + 3x_3y_2$

(ii) $4x_1y_1 + x_1y_2 + x_2y_1 - 2x_2y_2 - 4x_2y_3 - 4x_3y_2 + 7x_3y_3$.

Which of the above forms is symmetric ?

3. Determine the transformation matrices $\mathbf{P}$ and $\mathbf{Q}$ so that the bilinear form $\mathbf{X}^T \mathbf{A} \mathbf{Y} = x_1y_1 + x_1y_2 + 2x_1y_3 + x_2y_1 + 2x_2y_2 + 3x_2y_3 - x_3y_2 - x_3y_3$ is equivalent to a canonical form.

4. Obtain the matrices corresponding to the following quadratic forms

(i) $ax^2 + 2hxy + by^2$

(ii) $2x_1x_2 + 6x_1x_3 - 4x_2x_3$.

(iii) $x_1^2 + 5x_2^2 - 7x_3^2$

(iv) $2x_1^2 - 7x_3^2 + 4x_1x_2 - 6x_2x_3$.

5. Obtain the matrices corresponding to the following quadratic forms :

(i) $a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + 2a_{12}x_1x_2 + 2a_{23}x_2x_3 + 2a_{31}x_3x_1$.

(ii) $x_1^2 - 2x_2^2 + 4x_3^2 - 4x_4^2 - 2x_1x_2 + 3x_1x_3 + 4x_2x_3 - 5x_3x_4$.

(iii) $x_1x_2 + x_2x_3 + x_3x_1 + x_1x_4 + x_2x_4 + x_3x_4$.

(iv) $x_1^2 - 2x_2x_3 - x_3x_4$.

(v) $d_1x_1^2 + d_2x_2^2 + d_3x_3^2 + d_4x_4^2 + d_5x_5^2$.

6. Find the matrix of the quadratic form

$$x_1^2 - 2x_2^2 - 3x_3^2 + 4x_1x_2 + 6x_1x_3 - 8x_2x_3.$$

and verify that it can be written as a matrix product $\mathbf{X}'\mathbf{A}\mathbf{X}$.

7. Write down the quadratic forms corresponding to the following symmetric matrices :

(i)
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

(ii) $\text{diag.} [\lambda_1, \lambda_2, \dots, \lambda_n]$

8. Write down the quadratic form corresponding to the matrix

$$\begin{bmatrix} 0 & a & b & c \\ a & 0 & l & m \\ b & l & 0 & p \\ c & m & p & 0 \end{bmatrix}.$$

9. Write down the quadratic form associated with the matrix

$$\mathbf{B} = \begin{bmatrix} 2 & 1 & 2 \\ -3 & -3 & -1 \\ 4 & 1 & 3 \end{bmatrix}.$$

Rewrite the matrix $\mathbf{A}$ of the form so that it is symmetric.

Answers 1

1. (i) f is a bilinear form on $\mathbf{R}^2$

(ii) f is not a bilinear form on $\mathbf{R}^2$

2. (i) $\mathbf{A} = \begin{bmatrix} -5 & -1 \\ 2 & 0 \\ -1 & 3 \end{bmatrix}$. The given bilinear form is not symmetric

(ii) $\mathbf{A} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & -2 & -4 \\ 0 & -4 & 7 \end{bmatrix}$. The given bilinear form is symmetric

3. $\mathbf{P} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, $\mathbf{Q} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

4. (i) $\begin{bmatrix} a & h \\ h & b \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & -2 \\ 3 & -2 & 0 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -7 \end{bmatrix}$ (iv) $\begin{bmatrix} 2 & 2 & 0 \\ 2 & 0 & -3 \\ 0 & -3 & -7 \end{bmatrix}$

5. (i) $\begin{bmatrix} a_{11} & a_{12} & a_{31} \\ a_{12} & a_{22} & a_{23} \\ a_{31} & a_{23} & a_{33} \end{bmatrix}$. (ii) $\begin{bmatrix} 1 & -1 & 0 & \frac{3}{2} \\ -1 & -2 & 2 & 0 \\ 0 & 2 & 4 & -\frac{5}{2} \\ \frac{3}{2} & 0 & -\frac{5}{2} & -4 \end{bmatrix}$

(iii) $\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & 0 \end{bmatrix}$

(v) $\text{diag. } [d_1, d_2, d_3, d_4, d_5]$

6. $\begin{bmatrix} 1 & 2 & 3 \\ 2 & -2 & -4 \\ 3 & -4 & -3 \end{bmatrix}$

7. (i) $x_1^2 + x_3^2 + 4x_1x_2 + 6x_1x_3 + 6x_2x_3$

(ii) $\lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2$

$$8. \quad 2ax_1x_2 + 2bx_1x_3 + 2cx_1x_4 + 2lx_2x_3 + 2mx_2x_4 + 2px_3x_4$$

$$9. \quad 2x_1^2 - 3x_2^2 + 3x_3^2 - 2x_1x_2 + 6x_1x_3$$

$$A = \begin{bmatrix} 2 & -1 & 3 \\ -1 & -3 & 0 \\ 3 & 0 & 3 \end{bmatrix}$$

8.9 Linear Transformations

Suppose V_n is the vector space of all ordered n -tuples of the elements of a field F and let the vectors in V_n be written as column vectors. Let $\mathbf{P}$ be a matrix of order n over the field F . If $\mathbf{Y} = [y_1, y_2, \dots, y_n]^T$ is a vector in V_n , then $\mathbf{PY}$ is a matrix of the type $n \times 1$. Obviously $\mathbf{PY}$ is a matrix of the type $n \times 1$. Thus $\mathbf{PY}$ is also a vector in V_n . Let

$$\mathbf{PY} = \mathbf{X} = [x_1, x_2, \dots, x_n]^T.$$

The relation $\mathbf{PY} = \mathbf{X}$ thus gives a mapping from V_n into V_n .

Since $\mathbf{P}(a\mathbf{Y}_1 + b\mathbf{Y}_2) = a(\mathbf{PY}_1) + b(\mathbf{PY}_2)$, therefore this mapping is a linear transformation. If the matrix $\mathbf{P}$ is non-singular, then the linear transformation is also said to be non-singular. Also if the matrix $\mathbf{P}$ is non-singular, then the mapping $\mathbf{PY} = \mathbf{X}$ is one-one onto as shown below :

Mapping $\mathbf{P}$ is **one-one**. We have $\mathbf{PY}_1 = \mathbf{PY}_2$

$$\Rightarrow \mathbf{P}^{-1}(\mathbf{PY}_1) = \mathbf{P}^{-1}(\mathbf{PY}_2) \Rightarrow \mathbf{Y}_1 = \mathbf{Y}_2.$$

Therefore the mapping $\mathbf{P}$ is one-one.

Mapping $\mathbf{P}$ is **onto**. Let $\mathbf{Z}$ be any vector in V_n . Then $\mathbf{P}^{-1}\mathbf{Z}$ is also a vector of V_n and we have $\mathbf{P}(\mathbf{P}^{-1}\mathbf{Z}) = (\mathbf{PP}^{-1})\mathbf{Z} = \mathbf{IZ} = \mathbf{Z}$. Therefore the mapping $\mathbf{P}$ is onto.

If the linear transformation $\mathbf{PY} = \mathbf{X}$ is non-singular, then $\mathbf{PY} = \mathbf{O}$ if and only if $\mathbf{Y} = \mathbf{O}$.

If $\mathbf{Y} = \mathbf{O}$ then obviously $\mathbf{PY} = \mathbf{O}$.

$$\text{Conversely, } \mathbf{PY} = \mathbf{O} \Rightarrow \mathbf{P}^{-1}(\mathbf{PY}) = \mathbf{P}^{-1}\mathbf{O} \Rightarrow \mathbf{Y} = \mathbf{O}.$$

8.10 Congruence of Matrices

Definition: A square matrix $\mathbf{B}$ of order n over a field F is said to be congruent to another square matrix $\mathbf{A}$ of order n over F , if there exists a non-singular matrix $\mathbf{P}$ over F such that

$$\mathbf{B} = \mathbf{P}^T \mathbf{A} \mathbf{P}.$$

Theorem 1: The relation of 'congruence of matrices' is an equivalence relation in the set of all $n \times n$ matrices over a field F .

Proof: Reflexivity: Let $\mathbf{A}$ be any $n \times n$ matrix over a field F . Then $\mathbf{A} = \mathbf{I}^T \mathbf{A} \mathbf{I}$, where $\mathbf{I}$ is unit matrix of order n over F . Since $\mathbf{I}$ is non-singular, therefore $\mathbf{A}$ is congruent to itself.

Symmetry: Suppose $\mathbf{A}$ is congruent to $\mathbf{B}$. Then $\mathbf{A} = \mathbf{P}' \mathbf{B} \mathbf{P}$, where $\mathbf{P}$ is non-singular.

$$\therefore (\mathbf{P}')^{-1} \mathbf{A} \mathbf{P}^{-1} = (\mathbf{P}')^{-1} \mathbf{P}' \mathbf{B} \mathbf{P} \mathbf{P}^{-1} = \mathbf{B}$$

$$\Rightarrow (\mathbf{P}^{-1})' \mathbf{A} \mathbf{P}^{-1} = \mathbf{B} \quad [\because (\mathbf{P}')^{-1} = (\mathbf{P}^{-1})']$$

$\Rightarrow \mathbf{B}$ is congruent to $\mathbf{A}$.

Transitivity: Suppose $\mathbf{A}$ is congruent to $\mathbf{B}$ and $\mathbf{B}$ is congruent to $\mathbf{C}$. Then $\mathbf{A} = \mathbf{P}' \mathbf{B} \mathbf{P}$, $\mathbf{B} = \mathbf{Q}' \mathbf{C} \mathbf{Q}$, where $\mathbf{P}$ and $\mathbf{Q}$ are non-singular. Therefore

$$\mathbf{A} = \mathbf{P}' (\mathbf{Q}' \mathbf{C} \mathbf{Q}) \mathbf{P} = (\mathbf{P}' \mathbf{Q}') \mathbf{C} \mathbf{Q} \mathbf{P} = (\mathbf{Q} \mathbf{P})' \mathbf{C} (\mathbf{Q} \mathbf{P}).$$

Since $\mathbf{Q} \mathbf{P}$ is also a non-singular matrix, therefore $\mathbf{A}$ is congruent to $\mathbf{C}$.

Thus the relation of 'congruence of matrices' is reflexive, symmetric and transitive. So it is an equivalence relation.

Theorem 2 : Every matrix congruent to a symmetric matrix is a symmetric matrix.

Proof : Let a matrix $\mathbf{B}$ be congruent to a symmetric matrix $\mathbf{A}$.

Then there exists a non-singular matrix $\mathbf{P}$ such that $\mathbf{B} = \mathbf{P}' \mathbf{A} \mathbf{P}$.

$$\text{We have } \mathbf{B}' = (\mathbf{P}' \mathbf{A} \mathbf{P})' = \mathbf{P}' \mathbf{A}' (\mathbf{P}')' = \mathbf{P}' \mathbf{A}' \mathbf{P}$$

$$= \mathbf{P}' \mathbf{A} \mathbf{P} \quad [\because \mathbf{A} \text{ is symmetric } \Rightarrow \mathbf{A}' = \mathbf{A}]$$

$$= \mathbf{B}.$$

$\therefore \mathbf{B}$ is also a symmetric matrix.

Congruence operation on a square matrix or Congruence transformations of a square matrix.

A congruence operation of a square matrix is an operation of any one of the following three types :

- (i) Interchange of the i th and the j th rows as well as of the i th and the j th columns. Both should be applied simultaneously. Thus the operation $R_i \leftrightarrow R_j$ followed by $C_i \leftrightarrow C_j$ is a congruence operation.
- (ii) Multiplication of the i th row as well as the i th column by a non-zero number c i.e., $R_i \rightarrow cR_i$ followed by $C_i \rightarrow cC_i$.
- (iii) $R_i \rightarrow R_i + kR_j$ followed by $C_i \rightarrow C_i + kC_j$.

Now we shall show that each congruent transformation of a matrix consists of a pair of elementary transformations, one row and the other column, such that of the corresponding elementary matrices each is the transpose of the other.

- (a) Let $\mathbf{E}^*$, $\mathbf{E}$ be the elementary matrices corresponding to the elementary transformations $R_i \leftrightarrow R_j$ and $C_i \leftrightarrow C_j$ respectively. Then

$$\mathbf{E}^* = \mathbf{E} = \mathbf{E}'.$$

- (b) Let $\mathbf{E}^*$, $\mathbf{E}$ be the elementary matrices corresponding to the elementary transformations $R_i \rightarrow cR_i$ and $C_i \rightarrow cC_i$ respectively where $c \neq 0$. Then

$$\mathbf{E}^* = \mathbf{E} = \mathbf{E}'.$$

- (c) Let $\mathbf{E}^*$, $\mathbf{E}$ be the elementary matrices corresponding to the elementary transformations $R_i \rightarrow R_i + kR_j$ and $C_i \rightarrow C_i + kC_j$ respectively. Then

$$\mathbf{E}^* = \mathbf{E}'.$$

Now we know that every elementary row (column) transformation of a matrix can be brought about by pre-multiplication (post-multiplication) with the corresponding elementary matrix. Therefore if a matrix $\mathbf{B}$ has been obtained from $\mathbf{A}$ by a finite chain of congruent operations applied on $\mathbf{A}$, then there exist elementary matrices $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_s$ such that

$$\begin{aligned} \mathbf{B} &= \mathbf{E}_s' \dots \mathbf{E}_2' \mathbf{E}_1' \mathbf{A} \mathbf{E}_1 \mathbf{E}_2 \dots \mathbf{E}_s \\ &= (\mathbf{E}_1 \mathbf{E}_2 \dots \mathbf{E}_s)' \mathbf{A} (\mathbf{E}_1 \mathbf{E}_2 \dots \mathbf{E}_s) \\ &= \mathbf{P}' \mathbf{A} \mathbf{P}, \text{ where } \mathbf{P} = \mathbf{E}_1 \mathbf{E}_2 \dots \mathbf{E}_s \text{ is a non-singular matrix.} \end{aligned}$$

Therefore $\mathbf{B}$ is congruent to $\mathbf{A}$. Thus every matrix $\mathbf{B}$ obtained from any given matrix $\mathbf{A}$ by subjecting $\mathbf{A}$ to a finite chain of congruent operations is congruent to $\mathbf{B}$.

The converse is also true. If $\mathbf{B}$ is congruent to $\mathbf{A}$, then

$$\mathbf{B} = \mathbf{P}' \mathbf{A} \mathbf{P}$$

where $\mathbf{P}$ is a non-singular matrix. Now every non-singular matrix can be expressed as the product of elementary matrices. Therefore we can write $\mathbf{P} = \mathbf{E}_1 \mathbf{E}_2 \dots \mathbf{E}_s$ where $\mathbf{E}_1, \dots, \mathbf{E}_s$ are elementary matrices. Then $\mathbf{B} = \mathbf{E}_s' \dots \mathbf{E}_2' \mathbf{E}_1' \mathbf{A} \mathbf{E}_1 \mathbf{E}_2 \dots \mathbf{E}_s$. Therefore $\mathbf{B}$ is obtained from $\mathbf{A}$ by a finite chain of congruent operations applied on $\mathbf{A}$.

8.11 Congruence of Quadratic Forms or Equivalence of Quadratic Forms

Definition: Two quadratic forms $\mathbf{X}^T \mathbf{A} \mathbf{X}$ and $\mathbf{Y}^T \mathbf{B} \mathbf{Y}$ over a field F are said to be congruent or equivalent over F if their respective matrices $\mathbf{A}$ and $\mathbf{B}$ are congruent over F . Thus $\mathbf{X}^T \mathbf{A} \mathbf{X}$ is equivalent to $\mathbf{Y}^T \mathbf{B} \mathbf{Y}$ if there exists a non-singular matrix $\mathbf{P}$ over F such that $\mathbf{P}^T \mathbf{A} \mathbf{P} = \mathbf{B}$.

Since congruence of matrices is an equivalence relation, therefore equivalence of quadratic forms is also an equivalence relation.

Equivalence of Real Quadratic Forms:

Definition: Two real quadratic forms $\mathbf{X}^T \mathbf{A} \mathbf{X}$ and $\mathbf{Y}^T \mathbf{B} \mathbf{Y}$ are said to be real equivalent, orthogonally equivalent, or complex equivalent according as there exists a non-singular real, orthogonal, or a non-singular complex matrix $\mathbf{P}$ such that

$$\mathbf{B} = \mathbf{P}^T \mathbf{A} \mathbf{P}.$$

8.12 The Linear Transformation of a Quadratic Form

Consider a quadratic form

$$\mathbf{X}^T \mathbf{A} \mathbf{X} \quad \dots(1)$$

and a non-singular linear transformation

$$\mathbf{X} = \mathbf{P} \mathbf{Y} \quad \dots(2)$$

so that $\mathbf{P}$ is a non-singular matrix.

Putting $\mathbf{X} = \mathbf{P} \mathbf{Y}$ in (1), we get

$$\begin{aligned} \mathbf{X}^T \mathbf{A} \mathbf{X} &= (\mathbf{P} \mathbf{Y})^T \mathbf{A} (\mathbf{P} \mathbf{Y}) = \mathbf{Y}^T \mathbf{P}^T \mathbf{A} \mathbf{P} \mathbf{Y} \\ &= \mathbf{Y}^T \mathbf{B} \mathbf{Y}, \text{ where } \mathbf{B} = \mathbf{P}^T \mathbf{A} \mathbf{P}. \end{aligned}$$

Since $\mathbf{B}$ is congruent to a symmetric matrix $\mathbf{A}$, therefore $\mathbf{B}$ is also a symmetric matrix. Thus $\mathbf{Y}^T \mathbf{B} \mathbf{Y}$ is a quadratic form. It is called a linear transform of the form $\mathbf{X}^T \mathbf{A} \mathbf{X}$ by the non-singular matrix $\mathbf{P}$. The matrix of the quadratic form $\mathbf{Y}^T \mathbf{B} \mathbf{Y}$ is

$$\mathbf{B} = \mathbf{P}^T \mathbf{A} \mathbf{P}.$$

Thus the quadratic form $\mathbf{Y}^T \mathbf{B} \mathbf{Y}$ is congruent to $\mathbf{X}^T \mathbf{A} \mathbf{X}$.

Theorem: *The ranges of values of two congruent quadratic forms are the same.*

Proof: Let $\phi = \mathbf{X}' \mathbf{A} \mathbf{X}$ and $\psi = \mathbf{Y}' \mathbf{B} \mathbf{Y}$ be two congruent quadratic forms. Then there exists a non-singular matrix $\mathbf{P}$ such that $\mathbf{B} = \mathbf{P}^T \mathbf{A} \mathbf{P}$.

Consider the linear transformation $\mathbf{X} = \mathbf{P} \mathbf{Y}$.

Let $\phi = p$ when $\mathbf{X} = \mathbf{X}_1$. Then $p = \mathbf{X}_1' \mathbf{A} \mathbf{X}_1$. The value of ψ when $\mathbf{Y} = \mathbf{P}^{-1} \mathbf{X}_1$ is

$$\begin{aligned} &= (\mathbf{P}^{-1} \mathbf{X}_1)' \mathbf{B} (\mathbf{P}^{-1} \mathbf{X}_1) = \mathbf{X}_1' (\mathbf{P}^{-1})' \mathbf{P}' \mathbf{A} \mathbf{P} \mathbf{P}^{-1} \mathbf{X}_1 \\ &= \mathbf{X}_1' (\mathbf{P}')^{-1} \mathbf{P}' \mathbf{A} \mathbf{X}_1 = \mathbf{X}_1' \mathbf{A} \mathbf{X}_1 = p. \end{aligned}$$

Thus each value of ϕ is equal to some value of ψ .

Conversely let $\psi = q$ when $\mathbf{Y} = \mathbf{Y}_1$. Then $q = \mathbf{Y}_1' \mathbf{B} \mathbf{Y}_1$. The value of ϕ when $\mathbf{X} = \mathbf{P} \mathbf{Y}_1$ is

$$\begin{aligned} &= (\mathbf{P} \mathbf{Y}_1)' \mathbf{A} (\mathbf{P} \mathbf{Y}_1) = \mathbf{Y}_1' \mathbf{P}' \mathbf{A} \mathbf{P} \mathbf{Y}_1 \\ &= \mathbf{Y}_1' \mathbf{B} \mathbf{Y}_1 = q. \end{aligned}$$

Thus each value of ψ is equal to some value of ϕ .

Hence ϕ and ψ have the same ranges of values.

Corollary: *If the quadratic form $\mathbf{Y}' \mathbf{B} \mathbf{Y}$ is a linear transform of the quadratic form $\mathbf{X}' \mathbf{A} \mathbf{X}$ by a non-singular matrix $\mathbf{P}$, then the two forms are congruent and so have the same ranges of values.*

8.13 Congruent Reduction of a Symmetric Matrix

Theorem : *If $\mathbf{A}$ be any n -rowed non-zero symmetric matrix of rank r , over a field F , then there exists an n -rowed non-singular matrix $\mathbf{P}$ over F such that*

$$P'AP = \begin{bmatrix} A_1 & O \\ O & O \end{bmatrix}$$

where A_1 is a non-singular diagonal matrix of order r over F and each O is a null matrix of suitable size.

Or

Every symmetric matrix of rank r is congruent to a diagonal matrix, r , of whose diagonal elements only are non-zero.

Proof : We shall prove the theorem by induction on n , the order of the given matrix. If $n = 1$, the theorem is obviously true. Let us suppose that the theorem is true for all symmetric matrices of order $n - 1$. Then we shall show that it is also true for an $n \times n$ symmetric matrix A .

Let $A = [a_{ij}]_{n \times n}$ be a symmetric matrix of rank r over a field F . First we shall show that there exists a matrix $B = [b_{ij}]_{n \times n}$ over F congruent to A such that $b_{11} \neq 0$.

Case 1: If $a_{11} \neq 0$, then we take $B = A$.

Case 2: If $a_{11} = 0$, but some diagonal element of A , say, $a_{ii} \neq 0$, then applying the congruent operation $R_i \leftrightarrow R_1, C_i \leftrightarrow C_1$ to A , we obtain a matrix B congruent to A such that

$$b_{11} = a_{ii} \neq 0.$$

Case 3: Suppose that each diagonal element of A is 0. Since A is a non-zero matrix, let a_{ij} be a non-zero element of A . Then $a_{ij} = a_{ji} \neq 0$.

Applying the congruent operation $R_i \rightarrow R_i + R_j, C_i \rightarrow C_i + C_j$ to A , we obtain a matrix $D = [d_{ij}]_{n \times n}$ congruent to A such that $d_{ii} = a_{ij} + a_{ji} = 2a_{ij} \neq 0$.

Now applying the congruent operation $R_i \leftrightarrow R_1, C_i \leftrightarrow C_1$ to D we obtain a matrix

$$B = [b_{ij}]_{n \times n}$$

congruent to D and, therefore, also congruent to A such that $b_{11} = d_{ii} \neq 0$.

Thus there always exists a matrix $B = [b_{ij}]_{n \times n}$

congruent to a symmetric matrix, such that the leading element of B is not zero.

Since B is congruent to a symmetric matrix, therefore B itself is a symmetric matrix.

Since $b_{11} \neq 0$, therefore all elements in the first row and first column of B , except the leading element, can be made 0 by suitable congruent operations. We thus have a matrix

$$C = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & & & \\ \vdots & B_1 & & \\ 0 & & & \end{bmatrix},$$

congruent to B and, therefore, also congruent to A such that B_1 is a square matrix of order $n - 1$. Since C is congruent to a symmetric matrix A , therefore C is also a symmetric matrix and consequently B_1 is also a symmetric matrix. Thus B_1 is a symmetric matrix of order $n - 1$.

Therefore by our induction hypothesis it can be reduced to a diagonal matrix by congruent operations. If the congruent operations applied to $\mathbf{B}_1$ for this purpose be applied to $\mathbf{C}$, they will not affect the first row and the first column of $\mathbf{C}$. So $\mathbf{C}$ can be reduced to a diagonal matrix by congruent operations. Thus $\mathbf{A}$ is congruent to a diagonal matrix, say $\text{diag} [\lambda_1, \lambda_2, \dots, \lambda_k, 0, 0, \dots, 0]$. Thus there exists a non-singular matrix $\mathbf{P}$ such that

$$\mathbf{P}'\mathbf{A}\mathbf{P} = \text{diag} [\lambda_1, \dots, \lambda_k, 0, \dots, 0].$$

Since $\text{rank } \mathbf{A} = r$ and the rank of a matrix does not change on multiplication by a non-singular matrix, therefore rank of the matrix $\mathbf{P}'\mathbf{A}\mathbf{P} = \text{diag} [\lambda_1, \dots, \lambda_k, 0, \dots, 0]$ is also r . So precisely r elements of $\text{diag} [\lambda_1, \dots, \lambda_k, 0, \dots, 0]$ are non-zero. Therefore $k = r$ and thus $\mathbf{P}'\mathbf{A}\mathbf{P} = \text{diag} [\lambda_1, \dots, \lambda_r, 0, \dots, 0]$.

Thus $\mathbf{A}$ can be reduced to diagonal form by congruent operations.

The proof is now complete by induction.

Corollary: *Corresponding to every quadratic form $\mathbf{X}'\mathbf{A}\mathbf{X}$ over a field F , there exists a non-singular linear transformation $\mathbf{X} = \mathbf{P}\mathbf{Y}$ over F , such that the form $\mathbf{X}'\mathbf{A}\mathbf{X}$ transforms to a sum of r square terms $\lambda_1 y_1^2 + \dots + \lambda_r y_r^2$,*

where $\lambda_1, \dots, \lambda_r$ belong to the field F and r is the rank of the matrix $\mathbf{A}$.

Rank of a quadratic form: Definition: *Let $\mathbf{X}'\mathbf{A}\mathbf{X}$ be a quadratic form over a field F . The rank of the matrix $\mathbf{A}$ is called the rank of the quadratic form $\mathbf{X}'\mathbf{A}\mathbf{X}$.*

If $\mathbf{X}'\mathbf{A}\mathbf{X}$ is a quadratic form of rank r , then there exists a non-singular matrix $\mathbf{P}$ which will reduce the form $\mathbf{X}'\mathbf{A}\mathbf{X}$ to a sum of r square terms.

Working Rule for Numerical Problems:

We should transform the given symmetric matrix $\mathbf{A}$ to diagonal form by applying congruent operations. Then the application of corresponding column operations to the unit matrix $\mathbf{I}_n$ will give us a non-singular matrix $\mathbf{P}$ such that

$$\mathbf{P}'\mathbf{A}\mathbf{P} = \text{a diagonal matrix.}$$

The whole process will be clear from the following examples.

Illustrative Examples

Example 9: *Determine a non-singular matrix $\mathbf{P}$ such that $\mathbf{P}'\mathbf{A}\mathbf{P}$ is a diagonal matrix, where*

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

Interpret the result in terms of quadratic forms.

Solution: We write $\mathbf{A} = \mathbf{IAI}$

$$\text{i.e.,} \quad \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We shall reduce $\mathbf{A}$ to diagonal form by applying congruent operations. On the right hand side $\mathbf{IAI}$, the corresponding row operations will be applied on pre-factor $\mathbf{I}$ and the column operations will be applied on the post-factor $\mathbf{I}$. There is no need of actually applying column operations on post-factor $\mathbf{I}$ because at any stage the matrix obtained by applying column operations on post-factor $\mathbf{I}$ will be the transpose of the matrix obtained by applying row operations on pre-factor $\mathbf{I}$.

Performing the congruent operations

$$R_2 \rightarrow R_2 + \frac{1}{3} R_1, C_2 \rightarrow C_2 + \frac{1}{3} C_1$$

$$\text{and} \quad R_3 \rightarrow R_3 - \frac{1}{3} R_1, C_3 \rightarrow C_3 - \frac{1}{3} C_1,$$

we have

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{3} & \frac{7}{3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{1}{3} & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now performing the congruent operation

$$R_3 \rightarrow R_3 + \frac{1}{7} R_2, C_3 \rightarrow C_3 + \frac{1}{7} C_2, \text{ we get}$$

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & 0 \\ 0 & 0 & \frac{16}{7} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{2}{7} & \frac{1}{7} & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & \frac{1}{3} & -\frac{2}{7} \\ 0 & 1 & \frac{1}{7} \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus we obtain a non-singular matrix

$$\mathbf{P} = \begin{bmatrix} 1 & \frac{1}{3} & -\frac{2}{7} \\ 0 & 1 & \frac{1}{7} \\ 0 & 0 & 1 \end{bmatrix} \text{ such that } \mathbf{P}'\mathbf{A}\mathbf{P} = \text{diag.} \left[6, \frac{7}{3}, \frac{16}{7} \right].$$

The quadratic form corresponding to the matrix $\mathbf{A}$ is

$$\mathbf{X}'\mathbf{A}\mathbf{X} = 6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1. \quad \dots(1)$$

The non-singular transformation corresponding to the matrix $\mathbf{P}$ is given by $\mathbf{X} = \mathbf{P}\mathbf{Y}$
i.e.,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{3} & -\frac{2}{7} \\ 0 & 1 & \frac{1}{7} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

which is equivalent to

$$\left. \begin{aligned} x_1 &= y_1 + \frac{1}{3}y_2 - \frac{2}{7}y_3 \\ x_2 &= y_2 + \frac{1}{7}y_3 \\ x_3 &= y_3 \end{aligned} \right\} \dots(2)$$

The transformation (2) will reduce the quadratic form (1) to the diagonal form

$$\mathbf{Y}' \mathbf{P}' \mathbf{A} \mathbf{P} \mathbf{Y} = 6y_1^2 + \frac{7}{3}y_2^2 + \frac{16}{7}y_3^2.$$

The rank of the quadratic form $\mathbf{X}' \mathbf{A} \mathbf{X}$ is 3. So it has been reduced to a form which is a sum of three squares.

8.14 Reduction of a Real Quadratic Form

Definition: Diagonal and Unit Quadratic Forms: *If the matrix of a quadratic form is diagonal, then it is called a diagonal quadratic form.*

For example, $X^T \text{diag. } [a_1, a_2, \dots, a_n] X = a_1 x_1^2 + a_2 x_2^2 + \dots + a_n x_n^2$ is a diagonal quadratic form. In the diagonal quadratic form, some of the a_i 's may be zero.

If the matrix of a quadratic form is unit, then it is called a unit quadratic form.

For example, $X^T I_n X = x_1^2 + x_2^2 + \dots + x_n^2$ is a unit quadratic form.

Theorem 1: *If $\mathbf{A}$ be any n -rowed real symmetric matrix of rank r , then there exists a real non-singular matrix $\mathbf{P}$ such that*

$$\mathbf{P}' \mathbf{A} \mathbf{P} = \text{diag. } [1, 1, \dots, 1, -1, -1, \dots, -1, 0, \dots, 0]$$

so that 1, appears p times and -1 appears $r - p$ times.

Proof: $\mathbf{A}$ is a real symmetric matrix of rank r . Therefore there exists a non-singular real matrix $\mathbf{Q}$ such that $\mathbf{Q}' \mathbf{A} \mathbf{Q}$ is a diagonal matrix $\mathbf{D}$ with precisely r non-zero diagonal elements. Let

$$\mathbf{Q}' \mathbf{A} \mathbf{Q} = \mathbf{D} = \text{diag. } [\lambda_1, \lambda_2, \dots, \lambda_r, 0, \dots, 0].$$

Suppose that p of the non-zero diagonal elements are positive. Then $r - p$ are negative.

Since in a diagonal matrix the positions of the diagonal elements occurring in i^{th} and j^{th} rows can be interchanged by applying the congruent operation $R_i \leftrightarrow R_j, C_i \leftrightarrow C_j$, therefore without any loss of generality we can take $\lambda_1, \dots, \lambda_p$ to be positive and $\lambda_{p+1}, \dots, \lambda_r$ to be negative.

Let $\mathbf{S}$ be the $n \times n$ (real) diagonal matrix with diagonal elements

$$\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_p}}, \frac{1}{\sqrt{-\lambda_{p+1}}}, \dots, \frac{1}{\sqrt{-\lambda_r}}, 1, \dots, 1.$$

Then
$$\mathbf{S} = \text{diag} \left[\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_p}}, \frac{1}{\sqrt{(-\lambda_{p+1})}}, \dots, \frac{1}{\sqrt{(-\lambda_r)}}, 1, \dots, 1 \right]$$

is a real non-singular diagonal matrix and $\mathbf{S}' = \mathbf{S}$.

If we take $\mathbf{P} = \mathbf{QS}$, then $\mathbf{P}$ is also real non-singular matrix and we have

$$\begin{aligned} \mathbf{P}'\mathbf{A}\mathbf{P} &= (\mathbf{QS})'\mathbf{A}(\mathbf{QS}) = \mathbf{S}'\mathbf{Q}'\mathbf{A}\mathbf{Q}\mathbf{S} = \mathbf{S}'\mathbf{D}\mathbf{S} = \mathbf{S}\mathbf{D}\mathbf{S} \\ &= \text{diag. } [1, \dots, 1, -1, \dots, -1, 0, \dots, 0] \end{aligned}$$

so that 1 and -1 appear p and $r - p$ times respectively.

Corollary: If $\mathbf{X}'\mathbf{A}\mathbf{X}$ is a real quadratic form of rank r in n variables, then there exists a real non-singular linear transformation $\mathbf{X} = \mathbf{PY}$ which transforms $\mathbf{X}'\mathbf{A}\mathbf{X}$ to the form

$$\mathbf{Y}'\mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{Y} = y_1^2 + \dots + y_p^2 - y_{p+1}^2 - \dots - y_r^2.$$

Canonical or Normal form of a real quadratic form: Definition: If $\mathbf{X}'\mathbf{A}\mathbf{X}$ is a real quadratic form in n variables, then there exists real non-singular linear transformation $\mathbf{X} = \mathbf{PY}$ which transforms $\mathbf{X}'\mathbf{A}\mathbf{X}$ to the form

$$y_1^2 + \dots + y_p^2 - y_{p+1}^2 - \dots - y_r^2.$$

In the new form the given quadratic form has been expressed as a sum and difference of the squares of new variables. This latter expression is called the canonical form or normal form of the given quadratic form.

If $\phi = \mathbf{X}'\mathbf{A}\mathbf{X}$ is a real quadratic form of rank r , then $\mathbf{A}$ is a matrix of rank r . If the real non-singular linear transformation $\mathbf{X} = \mathbf{PY}$ reduces ϕ to normal form, then $\mathbf{P}'\mathbf{A}\mathbf{P}$ is a diagonal matrix having 1 and -1 as its non-zero diagonal elements. Since $\mathbf{P}'\mathbf{A}\mathbf{P}$ is also of rank r , therefore it will have precisely r non-zero diagonal elements. Thus the number of terms in each normal form of a given real quadratic form is the same. Now we shall prove that the number of positive terms in any two normal reductions of a real quadratic form is the same.

Theorem 2: The number of positive terms in any two normal reductions of a real quadratic form is the same.

Proof: Let $\phi = \mathbf{X}'\mathbf{A}\mathbf{X}$ be a real quadratic form of rank r in n variables. Suppose the real non-singular linear transformations

$$\mathbf{X} = \mathbf{PY} \quad \text{and} \quad \mathbf{X} = \mathbf{QZ}$$

transform ϕ to the normal forms

$$y_1^2 + \dots + y_p^2 - y_{p+1}^2 - \dots - y_r^2 \quad \dots(1)$$

$$\text{and} \quad z_1^2 + \dots + z_q^2 - z_{q+1}^2 - \dots - z_r^2 \quad \dots(2)$$

respectively.

To prove that $p = q$.

Let $p < q$. Obviously $y_1, \dots, y_n, z_1, \dots, z_n$ are linear homogeneous functions of $x_1, \dots, x_n$. Since $q > p$, therefore $q - p > 0$. So $n - (q - p)$ is less than n . Therefore $(n - q) + p$ is less than n .

Now $y_1 = 0, y_2 = 0, \dots, y_p = 0, z_{q+1} = 0, z_{q+2} = 0, \dots, z_n = 0$ are $(n - q) + p$ linear homogeneous equations in n unknowns $x_1, \dots, x_n$. Since the number of equations is less than the number of unknowns n , therefore these equations must possess a non-zero solution. Let $x_1 = a_1, \dots, x_n = a_n$ be a non-zero solution of these equations and let $\mathbf{X}_1 = [a_1, \dots, a_n]'$. Let $\mathbf{Y} = [b_1, \dots, b_n]' = \mathbf{Y}_1$ and $\mathbf{Z} = [c_1, \dots, c_n]'$ when $\mathbf{X} = \mathbf{X}_1$. Then $b_1 = 0, \dots, b_p = 0$ and $c_{q+1} = 0, \dots, c_n = 0$. Putting $\mathbf{Y} = [b_1, \dots, b_n]'$ in (1) and $\mathbf{Z} = [c_1, \dots, c_n]'$ in (2), we get two values of ϕ when $\mathbf{X} = \mathbf{X}_1$.

These must be equal. Therefore we have

$$\begin{aligned} & -b_{p+1}^2 - \dots - b_r^2 = c_1^2 + \dots + c_q^2 \\ \Rightarrow & b_{p+1} = 0, \dots, b_r = 0 \\ \Rightarrow & \mathbf{Y}_1 = \mathbf{0} \\ \Rightarrow & \mathbf{P}^{-1} \mathbf{X}_1 = \mathbf{0} \quad [\because \mathbf{X}_1 = \mathbf{P} \mathbf{Y}_1] \\ \Rightarrow & \mathbf{X}_1 = \mathbf{0}, \end{aligned}$$

which is a contradiction since $\mathbf{X}_1$ is a non-zero vector.

Thus we cannot have $p < q$. Similarly, we cannot have $q < p$. Hence we must have $p = q$.

Corollary: *The number of negative terms in any two normal reductions of a real quadratic form is the same. Also the excess of the number of positive terms over the number of negative terms in any two normal reductions of a real quadratic form is the same.*

Signature and index of a real quadratic form:

Definition: Let $y_1^2 + \dots + y_p^2 - y_{p+1}^2 - \dots - y_r^2$ be a normal form of a real quadratic form $\mathbf{X}' \mathbf{A} \mathbf{X}$ of rank r . The number of positive terms in a normal form of $\mathbf{X}' \mathbf{A} \mathbf{X}$ is called the index of the quadratic form. The excess of the number of positive terms over the number of negative terms in a normal form of

$$\mathbf{X}' \mathbf{A} \mathbf{X} \text{ i.e., } p - (r - p) = 2p - r$$

is called the signature of the quadratic form and is usually denoted by s .

Thus $s = 2p - r$.

In terms of signature theorem 2 may be stated as follows :

Theorem 3: Sylvester's Law of Inertia: *The signature of a real quadratic form is invariant for all normal reductions.*

Proof: For its proof give definition of signature and the proof of theorem 2.

Theorem 4: *Two real quadratic forms in n variables are real equivalent if and only if they have the same rank and index (or signature).*

Proof: Suppose $\mathbf{X}' \mathbf{A} \mathbf{X}$ and $\mathbf{Y}' \mathbf{B} \mathbf{Y}$ are two real quadratic forms in the same number of variables.

Let us first assume that the two forms are equivalent. Then there exists a real non-singular linear transformation $\mathbf{X} = \mathbf{P} \mathbf{Y}$ which transforms $\mathbf{X}' \mathbf{A} \mathbf{X}$ to $\mathbf{Y}' \mathbf{B} \mathbf{Y}$ i.e., $\mathbf{B} = \mathbf{P}' \mathbf{A} \mathbf{P}$.

Now suppose the real non-singular linear transformation $Y = QZ$ transforms $Y'BY$ to normal form $Z'CZ$. Then $C = Q'BQ$. Since P and Q are real non-singular matrices, therefore PQ is also a real non-singular matrix. The linear transformation $X = (PQ)Z$ will transform $X'AX$ to the form

$$(PQZ)'A(PQZ) = Z'Q'P'APQZ = Z'Q'BQZ.$$

Thus the two given quadratic forms have a common normal form. Hence they have the same rank and same index (or signature).

Conversely, suppose that the two forms have the same rank r and the same signature s . Then they have the same index p where $2p - r = s$. So they can be reduced to the same normal form

$$Z'CZ = z_1^2 + \dots + z_p^2 - z_{p+1}^2 - \dots - z_r^2$$

by real non-singular linear transformations, say, $X = PZ$ and $Y = QZ$ respectively. Then $P'AP = C$ and $Q'BQ = C$.

Therefore $Q'BQ = P'AP$. This gives

$$B = (Q')^{-1} P'APQ^{-1} = (Q^{-1})' P'APQ^{-1} = (PQ^{-1})' A (PQ^{-1}).$$

Therefore the real non-singular linear transformation $X = (PQ^{-1})Y$ transforms $X'AX$ to $Y'BY$. Hence the two given quadratic forms are real equivalent.

Reduction of a real quadratic form in the complex field:

Theorem 5: *If A be any n -rowed real symmetric matrix of rank r , there exists a non-singular matrix P whose elements may be any complex numbers such that*

$$P'AP = \text{diag. } [1, 1, \dots, 1, 0, \dots, 0] \text{ where } 1 \text{ appears } r \text{ times.}$$

Proof: A is a real symmetric matrix of rank r . Therefore there exists a non-singular real matrix Q such that $Q'AQ$ is a diagonal matrix D with precisely r non-zero diagonal elements. Let

$$Q'AQ = D = \text{diag. } [\lambda_1, \dots, \lambda_r, 0, \dots, 0].$$

The real numbers $\lambda_1, \dots, \lambda_r$ may be positive or negative or both.

Let S be the $n \times n$ (complex) diagonal matrix with diagonal elements

$$\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_r}}, 1, \dots, 1.$$

Then $S = \text{diag. } \left[\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_r}}, 1, \dots, 1 \right]$ is a complex non-singular diagonal matrix and $S' = S$.

If we take $P = QS$, then P is also a complex non-singular matrix and we have $P'AP = (QS)'A(QS) = S'Q'AQS = S'DS = SDS = \text{diag. } [1, 1, \dots, 1, 0, \dots, 0]$ so that 1 appears r times. Hence the result.

Corollary 1: *Every real quadratic form $X'AX$ is complex-equivalent to the form $z_1^2 + \dots + z_r^2$ where r is the rank of A .*

Corollary 2: Two real quadratic forms in n variables are complex equivalent if and only if they have the same rank.

Orthogonal reduction of a real quadratic form:

Theorem 6: If $\phi = \mathbf{X}'\mathbf{A}\mathbf{X}$ be a real quadratic form of rank r in n variables, then there exists a real orthogonal transformation $\mathbf{X} = \mathbf{P}\mathbf{Y}$ which transforms ϕ to the diagonal form

$$\lambda_1 y_1^2 + \dots + \lambda_r y_r^2,$$

where $\lambda_1, \dots, \lambda_r$ are the, r , non-zero eigenvalues of $\mathbf{A}$, $n - r$ eigenvalues of $\mathbf{A}$ being equal to zero.

Proof: Since $\mathbf{A}$ is a real symmetric matrix, therefore there exists a real orthogonal matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$, where $\mathbf{D}$ is a diagonal matrix whose diagonal elements are the eigenvalues of $\mathbf{A}$.

Since $\mathbf{A}$ is of rank r , therefore $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$ is also of rank r . So $\mathbf{D}$ has precisely r non-zero diagonal elements. Consequently $\mathbf{A}$ has exactly r non-zero eigenvalues, the remaining $n - r$ eigenvalues of $\mathbf{A}$ being zero. Let $\mathbf{D} = \text{diag.} [\lambda_1, \dots, \lambda_r, \dots, 0, \dots, 0]$.

Since $\mathbf{P}^{-1} = \mathbf{P}'$, therefore $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$

$$\Rightarrow \mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{D} \Rightarrow \mathbf{A} \text{ is congruent to } \mathbf{D}.$$

Now consider the real orthogonal transformation $\mathbf{X} = \mathbf{P}\mathbf{Y}$. We have

$$\begin{aligned} \mathbf{X}'\mathbf{A}\mathbf{X} &= (\mathbf{P}\mathbf{Y})'\mathbf{A}(\mathbf{P}\mathbf{Y}) = \mathbf{Y}'\mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{Y} = \mathbf{Y}'\mathbf{D}\mathbf{Y} \\ &= \lambda_1 y_1^2 + \dots + \lambda_r y_r^2. \end{aligned}$$

Hence the result.

Theorem 7: Every real quadratic form $\mathbf{X}'\mathbf{A}\mathbf{X}$ in n variables is real equivalent to the form

$$y_1^2 + \dots + y_p^2 - y_{p+1}^2 - \dots - y_r^2,$$

where r is the rank of $\mathbf{A}$ and p is the number of positive eigenvalues of $\mathbf{A}$.

Proof: $\mathbf{A}$ is a real symmetric matrix. Therefore there exists a real orthogonal matrix $\mathbf{Q}$ such that $\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \mathbf{Q}'\mathbf{A}\mathbf{Q} = \mathbf{D}$,

where $\mathbf{D}$ is a diagonal matrix whose diagonal elements are the eigenvalues of $\mathbf{A}$. Since $\mathbf{A}$ is of rank r , therefore $\mathbf{D}$ is also of rank r . So $\mathbf{D}$ has exactly r non-zero diagonal elements. Consequently $\mathbf{A}$ has exactly r non-zero eigenvalues, the remaining $n - r$ eigenvalues of $\mathbf{A}$ being zero. Let $\mathbf{D} = \text{diag.} [\lambda_1, \lambda_2, \dots, \lambda_r, 0, \dots, 0]$.

Let $\lambda_1, \dots, \lambda_p$ be positive and $\lambda_{p+1}, \dots, \lambda_r$ be negative. Let $\mathbf{S}$ be the $n \times n$ real diagonal matrix with diagonal elements

$$\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_p}}, \frac{1}{\sqrt{(-\lambda_{p+1})}}, \dots, \frac{1}{\sqrt{(-\lambda_r)}}, 1, \dots, 1.$$

Then $\mathbf{S}$ is non-singular and $\mathbf{S}' = \mathbf{S}$. If we take $\mathbf{P} = \mathbf{Q}\mathbf{S}$, then $\mathbf{P}$ is also a real non-singular matrix and we have

$$\begin{aligned} \mathbf{P}'\mathbf{A}\mathbf{P} &= (\mathbf{Q}\mathbf{S})'\mathbf{A}(\mathbf{Q}\mathbf{S}) = \mathbf{S}'\mathbf{Q}'\mathbf{A}\mathbf{Q}\mathbf{S} = \mathbf{S}\mathbf{D}\mathbf{S} \\ &= \text{diag.} [1, \dots, 1, -1, \dots, -1, 0, \dots, 0] \end{aligned}$$

so that 1 and -1 appear p and $r - p$ times respectively.

Now the real non-singular linear transformation $\mathbf{X} = \mathbf{P}\mathbf{Y}$ reduces $\mathbf{X}'\mathbf{A}\mathbf{X}$ to the form $\mathbf{Y}'\mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{Y}$ i.e.,

$$y_1^2 + \dots + y_p^2 - y_{p+1}^2 - \dots - y_r^2.$$

Hence the result.

Corollary: Two real quadratic forms $\mathbf{X}'\mathbf{A}\mathbf{X}$ and $\mathbf{Y}'\mathbf{B}\mathbf{Y}$ in the same number of variables are real equivalent if and only if $\mathbf{A}$ and $\mathbf{B}$ have the same number of positive and negative eigenvalues.

Important Note: If $\mathbf{X}'\mathbf{A}\mathbf{X}$ is a real quadratic form, then the number of non-zero eigenvalues of $\mathbf{A}$ is equal to the rank of $\mathbf{X}'\mathbf{A}\mathbf{X}$ and the number of positive eigenvalues of $\mathbf{A}$ is equal to the index of $\mathbf{X}'\mathbf{A}\mathbf{X}$.

Theorem 8: Two real quadratic forms $\mathbf{X}'\mathbf{A}\mathbf{X}$ and $\mathbf{Y}'\mathbf{B}\mathbf{Y}$ are orthogonally equivalent if and only if $\mathbf{A}$ and $\mathbf{B}$ have the same eigenvalues and these occur with the same multiplicities.

Proof: If $\mathbf{A}$ and $\mathbf{B}$ have eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ and $\mathbf{D}$ is a diagonal matrix with $\lambda_1, \lambda_2, \dots, \lambda_n$ as diagonal elements, then there exist orthogonal matrices $\mathbf{P}$ and $\mathbf{Q}$ such that $\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{D} = \mathbf{Q}'\mathbf{B}\mathbf{Q}$.

$$\begin{aligned} \text{Now} \quad & \mathbf{Q}'\mathbf{B}\mathbf{Q} = \mathbf{P}'\mathbf{A}\mathbf{P} \\ \Rightarrow \quad & \mathbf{B} = (\mathbf{Q}')^{-1} \mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{Q}^{-1} \\ & = (\mathbf{Q}^{-1})' \mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{Q}^{-1} \\ & = (\mathbf{P}\mathbf{Q}^{-1})' \mathbf{A} (\mathbf{P}\mathbf{Q}^{-1}). \end{aligned}$$

Since $\mathbf{P}\mathbf{Q}^{-1}$ is an orthogonal matrix, therefore $\mathbf{Y}'\mathbf{B}\mathbf{Y}$ is orthogonally equivalent to $\mathbf{X}'\mathbf{A}\mathbf{X}$.

Conversely, if the two forms are orthogonally equivalent, then there exists an orthogonal matrix $\mathbf{P}$ such that $\mathbf{B} = \mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. Therefore $\mathbf{A}$ and $\mathbf{B}$ are similar matrices and so have the same eigenvalues with the same multiplicities.

Illustrative Examples

Example 10: Reduce each of the following quadratic forms in three variables to real canonical form and find its rank and signature. Also write in each case the linear transformation which brings about the normal reduction.

- (i) $2x_1^2 + x_2^2 - 3x_3^2 - 8x_2x_3 - 4x_3x_1 + 12x_1x_2$.
(ii) $6x_1^2 + 3x_2^2 + 14x_3^2 + 4x_2x_3 + 18x_3x_1 + 4x_1x_2$.

Solution: (i) The matrix $\mathbf{A}$ of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 2 & 6 & -2 \\ 6 & 1 & -4 \\ -2 & -4 & -3 \end{bmatrix}.$$

We write $\mathbf{A} = \mathbf{IAI}$ i.e.,

$$\begin{bmatrix} 2 & 6 & -2 \\ 6 & 1 & -4 \\ -2 & -4 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now we shall reduce $\mathbf{A}$ to diagonal form by applying congruence operations on it. Performing $R_2 \rightarrow R_2 - 3R_1$, $C_2 \rightarrow C_2 - 3C_1$ and $R_3 \rightarrow R_3 + R_1$, $C_3 \rightarrow C_3 + C_1$, we get

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -17 & 2 \\ 0 & 2 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -3 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

[Note that we apply the row and column operations on $\mathbf{A}$ in two separate steps. But in order to save labour we should apply them in one step. For this we should not first write the first row. After changing R_2 and R_3 with the corresponding row operations we should simply write 0 in the second and third places of the first row and the first element of the first row should be kept unchanged].

Now performing $R_3 \rightarrow R_3 + \frac{2}{17} R_2$, $C_3 \rightarrow C_3 + \frac{2}{17} C_2$, we get

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -17 & 0 \\ 0 & 0 & -\frac{81}{17} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ \frac{11}{17} & \frac{2}{17} & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -3 & \frac{11}{17} \\ 0 & 1 & \frac{2}{17} \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_1 \rightarrow \frac{1}{\sqrt{2}} R_1$, $C_1 \rightarrow \frac{1}{\sqrt{2}} C_1$; $R_2 \rightarrow \frac{1}{\sqrt{17}} R_2$, $C_2 \rightarrow \frac{1}{\sqrt{17}} C_2$;

and $R_3 \rightarrow \sqrt{(17/81)} R_3$, $C_3 \rightarrow \sqrt{(17/81)} C_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} a & 0 & 0 \\ -3b & b & 0 \\ \frac{11}{17}c & \frac{2}{17}c & c \end{bmatrix} \mathbf{A} \begin{bmatrix} a & -3b & \frac{11}{17}c \\ 0 & b & \frac{2}{17}c \\ 0 & 0 & c \end{bmatrix},$$

where $a = 1/\sqrt{2}$, $b = 1/\sqrt{17}$, $c = \sqrt{(17/81)}$.

Thus the linear transformation $\mathbf{X} = \mathbf{PY}$

where

$$\mathbf{P} = \begin{bmatrix} a & -3b & \frac{11}{17}c \\ 0 & b & \frac{2}{17}c \\ 0 & 0 & c \end{bmatrix}, \quad \mathbf{X} = [x_1 \quad x_2 \quad x_3]', \quad \mathbf{Y} = [y_1 \quad y_2 \quad y_3]',$$

transforms the given quadratic form to the normal form

$$y_1^2 - y_2^2 - y_3^2. \quad \dots(1)$$

The rank r of the given quadratic form = the number of non-zero terms in its normal form (1) = 3.

The signature of the given quadratic form = the excess of the number of positive terms over the number of negative terms in its normal form = $1 - 2 = -1$.

The index of the given quadratic form = the number of positive terms in its normal form = 1.

The linear transformation $\mathbf{X} = \mathbf{PY}$ which brings about this normal reduction is given by

$$x_1 = ay_1 - 3by_2 + \frac{11}{7}cy_3, x_2 = by_2 + \frac{2}{17}cy_3, x_3 = cy_3.$$

(ii) The matrix of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 6 & 2 & 9 \\ 2 & 3 & 2 \\ 9 & 2 & 14 \end{bmatrix}.$$

$$\text{We write } \begin{bmatrix} 6 & 2 & 9 \\ 2 & 3 & 2 \\ 9 & 2 & 14 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing congruence operations

$$R_2 \rightarrow R_2 - \frac{1}{3}R_1, C_2 \rightarrow C_2 - \frac{1}{3}C_1;$$

$$\text{and } R_3 \rightarrow R_3 - \frac{3}{2}R_1, C_3 \rightarrow C_3 - \frac{3}{2}C_1,$$

we get

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & -1 \\ 0 & -1 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 \\ -\frac{3}{2} & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -\frac{1}{3} & -\frac{3}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_3 \rightarrow R_3 + \frac{3}{7}R_2, C_3 \rightarrow C_3 + \frac{3}{7}C_2$, we get

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & 0 \\ 0 & 0 & \frac{1}{14} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 \\ \frac{23}{14} & \frac{3}{7} & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -\frac{1}{3} & \frac{23}{14} \\ 0 & 1 & \frac{3}{7} \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_1 \rightarrow (1/\sqrt{6})R_1, C_1 \rightarrow (1/\sqrt{6})C_1; R_2 \rightarrow \sqrt{(3/7)}R_2, C_2 \rightarrow \sqrt{(3/7)}C_2;$
 $R_3 \rightarrow (1/14)R_3, C_3 \rightarrow (1/\sqrt{14})C_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a & 0 & 0 \\ -\frac{1}{3}b & b & 0 \\ \frac{23}{14}c & \frac{3}{7}c & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} a & -\frac{1}{3}b & \frac{23}{14}c \\ 0 & b & \frac{3}{7}c \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{where } a = \frac{1}{\sqrt{6}}, b = \sqrt{(3/7)}, c = \frac{1}{\sqrt{14}}.$$

Thus the linear transformation $\mathbf{X} = \mathbf{PY}$ where

$$\mathbf{P} = \begin{bmatrix} a & -\frac{1}{3}b & \frac{23}{14}c \\ 0 & b & \frac{3}{7}c \\ 0 & 0 & 1 \end{bmatrix}$$

transforms the given quadratic form to the normal form

$$y_1^2 + y_2^2 + y_3^2.$$

The rank of the given quadratic form is 3 and its signature is

$$3 - 0 = 3.$$

Example 11: Find an orthogonal matrix $\mathbf{P}$ that will diagonalize the real symmetric matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}.$$

Interpret the result in terms of quadratic forms.

Solution: The characteristic equation of the given matrix is

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

$$\text{i.e., } \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & -1 \\ 1 & -1 & -\lambda \end{vmatrix} = 0 \text{ i.e., } (\lambda - 1)^2 (\lambda + 2) = 0.$$

$\therefore$ the eigenvalues of $\mathbf{A}$ are 1, 1, -2.

Corresponding to the eigenvalue 1 we can find two mutually orthogonal eigenvectors of $\mathbf{A}$ by solving

$$(\mathbf{A} - \mathbf{I}) \mathbf{X} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{or } -x_1 + x_2 + x_3 = 0.$$

Two orthogonal solutions are

$$\mathbf{X}_1 = [1, 0, 1]' \text{ and } \mathbf{X}_2 = [1, 2, -1]'$$

An eigenvector corresponding to the eigenvalue -2, is found by solving

$$2x_1 + x_2 + x_3 = 0, x_1 + 2x_2 - x_3 = 0$$

to be $\mathbf{X}_3 = [-1, 1, 1]'$. The required matrix $\mathbf{P}$ is therefore a matrix whose columns are unit vectors which are scalar multiples of $\mathbf{X}_1$, $\mathbf{X}_2$ and $\mathbf{X}_3$.

$$\therefore \mathbf{P} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$

We have $\mathbf{P}' \mathbf{A} \mathbf{P} = \text{diag. } [1, 1, -2]$

The quadratic form corresponding to the symmetric matrix $\mathbf{A}$ is

$$\phi = 2x_1x_2 + 2x_1x_3 - 2x_2x_3.$$

The orthogonal linear transformation $\mathbf{X} = \mathbf{PY}$ will transform it to the diagonal form

$$y_1^2 + y_2^2 - 2y_3^2.$$

The rank of the quadratic form ϕ = the number of non-zero eigenvalues of its matrix $\mathbf{A} = 3$.

The signature of the quadratic form ϕ = the number of positive eigenvalues of $\mathbf{A}$ = the number of negative eigenvalues of $\mathbf{A} = 2 - 1 = 1$. The normal form is $z_1^2 + z_2^2 - z_3^2$.

Example 12: Reduce the quadratic form

$$6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 + 4x_1x_3 - 2x_2x_3$$

to the canonical form by an orthogonal transformation and hence find the rank and signature of the given quadratic form.

Solution: The matrix $\mathbf{A}$ of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

The characteristic equation of $\mathbf{A}$ is $|\mathbf{A} - \lambda \mathbf{I}| = 0$

$$\text{or} \quad \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\text{or} \quad \begin{vmatrix} 6-\lambda & -2 & 0 \\ -2 & 3-\lambda & 2-\lambda \\ 2 & -1 & 2-\lambda \end{vmatrix} = 0, C_3 \rightarrow C_3 + C_2$$

$$\text{or} \quad (2-\lambda) \begin{vmatrix} 6-\lambda & -2 & 0 \\ -2 & 3-\lambda & 1 \\ 2 & -1 & 1 \end{vmatrix} = 0$$

$$\text{or} \quad (2-\lambda) \begin{vmatrix} 6-\lambda & -2 & 0 \\ -4 & 4-\lambda & 0 \\ 2 & -1 & 1 \end{vmatrix} = 0, R_2 \rightarrow R_2 - R_3$$

$$\text{or} \quad (2-\lambda) [(6-\lambda)(4-\lambda) - 8] = 0$$

$$\text{or} \quad (2-\lambda)(\lambda^2 - 10\lambda + 16) = 0$$

$$\text{or} \quad (2-\lambda)(\lambda-2)(\lambda-8) = 0.$$

$\therefore$ the eigenvalues of $\mathbf{A}$ are 2, 2, 8.

The eigenvalue 8 is of algebraic multiplicity 1. So there will be only one linearly independent eigenvector corresponding to this value.

The eigenvectors corresponding to the eigenvalue 8 are given by the equation

$$(\mathbf{A} - 8\mathbf{I}) \mathbf{X} = \mathbf{O}$$

or

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since these equations have only one linearly independent solution, therefore the coefficient matrix of these equations is of rank 2 and its third row can be made zero

by elementary row operations. So in order to find an eigenvector $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

corresponding to the eigenvalue 8, it is sufficient to find x_1, x_2, x_3 satisfying the equations

$$-2x_1 - 2x_2 + 2x_3 = 0 \quad \dots(1)$$

$$\text{and} \quad -2x_1 - 5x_2 - x_3 = 0 \quad \dots(2)$$

Subtracting (2) from (1), we get

$$3x_2 + 3x_3 = 0.$$

$\therefore x_2 = 1, x_3 = -1, x_1 = -2$ is a solution

Thus $\mathbf{X}_1 = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$ is an eigenvector of $\mathbf{A}$ corresponding to the eigenvalue 8.

The eigenvalue 2 is of algebraic multiplicity 2. So we are to find two mutually orthogonal eigenvectors corresponding to it.

The eigenvectors $\mathbf{X}$ corresponding to the eigenvalue 2 are given by the equation

$$(\mathbf{A} - 2\mathbf{I}) \mathbf{X} = \mathbf{O}$$

or

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since these equations have two linearly independent solutions, therefore their coefficient matrix is of rank 1 and its second and third rows can be made zero by elementary row operations. So we should find two orthogonal solutions of the equation

$$4x_1 - 2x_2 + 2x_3 = 0$$

$$\text{i.e.,} \quad 2x_1 - x_2 + x_3 = 0. \quad \dots(3)$$

Obviously $x_1 = 0, x_2 = 1, x_3 = 1$ is a solution of (3).

$\therefore \mathbf{X}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ is an eigenvector of A corresponding to the eigenvalue 2.

Let $\mathbf{X}_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be another eigenvector of $\mathbf{A}$ corresponding to the eigenvalue 2

and let $\mathbf{X}_3$ be orthogonal to $\mathbf{X}_2$.

Then $2x - y + z = 0$ [$\because \mathbf{X}_3$ is a solution of (3)]

and $0 + y + z = 0$ [$\because \mathbf{X}_2$ and $\mathbf{X}_3$ are orthogonal]

Obviously $y = 1, z = -1, x = 1$ is a solution.

$$\therefore \mathbf{X}_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Lengths of the vectors $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ are $\sqrt{6}, \sqrt{2}, \sqrt{3}$ respectively.

$$\therefore \mathbf{P} = \left[\frac{1}{\sqrt{6}} \mathbf{X}_1, \frac{1}{\sqrt{2}} \mathbf{X}_2, \frac{1}{\sqrt{3}} \mathbf{X}_3 \right] = \begin{bmatrix} -2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{bmatrix}$$

is the required orthogonal matrix that will diagonalize $\mathbf{A}$. We have $\mathbf{P}^{-1} = \mathbf{P}^T$ and

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{P}^T \mathbf{A} \mathbf{P} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

The orthogonal linear transformation $\mathbf{X} = \mathbf{P}\mathbf{Y}$ will transform the given quadratic form to the diagonal form

$$8y_1^2 + 2y_2^2 + 2y_3^2.$$

The canonical form of the given quadratic form is

$$z_1^2 + z_2^2 + z_3^2.$$

The rank of the given quadratic form = the number of non-zero eigenvalues of its matrix $\mathbf{A} = 3$.

The signature of the given quadratic form = the number of positive eigenvalues of $\mathbf{A}$ – the number of negative eigenvalues of $\mathbf{A}$

$$= 3 - 0 = 3.$$

8.15 Lagrange's Reduction of a Real Quadratic Form

This method proceeds by forming squares successively by taking lesser and lesser number of variables among $x_1, x_2, \dots, x_n$. We shall illustrate this method by the following examples.

Illustrative Examples

Example 13: Reduce the quadratic form

$$q(x_1, x_2, x_3) = 4x_1^2 + 10x_2^2 + 11x_3^2 - 4x_1x_2 + 12x_1x_3 - 12x_2x_3$$

to diagonal form by using Lagrange's reduction.

Solution: We have

$$q(x_1, x_2, x_3) = 4x_1^2 + 10x_2^2 + 11x_3^2 - 4x_1x_2 + 12x_1x_3 - 12x_2x_3.$$

Collecting the terms containing x_1 and factoring out x_1 from the terms consisting of x_1x_2 and x_1x_3 , we have

$$q = 4\{x_1^2 - x_1(x_2 - 3x_3)\} + 10x_2^2 + 11x_3^2 - 12x_2x_3.$$

Completing a square on the term x_1 by adding and subtracting suitable terms as needed, we get

$$\begin{aligned} q &= 4\left\{x_1 - \frac{1}{2}(x_2 - 3x_3)\right\}^2 - (x_2 - 3x_3)^2 + 10x_2^2 + 11x_3^2 - 12x_2x_3 \\ &= 4\left(x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3\right)^2 + 9x_2^2 - 6x_2x_3 + 2x_3^2 \\ &= 4\left(x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3\right)^2 + 9\left(x_2^2 - \frac{2}{3}x_2x_3\right) + 2x_3^2. \end{aligned}$$

Now completing square on x_2 , we have

$$\begin{aligned} q &= 4\left(x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3\right)^2 + 9\left(x_2 - \frac{1}{3}x_3\right)^2 - x_3^2 + 2x_3^2 \\ &= 4\left(x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3\right)^2 + 9\left(x_2 - \frac{1}{3}x_3\right)^2 + x_3^2. \end{aligned}$$

Now we put

$$y_1 = x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3$$

$$y_2 = x_2 - \frac{1}{3}x_3$$

$$y_3 = x_3$$

which is of the form $\mathbf{Y} = \mathbf{P}\mathbf{X}$ with the transformation matrix

$$\mathbf{P} = \begin{bmatrix} 1 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix}$$

which is non-singular with determinant equal to 1.

With this transformation, the given quadratic form reduces to the diagonal form

$$q = 4y_1^2 + 9y_2^2 + y_3^2.$$

If we choose $z_1 = 2y_1$, $z_2 = 3y_2$, $z_3 = y_3$, then we have $q = z_1^2 + z_2^2 + z_3^2$, so that the given quadratic form is reduced to a unit quadratic form.

Example 14: Using Lagrange's reduction reduce the quadratic form

$$q(x_1, x_2, x_3) = x_1^2 + 2x_1x_2 + 3x_2^2 + 3x_2x_3$$

to a diagonal form.

Solution: Set

$$x_1 = y_1, \quad x_2 = y_1 + y_2, \quad x_3 = y_3.$$

The transformation $\mathbf{X} = \mathbf{PY}$ where $\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is non-singular because

$$\det \mathbf{P} = 1 \neq 0.$$

The quadratic form q now becomes

$$\begin{aligned} q &= y_1^2 + 2y_1(y_1 + y_2) + 3(y_1 + y_2)^2 + 3(y_1 + y_2)y_3 \\ &= y_1^2 + 2y_1y_2 + 5y_1^2 + 3y_2^2 + 3y_2y_3 \\ &= \{y_1^2 + y_1(y_2 + 5y_3)\} + 3y_2^2 + 3y_2y_3 \\ &= \left\{y_1 + \frac{1}{2}(y_2 + 5y_3)\right\}^2 - \frac{1}{4}(y_2^2 + 10y_2y_3 + 25y_3^2) + 3y_2^2 + 3y_2y_3 \\ &= \left(y_1 + \frac{1}{2}y_2 + \frac{5}{2}y_3\right)^2 - \frac{1}{4}(y_2^2 + 10y_2y_3 + 25y_3^2) + 3y_2^2 + 3y_2y_3 \\ &= \left(y_1 + \frac{1}{2}y_2 + \frac{5}{2}y_3\right)^2 - \frac{1}{4}(y_2^2 - 2y_2y_3) - \frac{25}{4}y_3^2 \\ &= \left(y_1 + \frac{1}{2}y_2 + \frac{5}{2}y_3\right)^2 - \frac{1}{4}(y_2 - y_3)^2 + \frac{1}{4}y_3^2 - \frac{25}{4}y_3^2 \\ &= \left(y_1 + \frac{1}{2}y_2 + \frac{5}{2}y_3\right)^2 - \frac{1}{4}(y_2 - y_3)^2 - 6y_3^2. \end{aligned}$$

Now put $z_1 = y_1 + \frac{1}{2}y_2 + \frac{5}{2}y_3 = \frac{1}{2}x_1 + \frac{1}{2}x_2 + \frac{5}{2}x_3$

$$z_2 = y_2 - y_3 = x_2 - x_1 - x_3$$

$$z_3 = y_3 = x_3$$

so that the non-singular transformation $\mathbf{Z} = \mathbf{QX}$ i.e.,

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{5}{2} \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

with $\det \mathbf{Q} = 1$ will reduce the given quadratic form to the diagonal form

$$q = z_1^2 - \frac{1}{4}z_2^2 - 6z_3^2.$$

Comprehensive Exercise 2

1. Write the matrix and find the rank of each of the following quadratic forms :

(i) $x_1^2 - 2x_1x_2 + 2x_2^2$

(ii) $4x_1^2 + x_2^2 - 8x_3^2 + 4x_1x_2 - 4x_1x_3 + 8x_2x_3$.

2. Determine a non-singular matrix $\mathbf{P}$ such that $\mathbf{P}'\mathbf{A}\mathbf{P}$ is a diagonal matrix, where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 2 & 3 & 0 \end{bmatrix}.$$

3. Reduce each of the following quadratic forms in three variables to real canonical form and find its rank and signature. Also write in each case the linear transformation which brings about the normal reduction.

(i) $x^2 - 2y^2 + 3z^2 - 4yz + 6zx$.

(ii) $x^2 + 2y^2 + 2z^2 - 2xy - 2yz + zx$.

4. Reduce the following quadratic form to canonical form and find its rank and signature :

$$x^2 + 4y^2 + 9z^2 + t^2 - 12yz + 6zx - 4xy - 2xt - 6zt.$$

5. Using Lagrange's reduction reduce the quadratic form

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 5 & -2 & 0 \\ -2 & 6 & 2 \\ 0 & 2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

to a diagonal form.

6. Using Lagrange's reduction reduce the quadratic form

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \begin{bmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

to a diagonal form.

7. Using Lagrange's reduction reduce the quadratic form

$$q(x_1, x_2, x_3) = (x_1 + x_2 + x_3)x_2 \text{ to a diagonal form.}$$

8. Using Lagrange's reduction reduce the quadratic form

$$q(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_3x_1 \text{ to a diagonal form.}$$

9. Using Lagrange's reduction reduce the quadratic form

$$q(x_1, x_2, x_3) = x_1^2 + 4x_2^2 + 16x_3^2 + 4x_1x_2 + 8x_1x_3 + 17x_2x_3$$

to a diagonal form.

Answers 2

$$1. \quad (i) \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \text{rank } 2 \qquad (ii) \begin{bmatrix} 4 & 2 & -2 \\ 2 & 1 & 4 \\ -2 & 4 & -8 \end{bmatrix}, \text{rank } 3$$

$$2. \quad \mathbf{P} = \begin{bmatrix} 1 & -\frac{1}{2} & -3 \\ 1 & \frac{1}{2} & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

3. (i) The linear transformation $\mathbf{X} = \mathbf{PY}$ where

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & -\frac{3}{2} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix}, \mathbf{X} = [x \ y \ z]', \mathbf{Y} = [y_1 \ y_2 \ y_3]'$$

transforms the given quadratic form to the normal form $y_1^2 - y_2^2 - y_3^2$

The rank of the given quadratic form is 3 and its signature is -1

(ii) The linear transformation $\mathbf{X} = \mathbf{PY}$ where

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1/\sqrt{6} \\ 0 & 0 & \sqrt{2/3} \end{bmatrix}$$

transforms the given quadratic form to the normal form $y_1^2 + y_2^2 + y_3^2$

Rank of the given quadratic form is 3 and its signature is 3

$$4. \quad y_1^2 - y_2^2 + y_4^2$$

Rank of the given quadratic form is 3 and its signature is 1

$$5. \quad \frac{1}{5} y_1^2 + \frac{10}{13} y_2^2 + \frac{81}{13} y_3^2$$

$$6. \quad \frac{1}{3} y_1^2 + \frac{8}{3} y_2^2 + \frac{1}{3} y_3^2 + \frac{8}{3} y_4^2$$

$$7. \quad \frac{1}{4} (y_1^2 - y_2^2)$$

$$8. \quad \frac{1}{4} (z_1^2 - z_2^2) - z_3^2$$

$$9. \quad z_1^2 + \frac{1}{4} (z_2^2 - z_3^2)$$

8.16 Value Class of a Real Quadratic Form. Definite, Semi-definite and Indefinite Real Quadratic Forms

Definitions: Let $\phi = \mathbf{X}' \mathbf{A} \mathbf{X}$ be a real quadratic form in n variables $x_1, \dots, x_n$. The form ϕ is said to be

(i) **Positive Definite (PD)** if $\phi \geq 0$ for all real values of the variables $x_1, \dots, x_n$ and $\phi = 0$ only if $\mathbf{X} = \mathbf{O}$ i.e., $\phi = 0 \Rightarrow x_1 = x_2 = \dots = x_n = 0$.

For example the quadratic form $x_1^2 - 4x_1 x_2 + 5x_2^2$ in two variables is positive definite because it can be written as

$$(x_1 - 2x_2)^2 + x_2^2,$$

which is ≥ 0 for all real values of x_1, x_2 and

$$(x_1 - 2x_2)^2 + x_2^2 = 0 \Rightarrow x_1 - 2x_2 = 0, x_2 = 0$$

$$\Rightarrow x_1 = 0, x_2 = 0.$$

Similarly the quadratic form $x_1^2 + x_2^2 + x_3^2$ in three variables is a positive definite form.

(ii) **Negative definite (ND)** if $\phi \leq 0$ for all real values of the variables $x_1, \dots, x_n$ and $\phi = 0$ only if $x_1 = x_2 = \dots = x_n = 0$.

For example $-x_1^2 - x_2^2 - x_3^2$ is a negative definite form in three variables.

(iii) **Positive semi-definite (PSD)** if $\phi \geq 0$ for all real values of the variables $x_1, \dots, x_n$ and $\phi = 0$ for some non-zero real vector $\mathbf{X}$ i.e., $\phi = 0$ for some real values of the variables $x_1, x_2, \dots, x_n$ not all zero.

For example the quadratic form

$$x_1^2 + x_2^2 + 2x_3^2 - 2x_1x_3 - 2x_2x_3$$

is positive semi-definite because it can be written in the form

$$(x_1 - x_3)^2 + (x_2 - x_3)^2,$$

which is ≥ 0 for all real values of x_1, x_2, x_3 but is zero for non-zero values also, for example, $x_1 = x_2 = x_3 = 1$.

Similarly the quadratic form $x_1^2 + x_2^2 + 0x_3^2$ in three variables x_1, x_2, x_3 is positive semi-definite. It is non-negative for all real values of x_1, x_2, x_3 and it is zero for values $x_1 = 0, x_2 = 0, x_3 = 2$ which are not all zero.

(iv) **Negative semi-definite (NSD)** if $\phi \leq 0$ for all real values of the variables $x_1, \dots, x_n$ and $\phi = 0$ for some values of the variables $x_1, \dots, x_n$ not all zero.

For example the quadratic form $-x_1^2 - x_2^2 - 0x_3^2$ in three variables x_1, x_2, x_3 is negative semi-definite.

(v) **Indefinite (I)** if ϕ takes positive as well as negative values for real values of the variables $x_1, \dots, x_n$.

For example the quadratic form $x_1^2 - x_2^2 + x_3^2$ in three variables is indefinite. It takes positive value 1 when $x_1 = 1, x_2 = 1, x_3 = 1$ and it takes negative value -1 when $x_1 = 0, x_2 = 1, x_3 = 0$.

Note I: The above five classes of real quadratic forms are mutually exclusive and are called **value classes of real quadratic forms**. Every real quadratic form must belong to one and only one value class.

Note 2: A form which is positive definite or negative definite is called *definite* and a form which is positive semi-definite or negative semi-definite is called *semi-definite*.

Non-negative definite quadratic form:

Definition: A real quadratic form $\phi = \mathbf{X}'\mathbf{A}\mathbf{X}$ in n variables $x_1, \dots, x_n$, is said to be non-negative definite if it takes only non-negative values for all real values of $x_1, \dots, x_n$.

Thus ϕ is non-negative definite if $\phi \geq 0$ for all real values of $x_1, \dots, x_n$. A non-negative definite quadratic form may be positive definite or positive semi-definite. It is positive definite if it takes the value 0 only when $x_1 = x_2 = \dots = x_n = 0$.

Classification of real-symmetric matrices:

Definite, semi-definite and indefinite real symmetric matrices:

Definition: A real symmetric matrix $\mathbf{A}$ is said to be *definite*, *semi-definite* or *indefinite* if the corresponding quadratic form $\mathbf{X}'\mathbf{A}\mathbf{X}$ is *definite*, *semi-definite* or *indefinite* respectively.

Positive definite real symmetric matrix:

Definition: A real symmetric matrix $\mathbf{A}$ is said to be *positive definite* if the corresponding form $\mathbf{X}'\mathbf{A}\mathbf{X}$ is *positive definite*.

Non-negative definite real symmetric matrix:

Definition: A real symmetric matrix $\mathbf{A}$ is said to be *non-negative definite* if the associated quadratic form $\mathbf{X}'\mathbf{A}\mathbf{X}$ is *non-negative definite*.

Theorem 1: All real equivalent real quadratic forms have the same value class.

Proof: Let $\phi = \mathbf{X}'\mathbf{A}\mathbf{X}$ and $\psi = \mathbf{Y}'\mathbf{B}\mathbf{Y}$ be two real equivalent real quadratic forms.

Then there exists a real non-singular matrix $\mathbf{P}$ such that $\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{B}$ and $(\mathbf{P}^{-1})'\mathbf{B}\mathbf{P}^{-1} = \mathbf{A}$. The real non-singular linear transformation $\mathbf{X} = \mathbf{P}\mathbf{Y}$ transforms the quadratic form ϕ into the quadratic form ψ and the inverse transformation $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}$ transforms the quadratic form ψ into the quadratic form ϕ . The two quadratic forms have the same ranges of values. The vectors $\mathbf{X}$ and $\mathbf{Y}$ for which ϕ and ψ have the same value are connected by the relations $\mathbf{X} = \mathbf{P}\mathbf{Y}$ and $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}$. Thus the vector $\mathbf{Y}$ for which ψ has the same value as ϕ has for the vector $\mathbf{X}$ is given by $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}$. Similarly the vector $\mathbf{X}$ for which ϕ has the same value as ψ has for the vector $\mathbf{Y}$ is given by $\mathbf{X} = \mathbf{P}\mathbf{Y}$.

Now we shall discuss the five cases separately.

Case I: ϕ is positive definite if and only if ψ is positive definite.

Suppose ϕ is positive definite.

Then $\phi \geq 0$ and $\phi = 0 \Rightarrow \mathbf{X} = \mathbf{O}$.

Since ϕ and ψ have same ranges of values, therefore

$$\phi \geq 0 \Rightarrow \psi \geq 0.$$

Also $\psi = 0 \Rightarrow \mathbf{Y}' \mathbf{B} \mathbf{Y} = 0$

$\Rightarrow (\mathbf{P}\mathbf{Y})' \mathbf{A} (\mathbf{P}\mathbf{Y}) = 0$ [$\because \phi$ has the same value for the vector $\mathbf{P}\mathbf{Y}$ as ψ has for the vector $\mathbf{Y}$]

$\Rightarrow \mathbf{P}\mathbf{Y} = \mathbf{O}$ [$\because \phi$ is positive definite means $\mathbf{X}' \mathbf{A} \mathbf{X} = 0 \Rightarrow \mathbf{X} = \mathbf{O}$]

$\Rightarrow \mathbf{P}^{-1} (\mathbf{P}\mathbf{Y}) = \mathbf{P}^{-1} \mathbf{O}$

$\Rightarrow \mathbf{Y} = \mathbf{O}.$

Thus ψ is also positive definite.

Conversely suppose that ψ is positive definite.

Then $\psi \geq 0$ and $\psi = 0 \Rightarrow \mathbf{Y} = \mathbf{O}.$

Since ϕ and ψ have the same ranges of values, therefore

$$\psi \geq 0 \Rightarrow \phi \geq 0.$$

Also $\phi = 0 \Rightarrow \mathbf{X}' \mathbf{A} \mathbf{X} = 0$

$\Rightarrow (\mathbf{P}^{-1} \mathbf{X})' \mathbf{B} (\mathbf{P}^{-1} \mathbf{X}) = 0$ [$\because \psi$ has the same value for the vector $\mathbf{P}^{-1} \mathbf{X}$ as ϕ has for the vector $\mathbf{X}$]

$\Rightarrow \mathbf{P}^{-1} \mathbf{X} = \mathbf{O}$ [$\because \psi$ is positive definite]

$\Rightarrow \mathbf{P} (\mathbf{P}^{-1} \mathbf{X}) = \mathbf{P} \mathbf{O}$

$\Rightarrow \mathbf{X} = \mathbf{O}.$

Thus ϕ is also positive definite.

Case II: ϕ is negative definite if and only if ψ is negative definite.

The proof is the same as in case I.

The only difference is that we are to replace the expressions $\phi \geq 0, \psi \geq 0$ by the expressions $\phi \leq 0, \psi \leq 0$.

Case III: ϕ is positive semi-definite if and only if ψ is positive semi-definite.

Since ϕ and ψ have the same ranges of values, therefore $\phi \geq 0$ if and only if $\psi \geq 0$.

Further since $\mathbf{P}$ is non-singular, therefore

$$\mathbf{X} \neq \mathbf{O} \Rightarrow \mathbf{Y} = \mathbf{P}^{-1} \mathbf{X} \neq \mathbf{O}$$

and $\mathbf{Y} \neq \mathbf{O} \Rightarrow \mathbf{X} = \mathbf{P} \mathbf{Y} \neq \mathbf{O}.$

Also the vectors $\mathbf{X}$ and $\mathbf{Y}$ for which ϕ and ψ have the same values are connected by the relations $\mathbf{X} = \mathbf{P}\mathbf{Y}$ and $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}$. Therefore $\phi = 0$ for some non-zero vector $\mathbf{X}$ if and only if $\psi = 0$ for some non-zero vector $\mathbf{Y}$.

Hence ϕ is positive semi-definite if and only if ψ is positive semi-definite.

Case IV: ϕ is negative semi-definite if and only if ψ is negative semi-definite.

For proof replace the expressions $\phi \geq 0, \psi \geq 0$ in case III by the expressions

$$\phi \leq 0, \psi \leq 0.$$

Case V: ϕ is indefinite if and only if ψ is indefinite. Since ϕ and ψ have the same ranges of values, therefore the result follows immediately.

Thus the proof of the theorem is complete.

Criterion for the value of a real quadratic form in terms of its rank and signature:

Theorem 2: Suppose r is the rank and s is the signature of a real quadratic form $\phi = \mathbf{X}' \mathbf{A} \mathbf{X}$ in n variables. Then ϕ is (i) positive definite if and only if $s = r = n$, (ii) negative definite if and only if $-s = r = n$, (iii) positive semi-definite if and only if $s = r < n$, (iv) negative semi-definite if and only if $-s = r < n$; and (v) indefinite if and only if $|s| \neq r$.

Proof: Let $\psi = y_1^2 + \dots + y_p^2 - y_{p+1}^2 - \dots - y_r^2$... (1)

be the real canonical form of the real **quadratic** form ϕ of rank r and signature s . Then $s = 2p - r$. Since ϕ and ψ are real equivalent real quadratic forms, therefore they have the same value class.

(i) Suppose $s = r = n$. Then $p = n$ and the real canonical form of ϕ becomes $y_1^2 + \dots + y_n^2$. But this is a positive definite quadratic form. So ϕ is also positive definite.

Conversely suppose that ϕ is positive definite. Then ψ is also a positive definite form in n variables. So we must have $\psi = y_1^2 + \dots + y_n^2$.

Hence $r = n, p = n, 2p - r = s = n$.

(ii) Suppose $-s = r = n$. Then $s = 2p - r$ gives $p = 0$. The real canonical form of ϕ becomes $-y_1^2 - \dots - y_n^2$ which is negative definite and so ϕ is also negative definite.

Conversely if ϕ is negative definite, then ψ is also negative definite and so we must have $\psi = -y_1^2 - \dots - y_n^2$.

Hence $r = n, p = 0, 2p - r = s = -n$ i.e., $-s = n$.

(iii) Suppose $s = r < n$. Then $s = 2p - r$ gives $p = r$ and the real canonical form of ϕ becomes $y_1^2 + \dots + y_r^2$ where $r < n$. But this is a positive semi-definite form in n variables. So ϕ is also positive semi-definite.

Conversely if ϕ is positive semi-definite, then ψ is also a positive semi-definite form in n variables. So we must have $\psi = y_1^2 + \dots + y_r^2$ where $r < n$.

Therefore $p = r < n$ and $s = 2p - r = r$. Thus $s = r < n$.

(iv) Suppose $-s = r < n$. Then $s = 2p - r$ gives $p = 0$ and the real canonical form of ϕ becomes $-y_1^2 - \dots - y_r^2$ where $r < n$. This is a negative semi-definite form in n variables. So ϕ is also negative semi-definite.

Conversely if ϕ is negative semi-definite, then ψ is also a negative semi-definite form in n variables. So we must have

$$\psi = -y_1^2 - \dots - y_r^2, \text{ where } r < n.$$

Therefore $p = 0$ and $s = 2p - r = -r$. Thus $-s = r < n$.

(v) Suppose $|s| \neq r$. Then $|2p - r| \neq r$. Therefore $p \neq 0$ and $p \neq r$ and so $0 < p < r$. Then in this case the canonical form of ϕ has positive as well as negative terms and so it is an indefinite form. Consequently ϕ is also indefinite.

Conversely if ϕ is indefinite, then ψ is also indefinite. So there must be positive as well as negative terms in ψ . Therefore $|s| \neq r$.

Criterion for the value class of a real quadratic form in terms of the eigenvalues of its matrix:

Suppose $\phi = \mathbf{X}'\mathbf{A}\mathbf{X}$ is a real quadratic form in n variables. Then $\mathbf{A}$ is a real symmetric matrix of order n . Suppose r is the number of non-zero eigenvalues of $\mathbf{A}$. Then $r = \text{rank of the quadratic form } \phi$. Further if $s = \text{the number of positive eigenvalues of } \mathbf{A} - \text{the number of negative eigenvalues of } \mathbf{A}$, then $s = \text{signature of } \phi$. Hence with the help of theorem 2, we arrive at the following conclusion.

Theorem 3: A real quadratic form $\phi = \mathbf{X}'\mathbf{A}\mathbf{X}$ in n variables is

- (i) positive definite if and only if all the eigenvalues of $\mathbf{A}$ are positive.
- (ii) negative definite if and only if all the eigenvalues of $\mathbf{A}$ are negative.
- (iii) positive semi-definite if and only if all the eigenvalues of $\mathbf{A}$ are ≥ 0 and at least one eigenvalue is 0.
- (iv) negative semi-definite if and only if all the eigenvalues of $\mathbf{A}$ are ≤ 0 and at least one eigenvalue of $\mathbf{A}$ is 0.
- (v) indefinite if and only if $\mathbf{A}$ has positive as well as negative eigenvalues.

On account of its importance we shall give an independent proof of case (i).

Theorem 4: A real symmetric matrix is positive definite if and only if all its eigenvalues are positive.

Proof: Let $\mathbf{A}$ be a real symmetric matrix of order n . Then there exists an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{D} = \text{diag} [\lambda_1, \lambda_2, \dots, \lambda_n]$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of $\mathbf{A}$.

Let $\mathbf{X}'\mathbf{A}\mathbf{X}$ be the real quadratic form corresponding to the matrix $\mathbf{A}$. Let us transform this quadratic form by the real non-singular linear transformation $\mathbf{X} = \mathbf{P}\mathbf{Y}$ where $\mathbf{Y} = [y_1, y_2, \dots, y_n]'$. Then

$$\mathbf{X}'\mathbf{A}\mathbf{X} = (\mathbf{P}\mathbf{Y})'\mathbf{A}(\mathbf{P}\mathbf{Y}) = \mathbf{Y}'\mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{Y} = \mathbf{Y}'\mathbf{D}\mathbf{Y}.$$

$$\text{Therefore } \mathbf{X}'\mathbf{A}\mathbf{X} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2. \quad \dots(1)$$

Now suppose that $\lambda_1, \lambda_2, \dots, \lambda_n$ are all positive. Then the right hand side of (1) ensures that $\mathbf{X}'\mathbf{A}\mathbf{X} \geq 0$ for all real vectors $\mathbf{X}$. Also

$$\begin{aligned} & \mathbf{X}'\mathbf{A}\mathbf{X} = 0 \\ \Rightarrow & \lambda_1 y_1^2 + \dots + \lambda_n y_n^2 = 0 \\ \Rightarrow & y_1 = y_2 = \dots = y_n = 0 \quad [\because \lambda_1, \dots, \lambda_n \text{ are all positive}] \\ \Rightarrow & \mathbf{Y} = \mathbf{O} \\ \Rightarrow & \mathbf{P}^{-1}\mathbf{X} = \mathbf{O} \quad [\because \mathbf{X} = \mathbf{P}\mathbf{Y} \Rightarrow \mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}] \\ \Rightarrow & \mathbf{P}(\mathbf{P}^{-1}\mathbf{X}) = \mathbf{P}\mathbf{O} \\ \Rightarrow & \mathbf{X} = \mathbf{O}. \end{aligned}$$

Thus if $\lambda_1, \lambda_2, \dots, \lambda_n$ are all positive, then $\mathbf{X}' \mathbf{A} \mathbf{X}$ is positive definite and so the matrix $\mathbf{A}$ is positive definite.

Conversely suppose that $\mathbf{A}$ is a positive definite matrix. Then the quadratic form $\mathbf{X}' \mathbf{A} \mathbf{X}$ is positive definite. So

$\mathbf{X}' \mathbf{A} \mathbf{X} \geq 0$ for all real vectors $\mathbf{X}$

$$\Rightarrow \lambda_1 y_1^2 + \dots + \lambda_n y_n^2 \geq 0 \text{ for all real vectors } \mathbf{Y}$$

$$\Rightarrow \lambda_1, \dots, \lambda_n \text{ are all } \geq 0.$$

Also $\mathbf{X}' \mathbf{A} \mathbf{X} = 0$ only if $\mathbf{X} = \mathbf{O}$

$$\Rightarrow \lambda_1 y_1^2 + \dots + \lambda_n y_n^2 = 0 \text{ only if } \mathbf{P} \mathbf{Y} = \mathbf{O}$$

$$\Rightarrow \lambda_1 y_1^2 + \dots + \lambda_n y_n^2 = 0 \text{ only if } \mathbf{Y} = \mathbf{O} \quad [\because \mathbf{P} \text{ is non-singular means } \mathbf{P} \mathbf{Y} = \mathbf{O} \text{ only if } \mathbf{Y} = \mathbf{O}]$$

$$\Rightarrow \lambda_1, \dots, \lambda_n \text{ are all not equal to zero.}$$

Therefore if $\mathbf{A}$ is positive definite, then $\lambda_1, \lambda_2, \dots, \lambda_n$ are all > 0 .

This completes the proof of the theorem.

Corollary 1: A positive definite real symmetric matrix is non-singular.

Proof: Suppose $\mathbf{A}$ is a positive definite real symmetric matrix. Then the eigenvalues of $\mathbf{A}$ are all positive. Also there exists an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{D},$$

where $\mathbf{D}$ is a diagonal matrix having the eigenvalues of $\mathbf{A}$ as its diagonal elements. So all diagonal elements of $\mathbf{D}$ are positive and thus $\mathbf{D}$ is non-singular.

Now $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1} \Rightarrow \mathbf{A}$ is non-singular.

Corollary 2: If the real quadratic form $\mathbf{X}^T \mathbf{A} \mathbf{X}$ is positive definite, then there exists a non-singular transformation $\mathbf{X} = \mathbf{P} \mathbf{Y}$ such that $\mathbf{X}^T \mathbf{A} \mathbf{X} = \mathbf{Y}^T \mathbf{Y}$.

Theorem 5: A real matrix $\mathbf{A}$ is symmetric and positive definite if and only if $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is symmetric and positive definite for any real and non-singular $\mathbf{P}$.

Proof: The condition is necessary. Suppose that $\mathbf{A}$ is real, symmetric and positive definite.

This implies that the real quadratic form $\mathbf{X}^T \mathbf{A} \mathbf{X}$ is positive definite. Put $\mathbf{X} = \mathbf{P} \mathbf{Y}$, where $\mathbf{Y} \neq \mathbf{O}$ if $\mathbf{X} \neq \mathbf{O}$ because $\mathbf{P}$ is non-singular. Hence from

$$\mathbf{X}^T \mathbf{A} \mathbf{X} = (\mathbf{P} \mathbf{Y})^T \mathbf{A} (\mathbf{P} \mathbf{Y}) = \mathbf{Y}^T (\mathbf{P}^T \mathbf{A} \mathbf{P}) \mathbf{Y}$$

we conclude that the right hand side is positive for all non-zero values of $\mathbf{Y}$.

So the quadratic form $\mathbf{Y}^T (\mathbf{P}^T \mathbf{A} \mathbf{P}) \mathbf{Y}$ is positive definite and hence the matrix $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is positive definite.

$$\text{Also } \mathbf{P}^T \mathbf{A} \mathbf{P} = \mathbf{P}^T \mathbf{A}^T \mathbf{P} \quad [\because \mathbf{A} \text{ is symmetric } \Rightarrow \mathbf{A}^T = \mathbf{A}]$$

$$= (\mathbf{P}^T \mathbf{A} \mathbf{P})^T.$$

$$\therefore \mathbf{P}^T \mathbf{A} \mathbf{P} \text{ is symmetric.}$$

Hence $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is symmetric and positive definite.

The condition is sufficient: Now suppose that $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is symmetric and positive definite for any real and non-singular $\mathbf{P}$. Putting $\mathbf{P} = \mathbf{I}$, we get

$$\mathbf{P}^T \mathbf{A} \mathbf{P} = \mathbf{I}^T \mathbf{A} \mathbf{I} = \mathbf{I} \mathbf{A} \mathbf{I} = \mathbf{A}.$$

Since $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is symmetric and positive definite, therefore $\mathbf{A}$ is symmetric and positive definite.

Theorem 6: A real symmetric matrix $\mathbf{A}$ is positive definite if and only if there exists a non-singular matrix $\mathbf{Q}$ such that

$$\mathbf{A} = \mathbf{Q}' \mathbf{Q}.$$

Proof: Suppose $\mathbf{A}$ is positive definite. Then all the eigenvalues of $\mathbf{A}$ are positive and we can find an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{P}' \mathbf{A} \mathbf{P} = \mathbf{D} = \text{diag. } [\lambda_1, \dots, \lambda_n]$$

where each $\lambda_i > 0$. Let $\mathbf{D}_1 = \text{diag. } [\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}]$. Then $\mathbf{D}_1^2 = \mathbf{D}$ and $\mathbf{D}_1' = \mathbf{D}_1$. We have

$$\begin{aligned} \mathbf{A} &= \mathbf{P} \mathbf{D} \mathbf{P}^{-1} = \mathbf{P} \mathbf{D}_1^2 \mathbf{P}^{-1} = \mathbf{P} \mathbf{D}_1 \mathbf{D}_1 \mathbf{P}' \\ &= (\mathbf{P} \mathbf{D}_1) (\mathbf{P} \mathbf{D}_1)' = \mathbf{Q}' \mathbf{Q} \text{ where } \mathbf{Q} = (\mathbf{P} \mathbf{D}_1)'. \end{aligned}$$

Clearly, $\mathbf{Q}$ is non-singular since $\mathbf{P}$ and $\mathbf{D}_1$ are non-singular.

Conversely suppose that $\mathbf{A} = \mathbf{Q}' \mathbf{Q}$ where $\mathbf{Q}$ is non-singular. We have for all real vectors $\mathbf{X}$,

$$\begin{aligned} \mathbf{X}' \mathbf{A} \mathbf{X} &= \mathbf{X}' \mathbf{Q}' \mathbf{Q} \mathbf{X} = (\mathbf{Q} \mathbf{X})' (\mathbf{Q} \mathbf{X}) \\ &= \mathbf{Y}' \mathbf{Y}, \text{ where } \mathbf{Y} = \mathbf{Q} \mathbf{X} \text{ is a real } n\text{-vector} \\ &\geq 0. \end{aligned}$$

$$\text{Also } \mathbf{X}' \mathbf{A} \mathbf{X} = 0 \Rightarrow \mathbf{Y}' \mathbf{Y} = 0$$

$$\Rightarrow \mathbf{Y} = \mathbf{0} \Rightarrow \mathbf{Q} \mathbf{X} = \mathbf{0}$$

$$[\because \mathbf{Y} = \mathbf{Q} \mathbf{X}]$$

$$\Rightarrow \mathbf{Q}^{-1} (\mathbf{Q} \mathbf{X}) = \mathbf{Q}^{-1} \mathbf{0} \Rightarrow \mathbf{X} = \mathbf{0}.$$

$\therefore \mathbf{X}' \mathbf{A} \mathbf{X}$ is a positive definite real quadratic form and so the symmetric matrix $\mathbf{A}$ is positive definite.

Theorem 7: Every real non-singular matrix $\mathbf{A}$ can be written as a product $\mathbf{A} = \mathbf{P} \mathbf{S}$, where $\mathbf{S}$ is a positive definite symmetric matrix and $\mathbf{P}$ is orthogonal.

Proof: Since $\mathbf{A}$ is non-singular, therefore by theorem 6, $\mathbf{A}' \mathbf{A}$ is a positive definite real symmetric matrix. Let $\mathbf{Q}$ be an orthogonal matrix such that

$$\mathbf{Q}^{-1} (\mathbf{A}' \mathbf{A}) \mathbf{Q} = \mathbf{Q}' (\mathbf{A}' \mathbf{A}) \mathbf{Q} = \mathbf{D} = \text{diag. } [\lambda_1, \dots, \lambda_n],$$

where $\lambda_1, \dots, \lambda_n$ are the positive real eigenvalues of $\mathbf{A}' \mathbf{A}$. Let

$$\mathbf{D}_1 = \text{diag. } [\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}].$$

$$\text{Then } \mathbf{D}_1^2 = \mathbf{D} \text{ and } \mathbf{D}_1' = \mathbf{D}_1.$$

Now let $\mathbf{S} = \mathbf{Q} \mathbf{D}_1 \mathbf{Q}'$. Clearly $\mathbf{S}' = \mathbf{S}$ and so $\mathbf{S}$ is symmetric. Moreover $\mathbf{S}$ is positive definite because it is similar to $\mathbf{D}_1$ which has positive eigenvalues.

$$\begin{aligned} \text{Also } \mathbf{S}^2 &= \mathbf{Q} \mathbf{D}_1 \mathbf{Q}' \mathbf{Q} \mathbf{D}_1 \mathbf{Q}' = \mathbf{Q} \mathbf{D}_1 \mathbf{Q}^{-1} \mathbf{Q} \mathbf{D}_1 \mathbf{Q}' \\ &= \mathbf{Q} \mathbf{D}_1^2 \mathbf{Q}^{-1} = \mathbf{Q} \mathbf{D} \mathbf{Q}^{-1} = \mathbf{A}' \mathbf{A}. \end{aligned}$$

Now let $\mathbf{P} = \mathbf{A}\mathbf{S}^{-1}$. Then $\mathbf{P}$ is orthogonal because

$$\begin{aligned}\mathbf{P}'\mathbf{P} &= (\mathbf{A}\mathbf{S}^{-1})' \mathbf{A}\mathbf{S}^{-1} = (\mathbf{S}^{-1})' \mathbf{A}'\mathbf{A}\mathbf{S}^{-1} \\ &= (\mathbf{S}^{-1})' \mathbf{S}^2 \mathbf{S}^{-1} && [\because \mathbf{A}'\mathbf{A} = \mathbf{S}^2] \\ &= (\mathbf{S}^{-1})' \mathbf{S} \mathbf{S} \mathbf{S}^{-1} = (\mathbf{S}^{-1})' \mathbf{S} \\ &= (\mathbf{S}')^{-1} \mathbf{S} = \mathbf{S}^{-1} \mathbf{S} && [\because \mathbf{S} \text{ is symmetric}] \\ &= \mathbf{I}.\end{aligned}$$

Thus $\mathbf{S} = \mathbf{Q}\mathbf{D}_1 \mathbf{Q}'$ is a positive definite real symmetric matrix and $\mathbf{P} = \mathbf{A}\mathbf{S}^{-1}$ is an orthogonal matrix and we have

$$\mathbf{P}\mathbf{S} = \mathbf{A}\mathbf{S}^{-1} \mathbf{S} = \mathbf{A}.$$

Hence the result.

Note: The decomposition $\mathbf{A} = \mathbf{P}\mathbf{S}$ obtained in theorem 7 is called the *polar factorization of A*.

8.17 Criterion for Positive-Definiteness of a Quadratic Form in Terms of Leading Principal Minors of its Matrix

Leading principal minors of a matrix. Definition : Let

$$\mathbf{A} = [a_{ij}]_{n \times n}$$

be a square matrix of order n . Then

$$\begin{aligned}A_1 &= a_{11}, \quad A_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \\ A_3 &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \dots, A_n = \begin{vmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} \end{vmatrix}\end{aligned}$$

are called the *leading principal minors of A*.

Before stating the main theorem we shall prove the following Lemma.

Lemma : If $\mathbf{A}$ is the matrix of a positive definite form, then

$$|\mathbf{A}| > 0.$$

Proof: If $\mathbf{X}'\mathbf{A}\mathbf{X}$ is a positive definite real quadratic form, then there exists a real non-singular matrix $\mathbf{P}$ such that

$$\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{I}.$$

$$\therefore |\mathbf{P}'\mathbf{A}\mathbf{P}| = |\mathbf{I}| = 1$$

$$\text{or } |\mathbf{P}'| |\mathbf{A}| |\mathbf{P}| = 1 \quad \text{or} \quad |\mathbf{A}| = 1/|\mathbf{P}|^2$$

$$[\because |\mathbf{P}| = |\mathbf{P}'| \neq 0]$$

Therefore $|\mathbf{A}|$ is positive.

Now we shall state and prove the main theorem.

Theorem: Sylvester's Criterion for Positive Definiteness: A necessary and sufficient condition for a real quadratic form $\mathbf{X}' \mathbf{A} \mathbf{X}$ to be positive definite is, that the leading principal minors of $\mathbf{A}$ are all positive.

Proof: The condition is necessary: Suppose $\mathbf{X}' \mathbf{A} \mathbf{X}$ is a positive definite quadratic form in n variables. Let k be any natural number such that $k \leq n$. Putting $x_{k+1} = 0, \dots, x_n = 0$ in the positive definite form $\mathbf{X}' \mathbf{A} \mathbf{X}$, we get a positive definite form in k variables $x_1, \dots, x_k$. The determinant of the matrix of this new quadratic form is the leading principal minor of order k of $\mathbf{A}$ and is positive by virtue of the lemma we have just proved. Thus every leading principal minor of the matrix of a positive definite quadratic form is positive.

The condition is sufficient: Now it is given that the leading principal minors of $\mathbf{A}$ are all positive and we are to prove that the form $\mathbf{X}' \mathbf{A} \mathbf{X}$ is positive definite. Here we shall use the principle of mathematical induction.

The result is true for quadratic forms in one variable since $a_{11} x^2$ is positive definite when a_{11} is positive.

Assume as our induction hypothesis that the theorem is true for quadratic forms in m variables. Then we shall prove that it is also true for quadratic forms in $(m+1)$ variables.

Now let $\mathbf{S}$ be any real symmetric matrix of order $(m+1)$ and let the leading principal minors of $\mathbf{S}$ be all positive. We partition $\mathbf{S}$ as follows :

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{B}_1 \\ \mathbf{B}_1' & \lambda \end{bmatrix},$$

where $\mathbf{B}$ is a real symmetric matrix of order m and $\mathbf{B}_1$ is an $m \times 1$ column matrix. By hypothesis the leading principal minors of $\mathbf{S}$ are all positive. Therefore $|\mathbf{S}|$ and the leading principal minors of $\mathbf{B}$ are all positive. Thus $\mathbf{B}$ is a real symmetric matrix of order m having all its leading principal minors positive. So by induction hypothesis the quadratic form corresponding to $\mathbf{B}$ is positive definite. Therefore there exists a non-singular matrix $\mathbf{P}$ of order m such that $\mathbf{P}' \mathbf{B} \mathbf{P} = \mathbf{I}_m$.

Since $|\mathbf{B}| > 0$, therefore $\mathbf{B}$ is non-singular. Let $\mathbf{C} = -\mathbf{B}^{-1} \mathbf{B}_1$. Then $\mathbf{C}$ is an $m \times 1$ column matrix. Also

$$\mathbf{C}' = -(\mathbf{B}^{-1} \mathbf{B}_1)' = -\mathbf{B}_1' (\mathbf{B}^{-1})' = -\mathbf{B}_1' (\mathbf{B}')^{-1} = -\mathbf{B}_1' \mathbf{B}^{-1},$$

since $\mathbf{B}' = \mathbf{B}$, $\mathbf{B}$ being symmetric. We have

$$\begin{aligned} \begin{bmatrix} \mathbf{P}' & \mathbf{O} \\ \mathbf{C}' & 1 \end{bmatrix} \mathbf{S} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} &= \begin{bmatrix} \mathbf{P}' & \mathbf{O} \\ \mathbf{C}' & 1 \end{bmatrix} \begin{bmatrix} \mathbf{B} & \mathbf{B}_1 \\ \mathbf{B}_1' & \lambda \end{bmatrix} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{P}' \mathbf{B} & \mathbf{P}' \mathbf{B}_1 \\ \mathbf{C}' \mathbf{B} + \mathbf{B}_1' & \mathbf{C}' \mathbf{B}_1 + \lambda \end{bmatrix} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{P}' \mathbf{B} \mathbf{P} & \mathbf{P}' \mathbf{B} \mathbf{C} + \mathbf{P}' \mathbf{B}_1 \\ \mathbf{C}' \mathbf{B} \mathbf{P} + \mathbf{B}_1' \mathbf{P} & \mathbf{C}' \mathbf{B} \mathbf{C} + \mathbf{B}_1' \mathbf{C} + \mathbf{C}' \mathbf{B}_1 + \lambda \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{O} & \mathbf{B}_1' \mathbf{C} + \lambda \end{bmatrix} [\because \mathbf{P}' \mathbf{B} \mathbf{P} = \mathbf{I}_m, \mathbf{C} = -\mathbf{B}^{-1} \mathbf{B}_1, \mathbf{C}' = -\mathbf{B}_1' \mathbf{B}^{-1}]$$

$$\text{Thus } \begin{bmatrix} \mathbf{P}' & \mathbf{O} \\ \mathbf{C}' & 1 \end{bmatrix} \mathbf{S} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{O} & \mathbf{B}_1' \mathbf{C} + \lambda \end{bmatrix}.$$

Taking determinants of both sides, we get

$$|\mathbf{P}'| \cdot |\mathbf{S}| \cdot |\mathbf{P}| = |\mathbf{I}_m| \cdot |\mathbf{B}_1' \mathbf{C} + \lambda| = |\mathbf{B}_1' \mathbf{C} + \lambda|$$

because $\mathbf{B}_1' \mathbf{C} + \lambda$ is an 1×1 matrix.

$$\therefore |\mathbf{P}|^2 \cdot |\mathbf{S}| = |\mathbf{B}_1' \mathbf{C} + \lambda| \quad [\because |\mathbf{P}| = |\mathbf{P}'|].$$

Since $|\mathbf{S}| > 0$ and $|\mathbf{P}| \neq 0$, therefore $\mathbf{B}_1' \mathbf{C} + \lambda$ is positive. Let $\mathbf{B}_1' \mathbf{C} + \lambda = \alpha^2$, where α is real. Then

$$\begin{bmatrix} \mathbf{P}' & \mathbf{O} \\ \mathbf{C}' & 1 \end{bmatrix} \mathbf{S} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{O} & \alpha^2 \end{bmatrix}.$$

Pre-multiplying and post-multiplying both sides with

$$\begin{bmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix}, \text{ we get}$$

$$\begin{bmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{P}' & \mathbf{O} \\ \mathbf{C}' & 1 \end{bmatrix} \mathbf{S} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix} = \mathbf{I}_{m+1}.$$

$$\text{Now let } \mathbf{Q} = \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix}.$$

Then $\mathbf{Q}$ is non-singular as it is the product of two non-singular matrices. Also

$$\mathbf{Q}' = \begin{bmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{P}' & \mathbf{O} \\ \mathbf{C}' & 1 \end{bmatrix}.$$

Therefore, we have

$$\mathbf{Q}' \mathbf{S} \mathbf{Q} = \mathbf{I}_{m+1}.$$

Thus the real symmetric matrix $\mathbf{S}$ of order $m+1$ is congruent to $\mathbf{I}_{m+1}$. So the quadratic form corresponding to $\mathbf{S}$ is positive definite.

The proof is now complete by induction.

Corollary: *The real quadratic form*

$$q(x_1, \dots, x_n) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

is negative definite if and only if

$$a_{11} < 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} < 0, \dots$$

Proof: For $q(x_1, \dots, x_n)$ to be negative definite, $-q(x_1, \dots, x_n)$ must be positive definite.

Applying Sylvester's criterion for positive definiteness of $-q$, we get successively

$$-a_{11} > 0, \begin{vmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \end{vmatrix} > 0, \begin{vmatrix} -a_{11} & -a_{12} & -a_{13} \\ -a_{21} & -a_{22} & -a_{23} \\ -a_{31} & -a_{32} & -a_{33} \end{vmatrix} > 0, \dots$$

From this follows the condition stated in the statement of the above corollary.

8.18 Some More Criteria to Check the Value Class of a Real Quadratic Form

Principal minors of a matrix:

Definition: Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n . Any minors of A obtained by deleting the corresponding rows and columns of A are called principal minors of A .

Consider the square matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

The principal minors of A of order 1 are a_{11}, a_{22}, a_{33} .

The principal minors of A of order 2 are

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}.$$

These have obtained respectively by deleting the third row and the third column, the second row and the second column and the first row and the first column.

The principal minor of A of order 3 is

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

Now we give some theorems without proof.

Theorem 1: A real quadratic form $\mathbf{X}^T A \mathbf{X}$ is positive definite if and only if all the principal minors of A are positive.

Theorem 2: A real quadratic form $\mathbf{X}^T A \mathbf{X}$ is negative definite if and only if all the principal minors of A of even order are positive and those of odd order are negative.

Theorem 3: A real quadratic form $\mathbf{X}^T A \mathbf{X}$ is positive semi-definite if and only if A is singular and all its principal minors are non-negative.

Theorem 4: A real quadratic form $\mathbf{X}^T A \mathbf{X}$ is negative semi-definite if and only if A is singular and all principal minors of even order of A are non-negative while those of odd order are non-positive.

Theorem 5: A real quadratic form $\mathbf{X}^T \mathbf{A} \mathbf{X}$ is positive semi-definite if the leading principal minors $A_1, \dots, A_{n-1}$ of $\mathbf{A}$ are positive and $\det \mathbf{A} = 0$.

Theorem 6: A real symmetric matrix $\mathbf{A}$ is indefinite if and only if at least one of the following conditions is satisfied :

(a) $\mathbf{A}$ has a negative principal minor of even order.

(b) $\mathbf{A}$ has a positive principal minor of odd order and a negative principal minor of odd order.

Illustrative Examples

Example 15: Prove that the quadratic form

$$6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$$

in three variables is positive definite.

Solution: The matrix $\mathbf{A}$ of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

The leading principal minors of $\mathbf{A}$ are

$$A_1 = 6, A_2 = \begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix} = 18 - 4 = 14,$$

$$A_3 = \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix} = \begin{vmatrix} 0 & 1 & -7 \\ 0 & 2 & 2 \\ 2 & -1 & 3 \end{vmatrix}, \text{ by } R_2 + R_3, R_1 - 3R_3$$

$$= 2(2 + 14) = \text{positive}.$$

Since the leading principal minors of $\mathbf{A}$ are all positive, therefore the given quadratic form is positive definite.

Example 16: Write the matrix $\mathbf{A}$ of the quadratic form

$$6x^2 + 35y^2 + 11z^2 + 4zx.$$

Find the eigenvalues of $\mathbf{A}$ and hence determine the value class of the given quadratic form.

Solution: The matrix $\mathbf{A}$ of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 6 & 0 & 2 \\ 0 & 35 & 0 \\ 2 & 0 & 11 \end{bmatrix}.$$

The characteristic equation of $\mathbf{A}$ is

$$\begin{vmatrix} 6-\lambda & 0 & 2 \\ 0 & 35-\lambda & 0 \\ 2 & 0 & 11-\lambda \end{vmatrix} = 0$$

$$\text{or} \quad (6 - \lambda)(35 - \lambda)(11 - \lambda) - 2 \times 2(35 - \lambda) = 0$$

$$\text{or} \quad (35 - \lambda)[(6 - \lambda)(11 - \lambda) - 4] = 0$$

$$\text{or} \quad (35 - \lambda)[\lambda^2 - 17\lambda + 62] = 0$$

$$\text{or} \quad (35 - \lambda) = 0, (\lambda^2 - 17\lambda + 62) = 0.$$

$$\therefore \quad \text{the eigenvalues of } \mathbf{A} \text{ are } 35, \frac{17 \pm \sqrt{41}}{2}.$$

Since the eigenvalues of $\mathbf{A}$ are all positive, therefore the quadratic form is positive definite.

Example 17: Show that the quadratic form

$$5x_1^2 + 26x_2^2 + 10x_3^2 + 4x_2x_3 + 14x_3x_1 + 6x_1x_2$$

in three variables is positive semi-definite and find a non-zero set of values of x_1, x_2, x_3 which makes the form zero.

Solution: The matrix of the given form is

$$A = \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix}.$$

$$\text{We write } \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now we shall reduce $\mathbf{A}$ to diagonal form by applying congruence operations on it. Performing

$$R_2 \rightarrow R_2 - \frac{3}{5} R_1, C_2 \rightarrow C_2 - \frac{3}{5} C_1; R_3 \rightarrow R_3 - \frac{7}{5} R_1, C_3 \rightarrow C_3 - \frac{7}{5} C_1,$$

$$\text{we get } \begin{bmatrix} 5 & 0 & 0 \\ 0 & 121/5 & -11/5 \\ 0 & -11/5 & 1/5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3/5 & 1 & 0 \\ -7/5 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -3/5 & -7/5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_3 \rightarrow R_3 + \frac{1}{11} R_2, C_3 \rightarrow C_3 + \frac{1}{11} C_2$, we get

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & 121/5 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3/5 & 1 & 0 \\ -16/11 & 1/11 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -3/5 & -16/11 \\ 0 & 1 & 1/11 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore the linear transformation

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & -3/5 & -16/11 \\ 0 & 1 & 1/11 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\text{i.e., } x_1 = y_1 - \frac{3}{5} y_2 - \frac{16}{11} y_3, x_2 = y_2 + \frac{1}{11} y_3, x_3 = y_3$$

transforms the given form to the diagonal form

$$5y_1^2 + \frac{121}{5}y_2^2 + 0y_3^2. \quad \dots(1)$$

But the quadratic form (1) in 3 variables is positive semi-definite and equivalent quadratic forms have the same value class. Therefore the given quadratic form is positive semi-definite.

The set of values $y_1 = 0, y_2 = 0, y_3 = 1$ makes (1) zero. Corresponding to this set of values, we have $x_3 = 1, x_2 = \frac{1}{11}, x_1 = -\frac{16}{11}$. This is a non-zero set of values of x_1, x_2, x_3 which makes the given quadratic form zero.

Example 18: Show that the form

$$x_1^2 + 2x_2^2 + 3x_3^2 + 2x_2x_3 - 2x_3x_1 + 2x_1x_2$$

in three variables is indefinite and find two sets of values of x_1, x_2, x_3 for which the form assumes positive and negative values.

Solution: The matrix of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 3 \end{bmatrix}.$$

$$\text{We write } \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now we shall reduce $\mathbf{A}$ to diagonal form by applying congruent operations on it. Performing

$R_2 \rightarrow R_2 - R_1, C_2 \rightarrow C_2 - C_1$; and $R_3 \rightarrow R_3 + R_1, C_3 \rightarrow C_3 + C_1$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_3 \rightarrow R_3 - 2R_2, C_3 \rightarrow C_3 - 2C_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

$\therefore$ the linear transformation

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\text{i.e., } \left. \begin{aligned} x_1 &= y_1 - y_2 + 3y_3 \\ x_2 &= y_2 - 2y_3 \\ x_3 &= y_3 \end{aligned} \right\} \quad \dots(\text{A})$$

transforms the given form to the diagonal form

$$y_1^2 + y_2^2 - 2y_3^2. \quad \dots(1)$$

The form (1) is indefinite and so the given quadratic form is also indefinite.

Obviously $y_1 = 0, y_2 = 0, y_3 = 1$ makes the form (1) negative and $y_1 = 0, y_2 = 1, y_3 = 0$ makes the form (1) positive. Substituting these values in the relations (A), we see that the sets of values $x_1 = 3, x_2 = -2, x_3 = 1; x_1 = -1, x_2 = 1, x_3 = 0$ respectively make the given form negative and positive.

Example 19: Show that every real non-singular matrix $\mathbf{A}$ can be expressed as

$$\mathbf{A} = \mathbf{QDR},$$

where $\mathbf{Q}$ and $\mathbf{R}$ are orthogonal and $\mathbf{D}$ is real diagonal.

Solution: Since $\mathbf{A}$ is a real non-singular matrix, therefore $\mathbf{A}'\mathbf{A}$ is a positive definite real symmetric matrix. Let $\mathbf{P}$ be an orthogonal matrix such that

$$\mathbf{P}^{-1} (\mathbf{A}'\mathbf{A}) \mathbf{P} = \mathbf{P}' (\mathbf{A}'\mathbf{A}) \mathbf{P} = \text{diag. } [\lambda_1, \dots, \lambda_n],$$

where $\lambda_1, \dots, \lambda_n$ are the positive real eigenvalues of the positive definite matrix $\mathbf{A}'\mathbf{A}$.

Let $\mathbf{D} = \text{diag. } [\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}]$. Then $\mathbf{D}$ is a real diagonal matrix and $\mathbf{D}' = \mathbf{D}$.

We have $\mathbf{D}'\mathbf{D} = \mathbf{D}^2 = \text{diag. } [\lambda_1, \dots, \lambda_n]$

$$\Rightarrow \mathbf{D}'\mathbf{D} = \mathbf{P}'\mathbf{A}'\mathbf{A}\mathbf{P}$$

$$\Rightarrow (\mathbf{P}')^{-1} \mathbf{D}' \mathbf{D} \mathbf{P}^{-1} = (\mathbf{P}')^{-1} \mathbf{P}'\mathbf{A}'\mathbf{A}\mathbf{P}\mathbf{P}^{-1}$$

$$\Rightarrow (\mathbf{P}')^{-1} \mathbf{D}' \mathbf{D} \mathbf{P}^{-1} = \mathbf{A}'\mathbf{A}$$

$$\Rightarrow (\mathbf{A}')^{-1} (\mathbf{P}')^{-1} \mathbf{D}' \mathbf{D} \mathbf{P}^{-1} \mathbf{A}^{-1} = \mathbf{I}$$

$$\Rightarrow (\mathbf{A}^{-1})' \mathbf{P} \mathbf{D}' \mathbf{D} \mathbf{P}' \mathbf{A}^{-1} = \mathbf{I} \quad [\because \mathbf{P} \text{ is orthogonal} \rightarrow \mathbf{P}' = \mathbf{P}^{-1}]$$

$$\Rightarrow (\mathbf{D}\mathbf{P}'\mathbf{A}^{-1})' (\mathbf{D}\mathbf{P}'\mathbf{A}^{-1}) = \mathbf{I}$$

$$\Rightarrow \mathbf{D}\mathbf{P}'\mathbf{A}^{-1} \text{ is orthogonal.}$$

Let $\mathbf{S} = \mathbf{D}\mathbf{P}'\mathbf{A}^{-1}$.

Then $\mathbf{S}$ is an orthogonal matrix.

Now let $\mathbf{Q} = \mathbf{S}^{-1}$; then $\mathbf{Q}$ is an orthogonal matrix. Also let $\mathbf{R} = \mathbf{P}'$. Then $\mathbf{R}$ is an orthogonal matrix.

We have $\mathbf{QDR} = (\mathbf{D}\mathbf{P}'\mathbf{A}^{-1})^{-1} \mathbf{D}\mathbf{P}' = \mathbf{A} (\mathbf{P}')^{-1} \mathbf{D}^{-1} \mathbf{D} \mathbf{P}' = \mathbf{A} (\mathbf{P}')^{-1} \mathbf{P}' = \mathbf{A}$.

Hence the result.

Example 20: If $\mathbf{A}$ is a positive definite real symmetric matrix, show that there exists a positive definite real symmetric matrix $\mathbf{B}$ such that $\mathbf{B}^2 = \mathbf{A}$.

Solution: Since $\mathbf{A}$ is a positive definite real symmetric matrix, therefore the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\mathbf{A}$ are all real and positive. Also there exists an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{D} = \text{diag. } [\lambda_1, \dots, \lambda_n].$$

Let $\mathbf{D}_1 = \text{diag.} [\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}]$. Then $\mathbf{D}_1^2 = \mathbf{D}$, $\mathbf{D}_1' = \mathbf{D}_1$, and the eigenvalues of $\mathbf{D}_1$ are all positive.

Now suppose that $\mathbf{B} = \mathbf{P} \mathbf{D}_1 \mathbf{P}^{-1} = \mathbf{P} \mathbf{D}_1 \mathbf{P}'$.

We have $\mathbf{B}' = (\mathbf{P} \mathbf{D}_1 \mathbf{P}')' = \mathbf{P} \mathbf{D}_1' \mathbf{P}' = \mathbf{P} \mathbf{D}_1 \mathbf{P}' = \mathbf{B}$.

$\therefore \mathbf{B}$ is a real symmetric matrix.

Also $\mathbf{B} = \mathbf{P} \mathbf{D}_1 \mathbf{P}^{-1} \Rightarrow \mathbf{B}$ is similar to $\mathbf{D}_1$. So $\mathbf{B}$ and $\mathbf{D}_1$ have the same eigenvalues.

Therefore the eigenvalues of $\mathbf{B}$ are all positive. So $\mathbf{B}$ is a positive definite real symmetric matrix.

Finally, we have

$$\mathbf{B}^2 = (\mathbf{P} \mathbf{D}_1 \mathbf{P}^{-1})^2 = \mathbf{P} \mathbf{D}_1 \mathbf{P}^{-1} \mathbf{P} \mathbf{D}_1 \mathbf{P}^{-1} = \mathbf{P} \mathbf{D}_1^2 \mathbf{P}^{-1} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1} = \mathbf{A}.$$

Hence the result.

Example 21: Determine the definiteness of the following quadratic form :

$$q(x_1, x_2, x_3) = [x_1, x_2, x_3] \begin{bmatrix} 2 & 0 & -2 \\ 1 & 5 & 2 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Solution: The given quadratic form is

$$\begin{aligned} q &= 2x_1^2 - 2x_1x_3 + x_2x_1 + 5x_2^2 + 2x_2x_3 - x_3x_1 + x_3x_2 + x_3^2 \\ &= 2x_1x_1 + \frac{1}{2}x_1x_2 - \frac{3}{2}x_1x_3 + \frac{1}{2}x_2x_1 + 5x_2x_2 \\ &\quad + \frac{3}{2}x_2x_3 - \frac{3}{2}x_3x_1 + \frac{3}{2}x_3x_2 + x_3x_3. \end{aligned}$$

The real symmetric matrix associated with the above quadratic form is

$$\mathbf{A} = \begin{bmatrix} 2 & \frac{1}{2} & -\frac{3}{2} \\ \frac{1}{2} & 5 & \frac{3}{2} \\ -\frac{3}{2} & \frac{3}{2} & 1 \end{bmatrix}.$$

The principal minors of $\mathbf{A}$ of order 1 are 2, 5, 1, those of order 2 are $\frac{39}{4}$, $-\frac{1}{4}$ and $\frac{11}{4}$;

the principal minor of order three is $-\frac{33}{4}$.

We see that $\mathbf{A}$ has a negative principal minor of even order, namely of order 2. Hence, by theorem 6 of 3.18, the matrix $\mathbf{A}$ is indefinite. Therefore, the given quadratic form is indefinite.

Example 22: Determine the value class of the real quadratic form

$$-x_1^2 - 2x_2^2 - 2x_3^2 + 2x_1x_2 + 2x_2x_3$$

in three variables.

Solution: The given quadratic form is

$$\begin{aligned} q(x_1, x_2, x_3) &= -x_1^2 - 2x_2^2 - 2x_3^2 + 2x_1x_2 + 2x_2x_3 \\ &= -x_1x_1 + x_1x_2 + 0x_1x_3 + x_2x_1 - 2x_2x_2 + x_2x_3 \\ &\quad + 0x_3x_1 + x_3x_2 - 2x_3x_3 \end{aligned}$$

$$= \mathbf{X}^T \mathbf{A} \mathbf{X}, \text{ where } \mathbf{A} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} \text{ is a symmetric matrix.}$$

The leading principal minors of $\mathbf{A}$ are

$$A_1 = -1, A_2 = \begin{vmatrix} -1 & 1 \\ 1 & -2 \end{vmatrix} = 2 - 1 = 1,$$

$$A_3 = \begin{vmatrix} -1 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{vmatrix} = \begin{vmatrix} -1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{vmatrix}, C_2 \rightarrow C_2 + C_1$$

$$= -1(2 - 1) = -1.$$

We see that $A_1 < 0$, $A_2 > 0$, $A_3 < 0$.

So the given quadratic form is negative definite.

Comprehensive Exercise 3

1. Prove that the quadratic form

$$6x_1^2 + 3x_2^2 + 14x_3^2 + 4x_2x_3 + 18x_3x_1 + 4x_1x_2$$

in three variables is positive definite.

2. Prove that the quadratic form

$$2x_1^2 + x_2^2 - 3x_3^2 - 8x_2x_3 - 4x_3x_1 + 12x_1x_2$$

in three variables is indefinite.

3. Prove that the quadratic form $6x^2 + 49y^2 + 51z^2 - 82yz + 20zx - 4xy$

in three variables is positive definite.

4. Show that the quadratic form $6x^2 + 17y^2 + 3z^2 - 20xy - 14yz + 8zx$

in three variables is positive semi-definite and find a non-zero set of values of x, y, z which makes the form zero.

5. Classify the following forms in three variables as definite, semi-definite and indefinite

(i) $2x^2 + 2y^2 + 3z^2 - 4yz - 4zx + 2xy$

(ii) $26x^2 + 20y^2 + 10z^2 - 4yz - 16zx - 36xy$

(iii) $x_1^2 + 4x_2^2 + x_3^2 - 4x_2x_3 + 2x_3x_1 - 4x_1x_2$.

6. Show that a real symmetric matrix is positive definite if and only if all its characteristic roots are positive.

7. Show that a real symmetric matrix $\mathbf{A}$ is positive definite iff $\mathbf{A}^{-1}$ exists and is positive definite.

8. Show that the quadratic form $2x^2 - 4xy + 3xz + 6y^2 + 6zy + 8z^2$ in three variables is positive definite.
9. Show that the quadratic form $y^2 + 2z^2 - 2yz + 2zx - 2xy$ in three variables is indefinite.
10. Determine the value class of the real quadratic form in three variables $x_1^2 + 3x_2^2 + 6x_3^2 - 2x_1x_2 + 4x_1x_3$.
11. Determine the value class of the real quadratic form in three variables $-2x_1^2 - 2x_2^2 - 2x_3^2 + 4x_2x_3$.
12. Determine the definiteness of the real quadratic form in three variables $4x_1^2 + x_2^2 + 9x_3^2 - 4x_1x_2 + 12x_1x_3$.

Answers 3

10. positive semi-definite. 11. negative semi-definite.
12. indefinite.

8.19 Hermitian Forms

Hermitian form: Definition :

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$ be any n -vector in the unitary space $\mathbf{C}^n$ and let $H = [h_{ij}]_{n \times n}$ be a Hermitian

matrix of order n over the complex field $\mathbf{C}$. Then the expression

$$h(x_1, x_2, \dots, x_n) = \mathbf{X}^\theta H \mathbf{X} = \sum_{i=1}^n \sum_{j=1}^n h_{ij} \bar{x}_i x_j$$

is called a Hermitian form of order n in the n complex variables $x_1, x_2, \dots, x_n$. The Hermitian matrix H is called the matrix of this Hermitian form.

If H is real, then a Hermitian form is called the *real Hermitian form*. Also, $\det H$ is defined as the *discriminant* of the Hermitian form, and a Hermitian form is called *singular* if its determinant is zero, otherwise it is called *non-singular*.

Note: A matrix H over the complex field $\mathbf{C}$ is called a Hermitian matrix if $H^\theta = H$, where H^θ denotes the conjugate transpose of the matrix H i.e.,

$$H^\theta = (\bar{H})' = (\overline{H'})$$

Some authors denote the conjugate transpose of a matrix H by the symbol H^* . A real symmetric matrix A is always a Hermitian matrix.

Theorem 1: A Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ assumes only real values for all complex n -vectors $\mathbf{X}$.

Proof: Suppose $\mathbf{X}^\theta H \mathbf{X}$ is a Hermitian form. Then H is a Hermitian matrix and so $H^\theta = H$.

Since $\mathbf{X}^\theta H \mathbf{X}$ is a 1×1 matrix, therefore it is symmetric and so

$$(\mathbf{X}^\theta H \mathbf{X})' = \mathbf{X}^\theta H \mathbf{X}.$$

$$\begin{aligned} \text{Now } \overline{(\mathbf{X}^\theta H \mathbf{X})} &= \overline{(\mathbf{X}^\theta H \mathbf{X})'} = (\mathbf{X}^\theta H \mathbf{X})^\theta \\ &= \mathbf{X}^\theta H^\theta (\mathbf{X}^\theta)^\theta = \mathbf{X}^\theta H \mathbf{X}. \end{aligned}$$

Thus $\mathbf{X}^\theta H \mathbf{X}$ and its conjugate are equal.

$\therefore \mathbf{X}^\theta H \mathbf{X}$ is a real 1×1 matrix.

Hence, $\mathbf{X}^\theta H \mathbf{X}$ has only real values.

Theorem 2 : The determinant and every leading principal minor of a Hermitian matrix H are real.

Proof : We have $|\overline{H}| = |\overline{H}| = |(\overline{H})'|$
 $= |H^\theta| = |H|. \quad [\because H^\theta = H, H \text{ being Hermitian}]$

$\therefore |H|$ is real.

Since every leading principal sub-matrix of a Hermitian matrix is Hermitian, we conclude that every leading principal minor of a Hermitian matrix is real.

This completes the proof of the theorem.

Theorem 3 : A Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ remains Hermitian by a non-singular linear transformation of coordinates defined by

$$\mathbf{X} = P \mathbf{Y}$$

where P is a non-singular matrix over the complex field $\mathbb{C}$.

Proof : Substituting $\mathbf{X} = P \mathbf{Y}$ in $h = \mathbf{X}^\theta H \mathbf{X}$, we get

$$h = (P \mathbf{Y})^\theta H (P \mathbf{Y}) = \mathbf{Y}^\theta (P^\theta H P) \mathbf{Y} = \mathbf{Y}^\theta Q \mathbf{Y},$$

where $Q = P^\theta H P$.

We have $Q^\theta = (P^\theta H P)^\theta = P^\theta H^\theta (P^\theta)^\theta = P^\theta H P = Q$.

$\therefore Q$ is Hermitian.

Hence, the transformed form $\mathbf{Y}^\theta Q \mathbf{Y}$ is Hermitian.

Remark: It can be easily seen that under this non-singular transformation, the rank of a Hermitian form remains invariant. We know that the rank of a matrix does not alter by pre-multiplication or post-multiplication by a non-singular matrix. Since P and P^θ are both non-singular, therefore $\text{rank } H = \text{rank } (P^\theta H P) = \text{rank } Q$.

Hence $\text{rank } \mathbf{X}^\theta H \mathbf{X} = \text{rank } \mathbf{Y}^\theta Q \mathbf{Y}$.

8.20 Unitary Reduction of a Hermitian Form

Diagonal and Unit Hermitian forms: Definition :

A Hermitian form represented by

$$h = c_1 \bar{x}_1 x_1 + c_2 \bar{x}_2 x_2 + \dots + c_n \bar{x}_n x_n$$

where the coefficients c_i 's are real as h is real, is referred to as diagonal Hermitian form.

If all c_i 's can be reduced to one, then a Hermitian form given by

$$h = \bar{x}_1 x_1 + \bar{x}_2 x_2 + \dots + \bar{x}_n x_n$$

is called a unit Hermitian form.

Theorem 1 : Every Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ of order n is unitarily equivalent (under a transformation $\mathbf{X} = P \mathbf{Y}$, P unitary) to the diagonal form

$$\lambda_1 \bar{y}_1 y_1 + \dots + \lambda_n \bar{y}_n y_n$$

where $\lambda_1, \dots, \lambda_n$ are eigenvalues of H .

Proof: Since H is a Hermitian matrix, therefore there exists a unitary matrix P such that

$$P^{-1} H P = P^\theta H P = D = \text{diag.} [\lambda_1, \dots, \lambda_n]$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of H .

Consider the unitary linear transformation $\mathbf{X} = P \mathbf{Y}$. We have

$$\begin{aligned} \mathbf{X}^\theta H \mathbf{X} &= (P \mathbf{Y})^\theta H (P \mathbf{Y}) = \mathbf{Y}^\theta P^\theta H P \mathbf{Y} = \mathbf{Y}^\theta D \mathbf{Y} \\ &= \lambda_1 \bar{y}_1 y_1 + \dots + \lambda_n \bar{y}_n y_n. \end{aligned}$$

Hence, the result.

Remark: If the rank of the Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ is r , then only r of its n eigenvalues $\lambda_1, \dots, \lambda_n$ will be non-zero and consequently only r terms in the above diagonal form will be non-zero.

8.21 Value Class of a Hermitian Form. Definite, Semi-definite and Indefinite Hermitian Forms

Definitions : A Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ is said to be

(i) **Positive definite** if $\mathbf{X}^\theta H \mathbf{X} \geq 0$ for all complex n -vectors $\mathbf{X}$ and $\mathbf{X}^\theta H \mathbf{X} = 0$ only if $\mathbf{X} = \mathbf{0}$ i.e., $x_1 = x_2 = \dots = x_n = 0$.

(ii) **Negative definite** if $\mathbf{X}^\theta H \mathbf{X} \leq 0$ for all complex n -vectors $\mathbf{X}$ and $\mathbf{X}^\theta H \mathbf{X} = 0$ only if $\mathbf{X} = \mathbf{0}$ i.e., $x_1 = x_2 = \dots = x_n = 0$.

(iii) **Positive semi-definite** if $\mathbf{X}^\theta H \mathbf{X} \geq 0$ for all complex n -vectors $\mathbf{X}$ and $\mathbf{X}^\theta H \mathbf{X} = 0$ for some non-zero complex n -vector $\mathbf{X}$ i.e., $\mathbf{X}^\theta H \mathbf{X} = 0$ for some complex values of the variables $x_1, x_2, \dots, x_n$ not all zero.

(iv) **Negative semi-definite** if $\mathbf{X}^{\theta} H \mathbf{X} \leq 0$ for all complex n -vectors $\mathbf{X}$ and $\mathbf{X}^{\theta} H \mathbf{X} = 0$ for some non-zero complex n -vector $\mathbf{X}$ i.e., $\mathbf{X}^{\theta} H \mathbf{X} = 0$ for some complex values of the variables $x_1, x_2, \dots, x_n$ not all zero.

(v) **Indefinite** if $\mathbf{X}^{\theta} H \mathbf{X}$ takes positive as well as negative values for some complex values of the variables $x_1, x_2, \dots, x_n$.

(vi) **Non-negative definite** if $\mathbf{X}^{\theta} H \mathbf{X} \geq 0$ for all complex n -vectors $\mathbf{X}$.

Non-negative definite and positive definite Hermitian matrices. Definition.

A Hermitian matrix H is called *non-negative definite* if the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is non-negative definite. A Hermitian matrix H is called *positive definite* if the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is positive definite.

Similar definitions are for other cases.

8.22 Some Results for Hermitian Forms Parallel to the Corresponding Results for Real Quadratic Forms

Index and Signature of a Hermitian form:

Definition : Let $\mathbf{X}^{\theta} H \mathbf{X}$ be a Hermitian form of order n . Let

$\lambda_1 \bar{z}_1 z_1 + \lambda_2 \bar{z}_2 z_2 + \dots + \lambda_p \bar{z}_p z_p - \lambda_{p+1} \bar{z}_{p+1} z_{p+1} - \dots - \lambda_r \bar{z}_r z_r$
be its diagonal form where $\lambda_1, \lambda_2, \dots, \lambda_p, \lambda_{p+1}, \dots, \lambda_r$ are all positive.

Here, r is the rank of the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$.

The number p of positive terms in this diagonal form is called the *index* of the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$. The excess of the number of positive terms over the number of negative terms in this diagonal form i.e., $p - (r - p) = 2p - r$ is called the *signature* of the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ and is denoted by s .

Theorem 1 : Sylvester's law of Inertia for Hermitian forms:

Under all non-singular linear transformations the rank r and the index p are invariant of a Hermitian form and consequently the signature s of a Hermitian form is also invariant under all non-singular linear transformations.

Equivalent Hermitian forms: Definition: Two Hermitian forms $\mathbf{X}^{\theta} H \mathbf{X}$ and $\mathbf{Y}^{\theta} Q \mathbf{Y}$ are said to be *equivalent* over the complex field $\mathbf{C}$ if one can be obtained from the other by a non-singular transformation over $\mathbf{C}$. Thus $\mathbf{X}^{\theta} H \mathbf{X}$ and $\mathbf{Y}^{\theta} Q \mathbf{Y}$ are equivalent if and only if there exists a non-singular transformation $\mathbf{X} = P \mathbf{Y}$ such that

$$\begin{aligned} \mathbf{X}^{\theta} H \mathbf{X} &= (P\mathbf{Y})^{\theta} H (P\mathbf{Y}) = \mathbf{Y}^{\theta} (P^{\theta} H P) \mathbf{Y} \\ &= \mathbf{Y}^{\theta} Q \mathbf{Y}, \text{ where } Q = P^{\theta} H P. \end{aligned}$$

Theorem 2 : Equivalence theorem of Hermitian forms: *Two Hermitian forms are equivalent if and only if they have the same rank and the same index.*

Theorem 3 : Let $\mathbf{X}^0 H \mathbf{X}$ be a Hermitian form of order n , rank r , and signature s . Then $\mathbf{X}^0 H \mathbf{X}$ is

- (i) positive definite if and only if $s = r = n$.
- (ii) negative definite if and only if $-s = r = n$.
- (iii) positive semi-definite if and only if $s = r < n$.
- (iv) negative semi-definite if and only if $-s = r < n$.
- (v) indefinite if and only if $|s| \neq r$ i.e., if and only if $|s| < r$.

Theorem 4 : If the Hermitian form $\mathbf{X}^0 H \mathbf{X}$ is positive definite, then there exists a non-singular transformation $\mathbf{X} = \mathbf{P} \mathbf{Y}$ such that $\mathbf{X}^0 H \mathbf{X} = \mathbf{Y}^0 \mathbf{Y}$.

Theorem 5 : Sylvester's criterion for positive definiteness of Hermitian forms:

A Hermitian form $\mathbf{X}^0 H \mathbf{X}$, where $H = [h_{ij}]_{n \times n}$, is positive definite if and only if all the leading principal minors of H are positive i.e., iff

$$h_{11}, \begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix}, \dots, \begin{vmatrix} h_{11} & \dots & h_{1n} \\ \dots & \dots & \dots \\ h_{n1} & \dots & h_{nn} \end{vmatrix} \text{ are all positive.}$$

Theorem 6 : A Hermitian form $\mathbf{X}^0 H \mathbf{X}$ is positive definite if and only if all the principal minors of H are positive.

Theorem 7 : The Hermitian form

$$h(x_1, \dots, x_n) = \mathbf{X}^0 H \mathbf{X} = \sum_{i=1}^n \sum_{j=1}^n h_{ij} \bar{x}_i x_j$$

is negative definite if and only if the leading principal minors of H are alternately negative and positive i.e., iff

$$h_{11} < 0, \begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix} > 0, \begin{vmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{vmatrix} < 0, \dots$$

Theorem 8 : A Hermitian form $\mathbf{X}^0 H \mathbf{X}$ is negative definite if and only if all the principal minors of H of even order are positive and those of odd order are negative.

Theorem 9 : A Hermitian form $\mathbf{X}^0 H \mathbf{X}$ is

- (i) Positive definite if and only if all the eigenvalues of H are positive.
 - (ii) Negative definite if and only if all the eigenvalues of H are negative.
 - (iii) Positive semi-definite iff all the eigenvalues of H are ≥ 0 with at least one zero eigenvalue.
 - (iv) Negative semi-definite if and only if all the eigenvalues of H are ≤ 0 with at least one zero eigenvalue.
 - (v) Indefinite iff H has at least one positive eigenvalue and one negative eigenvalue.
-

(vi) Non-negative definite iff the eigenvalues of H are all non-negative.

Theorem 10 : A Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is positive semi-definite iff H is singular and all its principal minors are non-negative.

Theorem 11 : A Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is negative semi-definite iff H is singular and the principal minors of even order of H are non-negative while those of odd order are non-positive.

Theorem 12 : A Hermitian matrix H is indefinite iff at least one of the following conditions is satisfied :

(i) H has a negative principal minor of even order.

(ii) H has a positive principal minor of odd order and a negative principal minor of odd order.

Illustrative Examples

Example 23 : Determine the definiteness of the following Hermitian forms $\mathbf{X}^{\theta} H \mathbf{X}$ in $\mathbb{C}^3$ where

$$(i) \quad H = \begin{bmatrix} 0 & -i & -i \\ i & 0 & -i \\ i & i & 0 \end{bmatrix} \quad (ii) \quad H = \begin{bmatrix} -2 & 1+2i & 0 \\ 1-2i & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Solution : (i) The characteristic equation of the matrix H is

$$|H - \lambda I| = 0$$

$$\text{i.e.,} \quad \begin{vmatrix} 0-\lambda & -i & -i \\ i & 0-\lambda & -i \\ i & i & 0-\lambda \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} -\lambda & -i & -i \\ i & -\lambda & -i \\ i & i & -\lambda \end{vmatrix} = 0$$

$$\text{or} \quad -\lambda (\lambda^2 + i^2) + i(-\lambda i + i^2) - i(i^2 + \lambda i) = 0$$

$$\text{or} \quad -\lambda (\lambda^2 - 1) - \lambda i^2 + i^3 - i^3 - \lambda i^2 = 0$$

$$\text{or} \quad -\lambda (\lambda^2 - 1) + \lambda + \lambda = 0$$

$$\text{or} \quad \lambda [-(\lambda^2 - 1) + 2] = 0 \quad \text{or} \quad \lambda (\lambda^2 - 3) = 0.$$

$\therefore$ the eigenvalues of H are $0, \sqrt{3}, -\sqrt{3}$.

Since the given Hermitian matrix H has a positive eigenvalue as well as a negative eigenvalue, therefore the given Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is indefinite.

Remark: The given Hermitian form reduces to the diagonal form

$$\sqrt{3} \bar{y}_1 y_1 - \sqrt{3} \bar{y}_2 y_2.$$

Its canonical form is $\bar{z}_1 z_1 - \bar{z}_2 z_2$.

The order of the given Hermitian form is 3. The rank r of the given Hermitian form = the number of non-zero terms in its diagonal form = 2.

The index p of the given Hermitian form = the number of positive terms in its diagonal form = 1.

The signature s of the given Hermitian form = the number of positive terms in its diagonal form – the number of negative terms in its diagonal form = $1 - 1 = 0$.

(ii) The characteristic equation of the matrix H of the given Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is $|\mathbf{H} - \lambda \mathbf{I}| = 0$

$$\text{or} \quad \begin{vmatrix} -2 - \lambda & 1 + 2i & 0 \\ 1 - 2i & -4 - \lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} = 0$$

$$\text{or} \quad -\lambda [(-2 - \lambda)(-4 - \lambda) - (1 - 2i)(1 + 2i)] = 0$$

$$\text{or} \quad -\lambda [(\lambda + 2)(\lambda + 4) - (1 + 4)] = 0$$

$$\text{or} \quad -\lambda (\lambda^2 + 6\lambda + 8 - 5) = 0 \quad \text{or} \quad \lambda (\lambda^2 + 6\lambda + 3) = 0.$$

$$\therefore \quad \lambda = 0, \frac{-6 \pm \sqrt{(36 - 12)}}{2} \text{ i.e., } \frac{-6 \pm \sqrt{24}}{2} \text{ i.e., } -3 \pm \sqrt{6}.$$

The eigenvalues $-3 + \sqrt{6}$ and $-3 - \sqrt{6}$ are both negative.

Thus the eigenvalues of the matrix H are all ≤ 0 and at least one of them is zero.

Hence the given Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is negative semi-definite.

Remark: Order of the given Hermitian form = order of the matrix $H = 3$.

Rank r of the given Hermitian form = 2.

Index p of the given Hermitian form = 0.

Signature s of the given Hermitian form = $0 - 2 = -2$.

A diagonal form of the given Hermitian form

$$= (3 - \sqrt{6}) \bar{y}_1 y_1 - (3 + \sqrt{6}) \bar{y}_2 y_2.$$

Its canonical form or normal form is

$$-\bar{z}_1 z_1 - \bar{z}_2 z_2.$$

Example 24 : Determine whether or not the following Hermitian forms in $\mathbf{C}^2$ are equivalent.

$$(i) \quad 2 \bar{x}_1 x_1 + 3i \bar{x}_2 x_1 - 3i \bar{x}_1 x_2 - \bar{x}_2 x_2, (2 + i) \bar{x}_2 x_1 + (2 - i) \bar{x}_1 x_2$$

$$(ii) \quad \bar{x}_1 x_1 + 2 \bar{x}_2 x_2 + (1 + 2i) \bar{x}_1 x_2 + (1 - 2i) \bar{x}_2 x_1, i \bar{x}_1 x_2 - i \bar{x}_2 x_1.$$

Solution: (i) Out of the two given Hermitian forms the first Hermitian form is

$$2 \bar{x}_1 x_1 - 3i \bar{x}_1 x_2 + 3i \bar{x}_2 x_1 - \bar{x}_2 x_2 \\ = \mathbf{X}^{\theta} H_1 \mathbf{X}, \text{ where } H_1 = \begin{bmatrix} 2 & -3i \\ 3i & -1 \end{bmatrix} \text{ is a Hermitian matrix.}$$

The characteristic equation of H_1 is $|\mathbf{H}_1 - \lambda \mathbf{I}| = 0$

$$\text{i.e.,} \quad \begin{vmatrix} 2 - \lambda & -3i \\ 3i & -1 - \lambda \end{vmatrix} = 0$$

$$\text{or} \quad (2 - \lambda)(-1 - \lambda) + 9i^2 = 0$$

$$\text{or} \quad -2 - \lambda + \lambda^2 - 9 = 0$$

$$\text{or} \quad \lambda^2 - \lambda - 11 = 0.$$

∴ the eigenvalues of H_1 are

$$\frac{1 \pm \sqrt{1+44}}{2} \text{ i.e., } \frac{1 \pm \sqrt{45}}{2} \text{ i.e., } \frac{1 \pm 3\sqrt{5}}{2}.$$

The eigenvalue $\frac{1+3\sqrt{5}}{2}$ is positive and the eigenvalue $\frac{1-3\sqrt{5}}{2}$ is negative.

The rank r of the Hermitian form $\mathbf{X}^{\theta} H_1 \mathbf{X}$ = the number of non-zero eigenvalues of $H_1 = 2$, the index p of the Hermitian form $\mathbf{X}^{\theta} H_1 \mathbf{X}$ = the number of positive eigenvalues of $H_1 = 1$.

The second Hermitian form is

$$0 \bar{x}_1 x_1 + (2-i) \bar{x}_1 x_2 + (2+i) \bar{x}_2 x_1 + 0 \bar{x}_2 x_2 \\ = \mathbf{X}^{\theta} H_2 \mathbf{X}, \text{ where } H_2 = \begin{bmatrix} 0 & 2-i \\ 2+i & 0 \end{bmatrix} \text{ is a Hermitian matrix.}$$

The characteristic equation of H_2 is

$$|H_2 - \lambda I| = 0 \quad \text{i.e.,} \quad \begin{vmatrix} -\lambda & 2-i \\ 2+i & -\lambda \end{vmatrix} = 0$$

$$\text{or} \quad \lambda^2 - (2+i)(2-i) = 0 \quad \text{or} \quad \lambda^2 - 5 = 0$$

$$\therefore \quad \lambda^2 = 5 \quad \text{or} \quad \lambda = \pm \sqrt{5}.$$

∴ the eigenvalues of H_2 are $+\sqrt{5}, -\sqrt{5}$.

The rank r of the Hermitian form $\mathbf{X}^{\theta} H_2 \mathbf{X}$ = the number of non-zero eigenvalues of $H_2 = 2$, the index p of the Hermitian form $\mathbf{X}^{\theta} H_2 \mathbf{X}$ = the number of positive eigenvalues of $H_2 = 1$.

Thus the Hermitian forms $\mathbf{X}^{\theta} H_1 \mathbf{X}$ and $\mathbf{X}^{\theta} H_2 \mathbf{X}$ have the same rank and the same index. Hence, they are equivalent.

(ii) Out of the two given Hermitian forms the first Hermitian form is

$$\bar{x}_1 x_1 + (1+2i) \bar{x}_1 x_2 + (1-2i) \bar{x}_2 x_1 + 2 \bar{x}_2 x_2 \\ = \mathbf{X}^{\theta} H_1 \mathbf{X}, \text{ where } H_1 = \begin{bmatrix} 1 & 1+2i \\ 1-2i & 2 \end{bmatrix} \text{ is a Hermitian matrix.}$$

The characteristic equation of H_1 is $|H_1 - \lambda I| = 0$

$$\text{i.e.,} \quad \begin{vmatrix} 1-\lambda & 1+2i \\ 1-2i & 2-\lambda \end{vmatrix} = 0$$

$$\text{or} \quad (1-\lambda)(2-\lambda) - (1-2i)(1+2i) = 0$$

$$\text{or} \quad \lambda^2 - 3\lambda + 2 - 5 = 0 \quad \text{or} \quad \lambda^2 - 3\lambda - 3 = 0.$$

$$\therefore \quad \lambda = \frac{3 \pm \sqrt{9+12}}{2} = \frac{3 \pm \sqrt{21}}{2}.$$

$$\therefore \text{ the eigenvalues of } H_1 \text{ are } \frac{3 + \sqrt{21}}{2}, \frac{3 - \sqrt{21}}{2}.$$

The eigenvalue $\frac{3 + \sqrt{21}}{2}$ is positive and the eigenvalue $\frac{3 - \sqrt{21}}{2}$ is negative.

The rank r of the Hermitian form $\mathbf{X}^{\theta} H_1 \mathbf{X}$ = the number of non-zero eigenvalues of $H_1 = 2$,

the index p of this Hermitian form = the number of positive eigenvalues of $H_1 = 1$.

The second Hermitian form is

$$0 \bar{x}_1 x_1 + i \bar{x}_1 x_2 - i \bar{x}_2 x_1 + 0 \bar{x}_2 x_2 \\ = \mathbf{X}^{\theta} H_2 \mathbf{X}, \text{ where } H_2 = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \text{ is a Hermitian matrix.}$$

The characteristic equation of H_2 is $|H_2 - \lambda I| = 0$

$$\text{i.e., } \begin{vmatrix} -\lambda & i \\ -i & -\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^2 + i^2 = 0 \quad \text{or} \quad \lambda^2 - 1 = 0.$$

$$\therefore \lambda^2 = 1 \quad \text{or} \quad \lambda = \pm 1.$$

$\therefore$ the eigenvalues of H_2 are 1, -1.

The rank r of the Hermitian form $\mathbf{X}^{\theta} H_2 \mathbf{X}$ = the number of non-zero eigenvalues of $H_2 = 2$ and its index p = the number of positive eigenvalues of $H_2 = 1$.

Thus the Hermitian forms $\mathbf{X}^{\theta} H_1 \mathbf{X}$ and $\mathbf{X}^{\theta} H_2 \mathbf{X}$ have the same rank and the same index. Hence, they are equivalent.

Example 25 : Reduce the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$, where

$$H = \begin{bmatrix} 2 & 1 - 2i \\ 1 + 2i & -2 \end{bmatrix},$$

to the canonical form by a unitary transformation and hence find the rank and signature of the given Hermitian form.

Solution : The characteristic equation of H is

$$|H - \lambda I| = 0 \quad \text{i.e., } \begin{vmatrix} 2 - \lambda & 1 - 2i \\ 1 + 2i & -2 - \lambda \end{vmatrix} = 0$$

$$\text{or } (\lambda - 2)(\lambda + 2) - (1 + 2i)(1 - 2i) = 0$$

$$\text{or } \lambda^2 - 4 - (1 + 4) = 0 \quad \text{or } \lambda^2 - 9 = 0.$$

$\therefore$ the eigenvalues of H are -3, 3.

The eigenvector $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ corresponding to the eigenvalue -3 is given by the equation

$$(H + 3I) \mathbf{X} = \mathbf{0} \quad \text{or} \quad \begin{bmatrix} 5 & 1 - 2i \\ 1 + 2i & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{i.e., } 5x_1 + (1 - 2i)x_2 = 0, \text{ and } (1 + 2i)x_1 + x_2 = 0.$$

Obviously $x_1 = 1 - 2i, x_2 = -5$ is a solution.

$\therefore \mathbf{X}_1 = \begin{bmatrix} 1-2i \\ -5 \end{bmatrix}$ is an eigenvector corresponding to the eigenvalue $\lambda = -3$.

Corresponding to the eigenvalue 3, the eigenvector is given by the equation

$$(H - 3I) \mathbf{X} = \mathbf{O}$$

$$\text{or} \quad \begin{bmatrix} -1 & 1-2i \\ 1+2i & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{i.e.,} \quad -x_1 + (1-2i)x_2 = 0,$$

$$\text{and} \quad (1+2i)x_1 - 5x_2 = 0.$$

Obviously $x_1 = 5, x_2 = 1+2i$ is a solution.

$\therefore \mathbf{X}_2 = \begin{bmatrix} 5 \\ 1+2i \end{bmatrix}$ is an eigenvector corresponding to $\lambda = 3$.

$$\begin{aligned} \text{Length of the vector } \mathbf{X}_1 &= \sqrt{|1-2i|^2 + |-5|^2} \\ &= \sqrt{5+25} = \sqrt{30} \end{aligned}$$

$$\begin{aligned} \text{and the length of the vector } \mathbf{X}_2 &= \sqrt{|5|^2 + |1+2i|^2} \\ &= \sqrt{25+5} = \sqrt{30}. \end{aligned}$$

$\therefore$ the unitary matrix P that will transform H to diagonal form is

$$\begin{aligned} P &= \left[\frac{1}{\sqrt{30}} \mathbf{X}_1, \frac{1}{\sqrt{30}} \mathbf{X}_2 \right] \\ &= \begin{bmatrix} (1-2i)/\sqrt{30} & 5/\sqrt{30} \\ -5/\sqrt{30} & (1+2i)/\sqrt{30} \end{bmatrix}. \end{aligned}$$

$$\text{Also} \quad P^{-1} = P^{\theta}$$

$$\text{and} \quad P^{-1} H P = P^{\theta} H P = \begin{bmatrix} -3 & 0 \\ 0 & 3 \end{bmatrix} = \text{diag.} [-3, 3].$$

The unitary transformation $\mathbf{X} = P \mathbf{Y}$ will transform the given Hermitian form to the equivalent diagonal form

$$\mathbf{Y}^{\theta} \text{diag.} \begin{bmatrix} -3 & 0 \\ 0 & 3 \end{bmatrix} \mathbf{Y} = -3 \bar{y}_1 y_1 + 3 \bar{y}_2 y_2.$$

The canonical form of the given Hermitian form is

$$- \bar{z}_1 z_1 + \bar{z}_2 z_2.$$

The rank of the given Hermitian form = the number of non-zero eigenvalues of its matrix $H = 2$.

The signature of the given Hermitian form = the number of positive eigenvalues of H - the number of negative eigenvalues of $H = 1 - 1 = 0$.

Comprehensive Exercise 4

1. Determine the definiteness of the following Hermitian forms $\mathbf{X}^{\theta} H \mathbf{X}$ in $\mathbb{C}^3$ where

$$(i) \quad H = \begin{bmatrix} 2 & 1-i & 2 \\ 1+i & 2 & 1+i \\ 2 & 1-i & 2 \end{bmatrix}$$

$$(ii) \quad H = \begin{bmatrix} -2 & 1-i & i \\ 1+i & -2 & 2+i \\ -i & 2-i & -6 \end{bmatrix}$$

$$(iii) \quad H = \begin{bmatrix} 2 & 1-i & i \\ 1+i & 4 & 2+i \\ -i & 2-i & 8 \end{bmatrix}.$$

2. Reduce the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$, where

$$H = \begin{bmatrix} 2 & i & 0 \\ -i & 1 & -i \\ 0 & i & 2 \end{bmatrix},$$

to the equivalent diagonal form by a unitary transformation and hence find the rank and signature of the given Hermitian form.

3. Show that a Hermitian matrix H is positive definite if and only if its eigenvalues are all positive and is non-negative definite if and only if its eigenvalues are all non-negative.
4. Show that a Hermitian matrix H is positive definite if and only if there exists a non-singular matrix Q such that $H = Q^{\theta} Q$.
5. If H is a positive definite or a positive semi-definite Hermitian matrix, show that there exists a Hermitian matrix B such that $B^2 = H$.

Answers 4

1. (i) Positive semi-definite. (ii) Negative definite.
(iii) Positive definite.
2. The unitary matrix P that will transform H to diagonal form is

$$P = \begin{bmatrix} -i/\sqrt{6} & 1/\sqrt{2} & i/\sqrt{3} \\ 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ -i/\sqrt{6} & -1/\sqrt{2} & i/\sqrt{3} \end{bmatrix}.$$

The unitary transformation $\mathbf{X} = P \mathbf{Y}$ will transform the given Hermitian form to the equivalent diagonal form $2 \bar{y}_2 y_2 + 3 \bar{y}_3 y_3$.

The rank of the given Hermitian form is 2 and its signature is 2.

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- A real quadratic form $\mathbf{X}^T A \mathbf{X}$ in three variables is equivalent to the diagonal form $3y_1^2 - 4y_2^2 + 5y_3^2$. Then the quadratic form $\mathbf{X}^T A \mathbf{X}$ is
 - Positive definite
 - Negative definite
 - Indefinite
 - Positive semi-definite.
- A real quadratic form $\mathbf{X}^T A \mathbf{X}$ in three variables is equivalent to the normal form $y_1^2 - y_2^2 - y_3^2$. If s is the signature of this quadratic form, then
 - $s = 1$
 - $s = -1$
 - $s = 2$
 - $s = 3$.
- A real quadratic form $\mathbf{X}^T A \mathbf{X}$ is positive semi-definite if and only if
 - the eigenvalues of the matrix A are all ≤ 0
 - the eigenvalues of the matrix A are all > 0
 - the eigenvalues of the matrix A are all negative
 - the eigenvalues of the matrix A are all ≥ 0 and at least one eigenvalue of A is zero.
- If n is the order, r is the rank and s is the signature of a real quadratic form in n variables, then the quadratic form is negative semi-definite if
 - $s = r = n$
 - $-s = r = n$
 - $s = r < n$
 - $-s = r < n$.
- A real quadratic form in three variables is equivalent to the diagonal form $6y_1^2 + 3y_2^2 + 0y_3^2$. Then the quadratic form is
 - Positive definite
 - Indefinite
 - Positive semi-definite
 - Negative.

Fill in the Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

- If the bilinear form

$$b(X, Y) = X^T AY = 2x_1y_1 + x_1y_2 - 2x_2y_1 + 3x_2y_2 - 3x_1y_3,$$

then the matrix $A = \dots\dots$

2. The bilinear form $b(X, Y) = X^T AY$ is said to be a symmetric bilinear form if the matrix A is a matrix.
3. An expression of the form $\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$, where a_{ij} 's are elements of a field F , is called a in the n variables $x_1, x_2, \dots, x_n$ over the field F .
4. There exists a one-to-one correspondence between the set of all quadratic forms in n variables over a field F and the set of all n -rowed matrices over F .
5. The matrix A corresponding to the quadratic form

$$d_1 x_1^2 + d_2 x_2^2 + d_3 x_3^2 + d_4 x_4^2 + d_5 x_5^2$$
 in five variables $x_1, x_2, \dots, x_5$ is $A = \dots\dots$.
6. The quadratic form corresponding to the symmetric matrix $\text{diag.} [\lambda_1, \lambda_2, \dots, \lambda_n]$ is
7. The quadratic form corresponding to the matrix

$$\begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & -2 \\ 5 & -2 & 4 \end{bmatrix}$$
 is
8. Two real quadratic forms in n variables are equivalent if and only if they have the same and the same index.
9. The number of positive terms in any two normal reductions of a real quadratic form is the
10. The rank of a real quadratic form $\mathbf{X}^T A \mathbf{X}$ is equal to the number of eigenvalues of the symmetric matrix A .
11. The real quadratic form $\mathbf{X}^T A \mathbf{X}$ is positive definite if and only if the eigenvalues of the matrix A are all
12. The real quadratic form $\mathbf{X}^T A \mathbf{X}$ is positive definite if $\mathbf{X}^T A \mathbf{X} \dots\dots 0$, for all non-zero vectors $\mathbf{X}$.
13. The eigenvalues of a Hermitian matrix are always
14. A Hermitian form $\mathbf{X}^H H \mathbf{X}$ always takes values for all complex n -vectors $\mathbf{X}$.
15. A Hermitian form $\mathbf{X}^H H \mathbf{X}$ is negative definite iff the eigenvalues of the matrix H are all

True or False

Write 'T' for true and 'F' for false statement.

1. The bilinear form

$3x_1y_1 + x_1y_2 + x_2y_1 - 2x_2y_2 - 4x_2y_3 - 4x_3y_2 + 3x_3y_3$
is symmetric.

2. The matrix corresponding to the quadratic form

$$d_1 x_1^2 + d_2 x_2^2 + d_3 x_3^2 + d_4 x_4^2$$

is a diagonal matrix.

3. Every real quadratic form over a field F in n variables $x_1, x_2, \dots, x_n$ can be expressed in the form $\mathbf{X}'B\mathbf{X}$ where $\mathbf{X}' = [x_1, x_2, \dots, x_n]'$ is a column vector and B is a skew-symmetric matrix of order n over the field F .
4. A real quadratic form in three variables is equivalent to the diagonal form

$$6y_1^2 + \frac{7}{3}y_2^2 - \frac{16}{7}y_3^2.$$

Then the rank of the quadratic form is 2.

5. A real quadratic form $\mathbf{X}^T A \mathbf{X}$ is positive definite iff the leading principal minors of the matrix A are all positive.
6. A real quadratic form $\mathbf{X}^T A \mathbf{X}$ is negative definite iff the leading principal minors of the matrix A are all negative.
7. A real quadratic form in three variables is equivalent to the diagonal form

$$4y_1^2 + 3y_2^2.$$

Then the quadratic form is positive definite.

8. A real quadratic form in four variables is equivalent to the normal form

$$y_1^2 + y_2^2 - y_3^2 - y_4^2.$$

Then the signature of the quadratic form is 0.

9. A Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ in three variables is equivalent to the diagonal form

$$4\bar{y}_1 y_1 + 5\bar{y}_2 y_2 + 7\bar{y}_3 y_3.$$

Then the Hermitian form is positive definite.

10. A Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ in three variables is equivalent to the diagonal form

$$\frac{3}{2}\bar{y}_1 y_1 + 5\bar{y}_2 y_2.$$

Then the Hermitian form is positive definite.

Answers

Multiple Choice Questions

- | | | |
|--------|--------|--------|
| 1. (c) | 2. (b) | 3. (d) |
| 4. (d) | 5. (c) | |

Fill in the Blank(s)

1. $\begin{bmatrix} 2 & 1 & -3 \\ -2 & 3 & 0 \end{bmatrix}$
2. symmetric
3. quadratic form
4. symmetric
5. diag. $[d_1, d_2, d_3, d_4, d_5]$
6. $\lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2$
7. $2x_1^2 + 3x_2^2 + 4x_3^2 + 2x_1x_2 + 10x_1x_3 - 4x_2x_3$
8. rank
9. same
10. non-zero
11. positive
12. $>$
13. real
14. real
15. negative

True or False

- | | | |
|---------|--------|--------|
| 1. T | 2. T | 3. F |
| 4. F | 5. T | 6. F |
| 7. F | 8. T | 9. T |
| 10. F | | |

